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Topic 1 - Single Topic

Question #1

Topic 1

A large mobile network operating company is building a machine learning model to predict customers who are likely to unsubscribe from the service. The company plans to offer an incentive for these customers as the cost of churn is far greater than the cost of the incentive. The model produces the following confusion matrix after evaluating on a test dataset of 100 customers:

n= 100	PREDICTED CHURN	
	Yes	No
ACTUAL Churn Yes	10	4
Actual No	10	76

Based on the model evaluation results, why is this a viable model for production?

- A. The model is 86% accurate and the cost incurred by the company as a result of false negatives is less than the false positives.
- B. The precision of the model is 86%, which is less than the accuracy of the model.
- C. The model is 86% accurate and the cost incurred by the company as a result of false positives is less than the false negatives.
- D. The precision of the model is 86%, which is greater than the accuracy of the model.

Correct Answer: A

Community vote distribution

C (52%)

A (48%)

  **joep21** Highly Voted 1 year, 7 months ago

Should it be C? Cost incurred by the company as a result of false positives (predicted churn, actual not churn) is less than the false negatives (predicted not churn, actual churn). Incentive cost < churn cost.

upvoted 34 times

  **felbuch** 1 year, 7 months ago

The question says "the cost of churn is far greater than the cost of the incentive", so we want to identify all the true churns, in order to do something about it. We don't want there to be any true churns we didn't see. This means we want false negatives as low as possible. So we want false negatives < false positives. So A.

upvoted 29 times

  **ExamTaker177** 1 year, 6 months ago

Exactly, Count of False Positive should be greater than count of False Negative.
In other words, cost / penalty for company is more when False Negative are predicted.
So, Answer C - Cost incurred by the company as a result of False Positives is less than the False Negatives.

upvoted 21 times

  **cpal012** 1 month, 1 week ago

No, the text clearly says the cost of churn is "far" greater - not equal to. One incident of churn could be higher than 10 incidents of incentive. Ans = A

upvoted 2 times

  **ShwetaGKT** 1 year, 5 months ago

prefect explanation
upvoted 1 times

  **tgaos** Highly Voted 11 months, 2 weeks ago

The Answer is A.
Reasons:
1. accurate is 86%
2. FN=4, FP= 10. The question is asking why this is a feasible model which means why this is working. So it is not asking the explanation of the unit cost of churn(FN) is greater than cost of incentive(FP). It is asking from the matrixs result, the number it self, FN(4) is less than FP(10). The model successfully keep a smaller number of FN regarding of FP.
upvoted 12 times

  **Lehidalg** Most Recent 11 hours, 8 minutes ago

The answer is C. Take a close look to phrases: "cost of churn is far greater than the cost of the incentive" ----> "cost incurred by the company as a result of false positives (False Predicted Churn, so receive incentive) is less than the false negatives (False Predicted No Churn, so Actual Churn, so more expensive than incentive)"
upvoted 1 times

  **brunokiyoshi** 1 month ago

Selected Answer: C

There are more FP's than FN's, however the costs of FN's are far larger than that of FP's. So:
 $\text{numberof(FP)} > \text{numberof(FN)}$, $\text{costperunit(FP)} \ll \text{costperunit(FN)}$. This itself could suggest that $\text{totalcosts(FP)} < \text{totalcosts(FN)}$, but would be somewhat subjective, since it is not stated how far the unitary costs are.

What is suggested, however, is that the model is indeed viable (question asks WHY the model is viable, and not WHETHER it's viable).

If the model didn't exist, there would be no way that there are FP's or FN's, but churns would still exist, which have the same cost as FN's.

So it means the total costs with FP's must be less than the total costs with FN's (churns).

upvoted 2 times

  **brunokiyoshi** 1 month, 1 week ago

The cost per unit of false negative is far larger than the cost of false positive. Even though there 6 false negatives more than false positives (slightly larger), the cost of a false positive being FAR larger than an undetected churn should make the costs with false positives less than the cost with false negatives.

upvoted 1 times

  **damaldon** 2 months ago

I passed the exam yesterday with 868/1000, and I have to tell you that ALL the question are here, you only need to follow the most voted answers and you will pass.

upvoted 3 times

  **Ajose0** 2 months, 1 week ago

Selected Answer: A

Based on the confusion matrix, the model has correctly classified 82 out of 100 customers who are likely to unsubscribe (true positives) and 13 out of 100 customers who are not likely to unsubscribe (true negatives). It has also misclassified 7 customers who are likely to unsubscribe as not likely to unsubscribe (false negatives) and 6 customers who are not likely to unsubscribe as likely to unsubscribe (false positives).

To determine why this is a viable model for production, we need to consider the specific needs and goals of the mobile network operating company. The company plans to offer an incentive to customers who are likely to unsubscribe, as the cost of churn is greater than the cost of the incentive. Therefore, the company is primarily concerned with identifying customers who are likely to unsubscribe (true positives) and minimizing the number of false negatives.

upvoted 2 times

  **Ajose0** 2 months, 1 week ago

Option A states that the model is 86% accurate and the cost incurred by the company as a result of false negatives is less than the false positives, which aligns with the company's goal of minimizing false negatives.

upvoted 1 times

  **uninit** 2 months, 4 weeks ago

Selected Answer: A

TP - 10: we predicted a customer will leave, and they actually leave.

TN - 76: we predicted a customer would not leave, and they actually did not leave.

FP - 10: We predicted a customer would leave, but they actually did not. (a not-that-bad thing)

FN - 4: We predicted a customer would not leave, but they actually left (a bad thing)

Accuracy = $(TP + TN) / (TP + TN + FP + FN) = 86\%$

Since, the cost of a customer leaving is greater than the cost of retention, any model that is viable for production (this is the key sentence, the question already states that the model is viable for production) would minimize the very bad thing, i.e. False Negatives, and would therefore have FN less than FP. Hence answer is A.

Also this - <https://medium.com/@datasci/to-churn-or-not-d9c4145bf29b>

upvoted 2 times

  **joe3232** 3 months ago

"cost of churn $[P = TP + FN]$ is far greater than the cost of the incentive $[PP = TP + FP]$."

$TP + FN > TP + FP$

$\Rightarrow FN > FP$

$\Rightarrow FP < FN$

C - "cost incurred by the company as a result of false positives $[FP]$ is less than the false negatives. $[FN]$ " $FP < FN$

upvoted 1 times

  **koakande** 3 months, 1 week ago

It is known that the cost of churn $>$ cost of incentive, as stated in the question. As a result, we would like to identify as many churns as possible and care less about misidentifying customers as churned (that is FP can be high since incentive cost is low). We care more about not seen churned customers (means FN should be minimized because it costs us a lot). Hence the desired goal is $FN < FP$ as stated in A.

Now from the model confusion matrix, we can see that model identify most churned 10 out of 14

upvoted 1 times

  **Tomatoteacher** 3 months, 2 weeks ago

Selected Answer: C

Definitely C. If you look at the same question in <https://aws.amazon.com/blogs/machine-learning/predicting-customer-churn-with-amazon-machine-learning/>. Same question, but the confusion matrix is flipped in this case(TP top left, Tn bottom right) . When you miss an actual churn (FN) this would cost the company more. Therefore the answer is C 100%. I will die on this hill. I spent 20 minutes researching this to be certain. Most people who put A are incorrectly saying FPs are actual churns that are stated as no churn.. that is what a FN is. You can trust me on this.

upvoted 3 times

-  **albu44** 3 months, 3 weeks ago

Selected Answer: C

C - "the cost of churn is far greater than the cost of the incentive"

upvoted 2 times
-  **dreswardev** 4 months ago

A, 4 is less than 76

upvoted 1 times
-  **vbal** 4 months ago

A make sense to me.

upvoted 1 times
-  **vbal** 4 months ago

Changing mine to C after realize that cost to company on False Positive = 10 = Predicted Churn Can't be > than cost to company on False Negative = 4 = Predicted NOT Churn.

upvoted 1 times
-  **rrshah83** 4 months, 1 week ago

Selected Answer: C

Business cost of incentive (FP) is less than cost of churn (FN)

upvoted 3 times
-  **kukreti18** 4 months, 1 week ago

C is correct

upvoted 1 times
-  **BethChen** 4 months, 3 weeks ago

Selected Answer: A

I agree with A

upvoted 1 times

A Machine Learning Specialist is designing a system for improving sales for a company. The objective is to use the large amount of information the company has on users' behavior and product preferences to predict which products users would like based on the users' similarity to other users.

What should the Specialist do to meet this objective?

- A. Build a content-based filtering recommendation engine with Apache Spark ML on Amazon EMR
- B. Build a collaborative filtering recommendation engine with Apache Spark ML on Amazon EMR.
- C. Build a model-based filtering recommendation engine with Apache Spark ML on Amazon EMR
- D. Build a combinative filtering recommendation engine with Apache Spark ML on Amazon EMR

Correct Answer: B

Many developers want to implement the famous Amazon model that was used to power the “People who bought this also bought these items” feature on Amazon.com. This model is based on a method called Collaborative Filtering. It takes items such as movies, books, and products that were rated highly by a set of users and recommending them to other users who also gave them high ratings. This method works well in domains where explicit ratings or implicit user actions can be gathered and analyzed.

Reference:
<https://aws.amazon.com/blogs/big-data/building-a-recommendation-engine-with-spark-ml-on-amazon-emr-using-zeppelin/>

Community vote distribution

B (100%)

- mlyu**

Highly Voted

1 year, 7 months ago

B

see https://en.wikipedia.org/wiki/Collaborative_filtering#Model-based

upvoted 18 times
- kalyanvarma**

Highly Voted

1 year, 5 months ago

Content-based filtering relies on similarities between features of items, whereas colloborative-based filtering relies on preferences from other users and how they respond to similar items.

upvoted 12 times
- brunokiyoshi**

Most Recent

1 month, 1 week ago

Selected Answer: B

Content-based recommendations rely on product similarity. If a user likes a product, products that are similar to that one will be recommended. Collaborative recommendations are based on user similarity. If you and other users have given similar reviews to a range of products, the model assumes it is likely that other products those other people have liked but that you haven't purchased should be a good recommendation for you.

upvoted 1 times
- Ajose0**

2 months, 1 week ago

Selected Answer: B

B. Build a collaborative filtering recommendation engine with Apache Spark ML on Amazon EMR.

Collaborative filtering is a technique used to recommend products to users based on their similarity to other users. It is a widely used method for building recommendation engines. Apache Spark ML is a distributed machine learning library that provides scalable implementations of collaborative filtering algorithms. Amazon EMR is a managed cluster platform that provides easy access to Apache Spark and other distributed computing frameworks.

upvoted 1 times
- solution123**

2 months, 3 weeks ago

Selected Answer: B

Build a collaborative filtering recommendation engine with Apache Spark ML on Amazon EMR. (TRUE)

Collaborative filtering is a commonly used method for recommendation systems that aims to predict the preferences of a user based on the behavior of similar users. In the case described, the objective is to use users' behavior and product preferences to predict which products they want, making collaborative filtering a good fit.

Apache Spark ML is a machine learning library that provides scalable, efficient algorithms for building recommendation systems, while Amazon EMR provides a cloud-based platform for running Spark applications.
You can find more detail in <https://www.udemy.com/course/aws-certified-machine-learning-specialty-2023>

upvoted 2 times
- dreswardev**

4 months ago

feature engineering is required, use model based

upvoted 1 times

  **ryuhei** 7 months, 2 weeks ago

Selected Answer: B

Answer is "B"

upvoted 1 times

  **roytruong** 1 year, 6 months ago

go for B

upvoted 2 times

  **cybe001** 1 year, 6 months ago

B is correct

<https://aws.amazon.com/blogs/big-data/building-a-recommendation-engine-with-spark-ml-on-amazon-emr-using-zeppelin/>

upvoted 6 times

A Mobile Network Operator is building an analytics platform to analyze and optimize a company's operations using Amazon Athena and Amazon S3.

The source systems send data in .CSV format in real time. The Data Engineering team wants to transform the data to the Apache Parquet format before storing it on Amazon S3.

Which solution takes the LEAST effort to implement?

- A. Ingest .CSV data using Apache Kafka Streams on Amazon EC2 instances and use Kafka Connect S3 to serialize data as Parquet
- B. Ingest .CSV data from Amazon Kinesis Data Streams and use Amazon Glue to convert data into Parquet.
- C. Ingest .CSV data using Apache Spark Structured Streaming in an Amazon EMR cluster and use Apache Spark to convert data into Parquet.
- D. Ingest .CSV data from Amazon Kinesis Data Streams and use Amazon Kinesis Data Firehose to convert data into Parquet.

Correct Answer: B

Community vote distribution

B (67%)

D (33%)

-   **DonaldCMLIN**

Highly Voted 

 1 year, 7 months ago

Answer is B
<https://github.com/ecloudvalley/Building-a-Data-Lake-with-AWS-Glue-and-Amazon-S3>
upvoted 24 times
-   **OmarSaadEldien** 1 year, 6 months ago

the Approve Of B
<https://aws.amazon.com/blogs/aws/new-serverless-streaming-etl-with-aws-glue/>
upvoted 7 times
-   **Antriksh** 1 year, 6 months ago

you cannot use AWS glue for streaming data. Clearly B is incorrect.
upvoted 2 times
-   **scuzzy2010** 1 year, 6 months ago

Even if the exam's answer is based on solution before AWS implemented the capability of AWS glue to process streaming data, this answer is still correct as Kinesis would output the data to S3 and Glue will pick it up from there and covert to parquet. Question does not say data must be converted to parquet in real time, it only says the csv data is received as a stream in real time.
upvoted 1 times
-   **GeeBeeEI** 1 year, 6 months ago

Actually question says "The source systems send data in CSV format in real time The Data Engineering team wants to transform the data to the Apache Parquet format before storing it on Amazon S3" same as saying data must be converted real time
upvoted 3 times
-   **zzeng** 1 year, 6 months ago

AWS Glue can do it now (2020 May)
<https://aws.amazon.com/jp/blogs/news/new-serverless-streaming-etl-with-aws-glue/>
upvoted 5 times
-   **hamimelon** 4 months, 1 week ago

This link is in Japanese
upvoted 2 times
-   **vetal**

Highly Voted 

 1 year, 7 months ago

D is wrong as kinesis firehose can convert from JSON to parquet but here we have CSV.
B is correct and here is another proof link: <https://medium.com/searce/convert-csv-json-files-to-apache-parquet-using-aws-glue-a760d177b45f>
upvoted 17 times
-   **zzeng** 1 year, 6 months ago

<https://docs.aws.amazon.com/firehose/latest/dev/record-format-conversion.html>
You are right.

<https://docs.aws.amazon.com/firehose/latest/dev/record-format-conversion.html>
If you want to convert an input format other than JSON, such as comma-separated values (CSV) or structured text, you can use AWS Lambda to transform it to JSON first
upvoted 8 times
-   **samy666** 12 months ago

But there is no Lambda in D

upvoted 2 times

  **AdolinKholin** 7 months ago

But there's a D in Lambda

upvoted 1 times

  **oso0348** Most Recent 4 weeks ago

Selected Answer: D

While Amazon Glue can be used to transform data from CSV to Parquet format, it requires setting up and configuring additional infrastructure. Specifically, you would need to set up an Amazon Glue Crawler to discover the schema of the incoming CSV data, create a Glue Job to transform the data into Parquet format, and configure a Glue Connection to write the transformed data to S3. These additional steps make this option more complex and potentially more costly to implement than Option D, which utilizes the fully managed Kinesis Data Firehose service to automatically convert data to Parquet format without requiring additional infrastructure setup.

upvoted 1 times

  **ZSun** 1 week, 6 days ago

Glue can convert CSV->Parquet; but Kinesis Firehouse need CSV->JSON (using lambda function), then JSON->Parquet, much complicated.

upvoted 1 times

  **brunokiyoshi** 1 month, 1 week ago

Amazon Kinesis Data Streams is commonly used to ingest large quantities .csv data in real time.

upvoted 1 times

  **expertguru** 3 months, 2 weeks ago

Amazon Kinesis Data Firehose can convert the format of your input data from JSON to Apache Parquet or Apache ORC before storing the data in Amazon S3. Parquet and ORC are columnar data formats that save space and enable faster queries compared to row-oriented formats like JSON. If you want to convert an input format other than JSON, such as comma-separated values (CSV) or structured text, you can use AWS Lambda to transform it to JSON first. For more information, see Amazon Kinesis Data Firehose Data Transformation. So there is additional work involved with firehose see below url

<https://docs.aws.amazon.com/firehose/latest/dev/record-format-conversion.html>

so answer should be B

upvoted 2 times

  **dreswardev** 4 months ago

B, for realtime and covert to catalog

upvoted 1 times

  **rrshah83** 4 months, 1 week ago

Selected Answer: B

firehose requires lambda for csv to parquet. Glue studio drag and drop, select source and dest s3 file formats; and glue studio auto-creates scripts... so least amount of effort with glue...

upvoted 2 times

  **chizeni** 8 months, 3 weeks ago

answer is D

can not use firehose to convert csv to parquet format.

<https://docs.aws.amazon.com/firehose/latest/dev/record-format-conversion.html>

ingest real time with kinesis and convert with glue

upvoted 1 times

  **matteocal** 9 months, 1 week ago

Selected Answer: B

The correct answer is B since Firehose cannot convert csv to parquet (only json):

<https://docs.aws.amazon.com/firehose/latest/dev/record-format-conversion.html>

upvoted 2 times

  **rjekstein** 9 months, 3 weeks ago

B is correct.

Kinesis firehose converts only Json to Parquet or ORC directly.

upvoted 1 times

  **ovokpus** 10 months, 1 week ago

Selected Answer: D

With MINIMUM amount of work, Kinesis firehose can convert CSV on the fly before storing it in S3. Glue cannot do this. Only Kinesis Firehose can.

upvoted 1 times

  **shammous** 4 months, 3 weeks ago

Kinesis firehose would call a lambda function for the conversion which requires more effort than using the predefined functions of Glue.

upvoted 1 times

  **passionatecricketer** 1 year, 5 months ago

In AWS, one of the generally accepted theories is that writing a Lambda function is considered to be least effort. In this problem, although B is definitely possible, it requires to configure a Glue job and do the data transformation mapping. I tend to think that AWS considers writing a one-liner Lambda is possibly lesser effort that configuring the Glue job, Hence I would go with D.

upvoted 2 times

  **samy666** 12 months ago

But the Lambda is not mentioned in the answer D

upvoted 2 times

  **StelSen** 1 year, 5 months ago

The Answer is B. LINK: <https://aws.amazon.com/glue/faqs/>
(Search 'real' in this FAQ and you will get an answer. Answer D required one more component called Lambda. So straightaway rejected)

upvoted 1 times

  **Vita_Rasta84444** 1 year, 6 months ago

D. Glue cannot work with live streams, while only Firehose could ingest the data to S3 and make transformation from csv to parquet.

upvoted 1 times

  **ahquiceno** 1 year, 6 months ago

Answer is D go to: <https://docs.aws.amazon.com/firehose/latest/dev/record-format-conversion.html>

upvoted 1 times

  **AmakamaxZanny** 1 year, 2 months ago

the format is .CSV firehose can only change JSON to parquet. or ORC.

upvoted 1 times

  **harmanbirstudy** 1 year, 6 months ago

A-- cannot be answer as Apache kafka S3 cannot write in parquet
B-- seems like a good answer but if the question is old , then at that time Glue did not had compatibility with Kinesis data streams
C--- cannot be the answer as Spark structured streaming need to read data from somewhere like Kafka topics , we cannot publish data directly to it and like other three options mention Apache kafka/Kinesis in it so this means it should have been in this one also if this was the correct answer.
D-- Seems like a good answer , but it needs lambda to convert CSV -- JSON and then Firehose's inbuilt ability to covert JSON to Parquet before storing to S3

Now the final analysis -- :)

-- The comments on this question started some time in Decmeber 2019/January2020 and the Glue capability to consume real time Kiensis data streams was annouounce much later on April 27 , 2020 which means this answer must be incorrect before this date

--Hence the likely choice is D , even though firehose Lambda thing is not mentioned in it specifically

Any counter comment ???

upvoted 6 times

  **cnethers** 1 year, 6 months ago

Very good points made, I would agree that the timing of the question needs to be taken into account. When the question was first posted Glue could not take real-time data and transform it, so that would make B a no starter and D the Answer however if you consider that there is now integration of Kinesis and Glue you can now do real-time processing so B is now possible and less effort than D. On balance, the question may not even be asked nowadays but it does demonstrate how answeres can change over time. Do your research and make sure you understand what capabilities a service has.

upvoted 3 times

  **Huy** 1 year, 5 months ago

Even Kinesis can process streaming data, B still take more effort than D. Converting CVS to JSON is a piece of cake in python. I think D is correct. Let's say if you have to design this system, what will you do? I will of-course use Lambda and Firehose first.

upvoted 1 times

  **mrsimoes** 1 year, 6 months ago

I think that the answer is B.

Please check:

<https://aws.amazon.com/about-aws/whats-new/2020/04/aws-glue-now-supports-serverless-streaming-etl/#:~:text=AWS%20Glue%20now%20supports%20streaming%20ETL.&text=Streaming%20ETL%20jobs%20in%20AWS,warehouses%2C%20or%20other%20data%20stores.>

upvoted 2 times

Question #4

Topic 1

A city wants to monitor its air quality to address the consequences of air pollution. A Machine Learning Specialist needs to forecast the air quality in parts per million of contaminates for the next 2 days in the city. As this is a prototype, only daily data from the last year is available. Which model is MOST likely to provide the best results in Amazon SageMaker?

- A. Use the Amazon SageMaker k-Nearest-Neighbors (kNN) algorithm on the single time series consisting of the full year of data with a predictor_type of regressor.
- B. Use Amazon SageMaker Random Cut Forest (RCF) on the single time series consisting of the full year of data.
- C. Use the Amazon SageMaker Linear Learner algorithm on the single time series consisting of the full year of data with a predictor_type of regressor.
- D. Use the Amazon SageMaker Linear Learner algorithm on the single time series consisting of the full year of data with a predictor_type of classifier.

Correct Answer: C

Reference:
<https://aws.amazon.com/blogs/machine-learning/build-a-model-to-predict-the-impact-of-weather-on-urban-air-quality-using-amazon-sagemaker/?ref=Welcome.AI>

Community vote distribution

C (80%)

A (20%)

- ozan11

Highly Voted

1 year, 7 months ago

answer should be C

upvoted 16 times
- roytruong

Highly Voted

1 year, 6 months ago

go for C

upvoted 6 times
- brunokiyoshi

Most Recent

1 month, 1 week ago

Selected Answer: C

Random cut forests in timeseries are used for anomaly detection, and not for forecasting. KNN's are classification algorithms. You would use the Linear Learner as a regressor, since forecasting falls into the domain of regression.

upvoted 2 times
- brunokiyoshi

1 month, 1 week ago

I mean, you could use KNN's for regression, but for forecasting I don't think so

upvoted 1 times
- Valcilio

1 month, 3 weeks ago

Selected Answer: C

KNN isn't for time series predicting, go for A!

upvoted 1 times
- Valcilio

1 month, 3 weeks ago

Im sorry, I wanted to say go for C!

upvoted 1 times
- rockyykrish

1 month, 3 weeks ago

Creating a machine learning model to predict air quality
To start small, we will follow the second approach, where we will build a model that will predict the NO2 concentration of any given day based on wind speed, wind direction, maximum temperature, pressure values of that day, and the NO2 concentration of the previous day. For this we will use the Linear Learner algorithm provided in Amazon SageMaker, enabling us to quickly build a model with minimal work.

Our model will consist of taking all of the variables in our dataset and using them as features of the Linear Learner algorithm available in Amazon SageMaker

upvoted 1 times
- AjoseO

2 months, 4 weeks ago

Selected Answer: A

Answer should be A.

k-Nearest-Neighbors (kNN) algorithm will provide the best results for this use case as it is a good fit for time series data, especially for predicting continuous values. The predictor_type of regressor is also appropriate for this task, as the goal is to forecast a continuous value (air quality in parts per million of contaminants). The other options are also viable, but may not provide as good of results as the kNN algorithm, especially with limited

data.

using the Amazon SageMaker Linear Learner algorithm with a predictor_type of regressor, may still provide reasonable results, but it assumes a linear relationship between the input features and the target variable (air quality), which may not always hold in practice, especially with complex time series data. In such cases, non-linear models like kNN may perform better. Furthermore, the kNN algorithm can handle irregular patterns in the data, which may be present in the air quality data, and provide more accurate predictions.

upvoted 1 times

  **hichemck** 5 months ago

Seen on Dec. 1 exam

upvoted 1 times

  **ryuhei** 7 months, 2 weeks ago

Selected Answer: C

Answer is "C" !!!

upvoted 1 times

  **yemauricio** 7 months, 3 weeks ago

answer C

upvoted 1 times

  **Huy** 1 year, 5 months ago

I go with A. Linear regression is not suitable for time series data. there is a library that implements knn for time-series <https://cran.r-project.org/web/packages/tsfknn/vignettes/tsfknn.html>

upvoted 1 times

  **Huy** 1 year, 5 months ago

I mean the air quality have many feature correlations that are not linear.

upvoted 1 times

  **syu31svc** 1 year, 6 months ago

If it's about forecasting then answer is C 100%; regression

upvoted 4 times

  **VB** 1 year, 6 months ago

C should be the answer.. D is not because it's not a classifier problem (..forecast the air quality in parts per million..).. it should be regression.

upvoted 3 times

  **rajs** 1 year, 7 months ago

It's a Regression problem so A)KNN B)RCF D) can be eliminated leaving us with the answer C

upvoted 4 times

  **Nahe** 1 year, 6 months ago

K-NN can also be used for regression. But it is not preferred here as we have very limited data of 2-3 days

upvoted 4 times

A Data Engineer needs to build a model using a dataset containing customer credit card information
How can the Data Engineer ensure the data remains encrypted and the credit card information is secure?

- A. Use a custom encryption algorithm to encrypt the data and store the data on an Amazon SageMaker instance in a VPC. Use the SageMaker DeepAR algorithm to randomize the credit card numbers.
- B. Use an IAM policy to encrypt the data on the Amazon S3 bucket and Amazon Kinesis to automatically discard credit card numbers and insert fake credit card numbers.
- C. Use an Amazon SageMaker launch configuration to encrypt the data once it is copied to the SageMaker instance in a VPC. Use the SageMaker principal component analysis (PCA) algorithm to reduce the length of the credit card numbers.
- D. Use AWS KMS to encrypt the data on Amazon S3 and Amazon SageMaker, and redact the credit card numbers from the customer data with AWS Glue.

Correct Answer: D

Community vote distribution

D (100%)

- vetal

Highly Voted

1 year, 7 months ago

Why not D? When the data encrypted on S3 and SageMaker uses the same AWS KMS key it can use encrypted data there.

upvoted 36 times
- Antriksh

1 year, 6 months ago

Answer would be D

upvoted 9 times
- WWODIN

1 year, 7 months ago

should be D

upvoted 12 times
- zzeng

1 year, 6 months ago

Should be D.

Use Glue to do ETL to Hash the card number

upvoted 8 times
- cybe001

Highly Voted

1 year, 7 months ago

D is correct

upvoted 8 times
- Valcilio

Most Recent

1 month, 3 weeks ago

Selected Answer: D

It's D, KMS key can be used for encrypting the data at rest!

upvoted 2 times
- ystotest

5 months ago

Selected Answer: D

agreed with D

upvoted 2 times
- jerto97

1 year, 5 months ago

IMHO, the problem with the question is that it is not clear whether the credit card number is used in the model. In that case discarding is never a good option. Hashing should be a safe option to keep it in the learning path

upvoted 1 times
- cloud_trail

1 year, 6 months ago

It's gotta be D but C is a clever fake answer. Use PCA to reduce the length of the credit card number? That's a clever joke, as if reducing the length of a character string is the same as reducing dimensionality in a feature set.

upvoted 3 times
- cnethers

1 year, 6 months ago

Can Glue do redaction?

upvoted 1 times
- cloud_trail

1 year, 5 months ago

Just have the Glue job remove the credit card column.

upvoted 1 times

  **syu31svc** 1 year, 6 months ago

Encryption on AWS can be done using KMS so D is the answer
upvoted 1 times

  **roytruong** 1 year, 6 months ago

D is correct
upvoted 1 times

  **PRC** 1 year, 6 months ago

D..KMS fully managed and other options are too whacky..
upvoted 4 times

  **AKT** 1 year, 6 months ago

D is correct
upvoted 1 times

  **bhavesh0124** 1 year, 7 months ago

Ans D is correct
upvoted 2 times

  **satya_alapati** 1 year, 7 months ago

is this really a viable reference? so misleading.
upvoted 2 times

  **JayK** 1 year, 7 months ago

I think it is D as using PCA to reduce the length of credit card numbers does not seem viable
upvoted 6 times

A Machine Learning Specialist is using an Amazon SageMaker notebook instance in a private subnet of a corporate VPC. The ML Specialist has important data stored on the Amazon SageMaker notebook instance's Amazon EBS volume, and needs to take a snapshot of that EBS volume. However, the ML Specialist cannot find the Amazon SageMaker notebook instance's EBS volume or Amazon EC2 instance within the VPC. Why is the ML Specialist not seeing the instance visible in the VPC?

- A. Amazon SageMaker notebook instances are based on the EC2 instances within the customer account, but they run outside of VPCs.
- B. Amazon SageMaker notebook instances are based on the Amazon ECS service within customer accounts.
- C. Amazon SageMaker notebook instances are based on EC2 instances running within AWS service accounts.
- D. Amazon SageMaker notebook instances are based on AWS ECS instances running within AWS service accounts.

Correct Answer: C

Reference:

<https://docs.aws.amazon.com/sagemaker/latest/dg/gs-setup-working-env.html>

Community vote distribution

C (67%)

A (33%)

  **mlyu**

Highly Voted 

 1 year, 7 months ago

I think the answer should be C
upvoted 22 times

  **dhs227**

Highly Voted 

 1 year, 7 months ago

The correct answer HAS TO be A

The instances are running in customer accounts but it's in an AWS managed VPC while exposing ENI to customer VPC if it was chosen. See explanation at <https://aws.amazon.com/blogs/machine-learning/understanding-amazon-sagemaker-notebook-instance-networking-configurations-and-advanced-routing-options/>
upvoted 17 times

  **mawsman** 1 year, 6 months ago

Actually your link says: The notebook instance is running in an Amazon SageMaker managed VPC as shown in the above diagram. That means the correct answer is C. An Amazon SageMaker managed VPC can only be created in an Amazon managed Account.
upvoted 16 times

  **scuzzy2010** 1 year, 6 months ago

Can't be A because A says "but they run outside of VPCs", which is not correct. They are attached to VPC, but it can either be AWS Service VPC or Customer VPC, or Both, as per the explanation url you provided.
upvoted 7 times

  **cloud_trail** 1 year, 5 months ago

This is exactly right. According to that document, if the notebook instance is not in a customer VPC, then it has to be in the Sagemaker managed VPC. See Option 1 in that document.
upvoted 1 times

  **Maaayaaa**

Most Recent 

 3 weeks, 3 days ago

Selected Answer: C

C, check the digram in <https://docs.aws.amazon.com/sagemaker/latest/dg/studio-notebooks-and-internet-access.html>
upvoted 2 times

  **oso0348** 4 weeks ago

Selected Answer: A

When a SageMaker notebook instance is launched in a VPC, it creates an Elastic Network Interface (ENI) in the subnet specified, but the underlying EC2 instance is not visible in the VPC. This is because the EC2 instance is managed by AWS, and it is outside of the VPC. The ENI acts as a bridge between the VPC and the notebook instance, allowing network connectivity between the notebook instance and other resources in the VPC. Therefore, the EBS volume of the notebook instance is also not visible in the VPC, and you cannot take a snapshot of the volume using VPC-based tools. Instead, you can create a snapshot of the EBS volume directly from the SageMaker console, AWS CLI, or SDKs.
upvoted 1 times

  **Valcilio** 1 month, 3 weeks ago

Selected Answer: C

Notebooks run inside a VPC not outside!
upvoted 1 times

  **krzyhoo** 2 months, 1 week ago

Selected Answer: C

Definitely C
upvoted 2 times

  **rrshah83** 4 months ago

Selected Answer: C

Sagemaker notebook instances run in AWS sagemaker service accounts:
<https://docs.aws.amazon.com/sagemaker/latest/dg/studio-notebooks-and-internet-access.html>

Additionally, A is wrong because sagemaker instances run in a VPC (not outside)... It is just in aws service account
upvoted 1 times

  **ovokpus** 10 months, 1 week ago

Selected Answer: A

AWS Managed services do not happen outside the customer account. So the Sagemaker Managed Instance is primarily attached to a Sagemaker managed VPC inside a customer account. There has never been a hint of cross-account operations taking place in basic sagemaker operations. And this will not be the only AWS managed service situated inside a customer AWS account.

A is the logical answer here. He cannot see the instance, because it will not show up in the EC2 console, especially if it is not attached to any customer VPC
upvoted 2 times

  **technoguy** 1 year, 5 months ago

C is correct
upvoted 4 times

  **AShahine21** 1 year, 5 months ago

I will go with C, as "but they run outside of VPCs" is wrong
upvoted 2 times

  **cloud_trail** 1 year, 6 months ago

Answer C. I've gone back and forth on this but reading the info at the link below, the customer notebook instance has to run either in a customer managed VPC or the Sagemaker service managed VPC. Either way, it's in a VPC. <https://aws.amazon.com/blogs/machine-learning/understanding-amazon-sagemaker-notebook-instance-networking-configurations-and-advanced-routing-options/>
upvoted 5 times

  **Joe_Zhang** 1 year, 6 months ago

the answer should be C: see this link: <https://docs.aws.amazon.com/sagemaker/latest/dg/studio-notebooks-and-internet-access.html>
upvoted 1 times

  **crispogioele** 1 year, 6 months ago

I think the answer is A. They are EC2 instance and they are attached to the customer VPC through an ENI (as shown in this link <https://aws.amazon.com/blogs/machine-learning/understanding-amazon-sagemaker-notebook-instance-networking-configurations-and-advanced-routing-options/>)
upvoted 2 times

  **cnethers** 1 year, 6 months ago

I would agree that A is the answer. That is the same ref article I have looked at
upvoted 1 times

  **harmanbirstudy** 1 year, 6 months ago

C is correct as it is Fully managed service so you cannot have OS level access to its resources.
A-- cannot be correct because it says Sagemakers EC2 instances run with VPC which is not correct , as with VPC means governed by its network policies incase of Sagemaker
upvoted 2 times

  **Achievement** 1 year, 6 months ago

C should be correct since the EC2 instance is not under customer manage
upvoted 2 times

  **AdityaB** 1 year, 6 months ago

A is the right answer .
upvoted 5 times

  **roytruong** 1 year, 6 months ago

go for C
upvoted 4 times

Question #7

Topic 1

A Machine Learning Specialist is building a model that will perform time series forecasting using Amazon SageMaker. The Specialist has finished training the model and is now planning to perform load testing on the endpoint so they can configure Auto Scaling for the model variant. Which approach will allow the Specialist to review the latency, memory utilization, and CPU utilization during the load test?

- A. Review SageMaker logs that have been written to Amazon S3 by leveraging Amazon Athena and Amazon QuickSight to visualize logs as they are being produced.
- B. Generate an Amazon CloudWatch dashboard to create a single view for the latency, memory utilization, and CPU utilization metrics that are outputted by Amazon SageMaker.
- C. Build custom Amazon CloudWatch Logs and then leverage Amazon ES and Kibana to query and visualize the log data as it is generated by Amazon SageMaker.
- D. Send Amazon CloudWatch Logs that were generated by Amazon SageMaker to Amazon ES and use Kibana to query and visualize the log data.

Correct Answer: B

Reference:

<https://docs.aws.amazon.com/sagemaker/latest/dg/monitoring-cloudwatch.html>

Community vote distribution

B (100%)

- mlyu

Highly Voted

1 year, 7 months ago

Agreed. Ans is B

upvoted 17 times
- Valcilio

Most Recent

1 month, 3 weeks ago

Selected Answer: B

It's B, even the resources that aren't visible in a first try are visible if you use cloudwatch agent.

upvoted 2 times
- DS2021

4 months ago

Selected Answer: B

Should be B

upvoted 2 times
- ystotest

5 months ago

Selected Answer: B

agreed with B

upvoted 2 times
- apprehensive_scar

1 year, 2 months ago

B is the ans

upvoted 1 times
- anttan

1 year, 5 months ago

Should be C right, as Cloudwatch does not have metrics for memory utilization.

upvoted 2 times
- anttan

1 year, 4 months ago

After further research, I think answer is B. While indeed true that Cloudwatch does not have metrics for memory utilization by default, you can achieve by installing CloudWatch agent on the EC2. The EC2 used by Sagemaker is pre-installed with Cloudwatch Agent.

upvoted 1 times
- passionatecricketer

1 year, 5 months ago

I do not think that CloudWatch, by default, logs memory utilization. It does log CPU utilization. If memory utilization is required, then a separate agent needs to be installed to watch for memory. Hence, in this case, we have to write an agent if the answer has to be B. Else, C looks to be a better solution.

upvoted 2 times
- Willnguyen22

1 year, 6 months ago

answer is B

upvoted 1 times
- syu31svc

1 year, 6 months ago

Answer is B 100%; very straightforward method
upvoted 1 times

  **scuzzy2010** 1 year, 6 months ago
B is correct. Don't need to use Kibana or QuickSight.
upvoted 1 times

  **roytruong** 1 year, 6 months ago
ans is B
upvoted 3 times

  **cybe001** 1 year, 7 months ago
B is correct
upvoted 3 times

A manufacturing company has structured and unstructured data stored in an Amazon S3 bucket. A Machine Learning Specialist wants to use SQL to run queries on this data.

Which solution requires the LEAST effort to be able to query this data?

- A. Use AWS Data Pipeline to transform the data and Amazon RDS to run queries.
- B. Use AWS Glue to catalogue the data and Amazon Athena to run queries.
- C. Use AWS Batch to run ETL on the data and Amazon Aurora to run the queries.
- D. Use AWS Lambda to transform the data and Amazon Kinesis Data Analytics to run queries.

Correct Answer: B

Community vote distribution

B (100%)

- cybe001

Highly Voted

1 year, 7 months ago

B is correct

upvoted 22 times
- dhs227

Highly Voted

1 year, 6 months ago

The correct answer HAS TO be B

Using Glue Use AWS Glue to catalogue the data and Amazon Athena to run queries against data on S3 are very typical use cases for those services.

D is not ideal, Lambda can surely do many things but it requires development/testing effort, and Amazon Kinesis Data Analytics is not ideal for ad-hoc queries.

upvoted 8 times
- Ajose0

Most Recent

2 months, 3 weeks ago

Selected Answer: B

AWS Glue is a fully managed ETL service that makes it easy to move data between data stores. It can automatically crawl, catalogue, and classify data stored in Amazon S3, and make it available for querying and analysis. With AWS Glue, you don't have to worry about the underlying infrastructure and can focus on your data.

Amazon Athena is an interactive query service that makes it easy to analyze data in Amazon S3 using standard SQL. It integrates with AWS Glue, so you can use the catalogued data directly in Athena without any additional data movement or transformation.

upvoted 2 times
- SK27

4 months, 3 weeks ago

Selected Answer: B

B is the easiest. We can use Glue crawler.

upvoted 2 times
- ryuhei

7 months, 2 weeks ago

Selected Answer: B

Answer B

upvoted 2 times
- vetaal

1 year, 3 months ago

Selected Answer: B

Querying data in S3 with SQL is almost always Athena.

upvoted 2 times
- gcpwhiz

1 year, 5 months ago

If AWS asks the question of querying unstructured data in an efficient manner, it is almost always Athena

upvoted 2 times
- cloud_trail

1 year, 5 months ago

B. I don't think that you even need Glue to transform anything. Just use Glue to define the schemas and then use Athena to query based on those schemas.

upvoted 2 times
- Willnguyen22

1 year, 6 months ago

answer is B

upvoted 1 times
- syu31svc

1 year, 6 months ago

SQL on S3 is Athena so answer is B for sure
upvoted 1 times

  **roytruong** 1 year, 6 months ago
B is right
upvoted 2 times

  **Jayraam** 1 year, 6 months ago
Answer is B.

Queries Against an Amazon S3 Data Lake
Data lakes are an increasingly popular way to store and analyze both structured and unstructured data. If you want to build your own custom Amazon S3 data lake, AWS Glue can make all your data immediately available for analytics without moving the data.

<https://aws.amazon.com/glue/>
upvoted 1 times

  **PRC** 1 year, 7 months ago
Correct Ans is D...Kinesis Data Analytics can use Lambda to transform and then run the SQL queries..
upvoted 1 times

  **Urban_Life** 1 year, 6 months ago
May I know why you are taking complex route?
upvoted 10 times

  **BigEv** 1 year, 7 months ago
Can Glue Crawler process unstructured data?
upvoted 1 times

  **mawsman** 1 year, 6 months ago
<https://aws.amazon.com/glue/> - See Use cases;
Queries against an Amazon S3 data lake
Data lakes are an increasingly popular way to store and analyze both structured and unstructured data. If you want to build your own custom Amazon S3 data lake, AWS Glue can make all your data immediately available for analytics without moving the data.
upvoted 3 times

  **ExamTaker123456789** 1 year, 6 months ago
You could use a glue job to transform your unstructured data into more appropriate formats. Also, depending on your data, you might be able to create a custom classifier in glue, which will be able to crawl your data - this works particularly well in semi-structured cases, say for log files.
upvoted 1 times

A Machine Learning Specialist is developing a custom video recommendation model for an application. The dataset used to train this model is very large with millions of data points and is hosted in an Amazon S3 bucket. The Specialist wants to avoid loading all of this data onto an Amazon SageMaker notebook instance because it would take hours to move and will exceed the attached 5 GB Amazon EBS volume on the notebook instance.

Which approach allows the Specialist to use all the data to train the model?

- A. Load a smaller subset of the data into the SageMaker notebook and train locally. Confirm that the training code is executing and the model parameters seem reasonable. Initiate a SageMaker training job using the full dataset from the S3 bucket using Pipe input mode.
- B. Launch an Amazon EC2 instance with an AWS Deep Learning AMI and attach the S3 bucket to the instance. Train on a small amount of the data to verify the training code and hyperparameters. Go back to Amazon SageMaker and train using the full dataset
- C. Use AWS Glue to train a model using a small subset of the data to confirm that the data will be compatible with Amazon SageMaker. Initiate a SageMaker training job using the full dataset from the S3 bucket using Pipe input mode.
- D. Load a smaller subset of the data into the SageMaker notebook and train locally. Confirm that the training code is executing and the model parameters seem reasonable. Launch an Amazon EC2 instance with an AWS Deep Learning AMI and attach the S3 bucket to train the full dataset.

Correct Answer: A

Community vote distribution

A (100%)

- JayK

Highly Voted

1 year, 6 months ago

Answer is A. The answer to this question is about Pipe mode from S3. The only options are A and C. As AWS Glue cannot be use to create models which is option C.
The correct answer is A
upvoted 27 times
- liangfb

Highly Voted

1 year, 6 months ago

Answer is A.
upvoted 13 times
- Valcilio

Most Recent

1 month, 3 weeks ago

Selected Answer: A

It's A, pipe mode is for dealing with very big data.
upvoted 2 times
- yemauricio

4 months, 1 week ago

Selected Answer: A

A, PIPE is to do that sort of modeling
upvoted 2 times
- Shailendraa

7 months, 3 weeks ago

When data is already in S3 and next it should move to Sagemaker.. so option A is suitable
upvoted 1 times
- Huy

1 year, 6 months ago

Answer is A. B, C & D can be dropped because there is no integration from/to Sage Maker train job (model).
upvoted 1 times
- cloud_trail

1 year, 6 months ago

Gotta be A. You need to use Pipe mode but Glue cannot train a model.
upvoted 2 times
- bobbylan1

1 year, 6 months ago

AAAAAa
upvoted 1 times
- Willnguyen22

1 year, 6 months ago

ans is A
upvoted 1 times
- GeeBeeEl

1 year, 6 months ago

Will you run AWS Deep Learning AMI for all cases where the data is very large in S3? Also what role is Glue playing here? Is there a transformation? These are the two issues for options B C and D. I believe they do not represent what is required to satisfy the requirements in the question. The answer definitely requires the pipe mode, but not with Glue. I go with A <https://aws.amazon.com/blogs/machine-learning/using-pipe-input-mode-for-amazon-sagemaker-algorithms/>

upvoted 3 times

  **roytruong** 1 year, 6 months ago

go for A

upvoted 2 times

  **PRC** 1 year, 6 months ago

Agree with A.

upvoted 2 times

  **cybe001** 1 year, 6 months ago

A is correct

upvoted 4 times

  **PhilipAWS** 1 year, 7 months ago

Answer is D not A

upvoted 2 times

  **hailiang** 1 year, 6 months ago

oh, come on

upvoted 7 times

  **cmm103** 1 year, 7 months ago

<https://aws.amazon.com/blogs/machine-learning/using-pipe-input-mode-for-amazon-sagemaker-algorithms/>

upvoted 9 times

A Machine Learning Specialist has completed a proof of concept for a company using a small data sample, and now the Specialist is ready to implement an end- to-end solution in AWS using Amazon SageMaker. The historical training data is stored in Amazon RDS. Which approach should the Specialist use for training a model using that data?

- A. Write a direct connection to the SQL database within the notebook and pull data in
- B. Push the data from Microsoft SQL Server to Amazon S3 using an AWS Data Pipeline and provide the S3 location within the notebook.
- C. Move the data to Amazon DynamoDB and set up a connection to DynamoDB within the notebook to pull data in.
- D. Move the data to Amazon ElastiCache using AWS DMS and set up a connection within the notebook to pull data in for fast access.

Correct Answer: B

Community vote distribution

B (100%)

- JayK

Highly Voted

1 year, 7 months ago

Answer is B as the data for a SageMaker notebook needs to be from S3 and option B is the only option that says it. The only thing with option B is that it is talking of moving data from MS SQL Server not RDS

upvoted 28 times
- mlyu

1 year, 7 months ago

<https://www.slideshare.net/AmazonWebServices/train-models-on-amazon-sagemaker-using-data-not-from-amazon-s3-aim419-aws-reinvent-2018>

upvoted 2 times
- HaiHN

1 year, 6 months ago

Please look at the slide 14 of that link, although the data source from DynamoDB or RDS, it is still need to use AWS Glue to move the data to S3 for SageMaker to use.

So, the right anwser should be B.

upvoted 2 times
- jasonsunbao

1 year, 6 months ago

I agree. As from the ML developer guide I just read, it is the MYSQL RDS that can be used as SQL datasource.

upvoted 2 times
- Valcilio

Most Recent

1 month, 3 weeks ago

Selected Answer: B

It's B, even if Microsoft SQL Server is a strange name for RDS, it's a possible database to use there and the data for sagemaker needs to be in S3!

upvoted 1 times
- AjoseO

2 months, 3 weeks ago

Selected Answer: B

In Option B approach, the Specialist can use AWS Data Pipeline to automate the movement of data from Amazon RDS to Amazon S3. This allows for the creation of a reliable and scalable data pipeline that can handle large amounts of data and ensure the data is available for training.

In the Amazon SageMaker notebook, the Specialist can then access the data stored in Amazon S3 and use it for training the model. Using Amazon S3 as the source of training data is a common and scalable approach, and it also provides durability and high availability of the data.

upvoted 1 times
- SophieSu

1 year, 6 months ago

B is the correct answer.

Official AWS Documentation:
"Amazon ML allows you to create a datasource object from data stored in a MySQL database in Amazon Relational Database Service (Amazon RDS). When you perform this action, Amazon ML creates an AWS Data Pipeline object that executes the SQL query that you specify, and places the output into an S3 bucket of your choice. Amazon ML uses that data to create the datasource."

upvoted 2 times
- cnethers

1 year, 6 months ago

While B is a valid answer, It is also possible to make a SQL connection in a notebook and create a data object so A could be a valid answer too

<https://stackoverflow.com/questions/36021385/connecting-from-python-to-sql-server>

<https://www.mssqltips.com/sqlservertip/6120/data-exploration-with-python-and-sql-server-using-jupyter-notebooks/>

upvoted 2 times
- gcpwhiz

1 year, 5 months ago

you need to choose the best answer, not any valid answer. Often, many of the answers are valid solutions, but are not best practice.

upvoted 2 times

  **scuzzy2010** 1 year, 6 months ago

B is correct. MS SQL Server is also under RDS.

upvoted 2 times

  **roytruong** 1 year, 6 months ago

B is right

upvoted 2 times

  **bhavesh0124** 1 year, 6 months ago

B it is

upvoted 1 times

  **cybe001** 1 year, 6 months ago

I'll go with B

upvoted 2 times

A Machine Learning Specialist receives customer data for an online shopping website. The data includes demographics, past visits, and locality information. The Specialist must develop a machine learning approach to identify the customer shopping patterns, preferences, and trends to enhance the website for better service and smart recommendations.

Which solution should the Specialist recommend?

- A. Latent Dirichlet Allocation (LDA) for the given collection of discrete data to identify patterns in the customer database.
- B. A neural network with a minimum of three layers and random initial weights to identify patterns in the customer database.
- C. Collaborative filtering based on user interactions and correlations to identify patterns in the customer database.
- D. Random Cut Forest (RCF) over random subsamples to identify patterns in the customer database.

Correct Answer: C

Community vote distribution

C (100%)

- WWODIN

Highly Voted

1 year, 7 months ago

answer should be C
Collaborative filtering is for recommendation, LDA is for topic modeling
upvoted 18 times
- syu31svc

Highly Voted

1 year, 6 months ago

In natural language processing, the latent Dirichlet allocation (LDA) is a generative statistical model that allows sets of observations to be explained by unobserved groups that explain why some parts of the data are similar.

Amazon SageMaker Random Cut Forest (RCF) is an unsupervised algorithm for detecting anomalous data points within a data set

Neural network is used for image detection

Answer is C
upvoted 8 times
- Valcilio

Most Recent

1 month, 3 weeks ago

Selected Answer: C

C, always when talk about recommendation you can think about collaborative patterns!
upvoted 2 times
- stjokerli

1 month, 3 weeks ago

A
LDA used before collaborative filtering is largely adopted.
1) the input data that we have doesn't lend itself to collaborative filtering - it requires a set of items and a set of users who have reacted to some of the items, which is NOT what we have
2) recommendation is just one thing that we want to do. What about trends?
3) collaborative filtering isn't one of the pre-built algorithms (weak argument, admittedly)
upvoted 1 times
- Shailendraa

7 months, 3 weeks ago

collaborative
upvoted 1 times
- apprehensive_scar

1 year, 2 months ago

C. Easy question.
upvoted 1 times
- technoguy

1 year, 6 months ago

its a appropriate use case of Collaborative filtering
upvoted 1 times
- roytruong

1 year, 6 months ago

this is C
upvoted 1 times
- sdsfsdsf

1 year, 6 months ago

I'm thinking that it is A because:
1) the input data that we have doesn't lend itself to collaborative filtering - it requires a set of items and a set of users who have reacted to some of the items, which is NOT what we have

2) recommendation is just one thing that we want to do. What about trends?
3) collaborative filtering isn't one of the pre-built algorithms (weak argument, admittedly)
upvoted 5 times

  **cybe001** 1 year, 6 months ago

Answer is C, demographics, past visits, and locality information data, LDA is appropriate
upvoted 3 times

  **cybe001** 1 year, 6 months ago
Collaborative filtering is appropriate
upvoted 3 times

  **DonaldCMLIN** 1 year, 7 months ago
Answer A might be more suitable than other

https://docs.aws.amazon.com/zh_tw/sagemaker/latest/dg/lda-how-it-works.html
upvoted 2 times

  **C10ud9** 1 year, 6 months ago
Also found this article which implemented LDA for analysing shopping trends, patterns.<https://arxiv.org/pdf/1810.08577.pdf>
upvoted 1 times

  **C10ud9** 1 year, 6 months ago
That said ..I'm split between LDA and Collaborative filtering(CF). One point is that CF is a method which will feed into Recommender system.
upvoted 1 times

  **rsimham** 1 year, 7 months ago
Not convinced with A. Answer C seems to be a better fit than A for recommendation model (LDA appears to be a topic-based model on unavailable data with similar patterns)
<https://aws.amazon.com/blogs/machine-learning/extending-amazon-sagemaker-factorization-machines-algorithm-to-predict-top-x-recommendations/>
upvoted 10 times

A Machine Learning Specialist is working with a large company to leverage machine learning within its products. The company wants to group its customers into categories based on which customers will and will not churn within the next 6 months. The company has labeled the data available to the Specialist.

Which machine learning model type should the Specialist use to accomplish this task?

- A. Linear regression
- B. Classification
- C. Clustering
- D. Reinforcement learning

Correct Answer: B

The goal of classification is to determine to which class or category a data point (customer in our case) belongs to. For classification problems, data scientists would use historical data with predefined target variables AKA labels (churner/non-churner) and answers that need to be predicted to train an algorithm. With classification, businesses can answer the following questions:

- Will this customer churn or not?
- Will a customer renew their subscription?
- Will a user downgrade a pricing plan?
- Are there any signs of unusual customer behavior?

Reference:

<https://www.kdnuggets.com/2019/05/churn-prediction-machine-learning.html>

Community vote distribution

B (100%)

rsimham

Highly Voted

1 year, 7 months ago

B seems to be okay

upvoted 13 times

ovokpus

Most Recent

10 months, 1 week ago

Selected Answer: B

This is definitely a classification problem

upvoted 3 times

Sivadharan

11 months, 3 weeks ago

Selected Answer: B

B is correct

upvoted 2 times

FabG

1 year, 5 months ago

B - it's a Binary Classification problem. Will the customer churn: Yes or No

upvoted 3 times

syu31svc

1 year, 5 months ago

100% is B since it is about labelled data

upvoted 1 times

eji

1 year, 6 months ago

i think the key is "the company has labeled the data" so this is classification, so it's B

upvoted 1 times

roytruong

1 year, 7 months ago

B is okey

upvoted 2 times

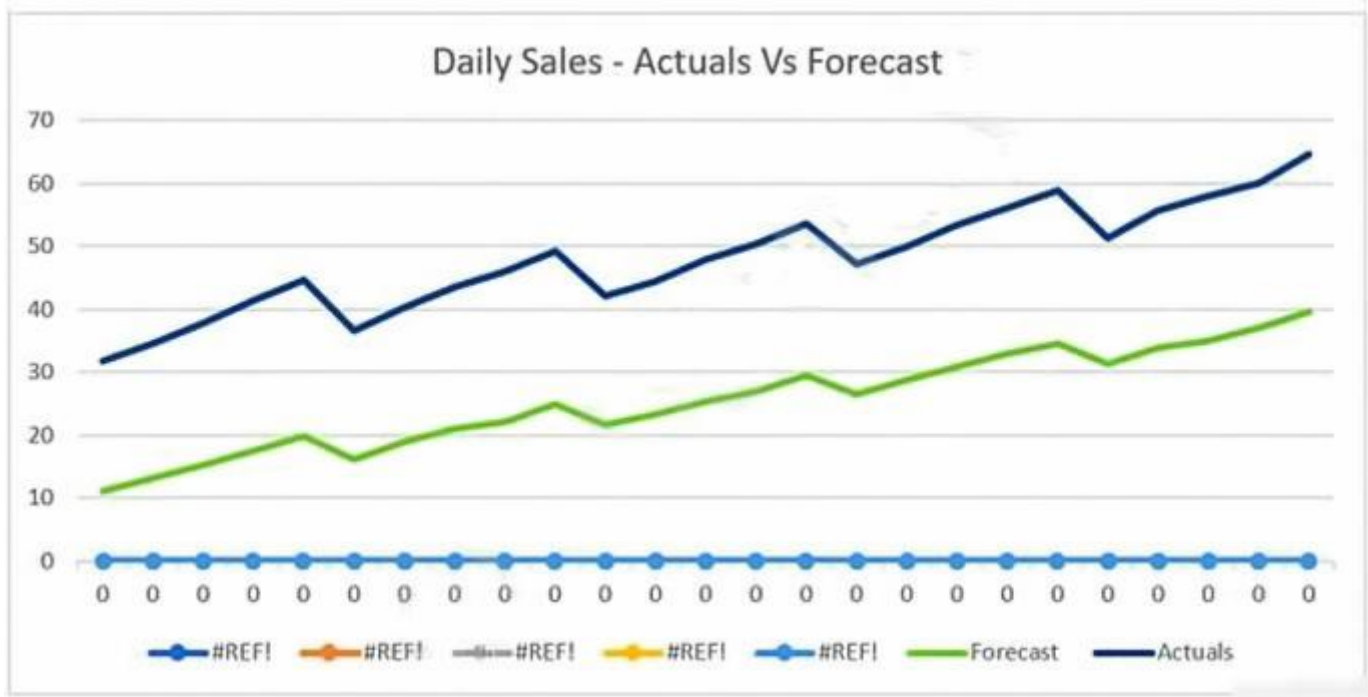
cybe001

1 year, 7 months ago

B is correct

upvoted 3 times

The displayed graph is from a forecasting model for testing a time series.



Considering the graph only, which conclusion should a Machine Learning Specialist make about the behavior of the model?

- A. The model predicts both the trend and the seasonality well
- B. The model predicts the trend well, but not the seasonality.
- C. The model predicts the seasonality well, but not the trend.
- D. The model does not predict the trend or the seasonality well.

Correct Answer: A

Community vote distribution

A (100%)

- jeetss1

Highly Voted

1 year, 6 months ago

A is correct answer.
Please Refer: <https://machinelearningmastery.com/decompose-time-series-data-trend-seasonality/>
upvoted 15 times
- Dr_Kiko

Highly Voted

1 year, 5 months ago

A; the problem is bias, not trends
upvoted 6 times
- Valcilio

Most Recent

1 month, 3 weeks ago

Selected Answer: A

The problem is Bias not trends or sesonality!
upvoted 1 times
- spamicho

1 year, 6 months ago

A is right, both trend (rising) and seasonality is there
upvoted 3 times
- btsql

1 year, 6 months ago

C is correct answer
upvoted 1 times
- btsql

1 year, 6 months ago

A is correct answer. Not C
upvoted 2 times
- Kuntazulu

1 year, 6 months ago

The trend is up, so isn't it correctly predicted? And the seasonality is also in sync, the amplitude is wrong.
upvoted 3 times
- georschi

1 year, 6 months ago

A is right. trend and seasonality are fine, level is the one the model gets wrong
upvoted 4 times
- NotAnMLProfessional

1 year, 7 months ago

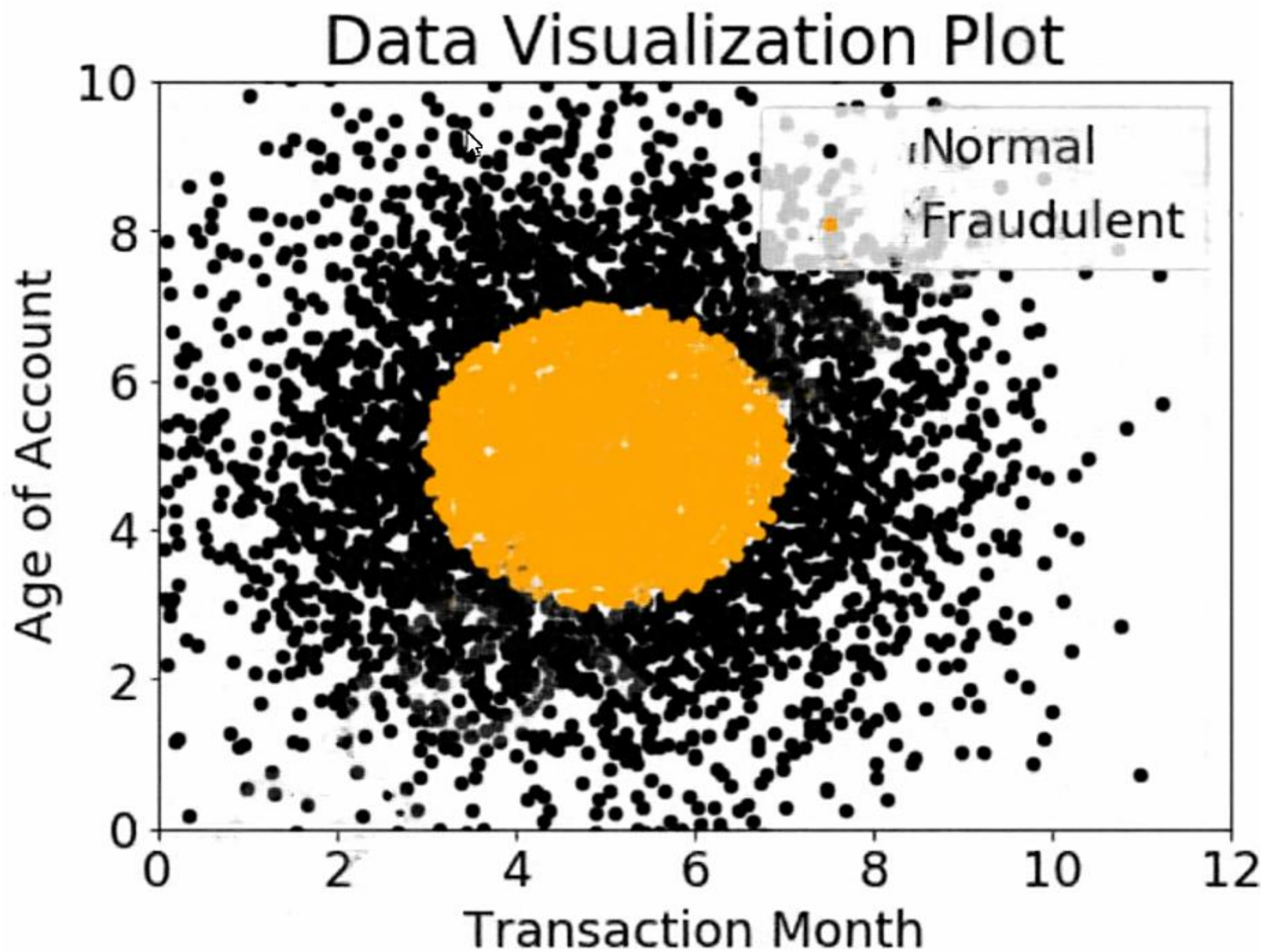
Should be C
upvoted 1 times

  **ashlash** 1 year, 7 months ago
Should be A
upvoted 2 times

Question #14

Topic 1

A company wants to classify user behavior as either fraudulent or normal. Based on internal research, a Machine Learning Specialist would like to build a binary classifier based on two features: age of account and transaction month. The class distribution for these features is illustrated in the figure provided.



Based on this information, which model would have the HIGHEST accuracy?

- A. Long short-term memory (LSTM) model with scaled exponential linear unit (SELU)
- B. Logistic regression
- C. Support vector machine (SVM) with non-linear kernel
- D. Single perceptron with tanh activation function

Correct Answer: C

Community vote distribution

C (100%)

mizuakari Highly Voted 1 year, 7 months ago

Answer is C. SVM sample use case is to put the dimensions into a higher hyperplane that can separates it. Seeing how separable it is, SVM can be used for it.
upvoted 18 times

Valcilio Most Recent 1 month, 3 weeks ago

Selected Answer: C
SVM for RBF mode is the answer!
upvoted 1 times

Broncomailo 1 year, 2 months ago

Selected Answer: C
Answer is C
upvoted 3 times

-  **Dr_Kiko** 1 year, 5 months ago
Textbook C
upvoted 3 times
-  **halfway** 1 year, 6 months ago
C. more reading for using non-linear kernel and separate samples with a hyperplane in a higher dimension space:
<https://medium.com/pursuitnotes/day-12-kernel-svm-non-linear-svm-5fdefe77836c>
upvoted 2 times
-  **spamicho** 1 year, 6 months ago
C seems right
upvoted 1 times
-  **Juka3lj** 1 year, 6 months ago
answer is C
upvoted 2 times
-  **omar_bahrain** 1 year, 7 months ago
Agree. The answer is A.

<https://www.surveypractice.org/article/2715-using-support-vector-machines-for-survey-research>
upvoted 1 times
-  **cnethers** 1 year, 7 months ago
This is a good explanation of SVM
<https://uk.mathworks.com/help/stats/support-vector-machines-for-binary-classification.html>
upvoted 3 times

Question #15

Topic 1

A Machine Learning Specialist at a company sensitive to security is preparing a dataset for model training. The dataset is stored in Amazon S3 and contains Personally Identifiable Information (PII). The dataset:

- ☞ Must be accessible from a VPC only.
- ☞ Must not traverse the public internet.

How can these requirements be satisfied?

- A. Create a VPC endpoint and apply a bucket access policy that restricts access to the given VPC endpoint and the VPC.
- B. Create a VPC endpoint and apply a bucket access policy that allows access from the given VPC endpoint and an Amazon EC2 instance.
- C. Create a VPC endpoint and use Network Access Control Lists (NACLs) to allow traffic between only the given VPC endpoint and an Amazon EC2 instance.
- D. Create a VPC endpoint and use security groups to restrict access to the given VPC endpoint and an Amazon EC2 instance

Correct Answer: A

Community vote distribution

A (80%)

B (20%)

  **rajs**

Highly Voted 

 1 year, 7 months ago

Important things to note here is that

1. "The Data in S3 Needs to be Accessible from VPC"
2. "Traffic should not Traverse internet"

To fulfill Requirement #2 we need a VPC endpoint
To RESTRICT the access to S3/Bucket
- Access allowed only from VPC via VPC Endpoint

Even though Sagemaker uses EC2 - we are NOT asked to secure the EC2 :)

So the answer is A
upvoted 32 times

  **sdsfsdsf**

Highly Voted 

 1 year, 6 months ago

Between A & B, the answer should be A. From here:
<https://docs.aws.amazon.com/vpc/latest/userguide/vpc-endpoints-s3.html#vpc-endpoints-s3-bucket-policies>
We can see that we restrict access using DENY if sourceVpce (vpc endpoint), or sourceVpc (vpc) is not equal to our VPCE/VPC. So we are using a DENY (choice A) and not an ALLOW policy (choice B).

Choices C, D we eliminate because they don't address S3 access at all.
upvoted 11 times

  **Ajose0**

Most Recent 

 2 months, 3 weeks ago

Selected Answer: A

In Option A, the Machine Learning Specialist would create a VPC endpoint for Amazon S3, which would allow traffic to flow directly between the VPC and Amazon S3 without traversing the public internet. Access to the S3 bucket containing PII can then be restricted to the VPC endpoint and the VPC using a bucket access policy. This would ensure that only instances within the VPC can access the data, and that the data does not traverse the public internet.

Option B and D, allowing access from an Amazon EC2 instance, would not meet the requirement of not traversing the public internet, as the EC2 instance would be accessible from the internet. Option C, using Network Access Control Lists (NACLs) to allow traffic between only the VPC endpoint and an EC2 instance, would also not meet the requirement of not traversing the public internet, as the EC2 instance would still be accessible from the internet.
upvoted 1 times

  **exam887** 11 months, 1 week ago

Selected Answer: A

The question do not mention EC2 at all, so should be A
upvoted 3 times

  **dunhill** 1 year, 4 months ago

I think it should be B. Traning instance is a EC2 instance and need to be set an endpoint to load the data from S3.
upvoted 1 times

- passionatecricketer** 1 year, 5 months ago

Selected Answer: B

AWS security is a conservative security model, which implies that access are denied by default rather than granted by default. We have to explicitly allow access to a AWS resource. Additionally, B talks about allowing access FROM the VPC to S3 while A talks about allowing access from S3 to VPC (which is not what we need).

So, B.

upvoted 1 times
- cpal012** 1 month, 1 week ago

Um, no. A VPC endpoint is outbound from the VPC to a supported AWS service.

upvoted 1 times
- technoguy** 1 year, 5 months ago

Will go with B

upvoted 1 times
- spamicho** 1 year, 5 months ago

Betting on B here, we should control access from VPC, not to VPC.

upvoted 1 times
- achiko** 1 year, 6 months ago

A!

Restricting access to a specific VPC endpoint

The following is an example of an Amazon S3 bucket policy that restricts access to a specific bucket, awsexamplebucket1, only from the VPC endpoint with the ID vpce-1a2b3c4d. The policy denies all access to the bucket if the specified endpoint is not being used. The aws:SourceVpce condition is used to specify the endpoint.

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/example-bucket-policies-vpc-endpoint.html>

upvoted 1 times
- senseikimoji** 1 year, 6 months ago

Can't be B. You simple cannot enable access to an endpoint to some selected instance. So A.

upvoted 1 times
- Huy** 1 year, 5 months ago

We shouldn't use private IP in bucket policy.

upvoted 1 times
- cloud_trail** 1 year, 6 months ago

B does not say enable access TO the VPC endpoint. It says to allow access FROM the endpoint. So B is the correct answer. A talks about restricting access TO the VPC endpoint, so that option is irrelevant. We're worried about access TO the S3 bucket, not access to the VPC. The question is not poorly-worded, but it is tricky and you need to read it carefully.

upvoted 1 times
- yeetusdeleetus** 1 year, 6 months ago

I also vote A.

upvoted 1 times
- Thai_Xuan** 1 year, 6 months ago

A

found here

"You can control which VPCs or VPC endpoints have access to your buckets by using Amazon S3 bucket policies. For examples of this type of bucket policy access control, see the following topics on restricting access."

<https://docs.aws.amazon.com/AmazonS3/latest/dev/example-bucket-policies-vpc-endpoint.html>

upvoted 2 times
- tff** 1 year, 6 months ago

agree with A

upvoted 1 times
- morara** 1 year, 6 months ago

A is the answer. No need to bring in EC2 coz the requirements defined are between S3 and the VPC.

upvoted 1 times
- roytruong** 1 year, 6 months ago

It is A

upvoted 1 times
- ac427** 1 year, 7 months ago

Instances in the VPC access the VPC EndPoint, not the VPC itself. So A is poorly worded like lots of AWS questions.

upvoted 1 times
- grandgale** 1 year, 7 months ago

A THE VPC access the s3 through the VPC endpoint without internet traffic.

upvoted 4 times

During mini-batch training of a neural network for a classification problem, a Data Scientist notices that training accuracy oscillates. What is the MOST likely cause of this issue?

- A. The class distribution in the dataset is imbalanced.
- B. Dataset shuffling is disabled.
- C. The batch size is too big.
- D. The learning rate is very high.

Correct Answer: D

Reference:
<https://towardsdatascience.com/deep-learning-personal-notes-part-1-lesson-2-8946fe970b95>

Community vote distribution

D (100%)

- gaku1016**

Highly Voted

1 year, 7 months ago

Answer is D.
Should the weight be increased or reduced so that the error is smaller than the current value? You need to examine the amount of change to know that. Therefore, we differentiate and check whether the slope of the tangent is positive or negative, and update the weight value in the direction to reduce the error. The operation is repeated over and over so as to approach the optimal solution that is the goal. The width of the update amount is important at this time, and is determined by the learning rate.

upvoted 16 times
- ozan11**

Highly Voted

1 year, 7 months ago

maybe D ?

upvoted 8 times
- Valcilio**

Most Recent

1 month, 3 weeks ago

Selected Answer: D

The big learning rating overshoot in true minima.

upvoted 2 times
- Ajose0**

2 months, 3 weeks ago

Selected Answer: D

A high learning rate can cause oscillations in the training accuracy because the optimizer makes large updates to the model parameters in each iteration, which can cause overshooting the optimal values. This can result in the model oscillating back and forth across the optimal solution.

upvoted 2 times
- Tomatoteacher**

3 months, 2 weeks ago

Selected Answer: D

D Learning rate is too high. Textbook example of learning rate being too high. Lower Learning_rate will take more iterations, or longer to train, but will settle in place.

upvoted 1 times
- Shailendraa**

7 months, 3 weeks ago

12-sep exam

upvoted 1 times
- Sam1610**

9 months, 4 weeks ago

D: per supuesto

upvoted 1 times
- missionml**

1 year, 1 month ago

A company sells thousands of products on a public website and wants to automatically identify products with potential durability problems. The company has 1.000 reviews with date, star rating, review text, review summary, and customer email fields, but many reviews are incomplete and have empty fields. Each review has already been labeled with the correct durability result. A machine learning specialist must train a model to identify reviews expressing concerns over product durability. The first model needs to be trained and ready to review in 2 days. What is the MOST direct approach to solve this problem within 2 days?

A.
Train a custom classifier by using Amazon Comprehend.

B.
Build a recurrent neural network (RNN) in Amazon SageMaker by using Gluon and Apache MXNet.

C.

Train a built-in BlazingText model using Word2Vec mode in Amazon SageMaker.
D.
Use a built-in seq2seq model in Amazon SageMaker.
upvoted 1 times

  **missionml** 1 year, 1 month ago

Is A valid option?
upvoted 1 times

  **btsql** 1 year, 5 months ago

D is correct. big batch size make local minia.
upvoted 1 times

  **jeetss1** 1 year, 6 months ago

it is a multiple answer question and answer should be both A and D
upvoted 1 times

  **syu31svc** 1 year, 6 months ago

Answer is D 100%; learning rate too high will cause such an event
upvoted 3 times

  **deep_n** 1 year, 6 months ago

The answer is D, from the Coursera deep learning specialization (course 2 - improving Deep NN)
upvoted 2 times

  **rickywck** 1 year, 6 months ago

I think D is the answer:

<https://www.jeremyjordan.me/nn-learning-rate/>
upvoted 2 times

  **emailtorajivk** 1 year, 7 months ago

If the learning rate is too small, it will take very long time to get to the bottom.If the learning rate is too big, it could get oscillate away from the bottom. If training a neural net and you find that the loss or accuracy is speeding to infinity the learning rate is too high.
upvoted 3 times

An employee found a video clip with audio on a company's social media feed. The language used in the video is Spanish. English is the employee's first language, and they do not understand Spanish. The employee wants to do a sentiment analysis.

What combination of services is the MOST efficient to accomplish the task?

- A. Amazon Transcribe, Amazon Translate, and Amazon Comprehend
- B. Amazon Transcribe, Amazon Comprehend, and Amazon SageMaker seq2seq
- C. Amazon Transcribe, Amazon Translate, and Amazon SageMaker Neural Topic Model (NTM)
- D. Amazon Transcribe, Amazon Translate and Amazon SageMaker BlazingText

Correct Answer: A

Community vote distribution

A (100%)

- DonaldCMLIN

Highly Voted

1 year, 7 months ago

the MOST efficient means to you don't need to coding, building infra
All of sevices are manage by AWS is good,
Transcribe, Amazon Translate, and Amazon Comprehend

Answer is A
upvoted 38 times
- WWODIN

1 year, 7 months ago

Agree, Answer is A
upvoted 9 times
- Phong

Highly Voted

1 year, 6 months ago

Definitely A.
upvoted 7 times
- Pg690

Most Recent

5 months ago

A is not 100% correct. You don't need to translate Spanish. Amazon Comprehend supports Spanish.
upvoted 5 times
- cpal012

1 month, 1 week ago

Arguably, you still need a translation since the person doesn't speak Spanish.
upvoted 1 times
- gnolam

7 months, 3 weeks ago

Selected Answer: A

A - confirmed by ACG
upvoted 2 times
- KM226

1 year, 4 months ago

Selected Answer: A

I agree that the answer is A
upvoted 1 times
- in4976

1 year, 4 months ago

Selected Answer: A

answer is a
upvoted 1 times
- Dr_Kiko

1 year, 5 months ago

A; D is wrong because The Amazon SageMaker BlazingText algorithm provides highly optimized implementations of the Word2vec and text classification algorithms.
upvoted 1 times
- harmanbirstudy

1 year, 6 months ago

The Question/Anwser is not poorly as someone mentioned.
--Even though Comprehend can do the analysis directly on Spanish (no need of translate) but if comprehend does analysis and the resulting words are still in spanish , it will no help the employee as he doesnt know Spanish.So the transalate after transcribe will help Employee understand what is being analyzed by Comprehend in next step.
So read the question carefully before jumping to conclusions. it will save you an Exam :)
upvoted 1 times

-   **senseikimoji** 1 year, 6 months ago

I don't get this question. Comprehend supports Spanish natively. There is no need for Translate, and translate would actually reduce effectiveness of sentimental analysis. However, BCD are all invalid choices.

upvoted 3 times
-   **ybad** 1 year, 6 months ago

A

because Comprehend can provide sentiment analysis

upvoted 2 times
-   **FastTrack** 1 year, 6 months ago

A,

<https://aws.amazon.com/getting-started/hands-on/analyze-sentiment-comprehend/>

upvoted 4 times
-   **syu31svc** 1 year, 6 months ago

Amazon Comprehend is needed for sure; answer is A 100%

upvoted 3 times
-   **hans1234** 1 year, 6 months ago

You actually do not need Translate. Comprehend can do sentiment analysis on spanish language. But in this case it is A.

upvoted 2 times
-   **Aanish** 1 year, 6 months ago

You are right but since the employee doesn't know Spanish it would be good if he has an insight into how well the sentiments were captured by comprehend.

upvoted 2 times
-   **Antriksh** 1 year, 6 months ago

These answers are clearly misleading. The correct answer is A

upvoted 3 times
-   **roytruong** 1 year, 6 months ago

it's A

upvoted 3 times
-   **AKT** 1 year, 6 months ago

Answer is A.

upvoted 3 times
-   **grandgale** 1 year, 6 months ago

should be D

<https://docs.aws.amazon.com/sagemaker/latest/dg/blazingtext.html>

upvoted 1 times
-   **h_sahu** 1 year, 6 months ago

It shouldn't be D as per me. Blazing text is used for classification. This can indirectly help in sentiment analysis no doubt. I believer A is the answer. Because, transcribe will be used to convert speech to text, translate will be used to translate those texts received from transcribe and Comprehend is a complete NLP tool from amazon out-of-the-box which is capable of sentiment analysis.

upvoted 2 times
-   **oji** 1 year, 6 months ago

A is more efficient because if using sagemaker we still need to build a model first, but amazon comprehen is pre-build model so A is more efficient. CMIIW

upvoted 6 times

A Machine Learning Specialist is packaging a custom ResNet model into a Docker container so the company can leverage Amazon SageMaker for training. The Specialist is using Amazon EC2 P3 instances to train the model and needs to properly configure the Docker container to leverage the NVIDIA GPUs.

What does the Specialist need to do?

- A. Bundle the NVIDIA drivers with the Docker image.
- B. Build the Docker container to be NVIDIA-Docker compatible.
- C. Organize the Docker container's file structure to execute on GPU instances.
- D. Set the GPU flag in the Amazon SageMaker CreateTrainingJob request body.

Correct Answer: A

Community vote distribution

B (100%)

- vetal

Highly Voted

1 year, 7 months ago

<https://docs.aws.amazon.com/sagemaker/latest/dg/sagemaker-dg.pdf>
page 55:
If you plan to use GPU devices, make sure that your containers are nvidia-docker compatible. Only the CUDA toolkit should be included on containers. Don't bundle NVIDIA drivers with the image. For more information about nvidia-docker, see NVIDIA/nvidia-docker.

So the answer is B
upvoted 40 times
- devsean

1 year, 7 months ago

Yeah, it's B. But the page in the developer guide is page number 201 (209 in pdf). Second bullet point at the top.
upvoted 5 times
- AKT

Highly Voted

1 year, 7 months ago

Answer is B. below is from AWS documentation,
If you plan to use GPU devices for model training, make sure that your containers are nvidia-docker compatible. Only the CUDA toolkit should be included on containers; don't bundle NVIDIA drivers with the image.
<https://docs.aws.amazon.com/sagemaker/latest/dg/your-algorithms-training-algo-dockerfile.html>
upvoted 12 times
- jackzhao

Most Recent

1 month, 2 weeks ago

B is correct!
upvoted 1 times
- Sylzys

1 month, 3 weeks ago

Selected Answer: B

As per aws documentation, answer is B, and A is even explicitly not recommended
upvoted 1 times
- bakarys

2 months, 1 week ago

Selected Answer: B

NVIDIA-Docker is a Docker container runtime plugin that allows the Docker container to access the GPU resources on the host machine. By building the Docker container to be NVIDIA-Docker compatible, the Docker container will have access to the NVIDIA GPU resources on the Amazon EC2 P3 instances, allowing for accelerated training of the ResNet model.
upvoted 1 times
- Ajose0

2 months, 3 weeks ago

Selected Answer: B

To leverage the NVIDIA GPUs on Amazon EC2 P3 instances for training with Amazon SageMaker, the Docker container must be built to be compatible with NVIDIA-Docker.

NVIDIA-Docker is a wrapper around Docker that makes it easier to use GPUs in containers by providing GPU-aware functionality.

To build a Docker container that is compatible with NVIDIA-Docker, the Specialist should install the NVIDIA GPU drivers in the Docker container and install the NVIDIA-Docker runtime on the EC2 instances.
upvoted 1 times
- Sorrybutnotsorry

1 year, 3 months ago

Selected Answer: B

As referred in other comments ans is B
upvoted 1 times

hussamS

1 year, 4 months ago

Selected Answer: B

ANS B
As mentioned byi other users
upvoted 1 times

sachin80

1 year, 6 months ago

As per me answer is B
upvoted 1 times

konradL

1 year, 6 months ago

The answer is for sure B - as mentioned by others. And this is clearly stated in the docs
upvoted 1 times

takahirokoyama

1 year, 6 months ago

Ans. is B.
upvoted 1 times

GeeBeeEI

1 year, 6 months ago

<https://docs.aws.amazon.com/sagemaker/latest/dg/sagemaker-dg.pdf> is clear "DO NOT BUNDLE NVIDIA DRIVERS WITH THE IMAGE" Details are found in <https://github.com/NVIDIA/nvidia-docker> A is wrong, C and D are out. Looks more like a B than anything
upvoted 4 times

Antriksh

1 year, 6 months ago

If you plan to use GPU devices for model training, make sure that your containers are nvidia-docker compatible. Only the CUDA toolkit should be included on containers; don't bundle NVIDIA drivers with the image. -- source: (<https://docs.aws.amazon.com/sagemaker/latest/dg/your-algorithms-training-algo-dockerfile.html>)
Correct answer is 'B'
upvoted 4 times

roytruong

1 year, 7 months ago

this is B
upvoted 1 times

heihei

1 year, 7 months ago

i think it's b
upvoted 5 times

A Machine Learning Specialist is building a logistic regression model that will predict whether or not a person will order a pizza. The Specialist is trying to build the optimal model with an ideal classification threshold.

What model evaluation technique should the Specialist use to understand how different classification thresholds will impact the model's performance?

- A. Receiver operating characteristic (ROC) curve
- B. Misclassification rate
- C. Root Mean Square Error (RMSE)
- D. L1 norm

Correct Answer: A

Reference:
<https://docs.aws.amazon.com/machine-learning/latest/dg/binary-model-insights.html>

Community vote distribution

A (100%)

-  **rsimham**

Highly Voted 

 1 year, 7 months ago

Ans. A is correct
upvoted 19 times
-  **AKT**

Highly Voted 

 1 year, 7 months ago

Answer is A.
An ROC curve (receiver operating characteristic curve) is a graph showing the performance of a classification model at all classification thresholds
upvoted 7 times
-  **Valcilio**

Most Recent 

 1 month, 3 weeks ago

Selected Answer: A

ROC curve is for defining the threshold.
upvoted 2 times
-  **spamicho** 1 year, 5 months ago

A surely
upvoted 1 times
-  **syu31svc** 1 year, 5 months ago

Question is about classification so confusion matrix would come into mind; A is the answer
upvoted 1 times
-  **GeeBeeEl** 1 year, 6 months ago

A is indeed correct see <https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc>
An ROC curve (receiver operating characteristic curve) is a graph showing the performance of a classification model at all classification thresholds.
This curve plots two parameters:
 - True Positive Rate
 - False Positive RateTrue Positive Rate (TPR) is a synonym for recall and is therefore defined as follows:
 $TPR = TP / TP + FN$
False Positive Rate (FPR) is defined as follows:
 $FPR = FP / FP + TN$
upvoted 3 times
-  **hans1234** 1 year, 6 months ago

It is A.
upvoted 1 times
-  **roytruong** 1 year, 6 months ago

obviously A
upvoted 1 times
-  **bitiyaha** 1 year, 7 months ago

Root Mean Square Error (RMSE) Ans. c
upvoted 1 times
-  **bzhao** 1 year, 6 months ago

I think RMSE is for regression model

upvoted 4 times

An interactive online dictionary wants to add a widget that displays words used in similar contexts. A Machine Learning Specialist is asked to provide word features for the downstream nearest neighbor model powering the widget.

What should the Specialist do to meet these requirements?

- A. Create one-hot word encoding vectors.
- B. Produce a set of synonyms for every word using Amazon Mechanical Turk.
- C. Create word embedding vectors that store edit distance with every other word.
- D. Download word embeddings pre-trained on a large corpus.

Correct Answer: A

Reference:
<https://aws.amazon.com/blogs/machine-learning/amazon-sagemaker-object2vec-adds-new-features-that-support-automatic-negative-sampling-and-speed-up-training/>

Community vote distribution

D (100%)

- JayK

Highly Voted

1 year, 6 months ago

the solution is word embedding. As it is a interactive online dictionary, we need pre-trained word embedding thus the answer is D. In addition, there is no mention that the online dictionary is unique and does not have a pre-trained word embedding.

Thus I strongly feel the answer is D

upvoted 28 times
- cybe001

Highly Voted

1 year, 6 months ago

D is correct. It is not a specialized dictionary so use the existing word corpus to train the model

upvoted 15 times
- Ajose0

Most Recent

2 months, 3 weeks ago

Selected Answer: D

D. Download word embeddings pre-trained on a large corpus.

Word embeddings are a type of dense representation of words, which encode semantic meaning in a vector form. These embeddings are typically pre-trained on a large corpus of text data, such as a large set of books, news articles, or web pages, and capture the context in which words are used. Word embeddings can be used as features for a nearest neighbor model, which can be used to find words used in similar contexts.

Downloading pre-trained word embeddings is a good way to get started quickly and leverage the strengths of these representations, which have been optimized on a large amount of data. This is likely to result in more accurate and reliable features than other options like one-hot encoding, edit distance, or using Amazon Mechanical Turk to produce synonyms.

upvoted 2 times
- Shailendraa

7 months, 3 weeks ago

12-sep exam

upvoted 1 times
- helpaws

8 months, 1 week ago

Selected Answer: D

DDDDDDDDDDDDDD

upvoted 3 times
- engomaradel

1 year, 5 months ago

D for sure

upvoted 2 times
- yeetusdeleetus

1 year, 5 months ago

Definitely D.

upvoted 3 times
- weslleylc

1 year, 6 months ago

A)It requires that document text be cleaned and prepared such that each word is one-hot encoded.

Ref:<https://machinelearningmastery.com/what-are-word-embeddings/>

upvoted 1 times
- syu31svc

1 year, 6 months ago

I don't see how one-hot encoding works; I would say D 100%

B & C are definitely wrong

upvoted 2 times

  **GeeBeeEI** 1 year, 6 months ago

took a look at <https://medium.com/@athif.shaffy/one-hot-encoding-of-text-b69124bef0a7> it appears this is useful at the point of running the model. See: <https://arxiv.org/abs/1705.08488> Word embeddings are dense, low-dimensional vector representations of words that are commonly used as input features in a variety of natural language processing (NLP) tasks [1]. In contrast to symbolic one-hot or hierarchical clustering-based representations, real-valued embedding vectors easily reflect varying degrees of similarity between words, and significantly reduce sparsity in linear algebra operations.

upvoted 1 times

  **GeeBeeEI** 1 year, 6 months ago

Appears D is the correct answer....

upvoted 2 times

  **Achievement** 1 year, 6 months ago

I think should be A, we need to provide features for next step

upvoted 1 times

  **Dr_Kiko** 1 year, 5 months ago

learn what one-hot means. each and every word will be unique, how will you work with that data? D is the answer

upvoted 1 times

  **andreylh** 1 year, 6 months ago

Why not A since it is required to pass words as features to a nearest neighbor model?

upvoted 3 times

  **roytruong** 1 year, 6 months ago

absolutely D

upvoted 3 times

  **AKT** 1 year, 6 months ago

answer is D

upvoted 2 times

  **WWODIN** 1 year, 7 months ago

Seems D?

upvoted 5 times

  **WWODIN** 1 year, 7 months ago

sorry, seems A is a necessary first step

upvoted 2 times

  **WWODIN** 1 year, 7 months ago

sorry again, should be C or D, but more towards D

upvoted 2 times

  **HaiHN** 1 year, 5 months ago

C is irrelevant as it only concern about the "edit distance", not the meaning of the word for "used in similar context"

Correct answer should be D

upvoted 2 times

  **cloud_trail** 1 year, 5 months ago

Correct. Edit distance or Levenshtein Distance concerns spelling and is used by spell checkers. Has nothing to do with context. D is the obvious answer. There's no need to create your own embedding when you can just download pre-trained ones.

upvoted 3 times

  **rsimham** 1 year, 7 months ago

C sounds to be right for me, not sure.

<https://medium.com/@athif.shaffy/one-hot-encoding-of-text-b69124bef0a7>

upvoted 2 times

  **vetal** 1 year, 7 months ago

Why not D?

upvoted 4 times

Question #21

Topic 1

A Machine Learning Specialist is configuring Amazon SageMaker so multiple Data Scientists can access notebooks, train models, and deploy endpoints. To ensure the best operational performance, the Specialist needs to be able to track how often the Scientists are deploying models, GPU and CPU utilization on the deployed SageMaker endpoints, and all errors that are generated when an endpoint is invoked. Which services are integrated with Amazon SageMaker to track this information? (Choose two.)

- A. AWS CloudTrail
- B. AWS Health
- C. AWS Trusted Advisor
- D. Amazon CloudWatch
- E. AWS Config

Correct Answer: AD

Reference:

<https://aws.amazon.com/sagemaker/faqs/>

Community vote distribution

AD (100%)

- rsimham

Highly Voted

1 year, 6 months ago

AD is correct
upvoted 24 times
- roytruong

Highly Voted

1 year, 6 months ago

cloudtrail and cloudwatch, no thinking
upvoted 5 times
- joe3232

Most Recent

3 months ago

Log Amazon Sagemaker API Calls with AWS CloudTrail - <https://docs.aws.amazon.com/sagemaker/latest/dg/logging-using-cloudtrail.html>
upvoted 1 times
- f4bi4n

10 months, 2 weeks ago

I wouldn't be so sure about CloudTrail, AWS Configs also tracks Sagemaker and the resource "AWS::Sagemaker::Model"
upvoted 1 times
- f4bi4n

10 months, 2 weeks ago

just seen, this was release 4 days ago...
<https://aws.amazon.com/about-aws/whats-new/2022/06/aws-config-15-new-resource-types/>
upvoted 2 times
- yogesh1

1 year, 2 months ago

Selected Answer: AD

A&D
CloudWatch and ClouTrail
upvoted 2 times

hess

1 year, 3 months ago

AD Are Correct.
upvoted 1 times

eji

1 year, 6 months ago

CloudTrail is use to track scientist how ofthe they deploy a model
CloudWatch for monitoring GPU and CPU
so answer is A & D
upvoted 4 times

Urban_Life

1 year, 6 months ago

absolutely
upvoted 1 times
- <https://www.examtopycs.com/exams/amazon/aws-certified-machine-learning-specialty/custom-view/>

44/406

A retail chain has been ingesting purchasing records from its network of 20,000 stores to Amazon S3 using Amazon Kinesis Data Firehose. To support training an improved machine learning model, training records will require new but simple transformations, and some attributes will be combined. The model needs to be retrained daily.

Given the large number of stores and the legacy data ingestion, which change will require the LEAST amount of development effort?

- A. Require that the stores to switch to capturing their data locally on AWS Storage Gateway for loading into Amazon S3, then use AWS Glue to do the transformation.
- B. Deploy an Amazon EMR cluster running Apache Spark with the transformation logic, and have the cluster run each day on the accumulating records in Amazon S3, outputting new/transformed records to Amazon S3.
- C. Spin up a fleet of Amazon EC2 instances with the transformation logic, have them transform the data records accumulating on Amazon S3, and output the transformed records to Amazon S3.
- D. Insert an Amazon Kinesis Data Analytics stream downstream of the Kinesis Data Firehose stream that transforms raw record attributes into simple transformed values using SQL.

Correct Answer: D

Community vote distribution

D (100%)

- cybe001

Highly Voted

1 year, 7 months ago

D is correct. Question has ""simple transformations, and some attributes will be combined" and Least development effort. Kinesis analytics can get data from Firehose, transform and write to S3
<https://docs.aws.amazon.com/kinesisanalytics/latest/java/examples-s3.html>
upvoted 42 times
- mawsman

1 year, 7 months ago

Best explanation here, kudos.
upvoted 3 times
- kakalotka

1 year, 6 months ago

I can't find any information that indicate Kinesis data analytics taking data from firehose
upvoted 1 times
- devsean

Highly Voted

1 year, 7 months ago

I think the answer is B. D would be correct if they didn't want to transform the legacy data from before the switch, but it seems like they do. Choosing D would mean that you'd have to use an EC2 instance or something else to transform the legacy data along with adding the Kinesis data analytics functionality. Also, there is no real-time requirement so daily transformation is fine.
upvoted 6 times
- ZSun

1 week, 5 days ago

If the question is "least cost" then B, but the question is "least developpe effort, then you want to keep original architerture. I agree that for daily ETL instead of real-time, and large dataset, B is better option.
upvoted 1 times
- HaiHN

1 year, 6 months ago

You can use Lambda instead of EC2. So D should be OK.
<https://aws.amazon.com/blogs/big-data/preprocessing-data-in-amazon-kinesis-analytics-with-aws-lambda/>
upvoted 1 times
- Valcilio

Most Recent

1 month, 3 weeks ago

Selected Answer: D

The best answer is to use a lambda, but the letter D can do it very good too in the absence of the lambda option.
upvoted 2 times

Huy

1 year, 5 months ago

The best way to transform data is before it arrives to S3 so D should be best answer. But D is not completed. It should have another Firehose to deliver results to S3.
upvoted 5 times

cloud_trail

1 year, 6 months ago

I go with D. A tough question, though. And C are definitely out. They key to the question is that it does not say that the transformed data needs to be stored again in S3. It just needs to be sent to the model for training after being transformed. So a Kinesis Data Analytics stream is appropriate to do the transformation.
upvoted 1 times

-   **harmanbirstudy** 1 year, 6 months ago

Legacy data -- Firehose -- Kinesis Analytics -- S3.This happens in near real time before the data ends up in S3.
--Legacy data -- Firehose -- S3 is already happening (mentioned in first line in question), adding Kinesis Data Analytics to do simple transformation joins using SQL on the incoming data is the LEAST amount of work needed.
Kinesis Data analytics can write o S3. here is the AWS link with working example.Even Though Udemy tutorial said it cannot write directly to S3 :) .

<https://docs.aws.amazon.com/kinesisanalytics/latest/java/examples-s3.html>
upvoted 3 times
-   **gamaX** 1 year, 6 months ago

It seems that LEAST developmnet effort:
<https://aws.amazon.com/fr/blogs/big-data/preprocessing-data-in-amazon-kinesis-analytics-with-aws-lambda/>

and GRETAST development effort:
<https://aws.amazon.com/fr/blogs/big-data/optimizing-downstream-data-processing-with-amazon-kinesis-data-firehose-and-amazon-emr-running-apache-spark/>
upvoted 1 times
-   **HaiHN** 1 year, 6 months ago

It's D

<https://aws.amazon.com/blogs/big-data/preprocessing-data-in-amazon-kinesis-analytics-with-aws-lambda/>

"In some scenarios, you may need to enhance your streaming data with additional information, before you perform your SQL analysis. Kinesis Analytics gives you the ability to use data from Amazon S3 in your Kinesis Analytics application, using the Reference Data feature. However, you cannot use other data sources from within your SQL query."
upvoted 1 times
-   **h_sahu** 1 year, 6 months ago

I believe, kinesis should be used only in case of live data stream and this is not the case here. So as per me D shouldn't be the answer. I think A should be the answer as AWS storage gateway is something which is used alongwith on premise applications to move data to s3. Then glue can be used to transform the data.
upvoted 1 times

  **cloud_trail** 1 year, 6 months ago

With option A, you would be changing the legacy data ingestion, a huge development effort. Remember, you're talking about 20,000 stores.
upvoted 2 times
-   **hans1234** 1 year, 6 months ago

It is D.
upvoted 1 times
-   **dikers** 1 year, 6 months ago

I think the answer is D, because require the LEAST amount of development effort.
upvoted 1 times
-   **roytruong** 1 year, 6 months ago

it's D, kinesis analytic can easily connect with firehose
upvoted 2 times
-   **dreemswang** 1 year, 7 months ago

why not A. it seems good to me
upvoted 2 times
-   **ExamTaker123456789** 1 year, 6 months ago

"require stores to capture data locally using S3 gateway" - for 20k stores this creates a HUUUGE operational overhead and development effort, definitely wrong
upvoted 2 times
-   **PRC** 1 year, 7 months ago

D is correct...rest all need some kind of manual intervention as well as they are not simple..Firehose allows transformation as well as moving into S3
upvoted 6 times
-   **devsean** 1 year, 7 months ago

I think the answer is B. D would be correct if they didn't want to transform the legacy data from before the switch, but it seems like they do. Choosing D would mean that you'd have to use an EC2 instance or something else to transform the legacy data along with adding the Kinesis data analytics functionality. Also, there is no real-time requirement so daily transformation is fine.
upvoted 2 times

  **hailiang** 1 year, 6 months ago

Its D, because with KDA you can transform the data with SQL while with EMR you need to write code, considering the requirement of "least development effort", so D
upvoted 3 times
-   **am7** 1 year, 7 months ago

can be B

upvoted 1 times

  **vetal** 1 year, 7 months ago

Amazon Kinesis Data Analytics can not send data to S3 directly - it needs something like Kinesis Data Firehose after it.

upvoted 3 times

A Machine Learning Specialist is building a convolutional neural network (CNN) that will classify 10 types of animals. The Specialist has built a series of layers in a neural network that will take an input image of an animal, pass it through a series of convolutional and pooling layers, and then finally pass it through a dense and fully connected layer with 10 nodes. The Specialist would like to get an output from the neural network that is a probability distribution of how likely it is that the input image belongs to each of the 10 classes.

Which function will produce the desired output?

- A. Dropout
- B. Smooth L1 loss
- C. Softmax
- D. Rectified linear units (ReLU)

Correct Answer: C

Community vote distribution

C (100%)

  **DonaldCMLIN** Highly Voted 1 year, 7 months ago

C might be much suitable
softmax is to turn numbers into probabilities.

<https://medium.com/data-science-bootcamp/understand-the-softmax-function-in-minutes-f3a59641e86d>
upvoted 29 times

  **rsimham** Highly Voted 1 year, 7 months ago

C is right. Softmax function is used for multi-class predictoins
upvoted 14 times

  **Valcilio** Most Recent 1 month, 3 weeks ago

Selected Answer: C
Multiclassification with probabilities is about softmax!
upvoted 1 times

  **vbal** 3 months, 3 weeks ago

Softmax is for probability distribution
upvoted 1 times

  **technoguy** 1 year, 5 months ago

it should be C. Softmax
Softmax converts outputs to Probabilites of each classification
upvoted 2 times

  **omar8024** 1 year, 6 months ago

absolutely C
upvoted 1 times

  **takahirokoyama** 1 year, 6 months ago

Absolute C.
upvoted 4 times

  **cloud_trail** 1 year, 6 months ago

This is as easy a question as you will likely see on the exam, Everyone has the right answer here.
upvoted 3 times

  **felbuch** 1 year, 6 months ago

C --> Softmax.

Let's go over the alternatives:
A. Dropout --> Not really a function, but rather a method to avoid overfitting. It consists of dropping some neurons during the training process, so that the performance of our algorithm does not become very dependent on any single neuron.
B. Smooth L1 loss --> It's a loss function, thus a function to be minimized by the entire neural network. It's not an activation function.
C. Softmax --> This is the traditional function used for multi-class classification problems (such as classifying an animal into one of 10 categories)
D. Rectified linear units (ReLU) --> This activation function is often used on the first and intermediate (hidden) layers, not on the final layer. In any case, it wouldn't make sense to use it for classification because its values can exceed 1 (and probabilities can't)
upvoted 8 times

-  **MOMoez** 1 year, 6 months ago

C, Softmax is the best suitable answer
Ref: The softmax function, also known as softargmax[1]:184 or normalized exponential function,[2]:198 is a generalization of the logistic function to multiple dimensions. It is used in multinomial logistic regression and is often used as the last activation function of a neural network to normalize the output of a network to a probability distribution over predicted output classes, based on Luce's choice axiom.
upvoted 1 times
-  **ybad** 1 year, 6 months ago

You guys are right, the answer is C since it automatically provides the output with a confidence interval...

Relu could be used as well but it needs to be coded in to provide the probabilities

<https://medium.com/@himanshuxd/activation-functions-sigmoid-relu-leaky-relu-and-softmax-basics-for-neural-networks-and-deep-8d9c70eed91e>
upvoted 1 times
-  **yeetusdeleetus** 1 year, 6 months ago

Definitely C
upvoted 1 times
-  **bidbs** 1 year, 6 months ago

Definitely softmax.
upvoted 1 times
-  **hans1234** 1 year, 6 months ago

Are you sure it is C?
The output should be "[the probability that] the input image belongs to each of the 10 classes." And not the most likely class with the highest probability, which would be the result of softmax layer.
upvoted 1 times
-  **hans1234** 1 year, 6 months ago

Yes, softmax returns indeed a vector of probabilities.
upvoted 1 times
-  **roytruong** 1 year, 6 months ago

C, everyone with basic knowledge in neural network can easily see that
upvoted 2 times
-  **deep_n** 1 year, 7 months ago

Not even debatable!
C!!!!
upvoted 3 times
-  **jasonsunbao** 1 year, 7 months ago

It is softmax. ReLu is used for activation, not for classification
upvoted 1 times
-  **ejj** 1 year, 6 months ago

softmax is also activation functioin
upvoted 2 times

A Machine Learning Specialist trained a regression model, but the first iteration needs optimizing. The Specialist needs to understand whether the model is more frequently overestimating or underestimating the target.

What option can the Specialist use to determine whether it is overestimating or underestimating the target value?

- A. Root Mean Square Error (RMSE)
- B. Residual plots
- C. Area under the curve
- D. Confusion matrix

Correct Answer: B

Community vote distribution

B (100%)

- vetal

Highly Voted

1 year, 7 months ago

RMSE says about the error value but not the sign of error. The question is to find whether the model overestimates or underestimates - I guess residual plots clearly show that

answer B

upvoted 35 times
- rsimham

Highly Voted

1 year, 7 months ago

Answer is B. Residual plot distribution indicates over or under-estimations

upvoted 14 times
- Valcilio

Most Recent

1 month, 3 weeks ago

Selected Answer: B

Residual plots shows mistake by mistake!

upvoted 1 times
- vetaal

1 year, 3 months ago

Selected Answer: B

B - Residual plots it is - <https://docs.aws.amazon.com/machine-learning/latest/dg/regression-model-insights.html>

upvoted 1 times
- felbuch

1 year, 5 months ago

Residual Plots (B).

AUC and Confusion Matrices are used for classification problems, not regression.

And RMSE does not tell us if the target is being over or underestimated, because residuals are squared! So we actually have to look at the residuals themselves. And that's B.

upvoted 4 times
- cnethers

1 year, 5 months ago

Root Mean Square Error (RMSE) is the standard deviation of the residuals (prediction errors). Residuals are a measure of how far from the regression line data points are; RMSE is a measure of how spread out these residuals are. In other words, it tells you how concentrated the data is around the line of best fit. Root mean square error is commonly used in climatology, forecasting, and regression analysis to verify experimental results.

1) Squaring the residuals.

2) Finding the average of the residuals.

3) Taking the square root of the result.

upvoted 1 times
- cnethers

1 year, 5 months ago

Residual Plots (B). would have to be my answer

upvoted 1 times
- Thai_Xuan

1 year, 5 months ago

residual plot

<https://docs.aws.amazon.com/machine-learning/latest/dg/regression-model-insights.html>

upvoted 5 times
- syu31svc

1 year, 6 months ago

<https://stattrek.com/statistics/dictionary.aspx?definition=residual%20plot#:~:text=A%20residual%20plot%20is%20a,nonlinear%20model%20is%20more%20appropriate.>

Answer is B
upvoted 1 times

  **Antriksh** 1 year, 6 months ago
without a second thought residual plot
upvoted 2 times

  **qururu** 1 year, 6 months ago
The answer is B. Refer to Exercise 7.2.1.A
[https://stats.libretexts.org/Bookshelves/Introductory_Statistics/Book%3A_OpenIntro_Statistics_\(Diez_et_al\)/07%3A_Introduction_to_Linear_Regression/7.02%3A_Line_Fitting%2C_Residuals%2C_and_Correlation](https://stats.libretexts.org/Bookshelves/Introductory_Statistics/Book%3A_OpenIntro_Statistics_(Diez_et_al)/07%3A_Introduction_to_Linear_Regression/7.02%3A_Line_Fitting%2C_Residuals%2C_and_Correlation)
upvoted 1 times

  **C10ud9** 1 year, 6 months ago
Residual plot it is Option B
upvoted 1 times

  **roytruong** 1 year, 6 months ago
Residual plot
upvoted 2 times

  **deep_n** 1 year, 6 months ago
B is the correct answer!!!!
RMSE has the S in it that is square... that vanishes the above below factor of the prediction.
Answers C and D are for other type of problems
upvoted 4 times

  **swagy** 1 year, 6 months ago
It should be B. The residual plot will be give whether the target value is overestimated or underestimated.
upvoted 1 times

  **Jayraam** 1 year, 6 months ago
Answer is C.

<https://www.youtube.com/watch?v=MrjWcywVEiU>
upvoted 2 times

  **ExamTaker123456789** 1 year, 6 months ago
Answer is B.
Your vid shows a technique that is useful for defining integrals and has NOTHING to do linear regression. Also, it over-/underestimates the area under the curve, NOT the target value.
upvoted 2 times

  **cloud_trail** 1 year, 5 months ago
Good grief, AUC is used for classification not regression.
upvoted 1 times

  **PRC** 1 year, 6 months ago
B..Residual helps to find out whether the model is underestimating or overestimating
upvoted 3 times

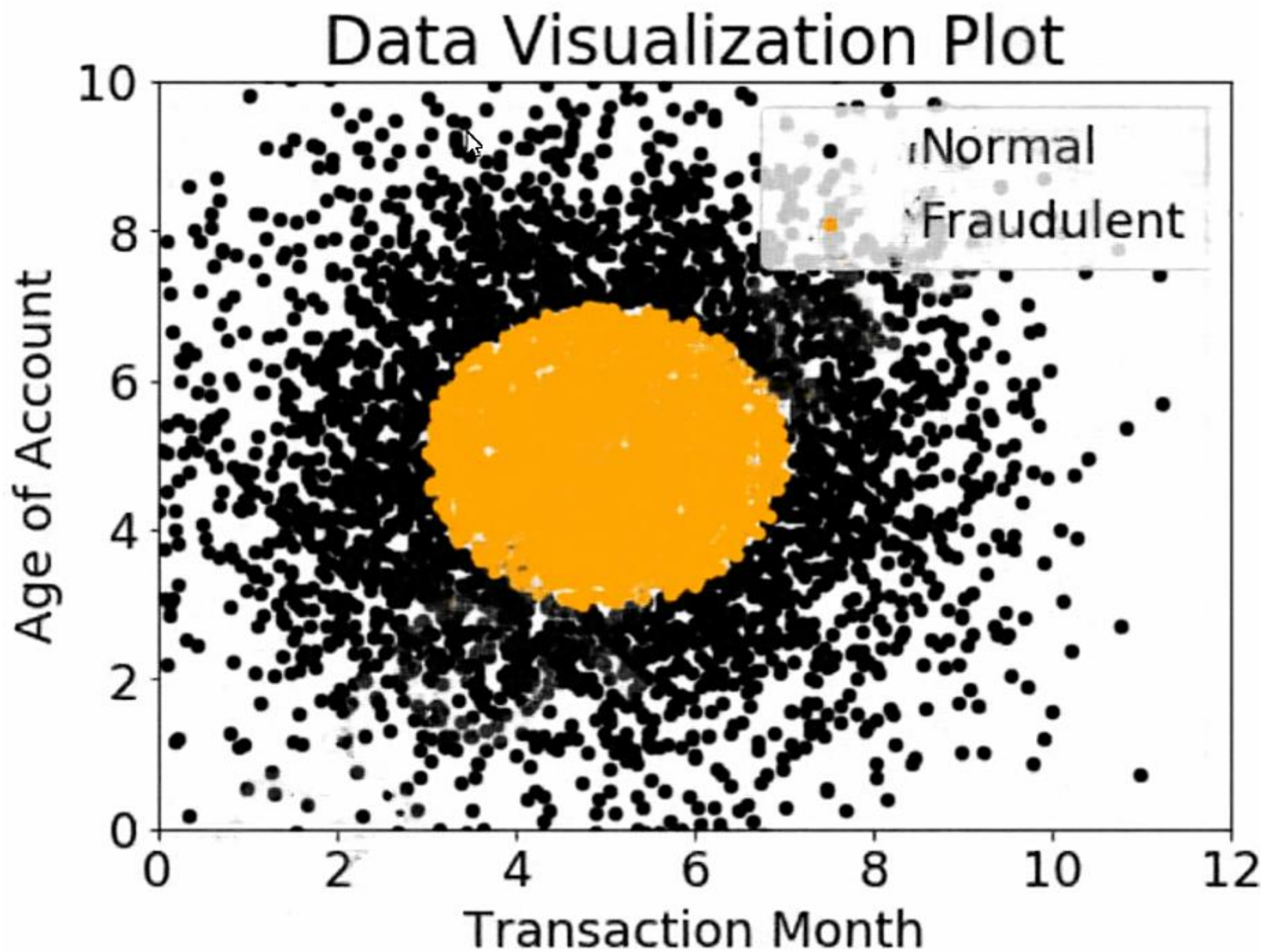
  **AKT** 1 year, 7 months ago
answer is B
upvoted 2 times

  **Phong** 1 year, 7 months ago
Go for B. Residual spots can handle this problem
upvoted 4 times

Question #25

Topic 1

A company wants to classify user behavior as either fraudulent or normal. Based on internal research, a Machine Learning Specialist would like to build a binary classifier based on two features: age of account and transaction month. The class distribution for these features is illustrated in the figure provided.



Based on this information, which model would have the HIGHEST recall with respect to the fraudulent class?

- A. Decision tree
- B. Linear support vector machine (SVM)
- C. Naive Bayesian classifier
- D. Single Perceptron with sigmoidal activation function

Correct Answer: C

Community vote distribution

C (86%)

14%

E_aws Highly Voted 1 year, 6 months ago

C is the correct answer because gaussian naive Bayes can do this nicely.
upvoted 12 times

E_aws 1 year, 6 months ago

of course it doesn't mention the gaussian here and refers to naive bayes in general, but I'm still positive with C.
upvoted 1 times

blubb Highly Voted 1 year, 6 months ago

Answer should be A:
B: LINEAR SVM is a linear classifier
-> All of these have a linear decision boundary (so it's just a line $y = mx + b$). This leads to a bad recall and so A must be the right choice.
upvoted 6 times

Ahmedhadi_ Most Recent 2 days ago

Selected Answer: A

it's A. Decision Trees can do this easily
upvoted 1 times

  **hug_c0sm0s** 2 months ago

Selected Answer: C

This is the answer given by Bing Chat:Based on the figure provided in the link and the information given in your message, it appears that a Naive Bayesian classifier would have the highest recall with respect to the fraudulent class. This is because a Naive Bayesian classifier can handle overlapping class distributions and can work well when there is a clear separation between classes as shown in the figure. However, it's important to note that model selection should also take into account other factors such as precision and overall accuracy.
upvoted 1 times

  **bakarys** 2 months, 1 week ago

Selected Answer: C

C. Naive Bayesian classifier.

Naive Bayes classifiers are based on Bayes' theorem and are particularly useful for binary classification problems where the features are categorical or numerical. They are known to perform well on small datasets and are relatively fast to train and predict.

In this case, there are two features, the age of the account and the transaction month, which can be used to classify user behavior as fraudulent or normal. Naive Bayes classifiers are suitable for this type of data as they work well with categorical features.
upvoted 1 times

  **vbal** 3 months, 3 weeks ago

No way the answer could be c in my opinion. It is A. Naive is used for text based and sentiment kind of problems and a best fit for problem like spam emails. In case of spam problem FN high is fine FP high is not good. if FN is high in the provided problem it would reduce your recall. $\text{recall} = \frac{tp}{tp+fn}$.
upvoted 1 times

  **mavios** 4 months ago

The trick in the sentence `Highest recall` which means the least amount of False Negatives. With decision trees, you need a tree of depth 4 to achieve 100% recall. So I will go with A. Naive Bayes might not guarantee that because it is probability based.
upvoted 2 times

  **mavios** 4 months ago

Actually, you can probably get 100% recall with decision tree of depth 2, and guarantee 100% recall with decision tree of depth 3. The idea is, your model can say it is Fraudulent even if it is not, but cannot say it is Normal when it is Fraudulent. A decision tree with two rules that checks either the age of the account is between 3 and 7 or the month of transaction is between 3 and 7 will give 0 False Negatives and 100% recall every single time. So answer should be A
upvoted 1 times

  **ckkobe24** 11 months, 3 weeks ago

Selected Answer: C

Decision tree need to see what is the implementation , so I will go for C
upvoted 1 times

  **geekgirl007** 1 year, 3 months ago

Selected Answer: C

answer is C. Naive Bayesian gives the highest recall, check out this article comparing it to decision trees:
https://thesai.org/Downloads/Volume4No11/Paper_5-Performance_Comparison_between_Na%C3%AFve_Bayes.pdf
upvoted 3 times

  **Dr_Kiko** 1 year, 6 months ago

B.
Recall: The ability of a model to find all the relevant cases within a data set.
Now open any textbook on SVM - it's a classic picture shown
upvoted 1 times

  **Tony_1406** 7 months, 1 week ago

But the available answer says Linear SVM so you are restricted linear kernels only.
upvoted 3 times

  **BeastX** 1 year, 6 months ago

I would say A
upvoted 2 times

  **INNN** 1 year, 6 months ago

Answer is correct. It has to be C. Check out there:
<https://stackoverflow.com/questions/21468469/logistic-regression-and-naive-bayes-for-this-dataset>
The variance in yellow is smaller and bigger in black. NB can make a perfect prediction.

On the other hand, tree models are known for inefficiency on round decision boundary, since you have to got thousand small rectangles to approximate the round shape.
upvoted 5 times

  **eeah** 1 year ago

Gaussian NB would be perfect here.
upvoted 1 times

  **yummytaco** 1 year, 6 months ago

I think it's decision tree since it's a non linear and Naive Bayes is a linear classifier based on:
<https://www.cs.cornell.edu/courses/cs4780/2018fa/lectures/lecturenote05.html>
upvoted 2 times

  **Vita_Rasta84444** 1 year, 6 months ago

It is B. Only SVM could create the most non-linear bound between fraudulent and non fraudulent and hence achieve the highest recall.
upvoted 1 times

  **gcpwhiz** 1 year, 6 months ago

B is linear SVM. To use SVM here you need non-linear w hyperplane
upvoted 5 times

  **jerto97** 1 year, 6 months ago

No, you'd need a kernel function to classify that boundary
upvoted 2 times

  **achiko** 1 year, 7 months ago

why not a decision tree? its also non linear
upvoted 3 times

  **rhuanca** 11 months, 1 week ago

Maybe it is a random forest but with a single tree no
upvoted 1 times

  **SophieSu** 1 year, 7 months ago

The point of this question is "Non-Linear" clustering. Naive Bayes classifier in general is not linear.
upvoted 4 times

  **cnethers** 1 year, 7 months ago

Real time Prediction: Naive Bayes is an eager learning classifier and it is sure fast. Thus, it could be used for making predictions in real time.
Multi class Prediction: This algorithm is also well known for multi class prediction feature. Here we can predict the probability of multiple classes of target variable.
Text classification/ Spam Filtering/ Sentiment Analysis: Naive Bayes classifiers mostly used in text classification (due to better result in multi class problems and independence rule) have higher success rate as compared to other algorithms. As a result, it is widely used in Spam filtering (identify spam e-mail) and Sentiment Analysis (in social media analysis, to identify positive and negative customer sentiments)
upvoted 1 times

A Machine Learning Specialist kicks off a hyperparameter tuning job for a tree-based ensemble model using Amazon SageMaker with Area Under the ROC Curve (AUC) as the objective metric. This workflow will eventually be deployed in a pipeline that retrains and tunes hyperparameters each night to model click-through on data that goes stale every 24 hours.

With the goal of decreasing the amount of time it takes to train these models, and ultimately to decrease costs, the Specialist wants to reconfigure the input hyperparameter range(s).

Which visualization will accomplish this?

- A. A histogram showing whether the most important input feature is Gaussian.
- B. A scatter plot with points colored by target variable that uses t-Distributed Stochastic Neighbor Embedding (t-SNE) to visualize the large number of input variables in an easier-to-read dimension.
- C. A scatter plot showing the performance of the objective metric over each training iteration.
- D. A scatter plot showing the correlation between maximum tree depth and the objective metric.

Correct Answer: B

Community vote distribution

D (75%)

C (25%)

  **cloud_trail** Highly Voted 1 year, 6 months ago

This is a very tricky question. The idea is to reconfigure the ranges of the hyperparameters. A refers to a feature, not a hyperparameter. A is out. C refers to training the model, not optimizing the range of hyperparameters. C is out. Now it gets tricky. D will let you find determine what the approximately best tree depth is. That's good. That's what you're trying to do but it's only one of many hyperparameters. It's the best choice so far. B is tricky. t-SNE does help you visualize multidimensional data but option B refers to input variables, not hyperparameters. For this very tricky question, I would do with D. It's the only one that accomplishes the task of limiting the range of a hyperparameter, even if it is only one of them.

upvoted 40 times

  **cnethers** 1 year, 6 months ago

It's good to see someone keeping a thoughtful and curious mind to this question. I too have the same conclusion, not an easy question.

upvoted 3 times

  **ovokpus** 10 months, 1 week ago

But, how do you optimize hyperparameters without training experiments? That is why C is the best option. You get a value for each unique combination of hyperparameters.

upvoted 1 times

  **Dr_Kiko** 1 year, 5 months ago

B is also wrong as t-SNE picture is not actionable - good visual but ... that's it. try pictures here <https://lvdmaaten.github.io/tsne/>

upvoted 1 times

  **AddiWei** 1 year, 2 months ago

When you are tuning hyperparameters you are literally training multiple models and searching for the best ones.

upvoted 2 times

  **heihei** Highly Voted 1 year, 7 months ago

B doesn't make sense

I think it's D

upvoted 14 times

  **Valcilio** Most Recent 1 month, 3 weeks ago

Selected Answer: D

It's about parameters, not about dimensionality.

upvoted 1 times

  **hug_c0sm0s** 2 months ago

Selected Answer: D

BING chat chooses this answer, and provides an explanation:The correct answer is D. A scatter plot showing the correlation between maximum tree depth and the objective metric. This visualization will help the Machine Learning Specialist to understand how changes in maximum tree depth affect the performance of the model with respect to the objective metric (AUC). By analyzing this relationship, the Specialist can adjust the range of maximum tree depth values used during hyperparameter tuning to decrease training time and costs.

upvoted 1 times

  **bakarys** 2 months, 1 week ago

Selected Answer: C

- C

This visualization will allow the Machine Learning Specialist to track the performance of the model over time as different hyperparameter configurations are tried. By analyzing this plot, the Specialist can identify which hyperparameters are leading to better model performance and adjust the input hyperparameter range(s) accordingly, with the goal of decreasing the time it takes to train the models and decrease costs.

upvoted 1 times

  **cpal012** 3 weeks, 6 days ago

Agree, this is my thought process too. I hope the exam is no where near this nebulous

upvoted 1 times

  **mavios** 4 months ago

The correct answer is definitely D. Because it is talking about Hyper Parameter, it also is requiring the specialist to reduce time and cost. For an ensemble of trees model, the only hyper parameters that affect cost is the number of trees and the maximum depth of trees. and only D is speaking to these two facts

upvoted 2 times

  **Peeking** 4 months, 3 weeks ago

I agree with D, just that it mentions only one hyperparameter.

upvoted 1 times

  **[Removed]** 10 months, 3 weeks ago

Selected Answer: D

d is correct

upvoted 1 times

  **John_Pongthorn** 1 year, 2 months ago

hyperparameter combination, this phrase should appear in one of the choices

upvoted 1 times

  **John_Pongthorn** 1 year, 2 months ago

I read form somewhere reliable.
Awnseer Choice shoud be
Use a scatter plot to visualize the results for each Area Under the Curve (AUC)-hyperparameter combination.

upvoted 1 times

  **AddiWei** 1 year, 2 months ago

Objective is "reduce the time it takes to train". You reduce the time by reducing training iterations.

upvoted 1 times

  **technoguy** 1 year, 5 months ago

Linear SVM is wrong. since the data cannot be separated with line

upvoted 1 times

  **Dr_Kiko** 1 year, 5 months ago

A makes no sense; B is visualization (educate yourself how a t-sne picture looks like - good luck using it for tuning). Leaning towards D

upvoted 1 times

  **Huy** 1 year, 6 months ago

B is answer. D is identify only one hyperparameter. I think there are several hyperparameters need to be combined to get the best AUC value. And that is the t-SNE used for.

upvoted 1 times

  **Huy** 1 year, 6 months ago

Ah, may be I was wrong. B is to "visualize the large number of input variables". D makes sense regarding that max_depth is to simplify the model and decrease the training time.

upvoted 1 times

  **SophieSu** 1 year, 6 months ago

Focus on 2 points: Tree Model and Hyperparameter tuning. Definitely D

upvoted 4 times

  **harmanbirstudy** 1 year, 6 months ago

I think its A ..
<https://www.analyticsvidhya.com/blog/2016/03/complete-guide-parameter-tuning-xgboost-with-codes-python/>

upvoted 1 times

  **ybad** 1 year, 6 months ago

B is right...

it actually does make sense to have B, it is used for dimensional reductions, but it does this by properly understanding the parameters of the model and visualizing the multi dimensional data into clusters, thus understanding what the model is doing to optimize itself,

Also, reducing the dimensionality of a model allows it to train faster, thus reducing the cost, which works for the required goal...
https://en.wikipedia.org/wiki/T-distributed_stochastic_neighbor_embedding

upvoted 1 times

Question #27

Topic 1

A Machine Learning Specialist is creating a new natural language processing application that processes a dataset comprised of 1 million sentences. The aim is to then run Word2Vec to generate embeddings of the sentences and enable different types of predictions. Here is an example from the dataset:

"The quck BROWN FOX jumps over the lazy dog."

Which of the following are the operations the Specialist needs to perform to correctly sanitize and prepare the data in a repeatable manner? (Choose three.)

- A. Perform part-of-speech tagging and keep the action verb and the nouns only.
- B. Normalize all words by making the sentence lowercase.
- C. Remove stop words using an English stopword dictionary.
- D. Correct the typography on "quck" to "quick."
- E. One-hot encode all words in the sentence.
- F. Tokenize the sentence into words.

Correct Answer: BCF

Community vote distribution

BCF (100%)

- ozan11

Highly Voted

1 year, 7 months ago

B C F should be correct.

upvoted 35 times
- BigEv

Highly Voted

1 year, 7 months ago

I will select B, C, F

1- Apply words stemming and lemmatization

2- Remove Stop words

3- Tokenize the sentences

<https://towardsdatascience.com/nlp-extracting-the-main-topics-from-your-dataset-using-lda-in-minutes-21486f5aa925>

upvoted 24 times
- Valcilio

Most Recent

1 month, 3 weeks ago

Selected Answer: BCF

Data need to be tokenized and cleaned!

upvoted 1 times
- Aninina

4 months ago

Selected Answer: BCF

B, C F is the correct

upvoted 2 times
- SophieSu

1 year, 6 months ago

BCF correct. D is not correct (Pay attention to "in a repeatable manner" in the question.)

upvoted 2 times
- cloud_trail

1 year, 6 months ago

B/C/F. D should not be performed because spell check is a subjective thing. You don't know for sure what the word was supposed to be if you have a typo.

upvoted 2 times
- harmanbirstudy

1 year, 6 months ago

I saw this exact question on "whizlabs" practice exam and correct options were B/C/F

upvoted 1 times
- GeeBeeEl

1 year, 6 months ago

<https://towardsdatascience.com/an-implementation-guide-to-word2vec-using-numpy-and-google-sheets-13445eebd281>

Data Preparation — Define corpus, clean, normalise and tokenise words

To begin, we start with the following corpus:

"natural language processing and machine learning is fun and exciting"

For simplicity, we have chosen a sentence without punctuation and capitalization. Also, we did not remove stop words "and" and "is".

In reality, text data are unstructured and can be "dirty". Cleaning them will involve steps such as

o removing stop words,

o removing punctuations,

o convert text to lowercase (actually depends on your use-case),
o replacing digits, etc.
o After preprocessing, we then move on to tokenising the corpus
Answer: B, C, F

upvoted 6 times

  **cnethers** 1 year, 6 months ago

BCF is 100% correct
upvoted 2 times

  **Antriksh** 1 year, 6 months ago

Correct answers are B, C and F
upvoted 2 times

  **TuanAnh** 1 year, 6 months ago

The correct answer is B, C and F
A: POS tagging has nothing to do with word2vec
D: fixing "quck" to "quick" only works for that specific word
F: word2vec can use CBOW or skipgram, so no need to have one-hot decoding here
upvoted 4 times

  **TuanAnh** 1 year, 6 months ago

sorry E: word2vec can use CBOW or skipgram, so no need to have one-hot decoding here
upvoted 4 times

  **PRC** 1 year, 6 months ago

BCF is correct
upvoted 2 times

  **AKT** 1 year, 7 months ago

B, C F correct
upvoted 2 times

  **Phong** 1 year, 7 months ago

B, C, and F are correct answers. I have done this question many times in many practice tests.
upvoted 12 times

  **tap123** 1 year, 7 months ago

B, C, F are my choice. D is also possible but not as widely used as others.
upvoted 3 times

  **cybe001** 1 year, 7 months ago

Why C is not included in the answer? ABCD, all are correct answers
upvoted 1 times

  **halfway** 1 year, 6 months ago

"choose three"
upvoted 1 times

A company is using Amazon Polly to translate plaintext documents to speech for automated company announcements. However, company acronyms are being mispronounced in the current documents.

How should a Machine Learning Specialist address this issue for future documents?

- A. Convert current documents to SSML with pronunciation tags.
- B. Create an appropriate pronunciation lexicon.
- C. Output speech marks to guide in pronunciation.
- D. Use Amazon Lex to preprocess the text files for pronunciation

Correct Answer: A

Reference:
<https://docs.aws.amazon.com/polly/latest/dg/ssml.html>

Community vote distribution

B (77%)

A (23%)

- VB

Highly Voted

1 year, 7 months ago

SSML is specific to that particular document, like W3C can be pronounced as "World Wide Web Consortium" using `<sub alias="World Wide Web Consortium">W3C</sub>` in that specific document and when you create a new document, you need to format again. But with LEXICONS, you can upload a lexicon file once and ALL the FUTURE documents can just have W3C and that will be pronounced as "World Wide Web Consortium".. so answer is B, because the question asks for "future" documents.

upvoted 38 times
- cloud_trail

1 year, 6 months ago

For the exact reason you state, the correct answer is A. For every different document, a particular acronym may mean something different so you must have a solution that is document-specific.

upvoted 3 times
- ovokpus

10 months, 1 week ago

It is the same business, so the acronyms are not expected to change from document to document

upvoted 3 times
- Madwyn

1 year, 6 months ago

A.The document section for "Pronouncing Acronyms and Abbreviations".
Source: <https://docs.aws.amazon.com/polly/latest/dg/supportedtags.html>

upvoted 3 times
- cybe001

Highly Voted

1 year, 7 months ago

I think the answer is B.
<https://docs.aws.amazon.com/polly/latest/dg/managing-lexicons.html>
<https://www.smashingmagazine.com/2019/08/text-to-speech-aws/>

upvoted 18 times
- Tony_1406

7 months, 1 week ago

Lifted from the above link - "Your text might include an acronym, such as W3C. You can use a lexicon to define an alias for the word W3C so that it is read in the full, expanded form (World Wide Web Consortium)."
Clearly this is the same use case.

upvoted 1 times
- Sylzys

Most Recent

1 month, 3 weeks ago

Selected Answer: B

Key here being "for future documents", answer should be B as SSML is for a specific document only

upvoted 2 times

Chelseajcole

2 months, 1 week ago

this should be multiple choice question which answer is a AND b

upvoted 1 times

bakarys

2 months, 1 week ago

Selected Answer: B

response B
- A pronunciation lexicon is a list of words and their correct phonetic pronunciation that can be used to improve the accuracy of text-to-speech conversion. In this case, the Machine Learning Specialist can create a custom lexicon for the company's acronyms and upload it to Amazon Polly. This will ensure that the acronyms are pronounced correctly in the future announcements.
- <https://www.examtopycs.com/exams/amazon/aws-certified-machine-learning-specialty/custom-view/>

60/406

upvoted 2 times

  **SK27** 4 months, 3 weeks ago

Selected Answer: B

Should be B

upvoted 2 times

  **masoa3b** 5 months, 3 weeks ago

Selected Answer: B

With Amazon Polly's custom lexicons or vocabularies, you can modify the pronunciation of particular words, such as company names, acronyms, foreign words, etc. To customize these pronunciations, you upload an XML file with lexical entries. {rpmimcoatopm ;exocpms enable you to customize the pronunciation of words. Amazon Polly provides API operations that you can use to store lexicons in an AWS region. Those lexicons are then specific to that particular region.

References:

<https://docs.aws.amazon.com/polly/latest/dg/managing-lexicons.html>

<https://aws.amazon.com/blogs/machine-learning/create-accessible-training-with-initiafy-and-amazon-polly/>

upvoted 3 times

  **uninit** 7 months, 1 week ago

Selected Answer: B

Ref: <https://docs.aws.amazon.com/polly/latest/dg/managing-lexicons.html>

Your text might include an acronym, such as W3C. You can use a lexicon to define an alias for the word W3C so that it is read in the full, expanded form (World Wide Web Consortium).

upvoted 2 times

  **gnolam** 7 months, 3 weeks ago

Selected Answer: A

According to A Cloud Guru practice exam, which has nearly identical question:

Using SSML-enhanced input text gives you additional control over how Amazon Polly generates speech from the text you provide. Using these tags allows you to substitute a different word (or pronunciation) for selected text such as an acronym or abbreviation. You can also create a dictionary lexicon to apply to any future tasks instead of apply SSML to each individual document.

upvoted 3 times

  **vanluigi** 11 months, 2 weeks ago

Lexicons give you additional control over how Amazon Polly pronounces words uncommon to the selected language. For example, you can specify the pronunciation using a phonetic alphabet.

Hence, the correct answer is: Use pronunciation lexicons.

The option that says: Use a viseme Speech Mark is incorrect as this feature is just used to synchronize speech with facial animation (lip-syncing) or to highlight written words as they're spoken.

The option that says: Convert the scripts into Speech Synthesis Markup Language (SSML) and use the pronunciation tag is incorrect because Amazon Polly does not support this SSML tag.

The option that says: Convert the documents into Speech Synthesis Markup Language (SSML) and use the emphasis tag to guide the pronunciation is incorrect as this type of tag is simply used to emphasize words by changing the speaking rate and volume of the speech.

upvoted 1 times

  **vanluigi** 11 months, 2 weeks ago

tutorials Dodjo answer

upvoted 2 times

  **ckkobe24** 1 year ago

Selected Answer: B

SSML Tags does not clearly state it can help with acronyms issue??

upvoted 3 times

  **clawo** 1 year, 3 months ago

B Lexicons all the way

upvoted 1 times

  **bnagaraja9099** 1 year, 3 months ago

Selected Answer: B

SSML is specific to that particular doument, like W3C an be pronounced as "World Wide Web Consortium" using _{W3C} in that specific document and when you create a new document, you need to format again. But with LEXICONS, you can upload a lexicon file once and ALL the FUTURE documents can just have W3C and that will be pronounced as "World Wide Web Consortium".. so answer is B, because the question asks for "future" documents.

upvoted 3 times

  **geekgirl007** 1 year, 3 months ago

Selected Answer: A

A is correct. because SSML supported tags for pronouncing acronyms: <https://docs.aws.amazon.com/polly/latest/dg/supportedtags.html>

upvoted 2 times

- 

 **ideapress** 1 year, 6 months ago

B
Your text might include an acronym, such as W3C. You can use a lexicon to define an alias for the word W3C so that it is read in the full, expanded form (World Wide Web Consortium).
<https://docs.aws.amazon.com/polly/latest/dg/managing-lexicons.html>
upvoted 2 times
- 

 **AShahine21** 1 year, 6 months ago

It should be B based on: <https://docs.aws.amazon.com/polly/latest/dg/managing-lexicons.html>
1. it will solve the current problem
2. it will be helpful for Future documents
upvoted 2 times
- 

 **j2_me** 1 year, 6 months ago

<https://gigaom.com/2017/04/24/give-your-products-the-power-of-speech-using-amazon-polly/>
A is the correct answer , reasons
1) convert to SSML document
2) Associate custom pronunciation lexicons to SSML
upvoted 1 times

An insurance company is developing a new device for vehicles that uses a camera to observe drivers' behavior and alert them when they appear distracted. The company created approximately 10,000 training images in a controlled environment that a Machine Learning Specialist will use to train and evaluate machine learning models.

During the model evaluation, the Specialist notices that the training error rate diminishes faster as the number of epochs increases and the model is not accurately inferring on the unseen test images.

Which of the following should be used to resolve this issue? (Choose two.)

- A. Add vanishing gradient to the model.
- B. Perform data augmentation on the training data.
- C. Make the neural network architecture complex.
- D. Use gradient checking in the model.
- E. Add L2 regularization to the model.

Correct Answer: *BE*

Community vote distribution

BE (100%)

- vetal**

Highly Voted

1 year, 7 months ago

The model must have been overfitted. Regularization helps to solve the overfitting problem in machine learning (as well as data augmentation). Correct answers should be BE.

upvoted 33 times
- jasonsunbao**

1 year, 7 months ago

agree on BE

upvoted 3 times
- rajs**

1 year, 7 months ago

Agreed 100%

upvoted 5 times
- benson2021**

Highly Voted

1 year, 6 months ago

Answer: BE

<https://www.kdnuggets.com/2019/12/5-techniques-prevent-overfitting-neural-networks.html>

5 techniques to prevent overfitting:

1. Simplifying the model

2. Early stopping

3. Use data argumentation

4. Use regularization

5. Use dropouts

upvoted 9 times
- John_Pongthorn**

Most Recent

1 year, 2 months ago

B. Perform data augmentation on the training data. (it should add validation data as well)

data should be distributed among train validation and test.

upvoted 1 times
- KM226**

1 year, 3 months ago

Selected Answer: BE

Answer B&E looks good

upvoted 2 times
- engomaradel**

1 year, 6 months ago

B & E is the correct ans

upvoted 1 times
- roytruong**

1 year, 6 months ago

BE is exact

upvoted 3 times
- stamarpadar**

1 year, 6 months ago

BE are the correct answers

upvoted 4 times

-  **VB**

1 year, 6 months ago

Looks like B and D are correct.. For D -> <https://www.youtube.com/watch?v=P6EtCVrvYPU>

upvoted 3 times
-  **C10ud9**

1 year, 6 months ago

gradient checking doesn't resolve the issue, but adding it will confirm / deny the issue. So, it helps to validate the issue but not resolve. I would say B, E are correct

upvoted 3 times
-  **VB**

1 year, 6 months ago

L2 regularization tries to reduce the possibility of overfitting by keeping the values of the weights and biases small.

upvoted 3 times
-  **hughhughhugh**

1 year, 6 months ago

why not because of vanishing gradient?

upvoted 1 times
-  **lt626**

3 months, 2 weeks ago

Vanishing gradients are a problem when training a NN. Answer A mentions that the solution should be to add that, which is not possible. Correct solution is BE.
<https://www.kdnuggets.com/2022/02/vanishing-gradient-problem.html>

upvoted 1 times
-  **PRC**

1 year, 7 months ago

This is L2 Regularization....Do you think this is the right answer?

upvoted 1 times
-  **WWODIN**

1 year, 7 months ago

agree BE

upvoted 3 times

When submitting Amazon SageMaker training jobs using one of the built-in algorithms, which common parameters **MUST** be specified? (Choose three.)

- A. The training channel identifying the location of training data on an Amazon S3 bucket.
- B. The validation channel identifying the location of validation data on an Amazon S3 bucket.
- C. The IAM role that Amazon SageMaker can assume to perform tasks on behalf of the users.
- D. Hyperparameters in a JSON array as documented for the algorithm used.
- E. The Amazon EC2 instance class specifying whether training will be run using CPU or GPU.
- F. The output path specifying where on an Amazon S3 bucket the trained model will persist.

Correct Answer: AEF

Community vote distribution

CEF (50%)

ACD (17%)

ACF (17%)

Other

-   **DonaldCMLIN**

Highly Voted

 1 year, 7 months ago

THE ANSWER SHOUD BE CEF
IAM ROLE, INSTANCE TYPE, OUTPUT PATH
upvoted 26 times
-   **HaiHN** 1 year, 6 months ago

Should be C, E, F

From the SageMaker notebook example:
https://github.com/aws/amazon-sagemaker-examples/blob/master/introduction_to_amazon_algorithms/semantic_segmentation_pascalvoc/semantic_segmentation_pascalvoc.ipynb
Create the sagemaker estimator object.

ss_model = sagemaker.estimator.Estimator(training_image,
role,
train_instance_count = 1,
train_instance_type = 'ml.p3.2xlarge',
train_volume_size = 50,
train_max_run = 360000,
output_path = s3_output_location,
base_job_name = 'ss-notebook-demo',
sagemaker_session = sess)
upvoted 9 times
-   **uninit** 2 months, 4 weeks ago

It says InstanceClass - CPU/GPU in the question, not InstanceType
upvoted 3 times
-   **hamimelon** 4 months, 1 week ago

Why not A? You don't need to tell Sagemaker where the training data is located?
upvoted 1 times
-   **ZSun** 2 weeks, 1 day ago

You need to specify the InputDataConfig, but it does not need to be "S3"
I think the reason why A and B are wrong, not because data location is not required, but because it doesn't need to be S3, it can be Amazon S3, EFS, or FSx location
upvoted 1 times
-   **VB**

Highly Voted

 1 year, 6 months ago

From here https://docs.aws.amazon.com/zh_tw/sagemaker/latest/dg/API_CreateTrainingJob.html .. the only "Required: Yes" attributes are:
1. AlgorithmSpecification (in this TrainingInputMode is Required - i.e. File or Pipe)
2. OutputDataConfig (in this S3OutputPath is Required - where the model artifacts are stored)
3. ResourceConfig (in this EC2 InstanceType and VolumeSizeInGB are required)
4. RoleArn (..The Amazon Resource Name (ARN) of an IAM role that Amazon SageMaker can assume to perform tasks on your behalf...the caller of this API must have the iam:PassRole permission.)
5. StoppingCondition
6. TrainingJobName (The name of the training job. The name must be unique within an AWS Region in an AWS account.)

From the given options in the questions.. we have 2, 3, and 4 above. so, the answer is CEF.
upvoted 20 times

-   **cloud_trail** 1 year, 6 months ago

This is the best explanation that CEF is the right answer, IMO. The document at that url is very informative. It also specifically states that InputDataConfig is NOT required. Having said that, I have no idea how the model will train if it doesn't know where to find the training data, but that is what the document says. If someone can explain that, I'd like to hear the explanation.

upvoted 6 times
-   **cloud_trail** 1 year, 6 months ago

If I see this question on the actual exam, I'm going with AEF. The model absolutely must know where the training data is. I have seen other documentation that does confirm that you need the location of the input data, the compute instance and location to output the model artifacts.

upvoted 3 times
-   **CloudGuru_ZA** 1 year, 6 months ago

but you also need to specify the service role sagemaker should use otherwise it will not be able to perform actions on your behalf like provisioning the training instances.

upvoted 1 times
-   **rafaelo** 1 year, 4 months ago

Perfect explanation. It is CEF

upvoted 1 times
-   **oso0348**

Most Recent

 4 weeks ago

Selected Answer: ACD

The three common parameters that MUST be specified when submitting Amazon SageMaker training jobs using one of the built-in algorithms are:
A. The training channel identifying the location of training data on an Amazon S3 bucket.
C. The IAM role that Amazon SageMaker can assume to perform tasks on behalf of the users.
D. Hyperparameters in a JSON array as documented for the algorithm used.

Note that parameters B, E, and F are optional parameters that depend on the specific needs of the training job.

upvoted 1 times
-   **hug_c0sm0s** 2 months ago

Selected Answer: ACF

According to the document:https://docs.aws.amazon.com/sagemaker/latest/APIReference/API_CreateTrainingJob.html, the required are AlgorithmSpecification, OutputDataConfig, ResourceConfig(The resources, including the ML compute instances and ML storage volumes, to use for model training), RoleArn, StoppingCondition, TrainingJobName. so The answer should be ACF.

upvoted 1 times
-   **bakarys** 2 months, 1 week ago

A. The training channel identifying the location of training data on an Amazon S3 bucket.
C. The IAM role that Amazon SageMaker can assume to perform tasks on behalf of the users.
D. Hyperparameters in a JSON array as documented for the algorithm used.

Note: The validation channel, instance class, and output path are not required but may be specified depending on the training job.

upvoted 1 times
-   **unit1** 2 months, 4 weeks ago

A, C, F
For all those advocating E - please read the option carefully. It says - "The Amazon EC2 instance class specifying whether training will be run using CPU or GPU.". Nowhere do you specify the instance class - CPU/GPU. You specify the instance type - https://docs.aws.amazon.com/sagemaker/latest/APIReference/API_CreateTrainingJob.html
So this option is automatically out.
And, of course, you do need to specify the path of the training dataset, so A must be included.

upvoted 2 times
-   **vbal** 3 months, 3 weeks ago

As per this URL : https://docs.aws.amazon.com/sagemaker/latest/APIReference/API_ResourceConfig.html
InstanceType is Not required so I think ACF make sense.
InputDataConfig
Type: Array of Channel objects
Array Members: Minimum number of 1 item. Maximum number of 20 items.

upvoted 1 times
-   **Ob1KN0B** 8 months ago

Selected Answer: CEF

Not A because: https://docs.aws.amazon.com/zh_tw/sagemaker/latest/APIReference/API_CreateTrainingJob.html
InputDataConfig is marked as "Required:No". So its C, E, F.

upvoted 2 times
-   **Abdelrahman_Omran** 1 year ago

Selected Answer: CEF

CEF https://docs.aws.amazon.com/zh_tw/sagemaker/latest/dg/API_CreateTrainingJob.html

upvoted 1 times
-   **TerrancePythonJava** 1 year, 1 month ago

Selected Answer: AEF

AEF not CEF
upvoted 1 times

- Freegirl

1 year, 4 months ago

Can anyone explain why D is wrong?

upvoted 1 times
- AShahine21

1 year, 5 months ago

After re-double checking the documentation (https://docs.aws.amazon.com/sagemaker/latest/APIReference/API_CreateTrainingJob.html):
I will go with C E F
A is not required, we can read more under InputDataConfig

upvoted 2 times
- orangechickencombo

1 year, 5 months ago

https://docs.aws.amazon.com/sagemaker/latest/APIReference/API_CreateTrainingJob.html
Cmd + F
search for 'Required: Yes'
CEF

upvoted 2 times
- gcpwhiz

1 year, 6 months ago

This is a very simple question, one of the first items provided in the Developer Guide. Stop looking for alternative answers and minute details when the answer is simple. There are 4 items needed for training jobs:
1. The URL of the Amazon Simple Storage Service (Amazon S3) bucket where you've stored the training data.
2. The compute resources that you want SageMaker to use for model training. Compute resources are ML compute instances that are managed by SageMaker.
3. The URL of the S3 bucket where you want to store the output of the job.
4. The Amazon Elastic Container Registry path where the training code is stored.

<https://docs.aws.amazon.com/sagemaker/latest/dg/how-it-works-training.html>

upvoted 1 times
- tmld

1 year, 6 months ago

CEF
<https://docs.aws.amazon.com/sagemaker/latest/dg/sagemaker-mkt-algo-train.html>

upvoted 1 times
- sonalev419

1 year, 6 months ago

CEF (CANT BE A or B BECAUSE INPUTS CAN BE ON LOCAL MODE inputpath = FILE://
<https://sagemaker.readthedocs.io/en/stable/api/training/estimators.html#sagemaker.estimator.EstimatorBase>

upvoted 1 times
- exploringaws

1 year, 6 months ago

I think it is about parameters required to "running a training job" vs "creating a training job". Question is focusing on "Submitting a training job" so answer should be A, E, F based on the AWS documentation.

upvoted 3 times

Question #31

Topic 1

A monitoring service generates 1 TB of scale metrics record data every minute. A Research team performs queries on this data using Amazon Athena. The queries run slowly due to the large volume of data, and the team requires better performance. How should the records be stored in Amazon S3 to improve query performance?

A. CSV files

B. Parquet files

C. Compressed JSON

D. RecordIO

Correct Answer: B

Community vote distribution

B (100%)

gaku1016

Highly Voted

1 year, 7 months ago

Answer is B. Athena is best in Parquet format.

upvoted 22 times

emailtorajivk

Highly Voted

1 year, 6 months ago

You can improve the performance of your query by compressing, partitioning, or converting your data into columnar formats. Amazon Athena supports open source columnar data formats such as Apache Parquet and Apache ORC. Converting your data into a compressed, columnar format lowers your cost and improves query performance by enabling Athena to scan less data from S3 when executing your query

upvoted 11 times

apprehensive_scar

Most Recent

1 year, 2 months ago

Selected Answer: B

B it is

upvoted 2 times

benson2021

1 year, 6 months ago

Answer is B. <https://aws.amazon.com/tw/blogs/big-data/top-10-performance-tuning-tips-for-amazon-athena/>

But why does this question relate to Machine Learning?

upvoted 2 times

AddiWei

1 year, 2 months ago

Because you must explore data very quickly using SQL in order to run EDA / analyze data for ML purposes. Those explorations can inform on selecting features that can be used for modeling purposes.

upvoted 4 times

https://www.examtopycs.com/exams/amazon/aws-certified-machine-learning-specialty/custom-view/

68/406

Question #32

Topic 1

Machine Learning Specialist is working with a media company to perform classification on popular articles from the company's website. The company is using random forests to classify how popular an article will be before it is published. A sample of the data being used is below.

Article_Title	Author	Top_Keywords	Day_Of_Week	URL_of_Article	Page_Views
Building a Big Data Platform	Jane Doe	Big Data, Spark, Hadoop	Tuesday	http://examplecorp.com/data_platform.html	1300456
Getting Started with Deep Learning	John Doe	Deep Learning, Machine Learning, Spark	Tuesday	http://examplecorp.com/started_deep_learning.html	1230661
MXNet ML Guide	Jane Doe	Machine Learning, MXNet, Logistic Regression	Thursday	http://examplecorp.com/mxnet_guide.html	937291
Intro to NoSQL Databases	Mary Major	NoSQL, Operations, Database	Monday	http://examplecorp.com/nosql_intro_guide.html	407812

Given the dataset, the Specialist wants to convert the Day_Of_Week column to binary values.

What technique should be used to convert this column to binary values?

- A. Binarization
- B. One-hot encoding
- C. Tokenization
- D. Normalization transformation

Correct Answer: B

- omar_bahrain

Highly Voted

1 year, 6 months ago

I choose b

upvoted 17 times
- Juka3lj

Highly Voted

1 year, 6 months ago

Correct answer is B.

Example:

Mon | Tue | Wed

1 0 0

0 1 0

upvoted 9 times
- Tomatoteacher

Most Recent

3 months, 2 weeks ago

Originally I put A, (believing to be able to format it as (0,1,2,3,4,5,6), or something as it mentioned it to convert the column, but later I realized Binarization is only designed for continuous or numerical data. Even though one-hot encoding will create 6 more columns it is correct. B is correct.

upvoted 1 times
- Peeking

4 months, 3 weeks ago

B

1000000 = Mon

0100000 = Tue

0010000 = Wed

0001000 = Thur

0000100 = Fri

0000010 = Sat

0000001=Sun

upvoted 1 times
- benliu1974

7 months, 2 weeks ago

why not A? 001 010, 011

upvoted 1 times
- apprehensive_scar

1 year, 2 months ago

B is the obvious answer

upvoted 2 times
- bitsplease

1 year, 3 months ago

Binary encoding would've been a correct answer but it is not here & Binarization is used for continuous variables. leaving w/ option B
upvoted 1 times

  **Zhubajie** 1 year, 5 months ago

B is wrong. You do not need to one hot encode the variable in random trees. If you do so, you tree must be very deep, which is not efficient. The correct answer is C!
upvoted 1 times

  **gmnk999** 1 year ago

"The Specialist want to convert the Day Of Week column in the dataset to binary values." You are misreading the question. The answer is B.
upvoted 4 times

  **zach288** 1 year, 5 months ago

Stop misleading people, the question already asked to convert the data into binary. C is not even remotely close to be correct
upvoted 12 times

A gaming company has launched an online game where people can start playing for free, but they need to pay if they choose to use certain features. The company needs to build an automated system to predict whether or not a new user will become a paid user within 1 year. The company has gathered a labeled dataset from 1 million users.

The training dataset consists of 1,000 positive samples (from users who ended up paying within 1 year) and 999,000 negative samples (from users who did not use any paid features). Each data sample consists of 200 features including user age, device, location, and play patterns. Using this dataset for training, the Data Science team trained a random forest model that converged with over 99% accuracy on the training set. However, the prediction results on a test dataset were not satisfactory

Which of the following approaches should the Data Science team take to mitigate this issue? (Choose two.)

- A. Add more deep trees to the random forest to enable the model to learn more features.
- B. Include a copy of the samples in the test dataset in the training dataset.
- C. Generate more positive samples by duplicating the positive samples and adding a small amount of noise to the duplicated data.
- D. Change the cost function so that false negatives have a higher impact on the cost value than false positives.
- E. Change the cost function so that false positives have a higher impact on the cost value than false negatives.

Correct Answer: CD

Community vote distribution

CD (100%)

- Phong

Highly Voted

1 year, 7 months ago

I think it should be CD
C: because we need a balance dataset
D: The number of positive samples is large so model tends to predict 0 (negative) for all cases leading to False Negative problem. We should minimize that.
My opinion
upvoted 27 times
- Phong

Highly Voted

1 year, 7 months ago

I think it should be CD
C: because we need a balance dataset
D: The number of negative samples is large so model tends to predict 0 (negative) for all cases leading to False Negative problem. We should minimize that.
My opinion
upvoted 18 times
- John_Pongthorn

Most Recent

1 year, 2 months ago

Selected Answer: CD

C,D is correct (percentage of the positive class is key to decide which case we are interested in)
This question, positive class (Pay) is 0.01% as compared to 99.99(not pay) , as a result, we have to pay attention to Pay because if we miss 0.01% out, we didn't get revenue. it is a false negative.

In contrast to these questions, it positive class (Pay) is 40% as compared to negative class (60% not pay), it is avoidable to emphasize on 40% (if model predict as payment but in reality customer neglect), we won't get revenue the amount from false positive)

upvoted 3 times
- apprehensive_scar

1 year, 2 months ago

I think is CD
upvoted 1 times
- cloud_trail

1 year, 5 months ago

C and D. Hopefully, no one honestly thinks that B is a good answer. Never expose test data to the training set or vice versa. C is right because of the highly imbalanced training set. D is right because you want to minimize false negatives, maximize true positives, maximize recall of the positive class. I'm not sure why anyone's worried about precision in this case.
upvoted 3 times
- felbuch

1 year, 5 months ago

CD

The model has 99% accuracy because it's simply predicting that everyone's a negative. Since almost everyone's a negative, it will get almost everyone right.
So we need to penalize the model for predicting that someone is a negative when it is not (i.e. penalize false negatives). So that's D.

Also, it would be really nice to have more positives -- one way to do that is to follow option C.

upvoted 4 times

  **engomaradel** 1 year, 6 months ago

CD 100%

upvoted 1 times

  **ybad** 1 year, 6 months ago

CD

C:imbalance of test (1000 positive, 999000 negative = 0.1% positive) thus C to increase that

D :also to reduce generalizing, since everyone says no, the model would generalize to no, but increasing the penalty of a false negative would reduce generalizing..

upvoted 1 times

  **Omar_Cascudo** 1 year, 6 months ago

It is needed to diminish the FP, because they are player predicted to pay and in reality will not pay. So FP should impact the cost metric more. CE should be the answer.

upvoted 2 times

  **bidbs** 1 year, 6 months ago

CD are correct for sure.

upvoted 3 times

  **hans1234** 1 year, 6 months ago

It is C,E... we want to find all paying customers, which are positives, so we have to punish incorrectly finding negatives, which is E

upvoted 2 times

  **Wira** 1 year, 6 months ago

CD

although i am worried about the noise being introduced as it could skew the data nevertheless no better answer is given

upvoted 2 times

  **aws_razor** 1 year, 6 months ago

CD

We need high recall so that we do not miss many Positive cases. In that case we need to have less False Negative(FN) therefore it should have high impact on cost function.

upvoted 3 times

  **roytruong** 1 year, 6 months ago

in my view, CD are answers

C: of course, handle the imbalanced dataset

D: right now, model accuracy is 99%, it means model predict everything is negative leading to FN problem, so we need to minimize it more in cost function

upvoted 3 times

  **wuha5086** 1 year, 6 months ago

CD, FN are valuable players, we should care more on FN

upvoted 7 times

  **VB** 1 year, 6 months ago

Is my assumption right here?

ACTUAL

P PAY NPAY

R -----

E PAY TP FP

D

I NPAY FN TN

C -----

upvoted 1 times

  **AKT** 1 year, 7 months ago

C,D correct

upvoted 5 times

Question #34

Topic 1

A Data Scientist is developing a machine learning model to predict future patient outcomes based on information collected about each patient and their treatment plans. The model should output a continuous value as its prediction. The data available includes labeled outcomes for a set of 4,000 patients. The study was conducted on a group of individuals over the age of 65 who have a particular disease that is known to worsen with age.

Initial models have performed poorly. While reviewing the underlying data, the Data Scientist notices that, out of 4,000 patient observations, there are 450 where the patient age has been input as 0. The other features for these observations appear normal compared to the rest of the sample population

How should the Data Scientist correct this issue?

- A. Drop all records from the dataset where age has been set to 0.
- B. Replace the age field value for records with a value of 0 with the mean or median value from the dataset
- C. Drop the age feature from the dataset and train the model using the rest of the features.
- D. Use k-means clustering to handle missing features

Correct Answer: B

Community vote distribution

D (100%)

  **rajs** Highly Voted 1 year, 7 months ago

Dropping the Age feature is a NOT ATOLL a good idea - as age plays a critical role in this disease as per the question

Dropping 10% of data is NOT a good idea considering the fact that the number of observations is already low.

The Mean or Median are a potential solutions

But the question says that "Disease worsens after age 65 so there is a correlation between age and other symptoms related feature" So that means that using Unsupervised Learning we can make pretty good prediction of "Age"

So the answer is D Use K-Means clustering

upvoted 28 times

  **L2007** 1 year, 7 months ago

<https://www.displayr.com/5-ways-deal-missing-data-cluster-analysis/>

B is correct

upvoted 3 times

  **vetal** Highly Voted 1 year, 7 months ago

Replacing the age with mean or median might bring a bias to the dataset. Use k-means clustering to estimate the missing age based on other features might get better results. Removing 10% available data looks odd. Why not D?

upvoted 17 times

  **Tomatoteacher** Most Recent 3 months, 2 weeks ago

Selected Answer: D

A lot of people in the comments insist that you cannot use k-means for computing missing values. If you see using this link you can do it. There are Deep learning techniques which would be better, but K-means is better than the resulting mean. So D is correct and B would be second best.

[https://link.springer.com/chapter/10.1007/978-3-642-14834-](https://link.springer.com/chapter/10.1007/978-3-642-14834-7_56#:~:text=We%20divide%20the%20data%20set,is%20used%20as%20impute%20value.)

[7_56#:~:text=We%20divide%20the%20data%20set,is%20used%20as%20impute%20value.](https://link.springer.com/chapter/10.1007/978-3-642-14834-7_56#:~:text=We%20divide%20the%20data%20set,is%20used%20as%20impute%20value.)

upvoted 3 times

  **dhyuk** 4 months, 1 week ago

i Think its A

upvoted 1 times

  **passionatecricketer** 1 year, 5 months ago

Selected Answer: D

The following is an excerpt from a Springer Verlag book called Contemporary Computing.

"...We propose an efficient missing value imputation method based on clustering with weighted distance. We divide the data set into clusters based on user specified value K. Then find a complete valued neighbor which is nearest to the missing valued instance. Then we compute the missing value by taking the average of the centroid value and the centroidal distance of the neighbor. This value is used as impute value. In our proposed approach we use K-means technique with weighted distance and show that our approach results in better performance..."

upvoted 7 times

  **richardxyz** 1 year, 5 months ago

KNN is good to get the values from nearest neighbours, but K-means also works - get the values from the clusters

upvoted 2 times

🗨️ 👤 **G1807** 1 year, 5 months ago

B will introduce strong bias to the data ignoring the relativity of the age feature with other features since the disease worsens with age.

D should be the correct answer since we can impute missing data using unsupervised learning

<https://medium.com/jungle-book/missing-data-filling-with-unsupervised-learning-b448964030d>

upvoted 1 times

🗨️ 👤 **Exia** 1 year, 5 months ago

B.

D. K-means cannot handle missing data since it is unsupervised learning. You cannot know the predicted values. Use KNN instead.

upvoted 5 times

🗨️ 👤 **hero67** 1 year, 5 months ago

Since the missing values are considerable fraction from the whole data set, yet they are crucial to the model (disease is highly correlated to those aged than 65) then there is a need for imputation. It seems odd to use clustering technique to do such a job. Hence, the only correct answer is impute using mean or median values. On the other hand, imputing using K-nearest neighbors or KNNs would be a viable solution but not K-means clustering.

upvoted 3 times

🗨️ 👤 **abdohanfi** 1 year, 6 months ago

i think k means clustering is unsupervised and he said that the data is labeled so doesnt need unsupervised algorithm so i think the mean or median for the 0s are good option

upvoted 2 times

🗨️ 👤 **Vita_Rasta84444** 1 year, 6 months ago

I think it is D. If we just impute the median we will lose the important information, and age is one of the predictors. With K-means, we can make clusters and then impute the cluster age mean for the cluster.

upvoted 1 times

🗨️ 👤 **omar_bahrain** 1 year, 6 months ago

searching the net, I have notice that even K-Means Clustering can be also used for imputing missing data so D is still a good candidate

upvoted 5 times

🗨️ 👤 **cloud_trail** 1 year, 6 months ago

Wow there are a lot of bad comments for this one. Age is a critically important column so you cannot just dump the whole column. Eliminate C. k-means is not used for imputation. KNN is. D is a deception option and incorrect. KNN imputation would have been the best option but it's not listed here. Because age is a critical predictor as stated in the question, you just cannot assign the mean or median to it. That will ruin your model. Eliminate B. A is the best option listed. Dropping 11% of your training examples is not optimal but it is acceptable when you still have thousands of samples.

upvoted 3 times

🗨️ 👤 **Huy** 1 year, 5 months ago

Should not give missing data > 10%. So not A. Impute mean/median make sense in this case.

B is correct. In fact, this doesn't ruin your model. You still need to validate and test.

upvoted 1 times

🗨️ 👤 **Huy** 1 year, 5 months ago

May be D is better than B.

upvoted 1 times

🗨️ 👤 **DzR** 1 year, 6 months ago

It will be my first time to handle missing values with K-Means, so I will go with B, choosing between the mean and the median looking at the distribution of the data.

upvoted 3 times

🗨️ 👤 **dd1024** 1 year, 6 months ago

Should be B. Using K-mean is OK, but D said "handle missing feature". those 450 samples have incorrect input feature instead of "missing" feature, so D is a misleading option.

upvoted 2 times

🗨️ 👤 **yeetusdeleetus** 1 year, 6 months ago

Correct answer is B.

The fact that they mentioned the data with missing values appears normal compared to the rest of the population (ie, normally distributed) means that the mean will not introduce very significant bias. If anything it will introduce noise, not bias.

Using k-means is clever but would not solve the problem unless you also imputed values, which the answer does not state.

I personally would prefer to drop rows with missing values and I think it gives better results, and if the resulting dataset is too small you should gather more data. But, in the context of the question where they mention the data with missing values is 'normal' compared to the remainder of data, the mean is what we are pointed to.

upvoted 3 times

🗨️ 👤 **Omar_Cascudo** 1 year, 6 months ago

B and D are correct paths to follow. Clustering the patients, as disease gets worst with age, and assign the mean or median age of the cluster should not introduce bias. I go with D.

upvoted 1 times

A Data Science team is designing a dataset repository where it will store a large amount of training data commonly used in its machine learning models. As Data Scientists may create an arbitrary number of new datasets every day, the solution has to scale automatically and be cost-effective. Also, it must be possible to explore the data using SQL.

Which storage scheme is MOST adapted to this scenario?

- A. Store datasets as files in Amazon S3.
- B. Store datasets as files in an Amazon EBS volume attached to an Amazon EC2 instance.
- C. Store datasets as tables in a multi-node Amazon Redshift cluster.
- D. Store datasets as global tables in Amazon DynamoDB.

Correct Answer: A

Community vote distribution

A (100%)

  **rsimham** Highly Voted 1 year, 7 months ago
Ans: A (S3) is most cost effective
upvoted 15 times

  **sonalev419** Highly Voted 1 year, 6 months ago
A : S3 cost effective + athena (not c redshift dont support unstructured data)
upvoted 7 times

  **Ajose0** Most Recent 2 months, 2 weeks ago
Selected Answer: A
The most appropriate storage scheme for this scenario is option A: Store datasets as files in Amazon S3.

Amazon S3 is a highly scalable and cost-effective object storage service that can store a large amount of data. S3 can scale automatically to accommodate a large number of datasets, making it a good option for storing the training data used in machine learning models. Additionally, S3 supports SQL querying through Amazon Athena or Amazon Redshift Spectrum, allowing data scientists to easily explore the data.
upvoted 1 times

  **harmanbirstudy** 1 year, 6 months ago
"store a large amount of training data commonly used in its machine learning models".. well it cannot be anything other than S3. Athena can query S3 cataloged data with SQL commands.
Anwser is A
upvoted 2 times

  **Stephen_C** 1 year, 6 months ago
Amazon Redshift is not cost-effective.
upvoted 1 times

  **syu31svc** 1 year, 6 months ago
I would say C
<https://docs.aws.amazon.com/redshift/latest/mgmt/working-with-clusters.html>
"For workloads that require ever-growing storage, managed storage lets you automatically scale your data warehouse storage capacity without adding and paying for additional nodes."
upvoted 3 times

  **kwangje** 1 year, 6 months ago
Amazon Redshift is a fast, fully managed data warehouse that makes it simple and cost-effective to analyze all your data using standard SQL and your existing Business Intelligence (BI) tools. It allows you to run complex analytic queries against petabytes of structured data using sophisticated query optimization, columnar storage on high-performance storage, and massively parallel query execution. Most results come back in seconds.
upvoted 1 times

  **HaiHN** 1 year, 6 months ago
Data warehouse is not needed. For exploring data using SQL, you can use Athena
upvoted 5 times

  **roytruong** 1 year, 7 months ago
s3 is right
upvoted 1 times

  **cybe001** 1 year, 7 months ago

A, S3 is most cost effective

upvoted 2 times

Question #36

Topic 1

A Machine Learning Specialist deployed a model that provides product recommendations on a company's website. Initially, the model was performing very well and resulted in customers buying more products on average. However, within the past few months, the Specialist has noticed that the effect of product recommendations has diminished and customers are starting to return to their original habits of spending less. The Specialist is unsure of what happened, as the model has not changed from its initial deployment over a year ago.

Which method should the Specialist try to improve model performance?

- A. The model needs to be completely re-engineered because it is unable to handle product inventory changes.
- B. The model's hyperparameters should be periodically updated to prevent drift.
- C. The model should be periodically retrained from scratch using the original data while adding a regularization term to handle product inventory changes
- D. The model should be periodically retrained using the original training data plus new data as product inventory changes.

Correct Answer: *D*

Community vote distribution

D (100%)

  **rsimham**

Highly Voted 

 1 year, 7 months ago

Ans: D

upvoted 25 times

  **Valcilio**

Most Recent 

 1 month, 3 weeks ago

Selected Answer: D

D is the answer!

upvoted 1 times

  **Peeking** 4 months, 3 weeks ago

D is the answer. There has been a data drift resulting from new customer segment visiting the site. So, the model needs to be updated periodically with new data from the website.

upvoted 3 times

  **apprehensive_scar** 1 year, 2 months ago

Selected Answer: D

DDDDD. D :D

upvoted 1 times

  **cloud_trail** 1 year, 5 months ago

Incremental training. D.

upvoted 3 times

  **gamaX** 1 year, 6 months ago

Periodically Re-Fit

D

upvoted 1 times

  **eji** 1 year, 6 months ago

agree with D

upvoted 3 times

  **C10ud9** 1 year, 6 months ago

D is correct

upvoted 3 times

Question #37

Topic 1

A Machine Learning Specialist working for an online fashion company wants to build a data ingestion solution for the company's Amazon S3-based data lake.

The Specialist wants to create a set of ingestion mechanisms that will enable future capabilities comprised of:

- ☞ Real-time analytics
- ☞ Interactive analytics of historical data
- ☞ Clickstream analytics
- ☞ Product recommendations

Which services should the Specialist use?

- A. AWS Glue as the data catalog; Amazon Kinesis Data Streams and Amazon Kinesis Data Analytics for real-time data insights; Amazon Kinesis Data Firehose for delivery to Amazon ES for clickstream analytics; Amazon EMR to generate personalized product recommendations
- B. Amazon Athena as the data catalog; Amazon Kinesis Data Streams and Amazon Kinesis Data Analytics for near-real-time data insights; Amazon Kinesis Data Firehose for clickstream analytics; AWS Glue to generate personalized product recommendations
- C. AWS Glue as the data catalog; Amazon Kinesis Data Streams and Amazon Kinesis Data Analytics for historical data insights; Amazon Kinesis Data Firehose for delivery to Amazon ES for clickstream analytics; Amazon EMR to generate personalized product recommendations
- D. Amazon Athena as the data catalog; Amazon Kinesis Data Streams and Amazon Kinesis Data Analytics for historical data insights; Amazon DynamoDB streams for clickstream analytics; AWS Glue to generate personalized product recommendations

Correct Answer: A

Community vote distribution

A (100%)

🗲️ 👤 **rsimham** Highly Voted 👍 1 year, 7 months ago

Ans: A seems to be reasonable
upvoted 36 times

🗲️ 👤 **cybe001** Highly Voted 👍 1 year, 7 months ago

A looks correct but it is missing for "Interactive analytics of historical data"
upvoted 12 times

🗲️ 👤 **ZSun** 1 week, 5 days ago

AWS Glue as data catalog, then you can analyze historical data, such as running sql with Athena.
upvoted 1 times

🗲️ 👤 **ejj** 1 year, 6 months ago

but C is missing for "real-time analythics"
upvoted 1 times

🗲️ 👤 **ejj** 1 year, 6 months ago

and also C is saying historical data analytics for Kinesis Data analytics which is real-time analytics not historical, so the answer might not C but the answer is A
upvoted 1 times

🗲️ 👤 **planhanasan** 1 year, 5 months ago

Once you insert real-time data to ES, you can see historical data from Kibana dashboard.
upvoted 1 times

🗲️ 👤 **Valcilio** Most Recent 🕒 1 month, 3 weeks ago

Selected Answer: A

All are bad options, but A can do it.
upvoted 1 times

🗲️ 👤 **hug_c0sm0s** 2 months ago

Selected Answer: A

AWS Glue is a fully managed extract, transform, and load (ETL) service that makes it easy to move data between data stores. It can be used as a data catalog to store metadata information about the data in the data lake. Amazon Kinesis Data Streams and Amazon Kinesis Data Analytics can be used together to collect, process, and analyze real-time streaming data. Amazon Kinesis Data Firehose can be used to deliver streaming data to destinations such as Amazon ES for clickstream analytics. Finally, Amazon EMR can be used to run big data frameworks such as Apache Spark and Apache Hadoop to generate personalized product recommendations.
upvoted 1 times

-  **gamaX** 1 year, 6 months ago

A or C
<https://aws.amazon.com/blogs/big-data/retaining-data-streams-up-to-one-year-with-amazon-kinesis-data-streams/>
upvoted 1 times
-  **harmanbirstudy** 1 year, 6 months ago

Athena can do Interactive analytics on Historical data, but here its only use is "Athena as the data catalog" and this is the work of Glue data catalog using its crawlers, so it cannot be B or D.
--So its either A or C
-- Now Kinesis data Streams/Analytics is know for real time data analytics but if it is reading from data already stored in S3 using DMS then we can say it is getting historical data.
-- Here I am not very clear if Kinesis part will happen on incoming data before S3 or After data persists to S3 and Kinesis reads it through S3-->DMS--Kinesis data stream -- Kinesis analytics-->Firehose.
But still insights are always on real-time/current data based on historical data trends , so the statement in C "Analytics for historical data insights" is in-correct in general .
Hence ANSWER is :A
upvoted 4 times
-  **ybad** 1 year, 6 months ago

A is correct,

for those asking the difference between A and D, D talks about using kinesis stream and data analytics to create historical analysis.... waste of money no?
upvoted 2 times
-  **Th3Dud3** 1 year, 6 months ago

Answer = A
upvoted 4 times
-  **C10ud9** 1 year, 6 months ago

A it is
upvoted 2 times
-  **roytruong** 1 year, 6 months ago

it's A, ES can perform clickstream analytics and EMR can handle spark job recomendation at scale
upvoted 3 times
-  **BigPlums** 1 year, 6 months ago

Only C and D mention interactive analytics of historical data.
Glue won't provide personalised recommendation so it is C
upvoted 1 times
-  **BigEv** 1 year, 6 months ago

What is the difference between the solution in A or C ????
upvoted 2 times
-  **JayK** 1 year, 6 months ago

A is real time data analytics with Kinesis Data analytics and C is saying historical data which is wrong
upvoted 6 times
-  **ComPah** 1 year, 7 months ago

Looks like C Amazon ES has Klbana which supports click stream
upvoted 2 times
-  **ComPah** 1 year, 7 months ago

A is Correct
upvoted 4 times

A company is observing low accuracy while training on the default built-in image classification algorithm in Amazon SageMaker. The Data Science team wants to use an Inception neural network architecture instead of a ResNet architecture.

Which of the following will accomplish this? (Choose two.)

- A. Customize the built-in image classification algorithm to use Inception and use this for model training.
- B. Create a support case with the SageMaker team to change the default image classification algorithm to Inception.
- C. Bundle a Docker container with TensorFlow Estimator loaded with an Inception network and use this for model training.
- D. Use custom code in Amazon SageMaker with TensorFlow Estimator to load the model with an Inception network, and use this for model training.
- E. Download and apt-get install the inception network code into an Amazon EC2 instance and use this instance as a Jupyter notebook in Amazon SageMaker.

Correct Answer: CD

Community vote distribution

CD (100%)

- DonaldCMLIN

Highly Voted

1 year, 7 months ago

You might be spent a lot of money for ask AWS A.CHANGE built-in image OR B.Create a support case.

The effectial way BOTH RELATIVE TO SageMaker Estimator
C.DOCKER
OR BRING YOUR CODE BY
D.SageMaker with TensorFlow Estimator

THE BEAUTYFUL ANSWER ARE C AND D
upvoted 29 times
- Phong

Highly Voted

1 year, 6 months ago

I will go for C & D
upvoted 10 times
- hug_c0sm0s

Most Recent

2 months ago

Selected Answer: CD

Option A is not possible because the built-in image classification algorithm cannot be customized. Option B is not feasible because it is not possible to change the default image classification algorithm through a support case. Option E is also not a recommended approach because it involves manually installing software on an EC2 instance rather than using the managed services provided by SageMaker.
upvoted 1 times
- sqavi

2 months, 3 weeks ago

Selected Answer: CD

The effectial way BOTH RELATIVE TO SageMaker Estimator
C.DOCKER
OR BRING YOUR CODE BY
D.SageMaker with TensorFlow Estimator
upvoted 1 times
- Huy

1 year, 6 months ago

This question ask for 2 ways not a set of actions. So may be confused.
upvoted 1 times
- ahquiceno

1 year, 6 months ago

Answers AD go to: <https://docs.aws.amazon.com/sagemaker/latest/dg/docker-containers.html>
upvoted 2 times
- ybad

1 year, 6 months ago

CD and also A says it but in a more general term....
upvoted 1 times
- jaydec

1 year, 6 months ago

<https://aws.amazon.com/blogs/machine-learning/transfer-learning-for-custom-labels-using-a-tensorflow-container-and-bring-your-own-algorithm-in-amazon-sagemaker/>
upvoted 1 times

 **Antriksh** 1 year, 6 months ago
C and D are correct
upvoted 5 times

 **DonaldCMLIN** 1 year, 7 months ago
https://docs.aws.amazon.com/zh_tw/sagemaker/latest/dg/your-algorithms.html

<https://aws.amazon.com/tw/blogs/machine-learning/transfer-learning-for-custom-labels-using-a-tensorflow-container-and-bring-your-own-algorithm-in-amazon-sagemaker/>
https://docs.aws.amazon.com/zh_tw/sagemaker/latest/dg/tf.html
upvoted 1 times

A Machine Learning Specialist built an image classification deep learning model. However, the Specialist ran into an overfitting problem in which the training and testing accuracies were 99% and 75%, respectively.

How should the Specialist address this issue and what is the reason behind it?

- A. The learning rate should be increased because the optimization process was trapped at a local minimum.
- B. The dropout rate at the flatten layer should be increased because the model is not generalized enough.
- C. The dimensionality of dense layer next to the flatten layer should be increased because the model is not complex enough.
- D. The epoch number should be increased because the optimization process was terminated before it reached the global minimum.

Correct Answer: D

Reference:

https://www.tensorflow.org/tutorials/keras/overfit_and_underfit

Community vote distribution

B (100%)

DonaldCMLIN

Highly Voted

1 year, 7 months ago

DROPOUT HELPS PREVENT OVERFITTING
<https://keras.io/layers/core/#dropout>

THE BEAUTIFUL ANSER SHOULD BE B.
upvoted 51 times

rsimham

1 year, 7 months ago

agree. it should be B
upvoted 9 times

user009

Most Recent

1 month ago

Option B is the correct answer because increasing the dropout rate at the flatten layer helps prevent overfitting by randomly dropping out units during training, effectively creating a more robust model that can generalize better to new data. Dropout is a regularization technique that helps prevent overfitting by forcing the model to learn redundant representations of the data. By increasing the dropout rate at the flatten layer, the model becomes more generalized, which should help to improve the testing accuracy.

upvoted 1 times

Ajose0

2 months, 2 weeks ago

Selected Answer: B

Overfitting occurs when a model is too complex and memorizes the training data instead of learning the underlying pattern. As a result, the model performs well on the training data but poorly on new, unseen data.

Increasing the dropout rate, a regularization technique, can help combat overfitting by randomly dropping out some neurons during training, which prevents the model from relying too heavily on any single feature.

upvoted 1 times

sqavi

2 months, 3 weeks ago

Selected Answer: B

Model is overfitting, I will go with option B, increasing epoch will cause more overfitting
upvoted 1 times

desperatestudent

3 months, 1 week ago

Selected Answer: B

it should answer B.
upvoted 1 times

Shailendraa

7 months, 3 weeks ago

12-sep exam
upvoted 2 times

Moulichintakunta

7 months, 4 weeks ago

The link which mentioned in answer section also talks about dropout only , so Option B is correct
upvoted 1 times

ovokpus

10 months, 1 week ago

Selected Answer: B

I vote B for all the reasons already mentioned. Dropouts help prevent overfitting

upvoted 3 times

  **theprismdata** 11 months, 3 weeks ago

Selected Answer: B

Dropout prevent overfitting, and Early stopping is useful, increasing epoch is not effect in this situation
https://en.wikipedia.org/wiki/Early_stopping

upvoted 2 times

  **jonsnow777** 1 year, 3 months ago

B. All the other answers do not prevent overfitting.

upvoted 1 times

  **eganilovic** 1 year, 6 months ago

Correct answer is B! Number of epochs shoud be decreased.

upvoted 4 times

  **gcpwhiz** 1 year, 6 months ago

The reference article listed by the website also says dropout is the correct answer...go with B

upvoted 2 times

  **Vita_Rasta84444** 1 year, 6 months ago

B is true answer. More epochs will not stop overfitting.

upvoted 2 times

  **littlewat** 1 year, 6 months ago

B!!!!!!!!!!!!!!

upvoted 2 times

  **DzR** 1 year, 6 months ago

I will go for b because the problem is the model is failing to generalize and hence dropout will solve the problem.

upvoted 1 times

  **harmanbirstudy** 1 year, 6 months ago

It cannot be because with 99% accuracy the testing does need epochs and nowhere it says about epoch getting Early Terminated in Question , so we cannot just assume it.

Among the rest of the list only "Increasing Dropout" helps preventing Overfitting.

ANSWER is B

upvoted 2 times

  **ybad** 1 year, 6 months ago

Agree with all it should be B, only solution to reduce overfitting,

increasing epochs would actually cause it to overfit more....

upvoted 1 times

A Machine Learning team uses Amazon SageMaker to train an Apache MXNet handwritten digit classifier model using a research dataset. The team wants to receive a notification when the model is overfitting. Auditors want to view the Amazon SageMaker log activity report to ensure there are no unauthorized API calls.

What should the Machine Learning team do to address the requirements with the least amount of code and fewest steps?

- A. Implement an AWS Lambda function to log Amazon SageMaker API calls to Amazon S3. Add code to push a custom metric to Amazon CloudWatch. Create an alarm in CloudWatch with Amazon SNS to receive a notification when the model is overfitting.
- B. Use AWS CloudTrail to log Amazon SageMaker API calls to Amazon S3. Add code to push a custom metric to Amazon CloudWatch. Create an alarm in CloudWatch with Amazon SNS to receive a notification when the model is overfitting.
- C. Implement an AWS Lambda function to log Amazon SageMaker API calls to AWS CloudTrail. Add code to push a custom metric to Amazon CloudWatch. Create an alarm in CloudWatch with Amazon SNS to receive a notification when the model is overfitting.
- D. Use AWS CloudTrail to log Amazon SageMaker API calls to Amazon S3. Set up Amazon SNS to receive a notification when the model is overfitting

Correct Answer: B

Community vote distribution

B (100%)

-   **DonaldCMLIN**

Highly Voted 

 1 year, 7 months ago

THE ANSWER SHOULD BE B.
YOU DON'T NEED TO THROUGH LAMBDA TO INTERGE CLOUDTRAIL

Log Amazon SageMaker API Calls with AWS CloudTrail
<https://docs.aws.amazon.com/sagemaker/latest/dg/logging-using-cloudtrail.html>
upvoted 40 times
-   **rajs**

Highly Voted 

 1 year, 7 months ago

Agreed B for the following reasons

CloudTrail logs captured in S3 without any code/lambda
The custom metrics can be published to Cloudwatch...in this case it would be a test for overfit on MXNET which will set off an alarm which can then be subscribed on SNS
upvoted 10 times
-   **Paolo991**

Most Recent 

 1 month, 1 week ago

I would consider D as well.

You can just setup a SNS that is triggered by a built-in action like here:
<https://docs.aws.amazon.com/sagemaker/latest/dg/debugger-built-in-actions.html>

You can see that overfitting is a built-in rule for MXNet from here:
<https://docs.aws.amazon.com/sagemaker/latest/dg/debugger-built-in-rules.html>

Not that B is not working. Maybe the question was prior to this new solution.
upvoted 1 times
-   **Valcilio** 1 month, 3 weeks ago

Selected Answer: B

It's B.
upvoted 1 times
-   **Ajose0** 1 month, 4 weeks ago

Selected Answer: B

AWS CloudTrail provides a history of AWS API calls made on the account. The Machine Learning team can use AWS CloudTrail to log Amazon SageMaker API calls to Amazon S3. They can then use CloudWatch to create alarms and receive notifications when the model is overfitting.

To ensure auditors can view the Amazon SageMaker log activity report, the team can add code to push a custom metric to Amazon CloudWatch. This provides a single place to view and analyze logs across all the services and resources in the environment.
upvoted 1 times
-   **sonalev419** 1 year, 6 months ago

B. cloudwatch + metrics from sagemaker + sns https://docs.aws.amazon.com/fr_fr/sagemaker/latest/dg/training-metrics.html#define-train-metrics
upvoted 2 times

-  **ybad**

1 year, 6 months ago

B requires the least amount of code and satisfies all conditions

upvoted 2 times
-  **tochiebby**

1 year, 6 months ago

What does this line do?

"Add code to push a custom metric to Amazon CloudWatch"

upvoted 1 times
-  **Omar_Cascudo**

1 year, 6 months ago

It creates a metric for overfitting (accuracy of training data and accuracy of test data).

upvoted 5 times
-  **jonclem**

1 year, 6 months ago

Its not B. Why would you use CloudTrail?

Having used Lambda for API calls I'm inclined to agree with the original answer, C.

upvoted 1 times
-  **Pja1**

1 year, 6 months ago

<https://docs.aws.amazon.com/sagemaker/latest/dg/logging-using-cloudtrail.html>

upvoted 3 times
-  **fhuadeen**

1 year, 6 months ago

Because that is the only job of CloudTrail - to log actions taken on your AWS account. So why need a Lambda function to trigger it?

upvoted 3 times
-  **Antriksh**

1 year, 6 months ago

B it is

upvoted 2 times
-  **C10ud9**

1 year, 6 months ago

B it is

upvoted 1 times
-  **lisiuyiu**

1 year, 6 months ago

Agree on B

upvoted 2 times
-  **ac427**

1 year, 7 months ago

ALL AWS Service's API calls are logged to CloudTrail automatically.

upvoted 3 times

A Machine Learning Specialist is building a prediction model for a large number of features using linear models, such as linear regression and logistic regression.

During exploratory data analysis, the Specialist observes that many features are highly correlated with each other. This may make the model unstable.

What should be done to reduce the impact of having such a large number of features?

- A. Perform one-hot encoding on highly correlated features.
- B. Use matrix multiplication on highly correlated features.
- C. Create a new feature space using principal component analysis (PCA)
- D. Apply the Pearson correlation coefficient.

Correct Answer: C

Community vote distribution

C (100%)

- cybe001

Highly Voted

1 year, 7 months ago

C is correct

upvoted 17 times
- Valcilio

Most Recent

1 month, 3 weeks ago

Selected Answer: C

PCA and T-SNE are for solving the curse of dimensionality mencioned here!

upvoted 1 times
- DS2021

3 months, 2 weeks ago

I assume PCA is for unsupervised learning!...and the scenario in the question looks like supervised learning

upvoted 1 times
- GiyeonShin

2 months, 2 weeks ago

data (x, y) --> (PCA) --> preprocessed data(x', y) --> learning

why not for supervised learning?

upvoted 1 times
- BethChen

4 months, 3 weeks ago

Selected Answer: C

Tricky. The sentence 'many features are highly correlated with each other' is no use.

upvoted 1 times
- Shailendraa

7 months, 3 weeks ago

Answer C: Read through this carefully "What should be done to reduce the impact of having such a large number of features?" only answer comes in mind PCA

upvoted 1 times
- syu31svc

1 year, 6 months ago

You want to reduce features/dimension so PCA is the answer

upvoted 4 times
- Urban_Life

1 year, 6 months ago

Of course, it's PCA.

upvoted 1 times
- C10ud9

1 year, 7 months ago

PCA is the solution. So, answer is C

upvoted 1 times

A Machine Learning Specialist is implementing a full Bayesian network on a dataset that describes public transit in New York City. One of the random variables is discrete, and represents the number of minutes New Yorkers wait for a bus given that the buses cycle every 10 minutes, with a mean of 3 minutes.

Which prior probability distribution should the ML Specialist use for this variable?

- A. Poisson distribution
- B. Uniform distribution
- C. Normal distribution
- D. Binomial distribution

Correct Answer: D

Community vote distribution

A (100%)

- ComPah

Highly Voted

1 year, 7 months ago

A

If you have information about the average (mean) number of things that happen in some given time period / interval,Poisson distribution can give you a way to predict the odds of getting some other value on a given future day

upvoted 50 times
- swagy

Highly Voted

1 year, 7 months ago

Ans: A

<https://brilliant.org/wiki/poisson-distribution/>

upvoted 7 times
- Nadia0012

Most Recent

1 month, 3 weeks ago

Selected Answer: A

ANSWER IS A

<https://www.investopedia.com/terms/d/discrete-distribution.asp>

upvoted 1 times
- bakarys

2 months, 1 week ago

Selected Answer: A

The Poisson distribution is commonly used for count data, which is the case here as we are interested in the number of minutes New Yorkers wait for a bus. The Poisson distribution is characterized by a single parameter, lambda, which represents the mean and variance of the distribution. In this case, the mean is 3 minutes, so we would set lambda to 3. The Poisson distribution assumes that events occur independently of each other, which is a reasonable assumption in this case since the waiting time for each individual is likely to be independent of the waiting time for others.

upvoted 2 times
- Ajose0

2 months, 2 weeks ago

Selected Answer: A

The Poisson distribution is a discrete probability distribution that is commonly used to model the number of events that occur in a fixed interval of time, given an average rate of occurrence.

Since the buses cycle every 10 minutes and the mean wait time is 3 minutes, it is reasonable to assume that the number of minutes New Yorkers wait for a bus can be modeled by a Poisson distribution.

upvoted 1 times
- Tomatoteacher

3 months, 2 weeks ago

Selected Answer: A

100% A, as discrete, while binomial has to be binary data (success or failure)

upvoted 1 times
- Sonoko

4 months, 2 weeks ago

Selected Answer: A

A is a discrete distribution

upvoted 1 times
- Peeking

4 months, 3 weeks ago

I do choose Poisson. A.

upvoted 1 times
- Shailendraa

7 months, 3 weeks ago

12-sep exam

upvoted 3 times

  **Shailendraa** 7 months, 3 weeks ago

Answer is A .. these types on footfalls ,etc ..answer always Poisson-distribution

upvoted 1 times

  **KM226** 1 year, 4 months ago

Selected Answer: A

I agree that the answer is A: because the Poisson distribution is a probability distribution that is used to show how many times that buses are likely to occur over a specified period.

upvoted 1 times

  **Icovarrubias** 1 year, 5 months ago

Definitely A, Poisson.

upvoted 1 times

  **tmlD** 1 year, 5 months ago

A!!!!!!!!

upvoted 1 times

  **littlewat** 1 year, 5 months ago

A!!!!!!!!

upvoted 2 times

  **takahirokoyama** 1 year, 6 months ago

Completely A.

Check this site.(<https://jakevdp.github.io/blog/2018/09/13/waiting-time-paradox/>)

upvoted 3 times

  **DzR** 1 year, 6 months ago

When ever you are given the mean occurrence of things in an interval, the best distribution to model such an event is Poisson. A is the correct answer.

upvoted 2 times

  **harmanbirstudy** 1 year, 6 months ago

Anwser is A, I am no expert here but if you just google " buses cycle every 10 minutes, with a mean of 3 minutes" , resulting page talk about Poisson distribution

upvoted 2 times

A Data Science team within a large company uses Amazon SageMaker notebooks to access data stored in Amazon S3 buckets. The IT Security team is concerned that internet-enabled notebook instances create a security vulnerability where malicious code running on the instances could compromise data privacy.

The company mandates that all instances stay within a secured VPC with no internet access, and data communication traffic must stay within the AWS network.

How should the Data Science team configure the notebook instance placement to meet these requirements?

- A. Associate the Amazon SageMaker notebook with a private subnet in a VPC. Place the Amazon SageMaker endpoint and S3 buckets within the same VPC.
- B. Associate the Amazon SageMaker notebook with a private subnet in a VPC. Use IAM policies to grant access to Amazon S3 and Amazon SageMaker.
- C. Associate the Amazon SageMaker notebook with a private subnet in a VPC. Ensure the VPC has S3 VPC endpoints and Amazon SageMaker VPC endpoints attached to it.
- D. Associate the Amazon SageMaker notebook with a private subnet in a VPC. Ensure the VPC has a NAT gateway and an associated security group allowing only outbound connections to Amazon S3 and Amazon SageMaker.

Correct Answer: C

Community vote distribution

C (100%)

- DonaldCMLIN

Highly Voted

1 year, 7 months ago

NAT gateway COULD GO OUT TO THE INTERNET AND DOWNLOAD BACK MALICIOUS D. IS NOT A GOOD ANSWER.

THE SAFE ONE IS ANSWER C. ASSOCIATE WITH VPC_ENDPOINT AND S3_ENDPOINT

upvoted 34 times
- BigEv

Highly Voted

1 year, 7 months ago

C is correct

We must use the VPC endpoint (either Gateway Endpoint or Interface Endpoint)to comply with this requirement "Data communication traffic must stay within the AWS network".

<https://docs.aws.amazon.com/sagemaker/latest/dg/notebook-interface-endpoint.html>

upvoted 20 times
- Ajose0

Most Recent

2 months, 2 weeks ago

Selected Answer: C

This configuration would meet the company's requirements for security, as the notebook instance would be placed within a private subnet in a VPC, and data communication traffic would stay within the AWS network through the use of VPC endpoints for S3 and Amazon SageMaker.

Additionally, the VPC would not have internet access, further reducing the security risk.

upvoted 2 times
- rb39

7 months, 2 weeks ago

C - "and data communication traffic must stay within the AWS network." that discards D

upvoted 1 times
- StelSen

1 year, 6 months ago

Answer should be C. Because, Security team don't want Internet Access, Option-D has NAT and will get to Internet somehow. Also connecting S3 and SageMaker EC2 instance via VPC endpoints is best way to secure the resources.

upvoted 4 times
- cloud_trail

1 year, 6 months ago

Using a NAT gateway is the old way to do it. Option C is the way to do it now. <https://cloudacademy.com/blog/vpc-endpoint-for-amazon-s3/#:~:text=Accessing%20S3%20the%20old%20way%20%28without%20VPC%20Endpoint%29,has%20no%20access%20to%20any%20outside%20public%20resources>

upvoted 2 times
- harmanbirstudy

1 year, 6 months ago

"and data communication traffic must stay within the AWS network" , NAT gateway will always go over the Internet to access S3.with NAT you can put your instances in private subnet and NAT itself in public subnet , but still in order to access S3 it will go over the internet. SO answer cannot be D.

-- C is the only correct option here , as S3 VPC endpoints is a real thing "google it" and it sole purpose is to create route from VPC endpoint to S3 , without going over the Internet.

upvoted 3 times

  **scuzzy2010** 1 year, 6 months ago

C is correct answer. D is only applicable -"If your model needs access to an AWS service that doesn't support interface VPC endpoints or to a resource outside of AWS, create a NAT gateway and configure your security groups to allow outbound connections. "
<https://docs.aws.amazon.com/sagemaker/latest/dg/host-vpc.html>

upvoted 3 times

  **v24143** 1 year, 6 months ago

D is correct
upvoted 1 times

  **krakow1234** 1 year, 6 months ago

Answer is D, read third paragraph <https://docs.aws.amazon.com/sagemaker/latest/dg/appendix-notebook-and-internet-access.html>
upvoted 1 times

  **Potato_Noodle** 1 year, 6 months ago

NAT is the way that a VPC connect to internet and other ASW service when there is NO INTERNET ACCESS FOR VPC. Thus the answer is D.
upvoted 1 times

  **Th3Dud3** 1 year, 6 months ago

"concerned that internet-enabled notebook instances create a security vulnerability where malicious code running on the instances could compromise data privacy." NAT Gateway does not mitigate this risk!
upvoted 2 times

  **yeetusdeleetus** 1 year, 6 months ago

This is the correct answer.

If this answer is confusing, study some of the associate exams before going for this one. VPC endpoint and NAT gateway are similar, but NAT gateway is for giving resources in the VPC the chance to initiate connections with the internet, whereas a VPC endpoint only allows it to go to other AWS services, which is the best solution for this question.

upvoted 2 times

  **Th3Dud3** 1 year, 6 months ago

C:
If you configure your VPC so that it doesn't have internet access, models that use that VPC do not have access to resources outside your VPC. If your model needs access to resources outside your VPC, provide access with one of the following options:

If your model needs access to an AWS service that supports interface VPC endpoints, create an endpoint to connect to that service. For a list of services that support interface endpoints, see VPC Endpoints in the Amazon VPC User Guide. For information about creating an interface VPC endpoint, see Interface VPC Endpoints (AWS PrivateLink) in the Amazon VPC User Guide.

If your model needs access to an AWS service that doesn't support interface VPC endpoints or to a resource outside of AWS, create a NAT gateway and configure your security groups to allow outbound connections. For information about setting up a NAT gateway for your VPC, see Scenario 2: VPC with Public and Private Subnets (NAT) in the Amazon Virtual Private Cloud User Guide.

upvoted 5 times

  **sebtac** 1 year, 6 months ago

what is the difference between A & C? are both answers OK?
upvoted 1 times

  **jrff** 6 months ago

It is not enough for sagemaker to communicate to S3 if both of them are inside the same VPC. Sagamaker inside a VPC needs to create a endpoint to connect to other AWS services which has endpoint too.
upvoted 1 times

  **syu31svc** 1 year, 6 months ago

I would say the answer is C
<https://docs.aws.amazon.com/vpc/latest/userguide/vpc-endpoints.html>
<https://docs.aws.amazon.com/sagemaker/latest/dg/interface-vpc-endpoint.html>
upvoted 1 times

  **AShahine21** 1 year, 6 months ago

S3 buckets don't live in a VPC
upvoted 1 times

  **msamory** 1 year, 5 months ago

You can connect to an S3 bucket through an S3 vpc endpoint
upvoted 1 times

  **GeeBeeEI** 1 year, 6 months ago

See <https://docs.aws.amazon.com/sagemaker/latest/dg/appendix-notebook-and-internet-access.html>
A is wrong, S3 buckets don't live in a VPC
B is wrong, you don't use IAM policies to grant access to SageMaker
C may be right depending on what "it" is --- attached to it If by it, this is talking about the notebook VPC, then yes
D talks of security group and NAT gateway which was mentioned in the link. D is definitely correct based on link
upvoted 2 times

  **haison8x** 1 year, 6 months ago

I think the answer is D, there are no such thing as S3 VPC endpoints and Amazon SageMaker
upvoted 1 times

  **Thai_Xuan** 1 year, 6 months ago

take a look at <https://docs.aws.amazon.com/sagemaker/latest/dg/train-vpc.html#train-vpc-s3>
upvoted 2 times

A Machine Learning Specialist has created a deep learning neural network model that performs well on the training data but performs poorly on the test data.

Which of the following methods should the Specialist consider using to correct this? (Choose three.)

- A. Decrease regularization.
- B. Increase regularization.
- C. Increase dropout.
- D. Decrease dropout.
- E. Increase feature combinations.
- F. Decrease feature combinations.

Correct Answer: BCF

Community vote distribution

BCF (80%)BCE (20%)

  **cybe001** Highly Voted 1 year, 7 months ago
Yes, answer is BCF
upvoted 24 times

  **Phong** Highly Voted 1 year, 6 months ago
Go for BCF
upvoted 14 times

  **jackzhao** Most Recent 1 month, 2 weeks ago
BCF is correct.
upvoted 2 times

  **Ajose0** 2 months, 2 weeks ago
Selected Answer: BCF
Increasing regularization helps to prevent overfitting by adding a penalty term to the loss function to discourage the model from learning the noise in the data.

Increasing dropout helps to prevent overfitting by randomly dropping out some neurons during training, which forces the model to learn more robust representations that do not depend on the presence of any single neuron.

Decreasing the number of feature combinations helps to simplify the model, making it less likely to overfit.
upvoted 4 times

  **Tomatoteacher** 3 months, 2 weeks ago
Selected Answer: BCE
I see all the comments for BCF, although when you look at F it just says decrease 'feature combinations', not features themselves. In one way to decrease feature combinations results in having more features (less feature engineering), which in turn will cause more overfitting. Unless the question is badly worded, saying less feature combinations just mean those combinations, which components will not be used, then it has to be BCE.
upvoted 1 times

  **cpal012** 4 weeks ago
Decrease feature combinations - too many irrelevant features can influence the model by drowning out the signal with noise
upvoted 1 times

  **Ajose0** 2 months, 2 weeks ago
Increasing the number of feature combinations can sometimes improve the performance of a model if the model is underfitting the data.

However, in this context, it is not likely to be a solution to overfitting.
upvoted 1 times

  **Shailendraa** 7 months, 3 weeks ago
BCF - Always remember in case of overfitting - reduce features, Add regularisation and increase dropouts.
upvoted 3 times

  **ahquiceno** 1 year, 6 months ago
BCE: The main objective of PCA (technic to feature combination) is to simplify your model features into fewer components to help visualize patterns in your data and to help your model run faster. Using PCA also reduces the chance of overfitting your model by eliminating features with

high correlation.
<https://towardsdatascience.com/dealing-with-highly-dimensional-data-using-principal-component-analysis-pca-fea1ca817fe6>
upvoted 2 times

  **uninit** 3 months ago
AWS Documentation explicitly mentions reducing feature combinations to prevent overfitting - <https://docs.aws.amazon.com/machine-learning/latest/dg/model-fit-underfitting-vs-overfitting.html>
It's B C F
upvoted 1 times

  **cloud_trail** 1 year, 6 months ago
B/C/F Easy peasy.
upvoted 1 times

  **apnu** 1 year, 6 months ago
BCF 100%
upvoted 1 times

  **obaidur** 1 year, 6 months ago
BCF
F
explained in AWS document:
Feature selection: consider using fewer feature combinations, decrease n-grams size, and decrease the number of numeric attribute bins.
Increase the amount of regularization used
<https://docs.aws.amazon.com/machine-learning/latest/dg/model-fit-underfitting-vs-overfitting.html>
upvoted 4 times

  **fhuadeen** 1 year, 6 months ago
It is BCE:
you increase regularisation to reduce overfitting
you increase dropout to reduce overfitting
and feature combination is the combination of the strength of multiple complementary features to yield a more powerful feature (which means, you are reducing number of features). Reducing number of features and using features with stronger information helps to reduce overfitting.
upvoted 3 times

  **sebtac** 1 year, 6 months ago
BCF

it is C not D as higher value = higher ratio of 0-valued weights = higher regularization = less overfitting
upvoted 1 times

  **syu31svc** 1 year, 6 months ago
BCF for sure
upvoted 2 times

  **algorithmish** 1 year, 6 months ago
overfitting.
B - increase regularisation helps to generalize
C - increase dropout (probability of a node to be turned off) also helps to generalize
F - decreasing feature combinations (removing features like mean or variance, artificial features) helps to generalize
upvoted 5 times

  **eji** 1 year, 6 months ago
i think BCE , E because if we increase features combination, we also decrease the number of feature
upvoted 2 times

  **tff** 1 year, 6 months ago
B, C, F for sure
upvoted 1 times

  **DScore** 1 year, 6 months ago
BCF is the combination to go.
for drop-out value of 1 means no output (or 100% filtering out). Features should definitely decrease as more features tends to overfit.
upvoted 2 times

A Data Scientist needs to create a serverless ingestion and analytics solution for high-velocity, real-time streaming data. The ingestion process must buffer and convert incoming records from JSON to a query-optimized, columnar format without data loss. The output datastore must be highly available, and Analysts must be able to run SQL queries against the data and connect to existing business intelligence dashboards.

Which solution should the Data Scientist build to satisfy the requirements?

- A. Create a schema in the AWS Glue Data Catalog of the incoming data format. Use an Amazon Kinesis Data Firehose delivery stream to stream the data and transform the data to Apache Parquet or ORC format using the AWS Glue Data Catalog before delivering to Amazon S3. Have the Analysts query the data directly from Amazon S3 using Amazon Athena, and connect to BI tools using the Athena Java Database Connectivity (JDBC) connector.
- B. Write each JSON record to a staging location in Amazon S3. Use the S3 Put event to trigger an AWS Lambda function that transforms the data into Apache Parquet or ORC format and writes the data to a processed data location in Amazon S3. Have the Analysts query the data directly from Amazon S3 using Amazon Athena, and connect to BI tools using the Athena Java Database Connectivity (JDBC) connector.
- C. Write each JSON record to a staging location in Amazon S3. Use the S3 Put event to trigger an AWS Lambda function that transforms the data into Apache Parquet or ORC format and inserts it into an Amazon RDS PostgreSQL database. Have the Analysts query and run dashboards from the RDS database.
- D. Use Amazon Kinesis Data Analytics to ingest the streaming data and perform real-time SQL queries to convert the records to Apache Parquet before delivering to Amazon S3. Have the Analysts query the data directly from Amazon S3 using Amazon Athena and connect to BI tools using the Athena Java Database Connectivity (JDBC) connector.

Correct Answer: A

Community vote distribution

A (75%)

B (25%)

- DonaldCMLIN

Highly Voted

1 year, 7 months ago

Kinesis Data Analytics NO PARQET FORMAT, BESIDES THAT JSON NO NEED TO STORE IN S3. RDS ISN'T serverless ingestion and analytics solution

ANSWER IS A.

upvoted 31 times
- georgeZ

Highly Voted

1 year, 7 months ago

I thinks it should be A please check <https://aws.amazon.com/blogs/big-data/analyzing-apache-parquet-optimized-data-using-amazon-kinesis-data-firehose-amazon-athena-and-amazon-redshift/>

upvoted 13 times
- Ajose0

Most Recent

2 months, 2 weeks ago

Selected Answer: A

This solution leverages AWS Glue to create a schema of the incoming data format, which helps to buffer and convert the records to a query-optimized, columnar format without data loss.

The Amazon Kinesis Data Firehose delivery stream is used to stream the data and transform it to Apache Parquet or ORC format using the AWS Glue Data Catalog, and the data is stored in Amazon S3, which is highly available. The Analysts can then query the data directly from Amazon S3 using Amazon Athena, and connect to BI tools using the Athena JDBC connector.

This solution provides a serverless, scalable, and cost-effective solution for real-time streaming data ingestion and analytics.

upvoted 1 times

sqavi

2 months, 3 weeks ago

Selected Answer: A

Since you want to buffer and convert data so A is correct answer. No other option is fulfilling this requirement

upvoted 1 times

Peeking

4 months, 3 weeks ago

Selected Answer: A

I go for A. However, I am not sure why AWS Glue is very important here given that Firehose can convert JSON to parquet.

upvoted 2 times

Tony_1406

1 week, 2 days ago

<https://www.examttopics.com/exams/amazon/aws-certified-machine-learning-specialty/custom-view/>

94/406

If I haven't remembered correctly. Athena requires a schema of the S3 object to perform SQL query. That's probably why we need Glue for the schema

upvoted 1 times

  **itallomd** 4 months, 3 weeks ago

Why AWS Glue is needed? Firehose could convert to parquet directly...

upvoted 2 times

  **Ccindy** 5 months ago

Selected Answer: B

Kinesis Data Analytics is near real-time, not real time

upvoted 1 times

  **ryuhei** 7 months, 1 week ago

Selected Answer: A

Answer is "A"

upvoted 1 times

  **ovokpus** 10 months, 1 week ago

Selected Answer: A

The difference between "real-time" and "near-real-time" is pretty semantic(60s). The fact that the data comes through kinesis data streams (real time) is implied as the only valid input to firehose.

upvoted 1 times

  **ovokpus** 10 months, 1 week ago

Mind you, "the ingestion process must buffer and transform incoming records from JSON to a query-optimized, columnar format"

That is exactly what kinesis firehose does.
"Kinesis Data Firehose buffers incoming data before delivering it to Amazon S3. You can configure the values for S3 buffer size (1 MB to 128 MB) or buffer interval (60 to 900 seconds), and the condition satisfied first triggers data delivery to Amazon S3."

See link: <https://aws.amazon.com/kinesis/data-firehose/faqs/#:~:text=Kinesis%20Data%20Firehose%20buffers%20incoming,data%20delivery%20to%20Amazon%20S3>.

upvoted 3 times

  **TerrancePythonJava** 1 year, 1 month ago

Selected Answer: B

Data Firehose is always Near Real Time not Real Time. The prompt clearly states that process must be done in Real Time.

upvoted 1 times

  **anttan** 1 year, 4 months ago

Why A? Firehose is near real-time, and not real-time which is a requirement

upvoted 1 times

  **cpal012** 1 month, 3 weeks ago

There is no requirement for real time processing. It says the data is in real time but the processing of that data should buffer

upvoted 1 times

  **harmanbirstudy** 1 year, 6 months ago

ANSWER is A -- and every statement in it is accurate.
Firehose does integrate with GLue data catalog and it also "Buffers" the data .
"When Kinesis Data Firehose processes incoming events and converts the data to Parquet, it needs to know which schema to apply." This is achived by glue data catalog and athena and it works on real-time data ingest.See link below.

<https://aws.amazon.com/blogs/big-data/analyzing-apache-parquet-optimized-data-using-amazon-kinesis-data-firehose-amazon-athena-and-amazon-redshift/>

upvoted 5 times

  **lightblue** 1 year, 6 months ago

<https://aws.amazon.com/blogs/aws/new-serverless-streaming-etl-with-aws-glue/>
A is the answer imo

upvoted 1 times

  **oMARKOo** 1 year, 6 months ago

Should be A.
<https://docs.aws.amazon.com/firehose/latest/dev/record-format-conversion.html>
it states that " Use AWS Glue to create a schema in the AWS Glue Data Catalog. Kinesis Data Firehose then references that schema and uses it to interpret your input data. "

upvoted 1 times

  **williamsuning** 1 year, 6 months ago

I was initially about to choose A, but I think the answer A would be better if using Kinesis Data stream before firehose, since data passed to firehose has to be loss-tolerant during transfer and conflicts with "must buffer" during ingestion process.

upvoted 1 times

-   **syu31svc** 1 year, 6 months ago
Answer is A
Following links support it:
<https://docs.aws.amazon.com/athena/latest/ug/connect-with-jdbc.html>
https://aws.amazon.com/about-aws/whats-new/2018/05/stream_real_time_data_in_apache_parquet_or_orc_format_using_firehose/
upvoted 2 times
-   **Urban_Life** 1 year, 6 months ago
Answer 'A' is confusing. Still don't understand rational behind answer 'B'. Answer C & D is not even close. Anybody else has 2nd opinion? Either A or B?
upvoted 2 times
-   **Achievement** 1 year, 6 months ago
because it's streaming data, so not choose B?
upvoted 1 times

An online reseller has a large, multi-column dataset with one column missing 30% of its data. A Machine Learning Specialist believes that certain columns in the dataset could be used to reconstruct the missing data.

Which reconstruction approach should the Specialist use to preserve the integrity of the dataset?

- A. Listwise deletion
- B. Last observation carried forward
- C. Multiple imputation
- D. Mean substitution

Correct Answer: C

Reference:

<https://worldwidescience.org/topicpages/i/imputing+missing+values.html>

Community vote distribution

C (100%)

  **rsimham** Highly Voted 1 year, 7 months ago

C looks correct since multiple imputation can be performed based on the related variable as given in the question
upvoted 25 times

  **harmanbirstudy** Highly Voted 1 year, 6 months ago

Multiple Imputation by Chained Equations or MICE, as per udemy this is always the best answer of all
upvoted 7 times

  **Ajose0** Most Recent 2 months, 2 weeks ago

Selected Answer: C

Multiple imputation is a statistical technique for handling missing data that involves generating multiple versions of the dataset with missing values filled in, and then combining the results to produce a single, complete dataset.

This approach takes into account the relationship between variables in the dataset, and uses statistical models to predict missing values based on the information in other columns. This helps to preserve the integrity of the dataset by avoiding the introduction of bias or systematic error into the results.

upvoted 2 times

  **passionatecricketer** 1 year, 5 months ago

I am trying to understand why Mean Substitution is not the solution. Imputation typically uses the mean if the missing data is random, implying the substitution is not biased.

upvoted 2 times

  **cpal012** 1 month, 3 weeks ago

Mean substitution is limited to the current column. In this case, the requirement is to impute missing data from other columns
upvoted 1 times

  **rhuanca** 11 months, 2 weeks ago

Reason is if you replace 30% of the missing values , likely you will bias the variable.
upvoted 1 times

  **syu31svc** 1 year, 6 months ago

If it's handling missing data then imputation comes into play
Answer is C 100%
upvoted 1 times

  **Wira** 1 year, 6 months ago

<https://www.countants.com/blogs/heres-how-you-can-configure-automatic-imputation-of-missing-data/> C
upvoted 1 times

  **roytruong** 1 year, 6 months ago

it's C
upvoted 1 times

  **dhs227** 1 year, 6 months ago

A common strategy used to impute missing values is to replace missing values with the mean or median value. It is important to understand your data before choosing a strategy for replacing missing values. <https://docs.aws.amazon.com/machine-learning/latest/dg/feature-processing.html>
upvoted 2 times

A company is setting up an Amazon SageMaker environment. The corporate data security policy does not allow communication over the internet. How can the company enable the Amazon SageMaker service without enabling direct internet access to Amazon SageMaker notebook instances?

- A. Create a NAT gateway within the corporate VPC.
- B. Route Amazon SageMaker traffic through an on-premises network.
- C. Create Amazon SageMaker VPC interface endpoints within the corporate VPC.
- D. Create VPC peering with Amazon VPC hosting Amazon SageMaker.

Correct Answer: A

Reference:

<https://docs.aws.amazon.com/sagemaker/latest/dg/sagemaker-dg.pdf>

(46)

Community vote distribution

C (100%)

  **DonaldCMLIN** Highly Voted 1 year, 7 months ago

NAT CLOUD GO OUT TO THE INTERNET, IT STILL CANNOT PREVENT DOWNLOAD MALICIOUS BY YOURSELF.

THE RIGHT ANSWER IS C.
C.INTERFACE VPC ENDPOINT

<https://docs.aws.amazon.com/sagemaker/latest/dg/sagemaker-dg.pdf> (516)
https://docs.aws.amazon.com/zh_tw/vpc/latest/userguide/vpc-endpoints.html
upvoted 42 times

  **rsimham** 1 year, 7 months ago

Not sure if C is correct in this particular scenario.
From <https://docs.aws.amazon.com/sagemaker/latest/dg/sagemaker-dg.pdf>
Page 202 of the SageMaker Guide has:
If you allowed access to resources from your VPC, enable direct internet access. For Direct internet access, choose Enable. Without internet access, you can't train or host models from notebooks on this notebook instance unless your VPC has a NAT gateway and your security group allows outbound connect
upvoted 2 times

  **rsimham** 1 year, 7 months ago

A may the right answer
upvoted 1 times

  **Selectron** 1 year, 6 months ago

There are two possible solutions, but the safer solution and easier is trough VPC endpoints.

You can connect to your notebook instance from your VPC through an interface endpoint in your Virtual Private Cloud (VPC) instead of connecting over the internet. When you use a VPC interface endpoint, communication between your VPC and the notebook instance is conducted entirely and securely within the AWS network. And there is not problem that the notebooks does not have public internet. Because Amazon SageMaker notebook instances support Amazon Virtual Private Cloud (Amazon VPC) interface endpoints that are powered by AWS PrivateLink. Each VPC endpoint is represented by one or more Elastic Network Interfaces (ENIs) with private IP addresses in your VPC subnets. Each VPC endpoint is represented by one or more Elastic Network Interfaces (ENIs) with private IP addresses in your VPC subnets...
so the Answer is C.
upvoted 4 times

  **tap123** Highly Voted 1 year, 7 months ago

C is correct. "The VPC interface endpoint connects your VPC directly to the Amazon SageMaker API or Runtime without an internet gateway, **NAT** device, VPN connection, or AWS Direct Connect connection." <https://docs.aws.amazon.com/sagemaker/latest/dg/interface-vpc-endpoint.html>
upvoted 16 times

  **jackzhao** Most Recent 1 month, 3 weeks ago

C is correct, NAT allow outband traffic pass through internet.
upvoted 1 times

  **Nadia0012** 1 month, 3 weeks ago

Selected Answer: C

To prevent SageMaker from providing internet access to your Studio notebooks, you can disable internet access by specifying the VPC only network access type when you the onboard to Studio or call CreateDomain API. As a result, you won't be able to run a Studio notebook unless your VPC has an interface endpoint to the SageMaker API and runtime, or a NAT gateway with internet access, and your security groups allow outbound connections.

upvoted 2 times

  **Nadia0012** 1 month, 3 weeks ago

If you want to allow internet access, you must use a example through an internet gateway. If you don't want to allow internet access, NAT gateway with access to the internet, for create interface VPC endpoints (AWS PrivateLink) to allow Studio to access the following services with the corresponding service names. You must also associate the security groups for your VPC with these endpoints.

upvoted 1 times

  **Nadia0012** 1 month, 3 weeks ago

To disable direct internet access, under Direct Internet access, simply choose Disable – use VPC only , and select the Create notebook instance button at the bottom. You are ready to go.
from: <https://aws.amazon.com/blogs/machine-learning/customize-your-amazon-sagemaker-notebook-instances-with-lifecycle-configurations-and-the-option-to-disable-internet-access/#:~:text=To%20disable%20direct%20internet%20access%2C%20under%20Direct%20Internet%20access%2C%20simply,running%2C%20without%20direct%20internet%20access.>

upvoted 1 times

  **AjoseO** 2 months, 2 weeks ago

Selected Answer: C

A VPC interface endpoint is a private connection between a VPC and Amazon SageMaker that is powered by AWS PrivateLink. With a VPC interface endpoint, traffic between the VPC and Amazon SageMaker never leaves the Amazon network.

upvoted 2 times

  **Ob1KN0B** 8 months ago

Selected Answer: C

Page 3438 of <https://docs.aws.amazon.com/sagemaker/latest/dg/sagemaker-dg.pdf>

upvoted 2 times

  **ovokpus** 10 months, 1 week ago

Selected Answer: C

VPC Interface endpoints

upvoted 3 times

  **gcpwhiz** 1 year, 5 months ago

If the question just had the last sentence, the answer would be A or C, per this page:<https://docs.aws.amazon.com/sagemaker/latest/dg/appendix-notebook-and-internet-access.html>. "To disable direct internet access, you can specify a VPC for your notebook instance. By doing so, you prevent SageMaker from providing internet access to your notebook instance. As a result, the notebook instance won't be able to train or host models unless your VPC has an interface endpoint (PrivateLink) or a NAT gateway, and your security groups allow outbound connections."

HOWEVER, the question has more context that internet access is not allowed by the corporate policy. ("When you use a VPC interface endpoint, communication between your VPC and the notebook instance is conducted entirely and securely within the AWS network.") Therefore, the answer must be ONLY C.

upvoted 4 times

  **scuzzy2010** 1 year, 5 months ago

Answer is C. From <https://docs.aws.amazon.com/sagemaker/latest/dg/interface-vpc-endpoint.html> ->
"The VPC interface endpoint connects your VPC directly to the SageMaker API or Runtime without an internet gateway, NAT device, VPN connection, or AWS Direct Connect connection. The instances in your VPC don't need public IP addresses to communicate with the SageMaker API or Runtime."

upvoted 2 times

  **cloud_trail** 1 year, 6 months ago

I see a lot of people employing pretzel logic to try to explain why they should be using NAT. The question states no internet communication. Period. No internet means no NAT. Answer is C.

upvoted 4 times

  **Omar_Cascudo** 1 year, 6 months ago

The point here is that SM notebooks need internet access to download updates and some open data. Although C is ok, it won't allow SM notebook to download such data. NAT GW will allow this action. I go with A.

upvoted 2 times

  **weslleylc** 1 year, 6 months ago

Notebooks are Internet-enabled by default, If disabled, your VPC needs an interface endpoint (PrivateLink) or NAT Gateway, and allow outbound connections, for training and hosting to work. A doesn't have an outbound connection allowed the correct answer must be C.

upvoted 2 times

  **syu31svc** 1 year, 6 months ago

Answer is C
<https://docs.aws.amazon.com/sagemaker/latest/dg/interface-vpc-endpoint.html>

upvoted 1 times

  **jonclem** 1 year, 6 months ago

I'd be inclined to say A myself. Having Architected and built environments using PrivateLink (VPC Endpoints) the fundamental reason behind them is to keep your infra accessible in a private capacity.

So, say you are connecting from corp office to the Cloud and don't want the end-user exposing any data over the public internet, you would utilise an Endpoint connection.

@Donald... take your finger off CAPS dude ! You can make your comment without "shouting" !
upvoted 5 times

  **ardisch** 1 year, 6 months ago

The right answer is C
<https://docs.aws.amazon.com/sagemaker/latest/dg/sagemaker-dg.pdf> (927)
upvoted 1 times

  **kumarvn** 1 year, 6 months ago

Both A & C are correct
upvoted 1 times

  **kumarvn** 1 year, 6 months ago

Notebooks are internet enabled by default. I fdisabled, VPC needs an interface enpoint or NAT gateway, and allow outbound connections, for training and hosting to work.
upvoted 1 times

  **C10ud9** 1 year, 6 months ago

Has to be C
upvoted 2 times

A Machine Learning Specialist is training a model to identify the make and model of vehicles in images. The Specialist wants to use transfer learning and an existing model trained on images of general objects. The Specialist collated a large custom dataset of pictures containing different vehicle makes and models.

What should the Specialist do to initialize the model to re-train it with the custom data?

- A. Initialize the model with random weights in all layers including the last fully connected layer.
- B. Initialize the model with pre-trained weights in all layers and replace the last fully connected layer.
- C. Initialize the model with random weights in all layers and replace the last fully connected layer.
- D. Initialize the model with pre-trained weights in all layers including the last fully connected layer.

Correct Answer: B

Community vote distribution

B (100%)

  **rsimham** Highly Voted 1 year, 7 months ago

Ans B sounds correct
upvoted 28 times

  **Ajose0** Most Recent 2 months, 2 weeks ago

Selected Answer: B

In transfer learning, a pre-trained model is used as a starting point to train a new model on a different task, typically using a smaller dataset. The pre-trained model contains weights that have been learned from a large amount of data on a related task, and these weights can be leveraged to train the new model more efficiently.

To re-train the model with the custom data, the Specialist should initialize the model with pre-trained weights in all layers, as these weights can provide a good starting point for the new task. The Specialist should then replace the last fully connected layer, which is responsible for making the final predictions, as this layer will likely need to be modified to reflect the new task. By keeping the pre-trained weights in the other layers, the Specialist can take advantage of the knowledge learned from the previous task, and potentially speed up the training process.

upvoted 2 times

  **Peeking** 4 months, 3 weeks ago

Selected Answer: B

The fully connected layer will need to be trained from scratch to incorporate the features of his domain problem (Car models)

upvoted 2 times

  **Shailendraa** 7 months, 3 weeks ago

12-sep exam
upvoted 2 times

  **chrisdavidi** 9 months, 2 weeks ago

D is the best - here is why
Question is not to design a final production with deep lense - it is to use it as a dev platform to comeup with a edge ML vs. dump load all to S3 - which is very wasteful! AWS did not mae deeplense as a toy for devs! it is meant to help companies experiment with edge ML And then copy and reuse the open hardware platfom

upvoted 1 times

  **ckkobe24** 11 months, 3 weeks ago

Selected Answer: B

one of the method to implement transfer learning

upvoted 2 times

  **DzR** 1 year, 6 months ago

I will go with B, we are mainly concerned with the output layer for us to get the desired results, hence we need to replace it.

upvoted 1 times

  **bobdylan1** 1 year, 6 months ago

B is correct

upvoted 2 times

  **sebtac** 1 year, 6 months ago

Actually, it should be NONE of IT!.... it should be like B with exception that 20-40% top layers should be retrained :) -- this is classic transfer learning setup, so B is the answer here.

upvoted 2 times

-   **syu31svc** 1 year, 6 months ago

Since it is transfer learning where you retain knowledge from a solved problem, weights are to be pre-trained. So A and C are wrong. Between B and D, D keeps the last layer but that is not what you want since the question mentions a change of general objects to more specific types. So answer is B

upvoted 3 times
-   **scuzzy2010** 1 year, 6 months ago

I feel the answer is "C" because it was not trained on images of "similar" objects, it was trained on images of "general" objects. If it were trained on similar objects, then answer would be "B".

upvoted 1 times
-   **yeetusdeleetus** 1 year, 6 months ago

Transfer learning doesn't work like that. All images are 'somewhat' similar, so a pretrained neural network that was trained on different types of images is still better than random weights. B.

upvoted 1 times
-   **cloud_trail** 1 year, 6 months ago

If you do option C, you're not doing transfer learning at all. You might just as well start over with your own architecture.

upvoted 2 times
-   **tff** 1 year, 6 months ago

<https://www.hackerearth.com/practice/machine-learning/transfer-learning/transfer-learning-intro/tutorial/>
answer is B.
Read the scenario:
The target dataset is large and similar to the base training dataset.
Since the target dataset is large, we have more confidence that we won't overfit if we try to fine-tune through the full network. Therefore, we:

1.Remove the last fully connected layer and replace with the layer matching the number of classes in the target dataset;

2.Randomly initialize the weights in the new fully connected layer;

3.Initialize the rest of the weights using the pre-trained weights, i.e., unfreeze the layers of the pre-trained network;

4.Retrain the entire neural network;

So, first look at 3, and then at 1.

upvoted 3 times
-   **fw** 1 year, 6 months ago

From what I read in <https://kharshit.github.io/blog/2018/08/10/transfer-learning>
Choice seems to be C.
Case III: Large dataset, Similar data

remove the last fully connected layer and replace with a layer matching the number of classes in the new data set

randomly initialize the weights in the new fully connected layer

initialize the rest of the weights using the pre-trained weights

re-train the entire neural network

Case IV: Large dataset, Different data

remove the last fully connected layer and replace with a layer matching the number of classes in the new data set

retrain the network from scratch with randomly initialized weights

alternatively, you could just use the same strategy as the "large and similar" data case

upvoted 1 times
-   **C10ud9** 1 year, 6 months ago

B is correct, replace only the last layer in transfer learning

upvoted 2 times
-   **roytruong** 1 year, 7 months ago

B is right

upvoted 2 times

An office security agency conducted a successful pilot using 100 cameras installed at key locations within the main office. Images from the cameras were uploaded to Amazon S3 and tagged using Amazon Rekognition, and the results were stored in Amazon ES. The agency is now looking to expand the pilot into a full production system using thousands of video cameras in its office locations globally. The goal is to identify activities performed by non-employees in real time

Which solution should the agency consider?

- A. Use a proxy server at each local office and for each camera, and stream the RTSP feed to a unique Amazon Kinesis Video Streams video stream. On each stream, use Amazon Rekognition Video and create a stream processor to detect faces from a collection of known employees, and alert when non-employees are detected.
- B. Use a proxy server at each local office and for each camera, and stream the RTSP feed to a unique Amazon Kinesis Video Streams video stream. On each stream, use Amazon Rekognition Image to detect faces from a collection of known employees and alert when non-employees are detected.
- C. Install AWS DeepLens cameras and use the DeepLens_Kinesis_Video module to stream video to Amazon Kinesis Video Streams for each camera. On each stream, use Amazon Rekognition Video and create a stream processor to detect faces from a collection on each stream, and alert when non-employees are detected.
- D. Install AWS DeepLens cameras and use the DeepLens_Kinesis_Video module to stream video to Amazon Kinesis Video Streams for each camera. On each stream, run an AWS Lambda function to capture image fragments and then call Amazon Rekognition Image to detect faces from a collection of known employees, and alert when non-employees are detected.

Correct Answer: D

Reference:

<https://aws.amazon.com/blogs/machine-learning/video-analytics-in-the-cloud-and-at-the-edge-with-aws-deeplens-and-kinesis-video-streams/>

Community vote distribution

A (75%)

D (25%)

-   **scuzzy2010**

Highly Voted 

 1 year, 6 months ago

Answer is "A". C and D are out as DeepLens is not offered as a commercial product. It is purely for developers to experiment with.
From <https://aws.amazon.com/deeplens/device-terms-of-use/>
" (i) you may use the AWS DeepLens Device for personal, educational, evaluation, development, and testing purposes, and not to process your production workloads;"

A is correct as it's will analyse live video streams instead of images.
From <https://aws.amazon.com/rekognition/video-features/>
"Amazon Rekognition Video can identify known people in a video by searching against a private repository of face images. "
upvoted 34 times
-   **Mezaji** 1 year, 5 months ago

Agreed
upvoted 2 times
-   **WWODIN**

Highly Voted 

 1 year, 7 months ago

Why not A?
DeepLens is for development purpose and much more expensive than just a camera.
They are referring to 1000 camera in production scale?
upvoted 12 times
-   **cybe001** 1 year, 7 months ago

C is the correct answer. We could use A, since it is for security service, DeepLens allows to notify the security (through aws lamda) immediately when it sees non employee at the office location. So C is more appropriate for the problem than A.
upvoted 5 times
-   **scuzzy2010** 1 year, 6 months ago

DeepLens is for developers only, it is not available as a commercial product.
upvoted 5 times
-   **sdsfsdsf** 1 year, 7 months ago

A bit off topic but yeah, how could you justify using deep lens for production. Cameras have viewing angles, weather proofing, network connectivity issues (Wifi only), infra red for low lighting conditions, no power over ethernet? Using Deeplens would be laughable for a full production system.
upvoted 8 times
-   **Valcilio**

Most Recent 

 1 month, 3 weeks ago

Selected Answer: D

D is the answer now, DeepLens is used for situations like this!

upvoted 1 times

  **cpal012** 1 month, 1 week ago

Maybe, its EOL Jan 2024

upvoted 1 times

  **expertguru** 3 months, 3 weeks ago

Think big picture - you tested something (let say code python) and ready to implement into prod will you move python code or java code! Here in this particular case, they tested with actual video camera and they did not say deeplense so answer is A! For knowledge sake if they say in real exam it is tested with deeplense ---then ideal solution should be model inference happening at deeplense itself with search against existing employees and send back model inference when it detect new faces who are not employees back to cloud may be S3

upvoted 1 times

  **Sivadharan** 11 months, 3 weeks ago

Selected Answer: A

Answer is "A".

As mentioned in below user comment, DeepLens is not offered as a commercial product.

<https://aws.amazon.com/deeplens/device-terms-of-use/>

upvoted 3 times

  **geekgirl007** 1 year, 3 months ago

Answer is C based on this exact same article answer: <https://docs.aws.amazon.com/rekognition/latest/dg/streaming-video.html>

Rekognition stream processor doesn't need Lambda.

upvoted 3 times

  **pp13** 1 year, 4 months ago

It cannot be option C & D as it is given in terms n conditions of using deeplens that its not for production purpose

upvoted 1 times

  **anttan** 1 year, 5 months ago

Is it D? By process of elimination, it has to be C & D. C is eliminated as it's to detect faces, and not really detect the faces of known employees

upvoted 1 times

  **Huy** 1 year, 5 months ago

A. <https://aws.amazon.com/blogs/machine-learning/easily-perform-facial-analysis-on-live-feeds-by-creating-a-serverless-video-analytics-environment-with-amazon-rekognition-video-and-amazon-kinesis-video-streams/>

upvoted 1 times

  **StelSen** 1 year, 6 months ago

I will go with 'A'. Because setting up DeepLens will cost S\$304.59. Whereas setting up proxy server using Raspberry PI (production equivalent) will take around S\$50 + Camera may be \$50. So overall solution is for A might be \$100. Company already did a pilot with some existing camera. So why intro DeepLens now. Use cost effective solution.

upvoted 2 times

  **srinu3054** 1 year, 6 months ago

Its C!

<https://docs.aws.amazon.com/rekognition/latest/dg/streaming-video.html>

upvoted 4 times

  **SophieSu** 1 year, 6 months ago

D is correct. Because you DEFINITELY need the Lambda Function to "CALL" the Amazon Rekognition to start the job (detect faces)!

upvoted 2 times

  **achiko** 1 year, 6 months ago

C. You don't need Lambda between video stream and rekognition

upvoted 2 times

  **SophieSu** 1 year, 6 months ago

D is the correct answer. The point is that the lambda function is necessary before the Amazon Rekognition. See "<https://docs.aws.amazon.com/rekognition/latest/dg/stored-video-lambda.html>". "You can use Lambda functions with Amazon Rekognition Video operations. For example, the following diagram shows a website that uses a Lambda function to automatically start analysis of a video when it's uploaded to an Amazon S3 bucket. When the Lambda function is triggered, it calls StartLabelDetection to start detecting labels in the uploaded video. For information about using Lambda to process event notifications from an Amazon S3 bucket, see Using AWS Lambda with Amazon S3 Events."

upvoted 3 times

  **Aashi22** 1 year, 6 months ago

Cofusion b/w option C & D. Answer should be option D as per link <https://aws.amazon.com/blogs/machine-learning/video-analytics-in-the-cloud-and-at-the-edge-with-aws-deeplens-and-kinesis-video-streams/>

upvoted 2 times

  **scuzzy2010** 1 year, 6 months ago

Answer is "A". C and D are out because DeepLens is NOT a commercial product, it's sold only to developers.

A mentions 'stream processor' , and as mentioned here is specifically used to detect and recognise faces in streaming video ->

(https://docs.aws.amazon.com/rekognition/latest/dg/API_CreateStreamProcessor.html) "Amazon Rekognition stream processor that you can use to detect and recognize faces in a streaming video."

upvoted 4 times

  **cloud_trail** 1 year, 6 months ago

Option D. I think that the goal for the company is to do the facial recognition at the edge, which is what DeepLens does. The pilot program cameras were not ML enabled so they had to upload all all images from all cameras to S3 and then use Rekognition. Not practical with 1000s of "dumb" cameras. DL can do object detection and the call Rekognition to identify non-employees and the send alerts.

upvoted 1 times

  **harmanbirstudy** 1 year, 6 months ago

Deep lens cannot be used at large scale so it cannot be C or D
--Between A and B , A mentions Amazon Rekognition Video which is used for detection on video data.
-- Hence ANSWER is A

upvoted 3 times

A Marketing Manager at a pet insurance company plans to launch a targeted marketing campaign on social media to acquire new customers. Currently, the company has the following data in Amazon Aurora:

- ⇒ Profiles for all past and existing customers
- ⇒ Profiles for all past and existing insured pets
- ⇒ Policy-level information
- ⇒ Premiums received
- ⇒ Claims paid

What steps should be taken to implement a machine learning model to identify potential new customers on social media?

- A. Use regression on customer profile data to understand key characteristics of consumer segments. Find similar profiles on social media
- B. Use clustering on customer profile data to understand key characteristics of consumer segments. Find similar profiles on social media
- C. Use a recommendation engine on customer profile data to understand key characteristics of consumer segments. Find similar profiles on social media.
- D. Use a decision tree classifier engine on customer profile data to understand key characteristics of consumer segments. Find similar profiles on social media.

Correct Answer: C

Community vote distribution

B (100%)

🗨️ 👤 **DonaldCMLIN** Highly Voted 1 year, 7 months ago

All of the questions in the preceding examples rely on having example data that includes answers. There are times that you don't need, or can't get, example data with answers. This is true for problems whose answers identify groups. For example:

"I want to group current and prospective customers into 10 groups based on their attributes. How should I group them? " You might choose to send the mailing to customers in the group that has the highest percentage of current customers. That is, prospective customers that most resemble current customers based on the same set of attributes. For this type of question, Amazon SageMaker provides the K-Means Algorithm.

<https://docs.aws.amazon.com/sagemaker/latest/dg/algos.html>

Clustering algorithms are unsupervised. In unsupervised learning, labels that might be associated with the objects in the training dataset aren't used.

<https://docs.aws.amazon.com/sagemaker/latest/dg/algo-kmeans-tech-notes.html>

THE ANSWER COULD BE B.clustering on customer profile data to understand key characteristic
upvoted 34 times

🗨️ 👤 **rsimham** 1 year, 7 months ago

Yes, Clustering seems to be more appropriate in this scenario than recommender system
upvoted 10 times

🗨️ 👤 **haison8x** 1 year, 6 months ago

<https://towardsdatascience.com/customer-segmentation-with-machine-learning-a0ac8c3d4d84>

B
upvoted 3 times

🗨️ 👤 **hughhughhugh** Highly Voted 1 year, 7 months ago

why bit D? marketing campaign mainly use classifiers
upvoted 5 times

🗨️ 👤 **Nadia0012** Most Recent 1 month, 3 weeks ago

I am still not sure between B & C, reading this post <https://medium.com/@daniilkorbut/recommendation-system-algorithms-ba67f39ac9a3> seems clustering is possible but there are a few things to consider: clustering is done on large dataset while here is not mentioned. also it's an unsupervised algorithm, here we have existing customer's data.
upvoted 1 times

🗨️ 👤 **expertguru** 3 months, 3 weeks ago

C should not be answer --- they didn't say any recommendation AWS AI/ML service. If answers shown with any of the below two services then answer should have been select multiple choices!

1) <https://docs.aws.amazon.com/personalize/latest/dg/what-is-personalize.html>

Creating a targeted marketing campaign – You can use Amazon Personalize to generate segments of users who will most likely interact with items in your catalog. Then you can use an AWS service or third party service to create a targeted marketing campaign that promotes different items to different user segments.

2) Amazon pinpoint segment
<https://docs.aws.amazon.com/pinpoint/latest/userguide/segments.html>
upvoted 1 times

🗨️ **chrisdavid** 9 months, 2 weeks ago

we need to figure out when to use the recommendation engine vs regression vs clustering this is helpful: <https://dzone.com/articles/decision-trees-vs-clustering-algorithms-vs-linear>
however, a recommendation engine uses clustering to make the suggestion so it sounds more comprehensive I would say C
upvoted 2 times

🗨️ **ovokpus** 10 months, 1 week ago

Selected Answer: B

Clustering is basically the modelling method for customer segmentation.
upvoted 5 times

🗨️ **Vita_Rasta84444** 1 year, 5 months ago

I agree, the Answer is B. We do not have a new uncategorised customers, on the system, but we should find similar customers on the web, which we can do by clustering, making profila of customer with insured pet.
upvoted 2 times

🗨️ **astonm13** 1 year, 5 months ago

Considering the quote "to understand characteristics of customer segments", I would say recommender engine does not fit to it. I would go for the clustering (answer B). It is not supervised, cause we don't have info on those people who where targeted but refused to become our customers.
upvoted 1 times

🗨️ **ss001maryam** 1 year, 5 months ago

Answer is B. recomendation enginbe is correct if we want to make decisions on current customers, here we want to make decision on new customers as well. So we need clustring.
upvoted 1 times

🗨️ **cloud_trail** 1 year, 6 months ago

Option C. This is not purely unsupervised, as clustering would be, because we have current and past customer profiles to go on. We want to find new customers by finding similar profiles on social media. So it is supervised to some extent. It's not a cluster problem; it is user-user collaborative filtering. The key is to recognize that this is not clustering. You're not blindly trying to group people. You have existing profiles that you are comparing them to.
upvoted 4 times

🗨️ **harmanbirstudy** 1 year, 6 months ago

Recommendation Engine would make sense it we are recommending someone coming on our website and we recommend something them. Here we need to go to social media, where we cannot apply our own machine learning model,which means we need understand the data before targetting new customers on the social media adds.
Hence among the list only option that make sense it Clustering , so Anwser is B
upvoted 3 times

🗨️ **R_cool** 1 year, 6 months ago

what is real answer
upvoted 3 times

🗨️ **syu31svc** 1 year, 6 months ago

Answer is B; recommendation engine does not identify new customers as it suggests products to be given based on profiles
upvoted 2 times

🗨️ **GeeBeeEl** 1 year, 7 months ago

C is not correct at all. A recommendation engine would probably be factorization machine. This is a grouping problem, a clustering problem.....it is not a recommendation engine! B is the correct answer!!!
upvoted 3 times

🗨️ **roytruong** 1 year, 7 months ago

Go for C, find new user on social media that similar with our current user, it's called user-user collaborative filtering
upvoted 2 times

🗨️ **DScore** 1 year, 7 months ago

sorry, but any kind of collaborative filtering needs an interaction data, which is not present, and for new user, collaborative filtering suffers from cold start problems since they don't have data beforehand. This is pure case of unsupervised market segmentation problem, since we do not have any labels. Thus, by far and wide, the answer should be Clustering, i.e. option B.
upvoted 10 times

🗨️ **lightblue** 1 year, 6 months ago

we have labels - past customers and present customers
upvoted 1 times

🗨️ **lightblue** 1 year, 6 months ago

D can be answer imo
upvoted 1 times

  **mirik** 1 year, 7 months ago

what is the difference between old and new customer? The premiums receive. So, here we can assume premiums as labels and have supervised learning. Please correct me if I'm mistaking.

upvoted 1 times

  **rickywck** 1 year, 7 months ago

I would pick C but honest speaking, the question content is not entirely clear.

Very often when we launch a marketing campaign, there is specific target(s) in mind, e.g. selling specific type of insurance in this case, and usually we then analyze the profile of existing customers whose have that product and then try to target those with similar profile. This what most recommendation engine does.

upvoted 4 times

A manufacturing company has a large set of labeled historical sales data. The manufacturer would like to predict how many units of a particular part should be produced each quarter.

Which machine learning approach should be used to solve this problem?

- A. Logistic regression
- B. Random Cut Forest (RCF)
- C. Principal component analysis (PCA)
- D. Linear regression

Correct Answer: B

Community vote distribution

D (80%)

B (20%)

- DonaldCMLIN

Highly Voted

1 year, 7 months ago

HOW MANY/MUCH, THOSE ARE REGRESSION TOPIC, LOGISTIC FOR 0/1,YES/NO

https://docs.aws.amazon.com/zh_tw/machine-learning/latest/dg/regression-model-insights.html

THE ANSWER SHOULD BE D.

upvoted 56 times
- rsimham

1 year, 7 months ago

agree. RCF is mostly used for anomaly detection or separate outliers

upvoted 9 times
- syu31svc

Highly Voted

1 year, 6 months ago

Amazon SageMaker Random Cut Forest (RCF) is an unsupervised algorithm for detecting anomalous data points within a data set

Answer is D 100%

upvoted 10 times
- jackzhao

Most Recent

1 month, 2 weeks ago

D is correct!

upvoted 1 times
- oso0348

1 month, 3 weeks ago

Selected Answer: D

D. Linear regression would be the appropriate machine learning approach to solve this problem of predicting the number of units of a particular part to be produced each quarter. Linear regression is a supervised learning algorithm used for predicting continuous variables based on input features. In this case, the historical sales data can be used as input features, and the number of units produced each quarter can be used as the continuous target variable.

upvoted 1 times
- Nadia0012

1 month, 3 weeks ago

Selected Answer: D

definitely D.

upvoted 1 times
- Ajose0

2 months, 2 weeks ago

Selected Answer: D

This is a regression problem where the goal is to predict a continuous outcome, which in this case is the number of units of a particular part that should be produced each quarter. Linear regression is a simple and commonly used approach to solve such problems, where a linear relationship is established between the independent variables (e.g., historical sales data) and the dependent variable (e.g., number of units of a part to be produced).

upvoted 1 times
- Tomatoteacher

3 months, 2 weeks ago

Selected Answer: D

D, RCF answers here just link one article where RCF is implemented to find outliers in time series, or are able to deduce trends, but here they mention already labelled data, RCF is unsupervised, so that data would go to waste.

upvoted 1 times
- hamimelon

4 months, 1 week ago

Honestly, i think these are all bad answers. It should be time series modeling methods.

upvoted 2 times

  **Peeking** 4 months, 3 weeks ago

Selected Answer: D

The answer is D. B is out of it as RCF is used for anomaly detection. Logistic Regression is for Classification mainly. Only linear regression can be used if Time series algorithms are not part of the options.

upvoted 1 times

  **Jeremy1** 5 months, 2 weeks ago

Selected Answer: B

Looks like regression, but here is is... Random Cut Forest <https://docs.aws.amazon.com/quicksight/latest/user/how-does-rcf-generate-forecasts.html>

upvoted 2 times

  **Moulichintakunta** 7 months, 4 weeks ago

Selected Answer: D

here talks about measuring count which is a supervised regression task

upvoted 2 times

  **ovokpus** 10 months, 1 week ago

Selected Answer: D

This should be obvious.

upvoted 1 times

  **jvpigozzo** 1 year, 6 months ago

RCF goes with anomaly detection. Answer is D

upvoted 1 times

  **boston12311** 1 year, 6 months ago

Someone is having fun, highlighting silly options like RCF here. It's lin regression

upvoted 1 times

  **AWS__Newbie** 1 year, 6 months ago

Where does the answer come from? RCF is for anomaly detection. How can it be the right answer?

upvoted 2 times

  **littlewat** 1 year, 6 months ago

D!!!!!!

upvoted 2 times

  **astonm13** 1 year, 6 months ago

Its a regression problem - for sure D!

upvoted 2 times

Question #52

Topic 1

A financial services company is building a robust serverless data lake on Amazon S3. The data lake should be flexible and meet the following requirements:

- ☞ Support querying old and new data on Amazon S3 through Amazon Athena and Amazon Redshift Spectrum.
- ☞ Support event-driven ETL pipelines
- ☞ Provide a quick and easy way to understand metadata

Which approach meets these requirements?

- A. Use an AWS Glue crawler to crawl S3 data, an AWS Lambda function to trigger an AWS Glue ETL job, and an AWS Glue Data catalog to search and discover metadata.
- B. Use an AWS Glue crawler to crawl S3 data, an AWS Lambda function to trigger an AWS Batch job, and an external Apache Hive metastore to search and discover metadata.
- C. Use an AWS Glue crawler to crawl S3 data, an Amazon CloudWatch alarm to trigger an AWS Batch job, and an AWS Glue Data Catalog to search and discover metadata.
- D. Use an AWS Glue crawler to crawl S3 data, an Amazon CloudWatch alarm to trigger an AWS Glue ETL job, and an external Apache Hive metastore to search and discover metadata.

Correct Answer: A

Community vote distribution

A (67%)

C (33%)

🗨️ 👤 **DonaldCMLIN** Highly Voted 👍 1 year, 7 months ago
BOTH A AND B ARE ANSWERS.

BUT external Apache Hive MIGHT BE NOT SERVERLESS SOLUTION.

The AWS Glue Data Catalog is your persistent metadata store. It is a managed service that lets you store, annotate, and share metadata in the AWS Cloud in the same way you would in an Apache Hive metastore.
The Data Catalog is a drop-in replacement for the Apache Hive Metastore

https://docs.aws.amazon.com/zh_tw/glue/latest/dg/components-overview.html

BEAUTIFUL ANSWER IS A.
upvoted 41 times

🗨️ 👤 **rsimham** 1 year, 7 months ago
I am thinking about Answer C, because events can be triggered by cloudwatch w/Glue metastore
upvoted 1 times

🗨️ 👤 **qwerty456** 1 year, 6 months ago
you can't schedule AWS Batch with CloudWatch
upvoted 3 times

🗨️ 👤 **qwerty456** 1 year, 6 months ago
srr, looks like you can apart from Cron, the argument should be AWS Batch aren't SERVERLESS
upvoted 2 times

🗨️ 👤 **kalyanvarma** 1 year, 5 months ago
We can schedule batch with cloud watch events.
upvoted 1 times

🗨️ 👤 **ComPah** 1 year, 7 months ago
if we use Flexible as key word ..Using Lambda might be a constraint
upvoted 4 times

🗨️ 👤 **cybe001** Highly Voted 👍 1 year, 6 months ago
Answer is A. Lamda is the preferred way of implementing event-driven ETL job with S3, when new data arrives in S3, it notifies lamda which can start the ETL job.
upvoted 17 times

🗨️ 👤 **rb39** 7 months, 2 weeks ago
agree, event-driven means Lambda, CloudWatch alarms are just to trigger alarms based on log analysis.
upvoted 2 times

- jackzhao

Most Recent

1 month, 3 weeks ago

I will chose A, I think C & D is wrong, you can use Amazon CloudWatch Event to trigger lambda but not CloudWatch alarm.

upvoted 1 times
- Valcilio

1 month, 3 weeks ago

Selected Answer: A

Batch is more for configurations and other kinds of things by scheduling than event driven and batch data processing with ETL, the answer is A.

upvoted 1 times
- Jeremy1

5 months, 2 weeks ago

Selected Answer: A

Found this supporting A - Lambda used to trigger ETL job after crawler completes. The crawler starts on schedules or events (files arriving).

upvoted 1 times
- Skychaser

9 months, 3 weeks ago

Selected Answer: A

Based on Majority discussion

upvoted 2 times
- exam887

11 months, 1 week ago

Selected Answer: C

Quite confused between A&C since they all workable solution. In below AWS Blog, even mix the CloudWatch + Lambda to use the Glue. For key word event trigger, prefer CloudWatch
<https://aws.amazon.com/blogs/big-data/build-and-automate-a-serverless-data-lake-using-an-aws-glue-trigger-for-the-data-catalog-and-etl-jobs/>
<https://docs.aws.amazon.com/glue/latest/dg/automating-awsglue-with-cloudwatch-events.html>

upvoted 2 times
- ZSun

1 week, 1 day ago

cloudwatch and lambda function can work together to trigger event. But AWS batch cannot independently conduct ETL and require other service. when it comes to ETL, glue is much easier choice than Batch

upvoted 1 times
- VinceCar

5 months, 1 week ago

Agreed. CloudWatch could trigger event to launch Lambda. Refer to: <https://docs.aws.amazon.com/lambda/latest/dg/services-cloudwatchevents.html>

upvoted 1 times
- syu31svc

1 year, 6 months ago

Answer is A 100%

upvoted 2 times
- halfway

1 year, 6 months ago

A is preferred. Lambda can trigger ETL pipelines: <https://aws.amazon.com/glue/>

upvoted 3 times
- PRC

1 year, 6 months ago

A is correct...Lambda is event driven and Glue is serverless as opposed to Hive

upvoted 4 times

A company's Machine Learning Specialist needs to improve the training speed of a time-series forecasting model using TensorFlow. The training is currently implemented on a single-GPU machine and takes approximately 23 hours to complete. The training needs to be run daily. The model accuracy is acceptable, but the company anticipates a continuous increase in the size of the training data and a need to update the model on an hourly, rather than a daily, basis. The company also wants to minimize coding effort and infrastructure changes. What should the Machine Learning Specialist do to the training solution to allow it to scale for future demand?

- A. Do not change the TensorFlow code. Change the machine to one with a more powerful GPU to speed up the training.
- B. Change the TensorFlow code to implement a Horovod distributed framework supported by Amazon SageMaker. Parallelize the training to as many machines as needed to achieve the business goals.
- C. Switch to using a built-in AWS SageMaker DeepAR model. Parallelize the training to as many machines as needed to achieve the business goals.
- D. Move the training to Amazon EMR and distribute the workload to as many machines as needed to achieve the business goals.

Correct Answer: B

Community vote distribution

B (80%)

A (20%)

- JayK

Highly Voted

1 year, 7 months ago

the answer is B. using Hovord distribution results in less coding effort

upvoted 37 times
- cybe001

Highly Voted

1 year, 7 months ago

Answer is B. "minimize coding effort and infrastructure changes" If we use DeepAR then the code and infra has to be changed to work with DeepAR.

upvoted 12 times
- ZSun

Most Recent

1 week, 3 days ago

A is better option even though B helps. Firstly, you only have One GPU, in this case distributed training Horovod doesn't help much; Secondly, the question is about minimize "coding effort" not minimize budget. adding distributed framework require much more coding, but increase gpu instance only require single click.

upvoted 1 times
- Valcilio

Selected Answer: B

1 month, 3 weeks ago

Horovod distribution is accepted by sagemaker, making easy to implement!

upvoted 1 times
- Ajose0

Selected Answer: B

2 months, 2 weeks ago

Hovord distribution will allow the Machine Learning Specialist to take advantage of Amazon SageMaker's built-in support for Horovod, which is a popular, open-source distributed deep learning framework.

Implementing Horovod in TensorFlow will allow the Specialist to parallelize the training across multiple GPUs or instances, which can significantly reduce the time it takes to train the model.

This will allow the company to meet its requirement to update the model on an hourly basis, and minimize coding effort and infrastructure changes as it leverages the existing TensorFlow code and infrastructure, along with the scalability and ease of use of Amazon SageMaker.

upvoted 1 times
- joe3232

3 months ago

Are is there a 23X differential between the weakest and strongest GPU in AWS? (and allow for future growth). I don' tthink so.

upvoted 1 times
- vbal

3 months, 2 weeks ago

Answer:C- built-in sagemaker DeepAR model. minimize coding & infra changes.

upvoted 1 times
- cpal012

1 month, 3 weeks ago

But they are happy with it - just want it to go faster. Not throw the whole thing out.

upvoted 1 times
- KingGuo

Selected Answer: B

9 months, 4 weeks ago

the answer is B. using Hovord distribution results in less coding effort

upvoted 2 times

  **John_Pongthorn** 1 year, 2 months ago

Selected Answer: A

Most likely , it is A because it is based on AWS teachnology, why we have to use open source

we exam AWS ML , the answer should be relevant to AWS technology inevitably

<https://aws.amazon.com/sagemaker/distributed-training/>

upvoted 1 times

  **cloud_trail** 1 year, 5 months ago

This one reminds me of an old saying by Yogi Berra: "When you come to a fork in the road, take it." If you see Horovod as an option in a question about scaling TF, take it. Answer is B.

upvoted 7 times

  **RaniaSayed** 1 year, 6 months ago

I Think it's B

<https://aws.amazon.com/blogs/machine-learning/launching-tensorflow-distributed-training-easily-with-horovod-or-parameter-servers-in-amazon-sagemaker/>

&

<https://aws.amazon.com/blogs/machine-learning/multi-gpu-and-distributed-training-using-horovod-in-amazon-sagemaker-pipe-mode/>

upvoted 2 times

  **harmanbirstudy** 1 year, 6 months ago

Seen similar question on udemy/whizlab , its always Horvord when Tensorflow needs scaling. ANWSER is B

upvoted 5 times

  **yeetusdeleetus** 1 year, 6 months ago

"Minimize code and infrastructure changes" and "training set size increasing continuously" are in conflict.

The latter is necessary, while the former is more optional. This rules out A, because it will not scale to continuously-increasing dataset size for daily training.

C requires a total rewrite, which does not minimize coding effort.

This leaves B and D, and the question will be which is easier to implement. EMR is not, especially, known for being easy to implement, whereas Horovod + SageMaker specifically are. Also SageMaker is a named AWS service and is extremely expensive, so of course that's what Amazon recommends. :)

upvoted 1 times

  **sebtac** 1 year, 6 months ago

correct answer is B; the expected probably is C as AWS wants us to use their own solutions :)

upvoted 2 times

  **syu31svc** 1 year, 6 months ago

I would say that the answer is A and this is just my reasoning:

Minimize infrastructure changes would mean saving costs. Having to increase the number of machines to "as many as possible" would increase costs. Also, from <https://d1.awsstatic.com/whitepapers/aws-power-ml-at-scale.pdf>:
If the training times on a GPU card are insufficient for your business needs, we recommend that you try a more powerful GPU before moving to multiple GPUs

upvoted 1 times

  **mawsman** 1 year, 6 months ago

Possible B or C because of scaling and future needs. But Switching TensorFlow to DeepAT for time-series forecasting would be more effort imho than re-coding to horovod. I would choose B.

upvoted 3 times

  **deep_n** 1 year, 6 months ago

Correct answer is B

upvoted 2 times

Which of the following metrics should a Machine Learning Specialist generally use to compare/evaluate machine learning classification models against each other?

- A. Recall
- B. Misclassification rate
- C. Mean absolute percentage error (MAPE)
- D. Area Under the ROC Curve (AUC)

Correct Answer: D

Community vote distribution

D (100%)

- DonaldCMLIN

Highly Voted

1 year, 7 months ago

RECALL IS ONE OF FACTOR IN CLASSIFY,

AUC IS MORE FACTORS TO COMPREHENSIVE JUDGEMENT

https://docs.aws.amazon.com/zh_tw/machine-learning/latest/dg/cross-validation.html

ANSWER MIGHT BE D.

upvoted 35 times
- DScore

1 year, 6 months ago

Not might be, but should be D

upvoted 5 times
- devsean

1 year, 7 months ago

AUC is to determine hyperparams in a single model, not compare different models.

upvoted 6 times
- Valcilio

Most Recent

1 month, 3 weeks ago

Selected Answer: D

AUC is the best metric.

upvoted 1 times
- Ajose0

2 months, 2 weeks ago

Selected Answer: D

Area Under the ROC Curve (AUC) is a commonly used metric to compare and evaluate machine learning classification models against each other. The AUC measures the model's ability to distinguish between positive and negative classes, and its performance across different classification thresholds. The AUC ranges from 0 to 1, with a score of 1 representing a perfect classifier and a score of 0.5 representing a classifier that is no better than random.

While recall is an important evaluation metric for classification models, it alone is not sufficient to compare and evaluate different models against each other. Recall measures the proportion of actual positive cases that are correctly identified as positive, but does not take into account the false positive rate.

upvoted 1 times
- cloud_trail

1 year, 6 months ago

D. AUC is always used to compare ML classification models. The others can all be misleading. Consider the cases where classes are highly imbalanced. In those cases accuracy, misclassification rate and the like are useless. Recall is only useful if used in combination with precision or specificity, which what AUC does.

upvoted 4 times
- harmanbirstudy

1 year, 6 months ago

AUC/ROC work well with special case of Binary Classification not in general

upvoted 4 times
- MohamedSharaf

1 year, 6 months ago

AUC is to compare different models in terms of their separation power. 0.5 is useless as it's the diagonal line. 1 is perfect. I would go with F1 Score if it was an option. However, taking Recall only as a metric for comparing between models, would be misleading.

upvoted 4 times
- harmanbirstudy

1 year, 6 months ago

Its Accuracy,Precision,Recall and F1 score , there is no metion of AUC/ROC for comparing models in many articles , so ANSWER is A

upvoted 1 times

-   **Thai_Xuan** 1 year, 6 months ago
D. AUC is scale- and threshold-invariant, enabling it compare models.
<https://towardsdatascience.com/how-to-evaluate-a-classification-machine-learning-model-d81901d491b1>
upvoted 1 times
-   **johnny_chick** 1 year, 6 months ago
Actually A, B and D seem to be correct
upvoted 1 times
-   **deep_n** 1 year, 6 months ago
Probably D
<https://towardsdatascience.com/metrics-for-evaluating-machine-learning-classification-models-python-example-59b905e079a5>
upvoted 1 times
-   **hughhughhugh** 1 year, 6 months ago
why not B?
upvoted 1 times
-   **PRC** 1 year, 6 months ago
Answer should be D..ROC is used to determine the diagnostic capability of classification model varying on threshold
upvoted 3 times
-   **Hypermasterd** 1 year, 6 months ago
Should be A. A is the only one that generally works for classifcation.
AUC only works with binary classification.
upvoted 4 times
-   **sebas10** 1 year, 6 months ago
Could be, you mean in a multiclass clasification problem. But in that con context recall directly can't be compare because first you have to decide recall of what of the classes, in a 3 classes problem we have 3 recalls or you suppose a weighted recall or average recall ?. Do you think in that ?
upvoted 2 times
-   **mrsimoes** 1 year, 6 months ago
Also in multi-class classification, if you follow an One-vs_Rest strategy you can still use AUC.
https://scikit-learn.org/stable/auto_examples/model_selection/plot_roc.html#sphx-glr-auto-examples-model-selection-plot-roc-py
upvoted 1 times
-   **oMARKOo** 1 year, 6 months ago
Actually AUC could be generalized for multi-class problem.
<https://www.datascienceblog.net/post/machine-learning/performance-measures-multi-class-problems/>
upvoted 1 times
-   **stamarpadar** 1 year, 7 months ago
Correct Answer is D. Another benefit of using AUC is that it is classification-threshold-invariant like log loss.
<https://towardsdatascience.com/the-5-classification-evaluation-metrics-you-must-know-aa97784ff226>
upvoted 3 times

A company is running a machine learning prediction service that generates 100 TB of predictions every day. A Machine Learning Specialist must generate a visualization of the daily precision-recall curve from the predictions, and forward a read-only version to the Business team. Which solution requires the LEAST coding effort?

- A. Run a daily Amazon EMR workflow to generate precision-recall data, and save the results in Amazon S3. Give the Business team read-only access to S3.
- B. Generate daily precision-recall data in Amazon QuickSight, and publish the results in a dashboard shared with the Business team.
- C. Run a daily Amazon EMR workflow to generate precision-recall data, and save the results in Amazon S3. Visualize the arrays in Amazon QuickSight, and publish them in a dashboard shared with the Business team.
- D. Generate daily precision-recall data in Amazon ES, and publish the results in a dashboard shared with the Business team.

Correct Answer: C

Community vote distribution

B (60%)

C (40%)

- rsimham

Highly Voted

1 year, 7 months ago

Ans C is reasonable

upvoted 26 times
- cloud_trail

Highly Voted

1 year, 5 months ago

Agree with C. Quicksight cannot handle 100TB each day.

upvoted 5 times
- Ahmedhadi_

Most Recent

2 weeks, 2 days ago

Selected Answer: C

c is correct answer

upvoted 2 times
- Ahmedhadi_

2 weeks, 3 days ago

Selected Answer: B

B is the correct answer according to this resource and even according to ChatGPT, GPT-4 and Claude+
<https://aws.amazon.com/blogs/big-data/diligent-enhances-customer-governance-with-automated-data-driven-insights-using-amazon-quicksight/>

upvoted 1 times
- Valcilio

1 month, 3 weeks ago

Selected Answer: B

Quicksight can do it now with customazable methods!

upvoted 2 times
- Valcilio

1 month, 3 weeks ago

I don't know if in the past quicksight wasn't serveless, but today it's and can analyze as much data as is needed!

upvoted 1 times
- drcok87

3 months ago

Seems like quicksight can do all of this through built-in capabilities <https://docs.aws.amazon.com/quicksight/latest/user/making-data-driven-decisions-with-ml-in-quicksight.html>

Earlier I would go with C but now I'm leaning towards B

upvoted 1 times
- muralee_xo

3 months, 1 week ago

Q: How much data can I analyze with Amazon QuickSight?
With Amazon QuickSight you don't need to worry about scale. You can seamlessly grow your data from a few hundred megabytes to many terabytes of data without managing any infrastructure.
<https://aws.amazon.com/quicksight/resources/faqs/>
Quicksight indeed can handle 100 TB data and i think answer should be B.

upvoted 2 times
- eussou

3 months, 1 week ago

Selected Answer: C

C is what works as Quick Sight alone could not do it with B

upvoted 2 times

-   **Tomatoteacher** 3 months, 2 weeks ago

Selected Answer: B

The question specifically states: Which solution requires the LEAST coding effort?
With that being the real question the answer will be B.
C requires a lot of coding and it is possible for Quicksight to handle 100 TB of predictions, but it would require a high performance configuration of the underlying resources (a lot). That is my thought but could be wrong.

upvoted 3 times
-   **bitsplease** 1 year, 3 months ago

why not D?

upvoted 1 times
-   **Wira** 1 year, 6 months ago

can someone explain why we need EMR here? Quicksight is capable of running calculations..

upvoted 4 times
-   **oief2oi3fj23ogi23g** 1 year, 6 months ago

100TB data to much four quick sight alone

upvoted 6 times
-   **Urban_Life** 1 year, 6 months ago

Large volume.....that's why EMR is good. you can't even proceed with EC2 instance, however, Quicksight is ligh weight BI tool. From my own project experience, we move away from Quicksight to Tableau in middle of the deployment.

upvoted 5 times
-   **PRC** 1 year, 7 months ago

C is correct...100 TB daily can be handled by EMR and Quicksight (no coding) is the right solution for providing read only access to business

upvoted 4 times
-   **rajs** 1 year, 7 months ago

Agree with C

BUT

B would have been more appropriate if CURVE was specified instead of data

Generate daily precision-recall CURVE (instead of data) in Amazon QuickSight, and publish the results in a dashboard shared with the Business team

upvoted 2 times
-   **qwerty456** 1 year, 5 months ago

B won't solve the "daily scheduled" requirement

upvoted 4 times

A Machine Learning Specialist is preparing data for training on Amazon SageMaker. The Specialist is using one of the SageMaker built-in algorithms for the training. The dataset is stored in .CSV format and is transformed into a numpy.array, which appears to be negatively affecting the speed of the training.

What should the Specialist do to optimize the data for training on SageMaker?

- A. Use the SageMaker batch transform feature to transform the training data into a DataFrame.
- B. Use AWS Glue to compress the data into the Apache Parquet format.
- C. Transform the dataset into the RecordIO protobuf format.
- D. Use the SageMaker hyperparameter optimization feature to automatically optimize the data.

Correct Answer: C

Community vote distribution

C (100%)

- rsimham

Highly Voted

1 year, 7 months ago

C is okay

upvoted 19 times
- stamarpadar

Highly Voted

1 year, 7 months ago

Anwer is C.

Most Amazon SageMaker algorithms work best when you use the optimized protobuf recordIO format for the training data.

<https://docs.aws.amazon.com/sagemaker/latest/dg/cdf-training.html>

upvoted 12 times
- Ajose0

Most Recent

2 months, 2 weeks ago

Selected Answer: C

The Specialist should transform the dataset into the RecordIO protobuf format. This format is optimized for use with SageMaker and has been shown to improve the speed and efficiency of training algorithms.

Using the RecordIO protobuf format is a best practice for preparing data for use with Amazon SageMaker, and it is specifically recommended for use with the built-in algorithms.

upvoted 1 times
- Jeremy1

5 months, 2 weeks ago

Selected Answer: C

I would assume the issue is the transformation. It can be nasty slow between pandas / csv / numpy. Go to protobuf.

upvoted 1 times
- C10ud9

1 year, 5 months ago

C is the best

upvoted 5 times
- PRC

1 year, 6 months ago

Agree with C

upvoted 6 times

A Machine Learning Specialist is required to build a supervised image-recognition model to identify a cat. The ML Specialist performs some tests and records the following results for a neural network-based image classifier:

Total number of images available = 1,000

Test set images = 100 (constant test set)

The ML Specialist notices that, in over 75% of the misclassified images, the cats were held upside down by their owners.

Which techniques can be used by the ML Specialist to improve this specific test error?

- A. Increase the training data by adding variation in rotation for training images.
- B. Increase the number of epochs for model training
- C. Increase the number of layers for the neural network.
- D. Increase the dropout rate for the second-to-last layer.

Correct Answer: B

Community vote distribution

A (100%)

👤

DonaldCMLIN

Highly Voted 👍

1 year, 7 months ago

NO CORRECT TRAINING DATA, MORE WORKS JUST WASTE TIME.

ONE OF THE REASONS FOR POOR ACCURACY COULD BE INSUFFICIENT DATA. THIS CAN BE OVERCOME BY IMAGE AUGMENTATION. IMAGE AUGMENTATION IS A TECHNIQUE OF INCREASING THE DATASET SIZE BY PROCESSING (MIRRORING, FLIPPING, ROTATING, INCREASING/DECREASING BRIGHTNESS, CONTRAST, COLOR) THE IMAGES.

HTTPS://MEDIUM.COM/DATADRIVENINVESTOR/AUTO-MODEL-TUNING-FOR-KERAS-ON-AMAZON-SAGEMAKER-PLANT-SEEDLING-DATASET-7B591334501E

ANSWER A. ADD MORE TRAINING DATA FOR ROTATION IMAGES COULD BE A WAY TO DEAL WITH ISSUE

upvoted 58 times

👤

Jeremy1

5 months, 2 weeks ago

Donald, your caps lock is on.

upvoted 6 times

👤

Nadia0012

1 month, 3 weeks ago

LOL :D

upvoted 1 times

👤

rsimham

1 year, 7 months ago

agree with A

upvoted 9 times

👤

tap123

1 year, 6 months ago

The key phrase might be "constant test set", so you can't increase training set by shrinking the size of test set. Thus the only feasible choice is to increase training time by increasing the number of epochs => answer B.

upvoted 2 times

👤

VB

1 year, 6 months ago

A . Increase the training data by adding variation in rotation for training images.

It never says to move the images from Test data set (because it is constant)... only variations are added to the images..so, A is correct.

upvoted 1 times

👤

mawsmann

1 year, 6 months ago

The problem is images are upside down and misclassified. If right side up then the model would classify correctly. This can only be fixed by rotating not by trying to recognise upside down cat more times.

upvoted 3 times

👤

Urban_Life

1 year, 6 months ago

What's your answer B?

upvoted 1 times

👤

oso0348

Most Recent 🕒

1 month, 3 weeks ago

Selected Answer: A

the Specialist can apply data augmentation techniques to increase the training data by adding variation in rotation for training images. This technique will allow the model to learn to recognize cats in various orientations, including upside down.

upvoted 1 times

  **Ajose0** 2 months, 2 weeks ago

Selected Answer: A

Adding more variation in rotation to the training data can help the model to learn how to classify cats in different orientations, including when they are held upside down. This can improve the model's ability to identify cats in this position and reduce the misclassification rate for images in which the cats are upside down.

By adding more rotation to the training data, the model can be trained to generalize better to new images, including those with cats in different orientations. This can help to reduce overfitting and improve the model's overall performance.

upvoted 1 times

  **Tomatoteacher** 3 months, 2 weeks ago

Selected Answer: A

Only logical answer 100% A.

upvoted 1 times

  **Jeremy1** 5 months, 2 weeks ago

Selected Answer: A

More data is a good answer. A

upvoted 1 times

  **ryuhei** 7 months, 1 week ago

Selected Answer: A

Answer is "A"

upvoted 1 times

  **Morsa** 9 months, 3 weeks ago

Answer is A

upvoted 1 times

  **ovokpus** 10 months, 1 week ago

Selected Answer: A

This is a clear case of Data Augumentation solution.

upvoted 1 times

  **yc1005** 10 months, 3 weeks ago

Selected Answer: A

Common step in CNN, Image augmentation. A.

upvoted 1 times

  **AmakamaxZanny** 1 year, 2 months ago

Selected Answer: A

image data should be augmented which includes rotation (A)

upvoted 1 times

  **apprehensive_scar** 1 year, 2 months ago

Selected Answer: A

A! How come all the answers this site gives out are incorrect?

upvoted 2 times

  **Asrivastava3** 1 year, 4 months ago

B is the answer. The model always goes from underfitting to Optimal to overfitting when you increase the epochs. The problem statement says that data is misclassified (not correctly classified) by the model. The question then arise how can you make the model correctly classify the data....and the answer is by passing it through the model more numbers of the time (epochs) so that it can see the correct orientation. No where it says the data is not correct. Another way to say this - by adding more data (choice A) we are not making model classify it correctly. But by increasing number of epochs you are.

upvoted 1 times

  **joe3232** 3 months ago

or just rotate the images and get a better model

upvoted 2 times

  **pp13** 1 year, 4 months ago

it should be A

upvoted 1 times

  **eganilovic** 1 year, 5 months ago

The answer is A!

upvoted 3 times

  **tmlld** 1 year, 6 months ago

My answer is A
upvoted 1 times

  **ahquiceno** 1 year, 6 months ago

Answer is A, model need variation of data that CNN need learn and then predict, in this case data augmentation is the recommended step.
upvoted 1 times

  **weslleylc** 1 year, 6 months ago

Since the test set is constant, to modify your training data based on what you have saw on your test set will overfitting your model. Answer is probably B.
upvoted 2 times

A Machine Learning Specialist needs to be able to ingest streaming data and store it in Apache Parquet files for exploration and analysis. Which of the following services would both ingest and store this data in the correct format?

- A. AWS DMS
- B. Amazon Kinesis Data Streams
- C. Amazon Kinesis Data Firehose
- D. Amazon Kinesis Data Analytics

Correct Answer: C

Community vote distribution

C (100%)

- JayK

Highly Voted

1 year, 7 months ago

the answer is C. as the main point of the question is data transformation to Parquet format which is done by Kinesis Data Firehose not Data Stream. Coming to the data store the data store in Kinesis Data Stream is only for couple of days so it does not serve the purpose here

upvoted 51 times
- Valcilio

Most Recent

1 month, 3 weeks ago

Selected Answer: C

Firehose can do it if the data is in JSON or ORC format initially!

upvoted 2 times
- DS2021

2 months ago

It should be KDS

upvoted 1 times
- Ajose0

2 months, 2 weeks ago

Selected Answer: C

Amazon Kinesis Data Firehose is a fully managed service that can automatically load streaming data into data stores and analytics tools.

It can ingest real-time streaming data such as application logs, website clickstreams, and IoT telemetry data, and then store it in the correct format, such as Apache Parquet files, for exploration and analysis.

This makes it a suitable option for the requirement described in the question.

upvoted 1 times
- eganilovic

1 year, 5 months ago

Firehose

upvoted 4 times
- Thai_Xuan

1 year, 5 months ago

B

<https://github.com/ravsau/aws-exam-prep/issues/10>

upvoted 2 times
- weslleylc

1 year, 6 months ago

B) Only Amazon Kinesis Data Streams can store and Ingest data. We don't need to apply any transformation; the question asks to ingest and store data in Apache Parquet format, There is no assumption that the data coming in a different format than parquet.

upvoted 3 times
- joe3232

3 months ago

KDS cant store to s3

<https://stackoverflow.com/questions/66097886/writing-to-s3-via-kinesis-stream-or-firehose>

upvoted 1 times
- In

1 year, 6 months ago

It is C with no doubt

https://aws.amazon.com/about-aws/whats-new/2018/05/stream_real_time_data_in_apache_parquet_or_orc_format_using_firehose/

upvoted 4 times
- GeeBeeEl

1 year, 6 months ago

It appears all agree that the answer is between Firehose and Analytics. Kinesis Firehose is used for ingestion. Both firehose and analytics can store, only firehose can ingest. <https://docs.aws.amazon.com/firehose/latest/dev/record-format-conversion.html> shows firehose can store parquet to S3

upvoted 3 times

-   **GeeBeeEI** 1 year, 6 months ago

It appears all agree that the answer is between Firehose and Analytics. Data Streams handle stuff like event data, clickstream etc. Its not interested in special format, the focus is speed. The question did not talk of transformation, only ingestion. Kinesis Firehose is used for ingestion. Both firehose and analytics can store, only firehose can ingest. <https://docs.aws.amazon.com/firehose/latest/dev/record-format-conversion.html> shows firehose can store parquet to S3

upvoted 1 times
-   **Urban_Life** 1 year, 6 months ago

Think just like this -- batch process Glue ETL and Streaming process Firehose ETLcovert to parquet or any other format.

upvoted 1 times
-   **CMMC** 1 year, 6 months ago

C for Firehose

upvoted 2 times
-   **Erso** 1 year, 7 months ago

Just in case https://acloud.guru/forums/aws-certified-big-data-specialty/discussion/-KhI3MgPEo-FY5rfgl3J/what_is_difference_between_kin

upvoted 2 times
-   **BigEv** 1 year, 7 months ago

Amazon Kinesis Data Firehose can convert the format of your input data from JSON to Apache Parquet or Apache ORC before storing the data in Amazon S3.
https://github.com/awsdocs/amazon-kinesis-data-firehose-developer-guide/blob/master/doc_source/record-format-conversion.md

upvoted 3 times
-   **rsimham** 1 year, 7 months ago

I would go with B. Kinesis data streams stores data, while Firehose not.

upvoted 3 times
-   **cloud_trail** 1 year, 5 months ago

It's the other way around. Firehouses stores data; data streams does not.

upvoted 1 times

A data scientist has explored and sanitized a dataset in preparation for the modeling phase of a supervised learning task. The statistical dispersion can vary widely between features, sometimes by several orders of magnitude. Before moving on to the modeling phase, the data scientist wants to ensure that the prediction performance on the production data is as accurate as possible.

Which sequence of steps should the data scientist take to meet these requirements?

- A. Apply random sampling to the dataset. Then split the dataset into training, validation, and test sets.
- B. Split the dataset into training, validation, and test sets. Then rescale the training set and apply the same scaling to the validation and test sets.
- C. Rescale the dataset. Then split the dataset into training, validation, and test sets.
- D. Split the dataset into training, validation, and test sets. Then rescale the training set, the validation set, and the test set independently.

Correct Answer: D

Reference:

<https://www.kdnuggets.com/2018/12/six-steps-master-machine-learning-data-preparation.html>

Community vote distribution

B (64%)

C (36%)

  **cron0001**

Highly Voted 

 1 year ago

Selected Answer: C

C would be my answer here. Rescaling each set independently could lead to strange skews. Training set, Test set and Evaluation set should be on the same scale

upvoted 17 times

  **GiyeonShin** 4 months, 1 week ago

You're right. test set and val set should be rescaled on the same scale.
But the scale value should be extracted by only statistical value from training data.
I think C means that the rescaling stage is affected by the values from the whole data (with val, test set)
So, I think B is correct

upvoted 5 times

  **masoa3b**

Highly Voted 

 6 months, 1 week ago

Selected Answer: B

<https://stackoverflow.com/questions/49444262/normalize-data-before-or-after-split-of-training-and-testing-data>

C also leads to data leakage. You are using the test data to scale everything. So part of the data in the test set is used to scale for when you build the model on the training and check against the validation set.

upvoted 10 times

  **kukreti18**

Most Recent 

 4 days, 5 hours ago

C definitely, as rescaling after split leads to data snooping.

upvoted 1 times

  **tbomez** 1 month, 1 week ago

for the sake of consistency, it makes most sense to perform rescaling, e.g. min-max scaling or standardisation, before splitting the data into training, validation and test sets

upvoted 1 times

  **oso0348** 1 month, 3 weeks ago

Selected Answer: C

It is essential to rescale the data before splitting it into training, validation, and test sets to avoid data leakage. Rescaling the data after splitting can lead to the validation and test sets being influenced by information from the training set. Therefore, it is important to rescale the data before splitting it into different sets.

upvoted 1 times

  **Ajose0** 1 month, 4 weeks ago

Selected Answer: B

This is because rescaling the entire dataset before splitting it can lead to data leakage, where information from the validation or test sets can influence the scaling of the training set. Rescaling only the training set and then applying the same scaling to the validation and test sets ensures that the model is trained on data that is scaled in the same way as the production data.

upvoted 3 times

  **Siyuan_Zhu** 2 months, 1 week ago

Selected Answer: B

It is 100% B here, scaling should only be based on the training data ALONE. The testing data should be transformed by the same scaler based on the training data.

C would introduce information from the testing data in the process of scaling the training data, therefore introducing information leakage, and that can lead to the model picking up information from the test data.

upvoted 5 times

  **Tomatoteacher** 3 months, 2 weeks ago

Selected Answer: C

Answer is C. What have people been doing? I have always been instructed, and seen scaling preformed before splitting the data. You must do that, if you scale the 3 groups independently then you allow some bias to skew the data (especially if the validation and test sets are small). Always normalize/ scale data before any splitting!

upvoted 3 times

  **Tomatoteacher** 3 months, 2 weeks ago

As even though Option B would prevent data leakage, but it's not the best option to ensure the best performance on unseen data.

upvoted 1 times

  **GiyeonShin** 4 months, 1 week ago

Selected Answer: B

B

When a machine learns features of data, the machine should consider only "training data", not val and test. So, It should be eliminated the mean, var or some statistical values from val, and test data.

Because we don't know the real statistical value of real data.

upvoted 3 times

  **hamimelon** 4 months, 1 week ago

You should rescale the data first, so that all the data are on the same scale, and then split them.

upvoted 1 times

  **Jeremy1** 5 months, 2 weeks ago

Selected Answer: B

I was going to suggest D, but ... you should perform feature normalisation over the training data. Then perform normalisation on testing instances as well, but this time using the mean and variance of training explanatory variables.

This would make sense as you are making a model and then in the real world it works on "something"... you only have the trained model and have to apply what you have learned to the real word.

upvoted 1 times

  **shailendra1508** 6 months, 2 weeks ago

B is the correct Answer.

upvoted 3 times

  **31Rishab** 7 months, 1 week ago

Selected Answer: B

<https://datascience.stackexchange.com/questions/39932/feature-scaling-both-training-and-test-data>

upvoted 1 times

  **ovokpus** 10 months, 1 week ago

Selected Answer: B

<https://stackoverflow.com/questions/49444262/normalize-data-before-or-after-split-of-training-and-testing-data>

AND

<https://stats.stackexchange.com/questions/375165/scaling-separately-in-train-and-test-set>

tell the same story.

Split the data
Fit the scaler on the train set and use the same fit to transform train, val and test sets.

upvoted 4 times

  **[Removed]** 10 months, 2 weeks ago

C is not correct. B is correct

upvoted 4 times

  **yc1005** 10 months, 3 weeks ago

Selected Answer: B

B, details see here
<https://stackoverflow.com/questions/49444262/normalize-data-before-or-after-split-of-training-and-testing-data>

upvoted 4 times

  **tgaos** 11 months ago

D is correct

upvoted 1 times

Question #60

Topic 1

A Machine Learning Specialist is assigned a TensorFlow project using Amazon SageMaker for training, and needs to continue working for an extended period with no Wi-Fi access.

Which approach should the Specialist use to continue working?

A. Install Python 3 and boto3 on their laptop and continue the code development using that environment.

B. Download the TensorFlow Docker container used in Amazon SageMaker from GitHub to their local environment, and use the Amazon SageMaker Python SDK to test the code.

C. Download TensorFlow from tensorflow.org to emulate the TensorFlow kernel in the SageMaker environment.

D. Download the SageMaker notebook to their local environment, then install Jupyter Notebooks on their laptop and continue the development in a local notebook.

Correct Answer: B

Community vote distribution

B (100%)

DonaldCMLIN

Highly Voted

1 year, 6 months ago

ANSWER B.

YOU COULD INSTALL DOCKER-COMPOSE (AND NVIDIA-DOCKER IF TRAINING WITH A GPU) FOR LOCAL TRAINING

HTTPS://SAGEMAKER.READTHEDOCS.IO/EN/STABLE/OVERVIEW.HTML#LOCAL-MODE
HTTPS://GITHUB.COM/AWSLABS/AMAZON-SAGEMAKER-EXAMPLES/BLOB/MASTER/SAGEMAKER-PYTHON-SDK/TENSORFLOW_DISTRIBUTED_MNIST/TENSORFLOW_LOCAL_MODE_MNIST.IPYNB

upvoted 40 times

sqavi

2 months, 3 weeks ago

None of these links are working

upvoted 1 times

VB

Highly Voted

1 year, 6 months ago

https://aws.amazon.com/blogs/machine-learning/use-the-amazon-sagemaker-local-mode-to-train-on-your-notebook-instance/

B

upvoted 8 times

Valcilio

Most Recent

1 month, 3 weeks ago

Selected Answer: B

It's B

upvoted 1 times

SriAkula

1 year, 1 month ago

Answer : D

upvoted 2 times

noblade

1 year, 4 months ago

why not D?

upvoted 1 times

AddiWei

1 year, 2 months ago

My assumption is that D there is no way to test the code. You need the Sagemaker SDK in order to utilize dockerized container of Tensorflow from Sagemaker is my best guess.

upvoted 2 times

Tomatoteacher

3 months, 2 weeks ago

Cannot be D. If you used Jupyter notebook, you are unable to use it without internet access.

upvoted 1 times

CMMC

1 year, 6 months ago

Agreed for B

upvoted 4 times

https://www.examtopycs.com/exams/amazon/aws-certified-machine-learning-specialty/custom-view/

128/406

A Machine Learning Specialist is working with a large cybersecurity company that manages security events in real time for companies around the world. The cybersecurity company wants to design a solution that will allow it to use machine learning to score malicious events as anomalies on the data as it is being ingested. The company also wants be able to save the results in its data lake for later processing and analysis. What is the MOST efficient way to accomplish these tasks?

- A. Ingest the data using Amazon Kinesis Data Firehose, and use Amazon Kinesis Data Analytics Random Cut Forest (RCF) for anomaly detection. Then use Kinesis Data Firehose to stream the results to Amazon S3.
- B. Ingest the data into Apache Spark Streaming using Amazon EMR, and use Spark MLlib with k-means to perform anomaly detection. Then store the results in an Apache Hadoop Distributed File System (HDFS) using Amazon EMR with a replication factor of three as the data lake.
- C. Ingest the data and store it in Amazon S3. Use AWS Batch along with the AWS Deep Learning AMLs to train a k-means model using TensorFlow on the data in Amazon S3.
- D. Ingest the data and store it in Amazon S3. Have an AWS Glue job that is triggered on demand transform the new data. Then use the built-in Random Cut Forest (RCF) model within Amazon SageMaker to detect anomalies in the data.

Correct Answer: B

Community vote distribution

A (100%)

- DonaldCMLIN

Highly Voted

1 year, 7 months ago

I WOULD LIKE TO CHOOSE ANSWER A.

<https://aws.amazon.com/tw/blogs/machine-learning/use-the-built-in-amazon-sagemaker-random-cut-forest-algorithm-for-anomaly-detection/>
upvoted 57 times
- hamimelon

4 months, 1 week ago

Donald, do you know your CAPS LOCK has been on the whole time?

upvoted 7 times
- Nadia0012

1 month, 3 weeks ago

I know why his caps lock has been on :D to enter the "I am not robot" code easier :D

upvoted 2 times
- JayK

Highly Voted

1 year, 7 months ago

Answer is A. As the word anamoly talks about Random Cut Forest in the exam and that can be done in a cost effective manner using Kinesis Data Analytics

upvoted 13 times
- Ajose0

Most Recent

2 months, 2 weeks ago

Selected Answer: A

Amazon Kinesis Data Firehose is a fully managed service for streaming real-time data to Amazon S3 and can handle the ingestion of large amounts of data in real time. Kinesis Data Analytics Random Cut Forest (RCF) is a fully managed service that can be used to perform anomaly detection on streaming data, making it well suited for this use case. The results of the anomaly detection can then be streamed to Amazon S3 using Kinesis Data Firehose, providing a scalable and cost-effective data lake for later processing and analysis.

upvoted 2 times
- OssamaAbdelatif

5 months ago

I would select A

upvoted 1 times
- ovokpus

10 months, 1 week ago

Selected Answer: A

B is too resource intensive for that use case. I choose A, but I think the data should be better ingested using Kinesis streams

upvoted 3 times
- EuTuXia

1 year, 2 months ago

Selected Answer: A

Ans is A, but I have a doubt: firehose is not real-time (it's near real time) so as the data are ingested using firehose, we are not meeting the requirements, can someone pls explain this? thx

upvoted 2 times
- apprehensive_scar

1 year, 2 months ago

Selected Answer: A

A is 100% correct.
upvoted 2 times

  **Huy** 1 year, 5 months ago
A. Only A address all requirements. <https://docs.aws.amazon.com/kinesisanalytics/latest/sqlref/sqlrf-random-cut-forest.html>
upvoted 2 times

  **btsql** 1 year, 5 months ago
B is possible But A is correct i think
upvoted 1 times

  **yeetusdeleetus** 1 year, 6 months ago
Most efficient, streaming = A
upvoted 2 times

  **KK3** 1 year, 6 months ago
The answer is B since the requirement was real time and KFH is near real time i.e it has 60 sec latency
upvoted 2 times

  **NotAnMLProfessional** 1 year, 6 months ago
"as it is being ingested" is not necessarily realtime.. it could be near realtime :)
upvoted 1 times

  **ogm1** 10 months, 1 week ago
Answer should be B <https://stackoverflow.com/questions/65617845/obtaining-k-means-centroids-and-outliers-in-python-pyspark>
upvoted 3 times

  **fhuadeen** 1 year, 6 months ago
A doesn't seem correct actually. There is no service in AWS that is called "Amazon Kinesis Data Analytics Random Cut Forest (RCF)" for anomaly detection. Pay close attention to that phrase, there is nothing separating them, it is like a full name for one service.
upvoted 2 times

  **RLai** 4 months, 2 weeks ago
There is RCF in KDA.

<https://docs.aws.amazon.com/kinesisanalytics/latest/sqlref/sqlrf-random-cut-forest.html>
upvoted 1 times

  **Predicare** 1 year, 6 months ago
There is actually. Please check

https://www.google.com/url?sa=t&source=web&rct=j&url=https://docs.aws.amazon.com/kinesisanalytics/latest/sqlref/sqlrf-random-cut-forest.html&ved=2ahUKEwj_x-_pzovsAhXFURUIHYIvCssQFjAAegQIDRAC&usg=AOvVaw2a7_hUr7fxkO40fd_2eF5P
upvoted 2 times

  **syu31svc** 1 year, 6 months ago
Answer is A 100%; stream the results to S3 would answer the part on "save the results in its data lake" from the question
upvoted 2 times

  **PRC** 1 year, 6 months ago
A is the answer..Anomaly RCF, Real time Kinesis Analytics..Data Lake - S3 via Firehose
upvoted 3 times

  **Phong** 1 year, 6 months ago
A is the best suitable answer
upvoted 4 times

  **devsean** 1 year, 6 months ago
It's A. B doesn't put the data into the datalake in the end, which was one of the requirements.
upvoted 4 times

  **vetal** 1 year, 7 months ago
It depends on what "most efficient" means. The simplest solution is A - and it still supports all the requirements.
upvoted 5 times

  **ComPah** 1 year, 7 months ago
Doesn't Key Word Large means it needs distributed architecture SPARK
upvoted 1 times

A Data Scientist wants to gain real-time insights into a data stream of GZIP files.
Which solution would allow the use of SQL to query the stream with the LEAST latency?

- A. Amazon Kinesis Data Analytics with an AWS Lambda function to transform the data.
- B. AWS Glue with a custom ETL script to transform the data.
- C. An Amazon Kinesis Client Library to transform the data and save it to an Amazon ES cluster.
- D. Amazon Kinesis Data Firehose to transform the data and put it into an Amazon S3 bucket.

Correct Answer: A

Reference:
<https://aws.amazon.com/big-data/real-time-analytics-featured-partners/>

Community vote distribution

A (80%)

D (20%)

-   **cybe001**

Highly Voted 

 1 year, 6 months ago

A is correct. Kinesis Data Analytics can use lamda to convert GZIP and can run SQL on the converted data.
<https://aws.amazon.com/about-aws/whats-new/2017/10/amazon-kinesis-analytics-can-now-pre-process-data-prior-to-running-sql-queries/>
upvoted 40 times
-   **VB**

Highly Voted 

 1 year, 6 months ago

A is correct:

<https://aws.amazon.com/about-aws/whats-new/2017/10/amazon-kinesis-analytics-can-now-pre-process-data-prior-to-running-sql-queries/>

"To get started, simply select an AWS Lambda function from the Kinesis Analytics application source page in the AWS Management console. Your Kinesis Analytics application will automatically process your raw data records using the Lambda function, and send transformed data to your SQL code for further processing.
Kinesis Analytics provides Lambda blueprints for common use cases like converting GZIP
..."
upvoted 13 times
-   **Nadia0012**

Most Recent 

 1 month, 3 weeks ago

Selected Answer: A

the term "lease latency" is the the hidden point. with Glue we can have near real-time but Kinesis data analytics will give you real-time transformation with internal lambda
upvoted 2 times
-   **Valcilio** 1 month, 3 weeks ago

Selected Answer: A

A is correct, with KDA you can run sql queries in the data during the streaming (real-time SQL queries).
upvoted 1 times
-   **bakarys** 2 months, 1 week ago

Selected Answer: D

D. Amazon Kinesis Data Firehose to transform the data and put it into an Amazon S3 bucket would be the best solution for allowing the use of SQL to query the stream with the least latency. Amazon Kinesis Data Firehose can be configured to transform the data before writing it to Amazon S3 in real-time. Once the data is in S3, it can be queried using SQL with Amazon Athena, which is a serverless query service that allows running standard SQL queries against data stored in Amazon S3. This approach provides the lowest latency compared to other options and requires minimal setup and maintenance.
upvoted 1 times
-   **OssamaAbdelatif** 5 months ago

Selected Answer: A

A is correct.
upvoted 1 times
-   **AddiWei** 1 year, 2 months ago

And somehow "transformation" is added to the answer as a requirement when it clearly was not part of the requirement from the question.
upvoted 2 times
-   **apprehensive_scar** 1 year, 2 months ago

AAAAAAA
upvoted 1 times

-  **HalloSpencer** 1 year, 5 months ago
what about "LEAST latency"?
upvoted 4 times
-  **Erso** 1 year, 6 months ago
A is correct. you can pre-process data prior to running SQL queries with Kinesis Data Analytics and Lambda (more or less) is always a best practice :)
upvoted 3 times
-  **JayK** 1 year, 7 months ago
Answer is B. Kinesis Data Analytics does not do any transformation, it is only for querying. Glue ETL can have scripts that can transform the data
upvoted 2 times
-  **am7** 1 year, 7 months ago
But we need to run SQL on real time stream data.
upvoted 1 times
-  **SophieSu** 1 year, 5 months ago
so you need lambda
upvoted 1 times

A retail company intends to use machine learning to categorize new products. A labeled dataset of current products was provided to the Data Science team. The dataset includes 1,200 products. The labeled dataset has 15 features for each product such as title dimensions, weight, and price. Each product is labeled as belonging to one of six categories such as books, games, electronics, and movies.

Which model should be used for categorizing new products using the provided dataset for training?

- A. AnXGBoost model where the objective parameter is set to multi:softmax
- B. A deep convolutional neural network (CNN) with a softmax activation function for the last layer
- C. A regression forest where the number of trees is set equal to the number of product categories
- D. A DeepAR forecasting model based on a recurrent neural network (RNN)

Correct Answer: B

Community vote distribution

A (100%)

- rsimham

Highly Voted

1 year, 7 months ago

Ans: A XGBoost multi class classification. <https://medium.com/@gabrielziegler3/multiclass-multilabel-classification-with-xgboost-66195e4d9f2d>

CNN is used for image classificaiton problems

upvoted 33 times
- JayK

Highly Voted

1 year, 7 months ago

Answer is A. This a classification problem thus XGBoost and the fact that there are six categories SOFTMAX is the right activation function

upvoted 14 times
- oso0348

Most Recent

1 month, 3 weeks ago

Selected Answer: A

The XGBoost algorithm is a popular and effective technique for multi-class classification. The objective parameter can be set to multi:softmax, which uses a softmax objective function for multi-class classification. This will train the model to predict the probability of each product belonging to each category, and the most probable category will be chosen as the final prediction.

A deep convolutional neural network (CNN) (B) is a powerful technique commonly used for image recognition tasks. However, it is less appropriate for tabular data like the dataset provided.

upvoted 1 times
- Konga98

3 months ago

Selected Answer: A

A, CNN is used for image classification. It would be suitable if we were classifying products using pictures of them.

upvoted 1 times
- yemauricio

4 months, 1 week ago

Selected Answer: A

<https://xgboost.readthedocs.io/en/stable/parameter.html>

upvoted 1 times
- GiyeonShin

4 months, 1 week ago

Selected Answer: A

B - CNN is used for dataset that have "local intermediate features" ex) images, or textCNN, etc

C - We need classification model, not regression model

D - RNN is used for dataset that have sequential features

A is correct

upvoted 2 times
- Peeking

4 months, 3 weeks ago

Selected Answer: A

A is the best option here. Only 1200 items and 6 classes are not enough data to involve a deep neural architecture for classification.

upvoted 1 times
- Shailendraa

7 months, 3 weeks ago

Ans- A ... For multiclassification - multi: SoftMax

upvoted 1 times
- Morsa

9 months, 2 weeks ago

Selected Answer: A

That is a classification problem so A is the answer
upvoted 1 times

  **apprehensive_scar** 1 year, 2 months ago

Selected Answer: A

Easy one. A is correct
upvoted 2 times

  **Kevinkoo** 1 year, 5 months ago

Selected Answer: A

A is correct
upvoted 3 times

  **stardustWu** 1 year, 6 months ago

Definitely A.
upvoted 1 times

  **syu31svc** 1 year, 6 months ago

100% is A; the the others are clearly wrong

Convolutional Neural Network (ConvNet or CNN) is a special type of Neural Network used effectively for image recognition and classification
Recurrent neural networks (RNN) are a class of neural networks that is powerful for modeling sequence data such as time series or natural language
upvoted 5 times

  **Antriksh** 1 year, 6 months ago

A is correct. XGBoost with softmax
upvoted 1 times

  **PRC** 1 year, 6 months ago

A should be the answer. Multi-classification problem and hence XGBoost...CNN is for image classification
upvoted 1 times

  **Phong** 1 year, 6 months ago

CNN is mostly used for images. A is the answer
upvoted 3 times

A Data Scientist is working on an application that performs sentiment analysis. The validation accuracy is poor, and the Data Scientist thinks that the cause may be a rich vocabulary and a low average frequency of words in the dataset.

Which tool should be used to improve the validation accuracy?

- A. Amazon Comprehend syntax analysis and entity detection
- B. Amazon SageMaker BlazingText cbow mode
- C. Natural Language Toolkit (NLTK) stemming and stop word removal
- D. Scikit-learn term frequency-inverse document frequency (TF-IDF) vectorizer

Correct Answer: D

Reference:
<https://monkeylearn.com/sentiment-analysis/>

Community vote distribution

D (50%)	C (38%)	8%
---------	---------	----

-   **tap123**

Highly Voted 

 1 year, 7 months ago

D is correct. Amazon Comprehend syntax analysis != Amazon Comprehend sentiment analysis. You need to read choices very carefully.
upvoted 32 times
-   **mawsman** 1 year, 7 months ago

We're looking only to improve the validation accuracy and Comprehend syntax analysis would help that because the word set is rich and the sentiment carying words infrequent. We're not looking to replace the sentiment analysis tool with Comprehend.
upvoted 3 times
-   **DonaldCMLIN**

Highly Voted 

 1 year, 7 months ago

AWS COMPREHEND IS A NATURAL LANGUAGE PROCESSING (NLP) SERVICE THAT USES MACHINE LEARNING TO DISCOVER INSIGHTS FROM TEXT. AMAZON COMPREHEND PROVIDES KEYPHRASE EXTRACTION, SENTIMENT ANALYSIS, ENTITY RECOGNITION, TOPIC MODELING, AND LANGUAGE DETECTION APIS SO YOU CAN EASILY INTEGRATE NATURAL LANGUAGE PROCESSING INTO YOUR APPLICATIONS.

HTTPS://AWS.AMAZON.COM/COMPREHEND/FEATURES/?NC1=H_LS

JUST THROUGH AMAZON COMPREHEND IS MUCH EASY THAN OTHER
THE MUCH MORE CONVENIENT ANSWER IS A.
upvoted 22 times
-   **ComPah** 1 year, 7 months ago

Agree Also Keyword is TOOL rest are frameworks
upvoted 2 times
-   **vassof95**

Most Recent 

 1 week, 5 days ago

Selected Answer: D

I would say since the buzzword "low average frequency" comes up, the safe choise would be the tfid vectorizer.

I go for D.
upvoted 1 times
-   **ParkXD** 3 weeks, 3 days ago

Selected Answer: D

The Scikit-learn term frequency-inverse document frequency (TF-IDF) vectorizer is a widely used tool to mitigate the high dimensionality of text data.
Option A, Amazon Comprehend syntax analysis, and entity detection, can help in extracting useful features from the text, but it does not address the issue of high dimensionality.

Option B, Amazon SageMaker BlazingText cbow mode, is a tool for training word embeddings, which can help to represent words in a lower dimensional space. However, it does not directly address the issue of high dimensionality and low frequency of words.

Option C, Natural Language Toolkit (NLTK) stemming and stop word removal, can reduce the dimensionality of the feature space, but it does not address the issue of low-frequency words that are important for sentiment analysis.
upvoted 3 times
-   **cpal012** 1 month, 3 weeks ago

Selected Answer: C

Emphasis is on the rich words - so stemming can help reduce these to more common words. Blazing Text in cbow mode doesnt seem relevant is about providing words given a context. And TF-IDF I'm not sure would do anything except highlight the problem you are already having?

upvoted 1 times

  **bakarys** 2 months, 1 week ago

Selected Answer: D

D. Scikit-learn term frequency-inverse document frequency (TF-IDF) vectorizer would be the best tool to use in this scenario. The TF-IDF vectorizer will give less weight to the less frequent words in the dataset, and allow the more informative and frequent words to have a greater impact on the sentiment analysis. This can help to improve the validation accuracy of the model.

upvoted 4 times

  **drcok87** 2 months, 1 week ago

Refer to https://sagemaker-examples.readthedocs.io/en/latest/introduction_to_amazon_algorithms/blazingtext_word2vec_subwords_text8/blazingtext_word2vec_subwords_text8.html

Phrase "This is a limitation, especially for languages with large vocabularies and many rare words"

SageMaker BlazingText can learn vector representations associated with character n-grams; representing words as the sum of these character n-grams representations [1]. This method enables BlazingText to generate vectors for out-of-vocabulary (OOV) words, as demonstrated in this notebook.

Also, I suspect the solution would be one of "aws provided" services or algorithms
b

upvoted 1 times

  **Ajose0** 2 months, 2 weeks ago

Selected Answer: C

C. Natural Language Toolkit (NLTK) stemming and stop word removal.

Stemming and stop word removal are two common text pre-processing techniques that can be used to improve the accuracy of sentiment analysis models. Stemming involves reducing words to their root form, which can help reduce the size of the vocabulary and increase the average frequency of words in the dataset. Stop word removal involves removing commonly used words such as "the," "and," and "a" that do not carry much meaning, again reducing the size of the vocabulary and increasing the average frequency of words.

Option D, Scikit-learn term frequency-inverse document frequency (TF-IDF) vectorizer, is a common technique for converting text into numerical representations, but it does not address the issue of a rich vocabulary and low average frequency of words in the dataset.

upvoted 1 times

  **sqavi** 2 months, 2 weeks ago

Selected Answer: D

TF-IDF gives larger values for less frequent words in the document corpus. TF-IDF value is high when both IDF and TF values are high i.e the word is rare in the whole document but frequent in a document.

upvoted 4 times

  **yemauricio** 4 months, 1 week ago

Selected Answer: A

A is the right answer

B is not possible because it's an implementation of word2vec <https://docs.aws.amazon.com/sagemaker/latest/dg/blazingtext.html>

upvoted 1 times

  **Pg690** 5 months ago

C is the correct answer https://www.bogotobogo.com/python/NLTK/Stemming_NLTK.php

upvoted 2 times

  **jason_wong6** 5 months, 1 week ago

Selected Answer: C

The answer should be C.

Note that the question said "...cause may be a rich vocabulary and a low average frequency of words". It is a data processing problem, not modeling problem!

A is irrelevant.

B is wrong. The question did not mention that he is using a deep learning model. You can't assume a word embedding layer can easily fit into his model.

C is correct because stemming and stop word removal can reduce noises. It solves the root problem in processing step without touching the model.

D is wrong. TF-IDF vectorizer does not solve the problem and you can't assume it fits with the model.

upvoted 7 times

  **milan_ml** 9 months, 1 week ago

Selected Answer: B

B

<https://docs.aws.amazon.com/sagemaker/latest/dg/blazingtext.html>

upvoted 2 times

  **f4bi4n** 10 months, 2 weeks ago

C is the only solution which takes on the root cause... removing noise and increasing the value of the dataset

upvoted 2 times

  **spamicho** 1 year, 6 months ago

Betting on B here, it's Amazon service doing text classification through text embeddings. We have low frequency problem and with this one we also recognizing word similarities into account.

upvoted 2 times

  **srinu3054** 1 year, 6 months ago

Its B!! Blazing text has out-of-vocabulary (OOV) feature which can embed the non vocabulary words.

<https://docs.aws.amazon.com/sagemaker/latest/dg/blazingtext.html>

upvoted 9 times

  **mlpcr** 1 year, 6 months ago

I think Answer is C. Problem statement - "validation accuracy is poor, and ... cause may be a rich vocabulary and a low average frequency of words in the dataset."

So to improve frequency of useful/meaningful words in given text we need to remove noisy words and for other purpose so stemming.

upvoted 3 times

Machine Learning Specialist is building a model to predict future employment rates based on a wide range of economic factors. While exploring the data, the Specialist notices that the magnitude of the input features vary greatly. The Specialist does not want variables with a larger magnitude to dominate the model.

What should the Specialist do to prepare the data for model training?

- A. Apply quantile binning to group the data into categorical bins to keep any relationships in the data by replacing the magnitude with distribution.
- B. Apply the Cartesian product transformation to create new combinations of fields that are independent of the magnitude.
- C. Apply normalization to ensure each field will have a mean of 0 and a variance of 1 to remove any significant magnitude.
- D. Apply the orthogonal sparse bigram (OSB) transformation to apply a fixed-size sliding window to generate new features of a similar magnitude.

Correct Answer: C

Reference:
<https://docs.aws.amazon.com/machine-learning/latest/dg/data-transformations-reference.html>

Community vote distribution

C (100%)

-   **rsimham**

Highly Voted 

 1 year, 7 months ago

Ans: C; Normalization is correct
upvoted 34 times
-   **gcpwhiz** 1 year, 5 months ago

Ans is not C. What is listed there is the definition of STANDARDIZATION. Normalization just scales and is not useful for reducing the effect of outliers
upvoted 4 times
-   **gcpwhiz** 1 year, 5 months ago

nevermind ignore this
upvoted 3 times
-   **Phong**

Highly Voted 

 1 year, 6 months ago

Guys, I passed the exam today. It is a tough one but there are many questions here. Good luck everyone! Thank examtopics
upvoted 12 times
-   **haison8x** 1 year, 6 months ago

Hi Phong!
Please add my skype: haison8x
upvoted 2 times
-   **OssamaAbdelatif**

Most Recent 

 5 months ago

Selected Answer: C

ans C is correct.
upvoted 1 times
-   **Deepsachin** 1 year, 5 months ago

ANS should be C as Normalization work best in case of amplitude diff
upvoted 1 times
-   **grandgale** 1 year, 6 months ago

Hi, guys,
First thanks this website for the information it provided.
However, the ML exam has updated most of the questions. only 20+ questions here are included in today's test. Anyway, it is still helpful.
GOOD LUCK EVERYONE!
upvoted 9 times
-   **joker34** 1 year, 6 months ago

So there are 40+ other questions on the exam that aren't included in Examtopics?
upvoted 2 times
-   **nez15** 1 year, 7 months ago

QUESTION 69

A large consumer goods manufacturer has the following products on sale:

- 34 different toothpaste variants
- 48 different toothbrush variants
- 43 different mouthwash variants

The entire sales history of all these products is available in Amazon S3. Currently, the company is using custom-built autoregressive integrated moving average (ARIMA) models to forecast demand for these products. The company wants to predict the demand for a new product that will soon be launched.

Which solution should a Machine Learning Specialist apply?

- A. Train a custom ARIMA model to forecast demand for the new product.
- B. Train an Amazon SageMaker DeepAR algorithm to forecast demand for the new product.
- C. Train an Amazon SageMaker k-means clustering algorithm to forecast demand for the new product.
- D. Train a custom XGBoost model to forecast demand for the new product.

Correct Answer: B

upvoted 4 times

  **VB** 1 year, 6 months ago

<https://aws.amazon.com/blogs/machine-learning/forecasting-time-series-with-dynamic-deep-learning-on-aws/>

Answer: B

upvoted 1 times

  **nez15** 1 year, 7 months ago

QUESTION 68

An agency collects census information within a country to determine healthcare and social program needs by province and city. The census form collects responses for approximately 500 questions from each citizen.

Which combination of algorithms would provide the appropriate insights? (Select TWO.)

- A. The factorization machines (FM) algorithm
- B. The Latent Dirichlet Allocation (LDA) algorithm
- C. The principal component analysis (PCA) algorithm
- D. The k-means algorithm
- E. The Random Cut Forest (RCF) algorithm

Correct Answer: CD

upvoted 5 times

  **cybe001** 1 year, 7 months ago

I think the answer is A and B.

The census question and answer will be in text. Use LDA (unsupervised algorithm) which takes the census question/answer and groups them into categories. Use the categorization to group the people and identify similar people.

Use the Factorization Machine to group the people. For each person identify if they answer a question or not. Find the total questions they answered and that will be the Target variable. Now the problem is similar to movie recommendation (consider each question a movie and the total number of questions answered will be the Rating). Based on the questions a Person answered, Factorization Machine groups the people.

Findings from both the algorithms can be used to compare and identify the people for the social programs.

upvoted 2 times

  **jasonsunbao** 1 year, 7 months ago

FM is mainly used in recommendation system to find hidden variables between two known variables to find correlation between two variables.

upvoted 1 times

  **VB** 1 year, 6 months ago

<https://aws.amazon.com/blogs/machine-learning/analyze-us-census-data-for-population-segmentation-using-amazon-sagemaker/>

Answer: C and D

upvoted 4 times

  **nez15** 1 year, 7 months ago

QUESTION 67

A. Use AWS Lambda to trigger an AWS Step Functions workflow to wait for dataset uploads to complete in Amazon S3. Use AWS Glue to join the datasets. Use an Amazon CloudWatch alarm to send an SNS notification to the Administrator in the case of a failure.

B. Develop the ETL workflow using AWS Lambda to start an Amazon SageMaker notebook instance. Use a lifecycle configuration script to join the datasets and persist the results in Amazon S3. Use an Amazon CloudWatch alarm to send an SNS notification to the Administrator in the case of a failure.

C. Develop the ETL workflow using AWS Batch to trigger the start of ETL jobs when data is uploaded to Amazon S3. Use AWS Glue to join the datasets in Amazon S3. Use an Amazon CloudWatch alarm to send an SNS notification to the Administrator in the case of a failure.

D. Use AWS Lambda to chain other Lambda functions to read and join the datasets in Amazon S3 as soon as the data is uploaded to Amazon S3. Use an Amazon CloudWatch alarm to send an SNS notification to the Administrator in the case of a failure.

Correct Answer: A

upvoted 6 times

  **nez15** 1 year, 7 months ago

QUESTION 67

A Machine Learning Specialist is developing a daily ETL workflow containing multiple ETL jobs. The workflow consists of the following processes:

- Start the workflow as soon as data is uploaded to Amazon S3.
- When all the datasets are available in Amazon S3, start an ETL job to join the uploaded datasets with multiple terabyte-sized datasets already stored in Amazon S3.
- Store the results of joining datasets in Amazon S3.
- If one of the jobs fails, send a notification to the Administrator.

Which configuration will meet these requirements?

upvoted 3 times

 **nez15** 1 year, 7 months ago

QUESTION 66

A Machine Learning Specialist must build out a process to query a dataset on Amazon S3 using Amazon Athena. The dataset contains more than 800,000 records stored as plaintext CSV files. Each record contains 200 columns and is approximately 1.5 MB in size. Most queries will span 5 to 10 columns only.

How should the Machine Learning Specialist transform the dataset to minimize query runtime?

- A. Convert the records to Apache Parquet format.
- B. Convert the records to JSON format.
- C. Convert the records to GZIP CSV format.
- D. Convert the records to XML format.

Correct Answer: A

upvoted 11 times

A Machine Learning Specialist must build out a process to query a dataset on Amazon S3 using Amazon Athena. The dataset contains more than 800,000 records stored as plaintext CSV files. Each record contains 200 columns and is approximately 1.5 MB in size. Most queries will span 5 to 10 columns only.

How should the Machine Learning Specialist transform the dataset to minimize query runtime?

- A. Convert the records to Apache Parquet format.
- B. Convert the records to JSON format.
- C. Convert the records to GZIP CSV format.
- D. Convert the records to XML format.

Correct Answer: A

Using compressions will reduce the amount of data scanned by Amazon Athena, and also reduce your S3 bucket storage. It's a Win-Win for your AWS bill.

Supported formats: GZIP, LZO, SNAPPY (Parquet) and ZLIB.

Reference:

<https://www.cloudforecast.io/blog/using-parquet-on-athena-to-save-money-on-aws/>

Community vote distribution

A (100%)

- Erso

Highly Voted

1 year, 6 months ago

Answer A seems correct...

upvoted 12 times
- Erso

1 year, 6 months ago

sorry, the link <https://aws.amazon.com/blogs/big-data/prepare-data-for-model-training-and-invoke-machine-learning-models-with-amazon-athena/>

upvoted 1 times
- sonalev419

Highly Voted

1 year, 6 months ago

A (Most queries will span 5 to 10 columns only)

upvoted 5 times
- exam_prep

Most Recent

11 months, 1 week ago

clue is: most queries will span 5 to 10 column while there are 200 columns. Indicating Data Warehouse means columner storage. Option A is correct.

upvoted 2 times
- edvardo

1 year, 5 months ago

Selected Answer: A

A. See <https://aws.amazon.com/blogs/big-data/analyzing-data-in-s3-using-amazon-athena/>

upvoted 4 times

A Machine Learning Specialist is developing a daily ETL workflow containing multiple ETL jobs. The workflow consists of the following processes:

- * Start the workflow as soon as data is uploaded to Amazon S3.
- * When all the datasets are available in Amazon S3, start an ETL job to join the uploaded datasets with multiple terabyte-sized datasets already stored in Amazon S3.
- * Store the results of joining datasets in Amazon S3.
- * If one of the jobs fails, send a notification to the Administrator.

Which configuration will meet these requirements?

- A. Use AWS Lambda to trigger an AWS Step Functions workflow to wait for dataset uploads to complete in Amazon S3. Use AWS Glue to join the datasets. Use an Amazon CloudWatch alarm to send an SNS notification to the Administrator in the case of a failure.
- B. Develop the ETL workflow using AWS Lambda to start an Amazon SageMaker notebook instance. Use a lifecycle configuration script to join the datasets and persist the results in Amazon S3. Use an Amazon CloudWatch alarm to send an SNS notification to the Administrator in the case of a failure.
- C. Develop the ETL workflow using AWS Batch to trigger the start of ETL jobs when data is uploaded to Amazon S3. Use AWS Glue to join the datasets in Amazon S3. Use an Amazon CloudWatch alarm to send an SNS notification to the Administrator in the case of a failure.
- D. Use AWS Lambda to chain other Lambda functions to read and join the datasets in Amazon S3 as soon as the data is uploaded to Amazon S3. Use an Amazon CloudWatch alarm to send an SNS notification to the Administrator in the case of a failure.

Correct Answer: A

Reference:

<https://aws.amazon.com/step-functions/use-cases/>

Community vote distribution

A (100%)

- HaiHN

Highly Voted

1 year, 7 months ago

A: Correct. S3 events can trigger AWS Lambda function.
B: Wrong. There's nothing to do with SageMaker in the provided context.
C: Wrong. AWS Batch cannot receive events from S3 directly.
D: Wrong. Will not meed the requirement: "When all the datasets are available in Amazon S3..."

<https://docs.aws.amazon.com/step-functions/latest/dg/tutorial-cloudwatch-events-s3.html>
upvoted 33 times
- scuzzy2010

1 year, 7 months ago

I agree. Step Functions can be used to implement a workflow. In this case, wait for all the datasets to be loaded before triggering the glue job.

upvoted 3 times
- cloud_trail

1 year, 6 months ago

Actually, I think that D does meet the requirement of waiting until all datasets are in S3, BUT you do need Glue to join the datasets. Answer is still A.

upvoted 4 times
- Valcilio

Most Recent

1 month, 3 weeks ago

Selected Answer: A

Batch isn't event driven, answer is A.

upvoted 1 times
- matteocal

9 months, 1 week ago

If EMR were present I would have chose that because of the size of dataset, else is Glue

upvoted 1 times
- ZSun

1 week, 5 days ago

exactly, this is also where I got confused. Since Glue is not good at handling such large dataset, multiple terabyte-sized datasets + multiple ETL jobs + daily

upvoted 1 times
- Huy

1 year, 5 months ago

A. The answer omits stuffs like Lambda functions and Event Bridge. <https://aws.amazon.com/blogs/big-data/orchestrate-multiple-etl-jobs-using-aws-step-functions-and-aws-lambda/>

upvoted 2 times



 **johnvik** 1 year, 6 months ago

<https://d1.awsstatic.com/r2018/a/product-page-diagram-aws-step-functions-use-case-aws-glue.bc69d97a332c2dd29abb724dd747fd82ae110352.png>

upvoted 2 times

Question #68

Topic 1

An agency collects census information within a country to determine healthcare and social program needs by province and city. The census form collects responses for approximately 500 questions from each citizen.

Which combination of algorithms would provide the appropriate insights? (Choose two.)

- A. The factorization machines (FM) algorithm
- B. The Latent Dirichlet Allocation (LDA) algorithm
- C. The principal component analysis (PCA) algorithm
- D. The k-means algorithm
- E. The Random Cut Forest (RCF) algorithm

Correct Answer: CD

The PCA and K-means algorithms are useful in collection of data using census form.

Community vote distribution

CD (100%)

- HaiHN

Highly Voted

1 year, 6 months ago

C: (OK) Use PCA for reducing number of variables. Each citizen's response should have answer for 500 questions, so it should have 500 variables
D: (OK) Use K-means clustering

A: (Not OK) Factorization Machines Algorithm is usually used for tasks dealing with high dimensional sparse datasets
B: (Not OK) The Latent Dirichlet Allocation (LDA) algorithm should be used for task dealing topic modeling in NLP
E: (Not OK) Random Cut Forest should be used for detecting anormal in data

upvoted 28 times
- hans1234

Highly Voted

1 year, 6 months ago

<https://aws.amazon.com/blogs/machine-learning/analyze-us-census-data-for-population-segmentation-using-amazon-sagemaker/>

Answer: C and D

upvoted 11 times
- Ajose0

Most Recent

2 months, 2 weeks ago

Selected Answer: CD

C. The principal component analysis (PCA) algorithm
D. The k-means algorithm

PCA is a dimensionality reduction technique that can be used to identify the underlying structure of the census data. This algorithm can help to identify the most important questions and provide an overview of the relationship between the questions and the responses.

K-means is an unsupervised learning algorithm that can be used to segment the population into different groups based on their responses to the census questions. This algorithm can help to determine the healthcare and social program needs by province and city based on the responses collected from each citizen.

These algorithms can help to provide insights into the patterns and relationships within the census data, which can inform decision making for healthcare and social program planning.

upvoted 1 times
- Peeking

4 months, 3 weeks ago

Selected Answer: CD

Reduce dimensionality and cluster subjects.

upvoted 2 times
- ac427

1 year, 7 months ago

This is the same question as Topic 2 Q3

upvoted 1 times
- muralee_xo

3 months, 1 week ago

how to reach Topic 2 every questions here seem to belong to topic 1

upvoted 1 times

A large consumer goods manufacturer has the following products on sale:

- * 34 different toothpaste variants
- * 48 different toothbrush variants
- * 43 different mouthwash variants

The entire sales history of all these products is available in Amazon S3. Currently, the company is using custom-built autoregressive integrated moving average (ARIMA) models to forecast demand for these products. The company wants to predict the demand for a new product that will soon be launched. Which solution should a Machine Learning Specialist apply?

A. Train a custom ARIMA model to forecast demand for the new product.

B. Train an Amazon SageMaker DeepAR algorithm to forecast demand for the new product.

C. Train an Amazon SageMaker k-means clustering algorithm to forecast demand for the new product.

D. Train a custom XGBoost model to forecast demand for the new product.

Correct Answer: B

The Amazon SageMaker DeepAR forecasting algorithm is a supervised learning algorithm for forecasting scalar (one-dimensional) time series using recurrent neural networks (RNN). Classical forecasting methods, such as autoregressive integrated moving average (ARIMA) or exponential smoothing (ETS), fit a single model to each individual time series. They then use that model to extrapolate the time series into the future.

Reference:
<https://docs.aws.amazon.com/sagemaker/latest/dg/deepar.html>

Community vote distribution

B (100%)

- HaiHN

Highly Voted

1 year, 6 months ago

B

<https://docs.aws.amazon.com/sagemaker/latest/dg/deepar.html>

"...When your dataset contains hundreds of related time series, DeepAR outperforms the standard ARIMA and ETS methods. You can also use the trained model to generate forecasts for new time series that are similar to the ones it has been trained on."

upvoted 16 times
- Valcilio

Most Recent

1 month, 3 weeks ago

Selected Answer: B

DeepAr for new products forever!

upvoted 2 times
- AjoseO

2 months, 2 weeks ago

Selected Answer: B

The DeepAR algorithm is a powerful time series forecasting algorithm that is designed to handle multiple time series data and can handle irregularly spaced time series data and missing values, making it a good fit for this task.

Additionally, the large amount of sales history data available in Amazon S3 makes the use of a deep learning algorithm like DeepAR more appropriate.

upvoted 2 times
- Shailendraa

8 months, 2 weeks ago

B <https://docs.aws.amazon.com/sagemaker/latest/dg/deepar.html>

upvoted 1 times
- hans1234

1 year, 6 months ago

It is B

upvoted 3 times
- ac427

1 year, 7 months ago

This is the same question as Topic 2 Q4

upvoted 1 times

A Machine Learning Specialist uploads a dataset to an Amazon S3 bucket protected with server-side encryption using AWS KMS. How should the ML Specialist define the Amazon SageMaker notebook instance so it can read the same dataset from Amazon S3?

- A. Define security group(s) to allow all HTTP inbound/outbound traffic and assign those security group(s) to the Amazon SageMaker notebook instance.
- B. I configure the Amazon SageMaker notebook instance to have access to the VPC. Grant permission in the KMS key policy to the notebook's KMS role.
- C. Assign an IAM role to the Amazon SageMaker notebook with S3 read access to the dataset. Grant permission in the KMS key policy to that role.
- D. Assign the same KMS key used to encrypt data in Amazon S3 to the Amazon SageMaker notebook instance.

Correct Answer: D

Reference:
<https://docs.aws.amazon.com/sagemaker/latest/dg/encryption-at-rest.html>

Community vote distribution

C (86%)	14%
---------	-----

-   **seanLu** Highly Voted 1 year, 6 months ago

Should be C.
"You don't need to specify the AWS KMS key ID when you download an SSE-KMS-encrypted object from an S3 bucket. Instead, you need the permission to decrypt the AWS KMS key.

When a user sends a GET request, Amazon S3 checks if the AWS Identity and Access Management (IAM) user or role that sent the request is authorized to decrypt the key associated with the object. If the IAM user or role belongs to the same AWS account as the key, then the permission to decrypt must be granted on the AWS KMS key's policy."
https://aws.amazon.com/premiumsupport/knowledge-center/decrypt-kms-encrypted-objects-s3/?nc1=h_ls
upvoted 28 times
-   **askaron** Highly Voted 1 year, 7 months ago

Should be C.
I think it is not possible to assign a key directly to a Sagemaker notebook instance like D suggests.
Normally in AWS in general, IAM roles are used to do so. So C.
upvoted 6 times
-   **oso0348** Most Recent 3 weeks, 5 days ago

Selected Answer: C

To allow an Amazon SageMaker notebook instance to read a dataset stored in an Amazon S3 bucket that is protected with server-side encryption using AWS KMS, the ML Specialist should assign an IAM role to the Amazon SageMaker notebook with S3 read access to the dataset. The IAM role should have permissions to access the S3 bucket and the KMS key that was used to encrypt the data. This role should be granted permission in the KMS key policy to allow it to decrypt the data.
upvoted 1 times
-   **Nadia0012** 1 month, 3 weeks ago

Selected Answer: D

To encrypt the machine learning (ML) storage volume that is attached to notebooks, processing jobs, training jobs, hyperparameter tuning jobs, batch transform jobs, and endpoints, you can pass a AWS KMS key to SageMaker. If you don't specify a KMS key, SageMaker encrypts storage volumes with a transient key and discards it immediately after encrypting the storage volume. For notebook instances, if you don't specify a KMS key, SageMaker encrypts both OS volumes and ML data volumes with a system-managed KMS key.
upvoted 1 times
-   **Nadia0012** 1 month, 3 weeks ago

I correct myself- Option C is correct:
Background
AWS Key Management Service (AWS KMS) enables Server-side encryption to protect your data at rest. Amazon SageMaker training works with KMS encrypted data if the IAM role used for S3 access has permissions to encrypt and decrypt data with the KMS key. Further, a KMS key can also be used to encrypt the model artifacts at rest using Amazon S3 server-side encryption. Additionally, a KMS key can also be used to encrypt the storage volume attached to training, endpoint, and transform instances. In this notebook, we demonstrate SageMaker encryption capabilities using KMS-managed keys.
resource: https://github.com/aws/amazon-sagemaker-examples/blob/main/advanced_functionality/handling_kms_encrypted_data/handling_kms_encrypted_data.ipynb
Option D is correct if sagemaker does the encryption, if you are dealing with encrypted data then C is 100% correct.
upvoted 2 times
-   **Ajose0** 2 months, 2 weeks ago

Selected Answer: C

C. Assign an IAM role to the Amazon SageMaker notebook with S3 read access to the dataset. Grant permission in the KMS key policy to that role.

To access the encrypted dataset in Amazon S3, the Amazon SageMaker notebook instance must have the appropriate permissions. This can be achieved by assigning an IAM role to the notebook with read access to the dataset in Amazon S3 and granting permission in the KMS key policy to that role. This ensures that the notebook has the necessary permissions to access the encrypted data in Amazon S3, while adhering to best practices for securing sensitive data.

upvoted 2 times

ystotest

5 months ago

Selected Answer: C

agreed with C

upvoted 3 times

AmakamaxZanny

1 year, 2 months ago

Answer is C : Open the IAM console. Add a policy to the IAM user that grants the permissions to upload and download from the bucket. You can use a policy that's similar to the following:

<https://aws.amazon.com/premiumsupport/knowledge-center/s3-bucket-access-default-encryption/>
(number 2)

upvoted 1 times

Deepsachin

1 year, 5 months ago

Seems to be D

<https://docs.aws.amazon.com/sagemaker/latest/dg/encryption-at-rest-nbi.html>

upvoted 2 times

Madwyn

1 year, 6 months ago

Not D as if you assign the key in the notebook, that's not secure, it will make the encryption ineffective. Instead, you assign the access permission by using IAM.

upvoted 1 times

Nickname_L

1 year, 6 months ago

Should be C based on these resources

<https://stackoverflow.com/questions/66692579/aws-sagemaker-permissionerror-access-denied-reading-data-from-s3-bucket>

<https://aws.amazon.com/premiumsupport/knowledge-center/s3-bucket-access-default-encryption/>

upvoted 1 times

Huy

1 year, 6 months ago

C. D is wrong because Encryption key lets you encrypt data on the ML storage volume attached to the notebook instance using an AWS Key Management Service (AWS KMS) key. If you plan to store sensitive information on the ML storage volume, consider encrypting the information.

upvoted 1 times

mona_mansour

1 year, 6 months ago

should be c

You don't need to specify the AWS KMS key ID when you download an SSE-KMS-encrypted object from an S3 bucket. Instead, you need the permission to decrypt the AWS KMS key.

When a user sends a GET request, Amazon S3 checks if the AWS Identity and Access Management (IAM) user or role that sent the request is authorized to decrypt the key associated with the object. If the IAM user or role belongs to the same AWS account as the key, then the permission to decrypt must be granted on the AWS KMS key's policy.

upvoted 3 times

ahquiceno

1 year, 7 months ago

C is correct go to: <https://docs.aws.amazon.com/sagemaker/latest/dg/sagemaker-roles.html>

<https://docs.aws.amazon.com/glue/latest/dg/create-an-iam-role-sagemaker-notebook.html>.

upvoted 4 times

joep21

1 year, 7 months ago

D seems reasonable based on the explanation: <https://docs.aws.amazon.com/sagemaker/latest/dg/encryption-at-rest.html>

upvoted 2 times

takahirokoyama

1 year, 7 months ago

I think so.

upvoted 1 times

Question #71

Topic 1

A Data Scientist needs to migrate an existing on-premises ETL process to the cloud. The current process runs at regular time intervals and uses PySpark to combine and format multiple large data sources into a single consolidated output for downstream processing. The Data Scientist has been given the following requirements to the cloud solution:

- Combine multiple data sources.
- Reuse existing PySpark logic.
- Run the solution on the existing schedule.
- Minimize the number of servers that will need to be managed.

Which architecture should the Data Scientist use to build this solution?

- A. Write the raw data to Amazon S3. Schedule an AWS Lambda function to submit a Spark step to a persistent Amazon EMR cluster based on the existing schedule. Use the existing PySpark logic to run the ETL job on the EMR cluster. Output the results to a `processed` location in Amazon S3 that is accessible for downstream use.
- B. Write the raw data to Amazon S3. Create an AWS Glue ETL job to perform the ETL processing against the input data. Write the ETL job in PySpark to leverage the existing logic. Create a new AWS Glue trigger to trigger the ETL job based on the existing schedule. Configure the output target of the ETL job to write to a `processed` location in Amazon S3 that is accessible for downstream use.
- C. Write the raw data to Amazon S3. Schedule an AWS Lambda function to run on the existing schedule and process the input data from Amazon S3. Write the Lambda logic in Python and implement the existing PySpark logic to perform the ETL process. Have the Lambda function output the results to a `processed` location in Amazon S3 that is accessible for downstream use.
- D. Use Amazon Kinesis Data Analytics to stream the input data and perform real-time SQL queries against the stream to carry out the required transformations within the stream. Deliver the output results to a `processed` location in Amazon S3 that is accessible for downstream use.

Correct Answer: D

Community vote distribution

B (83%)

A (17%)

Paul_NoName

Highly Voted

1 year, 7 months ago

B it is .
upvoted 29 times

joep21

1 year, 6 months ago

I agree, B is serverless and reuses Pyspark. Similar example shown here: <https://docs.aws.amazon.com/glue/latest/dg/aws-glue-programming-python-samples-medicaid.html>
upvoted 11 times

SophieSu

Highly Voted

1 year, 6 months ago

A is not correct because Minimize the number of servers that will need to be managed. EMR is not server-less.
B is correct. AWS Glue supports an extension of the PySpark Python dialect for scripting extract, transform, and load...
C is not correct because using Lambda for ETL you will not be able to Reuse existing PySpark logic
D is not correct because Kinesis is not server-less. And you can not Reuse existing PySpark logic
upvoted 9 times

Maaayaaa

Most Recent

2 weeks, 3 days ago

Selected Answer: B

B ticks all boxes. Minimize servers -> AWS managed services -> Glue.
upvoted 2 times

bakarys

2 months, 1 week ago

Selected Answer: A

Option A would be the best response for this scenario.

This solution allows the Data Scientist to reuse the existing PySpark logic while migrating the ETL process to the cloud. The raw data is written to Amazon S3, and a Lambda function is scheduled to trigger a Spark step on a persistent EMR cluster based on the existing schedule. The PySpark logic is used to run the ETL job on the EMR cluster, and the results are output to a processed location in Amazon S3 that is accessible for downstream use. This solution minimizes the number of servers that need to be managed, and it allows for a seamless migration of the existing ETL process to the cloud.
upvoted 1 times

sqavi

2 months, 2 weeks ago

Selected Answer: B

Option D is wrong it should be B
upvoted 1 times

  **Peeking** 4 months, 3 weeks ago

D cannot be answer as there is no streaming data or Realtime processing.
upvoted 2 times

  **salads** 8 months, 2 weeks ago

Selected Answer: B
the answer is b
upvoted 2 times

  **Nickname_L** 1 year, 6 months ago

Answer should be B. Serverless, on a regular schedule (no real time requirement), reuses PySpark code in Glue ETL script.
upvoted 4 times

  **gcpwhiz** 1 year, 6 months ago

Answer is B as they specifically ask about reusing existing PySpark, which can be done with Glue
upvoted 3 times

  **Aashi22** 1 year, 6 months ago

https://docs.aws.amazon.com/glue/latest/dg/creating_running_workflows.html
upvoted 1 times

  **astonm13** 1 year, 6 months ago

It is B. ! "Minimize number of servers to be managed". B is a Serverless solution which fulfils other requirements!
upvoted 2 times

  **cnethers** 1 year, 6 months ago

I like both A & B however with B you would need to rewrite the Pyspark code to account for the ETL process you are now introducing, so it would not be using the original code. Both A & B are using managed services. Answer B would also require a notebook instance to get set up as there is no direct integration of Pysprak in Glue so there are some assumptions being made.
On Balance, I am leaning towards answer A
upvoted 4 times

  **Joe_Zhang** 1 year, 7 months ago

The answer should be A.
upvoted 2 times

A Data Scientist is building a model to predict customer churn using a dataset of 100 continuous numerical features. The Marketing team has not provided any insight about which features are relevant for churn prediction. The Marketing team wants to interpret the model and see the direct impact of relevant features on the model outcome. While training a logistic regression model, the Data Scientist observes that there is a wide gap between the training and validation set accuracy.

Which methods can the Data Scientist use to improve the model performance and satisfy the Marketing team's needs? (Choose two.)

- A. Add L1 regularization to the classifier
- B. Add features to the dataset
- C. Perform recursive feature elimination
- D. Perform t-distributed stochastic neighbor embedding (t-SNE)
- E. Perform linear discriminant analysis

Correct Answer: BE

Community vote distribution

AC (100%)

bluer1 Highly Voted 1 year ago

AC - correct answer
upvoted 12 times

bakarys Most Recent 2 months, 1 week ago

Selected Answer: AC

A. Add L1 regularization to the classifier and C. Perform recursive feature elimination are the methods that can be used to improve the model performance and satisfy the Marketing team's needs.

Explanation:

A. Adding L1 regularization to the logistic regression classifier can help to improve the model performance and reduce overfitting. This can also help to highlight the relevant features for churn prediction as L1 regularization can shrink the coefficients of irrelevant features to zero.

C. Recursive feature elimination can be used to select the most relevant features for the model. This can help to improve the model performance and highlight the relevant features for churn prediction.

upvoted 2 times

AjoseO 2 months, 2 weeks ago

Selected Answer: AC

A. Adding L1 regularization can help to reduce overfitting by shrinking the coefficients of less important features towards zero, which can improve the model's generalization performance on the validation set.

C. Recursive feature elimination is a feature selection technique that removes the least important feature at each iteration and trains the model on the remaining features until a desired number of features is reached. This method can be used to identify the most relevant features for the prediction task and reduce the dimensionality of the dataset, leading to improved model performance and interpretability for the Marketing team.

upvoted 1 times

wisoxe8356 4 months, 4 weeks ago

AC -
Key: logistic regression model = non linear in terms of Odds and Probability, however it is linear in terms of Log Odds.
Key: Large gap between training & validation = overfitting
=> 5 techniques to prevent overfitting:
1. Simplifying the model | 2. Early stopping
3. Use data argumentation | 4. Use regularization | 5. Use dropouts

A - yes to avoid overfitting (although i am thinking it is talking about regressor)

Not B - add feature will lead to overfitting

C - feature elimination - prevent overfitting

Not D - t-SNE is a nonlinear dimensionality reduction technique

Not E - find feature correlation only - Linear discriminant analysis (LDA), normal discriminant analysis (NDA), or discriminant function analysis is a generalization of Fisher's linear discriminant, a method used in statistics and other fields, to find a linear combination of features that characterizes or separates two or more classes of objects or events.

upvoted 3 times

itallomd 5 months, 1 week ago

L1 won't do naturally the feature elimination?
I guess AB

upvoted 1 times

  **Atreides457** 8 months ago

why not A & D? or C & D?
does not t-SNE grant the marketing team's wish for visualization of relationships? or are we to presume that A&C are best as C (recursive feature elimination) grants us some visualization of feature importance.

upvoted 1 times

  **tgaos** 10 months, 3 weeks ago

Selected Answer: AC

AC is correct
upvoted 3 times

  **NeverMinda** 10 months, 4 weeks ago

Selected Answer: AC

overfitting: add regularization, remove features
upvoted 4 times

An aircraft engine manufacturing company is measuring 200 performance metrics in a time-series. Engineers want to detect critical manufacturing defects in near- real time during testing. All of the data needs to be stored for offline analysis. What approach would be the MOST effective to perform near-real time defect detection?

- A. Use AWS IoT Analytics for ingestion, storage, and further analysis. Use Jupyter notebooks from within AWS IoT Analytics to carry out analysis for anomalies.
- B. Use Amazon S3 for ingestion, storage, and further analysis. Use an Amazon EMR cluster to carry out Apache Spark ML k-means clustering to determine anomalies.
- C. Use Amazon S3 for ingestion, storage, and further analysis. Use the Amazon SageMaker Random Cut Forest (RCF) algorithm to determine anomalies.
- D. Use Amazon Kinesis Data Firehose for ingestion and Amazon Kinesis Data Analytics Random Cut Forest (RCF) to perform anomaly detection. Use Kinesis Data Firehose to store data in Amazon S3 for further analysis.

Correct Answer: B

Community vote distribution

D (86%)

14%

- Joe_Zhang

Highly Voted

1 year, 7 months ago

D near-real time

upvoted 41 times
- cnethers

Highly Voted

1 year, 7 months ago

Glad we are all in agreement D is the correct answer

upvoted 17 times
- oso0348

Most Recent

3 weeks, 5 days ago

Selected Answer: C

Option D is not necessarily incorrect, but it may not be the most effective approach to perform near-real time defect detection in this scenario. Here are some potential drawbacks of this approach:

Amazon Kinesis Data Firehose is primarily used for data ingestion and delivery to other services, and may not be the best choice for real-time analysis.

Using Amazon Kinesis Data Analytics for anomaly detection may be less flexible than using Amazon SageMaker, which provides a wide range of algorithms and models for anomaly detection.

Random Cut Forest (RCF) is a popular anomaly detection algorithm used for time-series data, and Amazon SageMaker provides an RCF implementation that can be used for anomaly detection in real-time or offline. While Amazon Kinesis Data Analytics also provides RCF, using Amazon SageMaker may be a better choice for scalability and flexibility.

upvoted 1 times
- oso0348

3 weeks, 5 days ago

Selected Answer: C

Yes, option C can provide near real-time defect detection. Amazon SageMaker's Random Cut Forest (RCF) algorithm is designed to work with streaming data and can detect anomalies in near real-time. It can process data in batches as small as a single data point, making it well-suited for real-time anomaly detection.

In this scenario, if the manufacturing process is generating data in real-time, it can be ingested into Amazon S3 and processed by Amazon SageMaker's RCF algorithm, allowing for near real-time detection of critical manufacturing defects during testing.

upvoted 1 times
- ZSun

1 week, 4 days ago

this is ridiculous. How can you store in s3 and then conduct real-time analysis?

upvoted 2 times
- Paolo991

1 month, 1 week ago

Selected Answer: D

Firehose to ingest in near real time and RCF to do Anomaly is the best approach

upvoted 2 times
- fez_2312

1 month, 2 weeks ago

This question confused me because A is the only solution which can be implemented offline, S3 is an online, so how can the data be analysed if it has to be onlone?

upvoted 2 times

-   **sqavi** 2 months, 2 weeks ago

Selected Answer: D

D is correct answer. When solution is available in AWS as a service, no need to build it with anything else

upvoted 3 times
-   **ovokpus** 10 months, 1 week ago

Selected Answer: D

Key word is "near real time", which will require streams, and that is what Kinesis provides

upvoted 3 times
-   **clawo** 1 year, 3 months ago

Selected Answer: D

D of course

upvoted 2 times
-   **geekgirl007** 1 year, 3 months ago

Selected Answer: D

D is the answer. it's near real time so Kinesis and also anomaly detection RCF.

upvoted 2 times
-   **passionatecricketer** 1 year, 5 months ago

Whenever there is a requirement for real-time data ingestion (time series data and otherwise) and perform data transformation and real-time analytics on the fly, then the combination of Firehose and Kinesis Analytics is the de-facto architecture pattern.

upvoted 2 times
-   **Nickname_L** 1 year, 6 months ago

Should be D (near real time, auto anomaly detection using RCF, uses Firehose later to store data for offline analysis)

upvoted 2 times
-   **kezzzzz** 1 year, 6 months ago

Should be A. Ingest, store, do real time analytics and AI

<https://aws.amazon.com/iot-analytics/?c=i&sec=srv>

upvoted 1 times
-   **kezzzzz** 1 year, 6 months ago

Why not A?

upvoted 1 times
-   **dunhill** 1 year, 4 months ago

I believe that the reason is "need near-real-time detection". IoT needs jupyter for detect and analyse for later. In the other way, Kinesis Analytics can handle IoT data.

upvoted 2 times
-   **tmlld** 1 year, 6 months ago

S3 is an object storage, not for data ingestion. B&C are wrong. I go with D

upvoted 3 times
-   **Vita_Rasta84444** 1 year, 6 months ago

Answer is D

upvoted 2 times
-   **astonm13** 1 year, 6 months ago

D is correct!

upvoted 2 times

A Machine Learning team runs its own training algorithm on Amazon SageMaker. The training algorithm requires external assets. The team needs to submit both its own algorithm code and algorithm-specific parameters to Amazon SageMaker.

What combination of services should the team use to build a custom algorithm in Amazon SageMaker? (Choose two.)

- A. AWS Secrets Manager
- B. AWS CodeStar
- C. Amazon ECR
- D. Amazon ECS
- E. Amazon S3

Correct Answer: CE

Community vote distribution

CE (100%)

-   **Paul_NoName**

Highly Voted 

 1 year, 7 months ago

CE is the right answer. ECR uses ECS internally while using SGM.

upvoted 11 times
-   **joep21** 1 year, 6 months ago

CE based on criteria and this documentation: <https://docs.aws.amazon.com/sagemaker/latest/dg/sagemaker-mkt-create-model-package.html>

"For Location of inference image, type the path to the image that contains your inference code. The image must be stored as a Docker container in Amazon ECR.

For Location of model data artifacts, type the location in S3 where your model artifacts are stored."

upvoted 6 times
-   **ZSun** 1 week, 1 day ago

the answer is correct but the explanation is completely wrong.

The question is about how to create your own algorithm using container, not "put the inference in market" (which is your resource link).

the right citation should be "Adapting your own training container": create s3 to store model articraft, and push code to ECR.

upvoted 2 times
-   **SophieSu**

Highly Voted 

 1 year, 6 months ago

CE IS THE CORRECT ANSWER 100%

upvoted 5 times
-   **jackzhao**

Most Recent 

 1 month, 2 weeks ago

CE is correct!

upvoted 1 times
-   **Valcilio** 1 month, 3 weeks ago

ECR for the code, S3 for the parameters!

upvoted 1 times
-   **Valcilio** 1 month, 3 weeks ago

Selected Answer: CE

C contain the algorithm's image and E contain algorithm's parameters.

upvoted 2 times
-   **randomnamer** 1 year, 6 months ago

The location of the model artifacts. Model artifacts can either be packaged in the same Docker container as the inference code or stored in Amazon S3. Not so sure.

upvoted 1 times
-   **cnethers** 1 year, 6 months ago

<https://aws.amazon.com/blogs/machine-learning/bringing-your-own-custom-container-image-to-amazon-sagemaker-studio-notebooks/>

If you wish to use your private VPC to securely bring your custom container, you also need the following:

A VPC with a private subnet

VPC endpoints for the following services:

Amazon Simple Storage Service (Amazon S3)

Amazon SageMaker

Amazon ECR

AWS Security Token Service (AWS STS)

CodeBuild for building Docker containers

Answer C+E
upvoted 3 times

  **ahquiceno** 1 year, 7 months ago
For me CD. needs storage and create a custom docker using ECR to store it.
upvoted 1 times

  **ahquiceno** 1 year, 6 months ago
Sorry, CE is correct.
upvoted 1 times

  **gcpwhiz** 1 year, 5 months ago
Sagemaker will spin up the instances needed with the right image. No need to use ECS. CE is right
upvoted 1 times

Question #75

Topic 1

A Machine Learning Specialist wants to determine the appropriate SageMakerVariantInvocationsPerInstance setting for an endpoint automatic scaling configuration. The Specialist has performed a load test on a single instance and determined that peak requests per second (RPS) without service degradation is about 20 RPS. As this is the first deployment, the Specialist intends to set the invocation safety factor to 0.5. Based on the stated parameters and given that the invocations per instance setting is measured on a per-minute basis, what should the Specialist set as the SageMakerVariantInvocationsPerInstance setting?

- A. 10
- B. 30
- C. 600
- D. 2,400

Correct Answer: C

Community vote distribution

C (100%)

  **Paul_NoName**

Highly Voted 

 1 year, 6 months ago
C is correct .
SageMakerVariantInvocationsPerInstance = (MAX_RPS * SAFETY_FACTOR) * 60
AWS recommended Saf_fac =0 .5
upvoted 13 times

  **ahquiceno**

Highly Voted 

 1 year, 6 months ago
Answer C: SageMakerVariantInvocationsPerInstance = (MAX_RPS * SAFETY_FACTOR) * 60
<https://docs.aws.amazon.com/sagemaker/latest/dg/endpoint-scaling-loadtest.html>
upvoted 6 times

  **jackzhao**

Most Recent 

 1 month, 2 weeks ago
SageMakerVariantInvocationsPerInstance = (MAX_RPS * SAFETY_FACTOR) * 60
upvoted 1 times

  **King_Chess1** 3 months ago
SageMakerVariantInvocationsPerInstance = (MAX_RPS * SAFETY_FACTOR) * 60
(20RPS * 0.5Safety Factor) * 60
(10)*60 = 600
Answer C
upvoted 1 times

  **Peeking** 4 months, 3 weeks ago

Selected Answer: C

Maximum request at peak time = 20 RPS = 20x60 = 1200RPM
Safety factor of 0.5 = 1200*0.5 = 600
Basic setting of parameter = 600 (requests per minutes)
upvoted 1 times

A company uses a long short-term memory (LSTM) model to evaluate the risk factors of a particular energy sector. The model reviews multi-page text documents to analyze each sentence of the text and categorize it as either a potential risk or no risk. The model is not performing well, even though the Data Scientist has experimented with many different network structures and tuned the corresponding hyperparameters. Which approach will provide the MAXIMUM performance boost?

- A. Initialize the words by term frequency-inverse document frequency (TF-IDF) vectors pretrained on a large collection of news articles related to the energy sector.
- B. Use gated recurrent units (GRUs) instead of LSTM and run the training process until the validation loss stops decreasing.
- C. Reduce the learning rate and run the training process until the training loss stops decreasing.
- D. Initialize the words by word2vec embeddings pretrained on a large collection of news articles related to the energy sector.

Correct Answer: C

Community vote distribution

D (100%)

- jiadong

Highly Voted

1 year, 7 months ago

I think the right answer is D

upvoted 24 times
- SophieSu

Highly Voted

1 year, 7 months ago

D is correct.

C is not the best the answer because the question states that tuning parameters doesn't help a lot. Transfer learning would be better solution!

upvoted 11 times
- Peeking

Most Recent

4 months, 3 weeks ago

Selected Answer: D

Transfer learning, in my experience, has been a good way to boost performance when hyperparameter tuning did not work.

upvoted 2 times
- Sidekick

11 months, 1 week ago

The case ask for predicting labels for sentences, the appropriate algo should be "Text Classification" Which, just as "wrod2vec,i part of Blazing Text.

upvoted 1 times
- julpeg

1 year ago

Selected Answer: D

The answer should be D. My reasoning is that by using a word embedding which is trained on domain specific material, the embeddings between two words are more domain specific. This means that relations (good or bad) are represented in a better way, which also means that the model should be able to predict the results in a more accurate way.

upvoted 3 times
- bitsplease

1 year, 3 months ago

both A & D "seem" correct, but word2vec takes ORDER of words into acc (to some extent)--while TF-IDF does not. Thus max boost is from D.

B,C are wrong because the DS has tried several network architectures (aka LSTM) and hyperparameter tuning (aka option C)

upvoted 6 times
- ahmedelbhy

1 year, 6 months ago

i think answer is A as The model reviews multi-page text documents

upvoted 1 times
- GiyeonShin

4 months, 1 week ago

I think that the general tf-idf vectors cannot be directly adapted to the deep learning model, because of the large dimension in vector values

upvoted 1 times
- puffpuff

1 year, 6 months ago

I think it should be B

A/D are false flags because the question doesn't specify what kind of data engineering is currently done on the inputs, as a baseline

Per wikipedia, for GRUs, "GRUs have been shown to exhibit better performance on certain smaller and less frequent datasets", which fits the context of a particular energy sector

upvoted 2 times
- ChanduPatil

1 year, 6 months ago

why not B??

upvoted 1 times

  **GiyeonShin** 4 months, 1 week ago

Early stopping can give the model better performance, but I think that the model needs more condition like patience value for early stopping. This is because the model doesn't always show the performance at its maximum when the validation loss stops decreasing.

upvoted 1 times

  **GiyeonShin** 4 months, 1 week ago

Generally, LSTM has the better performance then GRU in large datasets such as multi-page documents. GRU has advantages of memory allocation and training time.

upvoted 1 times

  **jkreddy** 1 year, 6 months ago

It cannot be C, because hyper parameter tuning didnt work as given in question. Also, A and D are same, however, word2vec model internally implements tf-idf much more efficiently. So answer got to be D

upvoted 4 times

  **YJ4219** 1 year, 6 months ago

but they need to classify the whole sentence i think for such a case we use object2vec not word2vec, but since it's not available in the answers, B is the only answer left.

upvoted 2 times

  **tmld** 1 year, 7 months ago

I go for C

upvoted 2 times

  **astonm13** 1 year, 7 months ago

Agree. D seems more reasonable, as word2vec provides content of the sentences which is very important for evaluation of the risk.

upvoted 3 times

A Machine Learning Specialist needs to move and transform data in preparation for training. Some of the data needs to be processed in near-real time, and other data can be moved hourly. There are existing Amazon EMR MapReduce jobs to clean and feature engineering to perform on the data.

Which of the following services can feed data to the MapReduce jobs? (Choose two.)

- A. AWS DMS
- B. Amazon Kinesis
- C. AWS Data Pipeline
- D. Amazon Athena
- E. Amazon ES

Correct Answer: AE

Community vote distribution

BC (100%)

- Joe_Zhang**

Highly Voted

1 year, 7 months ago

should be BC

upvoted 30 times
- joep21**

1 year, 6 months ago

Agreed, AWS Example: <https://docs.aws.amazon.com/datapipeline/latest/DeveloperGuide/what-is-datapipeline.html>

upvoted 5 times
- Khalil11**

Most Recent

3 weeks, 6 days ago

The right answer is BC

upvoted 1 times
- Ajose0**

2 months, 2 weeks ago

Selected Answer: BC

AWS Data Pipeline (Option C) can be used to move the hourly data, as it provides a way to move data from various sources to Amazon EMR for processing.

Amazon Kinesis (Option B) can be used to process data in near-real time, as it is a real-time data streaming service that can handle large amounts of incoming data from multiple sources. The data can be fed to Amazon EMR MapReduce jobs for processing.

upvoted 3 times
- DS2021**

4 months ago

Selected Answer: BC

should be BC

upvoted 2 times
- Peeking**

4 months, 3 weeks ago

Selected Answer: BC

Kinesis for near realtime data and pipeline for the other data moved hourly.

upvoted 3 times
- John_Pongthorn**

1 year, 2 months ago

Selected Answer: BC

AWS ES is an elastic search , it is nothing to do with this question.

upvoted 3 times
- scsas**

1 year, 2 months ago

Kinesis data into EMR: <https://docs.aws.amazon.com/emr/latest/ReleaseGuide/emr-kinesis.html>

upvoted 1 times
- apprehensive_scar**

1 year, 2 months ago

BC. easy

upvoted 2 times
- KM226**

1 year, 3 months ago

I believe the answer is BC

upvoted 1 times

-  **Madwyn**

1 year, 6 months ago

BD.
Data Pipeline is to orchestrate the workflow, how can that feed data to the MR jobs?

upvoted 2 times
-  **Vita_Rasta84444**

1 year, 6 months ago

Answer is B and C

upvoted 1 times
-  **astonm13**

1 year, 6 months ago

Answer is for sure BC

upvoted 3 times
-  **takahirokoyama**

1 year, 6 months ago

Ans is BC.
(<https://aws.amazon.com/jp/emr/?whats-new-cards.sort-by=item.additionalFields.postDateTime&whats-new-cards.sort-order=desc>)

upvoted 2 times

A Machine Learning Specialist previously trained a logistic regression model using scikit-learn on a local machine, and the Specialist now wants to deploy it to production for inference only.

What steps should be taken to ensure Amazon SageMaker can host a model that was trained locally?

- A. Build the Docker image with the inference code. Tag the Docker image with the registry hostname and upload it to Amazon ECR.
- B. Serialize the trained model so the format is compressed for deployment. Tag the Docker image with the registry hostname and upload it to Amazon S3.
- C. Serialize the trained model so the format is compressed for deployment. Build the image and upload it to Docker Hub.
- D. Build the Docker image with the inference code. Configure Docker Hub and upload the image to Amazon ECR.

Correct Answer: D

Community vote distribution

A (100%)

- arulrajayaraj

Highly Voted

1 year, 6 months ago

Ans : A Refer the below :
<https://sagemaker-workshop.com/custom/containers.html>
upvoted 18 times
- Paul_NoName

Highly Voted

1 year, 7 months ago

A
<https://sagemaker-workshop.com/custom/containers.html>
upvoted 7 times
- Khalil11

Most Recent

3 weeks, 6 days ago

Selected Answer: A

A is the right answer
upvoted 3 times
- Nadia0012

1 month, 3 weeks ago

Selected Answer: A

For SageMaker to run a container for training or hosting, it needs to be able to find the image hosted in the image repository, Amazon Elastic Container Registry (Amazon ECR). The three main steps to this process are building locally, tagging with the repository location, and pushing the image to the repository.
upvoted 3 times
- geekgirl007

1 year, 3 months ago

Selected Answer: A

A for sure.
upvoted 3 times
- astonm13

1 year, 6 months ago

Answer is A.
upvoted 3 times
- cnethers

1 year, 6 months ago

Docker Hub is a repository so ANS D makes no sense. Option A is the way to go.
upvoted 2 times

A trucking company is collecting live image data from its fleet of trucks across the globe. The data is growing rapidly and approximately 100 GB of new data is generated every day. The company wants to explore machine learning uses cases while ensuring the data is only accessible to specific IAM users.

Which storage option provides the most processing flexibility and will allow access control with IAM?

- A. Use a database, such as Amazon DynamoDB, to store the images, and set the IAM policies to restrict access to only the desired IAM users.
- B. Use an Amazon S3-backed data lake to store the raw images, and set up the permissions using bucket policies.
- C. Setup up Amazon EMR with Hadoop Distributed File System (HDFS) to store the files, and restrict access to the EMR instances using IAM policies.
- D. Configure Amazon EFS with IAM policies to make the data available to Amazon EC2 instances owned by the IAM users.

Correct Answer: C

Community vote distribution

B (76%)

C (24%)

- Paul_NoName

Highly Voted

1 year, 6 months ago

B is the right answer

upvoted 32 times
- clawo

Highly Voted

1 year, 3 months ago

Selected Answer: B

B to use as storage with policies

upvoted 7 times
- Nadia0012

Most Recent

1 month, 3 weeks ago

Selected Answer: B

I would say the answer is B not because of the cost on EMR,, that is also a current answer. however: "most processing flexibility" indicates that S3 is a better option. because all ML solutions and work flows integrate with S3. it hasn't spoken what the ML solution and which services so I take the safe side and go with S3

upvoted 2 times
- KlaudYu

9 months, 2 weeks ago

Selected Answer: B

C is not affordable because it is ephemeral storage. <https://docs.aws.amazon.com/emr/latest/ManagementGuide/emr-plan-file-systems.html> "HDFS is used by the master and core nodes. One advantage is that it's fast; a disadvantage is that it's ephemeral storage which is reclaimed when the cluster ends. It's best used for caching the results produced by intermediate job-flow steps."

upvoted 4 times
- ZSun

5 days, 3 hours ago

the question does not require long-term storage.

upvoted 1 times
- geekgirl007

1 year, 3 months ago

Selected Answer: C

C is correct. it says real time data and to be used for ml process so EMR more suitable. also S3 bucket policies not same as IAM users so B is not correct.

upvoted 4 times
- ovokpus

10 months, 1 week ago

Why will you need to spin up servers (EMR) just to store visual data for ML?

upvoted 5 times
- Abdo702

1 year, 5 months ago

I think Amazon EMR is more appropriate, as the data scheme stated is a big data scheme.

<https://aws.amazon.com/emr/?whats-new-cards.sort-by=item.additionalFields.postDateTime&whats-new-cards.sort-order=desc>

upvoted 3 times
- Sourabh1703

1 year, 3 months ago

IAM support is required for storage feature , that is not possible as per options described as IAM is supported for HDFS for the instance running on top of it, hence B should be correct

upvoted 1 times

  **Vita_Rasta84444** 1 year, 6 months ago

B is the right answer
upvoted 2 times

  **srinu3054** 1 year, 6 months ago

S3 is the easy, scalable and secure option to store the image data.
upvoted 1 times

  **astonm13** 1 year, 6 months ago

B is the right answer
upvoted 1 times

  **zzaibis** 1 year, 6 months ago

B is an appropriate choice
upvoted 2 times

A credit card company wants to build a credit scoring model to help predict whether a new credit card applicant will default on a credit card payment. The company has collected data from a large number of sources with thousands of raw attributes. Early experiments to train a classification model revealed that many attributes are highly correlated, the large number of features slows down the training speed significantly, and that there are some overfitting issues.

The Data Scientist on this project would like to speed up the model training time without losing a lot of information from the original dataset. Which feature engineering technique should the Data Scientist use to meet the objectives?

- A. Run self-correlation on all features and remove highly correlated features
- B. Normalize all numerical values to be between 0 and 1
- C. Use an autoencoder or principal component analysis (PCA) to replace original features with new features
- D. Cluster raw data using k-means and use sample data from each cluster to build a new dataset

Correct Answer: B

Community vote distribution

C (83%)

A (17%)

  **ahquiceno**

Highly Voted

 1 year, 7 months ago

Answer C. Need reduce the features preserving the information on it this is achieve using PCA.

upvoted 25 times

  **VinceCar** 5 months, 2 weeks ago

PCA helps to speed up the training

upvoted 4 times

  **Dr_Kiko** 1 year, 5 months ago

without losing a lot of information from the original dataset since when PCA retains information?

upvoted 3 times

  **joep21**

Highly Voted

 1 year, 6 months ago

Answer is A, because one must avoid information loss that PCA or autoencoders introduce through new features (<https://www.i2tutorials.com/what-are-the-pros-and-cons-of-the-pca/>). Otherwise, I would perform C.

upvoted 5 times

  **rodrigus** 1 month, 2 weeks ago

A doesn't have sense. Self-correlation is for times series data, not for pair correlation

upvoted 1 times

  **SophieSu** 1 year, 6 months ago

If you REMOVE highly correlated features(that means in pairs), the model lost a lot of information.

upvoted 4 times

  **oso0348**

Most Recent

 3 weeks, 5 days ago

Selected Answer: C

Using an autoencoder or PCA can help reduce the dimensionality of the dataset by creating new features that capture the most important information in the original dataset while discarding some of the noise and highly correlated features. This can help speed up the training time and reduce overfitting issues without losing a lot of information from the original dataset. Option A may remove too many features and may not capture all the important information in the dataset, while option B only rescales the data and does not address the issue of highly correlated features. Option D is not a feature engineering technique and may not be an effective way to reduce the dimensionality of the dataset.

upvoted 1 times

  **Paolo991** 1 month, 1 week ago

Selected Answer: C

PCA builds new features starting from high correlated ones. So it matches the question

upvoted 1 times

  **Sneep** 3 months, 3 weeks ago

It's C.

The Data Scientist should use principal component analysis (PCA) to replace the original features with new features. PCA is a technique that reduces the dimensionality of a dataset by projecting it onto a lower-dimensional space, while preserving as much of the original variation as possible. This can help to speed up the training time of the model and reduce overfitting issues, without losing a significant amount of information from the original dataset.

upvoted 1 times

  **Aninina** 3 months, 4 weeks ago

Selected Answer: C

C: PCA is the solution

upvoted 1 times

  **ovokpus** 10 months, 1 week ago

Selected Answer: C

Correction to C. Removing correlated features from hundreds of columns will be tedious and time consuming. PCA is the way to go here.

Apologies for the flip

upvoted 2 times

  **ovokpus** 10 months, 1 week ago

Selected Answer: A

Answer is A. Eliminate features that are highly correlated. This will not compromise the quality of the feature space as much as PCA would.

upvoted 1 times

  **Vita_Rasta84444** 1 year, 5 months ago

Answer is C

upvoted 1 times

  **SophieSu** 1 year, 6 months ago

C is the correct answer. PCA reduces the dimensionality, solves the overfitting, at the mean time does not cause information loss.

upvoted 2 times

  **astonm13** 1 year, 6 months ago

Answer is C

upvoted 2 times

  **wolf90** 1 year, 6 months ago

I think it should be C?

upvoted 2 times

  **Paul_NoName** 1 year, 6 months ago

C seems to be correct.

upvoted 3 times

  **takahirokoyama** 1 year, 7 months ago

Ans. is C.

upvoted 2 times

A Data Scientist is training a multilayer perception (MLP) on a dataset with multiple classes. The target class of interest is unique compared to the other classes within the dataset, but it does not achieve an acceptable recall metric. The Data Scientist has already tried varying the number and size of the MLP's hidden layers, which has not significantly improved the results. A solution to improve recall must be implemented as quickly as possible.

Which techniques should be used to meet these requirements?

- A. Gather more data using Amazon Mechanical Turk and then retrain
- B. Train an anomaly detection model instead of an MLP
- C. Train an XGBoost model instead of an MLP
- D. Add class weights to the MLP's loss function and then retrain

Correct Answer: C

Community vote distribution

D (100%)

- joep21

Highly Voted

1 year, 7 months ago

For me answer is D, adjust to higher weight for class of interest: <https://androidkt.com/set-class-weight-for-imbalance-dataset-in-keras/>. More data may/may not be available and a data labeling job will take time.

upvoted 34 times
- Ajose0

Most Recent

2 months, 2 weeks ago

Selected Answer: D

Adding class weights to the MLP's loss function balances the class frequencies in the cost function during training, so the optimization process focuses more on the underrepresented class, improving recall.

upvoted 3 times
- Tomatoteacher

3 months, 2 weeks ago

Selected Answer: D

I have done this before, class weights help with unbalanced data. Only logical solution that would help if not done, XGBoost could be different, but who knows, both NNs and XGBoost have comparable performance. Answer D!

upvoted 3 times
- hamuozi

7 months ago

Selected Answer: D

In this example, it is necessary to improve recall as soon as possible, so instead of creating additional datasets, it is effective to change the weight of each class during learning.

upvoted 4 times
- victorlifan

8 months, 2 weeks ago

C: 'distinct' indicates we can simplify this as a binary classification problem; then, NN is just overkill. plus, retraining a NN is much slower than training an XGboost model

upvoted 2 times
- exam_prep

11 months ago

I feel answer is B. Question says Target is different than the input data which is hint for anomaly detection.

upvoted 1 times
- rhuanca

11 months, 2 weeks ago

I believe is C, because we already made all changes possible in MLP hidden layers and the results have not improved then we must change model so XGBoot seems the best option

upvoted 4 times
- KM226

1 year, 4 months ago

I believe the answer is C because we need to use hyperparameters to improve model performance.

upvoted 2 times
- ksarda11

1 year, 6 months ago

In case of the quickest possible way, D seems fine. For XGBoost, it will take a bit of time to code again

upvoted 4 times
- ahquiceno

1 year, 7 months ago

For me Answer A. Why no other model instead xgBoost, the model need more labeled data to be trained and learn more positive examples.

upvoted 2 times

  **SophieSu** 1 year, 6 months ago

A is incorrect. Even if you hire Amazon Mechanical Turk, you won't have more data. This question is NOT asking about "labeling".
upvoted 2 times

A Machine Learning Specialist works for a credit card processing company and needs to predict which transactions may be fraudulent in near-real time.

Specifically, the Specialist must train a model that returns the probability that a given transaction may fraudulent.

How should the Specialist frame this business problem?

- A. Streaming classification
- B. Binary classification
- C. Multi-category classification
- D. Regression classification

Correct Answer: C

Community vote distribution

B (86%)14%

- ahquiceno

Highly Voted

1 year, 7 months ago

Answer B.

upvoted 26 times
- SophieSu

Highly Voted

1 year, 7 months ago

B IS NOT CORRECT! Return the probability. Not the 1 or 0. D IS THE CORRECT ANSWER.

upvoted 13 times
- seanLu

1 year, 7 months ago

Logistic regression will give the probability, and logistic regression is a binary classification algorithm.
<https://machinelearningmastery.com/logistic-regression-for-machine-learning/>

upvoted 8 times
- srinu3054

1 year, 7 months ago

there is nothing like regression classification. (instead it should have said logistic regression). It should be Binary. i.e., either fraud or non fraud. Even with probabilities, we have a threshold to decide the class.

upvoted 11 times
- rodrigus

Most Recent

1 month, 2 weeks ago

The correct solution obviously is binary classification. For the comment above that says that binary classification doesn't returns a probability (for example SVM(classification) only returns a class and logistic, RFClassifier, XGBoostClassifier gives a probability and also a class given a threshold), you should ask yourself if that a regressor model returns always a probability, that is, if there is a restriction in a regressor model to predict values only in [0,1].

upvoted 1 times
- Ajose0

2 months, 2 weeks ago

Selected Answer: B

The Specialist is trying to determine whether a given transaction is fraudulent or not, which is a binary outcome (yes or no). Therefore, the problem should be framed as binary classification.

The goal is to predict the probability of a transaction being fraudulent or not, and based on that, the Specialist can make a binary decision (fraudulent or not).

upvoted 1 times
- Tomatoteacher

3 months, 2 weeks ago

Selected Answer: B

This is just binary classification, I don't understand how it could be anything else

upvoted 2 times
- Sneep

3 months, 3 weeks ago

It's B.

This business problem can be framed as a binary classification problem, where the goal is to predict whether a given transaction is fraudulent (positive class) or not fraudulent (negative class). The model should output a probability for each transaction, indicating the likelihood that it is fraudulent.

upvoted 2 times
- DS2021

4 months ago

Selected Answer: D

should be D

upvoted 1 times

  **RLai** 4 months, 2 weeks ago

Logistic regression models the probability of the default class (e.g. the first class).

For example, if we are modeling people’s sex as male or female from their height, then the first class could be male and the logistic regression model could be written as the probability of male given a person’s height.

upvoted 1 times

  **theprismdata** 11 months, 3 weeks ago

I think the answer is B, fraud has various cases which hard to define.
So, Classification result will be fraud or not fraud.

If Multi-category classification, must define case of fraud in detaily

upvoted 1 times

  **theprismdata** 11 months, 3 weeks ago

More specifically, anomaly detection model will be needed

upvoted 1 times

  **KM226** 1 year, 4 months ago

Selected Answer: B

I believe the answer is B: it a binary classification problem because we are classifying an observation into one of two categories and the target variable in this problem is limited to two options: fraudulent or not fraudulent

upvoted 3 times

  **lesh3000** 1 year, 5 months ago

well, regression classification is bullshit, I hope they formulate their questions better on the real exam. binary classification gives probability between 0 and 1

upvoted 3 times

  **Huy** 1 year, 6 months ago

D for me. I think they want to talk about Logistic Regression, which is used as classification.

upvoted 2 times

  **AjithkumarSL** 1 year, 6 months ago

Looks like Answer is D, The model need to predict the probability and Regression can do that. If the second sentence of the question was not there, then binary makes sense, It would have an Yes or No Thing.

upvoted 2 times

  **MrCarter** 1 year, 6 months ago

Should D say Logistic Regression? That would make it the best choice for sure

upvoted 2 times

  **StelSen** 1 year, 6 months ago

Answer is B: Understand the inputs and outputs for Classification and Regression from here: <https://machinelearningmastery.com/classification-versus-regression-in-machine-learning/#:~:text=Fundamentally%2C%20classification%20is%20about%20predicting,is%20about%20predicting%20a%20quantity.&text=That%20classification%20is%20the%20problem,quantity%20output%20for%20an%20example.>

upvoted 1 times

  **cnethers** 1 year, 7 months ago

this is a classic Binary Classification so B is the ANS

upvoted 8 times

  **Paul_NoName** 1 year, 7 months ago

D seems to be right.

upvoted 2 times

Question #83

Topic 1

A real estate company wants to create a machine learning model for predicting housing prices based on a historical dataset. The dataset contains 32 features.

Which model will meet the business requirement?

A. Logistic regression

B. Linear regression

C. K-means

D. Principal component analysis (PCA)

Correct Answer: B

Community vote distribution

B (100%)

SophieSu

Highly Voted

1 year, 6 months ago

B is the correct answer.

upvoted 19 times

jonsnow777

Most Recent

1 year, 3 months ago

Selected Answer: B

Answer B

upvoted 2 times

ahquiceno

1 year, 7 months ago

Answer B.

upvoted 4 times

A Machine Learning Specialist is applying a linear least squares regression model to a dataset with 1,000 records and 50 features. Prior to training, the ML Specialist notices that two features are perfectly linearly dependent. Why could this be an issue for the linear least squares regression model?

- A. It could cause the backpropagation algorithm to fail during training
- B. It could create a singular matrix during optimization, which fails to define a unique solution
- C. It could modify the loss function during optimization, causing it to fail during training
- D. It could introduce non-linear dependencies within the data, which could invalidate the linear assumptions of the model

Correct Answer: C

Community vote distribution

B (73%)

D (27%)

Paul_NoName

Highly Voted

1 year, 6 months ago

B is correct answer .
upvoted 20 times

hamimelon

4 months ago

Agree, B.
upvoted 2 times

pravv

1 year, 6 months ago

why B is the correct answer and not C?
upvoted 1 times

hamimelon

4 months ago

For example. If you have two variables, X and Y, and you have two data points. You want to solve the problem: $aX_1 + bY_1 = Z_1$, $aX_2 + bY_2 = Z_2$. However, if $Y = 2X \rightarrow Y_1 = 2X_1, Y_2 = 2X_2$, then problem becomes: $aX_1 + bY_1 = Z_1$, $a \cdot 2X_1 + b \cdot 2Y_1 = Z_2 = 2 \cdot Z_1$. So you end up with only one function: $aX_1 + bY_1 = Z_1$, meaning there will be more than one answer for (a, b).

If you are familiar with linear algebra, it's easier to express the concept.
upvoted 3 times

SophieSu

1 year, 6 months ago

A square matrix is singular, that is, its determinant is zero, if it contains rows or columns which are proportionally interrelated; in other words, one or more of its rows (columns) is exactly expressible as a linear combination of all or some other its rows (columns), the combination being without a constant term.
upvoted 5 times

Ajose0

Most Recent

2 months, 2 weeks ago

Selected Answer: B

The presence of linearly dependent features means that they are redundant, and provide no additional information to the model.

This can result in a matrix that is not invertible, which is a requirement for solving a linear least squares regression problem. The presence of a singular matrix can also cause numerical instability and make it impossible to find an optimal solution to the optimization problem.
upvoted 1 times

Sneep

3 months, 3 weeks ago

B: If two features in the dataset are perfectly linearly dependent, it means that one feature can be expressed as a linear combination of the other. This can create a singular matrix during optimization, as the linear model would be trying to fit a linear equation to a dataset where one variable is fully determined by the other. This would lead to an ill-defined optimization problem, as there would be no unique solution that minimizes the sum of the squares of the residuals. This could lead to problems during training, as the model would not be able to find appropriate parameter values to fit the data.
upvoted 4 times

yemauricio

4 months, 3 weeks ago

Selected Answer: B

linera dependence creates singular matrix that causes problems at the moment we fit the modle
upvoted 4 times

wisoxe8356

4 months, 4 weeks ago

Selected Answer: B

<https://towardsdatascience.com/multi-collinearity-in-regression-fe7a2c1467ea>
B - two features are perfectly linearly dependent = singular matrix during optimization
Not D - Not100% correct (as Multicollinearity happens when independent variables in the regression model are highly correlated to each other)
they can still be independent variables
upvoted 3 times

  **ovokpus** 10 months, 1 week ago

Selected Answer: D

Consider one of the 5 assumptions of linear regression. This situation violates the assumption of "No multicollinearity between feature variables"

Hence, D
upvoted 3 times

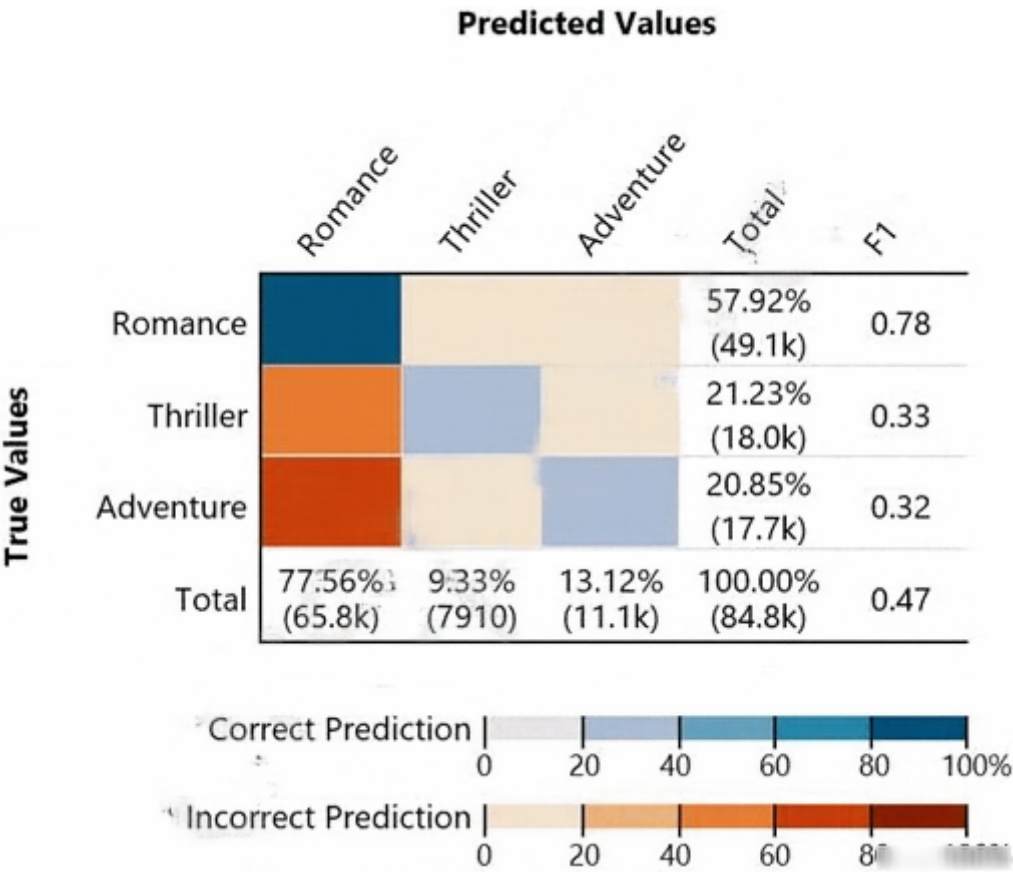
  **jerto97** 1 year, 6 months ago

B. See the multicollinearity problem in wikipedia <https://en.wikipedia.org/wiki/Multicollinearity> (second paragraph)
upvoted 4 times

  **takahirokoyama** 1 year, 6 months ago

This issue is overfitting.
upvoted 1 times

Given the following confusion matrix for a movie classification model, what is the true class frequency for Romance and the predicted class frequency for Adventure?



- A. The true class frequency for Romance is 77.56% and the predicted class frequency for Adventure is 20.85%
- B. The true class frequency for Romance is 57.92% and the predicted class frequency for Adventure is 13.12%
- C. The true class frequency for Romance is 0.78 and the predicted class frequency for Adventure is (0.47-0.32)
- D. The true class frequency for Romance is $77.56\% - 0.78$ and the predicted class frequency for Adventure is $20.85\% - 0.32$

Correct Answer: B

Community vote distribution

B (100%)

- SophieSu

Highly Voted

1 year, 6 months ago

B is the correct answer. Straightforward!

upvoted 17 times
- cnethers

Highly Voted

1 year, 6 months ago

<https://docs.aws.amazon.com/machine-learning/latest/dg/multiclass-model-insights.html>

upvoted 9 times
- Shailendraa

Most Recent

7 months, 3 weeks ago

12-sep exam

upvoted 1 times
- milan_ml

9 months, 1 week ago

Selected Answer: B

The image can be found here:
<https://vceguide.com/what-is-the-true-class-frequency-for-romance-and-the-predicted-class-frequency-for-adventure/>

upvoted 1 times
- obadazx

9 months, 4 weeks ago

No image is there!

upvoted 3 times
- SDikeman62

11 months, 3 weeks ago

WHy there is no image? Admin. Please fix it.

upvoted 6 times
- Juka3lj

1 year, 5 months ago

B is correct

upvoted 1 times

 **NotAnMLProfessional** 1 year, 6 months ago

A seems to be correct

upvoted 3 times

A Machine Learning Specialist wants to bring a custom algorithm to Amazon SageMaker. The Specialist implements the algorithm in a Docker container supported by Amazon SageMaker.

How should the Specialist package the Docker container so that Amazon SageMaker can launch the training correctly?

- A. Modify the bash_profile file in the container and add a bash command to start the training program
- B. Use CMD config in the Dockerfile to add the training program as a CMD of the image
- C. Configure the training program as an ENTRYPOINT named train
- D. Copy the training program to directory /opt/ml/train

Correct Answer: B

Community vote distribution

C (100%)

- 👤

Paul_NoName

Highly Voted 🍌

1 year, 7 months ago

C seems correct as per documentations.
upvoted 17 times
- 👤

joep21

Highly Voted 🍌

1 year, 7 months ago

I would answer C: <https://docs.aws.amazon.com/sagemaker/latest/dg/your-algorithms-training-algo-dockerfile.html>

"To configure a Docker container to run as an executable, use an ENTRYPOINT instruction in a Dockerfile. SageMaker overrides any default CMD statement in a container by specifying the train argument after the image name"
upvoted 14 times
- 👤

oso0348

Most Recent 🕒

3 weeks, 5 days ago

Selected Answer: C

To package a Docker container for use with Amazon SageMaker, the training program should be configured as an ENTRYPOINT named train in the Dockerfile. This means that the training program will be automatically executed when the container is launched by Amazon SageMaker, and it can be passed command-line arguments to specify hyperparameters or other training settings.
upvoted 1 times

👤

Ajose0

2 months, 2 weeks ago

Selected Answer: C

The recommended option to package the Docker container for Amazon SageMaker is to configure the training program as an ENTRYPOINT named train.

This is because ENTRYPOINT allows you to specify a command that will always be executed when the Docker container is run, ensuring that the training program will always run when the container is launched by Amazon SageMaker.

Additionally, naming the ENTRYPOINT "train" is a convention used by Amazon SageMaker to identify the main training script.
upvoted 1 times

👤

tsangckl

5 months ago

Selected Answer: C

It's C
upvoted 1 times

👤

gnolam

7 months, 3 weeks ago

C for sure
as per AWS docs:
> In your Dockerfile, use the exec form of the ENTRYPOINT instruction:
> ENTRYPOINT ["python", "k-means-algorithm.py"]
upvoted 1 times

👤

Juka3lj

1 year, 7 months ago

C is correct
upvoted 1 times

👤

Aashi22

1 year, 7 months ago

option C https://github.com/awsdocs/amazon-sagemaker-developer-guide/blob/master/doc_source/your-algorithms-training-algo-dockerfile.md
upvoted 1 times

<https://www.examtopycs.com/exams/amazon/aws-certified-machine-learning-specialty/custom-view/>

174/406

A Data Scientist needs to analyze employment data. The dataset contains approximately 10 million observations on people across 10 different features. During the preliminary analysis, the Data Scientist notices that income and age distributions are not normal. While income levels shows a right skew as expected, with fewer individuals having a higher income, the age distribution also shows a right skew, with fewer older individuals participating in the workforce.

Which feature transformations can the Data Scientist apply to fix the incorrectly skewed data? (Choose two.)

- A. Cross-validation
- B. Numerical value binning
- C. High-degree polynomial transformation
- D. Logarithmic transformation
- E. One hot encoding

Correct Answer: AB

Community vote distribution

BD (100%)

- seanLu

Highly Voted

1 year, 6 months ago

I would go with B,D. Refer to quantile binning and log transform below.
<https://towardsdatascience.com/understanding-feature-engineering-part-1-continuous-numeric-data-da4e47099a7b>
upvoted 29 times
- omar_bahrain

1 year, 6 months ago

agree B&D. both are strategies to eliminate the effect of skewing
upvoted 6 times
- OmarSaadEldien

1 year, 5 months ago

Agree with B &D
B binning for age
D for make income in normal dist
upvoted 7 times
- Joe_Zhang

Highly Voted

1 year, 7 months ago

SHOULD BE C,D
upvoted 10 times
- Shailendraa

Most Recent

7 months, 3 weeks ago

BD is correct
upvoted 1 times
- [Removed]

10 months, 3 weeks ago

A and E, it asks incorrectly
upvoted 1 times
- Sivadharan

11 months, 2 weeks ago

Selected Answer: BD

B & D. Reasonable explanation in below discussion.
upvoted 3 times
- angnam

1 year, 3 months ago

BD
With age, always do quantile binning
With skewed data, always use log.
upvoted 1 times
- Juka3lj

1 year, 6 months ago

B because we have skewed data with few exeptions
D log transform can change distribution of data
not C - because there is no indicaiton in the text, that data is following any of the HIGH DEGREE polynomial distribution like x^{10}
upvoted 5 times
- Vita_Rasta84444

1 year, 6 months ago

should be c and d
upvoted 4 times

-   **achiko** 1 year, 6 months ago
polynomial transformations can also be used for skewed data. <https://machinelearningmastery.com/polynomial-features-transforms-for-machine-learning/>
upvoted 3 times
-   **jiadong** 1 year, 6 months ago
It seems the ans are C,D
<https://anshikaaxena.medium.com/how-skewed-data-can-skrew-your-linear-regression-model-accuracy-and-transfromation-can-help-62c6d3fe4c53>
upvoted 5 times

A web-based company wants to improve its conversion rate on its landing page. Using a large historical dataset of customer visits, the company has repeatedly trained a multi-class deep learning network algorithm on Amazon SageMaker. However, there is an overfitting problem: training data shows 90% accuracy in predictions, while test data shows 70% accuracy only.

The company needs to boost the generalization of its model before deploying it into production to maximize conversions of visits to purchases. Which action is recommended to provide the HIGHEST accuracy model for the company's test and validation data?

- A. Increase the randomization of training data in the mini-batches used in training
- B. Allocate a higher proportion of the overall data to the training dataset
- C. Apply L1 or L2 regularization and dropouts to the training
- D. Reduce the number of layers and units (or neurons) from the deep learning network

Correct Answer: D

Community vote distribution

C (77%)

D (23%)

  **knightknt** Highly Voted 1 year ago

I think C will be answer, because we even don't know how many layers now, so apply L1,L2 and dropouts layer will be first resort to solve overfitting. If it still does not work, then to reduce layers

upvoted 10 times

  **Khalil11** Most Recent 3 weeks, 6 days ago

Selected Answer: C

C is the correct answer: The overfitting problem can be addressed by applying regularization techniques such as L1 or L2 regularization and dropouts. Regularization techniques add a penalty term to the cost function of the model, which helps to reduce the complexity of the model and prevent it from overfitting to the training data. Dropouts randomly turn off some of the neurons during training, which also helps to prevent overfitting.

upvoted 2 times

  **Valcilio** 1 month, 3 weeks ago

Selected Answer: C

D can work, but C is a better answer!

upvoted 1 times

  **drcok87** 2 months, 1 week ago

C and D both seems to be correct but, seems like removing layer is first step in to optimization

<https://www.kaggle.com/general/175912>

d

upvoted 1 times

  **Ajose0** 2 months, 2 weeks ago

Selected Answer: C

C. Apply L1 or L2 regularization and dropouts to the training" because regularization can help reduce overfitting by adding a penalty to the loss function for large weights, preventing the model from memorizing the training data.

Dropout is a regularization technique that randomly drops out neurons during the training process, further reducing the risk of overfitting.

upvoted 1 times

  **albu44** 3 months, 3 weeks ago

Selected Answer: D

"The first step when dealing with overfitting is to decrease the complexity of the model. To decrease the complexity, we can simply remove layers or reduce the number of neurons to make the network smaller."

<https://www.kdnuggets.com/2019/12/5-techniques-prevent-overfitting-neural-networks.html>

upvoted 1 times

  **Peeking** 4 months, 3 weeks ago

Selected Answer: D

Deep learning tuning order:

1. Number of layers
2. Number of neurons (indirectly implements dropout)
3. L1/L2 regularization
4. Dropout

upvoted 4 times

- Parth12

9 months, 2 weeks ago

Selected Answer: C

Here we are looking to reduce the Overfitting to improve the generalization. In order to do so, L1(or Lasso) regression has always been a good aide.

upvoted 3 times
- mtp1993

10 months, 1 week ago

Selected Answer: C

C, Regularization and dropouts should be the first attempt

upvoted 3 times
- ovokpus

10 months, 1 week ago

Selected Answer: C

Yes, C is right here. Regularization and Dropouts

upvoted 3 times
- Abdelrahman_Omran

1 year ago

Selected Answer: C

C is the answer

upvoted 4 times

A Machine Learning Specialist is given a structured dataset on the shopping habits of a company's customer base. The dataset contains thousands of columns of data and hundreds of numerical columns for each customer. The Specialist wants to identify whether there are natural groupings for these columns across all customers and visualize the results as quickly as possible.

What approach should the Specialist take to accomplish these tasks?

- A. Embed the numerical features using the t-distributed stochastic neighbor embedding (t-SNE) algorithm and create a scatter plot.
- B. Run k-means using the Euclidean distance measure for different values of k and create an elbow plot.
- C. Embed the numerical features using the t-distributed stochastic neighbor embedding (t-SNE) algorithm and create a line graph.
- D. Run k-means using the Euclidean distance measure for different values of k and create box plots for each numerical column within each cluster.

Correct Answer: B

Community vote distribution

A (91%)

9%

- ac71

Highly Voted

1 year, 7 months ago

A is correct. tSNE can do segmentation or grouping as well. Refer: <https://towardsdatascience.com/an-introduction-to-t-sne-with-python-example-5a3a293108d1>

upvoted 21 times
- SophieSu

Highly Voted

1 year, 6 months ago

A is definitely the correct answer.

Pay attention to what the question is asking:

"whether there are natural groupings for these columns across all customers and visualize the results as quickly as possible"

The key point is to visualize the "groupings"(exactly what t-SNE scatter plot does, it visualize high-dimensional data points on 2D space).

The question does not ask to visualize how many groups you would classify (K-Means Elbow Plot does not visualize the groupings, it is used to determine the optimal # of groups=K).

upvoted 16 times
- Sylzys

Most Recent

1 month, 2 weeks ago

Selected Answer: A

Elbow plot will not help visualize groups, only try to predict an optimal number of clusters.

I think A is a better choice here

upvoted 2 times
- AjoseO

2 months, 2 weeks ago

Selected Answer: A

A.

The t-SNE algorithm is a popular tool for visualizing high-dimensional datasets, as it can transform high-dimensional data into a 2D scatter plot, which makes it easier to visualize and understand the relationships between data points.

The scatter plot produced by t-SNE can be interpreted as a map that reveals the structure of the data, showing whether there are natural groupings or clusters within the data.

Option A is the quickest and simplest way to visualize the data in a meaningful way, allowing the Specialist to gain insights into the data more efficiently.

upvoted 2 times
- minkhant19

5 months, 1 week ago

A is correct

upvoted 1 times
- Shailendraa

7 months, 3 weeks ago

12-sep exam

upvoted 3 times
- Morsa

9 months, 2 weeks ago

Selected Answer: A

A as k-means elbow is erroneous. It does not helping here. Scatter plot and t-sne is the right answer

upvoted 2 times
- ovokpus

10 months ago

Selected Answer: A

An elbow plot (B) will not give you what the question is asking for. A scatter plot will, and t-SNE is first for visualizing before dimensionality reduction.

upvoted 2 times

  **Sadgamaya** 1 year, 1 month ago

A is correct as k means suffer from curse of dimensionality and t-she will be a better option.

upvoted 1 times

  **Mircuz** 1 year, 1 month ago

Selected Answer: A

The B,C,D plots are meaningless wrt the problem —> A

upvoted 2 times

  **Mircuz** 1 year, 1 month ago

Selected Answer: B

t-SNE suffers curse of dimensionality and is indicated for small datasets

upvoted 1 times

  **AddiWei** 1 year, 2 months ago

Additionally the numeric features don't require "embedding". I think they meant to write "standardize"

upvoted 1 times

  **apprehensive_scar** 1 year, 2 months ago

Rooting for A

upvoted 1 times

  **bitsplease** 1 year, 3 months ago

B & D are wrong--because data contains "thousands of columns" and using k-means with euclidean suffers from "curse of dimensionality"

Thus leaving A & C, you CANNOT viz clusters/groups/segments in a line graph so correct answer is A

upvoted 1 times

  **t_k_r** 1 year, 6 months ago

B is correct

<https://towardsdatascience.com/visualising-high-dimensional-datasets-using-pca-and-t-sne-in-python-8ef87e7915b>

According to this For tsne it is recommended to use other dimensional reduction methods to first reduce it to a reasonable amount when number of dimesions is too high

upvoted 1 times

  **orangechickencombo** 1 year, 6 months ago

B.

tSNE need to have grouped labels for visualization. Here looking for whether there is group or not

upvoted 1 times

  **Madwyn** 1 year, 5 months ago

tSNE, like PCA, is unsupervised, no labels are needed. It's used for high dimension data visualisation, which is for human to see the grouping.

So A.

upvoted 2 times

  **Juka3lj** 1 year, 6 months ago

A is correct answer

upvoted 1 times

A Machine Learning Specialist is planning to create a long-running Amazon EMR cluster. The EMR cluster will have 1 master node, 10 core nodes, and 20 task nodes. To save on costs, the Specialist will use Spot Instances in the EMR cluster.

Which nodes should the Specialist launch on Spot Instances?

- A. Master node
- B. Any of the core nodes
- C. Any of the task nodes
- D. Both core and task nodes

Correct Answer: A

Community vote distribution

C (100%)

- joep21

Highly Voted

1 year, 6 months ago

Answer is C. <https://docs.aws.amazon.com/emr/latest/ManagementGuide/emr-plan-instances-guidelines.html>

upvoted 23 times
- Sneep

3 months, 3 weeks ago

It's definitely C. The fact that this site indicates A is a clear sign that answers are just randomly selected, it would make zero sense to spot-instance the master node for an EMR cluster. Make sure you look at discussions for all of these questions.

upvoted 3 times
- SophieSu

Highly Voted

1 year, 6 months ago

C is the correct answer.

"Long-Running Clusters and Data Warehouses

If you are running a persistent Amazon EMR cluster that has a predictable variation in computational capacity, such as a data warehouse, you can handle peak demand at lower cost with Spot Instances. You can launch your master and core instance groups as On-Demand Instances to handle the normal capacity and launch task instance groups as Spot Instances to handle your peak load requirements."

upvoted 9 times
- Khalil11

Most Recent

3 weeks, 6 days ago

Selected Answer: C

The correct answer is C

upvoted 1 times
- Sylzys

1 month, 2 weeks ago

Selected Answer: C

I don't get why the wrong answer are still not updated after more than 1 year of everyone showing docs proving answer C..

upvoted 1 times
- Ajose0

2 months, 2 weeks ago

Selected Answer: C

Long-running clusters and data warehouses

If you are running a persistent Amazon EMR cluster that has a predictable variation in computational capacity, such as a data warehouse, you can handle peak demand at lower cost with Spot Instances.

You can launch your primary and core instance groups as On-Demand Instances to handle the normal capacity and launch the task instance group as Spot Instances to handle your peak load requirements.

upvoted 1 times
- SK27

4 months, 3 weeks ago

Selected Answer: C

Only task nodes can be deleted without losing data.

upvoted 1 times
- Twist3d

4 months, 3 weeks ago

C, If you want to cut cost on an EMR cluster in the most efficient way, use spot instances on the task nodes because it, task nodes do not store data so no risk of data loss

upvoted 1 times
- ovokpus

10 months, 1 week ago

Selected Answer: C

For Long running jobs, you do not want to compromise the Master node(sudden termination) or the core nodes (HDFS data loss).

Spot Instances on 20 task nodes are enough cost savings without compromising the job.

Hence, C
upvoted 3 times

  **Jump09** 10 months, 1 week ago

If your primary concern is the cost, then you can run the master node on spot instances.
upvoted 1 times

  **Jump09** 10 months, 1 week ago

Adding the related reference from the AWS documentation:
Master node on a Spot Instance
The master node controls and directs the cluster. When it terminates, the cluster ends, so you should only launch the master node as a Spot Instance if you are running a cluster where sudden termination is acceptable. This might be the case if you are testing a new application, have a cluster that periodically persists data to an external store such as Amazon S3, or are running a cluster where cost is more important than ensuring the cluster's completion.
upvoted 1 times

  **Jump09** 10 months, 1 week ago

In the question , there are no specific conditions mentioned except the concern with the COST, thus I think the answer should be A.
upvoted 1 times

  **benson2021** 1 year, 6 months ago

Answer: C. <https://aws.amazon.com/getting-started/hands-on/optimize-amazon-emr-clusters-with-ec2-spot/>
Amazon recommends using On-Demand instances for Master and Core nodes unless you are launching highly ephemeral workloads.
upvoted 5 times

  **xpada001** 1 year, 6 months ago

Answer should be C.
upvoted 3 times

  **ac71** 1 year, 6 months ago

Only master node is incorrect. Either use all on spot or only task or core on spot. As per:
<https://docs.aws.amazon.com/emr/latest/ManagementGuide/emr-plan-instances-guidelines.html>

Better to use only task node on spot for long running tasks/jobs
upvoted 3 times

  **MahEid** 1 year, 5 months ago

Answer is C
you should only run core nodes on Spot Instances /*when partial HDFS data loss is tolerable*/
Question is what "Should" be launched as spot instance
upvoted 1 times

A manufacturer of car engines collects data from cars as they are being driven. The data collected includes timestamp, engine temperature, rotations per minute (RPM), and other sensor readings. The company wants to predict when an engine is going to have a problem, so it can notify drivers in advance to get engine maintenance. The engine data is loaded into a data lake for training. Which is the MOST suitable predictive model that can be deployed into production?

- A. Add labels over time to indicate which engine faults occur at what time in the future to turn this into a supervised learning problem. Use a recurrent neural network (RNN) to train the model to recognize when an engine might need maintenance for a certain fault.
- B. This data requires an unsupervised learning algorithm. Use Amazon SageMaker k-means to cluster the data.
- C. Add labels over time to indicate which engine faults occur at what time in the future to turn this into a supervised learning problem. Use a convolutional neural network (CNN) to train the model to recognize when an engine might need maintenance for a certain fault.
- D. This data is already formulated as a time series. Use Amazon SageMaker seq2seq to model the time series.

Correct Answer: B

Community vote distribution

A (100%)

- ac71

Highly Voted

1 year, 7 months ago

This is a supervised problem and needs labels. Can't use clustering to find when faults can happen. CNN is for images not for timeseries data here. Hence, A seems appropriate.

upvoted 45 times
- astonm13

1 year, 6 months ago

Agree, the answer is A

upvoted 7 times
- youjun

1 year, 4 months ago

AGREE WITH YOU

upvoted 2 times
- Ajose0

Most Recent

2 months, 2 weeks ago

Selected Answer: A

A recurrent neural network (RNN) is a more suitable choice than a convolutional neural network (CNN) because the data collected from the engines is a sequence of values over time, and the goal is to predict a future event (an engine fault). RNNs are designed to handle sequential data and can learn patterns and dependencies over time, making them well-suited for time-series data like this.

On the other hand, CNNs are designed for image processing and are not ideal for sequential data.

upvoted 3 times
- spidy20

8 months, 2 weeks ago

Selected Answer: A

Answer should be A

upvoted 1 times
- Morsa

9 months, 3 weeks ago

Selected Answer: A

It can only be A. Agree with the comments before

upvoted 1 times
- irimala

11 months, 1 week ago

Selected Answer: A

Obviously A

upvoted 2 times
- apprehensive_scar

1 year, 2 months ago

Selected Answer: A

A - obviously.

upvoted 1 times
- bitsplease

1 year, 3 months ago

Seq2Seq also uses RNN under the hood, BUT option D. did not mention anything about "adding labels"--which is required here--hence --> A

upvoted 3 times

  **geekgirl007** 1 year, 3 months ago

Selected Answer: A

A is correct. CNN is for images and RNN is for timeseries.
upvoted 1 times

  **loyor94478** 1 year, 5 months ago

AAAAAAAAAAAAA
<https://towardsdatascience.com/how-to-implement-machine-learning-for-predictive-maintenance-4633cdbe4860>
upvoted 2 times

  **omar8024** 1 year, 6 months ago

I think A is correct
upvoted 1 times

  **Vita_Rasta84444** 1 year, 6 months ago

It is A
upvoted 1 times

A company wants to predict the sale prices of houses based on available historical sales data. The target variable in the company's dataset is the sale price. The features include parameters such as the lot size, living area measurements, non-living area measurements, number of bedrooms, number of bathrooms, year built, and postal code. The company wants to use multi-variable linear regression to predict house sale prices. Which step should a machine learning specialist take to remove features that are irrelevant for the analysis and reduce the model's complexity?

- A. Plot a histogram of the features and compute their standard deviation. Remove features with high variance.
- B. Plot a histogram of the features and compute their standard deviation. Remove features with low variance.
- C. Build a heatmap showing the correlation of the dataset against itself. Remove features with low mutual correlation scores.
- D. Run a correlation check of all features against the target variable. Remove features with low target variable correlation scores.

Correct Answer: D

Community vote distribution

D (73%)

B (27%)

  **puffpuff** Highly Voted 1 year, 6 months ago

D should be the more comprehensive answer. If it's not correlated, you can't make use of it in a linear regression
A lot of others say B, but low variance can also be due to the nature/typical magnitudes of the variable itself
upvoted 28 times

  **hamimelon** 4 months, 1 week ago

I think the problem with B is that what is considered "low variance"? The features are on different scales.
upvoted 1 times

  **V_B_** 8 months, 3 weeks ago

Correlation indicates only linear relation, but, there might be non linear as well. To exploit it in the Linear Regression, you can take the variables to some power or run some non linear preprocessing on it, and you don't have to change the algorithm for it.
So, answer B seem much more solid for me.
upvoted 1 times

  **ahquiceno** Highly Voted 1 year, 7 months ago

Answer B. Is not the best solucion prior can use other analysis. <https://community.dataquest.io/t/feature-selection-features-with-low-variance/2418>
If the variance is low or close to zero, then a feature is approximately constant and will not improve the performance of the model. In that case, it should be removed. Or if only a handful of observations differ from a constant value, the variance will also be very low.
upvoted 16 times

  **fshkkento** 1 year, 5 months ago

Low variance does not mean the feature is not important, right?
If variance of target true value is also small and the correlation between above feature and target, the feature can be important feature.
upvoted 7 times

  **rb39** 7 months, 3 weeks ago

it does. If feature and target are correlated and you expect the target to change, the feature must have some sort of variance. Otherwise it means feature is almost constant so does target.
upvoted 1 times

  **HunterZ9527** Most Recent 4 days, 23 hours ago

Selected Answer: D

I think D is the correct answer. If I remember correctly, Benjamini-Hochberg Method is essentially answer D if you consider the Hypothesis to be: the feature is powerfully influential to the target.
My problem with B is that the variance can be easily affected by the scale. In the question, the number of bedroom's variance is very low, while the sqrt of the house has a high variance, both of these could be very useful. Furthermore, zip codes are included, and it is safe to assume the variance of zip codes can be high, but the information is very limited, especially if you use them as numerical instead of categorical features.
upvoted 1 times

  **Valcilio** 1 month, 3 weeks ago

Selected Answer: D

B is correct but the answer in D is better.
upvoted 2 times

  **Ajose0** 2 months, 2 weeks ago

Selected Answer: D

D is preferred over C because the goal is to predict the sale price of houses, which is the target variable. By checking the correlation of each feature against the target variable, the machine learning specialist can identify which features are most relevant to the prediction of the sale price and which are less relevant. Removing features with low correlation to the target variable helps reduce the complexity of the model and potentially

improve its accuracy.

On the other hand, a heatmap showing the correlation of the dataset against itself (C) doesn't directly address the relevance of the features to the target variable, and so it's not as effective in reducing the complexity of the model.

upvoted 3 times

  **expertguru** 3 months, 3 weeks ago

Answer should be D, THIS is feature elimination /selection during feature Engineering. Choice c is so close just to confuse test takers to pick the wrong choice! See below C and D answers -- C should have been correct if the question asked about how to visualize correlation among independent variables! PROVIDED second sentence in C needs to be removed or to say which feature you will eliminate in such case then the one with low correlation against target out of those two.

- C. Build a heatmap showing the correlation of the dataset against itself. Remove features with low mutual correlation scores.
- D. Run a correlation check of all features against the target variable. Remove features with low target variable correlation scores.

upvoted 1 times

  **Ob1KN0B** 8 months, 2 weeks ago

Selected Answer: D

The multiple regression model is based on the following assumptions:

There is a linear relationship between the dependent variables and the independent variables
The independent variables are not too highly correlated with each other
yi observations are selected independently and randomly from the population
Residuals should be normally distributed with a mean of 0 and variance σ

upvoted 5 times

  **wakuwaku** 1 year, 2 months ago

I think the answer is D.
If the model is a decision tree or something like that, I don't think it is possible to make a decision based only on the direct correlation with the target variable.
But in multiple linear regression, the only thing that matters is the relationship between the target variable and the feature variable.
B, if the standard deviation is small but not zero, then we have information.

upvoted 3 times

  **apprehensive_scar** 1 year, 2 months ago

Selected Answer: B

B is correct.
upvoted 1 times

  **Peasfull** 1 year, 3 months ago

To eliminate extraneous information. So, the answer is D.
upvoted 2 times

  **Asrivastava3** 1 year, 4 months ago

Correct answer is D. The reason B is wrong because it is difficult to reason out why would you plot a histogram? Absolutely unnecessary step and distraction choice.
upvoted 4 times

  **passionatecricketer** 1 year, 5 months ago

Selected Answer: B

D is not the proper answer. Here is why:
It says that it is comparing with the target variable (dependent variable), which implies it is comparing the correlation between the dependent and independent variables. This type of comparison is usually done after a model is constructed in order to prevent assessing the predictive strength of the model. To compare the target label, the label you wish to predict, with the other variables before - is premature and will likely result in weakening your model.

Variables with low variance has very less information and the inclusion of which will likely weaken the model performance.

Hence, B.
upvoted 3 times

  **Mikky0** 1 year, 5 months ago

Answer is D.
<https://deep-r.medium.com/difference-between-variance-co-variance-and-correlation-ea0b7ddbba1>
upvoted 5 times

  **Huy** 1 year, 6 months ago

Answer C. Heatmaps is used to visualize for correlation matrix <https://towardsdatascience.com/better-heatmaps-and-correlation-matrix-plots-in-python-41445d0f2bec>
upvoted 1 times

  **mahmoudai** 1 year, 5 months ago

but is mentioned, "Remove features with low mutual correlation scores." which is wrong you should drop features with high correlation scores.
so Answer is D
upvoted 5 times

  **hero67** 1 year, 6 months ago

The problem with correlation tasks is it capture linear relations only. So, I would go with B
upvoted 1 times

  **YJ4219** 1 year, 6 months ago

I think the answer is D, because the correlation between each feature and the target, if this feature has low variance (as B suggests) this correlation calculation will be low and thus will be removed.
In short i think D is a more general and more accurate answer.
upvoted 4 times

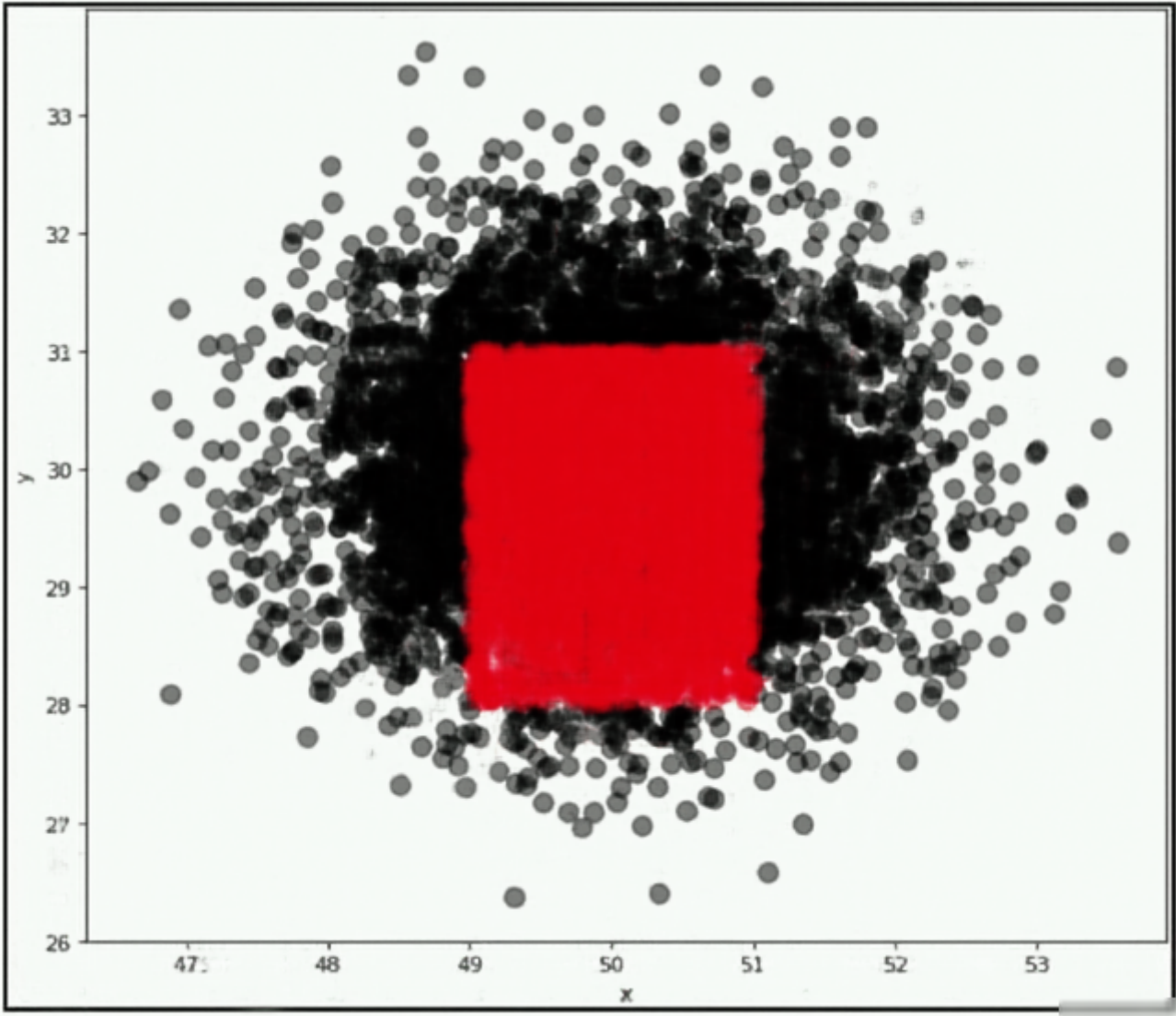
  **Vita_Rasta84444** 1 year, 6 months ago

It is D. We want to exclude the values with low predictive power. It is although more probable that variables with low variance have the lower predictive power, but it is not necessary so.
upvoted 3 times

Question #93

Topic 1

A company wants to classify user behavior as either fraudulent or normal. Based on internal research, a machine learning specialist will build a binary classifier based on two features: age of account, denoted by x , and transaction month, denoted by y . The class distributions are illustrated in the provided figure. The positive class is portrayed in red, while the negative class is portrayed in black.



Which model would have the HIGHEST accuracy?

- A. Linear support vector machine (SVM)
- B. Decision tree
- C. Support vector machine (SVM) with a radial basis function kernel
- D. Single perceptron with a Tanh activation function

Correct Answer: C

Community vote distribution

B (71%) C (29%)

- joep21** Highly Voted 1 year, 7 months ago

Due to straight angles, I would choose Decision tree. See https://scikit-learn.org/stable/auto_examples/classification/plot_classifier_comparison.html#sphx-glr-auto-examples-classification-plot-classifier-comparison-py
upvoted 23 times
- MrCarter** 1 year, 6 months ago

From your link it is obvious that the best answer is still SVM with RBF kernel. In your link the SVM-RBF got 88% accuracy on the 'square-like' dataset whereas the Decision tree achieved only 80%. Answer is SVM with RBF kernel
upvoted 12 times
- ttsun** 1 year, 5 months ago

note the data from sklearn link is shaped as a ball of mass not a square. the RBF kernel would be better but the question shows a square. Decision tree should be better fit for this problem.
upvoted 5 times
- SophieSu** Highly Voted 1 year, 7 months ago

B - Decision tree - is not the best answer. If you use decision tree to do clustering, every time you need to partition the space into 2 parts. Hence you will split the space into 3*3. The red points in the center box and the black points will fall into the 8 boxes around it. The black points will be identified as 8 different classes.

C is the correct answer. SVM with non-linear kernel is appropriate for non-linear clustering. Even if the shape is close to rectangular. SVM with non-linear kernel will be able to approximate the rectangular boundary shape.
upvoted 17 times

  **Madwyn** 1 year, 6 months ago

The tree works like this with this branch with 4 nodes:
Age > 49? Y
Age > 51? N
Transaction > 28? Y
Transaction > 31? N
Positive

Correct answer is B.
upvoted 6 times

  **Ajose0** Most Recent 2 months, 2 weeks ago

Selected Answer: B

A Decision tree produces a stepwise boundary that consists of rectilinear splits in the feature space that are perpendicular to the axes. The boundary is determined by the conditions specified in the tree.

Support Vector Machines (SVM) produce a non-linear boundary in the feature space that separates the classes. The boundary is defined as the maximum margin hyperplane, which is the line that maximally separates the classes while having the greatest margin between the classes. The boundary can be linear, polynomial, radial basis function (RBF) or other types of non-linear functions, depending on the choice of kernel and the configuration of the SVM.
upvoted 3 times

  **CertArvind** 3 months ago

Correct Answer is C
upvoted 1 times

  **ovokpus** 10 months, 1 week ago

Selected Answer: C

SVM with RBF kernel is the best answer here. KNN is also a solution, but it is not one of the options
upvoted 4 times

  **vetaal** 1 year, 3 months ago

Selected Answer: B

This answer is decision tree due to the Square decision boundaries.
upvoted 7 times

  **MrCarter** 1 year, 6 months ago

Answer is definitely C.
upvoted 4 times

  **gcpwhiz** 1 year, 6 months ago

answer is decision tree. Decision tree produces square boundary, SVM with radial function produces a circular boundary.
upvoted 5 times

  **Vita_Rasta84444** 1 year, 6 months ago

It is B. We do not use decision trees for clustering but for prediction. Straight lines provides that decision tree is most precise, while SVM is better with linear cernel.
upvoted 3 times

  **anttan** 1 year, 4 months ago

but isnt SVM with radical function meant for non-linear?
upvoted 1 times

A health care company is planning to use neural networks to classify their X-ray images into normal and abnormal classes. The labeled data is divided into a training set of 1,000 images and a test set of 200 images. The initial training of a neural network model with 50 hidden layers yielded 99% accuracy on the training set, but only 55% accuracy on the test set.

What changes should the Specialist consider to solve this issue? (Choose three.)

- A. Choose a higher number of layers
- B. Choose a lower number of layers
- C. Choose a smaller learning rate
- D. Enable dropout
- E. Include all the images from the test set in the training set
- F. Enable early stopping

Correct Answer: ADE

Community vote distribution

BDF (80%)

BCE (20%)

  **cnethers** Highly Voted 1 year, 6 months ago

when looking at an overfitting issue :
<https://www.kdnuggets.com/2019/12/5-techniques-prevent-overfitting-neural-networks.html>
1. Simplifying The Model (reduce number of layers)
2. Early Stopping
3. Use Data Augmentation
4. Use Regularization (L1 + L2)
5. Use Dropouts

So looking at the options:
B, D, F
upvoted 46 times

  **SophieSu** Highly Voted 1 year, 6 months ago

BDF !!!
upvoted 8 times

  **Aninina** Most Recent 3 months, 4 weeks ago

Selected Answer: BDF
BDF is the correct
upvoted 1 times

  **Peeking** 4 months, 3 weeks ago

Selected Answer: BDF
ADE is absolutely wrong. 50 layers is already overfitting the model. We cannot increase the number of layers again.
upvoted 1 times

  **GauravLahotiML** 5 months, 2 weeks ago

Selected Answer: BDF
BDF is the correct answer
upvoted 1 times

  **exam887** 11 months ago

Selected Answer: BDF
should be BDF
upvoted 1 times

  **NILKK** 1 year ago

One of the correct answer is showing as A. I wanted to understand how A(Choose Higher Number of Layers) is the correct Answer ?
upvoted 1 times

  **KM226** 1 year, 4 months ago

Selected Answer: BCE
I believe the answer is BCE because the model is overfitting.
upvoted 1 times

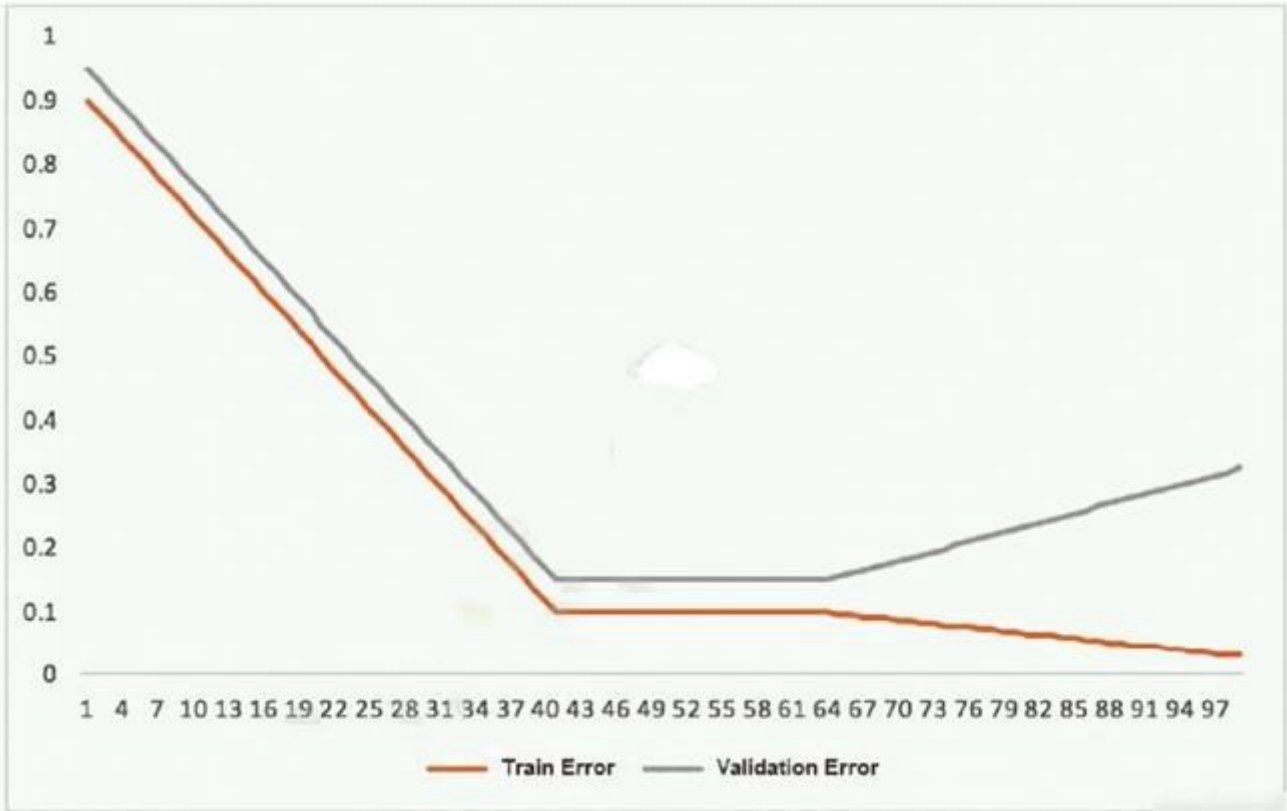
-  **johnvik** 1 year, 6 months ago
choose smaller learning rate c, d, f,
upvoted 1 times
-  **johnvik** 1 year, 6 months ago
ignore answer is correct BDF
upvoted 1 times
-  **Vita_Rasta84444** 1 year, 6 months ago
BDF!!!
upvoted 3 times
-  **astonm13** 1 year, 6 months ago
It is supposed to be BDF
upvoted 2 times
-  **joep21** 1 year, 6 months ago
I would choose B, D and E.
upvoted 1 times
-  **joep21** 1 year, 6 months ago
I meant to say B, D and F. Ignore above.
upvoted 4 times
-  **asdfkljh** 1 year, 7 months ago
is it BDF?
upvoted 1 times
-  **takahirokoyama** 1 year, 7 months ago
Ans. is B,,C,D.
Because this problem is overfitting.
upvoted 2 times
-  **Madwyn** 1 year, 5 months ago
Yes, it's overfitting and smaller learning rate won't help.
upvoted 2 times
-  **ahquiceno** 1 year, 7 months ago
Answers BDF
upvoted 1 times

This graph shows the training and validation loss against the epochs for a neural network.

The network being trained is as follows:

- Two dense layers, one output neuron
- 100 neurons in each layer
- 100 epochs

Random initialization of weights



Which technique can be used to improve model performance in terms of accuracy in the validation set?

- A. Early stopping
- B. Random initialization of weights with appropriate seed
- C. Increasing the number of epochs
- D. Adding another layer with the 100 neurons

Correct Answer: C

Community vote distribution

A (100%)

ahquiceno

Highly Voted

1 year, 7 months ago

Answer A.

upvoted 21 times

eganilovic

Highly Voted

1 year, 7 months ago

The answer is Early Stopping. Stopp the training before accuracy start do decrease.

upvoted 8 times

StelSen

1 year, 6 months ago

Appreciates your explanation. Cheers

upvoted 2 times

Peeking

Most Recent

4 months, 3 weeks ago

Selected Answer: A

Early stopping and not increasing epochs.

upvoted 1 times

ryuhei

7 months, 1 week ago

Selected Answer: A

Answer is "A"

upvoted 1 times

chrisabc

1 year, 6 months ago

Early Stopping can improve the model?

upvoted 3 times

  **Vita_Rasta84444** 1 year, 7 months ago

A is the answer
upvoted 2 times

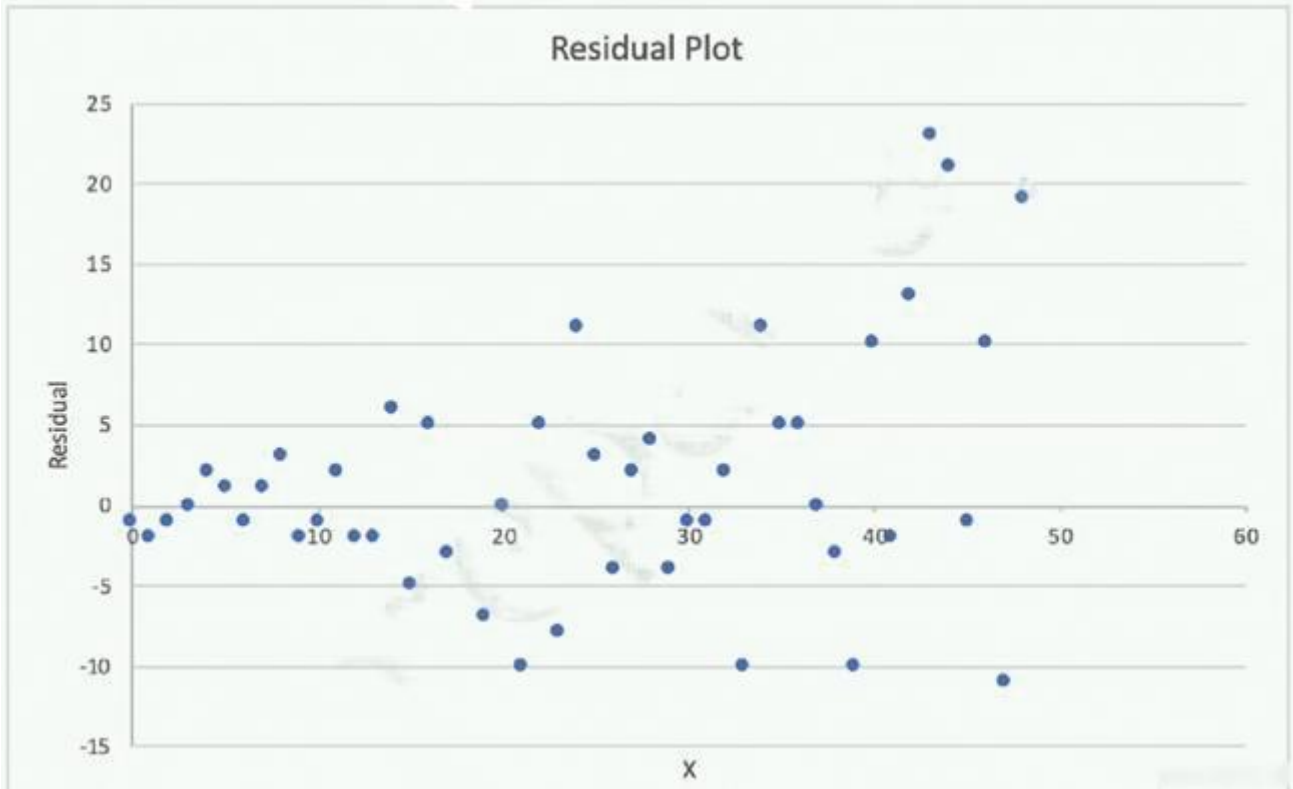
  **astonm13** 1 year, 7 months ago

I would go for A
upvoted 2 times

  **joep21** 1 year, 7 months ago

I would choose A.
upvoted 5 times

A Machine Learning Specialist is attempting to build a linear regression model.



Given the displayed residual plot only, what is the MOST likely problem with the model?

- A. Linear regression is inappropriate. The residuals do not have constant variance.
- B. Linear regression is inappropriate. The underlying data has outliers.
- C. Linear regression is appropriate. The residuals have a zero mean.
- D. Linear regression is appropriate. The residuals have constant variance.

Correct Answer: D

Community vote distribution

A (83%)

D (17%)

- joep21** Highly Voted 1 year, 7 months ago

I would choose A. See: <https://www.itl.nist.gov/div898/handbook/pmd/section4/pmd442.htm> and <https://blog.minitab.com/blog/the-statistics-game/checking-the-assumption-of-constant-variance-in-regression-analyses>
upvoted 23 times
- takahirokoyama** Highly Voted 1 year, 7 months ago

Ans. is A.
High-degree polynomial transformation.
upvoted 12 times
- TrekkingMachine** 1 year, 6 months ago

I think so too.
upvoted 2 times
- Nadia0012** Most Recent 1 month, 3 weeks ago

Selected Answer: A
<https://blog.minitab.com/en/the-statistics-game/checking-the-assumption-of-constant-variance-in-regression-analyses>
upvoted 1 times
- Tomatoteacher** 3 months, 2 weeks ago

Selected Answer: A
Kind of like heteroskedasticity, anyways it is A.
upvoted 1 times
- jrff** 6 months ago

Selected Answer: A
Answer is A. It does not has constant variance !
upvoted 3 times
- Shailendraa** 8 months ago

A is correct answer
upvoted 1 times

- ovokpus

10 months, 1 week ago

These images are broken. I cannot review the question properly!

upvoted 1 times
- John_Pongthorn

1 year, 2 months ago

Selected Answer: D

D is best answer
all x values are scattering as a whole , no matter what x is
<https://www.statisticshowto.com/residual-plot/>

if you take all x values to plot histogram , it will be bel-curv.

upvoted 1 times
- AddiWei

1 year, 2 months ago

And it does NOT mean linear regression is not appropriate. It means your linear regression model is biased due to several reasons.

upvoted 1 times
- eeah

1 year ago

yes, it does. One of the main assumptions is homoscedasticity.

upvoted 2 times
- AddiWei

1 year, 2 months ago

100% A

upvoted 1 times
- u404

1 year, 3 months ago

I will choose A , because the data is heteroscedastic. It violates a key assumption of linear regereession

upvoted 2 times
- Huy

1 year, 6 months ago

A. <https://www.originlab.com/doc/origin-help/residual-plot-analysis>

upvoted 2 times
- yummytaco

1 year, 6 months ago

Do not have content variance
<https://stats.stackexchange.com/questions/52089/what-does-having-constant-variance-in-a-linear-regression-model-mean>

upvoted 2 times
- Vita_Rasta84444

1 year, 6 months ago

Answer is A. As x raises, the residuals become higher and higher...

upvoted 1 times
- cnethers

1 year, 6 months ago

Some Good Reading https://www.andrew.cmu.edu/user/achoulde/94842/homework/regression_diagnostics.html

Ans is A

upvoted 7 times
- Sadgamaya

1 year ago

Thank you for sharing.

upvoted 1 times

A large company has developed a BI application that generates reports and dashboards using data collected from various operational metrics. The company wants to provide executives with an enhanced experience so they can use natural language to get data from the reports. The company wants the executives to be able ask questions using written and spoken interfaces. Which combination of services can be used to build this conversational interface? (Choose three.)

- A. Alexa for Business
- B. Amazon Connect
- C. Amazon Lex
- D. Amazon Polly
- E. Amazon Comprehend
- F. Amazon Transcribe

Correct Answer: *BEF*

Community vote distribution

CEF (50%)

CDF (25%)

CDE (18%)

7%

- astonm13

Highly Voted

1 year, 7 months ago

C - voice and text interface

E - understanding

F - Speech to text

upvoted 31 times
- hero67

1 year, 6 months ago

Why would I need to transcribe while I have Lex that do the NLU part? It would be more reasonable to select Either Connect (B) or Polly (D) if the specs to generate output speech.

upvoted 2 times
- Hariru

1 year, 6 months ago

E - is more to express the "feeling" or "mood". We would rather need something, that can speak to the customer. So my suggestion is c,d,f

upvoted 4 times
- eganilovic

Highly Voted

1 year, 6 months ago

If we need to build written and spoken interfaces we need :

F - Transcribe (speech to text)

D- Polly (text ot speech)

And for chatbot:

E - Lex

upvoted 21 times
- eganilovic

1 year, 6 months ago

*C - Lex

So C,D,F

upvoted 15 times
- weelz

1 year, 6 months ago

I second that, the keyword here is "conversational interface". so, no conversation without Amazon Lex

upvoted 1 times
- injoho

Most Recent

5 days, 15 hours ago

Selected Answer: CDF

Option C,D,F (old answer)

Option C (nowadays)

I do not think this question is still relevant for the exam.

Amazon Lex V2 has ASR(Automatic Speech to Text ~ Transcribe inside) & NTTS (Natural Text to Speech ~ Polly inside).

AWS Comprehend is needed to extract key phrases, or do sentiment analysis but in our case we asked to get insights from data and this is converted to implementing SQL providing text (What ChatGPT successfully do). So, Comprehend is redundant for the current task.

<https://aws.amazon.com/blogs/machine-learning/best-practices-for-creating-amazon-lex-interaction-models/>

<https://docs.aws.amazon.com/lexv2/latest/dg/what-is.html>

<https://aws.amazon.com/about-aws/whats-new/2021/11/amazon-lex-amazon-polly-neural-text-to-speech-ntts-voices/>
upvoted 2 times

  **injo** 5 days, 15 hours ago

Selected Answer: CDF

<https://aws.amazon.com/about-aws/whats-new/2021/11/amazon-lex-amazon-polly-neural-text-to-speech-ntts-voices/>
upvoted 1 times

  **SANDEEP_AWS** 1 month, 2 weeks ago

Selected Answer: CDE

The question asks: The company wants the executives to be able ask questions using written and spoken interfaces. So D might be important.
upvoted 1 times

  **Ajose0** 2 months, 2 weeks ago

Selected Answer: CEF

C - Amazon Lex for voice and text interface - note that Polly is inbuilt into Lex
E - Amazon Comprehend - understanding
F - Amazon Transcribe - Speech to text
upvoted 5 times

  **sqavi** 2 months, 2 weeks ago

Selected Answer: CDF

C. Amazon Lex Most Voted
D. Amazon Polly
F. Amazon Transcribe
E: Comprehend is not require in this case
upvoted 1 times

  **CertArvind** 3 months ago

Correct Answer is CEF
upvoted 1 times

  **Tomatoteacher** 3 months, 2 weeks ago

Selected Answer: CDF

This is a very strange question, I feel as if there would need to be one more sentence of explanation, anyways I go with C,D,F as dont need comprehend to find sentiment, classify documents, or whatnot, need conversational interface.
upvoted 1 times

  **expertguru** 3 months, 3 weeks ago

DEF if output needs via text ---the question confuses they are asking data but didn't exclusively say it is in which form ---conversational I see manager can ask questions via spoke --- but not sure conversational output they are asking ...,if it is conversational output I go with CEF
upvoted 1 times

  **hamimelon** 4 months, 1 week ago

I think it should be CDF
upvoted 1 times

  **ystotest** 5 months ago

Selected Answer: CDE

For speech inputs, Amazon Lex uses Amazon Transcribe behind the scenes to convert speech to text and then processes the text to understand the user's intent. Lex uses the knowledge learned from sample utterances provided during the training phase to detect the user intent and generate a response.
upvoted 1 times

  **aScientist** 5 months, 4 weeks ago

Selected Answer: CEF

Polly is incorrect because executives dont want answers read to them
Amazon connect is a contact center service
Alexa for business is an irrelevant option
upvoted 3 times

  **Ob1KN0B** 8 months, 2 weeks ago

Selected Answer: CEF

This is fairly an easy question but surprised to see the confusion.

"Allowing to get data from reports in natural language" - means speech to text - F. Transcribe, for conversion from speech to text AND E. Comprehend, for NLP text extraction.

"Be able to communicate using written or spoken" - means text or IVR which is C. Lex, only service which provides BOTH capabilities.
upvoted 4 times

  **KlaudYu** 9 months, 2 weeks ago

Selected Answer: CDF

Transcribe -> Lex -> Polly
upvoted 2 times

  **ovokpus** 10 months, 1 week ago

Selected Answer: CEF

Lex is suitable for office chat channels like slack to enhance written and spoken interfaces(chatbots)

Amazon Comprehend is suitable for text and Natural Language data.

Amazon Transcribe is suitable for speech to text.

Polly(text to speech) is not high priority in this use case
upvoted 2 times

  **ovokpus** 10 months, 1 week ago

Any answer without Lex is wrong
upvoted 1 times

A machine learning specialist works for a fruit processing company and needs to build a system that categorizes apples into three types. The specialist has collected a dataset that contains 150 images for each type of apple and applied transfer learning on a neural network that was pretrained on ImageNet with this dataset.

The company requires at least 85% accuracy to make use of the model.

After an exhaustive grid search, the optimal hyperparameters produced the following:

- 68% accuracy on the training set
- 67% accuracy on the validation set

What can the machine learning specialist do to improve the system's accuracy?

- A. Upload the model to an Amazon SageMaker notebook instance and use the Amazon SageMaker HPO feature to optimize the model's hyperparameters.
- B. Add more data to the training set and retrain the model using transfer learning to reduce the bias.
- C. Use a neural network model with more layers that are pretrained on ImageNet and apply transfer learning to increase the variance.
- D. Train a new model using the current neural network architecture.

Correct Answer: B

Community vote distribution

B (83%) Other

dolorez Highly Voted 11 months, 2 weeks ago

Selected Answer: B

the answer is B - the model is underfitting = high bias, so we want to reduce it
C is wrong because the intention is not to increase variance which equals overfitting (using a more complex model would be good, but to reduce bias not increase variance)
upvoted 10 times

cox1960 Most Recent 3 days, 13 hours ago

Selected Answer: A

450 images should be fine. HPO for me.
upvoted 1 times

expertguru 3 months, 3 weeks ago

BOTH VALIdation set and train set performing equally but performance not good. So the basic problem here is high bias (train error) and high variance (test error). Ideally we want both low, but there is trade-off need to be cautious to avoid overfitting. So this problem needs solution for Low bias first (so training performance improves with decent) for later to figure out whether that leads to overfit or not when you test it,! Answer choice B
upvoted 1 times

deng113jie 8 months, 3 weeks ago

why not A?
<https://aws.amazon.com/about-aws/whats-new/2022/07/amazon-sagemaker-automatic-model-tuning-supports-increased-limits-improve-accuracy-models/>
upvoted 2 times

[Removed] 10 months, 2 weeks ago

not B, c is corect
upvoted 1 times

edvardo 11 months, 2 weeks ago

Given that the model can't even fit the training set properly, it would be convenient to amplify the layers that are trained. If I understood the phrasing correctly, I would go with C.
upvoted 1 times

Istdanagan 1 year ago

Selected Answer: C

C, accuracy on training set is low, model not complex enough
upvoted 1 times

NeverMinda 10 months, 4 weeks ago

It only has 150 photos for training, more complex neural network won't help
upvoted 4 times

  **spaceexplorer** 1 year ago

B is more accurate, while adding more complexity for model is viable but you don't want to increase variance

upvoted 12 times

Question #99

Topic 1

A company uses camera images of the tops of items displayed on store shelves to determine which items were removed and which ones still remain. After several hours of data labeling, the company has a total of 1,000 hand-labeled images covering 10 distinct items. The training results were poor.

Which machine learning approach fulfills the company's long-term needs?

- A. Convert the images to grayscale and retrain the model
- B. Reduce the number of distinct items from 10 to 2, build the model, and iterate
- C. Attach different colored labels to each item, take the images again, and build the model
- D. Augment training data for each item using image variants like inversions and translations, build the model, and iterate.

Correct Answer: A

Community vote distribution

D (100%)

-   **ovokpus**

Highly Voted 

 10 months, 1 week ago

Selected Answer: D

Data Augumentation is the way to go here.

How does converting to grayscale help? What if the colors of the items are relevant in object identification???

upvoted 10 times
-   **Sylzys**

Most Recent 

 2 months, 1 week ago

D is of course the right answer, grayscale only won't help anything

upvoted 3 times
-   **PHTR** 3 months, 3 weeks ago

D is the CORRECT ANSWER

<https://research.aimultiple.com/data-augmentation/>

upvoted 1 times
-   **aScientist** 5 months, 4 weeks ago

Selected Answer: D

Data augmentation is correct. we need more samples

upvoted 1 times
-   **tgaos** 11 months ago

D is correct

upvoted 3 times
-   **cron0001** 1 year ago

Selected Answer: D

D is my answer for this. A can help but it'll need more than that.

upvoted 4 times
-   **Istdanagan** 1 year ago

Selected Answer: D

D, i guess

upvoted 4 times

Question #100

Topic 1

A Data Scientist is developing a binary classifier to predict whether a patient has a particular disease on a series of test results. The Data Scientist has data on 400 patients randomly selected from the population. The disease is seen in 3% of the population. Which cross-validation strategy should the Data Scientist adopt?

A. A k-fold cross-validation strategy with k=5

B. A stratified k-fold cross-validation strategy with k=5

C. A k-fold cross-validation strategy with k=5 and 3 repeats

D. An 80/20 stratified split between training and validation

Correct Answer: B

Community vote distribution

B (100%)

scuzzy2010

Highly Voted

1 year, 7 months ago

B - stratified k-fold cross-validation will enforce the class distribution in each split of the data to match the distribution in the complete training dataset.

upvoted 16 times

SophieSu

Highly Voted

1 year, 6 months ago

B is the correct answer. Use Stratified k-Fold Cross-Validation for Imbalanced Classification. Stratified train/test splits is an option too. But the question is specifically asking "cross-validation" strategy.

upvoted 9 times

Valcilio

Most Recent

1 month, 3 weeks ago

Selected Answer: B

Stratified cross validation is for unbalanced data like this!

upvoted 1 times

AWS__Newbie

1 year, 6 months ago

Why K=5?

upvoted 2 times

eeah

1 year ago

K=5 is just standard

upvoted 1 times

Vita_Rasta84444

1 year, 6 months ago

Yes, B...

upvoted 1 times

https://www.examtopycs.com/exams/amazon/aws-certified-machine-learning-specialty/custom-view/

201/406

A technology startup is using complex deep neural networks and GPU compute to recommend the company's products to its existing customers based upon each customer's habits and interactions. The solution currently pulls each dataset from an Amazon S3 bucket before loading the data into a TensorFlow model pulled from the company's Git repository that runs locally. This job then runs for several hours while continually outputting its progress to the same S3 bucket. The job can be paused, restarted, and continued at any time in the event of a failure, and is run from a central queue.

Senior managers are concerned about the complexity of the solution's resource management and the costs involved in repeating the process regularly. They ask for the workload to be automated so it runs once a week, starting Monday and completing by the close of business Friday. Which architecture should be used to scale the solution at the lowest cost?

- A. Implement the solution using AWS Deep Learning Containers and run the container as a job using AWS Batch on a GPU-compatible Spot Instance
- B. Implement the solution using a low-cost GPU-compatible Amazon EC2 instance and use the AWS Instance Scheduler to schedule the task
- C. Implement the solution using AWS Deep Learning Containers, run the workload using AWS Fargate running on Spot Instances, and then schedule the task using the built-in task scheduler
- D. Implement the solution using Amazon ECS running on Spot Instances and schedule the task using the ECS service scheduler

Correct Answer: C

Community vote distribution

A (67%)

B (25%)

8%

- jdstone

Highly Voted

1 year, 6 months ago

Answer is A
<https://aws.amazon.com/blogs/compute/gpu-workloads-on-aws-batch/>
upvoted 24 times
- Juka3lj

1 year, 6 months ago

Makes most sense
upvoted 2 times
- astonm13

Highly Voted

1 year, 6 months ago

I would go for D. As far as I know Fargate does not support GPU computing.
upvoted 6 times
- injoho

Most Recent

2 weeks, 6 days ago

Answer is A. (But the question is tricky)

A and D are both correct solutions but pay attention to the words - "Senior managers are concerned about the complexity of the solution's resource management and the costs". With Cost is everything simple - use Spot instances, with resource management - use higher abstraction servicr

AWS Batch is a management/abstraction layer on top of ECS and EC2 (and some other AWS resources). It does some things for you, like cost optimization, that can be difficult to do yourself. Think of it like Elastic Beanstalk for batch operations. It provides a management layer on top of lower-level AWS resources, but if you are comfortable managing those lower level resources yourself and want more control over them it is certainly an option to use those lower-level resources directly.
upvoted 2 times
- MIlb

3 weeks, 6 days ago

Selected Answer: C

Why not C? AWS Fargate is oriented to manage resources as you need.
upvoted 1 times
- austino

1 month ago

A looks good to me

<https://aws.amazon.com/blogs/compute/deep-learning-on-aws-batch/>
upvoted 1 times
- drcok87

2 months, 1 week ago

b: for those who think its B because of spot instance interruption, ready question phrase "The job can be paused, restarted, and continued at any time in the event of a failure, and is run from a central queue."
between a and c: at the time of this question i doubt if fargate supported GPU, even if it did I choose aws batch for job and fargate for services/apps that need to run all the time.

a is answer
upvoted 2 times

  **Ajose0** 2 months, 2 weeks ago

Selected Answer: A

Option A is the most cost-effective architecture as it uses GPU-compatible Spot Instance which is the lowest cost compute option for GPU instances in the AWS cloud.

AWS Batch is a fully managed service that schedules, runs, and manages the processing and analysis of batch workloads.

The use of AWS Deep Learning Containers enables the technology startup to use pre-built, optimized Docker containers for deep learning, which reduces the complexity of the solution's resource management and eliminates the need for repeated processing.

upvoted 1 times

  **Aninina** 4 months, 1 week ago

Selected Answer: A

Answer is A. Option B is similar to A, but it uses a low-cost GPU-compatible EC2 instance rather than a container, which may not be as flexible or scalable as using containers.

upvoted 4 times

  **matteocal** 9 months, 1 week ago

Selected Answer: A

Answer is A
<https://aws.amazon.com/blogs/compute/gpu-workloads-on-aws-batch/>

upvoted 2 times

  **ovokpus** 10 months, 1 week ago

Selected Answer: A

<https://aws.amazon.com/blogs/compute/gpu-workloads-on-aws-batch/>

There you have it

upvoted 1 times

  **ctrvol401** 10 months, 2 weeks ago

spot instance can be out of service any time, which makes it unsuitable for workflow on a regular basis. Answer is B

upvoted 1 times

  **TerrancePythonJava** 1 year, 1 month ago

Selected Answer: B

The answer is B. The AWS instance scheduler will allow the job to start Monday and end Friday. Secondly, spot instances would not make it through the week. A spot instance will be shut down as soon as the capacity is needed back.

upvoted 3 times

  **cpal012** 1 month, 3 weeks ago

AWS Batch works at the container level and will automatically recover failures and manage instances including spot GPU where available. It was born to do this task

upvoted 1 times

  **AmakamaxZanny** 1 year, 2 months ago

Answer D. The answer is not A because the question specifies it should run on a weekly schedule, there was not mention of how it would be scheduled from this answer. also AWS batch runs from job queue and not schedule :

AWS article:The AWS Batch scheduler evaluates when, where, and how to run jobs that have been submitted to a job queue. If you need to guarantee the order that jobs are run, use the dependsOn parameter to SubmitJob to specify the dependencies for each job.

upvoted 1 times

  **apprehensive_scar** 1 year, 2 months ago

A is correct

upvoted 1 times

  **Asrivastava3** 1 year, 4 months ago

A is incorrect because AWS batch is not used for scheduling and also deep learning containers are used with pre-installed framework - here the Tensorflow model is used from company repo. D is the correct answer that supports all requirements.

upvoted 1 times

  **vetaal** 1 year, 5 months ago

The question says "task be automated such that it runs once a week, beginning Monday and concluding by Friday's closing of business." Option A doesn't mention about scheduling, while option C is incorrect as Fargate doesn't support GPU.

It seems Option D would make most sense based on available choices and following link:

<https://aws.amazon.com/blogs/compute/powering-your-amazon-ecs-cluster-with-amazon-ec2-spot-instances/>

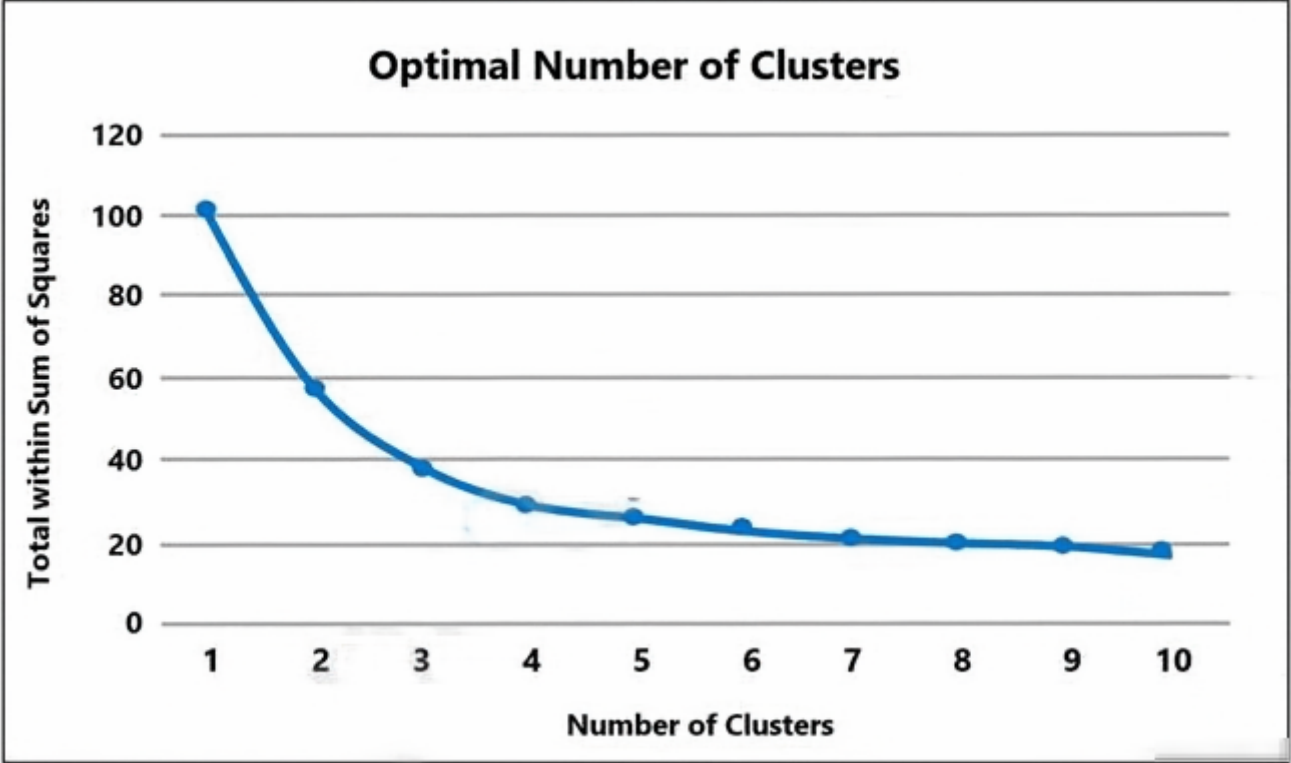
upvoted 3 times

  **anttan** 1 year, 5 months ago

Shouldnt be B, since the job will need to run many hours, and hence all the 3 spot-instances solution should be omitted?

upvoted 1 times

A Machine Learning Specialist prepared the following graph displaying the results of k-means for k = [1..10]:



Considering the graph, what is a reasonable selection for the optimal choice of k?

- A. 1
- B. 4
- C. 7
- D. 10

Correct Answer: C

Community vote distribution

B (100%)

- ksrivastavaSumit

Highly Voted

1 year, 7 months ago

B seems correct based on the elbow method

upvoted 14 times
- Juka3lj

1 year, 6 months ago

I agree, most likely B.

upvoted 4 times
- exam_prep

Highly Voted

11 months, 3 weeks ago

Hi all, I am not able to see the image. It is broken for me. Is it possible for someone to share the image.

upvoted 7 times
- Sylzys

Most Recent

2 months, 1 week ago

Selected Answer: B

Elbow is more visible at 4

upvoted 2 times
- rockingkiran

4 months, 1 week ago

Selected Answer: B

B seems correct

upvoted 1 times
- Peeking

4 months, 3 weeks ago

Selected Answer: B

<https://www.analyticsvidhya.com/blog/2021/01/in-depth-intuition-of-k-means-clustering-algorithm-in-machine-learning/>

upvoted 2 times
- Skychaser

9 months, 3 weeks ago

Selected Answer: B

Elbow method

upvoted 2 times

-   **Sadgamaya** 1 year, 1 month ago
. B seems correct.
upvoted 1 times
-   **mahmoudai** 1 year, 5 months ago
B seems better
upvoted 2 times
-   **Vita_Rasta84444** 1 year, 5 months ago
number 4 is the elbow of the hand, B is correct
upvoted 4 times
-   **cnethers** 1 year, 6 months ago
because the elbow method is a heuristic method its open to debate as to where the correct bend in the cluster is. It's a good tool to use with lower computation cost than computing the silhouette score. When looking at <https://www.youtube.com/watch?v=qs8nfzUsW5U> instead of eyeballing where the bend is, he calculates where the difference between scores is smaller than the 90th percentile
upvoted 2 times

A media company with a very large archive of unlabeled images, text, audio, and video footage wishes to index its assets to allow rapid identification of relevant content by the Research team. The company wants to use machine learning to accelerate the efforts of its in-house researchers who have limited machine learning expertise.

Which is the FASTEST route to index the assets?

- A. Use Amazon Rekognition, Amazon Comprehend, and Amazon Transcribe to tag data into distinct categories/classes.
- B. Create a set of Amazon Mechanical Turk Human Intelligence Tasks to label all footage.
- C. Use Amazon Transcribe to convert speech to text. Use the Amazon SageMaker Neural Topic Model (NTM) and Object Detection algorithms to tag data into distinct categories/classes.
- D. Use the AWS Deep Learning AMI and Amazon EC2 GPU instances to create custom models for audio transcription and topic modeling, and use object detection to tag data into distinct categories/classes.

Correct Answer: A

Community vote distribution

A (100%)

- SophieSu

Highly Voted

1 year, 7 months ago

A. Fastest route must Amazon Services.

upvoted 25 times
- AShahine21

1 year, 7 months ago

Amazon Mechanical Turk is an Amazon service

upvoted 2 times
- Flysun

Most Recent

1 month ago

I will choose B · <https://aws.amazon.com/cn/getting-started/hands-on/machine-learning-tutorial-label-training-data/>

upvoted 1 times
- Valcilio

1 month, 3 weeks ago

Selected Answer: A

Mechanical Turk is the most accurate, but the three services in letter A is the fastest!

upvoted 1 times
- Ajose0

2 months, 2 weeks ago

Selected Answer: A

A. Use Amazon Rekognition, Amazon Comprehend, and Amazon Transcribe to tag data into distinct categories/classes is the fastest route to index the assets. These AWS services provide pre-built machine learning models that can be used to tag the content in the archive without the need for building custom models from scratch.

This option would be faster than using custom models with the AWS Deep Learning AMI and Amazon EC2 GPU instances, or using Amazon Mechanical Turk for human labeling.

Additionally, the use of pre-built models reduces the need for machine learning expertise, aligning with the company's goal of accelerating efforts by its in-house researchers.

upvoted 2 times
- VinceCar

5 months, 2 weeks ago

A. Option B is for those without ML experience. But "researchers who have limited machine learning expertise", so A is better.

upvoted 1 times
- hess

1 year, 3 months ago

A. The most straight forward use of services.

upvoted 2 times
- YJ4219

1 year, 6 months ago

I would have said B, but in B it says "label footage" which means it ignored the rest of the data, so i'd go with A

upvoted 1 times
- Madwyn

1 year, 6 months ago

The question said "a very large archive" meaning a lot of money to pay for labour. B won't be as fast as machine, plus you only label the footage, ignored other stuff.

upvoted 2 times
- AjithkumarSL

1 year, 6 months ago

Would go for A
upvoted 1 times

  **AShahine21** 1 year, 7 months ago

I will go with B
upvoted 1 times

  **Juka3lj** 1 year, 7 months ago

Correct answer is A
upvoted 3 times

  **ksrivastavaSumit** 1 year, 7 months ago

B. as no one in-house is an expert and It probably is the fastest way to get there
upvoted 2 times

  **AhmedAbuMusa** 1 year, 6 months ago

Take into consideration that it is "a very large archive"
upvoted 1 times

A Machine Learning Specialist is working for an online retailer that wants to run analytics on every customer visit, processed through a machine learning pipeline.

The data needs to be ingested by Amazon Kinesis Data Streams at up to 100 transactions per second, and the JSON data blob is 100 KB in size.

What is the MINIMUM number of shards in Kinesis Data Streams the Specialist should use to successfully ingest this data?

- A. 1 shards
- B. 10 shards
- C. 100 shards
- D. 1,000 shards

Correct Answer: B

Community vote distribution

B (100%)

- joep21

Highly Voted

1 year, 6 months ago

Agreed, B it is. See <https://medium.com/slalom-data-analytics/amazon-kinesis-data-streams-auto-scaling-the-number-of-shards-105dc967bed5>

One shard can Ingest 1 MB/second or 1,000 records/second. So 100 KB * 100 = 10 MB (10 shards required)

upvoted 24 times
- PHTR

Most Recent

3 months, 3 weeks ago

B - Max. ingestion per shard = 1000 KB/s

--> 100 Records * 100 KB = 10.000 KB

--> 10.000 KB / 1000 KB/per Shard = 10 Shards

upvoted 2 times
- Shailendraa

8 months ago

10 should be correct.

upvoted 1 times
- tgaos

10 months, 4 weeks ago

Selected Answer: B

B is correct

upvoted 2 times
- mahmoudai

1 year, 5 months ago

100 kb * 100 t/second = 10000 kb = 10 mb

10mb / max_threshold_per_shard (1 mb) = 10 shards

upvoted 1 times
- benson2021

1 year, 6 months ago

Reference: <https://docs.aws.amazon.com/streams/latest/dev/service-sizes-and-limits.html>

upvoted 2 times

A Machine Learning Specialist is deciding between building a naive Bayesian model or a full Bayesian network for a classification problem. The Specialist computes the Pearson correlation coefficients between each feature and finds that their absolute values range between 0.1 to 0.95. Which model describes the underlying data in this situation?

- A. A naive Bayesian model, since the features are all conditionally independent.
- B. A full Bayesian network, since the features are all conditionally independent.
- C. A naive Bayesian model, since some of the features are statistically dependent.
- D. A full Bayesian network, since some of the features are statistically dependent.

Correct Answer: C

Community vote distribution

D (100%)

- joep21

Highly Voted

1 year, 7 months ago

I would say D, because of correlations and dependencies between features. See <https://towardsdatascience.com/basics-of-bayesian-network-79435e11ae7b> and <https://www.quora.com/Whats-the-difference-between-a-naive-Bayes-classifier-and-a-Bayesian-network?share=1>

upvoted 23 times
- Juka3lj

1 year, 6 months ago

I agree, makes moste sense

upvoted 1 times
- Vita_Rasta84444

Highly Voted

1 year, 6 months ago

It should be D. Naive Bayes is called naive because it assumes that each input variable is independent. This is a strong assumption and unrealistic for real data; however, the technique is very effective on a large range of complex problems.

upvoted 7 times
- Ajose0

Most Recent

2 months, 2 weeks ago

Selected Answer: D

In a full Bayesian network, features are connected to each other by edges that represent their conditional dependence relationships. A full Bayesian network is useful when the relationships between the features are complex, non-linear or when they are not conditionally independent.

In this situation, where the Pearson correlation coefficients range between 0.1 and 0.95, it suggests that there are dependencies between the features, indicating that a full Bayesian network would be appropriate to capture the relationships between the features and model the data.

upvoted 1 times
- ystotest

5 months ago

Selected Answer: D

distinction between Bayes theorem and Naive Bayes is that Naive Bayes assumes conditional independence where Bayes theorem does not. This means the relationship between all input features are independent . The Pearson correlation coefficient (r) is the most common way of measuring a linear correlation. It is a number between −1 and 1 that measures the strength and direction of the relationship between two variables.

upvoted 3 times
- Shailendraa

8 months ago

A naive Bayesian model, since some of the features, are statistically dependent.

upvoted 2 times
- SophieSu

1 year, 6 months ago

D. Naive bayes - features are independent given the class.

upvoted 6 times
- astonm13

1 year, 6 months ago

I would say, B. Naive Bayes assumes conditional independence and not statistical

upvoted 2 times
- abdohanfi

1 year, 5 months ago

you mean (a) naive bayes not (b)

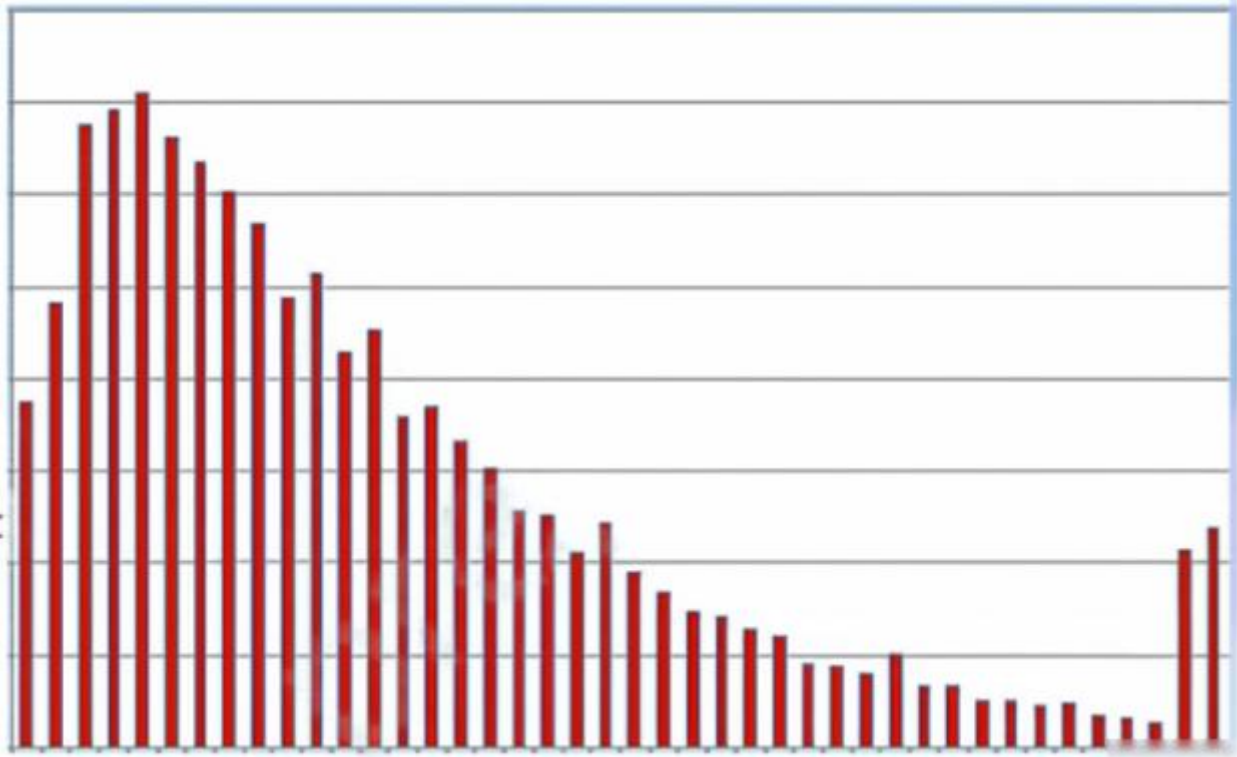
upvoted 1 times
- cnethers

1 year, 6 months ago

This is also a good source of information to help build your understanding <https://www.simplypsychology.org/correlation.html>

upvoted 1 times

A Data Scientist is building a linear regression model and will use resulting p-values to evaluate the statistical significance of each coefficient. Upon inspection of the dataset, the Data Scientist discovers that most of the features are normally distributed. The plot of one feature in the dataset is shown in the graphic.



What transformation should the Data Scientist apply to satisfy the statistical assumptions of the linear regression model?

- A. Exponential transformation
- B. Logarithmic transformation
- C. Polynomial transformation
- D. Sinusoidal transformation

Correct Answer: A

Community vote distribution

B (100%)

- astonm13

Highly Voted

1 year, 7 months ago

I would say B. Logarithmic transformation converts skewed distributions towards normal

upvoted 20 times
- cpal012

Most Recent

3 weeks, 6 days ago

'A' would make it considerably worse.

upvoted 1 times
- Zhechen0912

1 month ago

Selected Answer: B

Log Normal Distribution => Log() => Normal Distribution

upvoted 2 times
- sqavi

2 months, 2 weeks ago

Selected Answer: B

B is correct answer

upvoted 1 times
- vetaal

1 year, 3 months ago

Selected Answer: B

This is B, as this feature seems skewed while others have a regular distribution according to the question. The log transformation will reduce this features skewness.

upvoted 2 times
- YJ4219

1 year, 6 months ago

I think it's B.

reference: <https://corporatefinanceinstitute.com/resources/knowledge/other/positively-skewed-distribution/#:~:text=For%20positively%20skewed%20distributions%2C%20the,each%20value%20in%20the%20dataset.>

"For positively skewed distributions, the most popular transformation is the log transformation. The log transformation implies the calculations of the natural logarithm for each value in the dataset. The method reduces the skew of a distribution. Statistical tests are usually run only when the transformation of the data is complete."

upvoted 2 times

  **konradL** 1 year, 7 months ago

I would also go for B, as Log transformation is often mentioned, when we are talking about right (positive) skewness.
upvoted 3 times

A Machine Learning Specialist is assigned to a Fraud Detection team and must tune an XGBoost model, which is working appropriately for test data. However, with unknown data, it is not working as expected. The existing parameters are provided as follows.

```
param = {
    'eta': 0.05, # the training step for each iteration
    'silent': 1, # logging mode - quiet
    'n_estimators': 2000,
    'max_depth': 30,
    'min_child_weight': 3,
    'gamma': 0,
    'subsample': 0.8,
    'objective': 'multi:softprob', # error evaluation for multiclass training
    'num_class': 201} # the number of classes that exist in this dataset
num_round = 60 # the number of training iterations
```

Which parameter tuning guidelines should the Specialist follow to avoid overfitting?

- A. Increase the max_depth parameter value.
- B. Lower the max_depth parameter value.
- C. Update the objective to binary:logistic.
- D. Lower the min_child_weight parameter value.

Correct Answer: B

Community vote distribution

B (100%)

- SophieSu**

Highly Voted

1 year, 6 months ago

B lower max_depth is the correct answer.
D min_child_weight means something like "stop trying to split once your sample size in a node goes below a given threshold"
Lower min_child_weight, the tree becomes more deep and complex.
Increase min_child_weight, the tree will have less branches and less complexity.
upvoted 15 times
- Shailendraa**

Most Recent

7 months, 3 weeks ago

12-Sep Exam.
upvoted 1 times
- passionatecricketer**

1 year, 5 months ago

Selected Answer: B

When a model overfits, the solutions are:
1. Reduce model flexibility and complexity
2. Reduce the number of feature combinations
3. Decrease n-grams size
4. Decrease the number of numeric attribute bins
5. Increase the amount of regularization
6. Add dropout
upvoted 1 times
- Dr_Kiko**

1 year, 6 months ago

B. 30-deep tree is crazy; normally it's 6-7 no more
upvoted 1 times
- cnethers**

1 year, 6 months ago

A. Increase the max_depth parameter value. (This would increase the complexity resulting in overfitting)
B. Lower the max_depth parameter value. (This would reduce the complexity and minimize overfitting)
C. Update the objective to binary:logistic. it depends on what the target(s) generally you would have a binary classification for fraud detection but there is nothing to say you can't have a multi class so there is not enough information given.
D. Lower the min_child_weight parameter value. (This would reduce the complexity and minimize overfitting)

I find that there are 2 correct answers to this question which does not help B & D
upvoted 2 times
- arulrajayaraj**

1 year, 6 months ago

Ans : B , Lower values avoid over-fitting.
No for D - Larger values avoid over-fitting.
upvoted 8 times

  **cnethers** 1 year, 7 months ago

Thus, those parameters can be used to control the complexity of the trees. It is important to tune them together in order to find a good trade-off between model bias and variance

upvoted 2 times

  **cnethers** 1 year, 7 months ago

min_child_weight is the minimum weight (or number of samples if all samples have a weight of 1) required in order to create a new node in the tree. A smaller min_child_weight allows the algorithm to create children that correspond to fewer samples, thus allowing for more complex trees, but again, more likely to overfit.

upvoted 1 times

  **cnethers** 1 year, 7 months ago

max_depth is the maximum number of nodes allowed from the root to the farthest leaf of a tree. Deeper trees can model more complex relationships by adding more nodes, but as we go deeper, splits become less relevant and are sometimes only due to noise, causing the model to overfit.

upvoted 2 times

A data scientist is developing a pipeline to ingest streaming web traffic data. The data scientist needs to implement a process to identify unusual web traffic patterns as part of the pipeline. The patterns will be used downstream for alerting and incident response. The data scientist has access to unlabeled historic data to use, if needed.

The solution needs to do the following:

- ☞ Calculate an anomaly score for each web traffic entry.

Adapt unusual event identification to changing web patterns over time.

▪

Which approach should the data scientist implement to meet these requirements?

- A. Use historic web traffic data to train an anomaly detection model using the Amazon SageMaker Random Cut Forest (RCF) built-in model. Use an Amazon Kinesis Data Stream to process the incoming web traffic data. Attach a preprocessing AWS Lambda function to perform data enrichment by calling the RCF model to calculate the anomaly score for each record.
- B. Use historic web traffic data to train an anomaly detection model using the Amazon SageMaker built-in XGBoost model. Use an Amazon Kinesis Data Stream to process the incoming web traffic data. Attach a preprocessing AWS Lambda function to perform data enrichment by calling the XGBoost model to calculate the anomaly score for each record.
- C. Collect the streaming data using Amazon Kinesis Data Firehose. Map the delivery stream as an input source for Amazon Kinesis Data Analytics. Write a SQL query to run in real time against the streaming data with the k-Nearest Neighbors (kNN) SQL extension to calculate anomaly scores for each record using a tumbling window.
- D. Collect the streaming data using Amazon Kinesis Data Firehose. Map the delivery stream as an input source for Amazon Kinesis Data Analytics. Write a SQL query to run in real time against the streaming data with the Amazon Random Cut Forest (RCF) SQL extension to calculate anomaly scores for each record using a sliding window.

Correct Answer: A

Community vote distribution

D (100%)

- 🗨️

👤

jiadong

Highly Voted 🍎

1 year, 7 months ago

I think the answer is D - RCF works together with Data Analytics, and sliding window helped on new information

upvoted 23 times
- 🗨️

👤

SophieSu

1 year, 6 months ago

better to say "RCF is a built-in algorithm/function in Kinesis Data Analytics"

upvoted 3 times
- 🗨️

👤

Sidekick

Most Recent 🕒

8 months, 2 weeks ago

Answer is D

"The algorithm starts developing the machine learning model using current records in the stream when you start the application. The algorithm does not use older records in the stream for machine learning, nor does it use statistics from previous executions of the application."

<https://docs.aws.amazon.com/kinesisanalytics/latest/sqlref/sqlrf-random-cut-forest.html>

upvoted 4 times
- 🗨️

👤

apprehensive_scar

1 year, 2 months ago

Selected Answer: D

D it is. easy one

upvoted 2 times
- 🗨️

👤

vetaal

1 year, 3 months ago

Selected Answer: D

RCF is dynamic and adapts with time. D seems more appropriate.

upvoted 3 times
- 🗨️

👤

hess

1 year, 3 months ago

It is A. The only way to handle the historic data is using sagemaker and you can preprocess a data stream using a lambda.

upvoted 2 times
- 🗨️

👤

ZSun

1 week, 4 days ago

But, the question does not require using historical data. BTW, it only has unlabeled historic data, and unlabeled data is not really useful training a detection model.

upvoted 1 times

-   **AMEJack** 1 year, 5 months ago
Definitely D, Data Anytics is using RCF, Using window for selecting data with SQL
upvoted 1 times
-   **Huy** 1 year, 5 months ago
One more reason to select D, not A, is there is no Lambda function to preprocess record in Kinesis Data Stream.
upvoted 1 times
-   **gbrnq** 1 year, 6 months ago
“ Adapt unusual event identification to changing web patterns over time.” -> option A does not satisfy this, only mentions build the model once
upvoted 2 times
-   **randomnamer** 1 year, 6 months ago
The data scientist has access to unlabeled historic data to use, if needed. D has no mention of this. Also, A says the lambda function provides data enrichment. For me it's A.
upvoted 3 times
-   **seanLu** 1 year, 6 months ago
A and D both seems to works. But A does not satisfy requirement 2, adapt to patterns over time. Since the model is only trained on old data. So D may be better.
upvoted 3 times
-   **astonm13** 1 year, 7 months ago
It is definitely D
upvoted 1 times

A Data Scientist received a set of insurance records, each consisting of a record ID, the final outcome among 200 categories, and the date of the final outcome.

Some partial information on claim contents is also provided, but only for a few of the 200 categories. For each outcome category, there are hundreds of records distributed over the past 3 years. The Data Scientist wants to predict how many claims to expect in each category from month to month, a few months in advance.

What type of machine learning model should be used?

- A. Classification month-to-month using supervised learning of the 200 categories based on claim contents.
- B. Reinforcement learning using claim IDs and timestamps where the agent will identify how many claims in each category to expect from month to month.
- C. Forecasting using claim IDs and timestamps to identify how many claims in each category to expect from month to month.
- D. Classification with supervised learning of the categories for which partial information on claim contents is provided, and forecasting using claim IDs and timestamps for all other categories.

Correct Answer: D

Community vote distribution

C (100%)

- JBX2010

Highly Voted

1 year, 7 months ago

I think it should be C as the final outcome among 200 categories is already know. No need to build a classification model. It's pure forecasting problem.

upvoted 23 times
- abdohanfi

1 year, 6 months ago

he said for a few what about the unclassified many i think we need to make classification for the rest first as it will help us with forecasting later with month to month forecasting

upvoted 2 times
- SophieSu

Highly Voted

1 year, 7 months ago

C is my answer.

No need to do classification. Because you know whether the insurance has a claim or not in the dataset. The claim contents do not provide additional information.

upvoted 6 times
- Chelseajcole

Most Recent

2 months, 1 week ago

I would say no machine learning model needed at all. Just using count group by categories SQL is enough

upvoted 1 times
- drcok87

2 months, 2 weeks ago

FinalOutcome

1

2

.

200

RecordID, FinalOutcome, Date, ClaimContents

1

2

.

100000

Note: claim content has partial information, only for few of 200 categories

predict how many claims to expect in each category from month to month, a few months in advance

We dont need the claim contents, we have all we need from first 3 columns to train a forecast model

c

upvoted 1 times
- Ajose0

2 months, 2 weeks ago

Selected Answer: C

Forecasting using claim IDs and timestamps to identify how many claims in each category to expect from month to month.

The problem requires the prediction of the number of claims in each category for each month, which is a time series forecasting problem. The timestamps and record IDs can be used to model the underlying patterns in the data, and the model can be trained to predict the number of claims in each category for future months based on these patterns.

While the claim contents might provide additional information, the fact that partial information is only available for a few categories suggests that this information might not be enough to build a robust model, and that it might not be possible to apply supervised learning to all 200 categories. Instead, the model should be trained on the time series data (claim IDs and timestamps) for all categories, and the claim contents can be used to improve the accuracy of the model only for the categories for which such information is available.

upvoted 2 times

  **informatica** 3 months, 4 weeks ago

how can a forecasting/classification model can be based on the claim ID? (that should be unique)

upvoted 1 times

  **matteocal** 9 months, 1 week ago

Selected Answer: C

it's a forecasting problem, not a classification one

upvoted 2 times

  **Dr_Kiko** 1 year, 6 months ago

predict how many claims to expect in each category from month to month, a few months in advance

C is the only one mentioning forecasting

upvoted 2 times

  **kezzzzz** 1 year, 6 months ago

D is correct. Multi-label classification to impute the missing claim contents, then forecasting what we want. C is missing the imputation part.

upvoted 5 times

  **f4bi4n** 10 months, 2 weeks ago

The question is, can we get something useful out of the handful of 200s and will this impact the forecast as we could forecast the numbers without...

upvoted 1 times

  **randomnamer** 1 year, 6 months ago

It is true that the final outcome is known. But C does not use the partial information from the 200 categories. Reinforcement learning currently is state of the art in stock prediction and other time series. Why waste valuable information? For me it's B.

upvoted 2 times

  **cnethers** 1 year, 7 months ago

This is a supervised learning approach:

Supervised learning problems can be further grouped into regression and classification problems.

Classification: A classification problem is when the output variable is a category, such as "red" and "blue" or "disease" and "no disease."

Regression: A regression problem is when the output variable is a real value, such as "dollars" or "weight."

upvoted 2 times

A company that promotes healthy sleep patterns by providing cloud-connected devices currently hosts a sleep tracking application on AWS. The application collects device usage information from device users. The company's Data Science team is building a machine learning model to predict if and when a user will stop utilizing the company's devices. Predictions from this model are used by a downstream application that determines the best approach for contacting users.

The Data Science team is building multiple versions of the machine learning model to evaluate each version against the company's business goals. To measure long-term effectiveness, the team wants to run multiple versions of the model in parallel for long periods of time, with the ability to control the portion of inferences served by the models.

Which solution satisfies these requirements with MINIMAL effort?

- A. Build and host multiple models in Amazon SageMaker. Create multiple Amazon SageMaker endpoints, one for each model. Programmatically control invoking different models for inference at the application layer.
- B. Build and host multiple models in Amazon SageMaker. Create an Amazon SageMaker endpoint configuration with multiple production variants. Programmatically control the portion of the inferences served by the multiple models by updating the endpoint configuration.
- C. Build and host multiple models in Amazon SageMaker Neo to take into account different types of medical devices. Programmatically control which model is invoked for inference based on the medical device type.
- D. Build and host multiple models in Amazon SageMaker. Create a single endpoint that accesses multiple models. Use Amazon SageMaker batch transform to control invoking the different models through the single endpoint.

Correct Answer: D

Community vote distribution

B (88%)

13%

  **SophieSu**

Highly Voted 

 1 year, 6 months ago

B is the correct answer.

A/B testing with Amazon SageMaker is required in the Exam.

In A/B testing, you test different variants of your models and compare how each variant performs.

Amazon SageMaker enables you to test multiple models or model versions behind the `same endpoint` using `production variants`.

Each production variant identifies a machine learning (ML) model and the resources deployed for hosting the model.

To test multiple models by `distributing traffic` between them, specify the `percentage of the traffic` that gets routed to each model by specifying the `weight` for each `production variant` in the endpoint configuration.

upvoted 37 times

  **joep21**

Highly Voted 

 1 year, 6 months ago

I would answer B, it seems similar to this AWS example: <https://docs.aws.amazon.com/sagemaker/latest/dg/model-ab-testing.html#model-testing-target-variant>

upvoted 9 times

  **Ajose0**

Most Recent 

 2 months, 2 weeks ago

Selected Answer: B

This solution allows the Data Science team to build and host multiple models in Amazon SageMaker, which is a fully managed service for training, deploying, and managing machine learning models.

The team can then create an endpoint configuration with multiple production variants, which are different versions of the models. By programmatically updating the endpoint configuration, the team can control the portion of inferences served by the different models.

This allows them to evaluate the models against their business goals and measure their long-term effectiveness without having to make changes at the application layer.

upvoted 3 times

  **Morsa** 9 months, 2 weeks ago

Selected Answer: D

Answer D as it is said "the team intends to run numerous versions in parallel for extended periods of time," so batch transform

upvoted 1 times

  **cpal012** 3 weeks, 6 days ago

How can you create a single endpoint for batch transforms? this answer is nonsensical.

upvoted 1 times

- [Removed]

10 months, 2 weeks ago

it says,"host a sleep monitoring application", it is the host which means online, not batch, b is correct

upvoted 1 times
- John_Pongthorn

1 year, 2 months ago

Selected Answer: B

The possibility to alter the percentage of inferences supplied by the models.
Which method achieves these criteria with the LEAST amount of effort?

upvoted 4 times
- apprehensive_scar

1 year, 2 months ago

B. Easy

upvoted 2 times
- anttan

1 year, 4 months ago

Think anser is D, below is from the Sagemaker doc.

"https://docs.aws.amazon.com/sagemaker/latest/dg/batch-transform.html"

Use Batch Transform to Test Production Variants

To test different models or various hyperparameter settings, create a separate transform job for each new model variant and use a validation dataset. For each transform job, specify a unique model name and location in Amazon S3 for the output file. To analyze the results, use Inference Pipeline Logs and Metrics.

upvoted 4 times
- passionatecricketer

1 year, 4 months ago

The question talks about the LEAST amount of effort. In this case, there will be as many transform jobs required to be built as there are variants.
That may not be the least amount of effort.

upvoted 2 times

An agricultural company is interested in using machine learning to detect specific types of weeds in a 100-acre grassland field. Currently, the company uses tractor-mounted cameras to capture multiple images of the field as 10 × 10 grids. The company also has a large training dataset that consists of annotated images of popular weed classes like broadleaf and non-broadleaf docks.

The company wants to build a weed detection model that will detect specific types of weeds and the location of each type within the field. Once the model is ready, it will be hosted on Amazon SageMaker endpoints. The model will perform real-time inferencing using the images captured by the cameras.

Which approach should a Machine Learning Specialist take to obtain accurate predictions?

- A. Prepare the images in RecordIO format and upload them to Amazon S3. Use Amazon SageMaker to train, test, and validate the model using an image classification algorithm to categorize images into various weed classes.
- B. Prepare the images in Apache Parquet format and upload them to Amazon S3. Use Amazon SageMaker to train, test, and validate the model using an object- detection single-shot multibox detector (SSD) algorithm.
- C. Prepare the images in RecordIO format and upload them to Amazon S3. Use Amazon SageMaker to train, test, and validate the model using an object- detection single-shot multibox detector (SSD) algorithm.
- D. Prepare the images in Apache Parquet format and upload them to Amazon S3. Use Amazon SageMaker to train, test, and validate the model using an image classification algorithm to categorize images into various weed classes.

Correct Answer: C

Community vote distribution

C (100%)

  **SophieSu** Highly Voted 1 year, 6 months ago

C is my answer.
Pay attention that the question is asking for 2 things:
1. detect specific types of weeds
2. detect the location of each type within the field.

Image Classification can only classify images.
Object detection algorithm:
1.identifies all instances of objects within the image scene.
2.its location and scale in the image are indicated by a rectangular bounding box.

Data format for Computer Vision algorithms in SageMaker:
Recommend to use RecordIO.
upvoted 28 times

  **AjoseO** Most Recent 2 months, 2 weeks ago

Selected Answer: C

The goal is to detect specific types of weeds and their locations within a field, which is a task that requires object detection, rather than image classification. Object detection algorithms are designed to identify objects and their locations within an image, whereas image classification algorithms only categorize an entire image into various classes.

Single-shot multibox detectors (SSD) are a type of object detection algorithm that are well-suited for real-time inferencing and have been shown to be effective for a variety of object detection tasks.

By preparing the images in RecordIO format and using Amazon SageMaker, the company can easily train, test, and validate the model, making it easier to deploy the model in a scalable and secure environment.
upvoted 2 times

  **apprehensive_scar** 1 year, 2 months ago

C is the right answer. you need to detect location
upvoted 3 times

  **AShahine21** 1 year, 6 months ago

C
You can detect the type of weeds and the location within the field.
upvoted 1 times

  **cnethers** 1 year, 6 months ago

If they had an answer with "Faster R-CNN" then it would be different.
This is a good article talking about SSD, Faster R-CNN, R-FCN and others which is a good read.
<https://jonathan-hui.medium.com/object-detection-speed-and-accuracy-comparison-faster-r-cnn-r-fcn-ssd-and-yolo-5425656ae359>
upvoted 2 times

-   **cnethers** 1 year, 7 months ago
I would go with answer C ..
SSD are new architectures faster than the old CNN
<https://towardsdatascience.com/understanding-ssd-multibox-real-time-object-detection-in-deep-learning-495ef744fab>
upvoted 2 times
-   **joep21** 1 year, 7 months ago
I would select answer A, situation is very similar to this one: <https://aws.amazon.com/blogs/machine-learning/building-a-lawn-monitor-and-weed-detection-solution-with-aws-machine-learning-and-iot-services/>
upvoted 3 times
-   **crispogioele** 1 year, 6 months ago
I think it's better go with C, since the question also ask for the location of the weed on the field, while the example you posted is just a classifier.
upvoted 1 times
-   **attaraya** 1 year, 5 months ago
since field is divided in to 10x10 grid I felt A is more suitable.
upvoted 1 times
-   **cpal012** 3 weeks, 6 days ago
So you expect precisely one weed per grid (total 100) of an entire field? If a field is a hectare, then each grid would be 100m2
upvoted 1 times
-   **Bala1212081** 1 year, 6 months ago
The llink says clearly only classification and not talking about object detection. Answer should be C
upvoted 1 times

A manufacturer is operating a large number of factories with a complex supply chain relationship where unexpected downtime of a machine can cause production to stop at several factories. A data scientist wants to analyze sensor data from the factories to identify equipment in need of preemptive maintenance and then dispatch a service team to prevent unplanned downtime. The sensor readings from a single machine can include up to 200 data points including temperatures, voltages, vibrations, RPMs, and pressure readings.

To collect this sensor data, the manufacturer deployed Wi-Fi and LANs across the factories. Even though many factory locations do not have reliable or high- speed internet connectivity, the manufacturer would like to maintain near-real-time inference capabilities.

Which deployment architecture for the model will address these business requirements?

- A. Deploy the model in Amazon SageMaker. Run sensor data through this model to predict which machines need maintenance.
- B. Deploy the model on AWS IoT Greengrass in each factory. Run sensor data through this model to infer which machines need maintenance.
- C. Deploy the model to an Amazon SageMaker batch transformation job. Generate inferences in a daily batch report to identify machines that need maintenance.
- D. Deploy the model in Amazon SageMaker and use an IoT rule to write data to an Amazon DynamoDB table. Consume a DynamoDB stream from the table with an AWS Lambda function to invoke the endpoint.

Correct Answer: A

Community vote distribution

B (100%)

joep21

Highly Voted

1 year, 7 months ago

I would select B, based on the following AWS examples:
<https://aws.amazon.com/blogs/iot/industrial-iot-from-condition-based-monitoring-to-predictive-quality-to-digitize-your-factory-with-aws-iot-services/>
<https://aws.amazon.com/blogs/iot/using-aws-iot-for-predictive-maintenance/>
upvoted 22 times

SophieSu

Highly Voted

1 year, 7 months ago

B is my answer.

For latency-sensitive use cases and for use-cases that require analyzing large amounts of streaming data, it may not be possible to run ML inference in the cloud. Besides, cloud-connectivity may not be available all the time.

For these use cases, you need to deploy the ML model close to the data source.

SageMaker Neo + IoT GreenGrass

To design and push something to edge:

1. design something to do the job, say TF model
2. compile it for the edge device using SageMaker Neo, say Nvidia Jetson
3. run it on the edge using IoT GreenGrass

upvoted 16 times

Peeking

Most Recent

4 months, 3 weeks ago

Selected Answer: B

The described solution will be solved by an edge solution as internet relability is low. IoT Greengrass is the best solution for the edge inference.

upvoted 3 times

ovokpus

10 months, 1 week ago

Selected Answer: B

This is an edge solution, having as little traffic with AWS resources in regions. For this, start thinking IoT Greengrass and Sagemaker Neo, and you'll be halfway there.

Answer is B, no doubt

upvoted 3 times

apprehensive_scar

1 year, 2 months ago

B is the answer, obviously

upvoted 1 times

passionatecricketer

1 year, 5 months ago

Selected Answer: B

This solution requires edge capabilities and to be able to run the inference models in near real-time. SageMaker Neo is a deployable unit on the edge architecture (IoT Greengrass) which can host the runtime inference model.

upvoted 4 times

  **mahmoudai** 1 year, 6 months ago

A: not a complete solution a lot of details is missed
C: daily batch training is huge defect in this solution
D: writing to dynamoDB and invoking endpoint make this solution slower than using an IoT Green Grass

Answer: B

upvoted 1 times

  **Vita_Rasta84444** 1 year, 6 months ago

I would choose B because IoT reduce latency because they work on local machine
upvoted 1 times

  **astonm13** 1 year, 7 months ago

I would choose B
upvoted 1 times

A Machine Learning Specialist is designing a scalable data storage solution for Amazon SageMaker. There is an existing TensorFlow-based model implemented as a train.py script that relies on static training data that is currently stored as TFRecords.

Which method of providing training data to Amazon SageMaker would meet the business requirements with the LEAST development overhead?

- A. Use Amazon SageMaker script mode and use train.py unchanged. Point the Amazon SageMaker training invocation to the local path of the data without reformatting the training data.
- B. Use Amazon SageMaker script mode and use train.py unchanged. Put the TFRecord data into an Amazon S3 bucket. Point the Amazon SageMaker training invocation to the S3 bucket without reformatting the training data.
- C. Rewrite the train.py script to add a section that converts TFRecords to protobuf and ingests the protobuf data instead of TFRecords.
- D. Prepare the data in the format accepted by Amazon SageMaker. Use AWS Glue or AWS Lambda to reformat and store the data in an Amazon S3 bucket.

Correct Answer: D

Community vote distribution

B (100%)

  **joep21** Highly Voted 1 year, 6 months ago

I would select B. Based on the following AWS documentation it appears this is the right approach:
https://sagemaker.readthedocs.io/en/stable/frameworks/tensorflow/using_tf.html
<https://github.com/aws-samples/amazon-sagemaker-script-mode/blob/master/tf-horovod-inference-pipeline/train.py>
upvoted 20 times

  **SophieSu** Highly Voted 1 year, 6 months ago

B is my answer.
Reading Data
filenames = ["s3://bucketname/path/to/file1.tfrecord",
"s3://bucketname/path/to/file2.tfrecord"]
dataset = tf.data.TFRecordDataset(filenames)
upvoted 12 times

  **himanshu10k** Most Recent 1 week, 1 day ago

Internet connectivity issue: then how IOT can be a solution? (Correct answer should be A)
upvoted 1 times

  **AjoseO** 1 month, 1 week ago

Selected Answer: B

Amazon SageMaker script mode enables training a machine learning model using a script that you provide. By using the unchanged train.py script and putting the TFRecord data into an Amazon S3 bucket, you can easily point the Amazon SageMaker training invocation to the S3 bucket without reformatting the training data.

This option avoids the need to rewrite the train.py script or to prepare the data in a different format. It also leverages the scalability and cost-effectiveness of Amazon S3 for storing large amounts of data, which is important for training machine learning models.
upvoted 2 times

  **apprehensive_scar** 1 year, 2 months ago

B, obviously
upvoted 1 times

  **KM226** 1 year, 3 months ago

Selected Answer: B

I like answer B
upvoted 2 times

  **Zhubajie** 1 year, 5 months ago

Why not A? Why can't we train it from local?
upvoted 1 times

  **AddiWei** 1 year, 2 months ago

Sagemaker to my understanding requires the data to be in S3.
upvoted 5 times

  **Huy** 1 year, 5 months ago

B. <https://aws.amazon.com/about-aws/whats-new/2019/01/amazon-sagemaker-batch-transform-now-supports-tfrecord-format/>

upvoted 2 times

  **cnethers** 1 year, 6 months ago

Unfortunilty you can't use the script unchanged, there are some things that need to be added:
1. Make sure your script can handle --model_dir as an additional command line argument. If you did not specify a location when you created the TensorFlow estimator, an S3 location under the default training job bucket is used. Distributed training with parameter servers requires you to use the tf.estimator.train_and_evaluate API and to provide an S3 location as the model directory during training.
2. Load input data from the input channels. The input channels are defined when fit is called.

https://sagemaker.readthedocs.io/en/stable/frameworks/tensorflow/using_tf.html

Beause of the pre-rec Ans A and B are an easy disqualifcation.
There is no need to change the training format so option C is a red herring

Ans is D

Not the most obvious answer
upvoted 4 times

  **SophieSu** 1 year, 6 months ago

according your explanation, the correct answer should be B
upvoted 2 times

Question #114

Topic 1

The chief editor for a product catalog wants the research and development team to build a machine learning system that can be used to detect whether or not individuals in a collection of images are wearing the company's retail brand. The team has a set of training data. Which machine learning algorithm should the researchers use that BEST meets their requirements?

- A. Latent Dirichlet Allocation (LDA)
- B. Recurrent neural network (RNN)
- C. K-means
- D. Convolutional neural network (CNN)

Correct Answer: D

Community vote distribution

D (100%)

  **SophieSu** Highly Voted  1 year, 6 months ago

D. CNN - image
upvoted 34 times

  **ryuhei** Most Recent  7 months ago

Selected Answer: D

Answer is "D"
upvoted 1 times

A retail company is using Amazon Personalize to provide personalized product recommendations for its customers during a marketing campaign. The company sees a significant increase in sales of recommended items to existing customers immediately after deploying a new solution version, but these sales decrease a short time after deployment. Only historical data from before the marketing campaign is available for training. How should a data scientist adjust the solution?

- A. Use the event tracker in Amazon Personalize to include real-time user interactions.
- B. Add user metadata and use the HRNN-Metadata recipe in Amazon Personalize.
- C. Implement a new solution using the built-in factorization machines (FM) algorithm in Amazon SageMaker.
- D. Add event type and event value fields to the interactions dataset in Amazon Personalize.

Correct Answer: D

Community vote distribution

A (100%)

- SophieSu

Highly Voted

1 year, 6 months ago

A is the correct answer. Because in this case, it is not the problem with the existing historical data (event value, event type(click or not)), the sales do not keep growing and now you need to obtain more recent interactive data. An event tracker specifies a destination dataset group for new event data.

upvoted 35 times
- AjithkumarSL

1 year, 6 months ago

I agree.. A is the right choice.. The model need the real time data to adjust to create recommendations..

upvoted 2 times
- ovokpus

Highly Voted

10 months, 1 week ago

Selected Answer: A

here is the receipt:

https://docs.aws.amazon.com/personalize/latest/dg/maintaining-relevance.html

upvoted 5 times
- vingidost

Most Recent

1 month ago

A is the right choice.

upvoted 1 times
- AjoseO

2 months, 2 weeks ago

Selected Answer: A

A. Use the event tracker in Amazon Personalize to include real-time user interactions.

By using the event tracker in Amazon Personalize, the data scientist can collect real-time user interactions, including clicks, views, and purchases, and use these interactions to update the model and generate more accurate recommendations. This could help address the decrease in sales after deploying a new solution version, as the model can be updated to reflect the latest customer behavior. Additionally, including real-time user interactions can help the model better respond to changes in customer behavior and provide more relevant and personalized recommendations, which can help increase sales.

upvoted 2 times
- apprehensive_scar

1 year, 2 months ago

Selected Answer: A

easy one. A

upvoted 2 times
- Huy

1 year, 5 months ago

A.

https://docs.aws.amazon.com/personalize/latest/dg/recording-events.html

upvoted 3 times

A machine learning (ML) specialist wants to secure calls to the Amazon SageMaker Service API. The specialist has configured Amazon VPC with a VPC interface endpoint for the Amazon SageMaker Service API and is attempting to secure traffic from specific sets of instances and IAM users. The VPC is configured with a single public subnet.

Which combination of steps should the ML specialist take to secure the traffic? (Choose two.)

- A. Add a VPC endpoint policy to allow access to the IAM users.
- B. Modify the users' IAM policy to allow access to Amazon SageMaker Service API calls only.
- C. Modify the security group on the endpoint network interface to restrict access to the instances.
- D. Modify the ACL on the endpoint network interface to restrict access to the instances.
- E. Add a SageMaker Runtime VPC endpoint interface to the VPC.

Correct Answer: AC

Reference:

<https://aws.amazon.com/blogs/machine-learning/private-package-installation-in-amazon-sagemaker-running-in-internet-free-mode/>

Community vote distribution

AC (100%)

- mona_mansour

Highly Voted

1 year, 6 months ago

A&C...><https://aws.amazon.com/blogs/machine-learning/securing-all-amazon-sagemaker-api-calls-with-aws-privatelink/>
upvoted 15 times
- wisoxe8356

Highly Voted

4 months, 3 weeks ago

Selected Answer: AC

A - VPC endpoint policy can limit the access to specific group of user/roles
Not B - setting iam user policy can limit user access other aws service but not secure the traffic
C - "specific" sets of instances - means security rules in instance level
Not D - ACL (access control list) allows or denies specific inbound or outbound traffic at the subnet level.
Not E - VPC is configured with public subnet, adding interface without limit the traffic means not secure

upvoted 5 times
- venimus_vidimus_vicimus

Most Recent

5 months ago

Should be A & D n0?
We want to configure the endpoint - first to allow IAM users, second to control access to instances.
Since Security Groups are attached to instances (not VPCs) and only allow allow rules - it should be D.

upvoted 2 times
- exam_prep

11 months, 2 weeks ago

Yes, A & D are correct.
A> This will limit access to only names IAM users. It is like defining all for given principals as below:
{
"Statement": [
{
"Effect": "Allow",
"Principal": "*",
"Action": "*",
"Resource": "*" }
]
}
D-> To restrict access to certain instances or IP address you define deny rule at NACL level. Here VPC Interface endpoint is in subnet (the only subnet in VPC). So modify NACL configurations at this subnet level.
Security group are only for allowing the traffic not for deny so so C is incorrect.

upvoted 1 times
- yj123

1 year, 2 months ago

security group cannot restrict access explicitly, C?

upvoted 1 times
- yj123

1 year, 2 months ago

i mean A, D

upvoted 1 times
- passionatecricketer

1 year, 5 months ago

Selected Answer: AC

Security Group controls instance level access. The question requires instance level access.
The VPC endpoint is already set up. It needs a policy attachment for particular IAM Users. I would have preferred this to be IAM Roles instead of Users, as a more appropriate question. Nevertheless, answer is A & C.

upvoted 2 times

  **Madwyn** 1 year, 6 months ago

A say allow access TO the IAM users? That's wired, why to the IAM users? How do you access them?

upvoted 1 times

  **msamory** 1 year, 6 months ago

The VPC endpoint is already available waiting to be configured. No need to add one. A and E are out.
Furthermore if an IAM endpoint is not set, a default one will be provided and you can't have more than 1 IAM policy but can modify the one that's available.
-Restric access to only calls coming from the VPC, then modify the security group to give access to user group or roles that need access to that notebook.
I think the answer is B and C

upvoted 3 times

  **Madwyn** 1 year, 6 months ago

A says add a VPC endpoint policy, not add an endpoint.

upvoted 2 times

  **cnethers** 1 year, 6 months ago

<https://docs.aws.amazon.com/vpc/latest/userguide/vpc-endpoints-access.html>
<https://docs.aws.amazon.com/sagemaker/latest/dg/notebook-interface-endpoint.html#nbi-private-link-policy>
<https://docs.aws.amazon.com/vpc/latest/userguide/integrated-services-vpce-list.html>

upvoted 2 times

An e commerce company wants to launch a new cloud-based product recommendation feature for its web application. Due to data localization regulations, any sensitive data must not leave its on-premises data center, and the product recommendation model must be trained and tested using nonsensitive data only. Data transfer to the cloud must use IPsec. The web application is hosted on premises with a PostgreSQL database that contains all the data. The company wants the data to be uploaded securely to Amazon S3 each day for model retraining. How should a machine learning specialist meet these requirements?

- A. Create an AWS Glue job to connect to the PostgreSQL DB instance. Ingest tables without sensitive data through an AWS Site-to-Site VPN connection directly into Amazon S3.
- B. Create an AWS Glue job to connect to the PostgreSQL DB instance. Ingest all data through an AWS Site-to-Site VPN connection into Amazon S3 while removing sensitive data using a PySpark job.
- C. Use AWS Database Migration Service (AWS DMS) with table mapping to select PostgreSQL tables with no sensitive data through an SSL connection. Replicate data directly into Amazon S3.
- D. Use PostgreSQL logical replication to replicate all data to PostgreSQL in Amazon EC2 through AWS Direct Connect with a VPN connection. Use AWS Glue to move data from Amazon EC2 to Amazon S3.

Correct Answer: C

Reference:

https://docs.aws.amazon.com/dms/latest/userguide/CHAP_Source.PostgreSQL.html

Community vote distribution

A (100%)

  **cnethers** Highly Voted 1 year, 6 months ago

ASK : Extract Data over IPsec
So we need an ETL + Site to site VPN
GLUE is an ETL service but can it connect to PostgreSQL? yes
<https://docs.aws.amazon.com/glue/latest/dg/aws-glue-programming-etl-connect.html#aws-glue-programming-etl-connect-jdbc>
How to connect Glue to an on-site DB
<https://aws.amazon.com/blogs/big-data/how-to-access-and-analyze-on-premises-data-stores-using-aws-glue/>

My Answer would be A
Anser C only makes a 443 (SSL) connection so does not meet the IPsec requirement
upvoted 23 times

  **ksrivastavaSumit** Highly Voted 1 year, 7 months ago

A? IPsec needs to be covered as well
upvoted 7 times

  **StelSen** 1 year, 6 months ago

Yes. <https://aws.amazon.com/blogs/big-data/how-to-access-and-analyze-on-premises-data-stores-using-aws-glue/>. 'A' is the correct answer.
upvoted 3 times

  **Ajose0** Most Recent 2 months, 2 weeks ago

Selected Answer: A

A. Create an AWS Glue job to connect to the PostgreSQL DB instance. Ingest tables without sensitive data through an AWS Site-to-Site VPN connection directly into Amazon S3.

This solution meets the requirements of data localization regulations and secure data transfer. By creating an AWS Glue job to connect to the PostgreSQL DB instance, the machine learning specialist can extract tables without sensitive data. By using a Site-to-Site VPN connection, the data can be securely transferred from the on-premises data center to Amazon S3, where it can be used for model retraining. This solution ensures that any sensitive data remains in the on-premises data center, and that only non-sensitive data is uploaded to the cloud.
upvoted 1 times

  **matteocal** 9 months, 1 week ago

Selected Answer: A

IPSec means VPN
upvoted 2 times

  **geekgirl007** 1 year, 3 months ago

Answer is A. IPsec is not the same as SSL. Site to site VPN is for IPsec: <https://aws.amazon.com/vpn/site-to-site-vpn/>
Also Glue can directly connect to Postgres and upload to S3: <https://aws.amazon.com/blogs/big-data/how-to-access-and-analyze-on-premises-data-stores-using-aws-glue/>
upvoted 2 times

-   **Deepsachin** 1 year, 5 months ago
A is the answer
upvoted 1 times
-   **Dr_Kiko** 1 year, 6 months ago
Between A and C, I pick A because IPSec requires VPN; otherwise DMS is a better option
upvoted 3 times
-   **Dr_Kiko** 1 year, 6 months ago
Between A and C, I pick A because IPSec requires VPN; otherwise DMS is a better option
upvoted 1 times
-   **AMEJack** 1 year, 6 months ago
100% A, DMS is not supporting S3, but glue is. Also, Site to Site is supporting IPSec.
upvoted 1 times
-   **[Removed]** 1 year, 6 months ago
This is a very tricky question since DMS over VPN started to be supported Nov 2020
<https://aws.amazon.com/about-aws/whats-new/2020/11/now-privately-connect-to-aws-database-migration-service-from-amazon-virtual-private-cloud/>
So both A&C can work. If this question is older then A is the right answer, there have been a bit over 6 months since November 2020 so I would expect this question to become a bit clearer if it stays on the exam in its current form
upvoted 2 times
-   **StelSen** 1 year, 6 months ago
I was bit confused between C & A. But chose A. <https://aws.amazon.com/blogs/big-data/how-to-access-and-analyze-on-premises-data-stores-using-aws-glue/>
Because one of the requirement is to go through via IPSec. So, VPN required. DMS is not IPSec based although it's secured. AWS Glue supports on-premise
upvoted 3 times
-   **joep21** 1 year, 7 months ago
C it is, explanation is good.
upvoted 2 times

A logistics company needs a forecast model to predict next month's inventory requirements for a single item in 10 warehouses. A machine learning specialist uses Amazon Forecast to develop a forecast model from 3 years of monthly data. There is no missing data. The specialist selects the DeepAR+ algorithm to train a predictor. The predictor means absolute percentage error (MAPE) is much larger than the MAPE produced by the current human forecasters.

Which changes to the CreatePredictor API call could improve the MAPE? (Choose two.)

- A. Set PerformAutoML to true.
- B. Set ForecastHorizon to 4.
- C. Set ForecastFrequency to W for weekly.
- D. Set PerformHPO to true.
- E. Set FeaturizationMethodName to filling.

Correct Answer: *CD*

Reference:
<https://docs.aws.amazon.com/forecast/latest/dg/forecast.dg.pdf>

Community vote distribution

AD (100%)

-   **scuzzy2010**

Highly Voted 

 1 year, 6 months ago

I would choose A and D, however both of them is not possible at the same time. The question is ambiguous, it could mean which two options, but no necessarily both.
A - If you want Amazon Forecast to evaluate each algorithm and choose the one that minimizes the objective function, set PerformAutoML to true.
D - The following algorithms support HPO: - > DeepAR+.

upvoted 17 times
-   **Oscaaaar** 1 year, 6 months ago

If custom forecast types are specified, Forecast evaluates metrics at those specified forecast types, and takes the averages of those metrics to determine the optimal outcomes during HPO and AutoML.

For both AutoML and HPO, Forecast chooses the option that minimizes the average losses over the forecast types. During HPO, Forecast uses the first backtest window to find the optimal hyperparameter values. During AutoML, Forecast uses the averages across all backtest windows and the optimal hyperparameters values from HPO to find the optimal algorithm. <https://docs.aws.amazon.com/forecast/latest/dg/metrics.html>

upvoted 2 times
-   **Vita_Rasta84444**

Highly Voted 

 1 year, 6 months ago

It is A and D, there are no weekly data, they have only monthly data and can not switch horizon to 4

upvoted 7 times
-   **Ajose0**

Most Recent 

 2 months, 2 weeks ago

Selected Answer: AD

A. Set PerformAutoML to true.
D. Set PerformHPO to true.

Setting PerformAutoML to true will enable Amazon Forecast to automatically select the best algorithm and hyperparameters for your data and problem. This can help improve the MAPE by finding the optimal combination of algorithm and hyperparameters that minimize prediction error.

Setting PerformHPO to true will enable Amazon Forecast to perform a hyperparameter optimization search to find the best combination of hyperparameters that result in the best prediction performance. This can help improve the MAPE by finding the optimal combination of hyperparameters that minimize prediction error.

upvoted 1 times
-   **yemauricio** 4 months ago

Selected Answer: AD

A. Looking for better algorithms performance
D. Hyperparameters optimization

upvoted 1 times
-   **Shailendraa** 7 months, 3 weeks ago

12-sep exam

upvoted 3 times
-   **vanluigi** 11 months, 3 weeks ago

Why are not B and C? The question asks about modifications that increase MAPE (thats bad):

B - If FH is larger, error will increase

C - Data is based on months, change that will make erros on forecasting values

E - There is no data gap so is useless

A - Selec best between all should DECREASE MAPE

D - Tunning hyperparms will DECREASE MAPE

upvoted 5 times

  **mona_mansour** 1 year, 6 months ago

A&D...>By default, Amazon Forecast uses the 0.1 (P10), 0.5 (P50), and 0.9 (P90) quantiles for hyperparameter tuning during hyperparameter optimization (HPO) and for model selection during AutoML. If you specify custom forecast types when creating a predictor, Forecast uses those forecast types during HPO and AutoML.

If custom forecast types are specified, Forecast evaluates metrics at those specified forecast types, and takes the averages of those metrics to determine the optimal outcomes during HPO and AutoML.

For both AutoML and HPO, Forecast chooses the option that minimizes the average losses over the forecast types. During HPO, Forecast uses the first backtest window to find the optimal hyperparameter values. During AutoML, Forecast uses the averages across all backtest windows and the optimal hyperparameters values from HPO to find the optimal algorithm.

upvoted 4 times

  **SophieSu** 1 year, 6 months ago

C. ForecastFrequency

M- MONTHLY

W- WEEKLY

D. PerformHPO

Whether to perform hyperparameter optimization (HPO). HPO finds optimal hyperparameter values for your training data. The process of performing HPO is known as running a hyperparameter tuning job.

The default value is false. In this case, Amazon Forecast uses default hyperparameter values from the chosen algorithm.

E. FeaturizationMethodName

The name of the method. The "filling" method is the only supported method.

upvoted 2 times

  **seanLu** 1 year, 6 months ago

But for option C, according to the Developer Guide, The forecast frequency must be greater than or equal to the TARGET_TIME_SERIES dataset frequency. and the training data is monthly data, so ForecastFrequency can not be less than Monthly.

upvoted 6 times

  **SophieSu** 1 year, 7 months ago

ABE can be excluded. CD is my answer.

A. PerformAutoML

If you want Amazon Forecast to evaluate each algorithm and choose the one that minimizes the objective function, set PerformAutoML to true. The objective function is defined as the mean of the weighted losses over the forecast types. By default, these are the p10, p50, and p90 quantile losses. When AutoML is enabled, the following properties are disallowed:

AlgorithmArn

HPOConfig

PerformHPO

TrainingParameters

B. ForecastHorizon

Specifies the number of time-steps that the model is trained to predict. The forecast horizon is also called the prediction length.

For example, if you configure a dataset for daily data collection (using the DataFrequency parameter of the CreateDataset operation) and set the forecast horizon to 10, the model returns predictions for 10 days.

The maximum forecast horizon is the lesser of 500 time-steps or 1/3 of the TARGET_TIME_SERIES dataset length.

upvoted 3 times

A data scientist wants to use Amazon Forecast to build a forecasting model for inventory demand for a retail company. The company has provided a dataset of historic inventory demand for its products as a .csv file stored in an Amazon S3 bucket. The table below shows a sample of the dataset.

timestamp	item_id	demand	category	lead_time
2019-12-14	uni_000736	120	hardware	90
2020-01-31	uni_003429	98	hardware	30
2020-03-04	uni_000211	234	accessories	10

How should the data scientist transform the data?

- A. Use ETL jobs in AWS Glue to separate the dataset into a target time series dataset and an item metadata dataset. Upload both datasets as .csv files to Amazon S3.
- B. Use a Jupyter notebook in Amazon SageMaker to separate the dataset into a related time series dataset and an item metadata dataset. Upload both datasets as tables in Amazon Aurora.
- C. Use AWS Batch jobs to separate the dataset into a target time series dataset, a related time series dataset, and an item metadata dataset. Upload them directly to Forecast from a local machine.
- D. Use a Jupyter notebook in Amazon SageMaker to transform the data into the optimized protobuf recordIO format. Upload the dataset in this format to Amazon S3.

Correct Answer: B

Community vote distribution

A (100%)

- joep21

Highly Voted

1 year, 7 months ago

I would answer A. Target and metadata must be in two files and loaded from S3, based on documentation: <https://docs.aws.amazon.com/forecast/latest/dg/dataset-import-guidelines-troubleshooting.html>

upvoted 26 times
- ZSun

1 week, 5 days ago

1. I cannot find any evidence support the seperate file defination.
2. A,B,C all seperate datasets, this explanation is weak.

upvoted 1 times
- Ajose0

Most Recent

1 month, 1 week ago

Selected Answer: A

Amazon Forecast requires the input data to be separated into a target time series dataset and an item metadata dataset.

The target time series dataset should include the time series data that you want to use for forecasting, such as inventory demand in this case. The item metadata dataset should include the metadata that describes the items in the time series, such as product IDs, categories, and attributes.

Therefore, the data scientist should use ETL jobs in AWS Glue to separate the dataset into a target time series dataset and an item metadata dataset. Both datasets should be uploaded as .csv files to Amazon S3, which is a suitable storage option for input data to Amazon Forecast.

upvoted 4 times
- ystotest

5 months, 1 week ago

Selected Answer: A

I would vote for A

upvoted 3 times
- tgaos

11 months, 2 weeks ago

The answer is A.
According to the <https://docs.aws.amazon.com/forecast/latest/dg/forecast.dg.pdf> , page 51.
Target Time Series Dataset:
Required: timestamp, item_id, demand
Additional: lead_time

Item Metadata Dataset:
item_id, category
upvoted 2 times

tgaos

11 months ago

You can find the same question with the picture at <https://ccnav7.net/a-data-scientist-wants-to-use-amazon-forecast-to-build-a-forecasting-model-for-inventory-demand-for-a-retail-company/>
- <https://www.examtopycs.com/exams/amazon/aws-certified-machine-learning-specialty/custom-view/>

234/406

upvoted 1 times

  **DSJingguo** 1 year, 6 months ago

The correct answer is A
"Forecast supports only the comma-separated values (CSV) file format. You can't separate values using tabs, spaces, colons, or any other characters.

Guideline: Convert your dataset to CSV format (using only commas as your delimiter) and try importing the file again."

upvoted 1 times

  **achiko** 1 year, 6 months ago

lead time belongs to related time series, as its not a target variable

upvoted 1 times

A machine learning specialist is running an Amazon SageMaker endpoint using the built-in object detection algorithm on a P3 instance for real-time predictions in a company's production application. When evaluating the model's resource utilization, the specialist notices that the model is using only a fraction of the GPU.

Which architecture changes would ensure that provisioned resources are being utilized effectively?

- A. Redeploy the model as a batch transform job on an M5 instance.
- B. Redeploy the model on an M5 instance. Attach Amazon Elastic Inference to the instance.
- C. Redeploy the model on a P3dn instance.
- D. Deploy the model onto an Amazon Elastic Container Service (Amazon ECS) cluster using a P3 instance.

Correct Answer: D

Community vote distribution

B (100%)

joep21

Highly Voted

1 year, 7 months ago

B is correct. Redeploy with CPU and add elastic inference to reduce costs. See: <https://aws.amazon.com/machine-learning/elastic-inference/>
upvoted 25 times

Ajose0

Most Recent

2 months, 2 weeks ago

Selected Answer: B

Amazon Elastic Inference allows you to attach low-cost GPU-powered acceleration to EC2 and Sagemaker instances, to reduce the cost of running deep learning inference.

You can choose any CPU instance that is best suited to the overall compute and memory needs of your application, and then separately configure the right amount of GPU-powered inference acceleration. This would allow you to efficiently utilize resources and reduce costs.

upvoted 3 times

Peeking

4 months, 3 weeks ago

Selected Answer: B

Elastic inference enables GPU only when load increases. With 50% utilisation there is no need to deploy P3 as the base inference machine.

upvoted 1 times

ystotest

5 months, 1 week ago

Selected Answer: B

Agreed with B

upvoted 1 times

Shailendraa

7 months, 3 weeks ago

12-sep exam

upvoted 2 times

SriAkula

1 year, 1 month ago

Answer: B

Explanation: <https://aws.amazon.com/machine-learning/elastic-inference/>

upvoted 2 times

mahmoudai

1 year, 6 months ago

B: production mostly needs CPU with EI rather than GPU machines

upvoted 1 times

mona_mansour

1 year, 6 months ago

B..>Amazon Elastic Inference (EI) is a resource you can attach to your Amazon EC2 CPU instances to accelerate your deep learning (DL) inference workloads. Amazon EI accelerators come in multiple sizes and are a cost-effective method to build intelligent capabilities into applications running on Amazon EC2 instances.

upvoted 3 times

Vita_Rasta84444

1 year, 6 months ago

B is correct

upvoted 1 times

A data scientist uses an Amazon SageMaker notebook instance to conduct data exploration and analysis. This requires certain Python packages that are not natively available on Amazon SageMaker to be installed on the notebook instance.

How can a machine learning specialist ensure that required packages are automatically available on the notebook instance for the data scientist to use?

- A. Install AWS Systems Manager Agent on the underlying Amazon EC2 instance and use Systems Manager Automation to execute the package installation commands.
- B. Create a Jupyter notebook file (.ipynb) with cells containing the package installation commands to execute and place the file under the /etc/init directory of each Amazon SageMaker notebook instance.
- C. Use the conda package manager from within the Jupyter notebook console to apply the necessary conda packages to the default kernel of the notebook.
- D. Create an Amazon SageMaker lifecycle configuration with package installation commands and assign the lifecycle configuration to the notebook instance.

Correct Answer: B

Reference:
<https://towardsdatascience.com/automating-aws-sagemaker-notebooks-2dec62bc2c84>

Community vote distribution

D (92%)

8%

-   **joep21** Highly Voted 1 year, 7 months ago

I would select D. See AWS documentation: <https://docs.aws.amazon.com/sagemaker/latest/dg/nbi-add-external.html>
upvoted 18 times
-   **31Rishab** Most Recent 7 months, 1 week ago

Selected Answer: D
Even the link given suggest Option D
upvoted 3 times
-   **ovokpus** 10 months, 1 week ago

Selected Answer: D
Please ignore my previous comment, the answer is D
upvoted 2 times
-   **ovokpus** 10 months, 1 week ago

Selected Answer: B
Key word here is how can the developer "guarantee"?
He guarantees that by including the install commands as part of the notebook.

So, against the grain, I stand with B
upvoted 1 times
-   **ovokpus** 10 months, 1 week ago

Scratch that. The Answer is D
upvoted 1 times
-   **ayatkhrisat** 12 months ago

Selected Answer: D
D should be the answer
upvoted 3 times
-   **vetaal** 1 year, 3 months ago

Selected Answer: D
<https://docs.aws.amazon.com/sagemaker/latest/dg/nbi-add-external.html>
upvoted 1 times
upvoted 2 times
-   **geekgirl007** 1 year, 3 months ago

Selected Answer: D
D "automatically" is the key here and using lifecycle configuration <https://docs.aws.amazon.com/sagemaker/latest/dg/nbi-add-external.html>
upvoted 2 times

-   **rafaelo** 1 year, 4 months ago
It is D, although is not even the best answer in my opinion. Although by default conda packages are installed in ephemeral storage, you can change that default behaviour. I did that in my last project and we created our own conda environment that persisted between shutdowns.
upvoted 1 times
-   **abdohanfi** 1 year, 6 months ago
based on the refernce given under the answer its D not B
upvoted 1 times
-   **mona_mansour** 1 year, 6 months ago
You can install packages using the following methods:

1-Lifecycle configuration scripts
2-Notebooks – The following commands are supported.

%conda install

%pip install

3-The Jupyter terminal – You can install packages using pip and conda directly.
upvoted 2 times
-   **mona_mansour** 1 year, 6 months ago
NOT B ...>/etc/init contains configuration files used by Upstart.
ANS...>D
upvoted 1 times
-   **astonm13** 1 year, 6 months ago
Its for sure D
upvoted 1 times
-   **ksrivastavaSumit** 1 year, 6 months ago
D <https://docs.aws.amazon.com/sagemaker/latest/dg/nbi-add-external.html>
upvoted 3 times

A data scientist needs to identify fraudulent user accounts for a company's ecommerce platform. The company wants the ability to determine if a newly created account is associated with a previously known fraudulent user. The data scientist is using AWS Glue to cleanse the company's application logs during ingestion.

Which strategy will allow the data scientist to identify fraudulent accounts?

- A. Execute the built-in FindDuplicates Amazon Athena query.
- B. Create a FindMatches machine learning transform in AWS Glue.
- C. Create an AWS Glue crawler to infer duplicate accounts in the source data.
- D. Search for duplicate accounts in the AWS Glue Data Catalog.

Correct Answer: B

Reference:
<https://docs.aws.amazon.com/glue/latest/dg/machine-learning.html>

Community vote distribution
B (100%)

-   **mona_mansour**

Highly Voted 

 1 year, 6 months ago

B ,You can use the FindMatches transform to find duplicate records in the source data. A labeling file is generated or provided to help teach the transform.

upvoted 12 times
-   **joep21**

Highly Voted 

 1 year, 7 months ago

B it is. Reasonable explanation.

upvoted 6 times
-   **Valcilio**

Most Recent 

 1 month, 3 weeks ago

Selected Answer: B

It's B, how it's using Glue to clean the data the easiest way will be use Glue's ML FindMatches extension to do this too.

upvoted 1 times
-   **Tomatoteacher** 3 months, 2 weeks ago

Selected Answer: B

It is B.

upvoted 1 times
-   **omar_bahrain** 1 year, 7 months ago

Agree. Please refer to:
<https://aws.amazon.com/blogs/big-data/integrate-and-deduplicate-datasets-using-aws-lake-formation-findmatches/>

upvoted 2 times

A Data Scientist is developing a machine learning model to classify whether a financial transaction is fraudulent. The labeled data available for training consists of 100,000 non-fraudulent observations and 1,000 fraudulent observations.

The Data Scientist applies the XGBoost algorithm to the data, resulting in the following confusion matrix when the trained model is applied to a previously unseen validation dataset. The accuracy of the model is 99.1%, but the Data Scientist needs to reduce the number of false negatives.

	Predicted 0	Predicted 1
Actual 0	99,966	34
Actual 1	877	123

Which combination of steps should the Data Scientist take to reduce the number of false negative predictions by the model? (Choose two.)

- A. Change the XGBoost eval_metric parameter to optimize based on Root Mean Square Error (RMSE).
- B. Increase the XGBoost scale_pos_weight parameter to adjust the balance of positive and negative weights.
- C. Increase the XGBoost max_depth parameter because the model is currently underfitting the data.
- D. Change the XGBoost eval_metric parameter to optimize based on Area Under the ROC Curve (AUC).
- E. Decrease the XGBoost max_depth parameter because the model is currently overfitting the data.

Correct Answer: DE

Community vote distribution

BD (69%)BE (31%)

- LydiaGom

Highly Voted

11 months, 3 weeks ago

B and D

upvoted 11 times
- ovokpus

Highly Voted

10 months, 1 week ago

Selected Answer: BD

Compensate for imbalance and optimize on AUC. This is a class imbalance problem, not an overfitting problem.

upvoted 7 times
- rb39

7 months, 2 weeks ago

totally right, overfitting has nothing to do so there is no need to reduce tree depth

upvoted 1 times
- MIlb

Most Recent

4 weeks ago

Selected Answer: BE

I have some doubts about D and E.
Precision-Recall AUC is better than AUC curve in imbalanced classes. Then, I choose E

upvoted 2 times
- AjoseO

1 month, 1 week ago

Selected Answer: BD

Option A and Option E are unlikely to help reduce false negatives.

Option C, increasing max_depth, may lead to overfitting, which could make the model worse.

Option D, changing the eval_metric to optimize based on AUC, could help improve the model's ability to discriminate between the two classes.

Option B, increasing the scale_pos_weight parameter to adjust the balance of positive and negative weights, can help the model better handle imbalanced datasets, which is the case here. By increasing the weight of positive examples, the model will learn to prioritize correctly classifying them, which should reduce the number of false negatives.

upvoted 1 times
- Tomatoteacher

3 months, 2 weeks ago

Selected Answer: BD

BD, I have done this before, but it would be better to use Average Precision(AP) instead of AUC, but it is better than other answers.

upvoted 1 times
- Shailendraa

7 months, 3 weeks ago

12-sep exam

upvoted 1 times

  **ovokpus** 10 months, 1 week ago

Selected Answer: BE

Compensate for imbalance and overwriting.

upvoted 1 times

  **NeverMinda** 10 months, 4 weeks ago

Selected Answer: BE

B and E

upvoted 1 times

  **MLGuru** 11 months, 3 weeks ago

B. Increase the XGBoost scale_pos_weight parameter to adjust the balance of positive and negative weights is the correct answer.
According to https://docs.aws.amazon.com/sagemaker/latest/dg/xgboost_hyperparameters.html, scale_pos_weight controls the balance of positive and negative weights. It's useful for unbalanced classes.

upvoted 3 times

Question #124

Topic 1

A data scientist has developed a machine learning translation model for English to Japanese by using Amazon SageMaker's built-in seq2seq algorithm with 500,000 aligned sentence pairs. While testing with sample sentences, the data scientist finds that the translation quality is reasonable for an example as short as five words. However, the quality becomes unacceptable if the sentence is 100 words long. Which action will resolve the problem?

- A. Change preprocessing to use n-grams.
- B. Add more nodes to the recurrent neural network (RNN) than the largest sentence's word count.
- C. Adjust hyperparameters related to the attention mechanism.
- D. Choose a different weight initialization type.

Correct Answer: B

Community vote distribution

C (100%)

  **cnethers**

Highly Voted 

 1 year, 7 months ago

I agree with an answer of C
Attention mechanism. The disadvantage of an encoder-decoder framework is that model performance decreases as and when the length of the source sequence increases because of the limit of how much information the fixed-length encoded feature vector can contain. To tackle this problem, in 2015, Bahdanau et al. proposed the attention mechanism. In an attention mechanism, the decoder tries to find the location in the encoder sequence where the most important information could be located and uses that information and previously decoded words to predict the next token in the sequence.
upvoted 24 times

  **Ajose0**

Most Recent 

 2 months, 2 weeks ago

Selected Answer: C

C. Adjust hyperparameters related to the attention mechanism.

The seq2seq algorithm uses an attention mechanism to dynamically focus on relevant parts of the input sequence for each output sequence element. Increasing the attention mechanism's ability to learn dependencies between long input and output sequences might help improve the translation quality for long sentences.

The data scientist could try adjusting relevant hyperparameters such as attention depth or attention scale, or try a different attention mechanism such as scaled dot-product attention, to see if that improves the translation quality for long sentences.
upvoted 4 times

  **peterfish** 9 months, 2 weeks ago

Selected Answer: C

i go with C
upvoted 4 times

  **SriAkula** 1 year, 1 month ago

Ans: C
Explanation: <https://docs.aws.amazon.com/sagemaker/latest/dg/seq-2-seq-howitworks.html>
upvoted 2 times

  **AddiWei** 1 year, 2 months ago

This is such a niche question for a niche market. Geared towards someone who specializes in NLP.
upvoted 2 times

  **Juka3lj** 1 year, 6 months ago

c is correct
<https://docs.aws.amazon.com/sagemaker/latest/dg/seq-2-seq-howitworks.html>
upvoted 3 times

  **rajesriv** 1 year, 7 months ago

I believe the answer is C
<https://docs.aws.amazon.com/sagemaker/latest/dg/seq-2-seq-howitworks.html>
upvoted 3 times

A financial company is trying to detect credit card fraud. The company observed that, on average, 2% of credit card transactions were fraudulent. A data scientist trained a classifier on a year's worth of credit card transactions data. The model needs to identify the fraudulent transactions (positives) from the regular ones (negatives). The company's goal is to accurately capture as many positives as possible. Which metrics should the data scientist use to optimize the model? (Choose two.)

- A. Specificity
- B. False positive rate
- C. Accuracy
- D. Area under the precision-recall curve
- E. True positive rate

Correct Answer: AB

Community vote distribution

DE (71%)

BD (29%)

- littlewat

Highly Voted

1 year, 6 months ago

D, E is the answer. we need to make the recall rate(not precision) high.

upvoted 35 times
- joep21

Highly Voted

1 year, 7 months ago

To maximize detection of fraud in real-world, imbalanced datasets, D and E should always be applied.

https://en.wikipedia.org/wiki/Sensitivity_and_specificity

https://machinelearningmastery.com/roc-curves-and-precision-recall-curves-for-imbalanced-classification/

upvoted 13 times
- joep21

1 year, 7 months ago

Note, True positive rate = Sensitivity = Recall

upvoted 5 times
- cnethers

1 year, 7 months ago

that is not correct unfortunately

Recall is = Sensitivity = False Negative which is a Type II error

Precision = specificity = False Positive which is a Type I error

I do agree that in the real world you would focus on Recall/sensitivity ie. reducing type II errors.

However, in the question, they want to reduce the False Positives so you would need to focus on precision and specificity minimizing type I errors

upvoted 5 times
- cnethers

1 year, 7 months ago

My ANS would be AB

upvoted 6 times
- omar_bahrain

1 year, 6 months ago

I came a cross this article and I would say it might support choosing A&B as answers:

https://medium.com/datadriveninvestor/rethinking-the-right-metrics-for-fraud-detection-4edfb629c423

upvoted 1 times
- omar_bahrain

1 year, 6 months ago

specifity is different

than precision

upvoted 1 times
- Ajose0

Most Recent

2 months, 2 weeks ago

Selected Answer: DE

The goal is to accurately capture as many fraudulent transactions (positives) as possible. To optimize the model towards this goal, the data scientist should focus on metrics that emphasize the true positive rate and the area under the precision-recall curve.

True positive rate (TPR or sensitivity) is the proportion of actual positive cases that are correctly identified as positive by the model. A higher TPR means that more fraudulent transactions are being captured.

The precision-recall curve is a graph that shows the trade-off between precision and recall for different thresholds.

upvoted 1 times

  **Ajose0** 2 months, 2 weeks ago

Precision is the fraction of correctly identified positive instances among all instances the model has classified as positive. Recall, also known as the true positive rate, is the fraction of positive instances that are correctly identified as positive by the model. A higher area under the precision-recall curve indicates that the model is making fewer false positive predictions and more true positive predictions, which aligns with the goal of the financial company to accurately capture as many fraudulent transactions as possible.

upvoted 1 times

  **ystotest** 5 months ago

Selected Answer: DE

agreed with DE

upvoted 4 times

  **f4bi4n** 10 months, 2 weeks ago

Why not A and D?

- Specificity shows us how the FNR is
- AUC PR includes Precision and Recall which shows us the ratio of TP to TP/FP and TP to TP / FN

upvoted 2 times

  **SriAkula** 1 year, 1 month ago

Answer: D&E

upvoted 1 times

  **KM226** 1 year, 4 months ago

I meant say D&E not BD

upvoted 1 times

  **KM226** 1 year, 4 months ago

Selected Answer: BD

I believe the answer is B&D, which equals F1. F1 combines precision and Sensitivity.

upvoted 2 times

  **mahmoudai** 1 year, 5 months ago

D&E is the only choices that takes False Negatives into considration

upvoted 1 times

  **f4bi4n** 10 months, 2 weeks ago

TPR is already included in the AUC PR
TNR is not included in all others besides A

upvoted 1 times

  **AShahine21** 1 year, 5 months ago

Recall and TPR

D and E

upvoted 2 times

  **kawow** 1 year, 6 months ago

AB is the answer

upvoted 1 times

A machine learning specialist is developing a proof of concept for government users whose primary concern is security. The specialist is using Amazon SageMaker to train a convolutional neural network (CNN) model for a photo classifier application. The specialist wants to protect the data so that it cannot be accessed and transferred to a remote host by malicious code accidentally installed on the training container. Which action will provide the MOST secure protection?

- A. Remove Amazon S3 access permissions from the SageMaker execution role.
- B. Encrypt the weights of the CNN model.
- C. Encrypt the training and validation dataset.
- D. Enable network isolation for training jobs.

Correct Answer: D

Community vote distribution

D (100%)

- AShahine21

Highly Voted

1 year, 6 months ago

I will go with D, "cannot be accessed and transferred to a remote host by malicious code accidentally installed on the training container"

Based on the following link: <https://aws.amazon.com/blogs/security/secure-deployment-of-amazon-sagemaker-resources/EnableNetworkIsolation> – Set this to true when creating training, hyperparameter tuning, and inference jobs to prevent situations like malicious code being accidentally installed and transferring data to a remote host."

upvoted 17 times
- achiko

Highly Voted

1 year, 7 months ago

If you enable network isolation, the containers can't make any outbound network calls, even to other AWS services such as Amazon S3. Additionally, no AWS credentials are made available to the container runtime environment. In the case of a training job with multiple instances, network inbound and outbound traffic is limited to the peers of each training container. SageMaker still performs download and upload operations against Amazon S3 using your SageMaker execution role in isolation from the training or inference container.

upvoted 8 times
- Dr_Kiko

1 year, 5 months ago

ahaha this link literally contains the answer

For example, a malicious user or code that you accidentally install on the container (in the form of a publicly available source code library) could access your data and transfer it to a remote host.

upvoted 1 times
- Valcilio

Most Recent

1 month, 3 weeks ago

Selected Answer: D

It's D, not C because encrypted can be stole.

upvoted 3 times
- halfway

1 year, 5 months ago

I choose D. More document about it: <https://docs.aws.amazon.com/sagemaker/latest/dg/mkt-algo-model-internet-free.html>

upvoted 2 times
- benson2021

1 year, 6 months ago

Answer is D.

<https://aws.amazon.com/blogs/security/secure-deployment-of-amazon-sagemaker-resources/> search for 'isolation' and there is a security parameter : EnableNetworkIsolation talking about this.

upvoted 3 times
- Vita_Rasta84444

1 year, 6 months ago

I would choose C

upvoted 1 times

-   **omar_bahrain** 1 year, 7 months ago
most likely it is C.
<https://docs.aws.amazon.com/sagemaker/latest/dg/data-protection.html>
upvoted 5 times
-   **Dr_Kiko** 1 year, 5 months ago
incorrect; you CAN transfer encrypted files even w/o a key
D is a better option
upvoted 1 times

A medical imaging company wants to train a computer vision model to detect areas of concern on patients' CT scans. The company has a large collection of unlabeled CT scans that are linked to each patient and stored in an Amazon S3 bucket. The scans must be accessible to authorized users only. A machine learning engineer needs to build a labeling pipeline.

Which set of steps should the engineer take to build the labeling pipeline with the LEAST effort?

- A. Create a workforce with AWS Identity and Access Management (IAM). Build a labeling tool on Amazon EC2 Queue images for labeling by using Amazon Simple Queue Service (Amazon SQS). Write the labeling instructions.
- B. Create an Amazon Mechanical Turk workforce and manifest file. Create a labeling job by using the built-in image classification task type in Amazon SageMaker Ground Truth. Write the labeling instructions.
- C. Create a private workforce and manifest file. Create a labeling job by using the built-in bounding box task type in Amazon SageMaker Ground Truth. Write the labeling instructions.
- D. Create a workforce with Amazon Cognito. Build a labeling web application with AWS Amplify. Build a labeling workflow backend using AWS Lambda. Write the labeling instructions.

Correct Answer: B

Community vote distribution

C (100%)

  **joep21** Highly Voted 1 year, 7 months ago

I would answer C, because of the requirement that authorized users should only have access. These users will comprise the private workforce of AWS Ground Truth. See documentation: <https://docs.aws.amazon.com/sagemaker/latest/dg/sms-workforce-private.html>

upvoted 16 times

  **cnethers** 1 year, 7 months ago

agree C

upvoted 2 times

  **astonm13** 1 year, 6 months ago

Yes it is C

upvoted 1 times

  **CharlesChiang** 1 year, 6 months ago

Agree C

upvoted 3 times

  **benson2021** Highly Voted 1 year, 5 months ago

Answer is C. The question mentions that "to detect *areas* of concern on patients' CT scans", that can be achieved by bounding box instead of image classification.

bounding box: <https://docs.aws.amazon.com/sagemaker/latest/dg/sms-bounding-box.html>

image classification: <https://docs.aws.amazon.com/sagemaker/latest/dg/sms-image-classification.html>

upvoted 7 times

  **ZSun** 1 week ago

The concern here is not about "object detection" or "image classification". it is about using "ground truth" and "private workforce"

upvoted 1 times

  **Ajose0** Most Recent 2 months, 2 weeks ago

Selected Answer: C

This option would allow the medical imaging company to create a private workforce, which can ensure that only authorized users have access to the scans, and to use Amazon SageMaker Ground Truth to create a labeling job, which would simplify the labeling pipeline process.

upvoted 3 times

  **Shailendraa** 7 months, 3 weeks ago

12-sep exam

upvoted 1 times

  **ovokpus** 10 months, 1 week ago

Selected Answer: C

C - GroundTruth and privacy concerns

upvoted 2 times

A company is using Amazon Textract to extract textual data from thousands of scanned text-heavy legal documents daily. The company uses this information to process loan applications automatically. Some of the documents fail business validation and are returned to human reviewers, who investigate the errors. This activity increases the time to process the loan applications.

What should the company do to reduce the processing time of loan applications?

- A. Configure Amazon Textract to route low-confidence predictions to Amazon SageMaker Ground Truth. Perform a manual review on those words before performing a business validation.
- B. Use an Amazon Textract synchronous operation instead of an asynchronous operation.
- C. Configure Amazon Textract to route low-confidence predictions to Amazon Augmented AI (Amazon A2I). Perform a manual review on those words before performing a business validation.
- D. Use Amazon Rekognition's feature to detect text in an image to extract the data from scanned images. Use this information to process the loan applications.

Correct Answer: C

Community vote distribution

C (100%)

- joep21

Highly Voted

1 year, 6 months ago

I agree with C, given we are evaluating model inferences (predictions). See <https://aws.amazon.com/augmented-ai/> and <https://aws.amazon.com/blogs/machine-learning/automated-monitoring-of-your-machine-learning-models-with-amazon-sagemaker-model-monitor-and-sending-predictions-to-human-review-workflows-using-amazon-a2i/>

upvoted 19 times
- Dr_Kiko

1 year, 6 months ago

yeap, it literally says it there
Loan or mortgage applications, tax forms, and many other financial documents contain millions of data points which need to be processed and extracted quickly and effectively. Using Amazon Textract and Amazon A2I you can extract critical data from these forms

upvoted 5 times
- alp_ileri

Most Recent

1 month, 3 weeks ago

why not a?

upvoted 1 times
- ZSun

1 week ago

the differences rely on the function of these two service. Ground Truth is used for "labeling" typically, text or image label: if the service cannot automatically label the data, it send to ground truth and wait for human to label it.
but A2I is for validate prediction. The model already predict the results and human then add views to it.

upvoted 1 times
- alp_ileri

1 month, 3 weeks ago

i think ground truth can do same task instead of Augmented AI

upvoted 2 times
- Valcilio

1 month, 3 weeks ago

Selected Answer: C

The answer is C, Augmented AI is made for review ML predictions!

upvoted 2 times
- Ajose0

2 months, 2 weeks ago

Selected Answer: C

By routing the low-confidence predictions to Amazon Augmented AI, the company can reduce the time to process the loan applications by leveraging human intelligence to review and validate the predictions. This way, the company can quickly address any errors or mistakes that Amazon Textract might make, reducing the time to process loan applications.

upvoted 1 times
- Juka3lj

1 year, 6 months ago

correct is C

upvoted 2 times

A company ingests machine learning (ML) data from web advertising clicks into an Amazon S3 data lake. Click data is added to an Amazon Kinesis data stream by using the Kinesis Producer Library (KPL). The data is loaded into the S3 data lake from the data stream by using an Amazon Kinesis Data Firehose delivery stream. As the data volume increases, an ML specialist notices that the rate of data ingested into Amazon S3 is relatively constant. There also is an increasing backlog of data for Kinesis Data Streams and Kinesis Data Firehose to ingest. Which next step is MOST likely to improve the data ingestion rate into Amazon S3?

- A. Increase the number of S3 prefixes for the delivery stream to write to.
- B. Decrease the retention period for the data stream.
- C. Increase the number of shards for the data stream.
- D. Add more consumers using the Kinesis Client Library (KCL).

Correct Answer: C

Community vote distribution

C (67%)

A (33%)

- SophieSu

Highly Voted

1 year, 7 months ago

C is the correct answer. # of shard is determined by:
1. # of transactions per second times
2. data blob eg. 100 KB in size
3. One shard can Ingest 1 MB/second
upvoted 36 times
- dolorez

Highly Voted

11 months, 2 weeks ago

the answer should be A - the reason why shards are not the right answer is the lack of ProvisionedThroughputExceeded exceptions that occur when a KDS has too few shards. The scenario talks about a consistent pace of delivery into S3 and a rising backlog of data (which indicates KDS stream is still able to ingest data) in the stream, hence the S3 write limit per prefix is at fault:

<https://www.amazonaws.cn/en/kinesis/data-streams/faqs/#:~:text=Q%3A%20What%20happens%20if%20the%20capacity%20limits%20of%20a%20Kinesis%20stream%20are%20exceeded%20while%20the%20data%20producer%20adds%20data%20to%20the%20stream%3F>

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/optimizing-performance.html>
upvoted 9 times
- MIlb

Most Recent

3 weeks, 6 days ago

Selected Answer: C

To increase ingest
upvoted 3 times
- AjoseO

1 month, 1 week ago

Selected Answer: C

The increasing backlog of data for Kinesis Data Streams and Kinesis Data Firehose indicates that the ingestion rate is slower than the data production rate. Therefore, the next step to improve the data ingestion rate into Amazon S3 is to increase the capacity of Kinesis Data Streams by increasing the number of shards. This will increase the parallelism of data processing, allowing for a higher throughput rate. Option C is the correct answer.

Option A is incorrect because increasing the number of S3 prefixes for the delivery stream will not directly affect the ingestion rate into S3.
upvoted 4 times

Aninina

3 months, 3 weeks ago

Selected Answer: C

To improve the data ingestion rate into Amazon S3, the ML specialist should consider increasing the number of shards for the Kinesis data stream. A Kinesis data stream is made up of one or more shards, and each shard provides a fixed amount of capacity for ingesting and storing data. By increasing the number of shards, the specialist can increase the overall capacity of the data stream and improve the rate at which data is ingested.
upvoted 3 times

GauravLahotiML

5 months, 1 week ago

Selected Answer: C

C is the correct answer
upvoted 3 times

aScientist

5 months, 4 weeks ago

Selected Answer: C

https://www.examttopics.com/exams/amazon/aws-certified-machine-learning-specialty/custom-view/

249/406

Clearly S3 is a bottleneck. S3 has parallel performance across prefixes, thus increasing throughput
upvoted 3 times

  **niopio** 7 months, 2 weeks ago

Selected Answer: A

It seems S3 is the bottleneck. Adding more prefixes will help:
<https://docs.aws.amazon.com/AmazonS3/latest/userguide/optimizing-performance.html>
upvoted 5 times

  **Shailendraa** 7 months, 3 weeks ago

12-sep exam
upvoted 1 times

  **V_B_** 8 months, 3 weeks ago

The question seems to indicate the problem in the ability of S3 to load the data. Therefore, I think the answer is A.
<https://docs.aws.amazon.com/firehose/latest/dev/dynamic-partitioning.html>
upvoted 2 times

  **KlaudYu** 9 months, 2 weeks ago

Selected Answer: A

"As the amount of data rises, a machine learning professional sees that the pace at which data is fed into Amazon S3 remains rather consistent". It means there is a bottleneck in S3
upvoted 3 times

  **AddiWei** 1 year, 2 months ago

This question is a piece of sherd. I shadded trying to answer this question.
upvoted 6 times

A data scientist must build a custom recommendation model in Amazon SageMaker for an online retail company. Due to the nature of the company's products, customers buy only 4-5 products every 5-10 years. So, the company relies on a steady stream of new customers. When a new customer signs up, the company collects data on the customer's preferences. Below is a sample of the data available to the data scientist.

timestamp	user_id	product_id	preference_1	...	preference_10
2020-03-04	90	25	0.374	...	0.374
2020-03-04	90	61	0	...	0.374
2020-02-21	203	56	1	...	0.098

How should the data scientist split the dataset into a training and test set for this use case?

- A. Shuffle all interaction data. Split off the last 10% of the interaction data for the test set.
- B. Identify the most recent 10% of interactions for each user. Split off these interactions for the test set.
- C. Identify the 10% of users with the least interaction data. Split off all interaction data from these users for the test set.
- D. Randomly select 10% of the users. Split off all interaction data from these users for the test set.

Correct Answer: D

Community vote distribution

B (86%)

14%

- joep21

Highly Voted

1 year, 7 months ago

I would select B, straight from this AWS example: <https://aws.amazon.com/blogs/machine-learning/building-a-customized-recommender-system-in-amazon-sagemaker/>

upvoted 22 times
- ttsun

1 year, 5 months ago

the blog didn't mentioned anything about sample selection. how is B arrived?

upvoted 2 times
- NicZ1111

Highly Voted

1 year, 6 months ago

I think the answer is D because customers by only 4-5 products every 5-10 years so it doesn't make sense to get 10% interactions for each user as a test set.

upvoted 5 times
- VinceCar

5 months, 3 weeks ago

B. Recommendation should use the historcial to predict the furture action. B is using the older records to prediect the newer records. D is using 90% user to predict other 10%, 90% is irrelevant to other 10%.

upvoted 1 times
- jrff

6 months ago

Yes, agree. Answer should be D

upvoted 1 times
- cox1960

Most Recent

3 days, 7 hours ago

Selected Answer: D

You want different users in training and in testing datasets, which is C or D. In addition, B is wrong since you cannot take 10% of 4-5 transactions per customer. Actually, between B, C and D, only in D you can get exactly 10%.

upvoted 1 times
- Ajose0

2 months, 2 weeks ago

Selected Answer: B

This method is appropriate because it takes into account the unique buying behavior of each customer and is likely to reflect the latest preferences of the customer. It ensures that the test set contains a representative sample of the most recent customer preferences, which is important in this use case where customer preferences change infrequently over time.

upvoted 1 times
- aScientist

5 months, 4 weeks ago

Selected Answer: B

B makes the most business sense. Since customers buy products every 4-5 years, it makes sense to be able to predict future sales from really old data. splitting the test set to be only recent interactions is the best way to test model performance from historically 'recent' data

upvoted 1 times
- vetaal

1 year, 3 months ago

Selected Answer: B

It's a time sensitive dataset and hence B makes most sense as test set.
upvoted 1 times

  **gggsrs** 1 year, 3 months ago

Selected Answer: B

I think it is a problem of leakage, so B is the correct answer
<https://www.datarobot.com/wiki/target-leakage/>
upvoted 3 times

  **Juka3lj** 1 year, 6 months ago

B is correct, takes all users into consideration
D is wrong because it takes only 10% of total number of users.
upvoted 1 times

Question #131

Topic 1

A financial services company wants to adopt Amazon SageMaker as its default data science environment. The company's data scientists run machine learning (ML) models on confidential financial data. The company is worried about data egress and wants an ML engineer to secure the environment. Which mechanisms can the ML engineer use to control data egress from SageMaker? (Choose three.)

- A. Connect to SageMaker by using a VPC interface endpoint powered by AWS PrivateLink.
- B. Use SCPs to restrict access to SageMaker.
- C. Disable root access on the SageMaker notebook instances.
- D. Enable network isolation for training jobs and models.
- E. Restrict notebook presigned URLs to specific IPs used by the company.
- F. Protect data with encryption at rest and in transit. Use AWS Key Management Service (AWS KMS) to manage encryption keys.

Correct Answer: BDF

Community vote distribution

ADE (70%)

ADF (26%)

4%

- SophieSu

Highly Voted

1 year, 6 months ago

ADF - the concepts in ADF are explained in detail on the official Amazon Exam Readiness Exam Readiness: AWS Certified Machine Learning - Specialty. Amazon official materials do not mention other concepts in BCE.

upvoted 31 times
- scuzzy2010

1 year, 6 months ago

I agree with ADF. SCP is to control access to a service, it's not related to securing data.

upvoted 3 times
- rahulw230

Highly Voted

1 year, 6 months ago

As per official document only 4 ways to do data egress Enforcing deployment in VPC,Enforcing network isolation,Restricting notebook pre-signed URLs to IPs,Disabling internet access

Correct Ans - ADE

Read Controlling data egress section

Link - <https://aws.amazon.com/blogs/machine-learning/millennium-management-secure-machine-learning-using-amazon-sagemaker/>

upvoted 21 times
- mawsmann

Most Recent

2 weeks ago

Selected Answer: ADE

Not F because the que4stion is "to control data egress". F (encryption) is not egress control.

upvoted 2 times
- codehive

3 weeks, 2 days ago

Selected Answer: ADF

A, D, F are the mechanisms that the ML engineer can use to control data egress from SageMaker. B, C, and E do not directly control data egress from SageMaker. SCPs restrict access to AWS services, disabling root access on the SageMaker notebook instances improves security, and restricting notebook presigned URLs to specific IPs used by the company adds another layer of security, but none of these mechanisms control data egress from SageMaker.

upvoted 2 times
- MIlb

3 weeks, 6 days ago

Selected Answer: ADE

<https://aws.amazon.com/blogs/machine-learning/millennium-management-secure-machine-learning-using-amazon-sagemaker/>

upvoted 1 times
- MIlb

4 weeks ago

Selected Answer: ADF

Are the correct

upvoted 2 times
- aScientist

5 months, 4 weeks ago

Selected Answer: DEF

According to the subheadings in this case study:

<https://aws.amazon.com/blogs/machine-learning/millennium-management-secure-machine-learning-using-amazon-sagemaker/>

The relevant options are:
Controlling data egress:
Enforcing deployment in VPC: This does not require a VPN to be enabled
Enforcing network isolation
Restricting notebook pre-signed URLs to IPs
Disabling internet access

Enforcing encryption:
Enforcing job encryption: sagemaker:VolumeKmsKey
upvoted 1 times

  **DJiang** 11 months, 3 weeks ago

Selected Answer: ADF

I think option E "Restrict notebook presigned URLs to specific IPs used by the company. " is ambiguous. The official blog post at <https://aws.amazon.com/blogs/machine-learning/millennium-management-secure-machine-learning-using-amazon-sagemaker/> does indicate that the objective can be achieved by limiting access to the notebook by blocking some IPs. The point is not presigned URLs, but blocking IPs. The option E however puts more emphasis on presigned URLs.
upvoted 2 times

  **vetaal** 1 year, 3 months ago

Selected Answer: ADE

Per thr following link, it is ADE - see section on controlling data egress - <https://aws.amazon.com/blogs/machine-learning/millennium-management-secure-machine-learning-using-amazon-sagemaker/>
upvoted 5 times

  **geekgirl007** 1 year, 3 months ago

Selected Answer: ADE

the answer is ADE. see this doc "controlling data egress" section; <https://aws.amazon.com/blogs/machine-learning/millennium-management-secure-machine-learning-using-amazon-sagemaker/>
upvoted 7 times

  **hess** 1 year, 3 months ago

Selected Answer: ADE

SCP only restrict access to the service not the data.
upvoted 1 times

  **Dr_Kiko** 1 year, 5 months ago

C D E - read the article mentioned below
upvoted 2 times

  **Huy** 1 year, 6 months ago

BDE yes.
A is for Sagemaker access while F is for data encryption. They are not specific for the data egress control.
upvoted 4 times

  **btsql** 1 year, 6 months ago

ADF is myanswer. SCP is just secure copy.
upvoted 2 times

  **DSJingguo** 1 year, 6 months ago

ADF
F: Protect data with encryption at rest and in transit
A: Internetwork Traffic Privacy
D: Infrastructure Security
<https://docs.aws.amazon.com/sagemaker/latest/dg/data-protection.html>
upvoted 3 times

  **DonRichy** 1 year, 6 months ago

Answer is ADE, check below:
<https://aws.amazon.com/blogs/machine-learning/millennium-management-secure-machine-learning-using-amazon-sagemaker/>
upvoted 4 times

  **DonRichy** 1 year, 6 months ago

Sorry , its BDE and not ADE
upvoted 6 times

  **astonm13** 1 year, 6 months ago

I think it is ADE
upvoted 7 times

A company needs to quickly make sense of a large amount of data and gain insight from it. The data is in different formats, the schemas change frequently, and new data sources are added regularly. The company wants to use AWS services to explore multiple data sources, suggest schemas, and enrich and transform the data. The solution should require the least possible coding effort for the data flows and the least possible infrastructure management.

Which combination of AWS services will meet these requirements?

A.

- ☞ Amazon EMR for data discovery, enrichment, and transformation
- ☞ Amazon Athena for querying and analyzing the results in Amazon S3 using standard SQL
- ☞ Amazon QuickSight for reporting and getting insights

B.

- ☞ Amazon Kinesis Data Analytics for data ingestion
- ☞ Amazon EMR for data discovery, enrichment, and transformation
- ☞ Amazon Redshift for querying and analyzing the results in Amazon S3

C.

- ☞ AWS Glue for data discovery, enrichment, and transformation
- ☞ Amazon Athena for querying and analyzing the results in Amazon S3 using standard SQL
- ☞ Amazon QuickSight for reporting and getting insights

D.

- ☞ AWS Data Pipeline for data transfer
- ☞ AWS Step Functions for orchestrating AWS Lambda jobs for data discovery, enrichment, and transformation
- ☞ Amazon Athena for querying and analyzing the results in Amazon S3 using standard SQL
- ☞ Amazon QuickSight for reporting and getting insights

Correct Answer: A

🗉 👤 knightknt Highly Voted 1 year ago
I would choose C.
upvoted 34 times

🗉 👤 ovokpus Highly Voted 10 months, 1 week ago
Answer here is C. Glue, Athena and Quicksight are serverless and need little code (only SQL)
upvoted 10 times

🗉 👤 Ahmedhadi_ Most Recent 2 weeks ago
answer is c as data sources varies alot so requires glue crawler
upvoted 1 times

🗉 👤 mite_gvg 2 weeks, 2 days ago
C Is correct, you use Glue for ingestion
upvoted 2 times

🗉 👤 codehive 3 weeks, 2 days ago
Option C is the most suitable choice to meet the given requirements. AWS Glue is a fully managed extract, transform, and load (ETL) service that allows users to discover, enrich, and transform data easily, without the need for extensive coding. It supports different data sources, schema detection, and schema evolution, which makes it an ideal choice for the given scenario. Amazon Athena, a serverless interactive query service, allows users to run standard SQL queries against data stored in Amazon S3, which makes it easy to analyze the enriched and transformed data. Amazon QuickSight is a cloud-based business intelligence service that can connect to various data sources, including Amazon Athena, to create interactive dashboards and reports, which makes it a suitable choice for gaining insights from the data.
upvoted 1 times

🗉 👤 codehive 3 weeks, 2 days ago
Option A is not an ideal choice because Amazon EMR is a heavy-weight service and requires more infrastructure management than AWS Glue.
upvoted 1 times

🗉 👤 Siyuan_Zhu 2 months, 1 week ago
Go with C here
upvoted 1 times

🗉 👤 damaldon 2 months, 4 weeks ago
Answer is C
upvoted 1 times

🗉 👤 GauravLahotiML 5 months, 1 week ago

C is thr correct answer
upvoted 2 times

  **prathapsingh** 6 months, 1 week ago
I think A. EMR to handle volume and velocity and variety
upvoted 1 times

  **f4bi4n** 10 months, 2 weeks ago
A: too much infrastructure & code
B: Data intake with KDA is non-sense & too much infrastructure & code
C: glue, code but no infra admin, same for athena and quicksight
D: could work but multiple Lambda produces a code overhead and there is no data ingest in the question
upvoted 1 times

Question #133

Topic 1

A company is converting a large number of unstructured paper receipts into images. The company wants to create a model based on natural language processing (NLP) to find relevant entities such as date, location, and notes, as well as some custom entities such as receipt numbers. The company is using optical character recognition (OCR) to extract text for data labeling. However, documents are in different structures and formats, and the company is facing challenges with setting up the manual workflows for each document type. Additionally, the company trained a named entity recognition (NER) model for custom entity detection using a small sample size. This model has a very low confidence score and will require retraining with a large dataset.

Which solution for text extraction and entity detection will require the LEAST amount of effort?

- A. Extract text from receipt images by using Amazon Textract. Use the Amazon SageMaker BlazingText algorithm to train on the text for entities and custom entities.
- B. Extract text from receipt images by using a deep learning OCR model from the AWS Marketplace. Use the NER deep learning model to extract entities.
- C. Extract text from receipt images by using Amazon Textract. Use Amazon Comprehend for entity detection, and use Amazon Comprehend custom entity recognition for custom entity detection.
- D. Extract text from receipt images by using a deep learning OCR model from the AWS Marketplace. Use Amazon Comprehend for entity detection, and use Amazon Comprehend custom entity recognition for custom entity detection.

Correct Answer: C

Reference:
<https://aws.amazon.com/blogs/machine-learning/building-an-nlp-powered-search-index-with-amazon-textract-and-amazon-comprehend/>

Community vote distribution

C (100%)

  **exam_prep** Highly Voted  11 months, 1 week ago
C is the correct answer. You definitely need Amazon Textract service which eliminate options B & D. Between A & C - Comprehend will quicker.
upvoted 15 times

  **alp_ileri** Most Recent  1 month, 3 weeks ago
Selected Answer: C
I go for C
upvoted 2 times

  **Valcilio** 1 month, 3 weeks ago
Selected Answer: C
C is the best answer, textract is to extract data from documents and comprehend to understand the filling, objective or origin of a file.
upvoted 1 times

  **damaldon** 2 months, 4 weeks ago
C is correct, you can extract Entity information easily with Comprehend.
<https://aws.amazon.com/comprehend/features/>
upvoted 4 times

A company is building a predictive maintenance model based on machine learning (ML). The data is stored in a fully private Amazon S3 bucket that is encrypted at rest with AWS Key Management Service (AWS KMS) CMKs. An ML specialist must run data preprocessing by using an Amazon SageMaker Processing job that is triggered from code in an Amazon SageMaker notebook. The job should read data from Amazon S3, process it, and upload it back to the same S3 bucket.

The preprocessing code is stored in a container image in Amazon Elastic Container Registry (Amazon ECR). The ML specialist needs to grant permissions to ensure a smooth data preprocessing workflow.

Which set of actions should the ML specialist take to meet these requirements?

- A. Create an IAM role that has permissions to create Amazon SageMaker Processing jobs, S3 read and write access to the relevant S3 bucket, and appropriate KMS and ECR permissions. Attach the role to the SageMaker notebook instance. Create an Amazon SageMaker Processing job from the notebook.
- B. Create an IAM role that has permissions to create Amazon SageMaker Processing jobs. Attach the role to the SageMaker notebook instance. Create an Amazon SageMaker Processing job with an IAM role that has read and write permissions to the relevant S3 bucket, and appropriate KMS and ECR permissions.
- C. Create an IAM role that has permissions to create Amazon SageMaker Processing jobs and to access Amazon ECR. Attach the role to the SageMaker notebook instance. Set up both an S3 endpoint and a KMS endpoint in the default VPC. Create Amazon SageMaker Processing jobs from the notebook.
- D. Create an IAM role that has permissions to create Amazon SageMaker Processing jobs. Attach the role to the SageMaker notebook instance. Set up an S3 endpoint in the default VPC. Create Amazon SageMaker Processing jobs with the access key and secret key of the IAM user with appropriate KMS and ECR permissions.

Correct Answer: D

Community vote distribution

A (83%)

B (17%)

  **spaceexplorer**

Highly Voted 

 1 year ago

Selected Answer: A

A; IAM assigned to SageMaker Notebook instance can be passed to other SageMaker jobs like training; processing, automl, etc., upvoted 10 times

  **mawsmann**

Most Recent 

 2 weeks ago

Selected Answer: B

B is least privilege, since the notebook only needs access to sagemaker processing and the processing instance needs access to S3, KMS and ECR. upvoted 2 times

  **luckybme** 2 months, 3 weeks ago

Selected Answer: B

I think the answer is B. You need to pass a role to the DataProcessing job and the other options don't mention anything about passing a role to the call that submits the job. upvoted 1 times

  **Jerry84** 3 months, 1 week ago

Selected Answer: A

"Passing Roles
Actions like passing a role between services are a common function within SageMaker."

<https://docs.aws.amazon.com/sagemaker/latest/dg/sagemaker-roles.html>
upvoted 1 times

  **jrff** 6 months ago

D. Sagemaker need to access private S3. Only sagemaker with VPC endpoint is able to create a VPC link with S3 private link. upvoted 1 times

  **V_B_** 8 months, 3 weeks ago

Why not D? Seems that it should work
upvoted 1 times

  **tsangckl** 5 months ago

I think attach the endpoint in the default VPC is a problem for private
upvoted 1 times

- azi_2021** 11 months, 1 week ago

Selected Answer: A

Answer is A
upvoted 4 times
- exam_prep** 11 months, 2 weeks ago

Option A has its own merits by assigning required IAM permissions once to the notebook and then run different processing job without worrying about the permissions.
upvoted 1 times
- exam_prep** 11 months, 2 weeks ago

Why not B? What if you want to run multiple jobs from the same sagemaker notebook with different access requirements?
upvoted 1 times
- f4bi4n** 10 months, 2 weeks ago

I wanted to go also with B BUT the ECR could be the problem.
In this example (<https://docs.aws.amazon.com/sagemaker/latest/dg/build-your-own-processing-container.html>) they use also a role for the job.
But at the top they are writing:
"Ensure that your SageMaker IAM role can pull the image from Amazon ECR"
So probably A
upvoted 1 times

Question #135

Topic 1

A data scientist has been running an Amazon SageMaker notebook instance for a few weeks. During this time, a new version of Jupyter Notebook was released along with additional software updates. The security team mandates that all running SageMaker notebook instances use the latest security and software updates provided by SageMaker.

How can the data scientist meet this requirements?

- A. Call the CreateNotebookInstanceLifecycleConfig API operation
- B. Create a new SageMaker notebook instance and mount the Amazon Elastic Block Store (Amazon EBS) volume from the original instance
- C. Stop and then restart the SageMaker notebook instance
- D. Call the UpdateNotebookInstanceLifecycleConfig API operation

Correct Answer: C

Reference:
<https://docs.aws.amazon.com/sagemaker/latest/dg/nbi-software-updates.html>

Community vote distribution

C (100%)

- cron0001**

Highly Voted

 1 year ago

Selected Answer: C

This is correct according to official documentation.
<https://docs.aws.amazon.com/sagemaker/latest/dg/nbi-software-updates.html>
upvoted 14 times
- <https://www.examtopics.com/exams/amazon/aws-certified-machine-learning-specialty/custom-view/>

258/406

A library is developing an automatic book-borrowing system that uses Amazon Rekognition. Images of library members' faces are stored in an Amazon S3 bucket.

When members borrow books, the Amazon Rekognition CompareFaces API operation compares real faces against the stored faces in Amazon S3. The library needs to improve security by making sure that images are encrypted at rest. Also, when the images are used with Amazon Rekognition, they need to be encrypted in transit. The library also must ensure that the images are not used to improve Amazon Rekognition as a service. How should a machine learning specialist architect the solution to satisfy these requirements?

- A. Enable server-side encryption on the S3 bucket. Submit an AWS Support ticket to opt out of allowing images to be used for improving the service, and follow the process provided by AWS Support.
- B. Switch to using an Amazon Rekognition collection to store the images. Use the IndexFaces and SearchFacesByImage API operations instead of the CompareFaces API operation.
- C. Switch to using the AWS GovCloud (US) Region for Amazon S3 to store images and for Amazon Rekognition to compare faces. Set up a VPN connection and only call the Amazon Rekognition API operations through the VPN.
- D. Enable client-side encryption on the S3 bucket. Set up a VPN connection and only call the Amazon Rekognition API operations through the VPN.

Correct Answer: B

Community vote distribution

A (100%)

- knighktnt

Highly Voted

1 year ago

A Images passed to Amazon Rekognition API operations may be stored and used to improve the service unless you unless you have opted out by visiting the AI services opt-out policy page and following the process explained there
<https://docs.aws.amazon.com/rekognition/latest/dg/security-data-encryption.html>
upvoted 8 times
- BoroJohn

4 months, 1 week ago

https://docs.aws.amazon.com/organizations/latest/userguide/orgs_manage_policies_ai-opt-out.html
upvoted 1 times
- tgaos

11 months, 1 week ago

So the answer is A
upvoted 1 times
- ovokpus

Highly Voted

10 months ago

Selected Answer: A

Absolutely A. Rekognition API endpoints only support secure connections over HTTPS and all communication is encrypted in transit with TLS
upvoted 6 times
- blanco750

Most Recent

1 month, 2 weeks ago

According to Rekognition FAQs, You may opt out of having your image and video inputs used to improve or develop the quality of Amazon Rekognition and other Amazon machine-learning/artificial-intelligence technologies by using an AWS Organizations opt-out policy.
<https://aws.amazon.com/rekognition/faqs/>
upvoted 1 times
- mlcert1

4 months, 2 weeks ago

how is it A???
upvoted 1 times

A company is building a line-counting application for use in a quick-service restaurant. The company wants to use video cameras pointed at the line of customers at a given register to measure how many people are in line and deliver notifications to managers if the line grows too long. The restaurant locations have limited bandwidth for connections to external services and cannot accommodate multiple video streams without impacting other operations.

Which solution should a machine learning specialist implement to meet these requirements?

- A. Install cameras compatible with Amazon Kinesis Video Streams to stream the data to AWS over the restaurant's existing internet connection. Write an AWS Lambda function to take an image and send it to Amazon Rekognition to count the number of faces in the image. Send an Amazon Simple Notification Service (Amazon SNS) notification if the line is too long.
- B. Deploy AWS DeepLens cameras in the restaurant to capture video. Enable Amazon Rekognition on the AWS DeepLens device, and use it to trigger a local AWS Lambda function when a person is recognized. Use the Lambda function to send an Amazon Simple Notification Service (Amazon SNS) notification if the line is too long.
- C. Build a custom model in Amazon SageMaker to recognize the number of people in an image. Install cameras compatible with Amazon Kinesis Video Streams in the restaurant. Write an AWS Lambda function to take an image. Use the SageMaker endpoint to call the model to count people. Send an Amazon Simple Notification Service (Amazon SNS) notification if the line is too long.
- D. Build a custom model in Amazon SageMaker to recognize the number of people in an image. Deploy AWS DeepLens cameras in the restaurant. Deploy the model to the cameras. Deploy an AWS Lambda function to the cameras to use the model to count people and send an Amazon Simple Notification Service (Amazon SNS) notification if the line is too long.

Correct Answer: A

Community vote distribution

D (50%)

C (28%)

B (22%)

- spaceexplorer

Highly Voted

1 year ago

Selected Answer: D

Answer is D:
A is not correct since restaurant has limited bandwidth
B is not correct since cannot enable Rekognition service on DeepLens
C is not correct the same reason as A
upvoted 13 times
- dunhill

5 months, 3 weeks ago

I also agree with D. B is incorrect due to that it's no need to do "person is recognized". It just needs to count the number of people.
upvoted 2 times
- muralipr

9 months, 3 weeks ago

B is correct with Rekognition integrated with Deeplens and no extra configuration needed. (<https://aws.amazon.com/blogs/machine-learning/building-a-smart-garage-door-opener-with-aws-deeplens-and-amazon-rekognition/>)
upvoted 5 times
- RLai

4 months ago

In this blog, rekognition service is not running on Deeplens. It said "After you deploy the sample object detection project into AWS DeepLens, you need to change the inference (edge) Lambda function to upload image frames to Amazon S3. ... and Rekognition would do its work from the Cloud to image frames on S3."... It would still consume lots of bandwidth. So B is NOT correct.
upvoted 1 times
- wjohnny

Highly Voted

4 months, 1 week ago

Selected Answer: C

AWS will not recommend to use Deeplense in production. From <https://aws.amazon.com/deeplens/device-terms-of-use/>
upvoted 5 times
- alp_ileri

1 month, 3 weeks ago

aws doesn't allow use in production but in evaluation. can we accept counting number of people as an evaluation?
upvoted 1 times
- BTRYING

3 months ago

<https://aws.amazon.com/deeplens/device-terms-of-use/>
upvoted 2 times
- MIlb

Most Recent

4 weeks ago

Selected Answer: D

Deeplens + lambda + model inference
upvoted 3 times

  **fez_2312** 1 month, 2 weeks ago

After giving this some thought, I am thinking D. Tricky, my initial answer was C. But D is a better solution - given DeepLens and counting the number of people.
upvoted 1 times

  **SANDEEP_AWS** 1 month, 2 weeks ago

Selected Answer: B

<https://aws.amazon.com/ko/blogs/machine-learning/building-a-smart-garage-door-opener-with-aws-deeplens-and-amazon-rekognition/>
upvoted 1 times

  **Amit11011996** 1 month, 2 weeks ago

According to this link,
Answer should be D, because we can directly deploy model in Deep lense to count the number of people instead a use of rekognition.
upvoted 1 times

  **lizlizliz** 4 months, 2 weeks ago

<https://aws.amazon.com/blogs/machine-learning/optimize-workforce-in-your-store-using-amazon-rekognition/>
B
upvoted 1 times

  **Ob1KN0B** 8 months, 2 weeks ago

Selected Answer: B

B is the most suitable answer. A and C can be ignored because of requirement to use Amazon Video Streams which will not go well with low internet bandwidth. DeepLens is already compatible with Rekognition so better to use it rather than creating a custom model on SageMaker.
upvoted 4 times

  **Ob1KN0B** 8 months, 2 weeks ago

https://aws.amazon.com/deeplens/community-projects/Customer_Counter/
upvoted 1 times

  **KlaudYu** 9 months, 2 weeks ago

Selected Answer: B

B is possible. not stream only send detected.
<https://aws.amazon.com/ko/blogs/machine-learning/building-a-smart-garage-door-opener-with-aws-deeplens-and-amazon-rekognition/>
upvoted 2 times

  **f4bi4n** 10 months ago

Selected Answer: C

I would go with C. It seems that it not transfers the hole stream and all others make no sense. AWS will never recommend to use Deeplense in production. From <https://aws.amazon.com/deeplens/device-terms-of-use/>

a. Permitted Uses and Limited Licence. Subject to your compliance with this Agreement, and your payment of any applicable fees: (i) you may use the AWS DeepLens Device for personal, educational, evaluation, development, and testing purposes, and not to process your production workloads;
upvoted 4 times

  **rohit07cf** 12 months ago

I would go with C. Eliminate A because it won't be able to support/handle video streams. Eliminate DeepLens options due to limited bandwidth and no support for external connections. SIDE NOTE: AWS DeepLens integrates with Amazon Rekognition for advanced image analysis, Amazon SageMaker for training models, and with Amazon Polly to create speech-enabled projects. The device also connects securely to AWS IoT, Amazon SQS, Amazon SNS, Amazon S3, Amazon DynamoDB, and more.
upvoted 1 times

A company has set up and deployed its machine learning (ML) model into production with an endpoint using Amazon SageMaker hosting services. The ML team has configured automatic scaling for its SageMaker instances to support workload changes. During testing, the team notices that additional instances are being launched before the new instances are ready. This behavior needs to change as soon as possible. How can the ML team solve this issue?

- A. Decrease the cooldown period for the scale-in activity. Increase the configured maximum capacity of instances.
- B. Replace the current endpoint with a multi-model endpoint using SageMaker.
- C. Set up Amazon API Gateway and AWS Lambda to trigger the SageMaker inference endpoint.
- D. Increase the cooldown period for the scale-out activity.

Correct Answer: A

Reference:
<https://aws.amazon.com/blogs/machine-learning/configuring-autoscaling-inference-endpoints-in-amazon-sagemaker/>

Community vote distribution

D (100%)

- cron0001

Highly Voted

1 year ago

Selected Answer: D

I believe this is a problem to do with scaling out (increasing the number of instances), cooldown period should be increased.

<https://docs.aws.amazon.com/autoscaling/ec2/userguide/Cooldown.html>

upvoted 11 times
- Ajose0

Most Recent

2 months, 2 weeks ago

Selected Answer: D

The issue is related to scaling out, specifically the fact that new instances are being launched before the existing ones are ready.

To address this issue, the ML team could consider increasing the minimum number of instances, reducing the target value for CPU utilization, or increasing the warm-up time for the instances. These actions can help to ensure that new instances are not launched until the existing ones have reached a stable state, which can prevent performance issues and ensure the reliability of the service.

upvoted 2 times
- Ajose0

2 months, 2 weeks ago

Option D, which suggests increasing the cooldown period for the scale-out activity, could potentially help to address this issue by ensuring that the new instances are not launched too quickly.

Option A, which suggests decreasing the cooldown period for the scale-in activity and increasing the maximum capacity of instances, is not an appropriate solution to the problem described. Decreasing the cooldown period for scale-in activity would result in instances being terminated too quickly, and increasing the maximum capacity of instances would not necessarily prevent new instances from being launched too quickly.

upvoted 2 times
- ystotest

5 months, 1 week ago

Selected Answer: D

Agreed with D. should be increased not decreased

upvoted 1 times
- ryuhei

7 months, 1 week ago

Selected Answer: D

Answer is "D"

upvoted 1 times
- SDikeman62

11 months, 3 weeks ago

Selected Answer: D

Definitely D.

upvoted 2 times

A telecommunications company is developing a mobile app for its customers. The company is using an Amazon SageMaker hosted endpoint for machine learning model inferences.

Developers want to introduce a new version of the model for a limited number of users who subscribed to a preview feature of the app. After the new version of the model is tested as a preview, developers will evaluate its accuracy. If a new version of the model has better accuracy, developers need to be able to gradually release the new version for all users over a fixed period of time.

How can the company implement the testing model with the LEAST amount of operational overhead?

- A. Update the ProductionVariant data type with the new version of the model by using the CreateEndpointConfig operation with the InitialVariantWeight parameter set to 0. Specify the TargetVariant parameter for InvokeEndpoint calls for users who subscribed to the preview feature. When the new version of the model is ready for release, gradually increase InitialVariantWeight until all users have the updated version.
- B. Configure two SageMaker hosted endpoints that serve the different versions of the model. Create an Application Load Balancer (ALB) to route traffic to both endpoints based on the TargetVariant query string parameter. Reconfigure the app to send the TargetVariant query string parameter for users who subscribed to the preview feature. When the new version of the model is ready for release, change the ALB's routing algorithm to weighted until all users have the updated version.
- C. Update the DesiredWeightsAndCapacity data type with the new version of the model by using the UpdateEndpointWeightsAndCapacities operation with the DesiredWeight parameter set to 0. Specify the TargetVariant parameter for InvokeEndpoint calls for users who subscribed to the preview feature. When the new version of the model is ready for release, gradually increase DesiredWeight until all users have the updated version.
- D. Configure two SageMaker hosted endpoints that serve the different versions of the model. Create an Amazon Route 53 record that is configured with a simple routing policy and that points to the current version of the model. Configure the mobile app to use the endpoint URL for users who subscribed to the preview feature and to use the Route 53 record for other users. When the new version of the model is ready for release, add a new model version endpoint to Route 53, and switch the policy to weighted until all users have the updated version.

Correct Answer: D

Community vote distribution

C (59%)

A (41%)

- ayatkhrisat

Highly Voted

12 months ago

Selected Answer: A

<https://docs.aws.amazon.com/sagemaker/latest/dg/model-ab-testing.html>

upvoted 15 times
- RLai

4 months, 1 week ago

https://sagemaker-examples.readthedocs.io/en/latest/sagemaker_endpoints/a_b_testing/a_b_testing.html

Should be A

upvoted 1 times
- ayatkhrisat

11 months ago

after reviewing it maybe C not A

upvoted 7 times
- spaceexplorer

Highly Voted

1 year ago

Selected Answer: C

Answer is C, hosting two models under single endpoint has less operational overheads than two hosting endpoints

upvoted 10 times
- Zhechen0912

Most Recent

1 month ago

Selected Answer: C

I agree with C.

upvoted 2 times
- SANDEEP_AWS

1 month, 2 weeks ago

Selected Answer: C

Please see step 4: <https://docs.aws.amazon.com/sagemaker/latest/dg/model-ab-testing.html> & in option A it's mentioned that we set initial_weight to 0 which isn't true as the value should be 1.

upvoted 1 times

  **matteocal** 9 months, 1 week ago

Selected Answer: C

I did not found the InitialVariantWeight, only DesiredWeight, therefore is C:
https://docs.aws.amazon.com/sagemaker/latest/APIReference/API_DesiredWeightAndCapacity.html
upvoted 3 times

  **KlaudYu** 9 months, 2 weeks ago

Selected Answer: C

<https://docs.aws.amazon.com/sagemaker/latest/dg/model-ab-testing.html>
Step 4: Increase traffic to the best model
Now that we have determined that Variant2 performs better than Variant1, we shift more traffic to it. We can continue to use TargetVariant to invoke a specific model variant, but a simpler approach is to update the weights assigned to each variant by calling UpdateEndpointWeightsAndCapacities.
upvoted 6 times

  **VinceCar** 5 months, 3 weeks ago

Update should be the correct action to this change.
upvoted 1 times

A company offers an online shopping service to its customers. The company wants to enhance the site's security by requesting additional information when customers access the site from locations that are different from their normal location. The company wants to update the process to call a machine learning (ML) model to determine when additional information should be requested.

The company has several terabytes of data from its existing ecommerce web servers containing the source IP addresses for each request made to the web server. For authenticated requests, the records also contain the login name of the requesting user.

Which approach should an ML specialist take to implement the new security feature in the web application?

- A. Use Amazon SageMaker Ground Truth to label each record as either a successful or failed access attempt. Use Amazon SageMaker to train a binary classification model using the factorization machines (FM) algorithm.
- B. Use Amazon SageMaker to train a model using the IP Insights algorithm. Schedule updates and retraining of the model using new log data nightly.
- C. Use Amazon SageMaker Ground Truth to label each record as either a successful or failed access attempt. Use Amazon SageMaker to train a binary classification model using the IP Insights algorithm.
- D. Use Amazon SageMaker to train a model using the Object2Vec algorithm. Schedule updates and retraining of the model using new log data nightly.

Correct Answer: C

Community vote distribution

B (100%)

- knighstknt

Highly Voted

1 year ago

B? because ip insights algorithm is unsupervised learning that don't need label

upvoted 11 times
- spidy20

Most Recent

8 months, 2 weeks ago

Selected Answer: B

Answer should be B

upvoted 4 times
- Morsa

9 months, 2 weeks ago

Selected Answer: B

Agree with the comments beliw

upvoted 1 times
- tgaos

11 months, 1 week ago

B. <https://docs.aws.amazon.com/sagemaker/latest/dg/ip-insights.html>

upvoted 2 times
- spaceexplorer

1 year ago

Selected Answer: B

B; IP Insights for IP address anomaly detection

upvoted 4 times

A retail company wants to combine its customer orders with the product description data from its product catalog. The structure and format of the records in each dataset is different. A data analyst tried to use a spreadsheet to combine the datasets, but the effort resulted in duplicate records and records that were not properly combined. The company needs a solution that it can use to combine similar records from the two datasets and remove any duplicates.

Which solution will meet these requirements?

- A. Use an AWS Lambda function to process the data. Use two arrays to compare equal strings in the fields from the two datasets and remove any duplicates.
- B. Create AWS Glue crawlers for reading and populating the AWS Glue Data Catalog. Call the AWS Glue SearchTables API operation to perform a fuzzy- matching search on the two datasets, and cleanse the data accordingly.
- C. Create AWS Glue crawlers for reading and populating the AWS Glue Data Catalog. Use the FindMatches transform to cleanse the data.
- D. Create an AWS Lake Formation custom transform. Run a transformation for matching products from the Lake Formation console to cleanse the data automatically.

Correct Answer: D

Reference:
<https://aws.amazon.com/lake-formation/features/>

Community vote distribution

C (96%)

4%

- spaceexplorer

Highly Voted

1 year ago

Selected Answer: C

C; Glue can use FindMatches transformation to find duplicates
upvoted 19 times
- KlaudYu

11 months ago

It says "Each dataset contains records with a unique structure and format.", so C would not be correct.
upvoted 2 times
- f4bi4n

10 months, 2 weeks ago

but thats exactly the use of FindMatches:
The FindMatches transform enables you to identify duplicate or matching records in your dataset, even when the records do not have a common unique identifier and no fields match exactly
upvoted 4 times
- unitit

Most Recent

3 months ago

Selected Answer: C

It is C as described in the tutorial - <https://docs.aws.amazon.com/glue/latest/dg/machine-learning-transform-tutorial.html>

LakeFormation can also invoke a FindMatches algorithm (because it manages Data Ingestion through Glue), but we don't have a data lake in this example. No one would build a whole Data Lake - a process that takes days - only to find some matching records.
upvoted 4 times
- adisabeba

4 months, 3 weeks ago

Selected Answer: D

Lake Formation helps clean and prepare your data for analysis by providing a Machine Learning (ML) Transform called FindMatches for deduplication and finding matching records. For example, use FindMatches to find duplicate records in your database of restaurants, such as when one record lists "Joe's Pizza" at "121 Main St." and another shows "Joseph's Pizzeria" at "121 Main." You don't need to know anything about ML to do this. FindMatches will simply ask you to label sets of records as either "matching" or "not matching." The system will then learn your criteria for calling a pair of records a match and will build an ML Transform that you can use to find duplicate records within a database or matching records across two databases.
<https://aws.amazon.com/lake-formation/features/>
upvoted 1 times
- ogm1

10 months, 2 weeks ago

AWS Lake Formation FindMatches is a new machine learning (ML) transform that enables you to match records across different datasets as well as identify and remove duplicate records, with little to no human intervention
Ans is D
upvoted 2 times
- ovokpus

10 months, 1 week ago

Thing is, FindMatches is not a custom transformation in LakeFormation. And LakeFormation transforms are actually Glue jobs

upvoted 1 times

  **[Removed]** 10 months, 3 weeks ago
D is correct
upvoted 2 times

Question #142

Topic 1

A company provisions Amazon SageMaker notebook instances for its data science team and creates Amazon VPC interface endpoints to ensure communication between the VPC and the notebook instances. All connections to the Amazon SageMaker API are contained entirely and securely using the AWS network.

However, the data science team realizes that individuals outside the VPC can still connect to the notebook instances across the internet.

Which set of actions should the data science team take to fix the issue?

- A. Modify the notebook instances' security group to allow traffic only from the CIDR ranges of the VPC. Apply this security group to all of the notebook instances' VPC interfaces.
- B. Create an IAM policy that allows the sagemaker:CreatePresignedNotebookInstanceUrl and sagemaker:DescribeNotebookInstance actions from only the VPC endpoints. Apply this policy to all IAM users, groups, and roles used to access the notebook instances.
- C. Add a NAT gateway to the VPC. Convert all of the subnets where the Amazon SageMaker notebook instances are hosted to private subnets. Stop and start all of the notebook instances to reassign only private IP addresses.
- D. Change the network ACL of the subnet the notebook is hosted in to restrict access to anyone outside the VPC.

Correct Answer: B

Reference:
<https://gmoein.github.io/files/Amazon%20SageMaker.pdf>



  **cron0001** Highly Voted 1 year ago
Selected Answer: B
B appears to be correct according to the official source.
<https://docs.aws.amazon.com/sagemaker/latest/dg/notebook-interface-endpoint.html#notebook-private-link-restrict>
upvoted 13 times

  **ZSun** Most Recent 6 days, 21 hours ago
The problem about a is that "You can specify allow rules, but not deny rules." <https://docs.aws.amazon.com/vpc/latest/userguide/security-group-rules.html#security-group-rule-characteristics>
Therefore, you cannot restrict the unauthorized access
upvoted 1 times

  **Chelseajcole** 1 month, 4 weeks ago
Selected Answer: A
Should be security group thing
upvoted 3 times

  **Siyuan_Zhu** 2 months, 1 week ago
Selected Answer: A
The answer should be A according to this source
"Select the Subnet and Security group(s) as part of the VPC setting. To disable direct internet access, under Direct Internet access, simply choose Disable – use VPC only"

<https://aws.amazon.com/blogs/machine-learning/customize-your-amazon-sagemaker-notebook-instances-with-lifecycle-configurations-and-the-option-to-disable-internet-access/>
upvoted 2 times

  **damaldon** 2 months, 4 weeks ago
Answer is A, as per the following link
<https://aws.amazon.com/blogs/machine-learning/customize-your-amazon-sagemaker-notebook-instances-with-lifecycle-configurations-and-the-option-to-disable-internet-access/>
upvoted 2 times

A company will use Amazon SageMaker to train and host a machine learning (ML) model for a marketing campaign. The majority of data is sensitive customer data. The data must be encrypted at rest. The company wants AWS to maintain the root of trust for the master keys and wants encryption key usage to be logged.

Which implementation will meet these requirements?

- A. Use encryption keys that are stored in AWS Cloud HSM to encrypt the ML data volumes, and to encrypt the model artifacts and data in Amazon S3.
- B. Use SageMaker built-in transient keys to encrypt the ML data volumes. Enable default encryption for new Amazon Elastic Block Store (Amazon EBS) volumes.
- C. Use customer managed keys in AWS Key Management Service (AWS KMS) to encrypt the ML data volumes, and to encrypt the model artifacts and data in Amazon S3.
- D. Use AWS Security Token Service (AWS STS) to create temporary tokens to encrypt the ML storage volumes, and to encrypt the model artifacts and data in Amazon S3.

Correct Answer: C

Community vote distribution

C (100%)

  **exam_prep** Highly Voted  11 months, 1 week ago

C is correct answer. Straight forward to use KMS.

upvoted 8 times

  **Ajose0** Most Recent  2 months, 1 week ago

Selected Answer: C

Using customer managed keys in AWS KMS will allow the company to maintain the root of trust for the master keys, and AWS KMS will log key usage. This ensures that the encryption keys used to encrypt the ML data volumes and model artifacts are properly managed and secured. Additionally, using customer managed keys allows the company to have greater control over the encryption process.

upvoted 3 times

  **Jerry84** 2 months, 4 weeks ago

Selected Answer: C

<https://docs.aws.amazon.com/kms/latest/developerguide/security-logging-monitoring.html>

upvoted 1 times

A machine learning specialist stores IoT soil sensor data in Amazon DynamoDB table and stores weather event data as JSON files in Amazon S3. The dataset in DynamoDB is 10 GB in size and the dataset in Amazon S3 is 5 GB in size. The specialist wants to train a model on this data to help predict soil moisture levels as a function of weather events using Amazon SageMaker. Which solution will accomplish the necessary transformation to train the Amazon SageMaker model with the LEAST amount of administrative overhead?

- A. Launch an Amazon EMR cluster. Create an Apache Hive external table for the DynamoDB table and S3 data. Join the Hive tables and write the results out to Amazon S3.
- B. Crawl the data using AWS Glue crawlers. Write an AWS Glue ETL job that merges the two tables and writes the output to an Amazon Redshift cluster.
- C. Enable Amazon DynamoDB Streams on the sensor table. Write an AWS Lambda function that consumes the stream and appends the results to the existing weather files in Amazon S3.
- D. Crawl the data using AWS Glue crawlers. Write an AWS Glue ETL job that merges the two tables and writes the output in CSV format to Amazon S3.

Correct Answer: C

Community vote distribution

D (100%)

- cron0001

Highly Voted

1 year ago

Selected Answer: D

D. AWS Glue can connect with DynamoDB and join both data sets together via Glue Studio. Requiring minimal overheads

upvoted 16 times
- injoho

Most Recent

1 week, 6 days ago

D.

<https://aws.amazon.com/blogs/big-data/accelerate-amazon-dynamodb-data-access-in-aws-glue-jobs-using-the-new-aws-glue-dynamodb-elt-connector/>

upvoted 2 times
- Shailendraa

7 months, 3 weeks ago

12-sep exam

upvoted 1 times

A company sells thousands of products on a public website and wants to automatically identify products with potential durability problems. The company has 1.000 reviews with date, star rating, review text, review summary, and customer email fields, but many reviews are incomplete and have empty fields. Each review has already been labeled with the correct durability result.

A machine learning specialist must train a model to identify reviews expressing concerns over product durability. The first model needs to be trained and ready to review in 2 days.

What is the MOST direct approach to solve this problem within 2 days?

- A. Train a custom classifier by using Amazon Comprehend.
- B. Build a recurrent neural network (RNN) in Amazon SageMaker by using Gluon and Apache MXNet.
- C. Train a built-in BlazingText model using Word2Vec mode in Amazon SageMaker.
- D. Use a built-in seq2seq model in Amazon SageMaker.

Correct Answer: B

Community vote distribution

A (68%) C (20%) 12%

- ayatkhrisat

Highly Voted

11 months, 3 weeks ago

Selected Answer: A

A should be the answer
upvoted 13 times
- rb39

7 months, 2 weeks ago

it's a sentiment analysis problem => comprehend
upvoted 3 times
- ckkobe24

Highly Voted

11 months, 3 weeks ago

Selected Answer: C

BlazingText can also do supervised text classification
upvoted 5 times
- MIlb

Most Recent

4 weeks ago

Selected Answer: A

Comprehend can do Custom Classification
upvoted 2 times
- MIlb

4 weeks ago

Selected Answer: A

Comprehend can do Sentiment Analysis
upvoted 1 times
- fez_2312

1 month, 2 weeks ago

The answer is C, because of the amount of data, and the time constraint. C is the most efficient solution. Conventionally A would be the right answer, but given the time constraint the answer is C.
upvoted 1 times
- alp_ileri

1 month, 3 weeks ago

I would say blaze text. Cuz comprehend needs custom code, so we have only 2 days.
upvoted 2 times
- ystotest

5 months ago

Selected Answer: A

<https://docs.aws.amazon.com/comprehend/latest/dg/how-document-classification.html>
upvoted 1 times
- cron0001

1 year ago

Selected Answer: D

If the problem needs to be solved in 2 days I would avoid going with any customised solution which would eliminate A and B. As the data is labelled already we don't need an unsupervised algorithm therefore eliminating C. Which leaves us with D
upvoted 3 times
- f4bi4n

10 months, 2 weeks ago

its exactly the opposite, because its needs to be ready in 2 day I would use Comprehend ;) You don't need to write code, you have the data already available, so its faster then D

upvoted 7 times

Question #146

Topic 1

A company that runs an online library is implementing a chatbot using Amazon Lex to provide book recommendations based on category. This intent is fulfilled by an AWS Lambda function that queries an Amazon DynamoDB table for a list of book titles, given a particular category. For testing, there are only three categories implemented as the custom slot types: "comedy," "adventure," and "documentary."

A machine learning (ML) specialist notices that sometimes the request cannot be fulfilled because Amazon Lex cannot understand the category spoken by users with utterances such as "funny," "fun," and "humor." The ML specialist needs to fix the problem without changing the Lambda code or data in DynamoDB.

How should the ML specialist fix the problem?

- A. Add the unrecognized words in the enumeration values list as new values in the slot type.
- B. Create a new custom slot type, add the unrecognized words to this slot type as enumeration values, and use this slot type for the slot.
- C. Use the AMAZON.SearchQuery built-in slot types for custom searches in the database.
- D. Add the unrecognized words as synonyms in the custom slot type.

Correct Answer: C

Community vote distribution

D (100%)

  **ovokpus** Highly Voted 10 months, 1 week ago

Selected Answer: D

D is the answer.

The unrecognized words are synonyms for "comedy", so they should be added as synonyms under the comedy slot type

see the excerpt:

"For each intent, you can specify parameters that indicate the information that the intent needs to fulfill the user's request. These parameters, or slots, have a type. A slot type is a list of values that Amazon Lex uses to train the machine learning model to recognize values for a slot. For example, you can define a slot type called "Genres." Each value in the slot type is the name of a genre, "comedy," "adventure," "documentary," etc. You can define a synonym for a slot type value. For example, you can define the synonyms "funny" and "humorous" for the value "comedy."

<https://docs.aws.amazon.com/lex/latest/dg/howitworks-custom-slots.html>

upvoted 10 times

  **knightknt** Highly Voted 1 year ago

D? can not be C.Amazon Lex doesn't support the AMAZON.LITERAL or the AMAZON.SearchQuery built-in slot types.

<https://docs.aws.amazon.com/lex/latest/dg/howitworks-builtins-slots.html>

upvoted 9 times

  **cognito_22** 11 months, 4 weeks ago

<https://docs.aws.amazon.com/lex/latest/dg/howitworks-custom-slots.html>

upvoted 5 times

  **Ajose0** Most Recent 2 months, 1 week ago

Selected Answer: D

The ML specialist should add the unrecognized words as synonyms in the custom slot type. This will allow Amazon Lex to understand the user's intent even if they use synonyms for the predefined slot values. By adding the synonyms, Amazon Lex will recognize them as variations of the predefined slot values and map them to the appropriate slot value. This approach can be a quick and effective way to improve the accuracy of the chatbot's understanding of user requests without having to change the Lambda code or the data in DynamoDB.

upvoted 1 times

  **[Removed]** 10 months, 2 weeks ago

B is correct

upvoted 2 times

A manufacturing company uses machine learning (ML) models to detect quality issues. The models use images that are taken of the company's product at the end of each production step. The company has thousands of machines at the production site that generate one image per second on average.

The company ran a successful pilot with a single manufacturing machine. For the pilot, ML specialists used an industrial PC that ran AWS IoT Greengrass with a long-running AWS Lambda function that uploaded the images to Amazon S3. The uploaded images invoked a Lambda function that was written in Python to perform inference by using an Amazon SageMaker endpoint that ran a custom model. The inference results were forwarded back to a web service that was hosted at the production site to prevent faulty products from being shipped.

The company scaled the solution out to all manufacturing machines by installing similarly configured industrial PCs on each production machine. However, latency for predictions increased beyond acceptable limits. Analysis shows that the internet connection is at its capacity limit.

How can the company resolve this issue MOST cost-effectively?

- A. Set up a 10 Gbps AWS Direct Connect connection between the production site and the nearest AWS Region. Use the Direct Connect connection to upload the images. Increase the size of the instances and the number of instances that are used by the SageMaker endpoint.
- B. Extend the long-running Lambda function that runs on AWS IoT Greengrass to compress the images and upload the compressed files to Amazon S3. Decompress the files by using a separate Lambda function that invokes the existing Lambda function to run the inference pipeline.
- C. Use auto scaling for SageMaker. Set up an AWS Direct Connect connection between the production site and the nearest AWS Region. Use the Direct Connect connection to upload the images.
- D. Deploy the Lambda function and the ML models onto the AWS IoT Greengrass core that is running on the industrial PCs that are installed on each machine. Extend the long-running Lambda function that runs on AWS IoT Greengrass to invoke the Lambda function with the captured images and run the inference on the edge component that forwards the results directly to the web service.

Correct Answer: D

Community vote distribution

D (100%)

  **cron0001** Highly Voted  1 year ago

Selected Answer: D

D is correct according to official documentation.
<https://docs.aws.amazon.com/greengrass/v1/developerguide/ml-inference.html>
upvoted 12 times

  **rb39** 7 months, 2 weeks ago
A-C: excluded out, Direct Connect is expensive
upvoted 1 times

  **Ajose0** Most Recent  2 months, 1 week ago

Selected Answer: D

Option D eliminates the need for internet connection since the inference is done on the edge component, and the results are directly forwarded to the web service.

This approach also reduces the need for larger instances and direct connect connections, thus being the most cost-effective solution.
upvoted 3 times

A data scientist is using an Amazon SageMaker notebook instance and needs to securely access data stored in a specific Amazon S3 bucket. How should the data scientist accomplish this?

- A. Add an S3 bucket policy allowing GetObject, PutObject, and ListBucket permissions to the Amazon SageMaker notebook ARN as principal.
- B. Encrypt the objects in the S3 bucket with a custom AWS Key Management Service (AWS KMS) key that only the notebook owner has access to.
- C. Attach the policy to the IAM role associated with the notebook that allows GetObject, PutObject, and ListBucket operations to the specific S3 bucket.
- D. Use a script in a lifecycle configuration to configure the AWS CLI on the instance with an access key ID and secret.

Correct Answer: C

Community vote distribution

C (69%)

A (23%)

8%

  **salads**

Highly Voted 

 8 months, 2 weeks ago

Selected Answer: C

c is the right answer
upvoted 8 times

  **tgaos**

Highly Voted 

 11 months, 1 week ago

Agree with the Answer C. Attach the policy to the IAM roal associated with the notebook.
upvoted 7 times

  **ZSun**

Most Recent 

 2 weeks ago

AC are both correct answer, but A is better than C, mostly due to the limitation of IAM policy.
IAM policies: The maximum size of an IAM policy document is 6,144 characters. You can attach up to 10 policies to an IAM user, role, or group.
upvoted 1 times

  **Ajose0** 2 months, 1 week ago

Selected Answer: C

Option C ensures that the notebook instance is granted permission to access the S3 bucket without the need to provide credentials.

Option A is incorrect because it suggests adding a bucket policy that grants permission to a specific IAM principal, which is less secure than granting permission to an IAM role.
upvoted 1 times

  **ZSun** 5 days, 2 hours ago

I dont agree with this. Restrict bucket access only to limited principal is much secure than grant specific IAM prinicap. Restrict specific principal eliminate other visits, but grant specific IAM user permission does not exclude other visit.
upvoted 1 times

  **Shailendraa** 7 months, 3 weeks ago

12-sep exam
upvoted 4 times

  **[Removed]** 10 months, 3 weeks ago

C is correct
upvoted 4 times

  **edvardo** 11 months, 2 weeks ago

Selected Answer: A

Quoting the book "Data Science on AWS":
"Generally, we would use IAM identity-based policies if we need to define permissions for more than just S3, or if we have a number of S3 buckets, each with different permissions requirements. We might want to keep access control policies in the IAM environment.

We would use S3 bucket policies if we need a simple way to grant cross-account access to our S3 environment without using IAM roles, or if we reach the size limit for our IAM policy. We might want to keep access control policies in the S3 environment."

A would be the choice then.
upvoted 3 times

  **VinceCar** 5 months, 1 week ago

For A, only some operations are allowed, no specified users or roles have been granted this permission for these operations.
upvoted 1 times

  **dunhill** 5 months, 3 weeks ago

I am not sure but in question we don't have cross-account situation?
upvoted 1 times

  **colin1919** 6 months ago

Based on this logic indeed A would be better.
upvoted 1 times

  **ayatkhrisat** 11 months, 3 weeks ago

Selected Answer: B

B is the answer
upvoted 1 times

  **VinceCar** 5 months, 2 weeks ago

Only "securely access" is required, not encryption.
upvoted 1 times

  **bluer1** 12 months ago

A - for me
upvoted 2 times

A company is launching a new product and needs to build a mechanism to monitor comments about the company and its new product on social media. The company needs to be able to evaluate the sentiment expressed in social media posts, and visualize trends and configure alarms based on various thresholds.

The company needs to implement this solution quickly, and wants to minimize the infrastructure and data science resources needed to evaluate the messages.

The company already has a solution in place to collect posts and store them within an Amazon S3 bucket.

What services should the data science team use to deliver this solution?

- A. Train a model in Amazon SageMaker by using the BlazingText algorithm to detect sentiment in the corpus of social media posts. Expose an endpoint that can be called by AWS Lambda. Trigger a Lambda function when posts are added to the S3 bucket to invoke the endpoint and record the sentiment in an Amazon DynamoDB table and in a custom Amazon CloudWatch metric. Use CloudWatch alarms to notify analysts of trends.
- B. Train a model in Amazon SageMaker by using the semantic segmentation algorithm to model the semantic content in the corpus of social media posts. Expose an endpoint that can be called by AWS Lambda. Trigger a Lambda function when objects are added to the S3 bucket to invoke the endpoint and record the sentiment in an Amazon DynamoDB table. Schedule a second Lambda function to query recently added records and send an Amazon Simple Notification Service (Amazon SNS) notification to notify analysts of trends.
- C. Trigger an AWS Lambda function when social media posts are added to the S3 bucket. Call Amazon Comprehend for each post to capture the sentiment in the message and record the sentiment in an Amazon DynamoDB table. Schedule a second Lambda function to query recently added records and send an Amazon Simple Notification Service (Amazon SNS) notification to notify analysts of trends.
- D. Trigger an AWS Lambda function when social media posts are added to the S3 bucket. Call Amazon Comprehend for each post to capture the sentiment in the message and record the sentiment in a custom Amazon CloudWatch metric and in S3. Use CloudWatch alarms to notify analysts of trends.

Correct Answer: A

Community vote distribution

D (100%)

- cron0001

Highly Voted

1 year ago

Selected Answer: D

D is the correct answer.
Following from the previous comment. The company wants to minimize the infrastructure and data science resources needed to evaluate the messages. Therefore any custom services would be eliminated (A and B). Similarly DynamoDB would add complexity to the infrastructure there C is eliminated. leaving D
upvoted 11 times
- uninit

Most Recent

3 months ago

Selected Answer: D

Amazingly D is possible - <https://catalog.us-east-1.prod.workshops.aws/workshops/4faab440-8c3a-4527-bd11-0c88a6e6213c/en-US/30-build-the-application/400-send-sentiment-to-cloudwatch>
I was so sure of option C, because sending a sentiment to a custom CloudWatch metric just didn't make any sense. But you learn something new everyday.
upvoted 3 times
- dolorez

11 months, 2 weeks ago

This is a puzzling question, as both answers C and D miss essential steps:
C is missing DynamoDB Streams to capture new records
D is missing a notification mechanism like SNS, as CloudWatch Alarms alone can only be used as a trigger, but are not sufficient for notification
I agree that A and B should be eliminated for requiring data science development
upvoted 1 times
- NILKK

1 year ago

I also do agree that D is correct answer. In A, why we are adding extra dependency of Dynamo DB.
upvoted 4 times
- knightknt

1 year ago

D, blazing text is not for sentiment analysis.
The Amazon SageMaker BlazingText algorithm provides highly optimized implementations of the Word2vec and text classification algorithms. The Word2vec algorithm is useful for many downstream natural language processing (NLP) tasks, such as sentiment analysis, named entity recognition, machine translation, etc. Text classification is an important task for applications that perform web searches, information retrieval, ranking, and document classification.

upvoted 4 times

  **Maaayaaa** 6 days, 23 hours ago

BlazingText can do sentiment analysis:
<https://docs.aws.amazon.com/sagemaker/latest/dg/blazingtext.html>

upvoted 1 times

A bank wants to launch a low-rate credit promotion. The bank is located in a town that recently experienced economic hardship. Only some of the bank's customers were affected by the crisis, so the bank's credit team must identify which customers to target with the promotion. However, the credit team wants to make sure that loyal customers' full credit history is considered when the decision is made.

The bank's data science team developed a model that classifies account transactions and understands credit eligibility. The data science team used the XGBoost algorithm to train the model. The team used 7 years of bank transaction historical data for training and hyperparameter tuning over the course of several days.

The accuracy of the model is sufficient, but the credit team is struggling to explain accurately why the model denies credit to some customers. The credit team has almost no skill in data science.

What should the data science team do to address this issue in the MOST operationally efficient manner?

- A. Use Amazon SageMaker Studio to rebuild the model. Create a notebook that uses the XGBoost training container to perform model training. Deploy the model at an endpoint. Enable Amazon SageMaker Model Monitor to store inferences. Use the inferences to create Shapley values that help explain model behavior. Create a chart that shows features and SHapley Additive exPlanations (SHAP) values to explain to the credit team how the features affect the model outcomes.
- B. Use Amazon SageMaker Studio to rebuild the model. Create a notebook that uses the XGBoost training container to perform model training. Activate Amazon SageMaker Debugger, and configure it to calculate and collect Shapley values. Create a chart that shows features and SHapley Additive exPlanations (SHAP) values to explain to the credit team how the features affect the model outcomes.
- C. Create an Amazon SageMaker notebook instance. Use the notebook instance and the XGBoost library to locally retrain the model. Use the `plot_importance()` method in the Python XGBoost interface to create a feature importance chart. Use that chart to explain to the credit team how the features affect the model outcomes.
- D. Use Amazon SageMaker Studio to rebuild the model. Create a notebook that uses the XGBoost training container to perform model training. Deploy the model at an endpoint. Use Amazon SageMaker Processing to post-analyze the model and create a feature importance explainability chart automatically for the credit team.

Correct Answer: C

Community vote distribution

B (65%)

C (26%)

9%

-   **spaceexplorer**

Highly Voted 

 1 year ago

Selected Answer: B

B, SageMaker Model Debugger is used to generate SHAP values

upvoted 12 times
-   **siju13** 11 months, 3 weeks ago

<https://aws.amazon.com/blogs/machine-learning/ml-explainability-with-amazon-sagemaker-debugger/>

upvoted 7 times
-   **V_B_**

Highly Voted 

 8 months, 3 weeks ago

Selected Answer: C

I believe C is the right answer, it is simpler and more accurate than B.

upvoted 5 times
-   **Ahmedhadi_**

Most Recent 

 1 week, 6 days ago

Selected Answer: A

SageMaker Model Monitor is a tool that helps monitor the quality of model predictions over time by analyzing data inputs and outputs during inference. It can detect and alert when data drift or concept drift occurs, and can identify features that are most responsible for the changes in model behavior. Model Monitor can be used to continuously monitor and improve model performance, and can be integrated with SageMaker endpoints or SageMaker Pipelines.

SageMaker Debugger is a tool that helps debug machine learning models during training by analyzing the internal states of the model, such as weights and gradients, as well as the data inputs and outputs during training. It can detect and alert when common training issues occur, such as overfitting or underfitting, and can identify the root causes of these issues. Debugger can be used to improve model accuracy and convergence, and can be integrated with SageMaker training jobs.

upvoted 1 times
-   **Ahmedhadi_** 1 week ago

After reconsideration, it is actually B.

<https://aws.amazon.com/blogs/machine-learning/ml-explainability-with-amazon-sagemaker-debugger/>

upvoted 1 times

- MIlb**

4 weeks ago

Selected Answer: B

Debugger because we are in the context of "training data"

upvoted 1 times
- Amit11011996**

1 month, 2 weeks ago

Selected Answer: C

C is the straight forward and simpler.

upvoted 1 times
- Amit11011996**

1 month, 3 weeks ago

Why not C?

'C' is the most easiest way to find out.!

upvoted 1 times
- Chelseajcole**

1 month, 4 weeks ago

Selected Answer: B

This is debugger’s work

upvoted 2 times
- Ajose0**

2 months, 1 week ago

Selected Answer: A

Option A suggests using Amazon SageMaker Model Monitor to store inferences and create Shapley values that can help explain the model's behavior. This option can be more operationally efficient because it doesn't require the credit team to understand the complexities of Shapley values, and it doesn't necessarily slow down the model's inference time.

upvoted 1 times
- Ajose0**

2 months, 1 week ago

After a review, I go with option B

upvoted 1 times

A data science team is planning to build a natural language processing (NLP) application. The application's text preprocessing stage will include part-of-speech tagging and key phrase extraction. The preprocessed text will be input to a custom classification algorithm that the data science team has already written and trained using Apache MXNet.

Which solution can the team build MOST quickly to meet these requirements?

- A. Use Amazon Comprehend for the part-of-speech tagging, key phrase extraction, and classification tasks.
- B. Use an NLP library in Amazon SageMaker for the part-of-speech tagging. Use Amazon Comprehend for the key phrase extraction. Use AWS Deep Learning Containers with Amazon SageMaker to build the custom classifier.
- C. Use Amazon Comprehend for the part-of-speech tagging and key phrase extraction tasks. Use Amazon SageMaker built-in Latent Dirichlet Allocation (LDA) algorithm to build the custom classifier.
- D. Use Amazon Comprehend for the part-of-speech tagging and key phrase extraction tasks. Use AWS Deep Learning Containers with Amazon SageMaker to build the custom classifier.

Correct Answer: B

Community vote distribution

D (81%)

A (19%)

- exam_prep

Highly Voted

11 months, 1 week ago

I will go with A. Refer to link : <https://aws.amazon.com/comprehend/features/>
upvoted 15 times
- tgaos

11 months, 1 week ago

Agree. A is my answer.
1. part of speech tagging : https://docs.aws.amazon.com/comprehend/latest/dg/API_PartOfSpeechTag.html
2. Key phas extraction
<https://docs.aws.amazon.com/comprehend/latest/dg/how-key-phrases.html>
3. custum classification algorithm
<https://docs.aws.amazon.com/comprehend/latest/dg/how-document-classification.html>
upvoted 6 times
- ovokpus

Highly Voted

10 months, 1 week ago

Selected Answer: D

D is the answer. Using Apache MXNet rules out Comprehend from making the classification task
upvoted 13 times
- VinceCar

5 months, 2 weeks ago

any reference?
upvoted 1 times
- VinceCar

5 months, 2 weeks ago

"Automatically improve performance with optimized model training for popular frameworks like TensorFlow, PyTorch, and Apache MXNet."

<https://aws.amazon.com/cn/machine-learning/containers/>
upvoted 2 times
- MIlb

Most Recent

4 weeks, 1 day ago

Selected Answer: D

The other mxnet model is the key
upvoted 2 times
- AjoseO

2 months, 1 week ago

Selected Answer: D

option D is the most appropriate answer, given that the team has already written and trained a custom classification algorithm using Apache MXNet.

Option D allows the team to use Amazon Comprehend for part-of-speech tagging and key phrase extraction, while also using AWS Deep Learning Containers with Amazon SageMaker to build and deploy the custom classifier.
upvoted 3 times
- wolfsong

3 months ago

D for me. Question says "The preprocessed text WILL be input to a custom classification algorithm that the data science team has already written and trained using Apache MXNet". So for some reason they want to use MXNet to do the classification, not Amazon Comprehend. So using MXNet

for classification is a part of their requirement. How do we meet these requirements quickly? Well, use Amazon Comprehend for part-of-speech and key phrase tasks; and use container for the MXNet stuff.

upvoted 4 times

  **drcok87** 2 months, 2 weeks ago

I had selected "A" in my first go, thanks for understanding the question. Although, comprehend does all three, since they have already built custom classification, we only need to provide solution for first two.

D for me too.

upvoted 2 times

  **hamimelon** 3 months, 4 weeks ago

The question did not make it clear whether the new solution has to use the custom model that the team built or not.

upvoted 2 times

  **tsangckl** 5 months ago

Selected Answer: A

A for me

upvoted 1 times

  **ystotest** 5 months ago

Selected Answer: A

Agreed with A, Comprehend 3 functions

upvoted 1 times

  **HerbertK** 7 months, 1 week ago

Selected Answer: A

A for me.

<https://docs.aws.amazon.com/comprehend/latest/dg/how-document-classification.html>

upvoted 3 times

  **ckkobe24** 11 months, 3 weeks ago

Selected Answer: D

D for me

upvoted 4 times

A machine learning (ML) specialist must develop a classification model for a financial services company. A domain expert provides the dataset, which is tabular with 10,000 rows and 1,020 features. During exploratory data analysis, the specialist finds no missing values and a small percentage of duplicate rows. There are correlation scores of > 0.9 for 200 feature pairs. The mean value of each feature is similar to its 50th percentile.

Which feature engineering strategy should the ML specialist use with Amazon SageMaker?

- A. Apply dimensionality reduction by using the principal component analysis (PCA) algorithm.
- B. Drop the features with low correlation scores by using a Jupyter notebook.
- C. Apply anomaly detection by using the Random Cut Forest (RCF) algorithm.
- D. Concatenate the features with high correlation scores by using a Jupyter notebook.

Correct Answer: C

Community vote distribution

A (100%)

- ovokpus

Highly Voted

10 months, 1 week ago

Selected Answer: A

Dimensions are too high. Use PCA

upvoted 8 times
- LydiaGom

Highly Voted

11 months, 3 weeks ago

A should be the answer to avoid the curse of dimensionality

upvoted 6 times
- Gaby999

Most Recent

4 days, 18 hours ago

Selected Answer: A

Given that the dataset has 1,020 features and 200 of them are highly correlated, it is likely that the dataset suffers from multicollinearity. In such cases, dimensionality reduction techniques like principal component analysis (PCA) can be used to transform the data into a lower dimensional space without losing much information. Therefore, option A, "Apply dimensionality reduction by using the principal component analysis (PCA) algorithm" is the most appropriate feature engineering strategy for the ML specialist to use with Amazon SageMaker. This would help reduce the computational complexity of the model, improve model performance, and help to avoid overfitting.

upvoted 1 times
- Ajose0

2 months, 1 week ago

Selected Answer: A

A. Apply dimensionality reduction by using the principal component analysis (PCA) algorithm.

Since the dataset has many features, and a significant number of them have high correlation scores, the model may suffer from the curse of dimensionality. To reduce the dimensionality of the dataset, the specialist can use a technique like PCA, which reduces the number of features while still retaining the maximum amount of information. PCA can help remove redundant features and improve the model's performance by reducing the chances of overfitting. Additionally, since there are no missing values and a small percentage of duplicate rows, no data cleaning techniques like anomaly detection or dropping the features are required. Concatenating features with high correlation scores is not an appropriate strategy since it may lead to collinearity issues.

upvoted 1 times
- drcok87

2 months, 2 weeks ago

A PCA: PCA is a linear dimensionality reduction technique (algorithm) that transforms a set of correlated variables (p) into a smaller k (k<p) number of uncorrelated variables called principal components while retaining as much of the variation in the original dataset as possible

upvoted 1 times
- Peeking

4 months, 3 weeks ago

Selected Answer: A

Choosing C is answer by ExamTopics is completely laughable.

upvoted 1 times
- DJiang

11 months, 3 weeks ago

Selected Answer: A

I think it's A.

upvoted 4 times

A manufacturing company asks its machine learning specialist to develop a model that classifies defective parts into one of eight defect types. The company has provided roughly 100,000 images per defect type for training. During the initial training of the image classification model, the specialist notices that the validation accuracy is 80%, while the training accuracy is 90%. It is known that human-level performance for this type of image classification is around 90%.

What should the specialist consider to fix this issue?

- A. A longer training time
- B. Making the network larger
- C. Using a different optimizer
- D. Using some form of regularization

Correct Answer: D

Reference:
<https://acloud.guru/forums/aws-certified-machine-learning-specialty/discussion/-MGdBUKmq02zC3uOq4VL/AWS%20Exam%20Machine%20Learning>

Community vote distribution

D (100%)

-   **bluer1**

Highly Voted 

 1 year ago

D - over fitting problem.
upvoted 16 times
-   **Ajose0**

Most Recent 

 2 months, 1 week ago

Selected Answer: D

The specialist should consider using some form of regularization to fix this issue. Regularization techniques such as dropout or L2 regularization can help prevent overfitting, which can occur when the model performs well on the training data but poorly on the validation data.

Option A, a longer training time, might not necessarily fix the issue and could lead to overfitting if the model is already performing well on the training data.

Option B, making the network larger, could also lead to overfitting and may not be necessary if the current network architecture is sufficient to perform the classification task.

Option C, using a different optimizer, might not necessarily fix the issue and could lead to slower convergence or worse performance.

Therefore, option D, using some form of regularization, is the most appropriate solution to consider in this situation.

upvoted 3 times
-   **ystotest** 5 months, 1 week ago

Selected Answer: D

typical overfitting problem
upvoted 1 times
-   **DD4** 7 months, 1 week ago

C - It is not a overfitting problem as the training accuracy stands at 90%, which is at same level of human performance. That means the algorithm used is not optimized for this problem. So, some other algorithm should applied for this problem.
upvoted 2 times
-   **KlaudYu** 10 months ago

I'd go A. Regularization could not guarantee higher validation accuracy.
upvoted 2 times
-   **rhuanca** 10 months, 3 weeks ago

I believe answer is B , because clearly it is a overfiting problem , if we reduce complexity the accurate will reduce close to 80% ... But human works can reach up to 90% .
upvoted 1 times
-   **rhuanca** 10 months, 3 weeks ago

I mean looks like a overfitting problem....
upvoted 2 times

A machine learning specialist needs to analyze comments on a news website with users across the globe. The specialist must find the most discussed topics in the comments that are in either English or Spanish.

What steps could be used to accomplish this task? (Choose two.)

- A. Use an Amazon SageMaker BlazingText algorithm to find the topics independently from language. Proceed with the analysis.
- B. Use an Amazon SageMaker seq2seq algorithm to translate from Spanish to English, if necessary. Use a SageMaker Latent Dirichlet Allocation (LDA) algorithm to find the topics.
- C. Use Amazon Translate to translate from Spanish to English, if necessary. Use Amazon Comprehend topic modeling to find the topics.
- D. Use Amazon Translate to translate from Spanish to English, if necessary. Use Amazon Lex to extract topics form the content.
- E. Use Amazon Translate to translate from Spanish to English, if necessary. Use Amazon SageMaker Neural Topic Model (NTM) to find the topics.

Correct Answer: B

Reference:

https://docs.aws.amazon.com/sagemaker/latest/dg/lda.html

Community vote distribution

C (45%)

B (35%)

E (15%)

5%

- LydiaGom

Highly Voted

11 months, 4 weeks ago

C and E
B needs to build custom model
upvoted 22 times
- hamimelon

4 months, 1 week ago

The question did not say you cannot build a custom model. They have a ML specialist, so building a custom model shouldn't be a problem.
upvoted 1 times
- tgaos

11 months, 1 week ago

The SageMaker seq2seq algorithm is a supervised learning algorithm. And it needs to train then translate. translate can directly use to translate from Spanish to English
upvoted 4 times
- NILKK

Highly Voted

1 year ago

It asked 2 answers, but I can see only one answer. Please advise. Thanks!
upvoted 7 times
- daidaidai

Most Recent

2 days, 9 hours ago

Selected Answer: C

C and E, The most common topic modeling algorithm is called Latent Dirichlet Allocation (LDA), Amazon Comprehend uses a Latent dirichlet allocation-based learning model to determine the topics in a set of documents. https://docs.aws.amazon.com/comprehend/latest/dg/topic-modeling.html.
Despite its popularity and success, one limitation of this algorithm is that it makes an assumption that the distribution of words in a document follows a Dirichlet distribu- tion.The NTM algorithm relaxes this assumption and aims to learn a latent representa- tion without prior assumptions.
upvoted 1 times
- AjoseO

2 months, 1 week ago

Option A suggests using an unsupervised learning algorithm, BlazingText, to find the topics independently of language. This can be a good approach if the content is in multiple languages and we don't want to translate them first.

Option C suggests using Amazon Comprehend, which can detect the language and then perform topic modeling on the content in that language. If any comments are in Spanish, they will be translated to English using Amazon Translate before being fed to Amazon Comprehend.

Together, these options cover both cases - comments in English and comments in Spanish, without having to translate everything.
upvoted 1 times
- GiyeonShin

2 months ago

In Option A, BlazingText has two modes: word2vec, text classification.
https://docs.aws.amazon.com/sagemaker/latest/dg/blazingtext.html
Word2Vec is an unsupervised learning algorithm, but it is for word embeddings, so it is not appropriate to find the topics from corpus.
Text classification isn't also appropriate to find the topics, because there is no given specified topics. So Option A is not an answer
upvoted 1 times
- CertArvind

3 months ago

Answer should be B and C [B,C]
upvoted 3 times

  **BTRYING** 3 months, 1 week ago

Selected Answer: C

CE - <https://docs.aws.amazon.com/comprehend/latest/dg/topic-modeling.html>
upvoted 3 times

  **Tomatoteacher** 3 months, 2 weeks ago

Selected Answer: E

C and E, Comprehend uses LDA, and manually creating seq2seq is unnecessary here.
Tis truly C and E. Please change options so you can choose 2 answers pls.
upvoted 3 times

  **yemauricio** 3 months, 2 weeks ago

Selected Answer: C

best answer
upvoted 2 times

  **ystotest** 5 months ago

Selected Answer: C

Amazon Comprehend uses a Latent dirichlet allocation-based learning model to determine the topics in a set of documents. It examines each document to determine the context and meaning of a word. The set of words that frequently belong to the same context across the entire document set make up a topic.
upvoted 3 times

  **Morsa** 9 months, 3 weeks ago

B and D as two answers were expected. LDA and NTM are algorithms that build topics
upvoted 1 times

  **ovokpus** 10 months, 1 week ago

Selected Answer: B

B or E, changed my mind, because Comprehend is supervised while LDA and NTM are unsupervised. There is no mention of training and labeling in the question
upvoted 2 times

  **rjekstein** 9 months, 3 weeks ago

C and E Amazon comprehend topic modeling can be unspervised.
<https://docs.aws.amazon.com/comprehend/latest/dg/topic-modeling.html>

You can use Amazon Comprehend to examine the content of a collection of documents to determine common themes. For example, you can give Amazon Comprehend a collection of news articles, and it will determine the subjects, such as sports, politics, or entertainment. The text in the documents doesn't need to be annotated.
upvoted 7 times

  **ovokpus** 10 months, 1 week ago

C & E seems the best choices here
upvoted 5 times

  **[Removed]** 10 months, 2 weeks ago

B and E
upvoted 2 times

  **siju13** 11 months, 3 weeks ago

Selected Answer: B

B and C
<https://docs.aws.amazon.com/comprehend/latest/dg/topic-modeling.html>
upvoted 4 times

  **DJiang** 11 months, 4 weeks ago

Selected Answer: D

I would choose B and D. Amazon Lex is able to extract topic.
upvoted 1 times

  **rohit07cf** 12 months ago

Selected Answer: B

B and E
upvoted 1 times

A machine learning (ML) specialist is administering a production Amazon SageMaker endpoint with model monitoring configured. Amazon SageMaker Model Monitor detects violations on the SageMaker endpoint, so the ML specialist retrains the model with the latest dataset. This dataset is statistically representative of the current production traffic. The ML specialist notices that even after deploying the new SageMaker model and running the first monitoring job, the SageMaker endpoint still has violations.

What should the ML specialist do to resolve the violations?

- A. Manually trigger the monitoring job to re-evaluate the SageMaker endpoint traffic sample.
- B. Run the Model Monitor baseline job again on the new training set. Configure Model Monitor to use the new baseline.
- C. Delete the endpoint and recreate it with the original configuration.
- D. Retrain the model again by using a combination of the original training set and the new training set.

Correct Answer: B

Community vote distribution

B (100%)

-   **edvardo**

Highly Voted 

11 months, 3 weeks ago

I would go with B:
<https://docs.aws.amazon.com/sagemaker/latest/dg/model-monitor-create-baseline.html>
upvoted 6 times
-   **tgaos**

11 months, 1 week ago

Agree, the answer is B.
From the document, the violation file contains several checks and "The violations file is generated as the output of a MonitoringExecution" .
<https://docs.aws.amazon.com/sagemaker/latest/dg/model-monitor-interpreting-violations.html>.
upvoted 3 times
-   **Ajose0**

Most Recent 

2 months, 1 week ago

Selected Answer: B

Running the Model Monitor baseline job again with the new training set and configuring Model Monitor to use the new baseline is a valid option to resolve the violations.

By running the baseline job with the new training set, a new baseline is created, which can be used to compare with the new data to detect any drifts in the data distribution. Then, the updated baseline can be set as the new baseline for monitoring the endpoint.

So, option B is also a valid solution to resolve the violations.
upvoted 3 times

A company supplies wholesale clothing to thousands of retail stores. A data scientist must create a model that predicts the daily sales volume for each item for each store. The data scientist discovers that more than half of the stores have been in business for less than 6 months. Sales data is highly consistent from week to week. Daily data from the database has been aggregated weekly, and weeks with no sales are omitted from the current dataset. Five years (100 MB) of sales data is available in Amazon S3.

Which factors will adversely impact the performance of the forecast model to be developed, and which actions should the data scientist take to mitigate them?

(Choose two.)

- A. Detecting seasonality for the majority of stores will be an issue. Request categorical data to relate new stores with similar stores that have more historical data.
- B. The sales data does not have enough variance. Request external sales data from other industries to improve the model's ability to generalize.
- C. Sales data is aggregated by week. Request daily sales data from the source database to enable building a daily model.
- D. The sales data is missing zero entries for item sales. Request that item sales data from the source database include zero entries to enable building the model.
- E. Only 100 MB of sales data is available in Amazon S3. Request 10 years of sales data, which would provide 200 MB of training data for the model.

Correct Answer: AB

Reference:
<https://towardsdatascience.com/sales-forecasting-from-time-series-to-deep-learning-5d115514bfac>
<https://arxiv.org/ftp/arxiv/papers/1302/1302.6613.pdf>

Community vote distribution

AC (90%)

10%

-   **cron0001**

Highly Voted 

 1 year ago

Selected Answer: AC

AC would be my answer. As half the stores have only been open for 6 months, no seasonality would be captured. The aggregation of the daily also removes trends we see during the week which is also not great when we are looking for the daily predicated sales figure

upvoted 25 times
-   **rb39** 7 months, 2 weeks ago

B - no reason to assume there is not enough variance

D - missing data can be assumed to be 0, no need to ask for empty data

E - no reason to ask for two years of data having one already

upvoted 10 times
-   **Ajose0**

Most Recent 

 2 months, 1 week ago

Selected Answer: AC

A. Since more than half of the stores have been in business for less than 6 months, it will be challenging to detect seasonality patterns for these new stores. Therefore, one solution is to request categorical data to relate new stores with similar stores that have more historical data. This will help the model to identify common patterns and accurately forecast sales for new stores.

C. Since the sales data is aggregated by week, it may not be possible to identify daily patterns or trends. Hence, one solution is to request daily sales data from the source database to enable building a daily model. This will help the model to identify daily patterns and improve its forecasting accuracy.

upvoted 2 times
-   **peterfish** 9 months, 2 weeks ago

Selected Answer: CD

I go with CD. How could we ignore the days with 0 sales? The model should be trained so that it can predict 0 sales days as well.

upvoted 3 times
-   **KlaudYu** 10 months, 3 weeks ago

B, C, D are possible. A couldn't be an answer because the model must predict daily sales volumes while A says 'Request categorical data'.

upvoted 1 times

An ecommerce company is automating the categorization of its products based on images. A data scientist has trained a computer vision model using the Amazon SageMaker image classification algorithm. The images for each product are classified according to specific product lines. The accuracy of the model is too low when categorizing new products. All of the product images have the same dimensions and are stored within an Amazon S3 bucket. The company wants to improve the model so it can be used for new products as soon as possible.

Which steps would improve the accuracy of the solution? (Choose three.)

- A. Use the SageMaker semantic segmentation algorithm to train a new model to achieve improved accuracy.
- B. Use the Amazon Rekognition DetectLabels API to classify the products in the dataset.
- C. Augment the images in the dataset. Use open source libraries to crop, resize, flip, rotate, and adjust the brightness and contrast of the images.
- D. Use a SageMaker notebook to implement the normalization of pixels and scaling of the images. Store the new dataset in Amazon S3.
- E. Use Amazon Rekognition Custom Labels to train a new model.
- F. Check whether there are class imbalances in the product categories, and apply oversampling or undersampling as required. Store the new dataset in Amazon S3.

Correct Answer: BCE

Reference:
<https://docs.aws.amazon.com/rekognition/latest/dg/how-it-works-types.html> <https://towardsdatascience.com/image-processing-techniques-for-computer-vision-11f92f511e21> <https://docs.aws.amazon.com/rekognition/latest/customlabels-dg/training-model.html>

Community vote distribution

CEF (67%)CDF (22%)11%

-   **tgaos**

Highly Voted 

 11 months ago

B CE is correct.
upvoted 7 times
-   **cox1960**

Most Recent 

 3 days, 3 hours ago

Selected Answer: BCE

you fix images (C), train Rekognition with these images (E) and finally infer to get the classes (B)

upvoted 1 times
-   **Ahmedhadi_** 1 week ago

Selected Answer: CDF

Amazon Rekognition custom label model requires time, expertise, and resources, often taking months to complete. Additionally, it often requires thousands or tens-of-thousands of hand-labeled images to provide the model with enough data to accurately make decisions. The solution must be quick.
<https://aws.amazon.com/rekognition/custom-labels-features/>

upvoted 1 times

  **Ahmedhadi_** 4 days, 11 hours ago

Sorry, never mind my answer it's actually CEF.
upvoted 1 times

  **MIlb** 4 weeks, 1 day ago

Selected Answer: CEF

CEF is correct

upvoted 2 times

  **Chelseajcole** 1 month, 3 weeks ago

Selected Answer: CEF

D seems doing nothing with new product

upvoted 2 times

  **Siyuan_Zhu** 2 months, 1 week ago

Selected Answer: CEF

CEF is correct in my opinion.
D - The normalization and scaling would not provide the most dramatic improvement of the model.
E - By using transfer learning, we are able to utilize a pre-trained model that can recognize a lot of the lower-level features and retain only the last few layers for this specific purpose.
- <https://www.examtopics.com/exams/amazon/aws-certified-machine-learning-specialty/custom-view/>

287/406

upvoted 2 times

  **hichemck** 5 months ago

Selected Answer: CDF

CDF is correct. IMO

upvoted 1 times

  **dunhill** 5 months, 3 weeks ago

I would go CEF.
A. This is for a different kind of problem. No need to use it
B. No, because it is not pretrained on our data and won't improve it.
D. Usually we don't use SageMaker notebook to implement the normalization of pixels and scaling of the images as the tool...
upvoted 2 times

  **vanluigi** 11 months, 2 weeks ago

IMO BDE are correct:
A - Its not a segmentation problem, we just know the item
B - Its a prebuilt model so is quick
C - Yes
D - Create a sagemaker notebook is not worth
E - Yes is an amazon prebuilt model
F - Its not worth on future, will u check it after new item is added? Its not worth
upvoted 4 times

  **edvardo** 12 months ago

A. This is for a different kind of problem. No need to use it
B. No, because it is not pretrained on our data and won't improve it.
C. YES.
D. Maybe? It is a quick and dirty solution doing on a notebook, but it is quick and the normalization helps for this tasks.
E. Maybe? But we have trained a model, so we might suspect this option won't be much better than our solution as is.
F. YES.

I would go with CDF
upvoted 4 times

  **ckkobe24** 1 year ago

D&F are wrong due to "new dataset"?
upvoted 3 times

A data scientist is training a text classification model by using the Amazon SageMaker built-in BlazingText algorithm. There are 5 classes in the dataset, with 300 samples for category A, 292 samples for category B, 240 samples for category C, 258 samples for category D, and 310 samples for category E.

The data scientist shuffles the data and splits off 10% for testing. After training the model, the data scientist generates confusion matrices for the training and test sets.

Training data confusion matrix

		Predicted class					Total
		A	B	C	D	E	
True class	A	270	0	0	0	0	270
	B	1	260	0	0	2	263
	C	0	0	111	100	5	216
	D	4	3	132	92	1	232
	E	0	0	2	3	274	279
	Total	275	263	245	195	282	1260

Test data confusion matrix

		Predicted class					Total
		A	B	C	D	E	
True class	A	9	1	0	0	0	10
	B	2	25	0	2	0	29
	C	10	2	11	10	1	34
	D	1	0	12	14	0	27
	E	9	1	4	1	25	40
	Total	31	29	27	27	26	140

What could the data scientist conclude from these results?

- A. Classes C and D are too similar.
- B. The dataset is too small for holdout cross-validation.
- C. The data distribution is skewed.
- D. The model is overfitting for classes B and E.

Correct Answer: B

Community vote distribution

A (100%)

- LydiaGom

Highly Voted

11 months, 3 weeks ago

Isn't it A? the model doesn't classify C & D well.

upvoted 7 times
- dolorez

Highly Voted

11 months, 1 week ago

Selected Answer: A

the correct answer should be A, the model is clearly unable to tell C and D apart

the reason why B is incorrect is subtle - there is holdout validation or cross-validation, but not holdout cross-validation; while I think it would be more reasonable to use CV with such a small dataset rather than holdout, the answer is mixing terms and therefore should be wrong

also, the test set confusion matrix is still pretty comparable to the train set one, so I wouldn't say there is objective evidence to claim holdout is a wrong choice here

upvoted 5 times

DD4

Most Recent

7 months, 1 week ago

D - Training accuracies of B and E are higher than those of test, whereas A has similar accuracy in both. For C and D, test accuracy has actually improved.

upvoted 3 times

tgaos

11 months, 1 week ago

I think the answer is A. The model doesn't perform well on class C and D in both training and testing dataset. I don't think B is relevant to the question(cross-validation is not mentioned in the question)

upvoted 3 times

exam887

11 months, 1 week ago

Selected Answer: A

What means holdout cross validation. There should be holdout validation vs cross validation

upvoted 1 times

bluer1

12 months ago

B should be the correct answer

upvoted 2 times

A company that manufactures mobile devices wants to determine and calibrate the appropriate sales price for its devices. The company is collecting the relevant data and is determining data features that it can use to train machine learning (ML) models. There are more than 1,000 features, and the company wants to determine the primary features that contribute to the sales price.

Which techniques should the company use for feature selection? (Choose three.)

- A. Data scaling with standardization and normalization
- B. Correlation plot with heat maps
- C. Data binning
- D. Univariate selection
- E. Feature importance with a tree-based classifier
- F. Data augmentation

Correct Answer: CDF

Reference:

<https://towardsdatascience.com/an-overview-of-data-preprocessing-features-enrichment-automatic-feature-selection-60b0c12d75ad>

[https://towardsdatascience.com/feature-selection-using-python-for-classification-problem-b5f00a1c7028#:~:text=Univariate%20feature%20selection%20works%20by,analysis%20of%20variance%20\(ANOVA\).&text=That%20is%20why%20it%20is%20called%20'univariate](https://towardsdatascience.com/feature-selection-using-python-for-classification-problem-b5f00a1c7028#:~:text=Univariate%20feature%20selection%20works%20by,analysis%20of%20variance%20(ANOVA).&text=That%20is%20why%20it%20is%20called%20'univariate)

[https://towardsdatascience.com/feature-selection-using-python-for-classification-problem-b5f00a1c7028#:~:text=Univariate%20feature%20selection%20works%20by,analysis%20of%20variance%20\(ANOVA\).&text=That%20is%20why%20it%20is%20called%20'univariate](https://towardsdatascience.com/feature-selection-using-python-for-classification-problem-b5f00a1c7028#:~:text=Univariate%20feature%20selection%20works%20by,analysis%20of%20variance%20(ANOVA).&text=That%20is%20why%20it%20is%20called%20'univariate)

[https://towardsdatascience.com/feature-selection-using-python-for-classification-problem-b5f00a1c7028#:~:text=Univariate%20feature%20selection%20works%20by,analysis%20of%20variance%20\(ANOVA\).&text=That%20is%20why%20it%20is%20called%20'univariate](https://towardsdatascience.com/feature-selection-using-python-for-classification-problem-b5f00a1c7028#:~:text=Univariate%20feature%20selection%20works%20by,analysis%20of%20variance%20(ANOVA).&text=That%20is%20why%20it%20is%20called%20'univariate)

[https://towardsdatascience.com/feature-selection-using-python-for-classification-problem-b5f00a1c7028#:~:text=Univariate%20feature%20selection%20works%20by,analysis%20of%20variance%20\(ANOVA\).&text=That%20is%20why%20it%20is%20called%20'univariate](https://towardsdatascience.com/feature-selection-using-python-for-classification-problem-b5f00a1c7028#:~:text=Univariate%20feature%20selection%20works%20by,analysis%20of%20variance%20(ANOVA).&text=That%20is%20why%20it%20is%20called%20'univariate)

<https://arxiv.org/abs/2101.04530>

Community vote distribution

BDE (100%)

  **ckkobe24** Highly Voted 1 year ago

Selected Answer: BDE

i will go for B, D and E. B and D for me are like doing partial regression and corr plot can actually tell you briefly how well the univariate is correlated with your target and i guess that also apply for D.. and E , feature importance ranking that's what feature selection strategy want from my POV. And for Data Binning is data enrichment just like augmentations , but then the question was saying they want to do feature selection over 1k+ variables which implies they actually care more about which variable(s) can contribute more on determining the price ?

upvoted 17 times

  **Ajose0** Most Recent 2 months, 1 week ago

Selected Answer: BDE

B. Correlation plot with heat maps: This technique can be used to identify the relationship between each feature and the target variable (sales price). By creating a correlation plot with heat maps, the company can quickly visualize the strength and direction of the relationship between each feature and the target variable.

D. Univariate selection: This technique can be used to select the features that have the strongest relationship with the target variable. It involves analyzing each feature independently and selecting the ones that have the highest correlation with the target variable.

E. Feature importance with a tree-based classifier: This technique can be used to determine the most important features that contribute to the target variable. By using a tree-based classifier such as Random Forest or Gradient Boosting, the company can rank the importance of each feature and select the ones that have the highest importance.

upvoted 3 times

  **wolfsong** 2 months, 2 weeks ago

BDE as stated here: <https://towardsdatascience.com/feature-selection-techniques-in-machine-learning-with-python-f24e7da3f36e>

upvoted 2 times

  **Morsa** 9 months, 3 weeks ago

BDE for me

upvoted 4 times

  **ovokpus** 10 months, 1 week ago

Selected Answer: BDE

throwing my weight behind B D E. Correlation with heatmaps help us eliminate multicollinearity, Univariate testing helps us see which ones are correlated with the target, same as feature importances of tree-based algorithms.

upvoted 3 times

  **bluer1** 1 year ago

CDF for me

upvoted 1 times

Question #160

Topic 1

A power company wants to forecast future energy consumption for its customers in residential properties and commercial business properties. Historical power consumption data for the last 10 years is available. A team of data scientists who performed the initial data analysis and feature selection will include the historical power consumption data and data such as weather, number of individuals on the property, and public holidays. The data scientists are using Amazon Forecast to generate the forecasts. Which algorithm in Forecast should the data scientists use to meet these requirements?

- A. Autoregressive Integrated Moving Average (AIRMA)
- B. Exponential Smoothing (ETS)
- C. Convolutional Neural Network - Quantile Regression (CNN-QR)
- D. Prophet

Correct Answer: B

Reference:

<https://jesit.springeropen.com/articles/10.1186/s43067-020-00021-8>

Community vote distribution

C (89%) 11%

  **spaceexplorer** Highly Voted 1 year ago

Selected Answer: C

Answer is C, CNN-QR and DeepAR accepts related time series data (weather data, number of people on property, etc.,) upvoted 15 times

  **Ajose0** Most Recent 2 months, 1 week ago

Selected Answer: C

CNN-QR is a deep learning algorithm that can model complex relationships between the inputs and outputs, such as the weather and public holidays, with historical power consumption data. CNN-QR has been shown to be effective in generating accurate predictions in many different types of forecasting use cases, including demand forecasting.

ETS (Exponential Smoothing) is a classical time series algorithm that is often used for forecasting. It can be effective for simple time series data that have regular patterns, but may not be sufficient to handle the complexity of the given data.

ARIMA (Autoregressive Integrated Moving Average) is another classical time series algorithm that can model complex patterns in data. However, it may be difficult to use in cases where there are many different inputs and the relationships between the inputs and outputs are complex. upvoted 2 times

  **aScientist** 5 months, 3 weeks ago

Selected Answer: C

ARIMA & ES are both base time series algos that are available. DeeoAR+ & CNN-QR are refined and able to utilize external data as well to complement the time series data available upvoted 3 times

  **matteocal** 9 months, 1 week ago

Selected Answer: C

C, as explained here: <https://docs.aws.amazon.com/forecast/latest/dg/aws-forecast-choosing-recipes.html> upvoted 4 times

  **ovokpus** 10 months, 1 week ago

Selected Answer: A

According to the link below, it is either ARIMA or DeepAR.

So A is the answer here

<https://aws.amazon.com/blogs/machine-learning/making-accurate-energy-consumption-predictions-with-amazon-forecast/> upvoted 3 times

  **edvardo** 12 months ago

Given the provided data, I would discard A and B.

Amazon Forecast CNN-QR, Convolutional Neural Network - Quantile Regression, is a proprietary machine learning algorithm for forecasting scalar (one-dimensional) time series

I would choose D, Prophet. <https://docs.aws.amazon.com/forecast/latest/dg/aws-forecast-recipe-prophet.html>

How Prophet Works
Prophet is especially useful for datasets that:
Contain an extended time period (months or years) of detailed historical observations (hourly, daily, or weekly)
Have multiple strong seasonalities
Include previously known important, but irregular, events
Have missing data points or large outliers
Have non-linear growth trends that are approaching a limit
Prophet is an additive regression model with a piecewise linear or logistic growth curve trend. It includes a yearly seasonal component modeled using Fourier series and a weekly seasonal component modeled using dummy variables.
upvoted 2 times

  **aScientist** 5 months, 3 weeks ago
Prophet wont be able to use the additional data that is available in the question
upvoted 1 times

  **f4bi4n** 10 months, 2 weeks ago
Prophet doesn't accept historical-related time series, so it won't work here
<https://docs.aws.amazon.com/forecast/latest/dg/aws-forecast-choosing-recipes.html#comparing-algos>
upvoted 5 times

A company wants to use automatic speech recognition (ASR) to transcribe messages that are less than 60 seconds long from a voicemail-style application. The company requires the correct identification of 200 unique product names, some of which have unique spellings or pronunciations. The company has 4,000 words of Amazon SageMaker Ground Truth voicemail transcripts it can use to customize the chosen ASR model. The company needs to ensure that everyone can update their customizations multiple times each hour. Which approach will maximize transcription accuracy during the development phase?

- A. Use a voice-driven Amazon Lex bot to perform the ASR customization. Create customer slots within the bot that specifically identify each of the required product names. Use the Amazon Lex synonym mechanism to provide additional variations of each product name as mis-transcriptions are identified in development.
- B. Use Amazon Transcribe to perform the ASR customization. Analyze the word confidence scores in the transcript, and automatically create or update a custom vocabulary file with any word that has a confidence score below an acceptable threshold value. Use this updated custom vocabulary file in all future transcription tasks.
- C. Create a custom vocabulary file containing each product name with phonetic pronunciations, and use it with Amazon Transcribe to perform the ASR customization. Analyze the transcripts and manually update the custom vocabulary file to include updated or additional entries for those names that are not being correctly identified.
- D. Use the audio transcripts to create a training dataset and build an Amazon Transcribe custom language model. Analyze the transcripts and update the training dataset with a manually corrected version of transcripts where product names are not being transcribed correctly. Create an updated custom language model.

Correct Answer: A

Reference:

<https://docs.aws.amazon.com/lex/latest/dg/lex-dg.pdf>

Community vote distribution

C (75%)

D (25%)

- siju13

Highly Voted

11 months, 2 weeks ago

Selected Answer: C

Answer is C.

<https://aws.amazon.com/blogs/machine-learning/build-a-custom-vocabulary-to-enhance-speech-to-text-transcription-accuracy-with-amazon-transcribe/>

upvoted 9 times
- AjoseO

Most Recent

2 months, 1 week ago

Selected Answer: D

Option D involves using the available audio transcripts to create a training dataset and building a custom language model with Amazon Transcribe.

This approach provides a high degree of control over the transcription process and the ability to fine-tune the model to the specific vocabulary and pronunciation requirements of the company. Analyzing the transcripts and updating the training dataset with corrected versions is a crucial step in improving transcription accuracy.

It enables the model to learn from mistakes and to incorporate the unique spelling and pronunciation of the 200 required product names.

upvoted 3 times
- drcok87

2 months ago

D is an ideal answer however, the question ask for "The company needs to ensure that everyone can update their customizations multiple times each hour".

To retrain custom model each hour when we have changes, will be tedious and time consuming.

I go with c, where we can ask everyone to just update the config file.

upvoted 2 times
- Siyuan_Zhu

2 months, 1 week ago

Thank you AjoseO for all these detailed explanations! They are very useful!

upvoted 1 times
- DeepakPg

4 months, 1 week ago

Why not D though?

upvoted 1 times
- Yongs

4 months, 3 weeks ago

I think C is correct.

upvoted 2 times

  **bluer1** 12 months ago

A? any thought?

upvoted 1 times

A company is building a demand forecasting model based on machine learning (ML). In the development stage, an ML specialist uses an Amazon SageMaker notebook to perform feature engineering during work hours that consumes low amounts of CPU and memory resources. A data engineer uses the same notebook to perform data preprocessing once a day on average that requires very high memory and completes in only 2 hours. The data preprocessing is not configured to use GPU. All the processes are running well on an ml.m5.4xlarge notebook instance. The company receives an AWS Budgets alert that the billing for this month exceeds the allocated budget. Which solution will result in the MOST cost savings?

- A. Change the notebook instance type to a memory optimized instance with the same vCPU number as the ml.m5.4xlarge instance has. Stop the notebook when it is not in use. Run both data preprocessing and feature engineering development on that instance.
- B. Keep the notebook instance type and size the same. Stop the notebook when it is not in use. Run data preprocessing on a P3 instance type with the same memory as the ml.m5.4xlarge instance by using Amazon SageMaker Processing.
- C. Change the notebook instance type to a smaller general purpose instance. Stop the notebook when it is not in use. Run data preprocessing on an ml.r5 instance with the same memory size as the ml.m5.4xlarge instance by using Amazon SageMaker Processing.
- D. Change the notebook instance type to a smaller general purpose instance. Stop the notebook when it is not in use. Run data preprocessing on an R5 instance with the same memory size as the ml.m5.4xlarge instance by using the Reserved Instance option.

Correct Answer: B

Community vote distribution

C (60%)

D (40%)

  **spaceexplorer** Highly Voted 1 year ago

Selected Answer: C

B is wrong as it says it doesn't take advantage of GPUs
upvoted 11 times

  **exam_prep** Highly Voted 11 months, 3 weeks ago

I believe answer should be C. a) Initial processing needs less cpu and memory so that can be done on a smaller instance. b) Second operation is memory intensive so instance type should be changed to R5 type instance.
upvoted 5 times

  **MIlb** Most Recent 4 weeks, 1 day ago

Selected Answer: D

R instance with processes that uses lot of memory. Reserved instances for less cost
upvoted 1 times

  **Ajose0** 2 months, 1 week ago

Selected Answer: D

D over C because if the EC2 instance is being used consistently for the same two hours each day, customers could consider using a Reserved Instance with a term of 1 or 3 years and payment option that aligns with their usage pattern.

This would provide significant cost savings compared to On-Demand pricing for those two hours each day.
upvoted 1 times

  **jhonivy** 3 months ago

B.
C is wrong as ml.r5 is not stopped when not in use
upvoted 1 times

  **DeepakPg** 4 months, 1 week ago

Selected Answer: D

IMO D is correct. The reserved instance option for an R5 instance, as in Option D, would provide the greatest cost savings, as reserved instances offer a discounted hourly rate in exchange for a one-time payment for a committed usage term.
upvoted 3 times

  **ryuhei** 7 months ago

Selected Answer: C

I think C is correct.
It only runs for 2 hours once a day, so RI is wasted. So I think D is wrong.
upvoted 4 times

  **Tony_1406** 2 days, 17 hours ago

"Scheduled RIs: These are available to launch within the time windows you reserve. This option allows you to match your capacity reservation to a predictable recurring schedule that only requires a fraction of a day, a week, or a month."
You have the option to reserve for a fraction of a day. Since the question specify precisely how long the job is, it makes it suitable.
<https://aws.amazon.com/ec2/pricing/reserved-instances/>
upvoted 1 times

  **Shailendraa** 7 months, 3 weeks ago
12-sep exam
upvoted 2 times

  **rafael_teste** 8 months, 3 weeks ago

Selected Answer: D

I think it is D. Using RIs the customer can have the greatest cost savings, as stated by the question
upvoted 2 times

  **DJiang** 1 year ago

Selected Answer: D

I think the answer should be D: R5 instance is cheaper and appropriate for data processing under same memory size; reserved instance make it cheaper.
upvoted 3 times

  **exam887** 11 months, 1 week ago
only use 2 hours per day, so no need for RI
upvoted 8 times

A machine learning specialist is developing a regression model to predict rental rates from rental listings. A variable named Wall_Color represents the most prominent exterior wall color of the property. The following is the sample data, excluding all other variables:

Property_ID	Wall_Color
1000	Red
1001	White
1002	Green

The specialist chose a model that needs numerical input data.

Which feature engineering approaches should the specialist use to allow the regression model to learn from the Wall_Color data? (Choose two.)

- A. Apply integer transformation and set Red = 1, White = 5, and Green = 10.
- B. Add new columns that store one-hot representation of colors.
- C. Replace the color name string by its length.
- D. Create three columns to encode the color in RGB format.
- E. Replace each color name by its training set frequency.

Correct Answer: AD

Community vote distribution

BE (64%)

BD (27%)

9%

edvardo

Highly Voted

12 months ago

B, and E (frequency encoding)

upvoted 11 times

bluer1

Highly Voted

12 months ago

BD ? any thought?

upvoted 8 times

vanluigi

11 months, 3 weeks ago

I think D cannot be because distances in RGB format are not representative of points. CIELAB correlates numerical color values consistently with human visual perception.

upvoted 1 times

WilianCB

Most Recent

1 day, 9 hours ago

Selected Answer: BD

B for sure

D. This approach involves breaking down each color into its Red, Green, and Blue components and creating separate columns for each component. This allows the model to capture the information about the intensity of each color component, which can be useful in predicting the target variable.

A, C, and E are not suitable for encoding color data in a way that can be used by a regression model. The integer transformation approach in option A arbitrarily assigns values to colors without any meaningful relationship between them. The approach in option C replaces the color names with their length, which does not provide any useful information for the model. Option E replaces each color name with its frequency in the training set, which does not capture any information about the color itself.

upvoted 1 times

alp_ileri

1 month, 3 weeks ago

I think frequency encoding cannot be. What if some colors have same amount of frequency?

upvoted 2 times

Ajose0

2 months, 1 week ago

Selected Answer: BE

B. Add new columns that store one-hot representation of colors.

One-hot encoding is a common approach to represent categorical variables as numerical values. This approach creates new binary variables for each category and assigns a value of 1 to the corresponding category and 0 to the others. In this case, the specialist can create three new binary variables, one for each color (Red, White, and Green) and use them as input to the regression model.

E. Replace each color name by its training set frequency.

Another approach to convert categorical variables into numerical ones is to replace each category with its frequency of occurrence in the training set. In this case, the specialist can replace the color names with their respective frequencies (1/3 for Red, 1/3 for White, and 1/3 for Green) to represent them numerically.

upvoted 4 times

Ajose0

2 months, 1 week ago

Frequency encoding is a feature engineering technique used to convert categorical variables into numerical ones by replacing each category with the frequency of its occurrence in the training set. This approach can be useful when dealing with high-cardinality categorical variables, which are categorical variables with a large number of distinct categories.

upvoted 2 times

  **maxkm** 3 months ago

Selected Answer: BD

These are the only options preserving what "color" is. One-hot encoding is a default standard for any categorical data to be fed to a model that takes in numeric input. RGB format is a good numeric representation of any color by preserving its nature

upvoted 2 times

  **Sonoko** 4 months, 2 weeks ago

Selected Answer: AB

A&B
<https://victorzhou.com/blog/one-hot/#:~:text=One-Hot%20Encoding%20takes%20a%20single%20integer%20and%20produces,of%20colors%20are%20possible%3A%20red%2C%20green%2C%20or%20blue.>

upvoted 1 times

  **wolfsong** 2 months, 2 weeks ago

B & E.
It cannot be A because your URL specifically states that:
"This is known as integer encoding. For Machine Learning, this encoding can be problematic - in this example, we're essentially saying "green" is the average of "red" and "blue", which can lead to weird unexpected outcomes."

upvoted 2 times

  **ovokpus** 10 months, 1 week ago

Selected Answer: BE

Frequency encoding

upvoted 3 times

  **[Removed]** 10 months, 2 weeks ago

B and E

upvoted 3 times

  **tgaos** 11 months ago

BE is correct.
For e, please refer: <https://medium.com/analytics-vidhya/different-type-of-feature-engineering-encoding-techniques-for-categorical-variable-encoding-214363a016fb#:~:text=One%20Hot%20Encoding%3A%20%E2%80%94%20In%20this,slows%20down%20the%20learning%20significantly.>

upvoted 3 times

Question #164

Topic 1

A data scientist is working on a public sector project for an urban traffic system. While studying the traffic patterns, it is clear to the data scientist that the traffic behavior at each light is correlated, subject to a small stochastic error term. The data scientist must model the traffic behavior to analyze the traffic patterns and reduce congestion.

How will the data scientist MOST effectively model the problem?

- A. The data scientist should obtain a correlated equilibrium policy by formulating this problem as a multi-agent reinforcement learning problem.
- B. The data scientist should obtain the optimal equilibrium policy by formulating this problem as a single-agent reinforcement learning problem.
- C. Rather than finding an equilibrium policy, the data scientist should obtain accurate predictors of traffic flow by using historical data through a supervised learning approach.
- D. Rather than finding an equilibrium policy, the data scientist should obtain accurate predictors of traffic flow by using unlabeled simulated data representing the new traffic patterns in the city and applying an unsupervised learning approach.

Correct Answer: D

Reference:

<https://www.hindawi.com/journals/jat/2021/8878011/>

Community vote distribution

A (100%)

  **DD4** Highly Voted 8 months ago

answer : A, because the setting needs multi agents and is constrained with traffic light correlation.
upvoted 12 times

  **Skychaser** Highly Voted 9 months, 3 weeks ago

https://www.researchgate.net/publication/221456376_Multi-Agent_Reinforcement_Learning_for_Simulating_Pedestrian_Navigation
upvoted 5 times

  **Ajose0** Most Recent 2 months, 1 week ago

Selected Answer: A

A. The data scientist should obtain a correlated equilibrium policy by formulating this problem as a multi-agent reinforcement learning problem.

In this scenario, where the traffic behavior at each light is correlated, a multi-agent reinforcement learning (MARL) approach is well-suited to model the problem. In MARL, multiple agents interact with each other and the environment, and their behavior is influenced by the behavior of other agents. This approach is particularly useful in modeling traffic systems, where the behavior of each vehicle is affected by the behavior of other vehicles and traffic lights.

Formulating the problem as a MARL problem can help the data scientist obtain a correlated equilibrium policy, which can optimize traffic flow across multiple traffic lights by taking into account the correlations between them. By optimizing traffic flow across all traffic lights in a correlated way, it may be possible to reduce congestion and improve overall traffic efficiency.

upvoted 3 times

  **damaldon** 2 months, 3 weeks ago

Answer is A
upvoted 1 times

A data scientist is using the Amazon SageMaker Neural Topic Model (NTM) algorithm to build a model that recommends tags from blog posts. The raw blog post data is stored in an Amazon S3 bucket in JSON format. During model evaluation, the data scientist discovered that the model recommends certain stopwords such as "a," "an," and "the" as tags to certain blog posts, along with a few rare words that are present only in certain blog entries. After a few iterations of tag review with the content team, the data scientist notices that the rare words are unusual but feasible. The data scientist also must ensure that the tag recommendations of the generated model do not include the stopwords. What should the data scientist do to meet these requirements?

- A. Use the Amazon Comprehend entity recognition API operations. Remove the detected words from the blog post data. Replace the blog post data source in the S3 bucket.
- B. Run the SageMaker built-in principal component analysis (PCA) algorithm with the blog post data from the S3 bucket as the data source. Replace the blog post data in the S3 bucket with the results of the training job.
- C. Use the SageMaker built-in Object Detection algorithm instead of the NTM algorithm for the training job to process the blog post data.
- D. Remove the stopwords from the blog post data by using the CountVectorizer function in the scikit-learn library. Replace the blog post data in the S3 bucket with the results of the vectorizer.

Correct Answer: D

Reference:

<https://towardsdatascience.com/natural-language-processing-count-vectorization-with-scikit-learn-e7804269bb5e>

Community vote distribution

D (83%)

A (17%)

exam_prep

Highly Voted

11 months ago

D: Option A, B & C don't make sense. D removes the stop words and help in count vectors

upvoted 16 times

Valcilio

Most Recent

1 month, 3 weeks ago

Selected Answer: D

Needs to remove stopwords and the rare worlds are feasible.

upvoted 2 times

Peeking

4 months, 3 weeks ago

Selected Answer: D

The stop words need to be removed. The rare words don't need to be removed because it has been found that they are feasible tags.

upvoted 3 times

HerbertK

7 months, 1 week ago

Selected Answer: A

Why not A?

upvoted 1 times

VinceCar

5 months, 3 weeks ago

check the requirement in question: "the generated model do not include the stopwords"

upvoted 1 times

A company wants to create a data repository in the AWS Cloud for machine learning (ML) projects. The company wants to use AWS to perform complete ML lifecycles and wants to use Amazon S3 for the data storage. All of the company's data currently resides on premises and is 40 TB in size.

The company wants a solution that can transfer and automatically update data between the on-premises object storage and Amazon S3. The solution must support encryption, scheduling, monitoring, and data integrity validation.

Which solution meets these requirements?

- A. Use the S3 sync command to compare the source S3 bucket and the destination S3 bucket. Determine which source files do not exist in the destination S3 bucket and which source files were modified.
- B. Use AWS Transfer for FTPS to transfer the files from the on-premises storage to Amazon S3.
- C. Use AWS DataSync to make an initial copy of the entire dataset. Schedule subsequent incremental transfers of changing data until the final cutover from on premises to AWS.
- D. Use S3 Batch Operations to pull data periodically from the on-premises storage. Enable S3 Versioning on the S3 bucket to protect against accidental overwrites.

Correct Answer: C

Configure DataSync to make an initial copy of your entire dataset, and schedule subsequent incremental transfers of changing data until the final cut-over from on- premises to AWS.

Reference:

<https://aws.amazon.com/datasync/faqs/>

Community vote distribution

C (91%)

9%

- Sivadharan

Highly Voted

11 months, 3 weeks ago

Selected Answer: C

<https://aws.amazon.com/datasync/faqs/>
Based on answer for the question - "How do I use AWS DataSync to migrate data to AWS?"
upvoted 13 times
- matteocal

Highly Voted

9 months, 1 week ago

Selected Answer: C

it's DataSync
upvoted 5 times
- Ajose0

Most Recent

2 months, 1 week ago

Selected Answer: C

AWS DataSync is a service that can be used to transfer large amounts of data between on-premises storage and Amazon S3, EFS, or FSx for Windows File Server. DataSync is optimized for fast, automated, and secure transfers of large amounts of data, and it supports scheduling, monitoring, and data integrity validation.

In this scenario, the company wants a solution that can transfer and automatically update data between the on-premises object storage and Amazon S3, with support for encryption, scheduling, monitoring, and data integrity validation. DataSync meets all of these requirements, as it can transfer data using secure network connections, schedule data transfers, verify data integrity, and encrypt data in transit and at rest.

upvoted 2 times
- edvardo

11 months, 3 weeks ago

Selected Answer: A

A maybe?

<https://aws.amazon.com/datasync/faqs/>
upvoted 2 times
- edvardo

11 months, 3 weeks ago

I meant C

upvoted 4 times

A company has video feeds and images of a subway train station. The company wants to create a deep learning model that will alert the station manager if any passenger crosses the yellow safety line when there is no train in the station. The alert will be based on the video feeds. The company wants the model to detect the yellow line, the passengers who cross the yellow line, and the trains in the video feeds. This task requires labeling. The video data must remain confidential.

A data scientist creates a bounding box to label the sample data and uses an object detection model. However, the object detection model cannot clearly demarcate the yellow line, the passengers who cross the yellow line, and the trains.

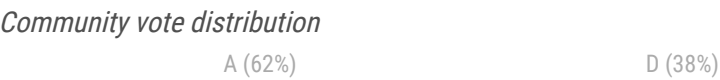
Which labeling approach will help the company improve this model?

- A. Use Amazon Rekognition Custom Labels to label the dataset and create a custom Amazon Rekognition object detection model. Create a private workforce. Use Amazon Augmented AI (Amazon A2I) to review the low-confidence predictions and retrain the custom Amazon Rekognition model.
- B. Use an Amazon SageMaker Ground Truth object detection labeling task. Use Amazon Mechanical Turk as the labeling workforce.
- C. Use Amazon Rekognition Custom Labels to label the dataset and create a custom Amazon Rekognition object detection model. Create a workforce with a third-party AWS Marketplace vendor. Use Amazon Augmented AI (Amazon A2I) to review the low-confidence predictions and retrain the custom Amazon Rekognition model.
- D. Use an Amazon SageMaker Ground Truth semantic segmentation labeling task. Use a private workforce as the labeling workforce.

Correct Answer: B

Reference:

<https://docs.aws.amazon.com/sagemaker/latest/dg/sms-workforce-management-public.html>



- LydiaGom

Highly Voted

11 months, 3 weeks ago

A : The data has label. So what we need to do is to enforce accuracy by reviewing low confidence ones internally

upvoted 15 times
- VinceCar

4 months, 3 weeks ago

Not A, bounding box should be a feature of Ground Truth.

https://docs.aws.amazon.com/zh_cn/sagemaker/latest/dg/sms-bounding-box.html

upvoted 1 times
- wolfsong

2 months, 2 weeks ago

It's A.

See <https://aws.amazon.com/rekognition/custom-labels-features/>.

It says "The Rekognition Custom Labels console provides a visual interface to make labeling your images fast and simple. The interface allows you to apply a label to the entire image or to identify and label specific objects in images using bounding boxes with a simple click-and-drag interface."

We are not using semantic segmentation, as it applies a label to every pixel. We don't want that, we want labels to bounding boxes.

upvoted 1 times
- spaceexplorer

Highly Voted

1 year ago

Selected Answer: D

D; B is using MTurk which uses public workforce which violates the requirements that videos need to be kept private

upvoted 13 times
- VinceCar

5 months, 3 weeks ago

Agreed. Option A used Amazon Augmented AI (Amazon A2I), not a good way to review confidential data.

upvoted 1 times
- Sidekick

8 months, 3 weeks ago

The Problem with Answer D though is that there is no "semantic segmentation labeling task" within the very limited list of Ground Truth type of jobs it offers.

There is a video classification Job type, and there is Video Frame labeling job type, which includes "video frame object detection job" and "video frame object tracking job. However, there is no Semantic Segmentation Labeling Job"

upvoted 2 times
- unitit

2 months, 4 weeks ago

There is a Ground Truth for Semantic Segmentation labelling task - <https://docs.aws.amazon.com/sagemaker/latest/dg/sms-semantic-segmentation.html>.

Therefore, D is correct.

upvoted 3 times

  **Sidekick** 8 months, 3 weeks ago

And Semantic segmentation is object classification done at the pixel level. Isn't that something only machines can do? Unless labelers are directing a machine to do the semantic segmentation for them, I think labelers are no use for it.

upvoted 1 times

  **Sidekick** 8 months, 3 weeks ago

<https://docs.aws.amazon.com/sagemaker/latest/dg/sms-video.html>

upvoted 1 times

  **Sidekick** 8 months, 3 weeks ago

A quick google search on SageMaker ground truth will show you that you can indeed create your own private labelers workforce and send them labelling jobs through GroundTruth.

"You have options to work with labelers inside and outside your organization. For example, you can send labeling jobs to your own labelers, or you can access a workforce of over 500,000 independent contractors who are already performing ML-related tasks through Amazon Mechanical Turk. If your data requires confidentiality or special skills, you can also use vendors that are pre-screened by AWS for quality and security procedures."

upvoted 1 times

  **Sidekick** 8 months, 3 weeks ago

Nevermind My bad. Just realized that you are referring particularly to Mechanical Turk being used as the labelers force for Ground Truth, which is what answer B referring to.

upvoted 1 times

  **blanco750** Most Recent 1 month, 1 week ago

Selected Answer: A

A looks a better option. as pointed by @tagos <https://aws.amazon.com/blogs/machine-learning/using-amazon-rekognition-custom-labels-and-amazon-a2i-for-detecting-pizza-slices-and-augmenting-predictions/>

Also some ppl mentioned in comments that A2I can't be used for confidential data. From FAQs

Q: Can Amazon Augmented AI third-party service providers process customer confidential data?

A: Yes, Amazon Augmented AI service providers can process customer confidential data. The Standard Service Agreement between AWS customers and the third-party service provider contains clauses to protect your confidential

upvoted 4 times

  **Amit11011996** 1 month, 2 weeks ago

Selected Answer: D

D is correct, we need private workforce to label the dataset. A2I won't help with confidential data.

upvoted 1 times

  **Chelseajcole** 1 month, 3 weeks ago

Selected Answer: A

It says yellow line so semantic segmentation not helpful

upvoted 1 times

  **Ajose0** 2 months, 1 week ago

Selected Answer: A

Since the object detection model cannot clearly demarcate the yellow line, passengers who cross the yellow line, and the trains, it requires further fine-tuning with labeled data. Amazon Rekognition Custom Labels can help the company to label the dataset with bounding boxes and create a custom object detection model. Creating a private workforce will ensure the data privacy of the confidential video feeds. Amazon A2I can be used to review the low-confidence predictions and improve the model. By using this approach, the company can improve the accuracy of the model and address the specific requirements of the task.

upvoted 4 times

  **Jerry84** 2 months, 4 weeks ago

Selected Answer: A

There are two key information in the question.

1. "bounding box to label the sample data and uses an object detection model."
2. "The video data must remain confidential"

For key point 1, we could rule out D as "semantic segmentation labeling" is a totally different labeling task from object detection labeling task. (see details in <https://docs.aws.amazon.com/sagemaker/latest/dg/sms-label-images.html>)

For key point 2, we could rule out B(Amazon Mechanical Turk), C(workforce with a third-party AWS Marketplace vendor). (see details in <https://docs.aws.amazon.com/sagemaker/latest/dg/sms-custom-templates-step1.html>)

upvoted 3 times

  **Peeking** 4 months, 3 weeks ago

Selected Answer: D

Private workforce is required. In A, Rekognition Custom Label is not for object labeling but for training.

upvoted 1 times

  **aScientist** 5 months, 3 weeks ago

Selected Answer: A

Rekognition custom labels allows you to identify the objects and scenes in images that are specific to your business needs. For example, you can find your logo in social media posts, identify your products on store shelves etc.
<https://aws.amazon.com/rekognition/custom-labels-features/>
upvoted 2 times

  **31Rishab** 6 months, 3 weeks ago

Link clearly states Mechanical Turk should not be used fo confidential data.
<https://docs.aws.amazon.com/sagemaker/latest/dg/sms-workforce-management-public.html>
upvoted 1 times

  **example_** 7 months, 2 weeks ago

<https://docs.aws.amazon.com/sagemaker/latest/dg/sms-semantic-segmentation.html>
upvoted 1 times

  **ovokpus** 10 months, 1 week ago

Selected Answer: A

I say A because:
Augmented AI can, but will not access Mechanical Turk in this instance. They used a private workforce instead.

Mechanical Turk is public, which disqualifies B. This also disqualifies C, which wants to use a 3rd party AWS vendor (not private).

D is out because the problem is not one of semantic segmentation, it is object detection.
upvoted 10 times

  **tgaos** 11 months, 1 week ago

Answer A. From the example: <https://aws.amazon.com/blogs/machine-learning/using-amazon-rekognition-custom-labels-and-amazon-a2i-for-detecting-pizza-slices-and-augmenting-predictions/>
upvoted 7 times

A data engineer at a bank is evaluating a new tabular dataset that includes customer data. The data engineer will use the customer data to create a new model to predict customer behavior. After creating a correlation matrix for the variables, the data engineer notices that many of the 100 features are highly correlated with each other.

Which steps should the data engineer take to address this issue? (Choose two.)

- A. Use a linear-based algorithm to train the model.
- B. Apply principal component analysis (PCA).
- C. Remove a portion of highly correlated features from the dataset.
- D. Apply min-max feature scaling to the dataset.
- E. Apply one-hot encoding category-based variables.

Correct Answer: *BD*

Reference:
<https://royalsocietypublishing.org/doi/10.1098/rsta.2015.0202> https://scikit-learn.org/stable/auto_examples/preprocessing/plot_all_scaling.html

Community vote distribution

BC (100%)

-   **ayatkhrisat**

Highly Voted 

 11 months, 4 weeks ago

Selected Answer: BC

B,C should be the answer
upvoted 15 times
-   **Ajose0**

Most Recent 

 2 months, 1 week ago

Selected Answer: BC

PCA is a widely used technique for reducing the dimensionality of high-dimensional datasets while retaining as much of the original variability as possible. It is particularly useful when dealing with highly correlated features.

Removing a portion of highly correlated features can be another effective way to address the issue of high correlation. By removing some of the correlated features, the model can become less complex and less prone to overfitting.
upvoted 2 times
-   **ovokpus** 10 months, 1 week ago

Selected Answer: BC

MinMax scaling does nothing to fix the issue here
upvoted 3 times
-   **ckkobe24** 11 months, 4 weeks ago

Selected Answer: BC

BC for me, minmax scaler cannot remove multicollinearity?
upvoted 4 times

A company is building a new version of a recommendation engine. Machine learning (ML) specialists need to keep adding new data from users to improve personalized recommendations. The ML specialists gather data from the users' interactions on the platform and from sources such as external websites and social media.

The pipeline cleans, transforms, enriches, and compresses terabytes of data daily, and this data is stored in Amazon S3. A set of Python scripts was coded to do the job and is stored in a large Amazon EC2 instance. The whole process takes more than 20 hours to finish, with each script taking at least an hour. The company wants to move the scripts out of Amazon EC2 into a more managed solution that will eliminate the need to maintain servers.

Which approach will address all of these requirements with the LEAST development effort?

- A. Load the data into an Amazon Redshift cluster. Execute the pipeline by using SQL. Store the results in Amazon S3.
- B. Load the data into Amazon DynamoDB. Convert the scripts to an AWS Lambda function. Execute the pipeline by triggering Lambda executions. Store the results in Amazon S3.
- C. Create an AWS Glue job. Convert the scripts to PySpark. Execute the pipeline. Store the results in Amazon S3.
- D. Create a set of individual AWS Lambda functions to execute each of the scripts. Build a step function by using the AWS Step Functions Data Science SDK. Store the results in Amazon S3.

Correct Answer: B

Reference:

<https://docs.aws.amazon.com/lambda/latest/dg/with-s3-example.html>

Community vote distribution

C (95%)5%

  **spaceexplorer**

Highly Voted 

 1 year ago

Selected Answer: C

C; Lambda execution time has hard limit of 15 mins which might not be enough for data processing
upvoted 13 times

  **ckkobe24** 11 months, 4 weeks ago

but C requires some coding efforts
upvoted 1 times

  **Ajose0**

Most Recent 

 2 months, 1 week ago

Selected Answer: C

The data pipeline involves cleaning, transforming, enriching, and compressing terabytes of data and storing the data in Amazon S3.

AWS Glue is an ETL service that makes it easy to move data between data stores. The Glue job allows you to use PySpark scripts to perform ETL tasks. With AWS Glue, you do not need to provision and manage servers, which eliminates the need to maintain servers, as required by the company.

Therefore, AWS Glue would address all of the company's requirements with the least development effort.
upvoted 3 times

  **maxkm** 3 months ago

Selected Answer: D

1) eliminate the need to maintain servers - Lambda is serverless 2) the least development effort - python scripts do no need to be rewritten for Lambda function
upvoted 1 times

  **GiyeonShin** 2 months, 1 week ago

"with each script taking at least an hour" - lambda would be time-out during taking job.
upvoted 2 times

  **milan_ml** 9 months, 1 week ago

Selected Answer: C

C as for Redshift I need to build a new pipeline
upvoted 1 times

  **ovokpus** 10 months, 1 week ago

Selected Answer: C

Converting python scripts to pyspark is less coding effort than writing up SQL, which is somewhat limited in the types of transformations it can do.

Lambda function responses are a deadend for reason already given (timeout)
upvoted 2 times

A retail company is selling products through a global online marketplace. The company wants to use machine learning (ML) to analyze customer feedback and identify specific areas for improvement. A developer has built a tool that collects customer reviews from the online marketplace and stores them in an Amazon S3 bucket. This process yields a dataset of 40 reviews. A data scientist building the ML models must identify additional sources of data to increase the size of the dataset.

Which data sources should the data scientist use to augment the dataset of reviews? (Choose three.)

- A. Emails exchanged by customers and the company's customer service agents
- B. Social media posts containing the name of the company or its products
- C. A publicly available collection of news articles
- D. A publicly available collection of customer reviews
- E. Product sales revenue figures for the company
- F. Instruction manuals for the company's products

Correct Answer: BDF

Community vote distribution

ABD (100%)

  **spaceexplorer** Highly Voted 1 year ago

Selected Answer: ABD

ABD; Email exchange between customer and customer service would be valuable data source.
upvoted 25 times

  **Ajose0** Most Recent 2 months, 1 week ago

Selected Answer: ABD

A: Emails exchanged by customers and the company's customer service agents can provide additional customer feedback and opinions about the products or services. This data can be used to improve the ML model.

B: Social media posts containing the name of the company or its products can provide additional customer feedback and opinions about the products or services, which can be used to improve the ML model.

D: A publicly available collection of customer reviews can be used to augment the existing dataset of reviews and increase the size of the dataset. This can help to improve the accuracy of the ML model.
upvoted 3 times

  **Ajose0** 2 months, 1 week ago

C: A publicly available collection of news articles and F: Instruction manuals for the company's products are not directly related to customer feedback and may not be relevant for improving the ML model in this context.

E: Product sales revenue figures for the company can provide valuable insights into the company's financial performance, but this data is not directly related to customer feedback and may not be useful for improving the ML model in this context.
upvoted 1 times

  **f4bi4n** 10 months, 2 weeks ago

I think ABE or maybe(!) ABF
A & B -> for sure
C -> No glue how this should help
D -> We have already reviews!?
E -> Could help to find correlations between negative / positive reviews and sales
F -> non-sense in the first moment and second as well, but maybe it could help to combine the information of Email, Social and Reviews to some problems.
upvoted 3 times

  **Sonoko** 4 months, 2 weeks ago

ABE makes more sense here.
upvoted 1 times

  **AdolinKholin** 6 months, 3 weeks ago

That's exactly the point though, we want more volume of reviews, or anything resembling reviews as is the case with email exchanges.
upvoted 1 times

  **bluer1** 1 year ago

BDF - correct answer
upvoted 1 times

A machine learning (ML) specialist wants to create a data preparation job that uses a PySpark script with complex window aggregation operations to create data for training and testing. The ML specialist needs to evaluate the impact of the number of features and the sample count on model performance.

Which approach should the ML specialist use to determine the ideal data transformations for the model?

- A. Add an Amazon SageMaker Debugger hook to the script to capture key metrics. Run the script as an AWS Glue job.
- B. Add an Amazon SageMaker Experiments tracker to the script to capture key metrics. Run the script as an AWS Glue job.
- C. Add an Amazon SageMaker Debugger hook to the script to capture key parameters. Run the script as a SageMaker processing job.
- D. Add an Amazon SageMaker Experiments tracker to the script to capture key parameters. Run the script as a SageMaker processing job.

Correct Answer: B

Reference:

<https://docs.aws.amazon.com/sagemaker/latest/dg/experiments.html>

Community vote distribution

D (70%)

B (30%)

  **dolorez** Highly Voted 11 months, 1 week ago

Selected Answer: D

while I agree that Sagemaker Experiments is the way to go, it only supports Training, Processing, and Transform jobs, so the right answer is to run the job as a processing job, hence D not B

<https://docs.aws.amazon.com/sagemaker/latest/dg/experiments-create.html#:~:text=CreateTrainingJob-,Processing,-Processor.run>

upvoted 11 times

  **Jerry84** 2 months, 1 week ago

Run PySpark script in SageMaker processing job

https://sagemaker.readthedocs.io/en/stable/amazon_sagemaker_processing.html

upvoted 1 times

  **Jerry84** 3 months, 2 weeks ago

"Generally, you use load_run with no arguments to track metrics, parameters, and artifacts within a SageMaker training or processing job script."

<https://docs.aws.amazon.com/sagemaker/latest/dg/experiments-create.html>

upvoted 2 times

  **bluer1** Highly Voted 1 year ago

B - <https://docs.aws.amazon.com/sagemaker/latest/dg/experiments.html>

upvoted 7 times

  **KlaudYu** 10 months, 2 weeks ago

But It doesn't describe glue job.

upvoted 3 times

  **Mlb** Most Recent 4 weeks, 1 day ago

Selected Answer: B

Key metrics is the "key". Then D is not a correct answer

upvoted 1 times

  **ZSun** 1 week, 5 days ago

what is the difference between key metrics and key parameteres? why we care about key metrics, because we can compare the key metrics of different parametes and then find impact of the number of features.

so the key is "glue" or "SageMaker processing"

upvoted 1 times

  **blanco750** 1 month, 1 week ago

Selected Answer: D

D looks the right answer

upvoted 2 times

  **SANDEEP_AWS** 1 month, 2 weeks ago

Selected Answer: B

<https://docs.aws.amazon.com/sagemaker/latest/dg/experiments.html> ---- Use SageMaker Experiments to view, manage, analyze, and compare both custom experiments that you programmatically create and experiments automatically created from SageMaker jobs.

upvoted 1 times

  **jhonivy** 3 months ago

B: Glue job goes with window aggregation operations

upvoted 1 times

  **aScientist** 5 months, 3 weeks ago

Selected Answer: D

https://sagemaker-examples.readthedocs.io/en/latest/sagemaker_processing/spark_distributed_data_processing/sagemaker-spark-processing.html

upvoted 3 times

  **ovokpus** 10 months, 1 week ago

Selected Answer: B

here:

<https://aws.amazon.com/about-aws/whats-new/2018/10/aws-glue-now-supports-connecting-amazon-sagemaker-notebooks-to-development-endpoints/#:~:text=AWS%20Glue%20now%20supports%20connecting%20Amazon%20SageMaker%20notebooks%20to%20development%20endp oints,-Posted%20On%3A%20Oct&text=You%20can%20now%20create%20an,an%20AWS%20Glue%20development%20endpoint.>

upvoted 5 times

A data scientist has a dataset of machine part images stored in Amazon Elastic File System (Amazon EFS). The data scientist needs to use Amazon SageMaker to create and train an image classification machine learning model based on this dataset. Because of budget and time constraints, management wants the data scientist to create and train a model with the least number of steps and integration work required. How should the data scientist meet these requirements?

- A. Mount the EFS file system to a SageMaker notebook and run a script that copies the data to an Amazon FSx for Lustre file system. Run the SageMaker training job with the FSx for Lustre file system as the data source.
- B. Launch a transient Amazon EMR cluster. Configure steps to mount the EFS file system and copy the data to an Amazon S3 bucket by using S3DistCp. Run the SageMaker training job with Amazon S3 as the data source.
- C. Mount the EFS file system to an Amazon EC2 instance and use the AWS CLI to copy the data to an Amazon S3 bucket. Run the SageMaker training job with Amazon S3 as the data source.
- D. Run a SageMaker training job with an EFS file system as the data source.

Correct Answer: A

Reference:

<https://aws.amazon.com/about-aws/whats-new/2019/08/amazon-sagemaker-works-with-amazon-fsx-lustre-amazon-efs-model-training/>

Community vote distribution

D (100%)

- cron0001

Highly Voted

1 year ago

Selected Answer: D

Should be D according to the following article
<https://aws.amazon.com/blogs/machine-learning/speed-up-training-on-amazon-sagemaker-using-amazon-efs-or-amazon-fsx-for-lustre-file-systems/>
upvoted 10 times

MIlb

Most Recent

4 weeks, 1 day ago

Selected Answer: D

Less effort, then D
upvoted 3 times

Jerry84

3 months, 2 weeks ago

Selected Answer: D

“When you create a training job, you specify the location of a training dataset and an input mode for accessing the dataset. For data location, Amazon SageMaker supports Amazon Simple Storage Service (Amazon S3), Amazon Elastic File System (Amazon EFS), and Amazon FSx for Lustre. ”

<https://docs.aws.amazon.com/sagemaker/latest/dg/model-access-training-data.html>
upvoted 2 times

srikanth923

6 months, 2 weeks ago

Selected Answer: D

<https://aws.amazon.com/blogs/machine-learning/mount-an-efs-file-system-to-an-amazon-sagemaker-notebook-with-lifecycle-configurations/>
upvoted 3 times

ckkobe24

11 months, 4 weeks ago

Amazon SageMaker now supports Amazon Elastic File System (Amazon EFS) and Amazon FSx for Lustre file systems as data sources for training machine learning models on SageMaker. then why not select D ??
upvoted 4 times

jhonivy

3 months ago

Time constraints. A is the right answer for this question
upvoted 1 times

wolfsong

2 months, 2 weeks ago

A is not the right answer. It's D.
- A requires this setup: EFS -> Lustre -> Sagemaker.
- D requires this setup: EFS -> Sagemaker
It's obviously not A.
upvoted 2 times

<https://www.examtopycs.com/exams/amazon/aws-certified-machine-learning-specialty/custom-view/>

314/406

A retail company uses a machine learning (ML) model for daily sales forecasting. The company's brand manager reports that the model has provided inaccurate results for the past 3 weeks.

At the end of each day, an AWS Glue job consolidates the input data that is used for the forecasting with the actual daily sales data and the predictions of the model. The AWS Glue job stores the data in Amazon S3. The company's ML team is using an Amazon SageMaker Studio notebook to gain an understanding about the source of the model's inaccuracies.

What should the ML team do on the SageMaker Studio notebook to visualize the model's degradation MOST accurately?

- A. Create a histogram of the daily sales over the last 3 weeks. In addition, create a histogram of the daily sales from before that period.
- B. Create a histogram of the model errors over the last 3 weeks. In addition, create a histogram of the model errors from before that period.
- C. Create a line chart with the weekly mean absolute error (MAE) of the model.
- D. Create a scatter plot of daily sales versus model error for the last 3 weeks. In addition, create a scatter plot of daily sales versus model error from before that period.

Correct Answer: C

Reference:

<https://machinelearningmastery.com/time-series-forecasting-performance-measures-with-python/>

Community vote distribution

C (100%)

exam_prep

Highly Voted

11 months, 1 week ago

C is the correct answer.

upvoted 12 times

Valcilio

Most Recent

1 month, 3 weeks ago

Selected Answer: C

C is the answer, line plots are good solutions for time series analysis.

upvoted 3 times

Jerry84

3 months, 2 weeks ago

Selected Answer: C

C is correct. We could view the "Degradation" as a trend.

Line charts are usually very helpful to show if there is any trend in the data over the period of time under analysis.

Histogram is normally used to visualizing distributions in your data.

upvoted 3 times

DD4

8 months ago

Should be A because it is daily forecasting and histograms before and after will show the comparable degradation.

upvoted 1 times

venimus_vidimus_vicimus

4 months, 2 weeks ago

It only states to plot daily sales.. how should that help with the error?

You need a plot of the actual and predicted values or or the errors - def not A

upvoted 1 times

weixing

10 months, 4 weeks ago

B, I guess

upvoted 3 times

wolfsong

2 months, 2 weeks ago

No, it's neither A or B as they use a histogram which would plot the distribution of errors. It will not tell you anything about how the model degrades over time, as the histogram will have no time component. You need a line chart for this. So it's C.

upvoted 1 times

An ecommerce company sends a weekly email newsletter to all of its customers. Management has hired a team of writers to create additional targeted content. A data scientist needs to identify five customer segments based on age, income, and location. The customers' current segmentation is unknown. The data scientist previously built an XGBoost model to predict the likelihood of a customer responding to an email based on age, income, and location.

Why does the XGBoost model NOT meet the current requirements, and how can this be fixed?

- A. The XGBoost model provides a true/false binary output. Apply principal component analysis (PCA) with five feature dimensions to predict a segment.
- B. The XGBoost model provides a true/false binary output. Increase the number of classes the XGBoost model predicts to five classes to predict a segment.
- C. The XGBoost model is a supervised machine learning algorithm. Train a k-Nearest-Neighbors (kNN) model with K = 5 on the same dataset to predict a segment.
- D. The XGBoost model is a supervised machine learning algorithm. Train a k-means model with K = 5 on the same dataset to predict a segment.

Correct Answer: C

Community vote distribution

D (100%)

  **spaceexplorer**

Highly Voted 

 1 year ago

Selected Answer: D

Answer is D! K-means used for customer segmentation
upvoted 14 times

  **Omijh** 11 months, 1 week ago

well, both are used for customer segmentation Knn & kmeans but kmeans is for unsupervised learning and knn is for supervised learning. since we have the data it's better to use supervised learning in this case. Ref: https://rstudio-pubs-static.s3.amazonaws.com/599866_59be74824ca7482ba99dbc8466dc36a0.html#:~:text=The%20difference%20between%20the%20two,to%20predict%20the%20unlabelled%20data.
upvoted 2 times

  **tgaos**

Highly Voted 

 11 months, 1 week ago

The answer is D.
1. "The current segmentation of consumers is unclear." so it is unsupervised learning.
2. Then K-means is for unsupervised learning.
upvoted 9 times

  **Peeking**

Most Recent 

 4 months, 3 weeks ago

Selected Answer: D

KNN has no k parameter in its input. C is not the answer.
upvoted 1 times

  **drcok87** 2 months, 3 weeks ago

in K-means also there is no input parameter "K". What i mean to say here is in knn the k is nothing but "kNN classifier identifies the class of a data point using the majority voting principle. If k is set to 5, the classes of 5 nearest points are examined."
upvoted 1 times

  **matteocal** 9 months, 1 week ago

D
The key work is that the classification is "unclear", therefore k-means
upvoted 3 times

A global financial company is using machine learning to automate its loan approval process. The company has a dataset of customer information. The dataset contains some categorical fields, such as customer location by city and housing status. The dataset also includes financial fields in different units, such as account balances in US dollars and monthly interest in US cents.

The company's data scientists are using a gradient boosting regression model to infer the credit score for each customer. The model has a training accuracy of 99% and a testing accuracy of 75%. The data scientists want to improve the model's testing accuracy.

Which process will improve the testing accuracy the MOST?

- A. Use a one-hot encoder for the categorical fields in the dataset. Perform standardization on the financial fields in the dataset. Apply L1 regularization to the data.
- B. Use tokenization of the categorical fields in the dataset. Perform binning on the financial fields in the dataset. Remove the outliers in the data by using the z- score.
- C. Use a label encoder for the categorical fields in the dataset. Perform L1 regularization on the financial fields in the dataset. Apply L2 regularization to the data.
- D. Use a logarithm transformation on the categorical fields in the dataset. Perform binning on the financial fields in the dataset. Use imputation to populate missing values in the dataset.

Correct Answer: B

Community vote distribution

A (100%)

ckkobe24

Highly Voted

11 months, 4 weeks ago

Selected Answer: A

agree it's A for me
upvoted 14 times

spaceexplorer

Highly Voted

1 year ago

A: it's overfitting so regularization is needed, need apply scaling on financial data fields as it's for regression problem; one hot encoding for city of the house field.
upvoted 10 times

Tony_1406

Most Recent

2 days, 15 hours ago

Selected Answer: A

Agree with A, but I think the answer is slightly inaccurate. L1 regularization within the model and to the loss function. As a result, some features will be removed in the model. The answer suggest L1 regularization is applied to the dataset directly.
upvoted 1 times

Ajose0

2 months, 1 week ago

Selected Answer: A

Option A is the most appropriate approach to improve the testing accuracy of the model.

One-hot encoding can effectively represent categorical variables in a numeric format that is suitable for machine learning models. Standardizing the financial fields can make the data more comparable and improve the model's performance. L1 regularization can help in feature selection and avoid overfitting by reducing the number of features.
upvoted 2 times

Peeking

4 months, 3 weeks ago

Why are most of the chosen answers by ExamTopics mostly obviously wrong? There is nothing like tokenisation of categorical variable and B should be obviously wrong.
upvoted 1 times

Shailendraa

7 months, 3 weeks ago

12-sep exam
upvoted 2 times

A machine learning (ML) specialist needs to extract embedding vectors from a text series. The goal is to provide a ready-to-ingest feature space for a data scientist to develop downstream ML predictive models. The text consists of curated sentences in English. Many sentences use similar words but in different contexts. There are questions and answers among the sentences, and the embedding space must differentiate between them.

Which options can produce the required embedding vectors that capture word context and sequential QA information? (Choose two.)

- A. Amazon SageMaker seq2seq algorithm
- B. Amazon SageMaker BlazingText algorithm in Skip-gram mode
- C. Amazon SageMaker Object2Vec algorithm
- D. Amazon SageMaker BlazingText algorithm in continuous bag-of-words (CBOW) mode
- E. Combination of the Amazon SageMaker BlazingText algorithm in Batch Skip-gram mode with a custom recurrent neural network (RNN)

Correct Answer: AC

Reference:
<https://aws.amazon.com/blogs/machine-learning/create-a-word-pronunciation-sequence-to-sequence-model-using-amazon-sagemaker/>
<https://docs.aws.amazon.com/sagemaker/latest/dg/object2vec.html>

Community vote distribution

AC (42%)	BC (21%)	BD (21%)	Other
----------	----------	----------	-------

- ovokpus** Highly Voted 10 months, 1 week ago

Selected Answer: AC

seq2seq and object2vec take care of more than just the words. Any response with blazingText is wrong because blazingText just uses a cbow (continuous bag of words), working only on individual words

upvoted 14 times
- Jerry84** 3 months, 2 weeks ago

“One of the well-known embedding techniques is Word2Vec, which provides embeddings for words.” “In addition to word embeddings, there are also use cases where we want to learn the embeddings of more general-purpose objects such as sentences, customers, and products. This is so we can build practical applications for information retrieval, product search, item matching, customer profiling based on similarity or as inputs for other supervised tasks. This is where Amazon SageMaker Object2Vec comes in.”

<https://aws.amazon.com/blogs/machine-learning/introduction-to-amazon-sagemaker-object2vec/>

upvoted 1 times
- peterfish** Highly Voted 9 months, 2 weeks ago

Selected Answer: BD

It should be B an D. The objective is to create a latent space/word embedding that puts similar words closer to each other for other purposes. Thus, we should use Sagemaker Blazing Text in unsupervised mode (Word2Vec mode). cbow, skip-grams, and batch skip-grams are the 3 algorithms for this. However, since we do not need to do the later part of E, E is not correct. The ans should be B and D.

upvoted 7 times
- WilianCB** Most Recent 23 hours, 59 minutes ago

Selected Answer: BC

I've never heard about using seq2seq to generate embedding vectors.

B = In skip gram, the order of the words does matter

C = It's made for these type of things, generate embeddings of "objects" (sentences or what have you).

upvoted 1 times
- MIlb** 4 weeks, 1 day ago

Selected Answer: BC

Word2Vec in n-gram (context sensitive)

Object2Vec: two tokens closer as positive class and two tokens away as negative class

upvoted 2 times
- blanco750** 1 month, 1 week ago

Selected Answer: BE

Question is looking for 2 things

1. Words embedding

2. sequential QA information(questions and answers among the sentences,)

For 1, we have Blazing text skip-gram , batch skip-gram and Object2vec. very difficult to choose

2. However there is part 2 which mentions sequential QA information. @Chelseajcole posted response from chatgpt :) which highlights RNN for

this. Interesting.
so it seems either B/C and E ?
Very difficult. i have already spent 1.5 hrs researching on this question alone :(
upvoted 1 times

  **blanco750** 1 month, 1 week ago

Actually I would change it to B and C. Please ignore the above comment.
upvoted 1 times

  **Chelseajcole** 2 months ago

ChatGPT answer:

Based on the requirements mentioned in the question, the two options that can produce the required embedding vectors that capture word context and sequential QA information are:

B. Amazon SageMaker BlazingText algorithm in Skip-gram mode - This algorithm is specifically designed for generating high-quality word embeddings and can capture the context of words within a sentence. The skip-gram mode is suitable for capturing the relationship between words that appear within a certain distance from each other in the text.

E. Combination of the Amazon SageMaker BlazingText algorithm in Batch Skip-gram mode with a custom recurrent neural network (RNN) - This option allows for the incorporation of additional information from the sequence of sentences by using an RNN. The batch skip-gram mode can be used to capture the relationships between words in a larger context, while the RNN can capture the sequential information and dependencies between the sentences.

Therefore, options B and E can produce the required embedding vectors that capture word context and sequential QA information.
upvoted 2 times

  **blanco750** 1 month, 1 week ago

Oh. this looks really compelling. The question mentions sequential QA information and RNN can be used to capture sequential information !
upvoted 1 times

  **drcok87** 2 months, 2 weeks ago

C is obvious
B or d
Could be d because CBOW->given context, predict word embeddings: "Which options can produce the required embedding vectors that capture word context"
upvoted 2 times

  **damaldon** 2 months, 3 weeks ago

Answer BD. Even option E could be, but custom RNN is not necessary.
<https://aws.amazon.com/blogs/machine-learning/enhanced-text-classification-and-word-vectors-using-amazon-sagemaker-blazingtext/>
upvoted 1 times

  **rrshah83** 4 months ago

Selected Answer: BC

Seq2Seq is for translations or sequences of some kind. They don't return word embedding.

BlazingText and Object2Vec provide word embedding.

Choosing the blazingtext mode: <https://www.mlexam.com/blazingtext-algorithm/>
Read: In skip-gram mode you supply a word and the model returns the context of the word. With CBOW you provide the context and a predicted word is returned.
upvoted 4 times

  **rjekstein** 9 months, 4 weeks ago

Should be A and B
A Seq2seq can generate embedding vectors
B Sagemaker Blazing Text in Skip-gram mode can return the context of word.
upvoted 2 times

  **f4bi4n** 10 months ago

Selected Answer: CD

my best guess -> CD
Both generate Vectors as response and cbow is used for context usecases
upvoted 3 times

  **phani1989** 10 months ago

I believe it should be C and E
upvoted 4 times

  **DJiang** 11 months, 3 weeks ago

Selected Answer: AE

I think answer is A,E: "identical terms but in distinct situations" refers to the typical question of mapping contexts to embedding, and therefore models that are able to handle contexts should be considered.
upvoted 1 times

  **thafa** 10 months, 1 week ago

Why is seq2seq relevant here? Isnt it for machine translation?
upvoted 1 times

A retail company wants to update its customer support system. The company wants to implement automatic routing of customer claims to different queues to prioritize the claims by category.

Currently, an operator manually performs the category assignment and routing. After the operator classifies and routes the claim, the company stores the claim's record in a central database. The claim's record includes the claim's category.

The company has no data science team or experience in the field of machine learning (ML). The company's small development team needs a solution that requires no ML expertise.

Which solution meets these requirements?

- A. Export the database to a .csv file with two columns: claim_label and claim_text. Use the Amazon SageMaker Object2Vec algorithm and the .csv file to train a model. Use SageMaker to deploy the model to an inference endpoint. Develop a service in the application to use the inference endpoint to process incoming claims, predict the labels, and route the claims to the appropriate queue.
- B. Export the database to a .csv file with one column: claim_text. Use the Amazon SageMaker Latent Dirichlet Allocation (LDA) algorithm and the .csv file to train a model. Use the LDA algorithm to detect labels automatically. Use SageMaker to deploy the model to an inference endpoint. Develop a service in the application to use the inference endpoint to process incoming claims, predict the labels, and route the claims to the appropriate queue.
- C. Use Amazon Textract to process the database and automatically detect two columns: claim_label and claim_text. Use Amazon Comprehend custom classification and the extracted information to train the custom classifier. Develop a service in the application to use the Amazon Comprehend API to process incoming claims, predict the labels, and route the claims to the appropriate queue.
- D. Export the database to a .csv file with two columns: claim_label and claim_text. Use Amazon Comprehend custom classification and the .csv file to train the custom classifier. Develop a service in the application to use the Amazon Comprehend API to process incoming claims, predict the labels, and route the claims to the appropriate queue.

Correct Answer: C

Reference:
<https://aws.amazon.com/blogs/machine-learning/intelligently-split-multi-form-document-packages-with-amazon-textract-and-amazon-comprehend/>

Community vote distribution

D (96%)

4%

-   **cron0001** Highly Voted 1 year ago

Selected Answer: D

I would say D. We shouldn't need Textract to extract columns from a database
upvoted 16 times
-   **Omijh** 11 months, 1 week ago

I think D because Textract doesn't support CSV but only PNG, JPEG, TIFF, and PDF formats
upvoted 6 times
-   **ovokpus** Highly Voted 10 months, 1 week ago

Selected Answer: D

D because it does not require heavy machine learning expertise
upvoted 5 times
-   **Ajose0** Most Recent 2 months, 1 week ago

Selected Answer: D

Option D meets the requirements. The solution requires no ML expertise, and the small development team can use the Amazon Comprehend custom classification API to train a model to automatically detect claim categories.

The company can export the database to a .csv file with two columns: claim_label and claim_text. Then, the development team can use the .csv file to train the custom classifier. Finally, the team can develop a service in the application to use the Amazon Comprehend API to process incoming claims, predict the labels, and route the claims to the appropriate queue.

This solution is straightforward, does not require extensive expertise, and can be implemented quickly.
upvoted 1 times
-   **DD4** 8 months ago

Selected Answer: A

It should A because Object2Vec is meant for text classification. The problem is to categorize the text based on the content.
upvoted 1 times

  **s12201812** 8 months ago

B. LDA is for topic modelling based on categories. Comprehend is for extracting the entities related to sentiments etc.
upvoted 1 times

  **GiyeonShin** 2 months, 1 week ago

Comprehend can be used for custom classification of NLP too(<https://aws.amazon.com/ko/comprehend/features/>).
LDA can find document topics and word distribution for topics, but it is necessary to manually link the topics with predefined customer category.
upvoted 1 times

  **hamimelon** 4 months, 1 week ago

But it says the solution should not require ML expertise. LDA requires ML expertise.
upvoted 1 times

  **Morsa** 9 months, 2 weeks ago

The firm needs a solution for their manual process and this means among others the routing of their client orders. To do that you will need textract
upvoted 1 times

A machine learning (ML) specialist is using Amazon SageMaker hyperparameter optimization (HPO) to improve a model's accuracy. The learning rate parameter is specified in the following HPO configuration:

```
{
  "Name": "learning_rate",
  "MaxValue" : "0.0001",
  "MinValue": "0.1"
}
```

During the results analysis, the ML specialist determines that most of the training jobs had a learning rate between 0.01 and 0.1. The best result had a learning rate of less than 0.01. Training jobs need to run regularly over a changing dataset. The ML specialist needs to find a tuning mechanism that uses different learning rates more evenly from the provided range between MinValue and MaxValue.

Which solution provides the MOST accurate result?

A. Modify the HPO configuration as follows:

```
{
  "Name": "learning_rate",
  "MaxValue" : "0.0001",
  "MinValue": "0.1"
}
```

Select the most

accurate hyperparameter configuration form this HPO job.

B. Run three different HPO jobs that use different learning rates form the following intervals for MinValue and MaxValue while using the same number of training jobs for each HPO job: [0.01, 0.1] [0.001, 0.01] [0.0001, 0.001] Select the most accurate hyperparameter configuration form these three HPO jobs.

C. Modify the HPO configuration as follows:

```
{
  "Name": "learning_rate",
  "MaxValue" : "0.0001",
  "MinValue": "0.1"
}
```

Select the most accurate

hyperparameter configuration form this training job.

D. Run three different HPO jobs that use different learning rates form the following intervals for MinValue and MaxValue. Divide the number of training jobs for each HPO job by three: [0.01, 0.1] [0.001, 0.01] [0.0001, 0.001] ▪ Select the most accurate hyperparameter configuration form these three HPO jobs.

Correct Answer: C

Community vote distribution

C (86%)

14%

- ovokpus

Highly Voted

10 months, 1 week ago

Selected Answer: C

"Choose logarithmic scaling when you are searching a range that spans several orders of magnitude. For example, if you are tuning a Tune a linear learner model model, and you specify a range of values between .0001 and 1.0 for the learning_rate hyperparameter, searching uniformly on a logarithmic scale gives you a better sample of the entire range than searching on a linear scale would, because searching on a linear scale would, on average, devote 90 percent of your training budget to only the values between .1 and 1.0, leaving only 10 percent of your training budget for the values between .0001 and .1."

based on the above from this link

<https://docs.aws.amazon.com/sagemaker/latest/dg/automatic-model-tuning-define-ranges.html>

C is clearly the answer

upvoted 6 times
- edvardo

Highly Voted

12 months ago

Selected Answer: C

I would choose C: <https://docs.aws.amazon.com/sagemaker/latest/dg/automatic-model-tuning-define-ranges.html>

upvoted 5 times
- DJiang

11 months, 3 weeks ago

But according to the doc you gave, "Logarithmic scaling works only for ranges that have only values greater than 0."
I think choosing ScalingType=Linear is the best fit, but there's no such option.
upvoted 2 times

  **cox1960** Most Recent 2 days, 4 hours ago

Selected Answer: C

not A since you choose reverse logarithmic scaling when you are searching a range that is highly sensitive to small changes that are very close to 1.
upvoted 1 times

  **MIlb** 4 weeks, 1 day ago

Selected Answer: B

It should be B.
In logarithmic parameters the min value is the maximum value. This is the reason that C is not correct
upvoted 2 times

  **ovokpus** 10 months, 1 week ago

"Choose logarithmic scaling when you are searching a range that spans several orders of magnitude. For example, if you are tuning a Tune a linear learner model model, and you specify a range of values between .0001 and 1.0 for the learning_rate hyperparameter, searching uniformly on a logarithmic scale gives you a better sample of the entire range than searching on a linear scale would, because searching on a linear scale would, on average, devote 90 percent of your training budget to only the values between .1 and 1.0, leaving only 10 percent of your training budget for the values between .0001 and .1."
upvoted 3 times

  **rhuanca** 11 months, 3 weeks ago

B looks better , because learning rates were splited up base on a previous experience (0.1 - 0,01) in this case we are changing the structure . On the other hand A and B only change scaletype and this means no real changes
upvoted 1 times

A manufacturing company wants to use machine learning (ML) to automate quality control in its facilities. The facilities are in remote locations and have limited internet connectivity. The company has 20 1çI’ of training data that consists of labeled images of defective product parts. The training data is in the corporate on- premises data center.

The company will use this data to train a model for real-time defect detection in new parts as the parts move on a conveyor belt in the facilities. The company needs a solution that minimizes costs for compute infrastructure and that maximizes the scalability of resources for training. The solution also must facilitate the company's use of an ML model in the low-connectivity environments.

Which solution will meet these requirements?

- A. Move the training data to an Amazon S3 bucket. Train and evaluate the model by using Amazon SageMaker. Optimize the model by using SageMaker Neo. Deploy the model on a SageMaker hosting services endpoint.
- B. Train and evaluate the model on premises. Upload the model to an Amazon S3 bucket. Deploy the model on an Amazon SageMaker hosting services endpoint.
- C. Move the training data to an Amazon S3 bucket. Train and evaluate the model by using Amazon SageMaker. Optimize the model by using SageMaker Neo. Set up an edge device in the manufacturing facilities with AWS IoT Greengrass. Deploy the model on the edge device.
- D. Train the model on premises. Upload the model to an Amazon S3 bucket. Set up an edge device in the manufacturing facilities with AWS IoT Greengrass. Deploy the model on the edge device.

Correct Answer: A

Reference:
<https://docs.aws.amazon.com/sagemaker/latest/dg/how-it-works-deployment.html>

Community vote distribution

C (100%)

-   **spaceexplorer**

Highly Voted 

 1 year ago

Selected Answer: C

C; Using S3 for scalable training and SageMaker Neo for compiling model for edge devices
upvoted 12 times
-   **Ajose0**

Most Recent 

 2 months, 1 week ago

Selected Answer: C

Moving the training data to an Amazon S3 bucket and training and evaluating the model by using Amazon SageMaker will reduce the company's compute infrastructure costs and maximize the scalability of resources for training.

Optimizing the model by using SageMaker Neo will further reduce costs by allowing the model to run on inexpensive edge devices.

Setting up an edge device in the manufacturing facilities with AWS IoT Greengrass and deploying the model on the edge device will enable the company to use the ML model in the low-connectivity environments.

This solution provides a complete end-to-end solution for the company's needs, from data storage to model deployment, while minimizing costs and providing scalability and offline capabilities.
upvoted 1 times
-   **Peeking** 4 months, 3 weeks ago

Selected Answer: C

C best satisfies the options of minimising cost, and taking care of lack of connectivity through edge deployment.
upvoted 1 times
-   **Morsa** 9 months, 2 weeks ago

Selected Answer: C

Same arguments as belie
upvoted 2 times
-   **exam_prep** 11 months, 2 weeks ago

C: is the correct answer. Upload 20 years of massive data to S3 for model training. Sagemaker for creating and training a model. Once ready, deploy at edge using IOT Greengrass (this takes care of poor internet connectivity issue which is not addressed by option A)
upvoted 3 times

A company has an ecommerce website with a product recommendation engine built in TensorFlow. The recommendation engine endpoint is hosted by Amazon SageMaker. Three compute-optimized instances support the expected peak load of the website. Response times on the product recommendation page are increasing at the beginning of each month. Some users are encountering errors. The website receives the majority of its traffic between 8 AM and 6 PM on weekdays in a single time zone. Which of the following options are the MOST effective in solving the issue while keeping costs to a minimum? (Choose two.)

- A. Configure the endpoint to use Amazon Elastic Inference (EI) accelerators.
- B. Create a new endpoint configuration with two production variants.
- C. Configure the endpoint to automatically scale with the InvocationsPerInstance metric.
- D. Deploy a second instance pool to support a blue/green deployment of models.
- E. Reconfigure the endpoint to use burstable instances.

Correct Answer: BD

Reference:

https://docs.aws.amazon.com/sagemaker/latest/APIReference/API_ProductionVariant.html <https://www.redhat.com/en/topics/devops/what-is-blue-green-deployment>

Community vote distribution

AC (79%)

AE (17%)

Sivadharan

Highly Voted

11 months, 3 weeks ago

Selected Answer: AC

AC - for me
<https://aws.amazon.com/machine-learning/elastic-inference/>
<https://aws.amazon.com/blogs/machine-learning/configuring-autoscaling-inference-endpoints-in-amazon-sagemaker/>
upvoted 13 times

Jerry84

2 months, 4 weeks ago

Agree.

The problem with E is that it mentioned "majority of its traffic between 8 AM and 6 PM on weekdays", therefore it is not cost effective during period from 6PM to 8AM weekdays and weekends. Whereas the auto-scaling(C) could save money during all the time.
upvoted 1 times

rohit07cf

Highly Voted

12 months ago

AC looks correct
upvoted 9 times

blanco750

Most Recent

1 month, 1 week ago

Selected Answer: AC

A and C are the right answers
upvoted 3 times

luckybme

2 months, 1 week ago

Interesting if option A will be relevant anymore, as AWS is discontinuing Elastic Inference starting Apr 15.
<https://docs.aws.amazon.com/sagemaker/latest/dg/ei.html>. Wonder if they change the option to include Inf instane type
upvoted 3 times

wjohnny

4 months, 2 weeks ago

Selected Answer: CE

Why elastic inference given that GPU is not necessary? I guess C and E (IMO)
upvoted 1 times

Ob1KN0B

8 months, 2 weeks ago

Selected Answer: AE

A - This is using Tensorflow which means Elastic inference can be used to save costs for GPU, thereby reducing the compute time.
E - Since the load is not uniform, it will help to use burstable instances to operate above the threshold when the situation demands.
C is not at all cost effective.
upvoted 3 times

matteocal

9 months, 1 week ago

Selected Answer: AC

AC is the most correct
upvoted 7 times

  **KlaudYu** 10 months, 2 weeks ago

Selected Answer: AE

I'd go AE if it requests for cost minimum
upvoted 2 times

  **tgaos** 11 months, 1 week ago

AC is correct
upvoted 4 times

  **bluer1** 1 year ago

AC - for me
upvoted 6 times

A real-estate company is launching a new product that predicts the prices of new houses. The historical data for the properties and prices is stored in .csv format in an Amazon S3 bucket. The data has a header, some categorical fields, and some missing values. The company's data scientists have used Python with a common open-source library to fill the missing values with zeros. The data scientists have dropped all of the categorical fields and have trained a model by using the open-source linear regression algorithm with the default parameters. The accuracy of the predictions with the current model is below 50%. The company wants to improve the model performance and launch the new product as soon as possible.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Create a service-linked role for Amazon Elastic Container Service (Amazon ECS) with access to the S3 bucket. Create an ECS cluster that is based on an AWS Deep Learning Containers image. Write the code to perform the feature engineering. Train a logistic regression model for predicting the price, pointing to the bucket with the dataset. Wait for the training job to complete. Perform the inferences.
- B. Create an Amazon SageMaker notebook with a new IAM role that is associated with the notebook. Pull the dataset from the S3 bucket. Explore different combinations of feature engineering transformations, regression algorithms, and hyperparameters. Compare all the results in the notebook, and deploy the most accurate configuration in an endpoint for predictions.
- C. Create an IAM role with access to Amazon S3, Amazon SageMaker, and AWS Lambda. Create a training job with the SageMaker built-in XGBoost model pointing to the bucket with the dataset. Specify the price as the target feature. Wait for the job to complete. Load the model artifact to a Lambda function for inference on prices of new houses.
- D. Create an IAM role for Amazon SageMaker with access to the S3 bucket. Create a SageMaker AutoML job with SageMaker Autopilot pointing to the bucket with the dataset. Specify the price as the target attribute. Wait for the job to complete. Deploy the best model for predictions.

Correct Answer: A

Reference:

<https://docs.aws.amazon.com/deep-learning-containers/latest/devguide/deep-learning-containers-ecs-setup.html>

Community vote distribution

D (100%)

-   **spaceexplorer**

Highly Voted 

 1 year ago

Selected Answer: D

D; A involves too much effort and management overhead.

upvoted 10 times
-   **LydiaGom** 11 months, 4 weeks ago

Agree. but A has feature engineering which is the problem of the current model... confusing

upvoted 1 times
-   **Raja3sa** 9 months, 1 week ago

we don't use Logistic Regression to predict price.

upvoted 2 times
-   **Shailendraa**

Most Recent 

 8 months, 2 weeks ago

D is Correct: trick to eliminate is A can not as Logistic is classification algo which gives binary outcome.B &C seems a lot of work .

upvoted 2 times
-   **Morsa** 9 months, 2 weeks ago

Selected Answer: D

The problem is not a classification problem so A is incorrect as logistic regression is used for binary problems. D is the correct solution

upvoted 3 times
-   **mk5183** 11 months ago

The problem is not which model you chose but primitive level feature engineering therefore correct answer should be "B"

upvoted 2 times
-   **edvardo** 11 months, 3 weeks ago

Selected Answer: D

D.

<https://aws.amazon.com/sagemaker/autopilot/>

Supports missing values, categorical features, etc.

The simplest solution for this case
upvoted 3 times

  **ckkobe24** 12 months ago

why not B ?
upvoted 2 times

A data scientist is reviewing customer comments about a company's products. The data scientist needs to present an initial exploratory analysis by using charts and a word cloud. The data scientist must use feature engineering techniques to prepare this analysis before starting a natural language processing (NLP) model.

Which combination of feature engineering techniques should the data scientist use to meet these requirements? (Choose two.)

- A. Named entity recognition
- B. Coreference
- C. Stemming
- D. Term frequency-inverse document frequency (TF-IDF)
- E. Sentiment analysis

Correct Answer: DE

Reference:
<https://www.analyticsvidhya.com/blog/2020/04/beginners-guide-exploratory-data-analysis-text-data/>

Community vote distribution

CD (96%)

4%

- DJiang

Highly Voted

12 months ago

Selected Answer: CD

Sentiment analysis is the result of analysis, not feature engineering. I think this answer should be C & D.
upvoted 11 times
- Shailendraa

Most Recent

7 months, 3 weeks ago

12-sep exam
upvoted 2 times
- Ob1KN0B

8 months, 2 weeks ago

Selected Answer: DE

<https://towardsdatascience.com/text-analysis-feature-engineering-with-nlp-502d6ea9225d>

Sentiment analysis IS a part of feature engg in NLP.
upvoted 1 times
- matteocal

9 months, 1 week ago

Selected Answer: CD

sentiment analysis is not part of feature engineering
upvoted 2 times
- Morsa

9 months, 3 weeks ago

Selected Answer: CD

I agree ABE are not feature engineering
upvoted 3 times
- edvardo

12 months ago

Selected Answer: CD

A, B, E are not feature engineering
upvoted 4 times
- ckkobe24

12 months ago

Selected Answer: CD

why not C & D
upvoted 4 times

A data scientist is evaluating a GluonTS on Amazon SageMaker DeepAR model. The evaluation metrics on the test set indicate that the coverage score is 0.489 and 0.889 at the 0.5 and 0.9 quantiles, respectively.

What can the data scientist reasonably conclude about the distributional forecast related to the test set?

- A. The coverage scores indicate that the distributional forecast is poorly calibrated. These scores should be approximately equal to each other at all quantiles.
- B. The coverage scores indicate that the distributional forecast is poorly calibrated. These scores should peak at the median and be lower at the tails.
- C. The coverage scores indicate that the distributional forecast is correctly calibrated. These scores should always fall below the quantile itself.
- D. The coverage scores indicate that the distributional forecast is correctly calibrated. These scores should be approximately equal to the quantile itself.

Correct Answer: D

Reference:

<https://aws.amazon.com/blogs/machine-learning/amazon-forecast-now-supports-the-generation-of-forecasts-at-a-quantile-of-your-choice/>

Community vote distribution

D (73%)

C (27%)

cognito_22

Highly Voted

11 months, 4 weeks ago

Selected Answer: D

https://ts.gluon.ai/tutorials/forecasting/quick_start_tutorial.html
upvoted 8 times

example_

7 months, 2 weeks ago

https://ts.gluon.ai/stable/tutorials/forecasting/quick_start_tutorial.html
upvoted 3 times

blanco750

Most Recent

1 month, 1 week ago

Selected Answer: C

Thanks to ChatGPT
Given the coverage score results, the data scientist can conclude that the distributional forecast related the test set is well calibrated. Specifically, when the model predicts quantiles, around % of the true values should fall within the 0.5 quantile range, and around90% of the true values should fall within the 0.9 quantile range., the GluonTS on Amazon SageMaker DeepAR model performance on the test set was concerning the coverage of the predicted quantiles.
upvoted 1 times

SANDEEP_AWS

1 month, 3 weeks ago

Selected Answer: C

Scores should always fall below the quantile itself. Ref: <https://d1.awsstatic.com/asset-repository/Amazon%20Forecast%20Technical%20Guide%20to%20Time-Series%20Forecasting%20Principles.pdf> -- Pg 18
upvoted 2 times

Tony_1406

2 days, 14 hours ago

<https://docs.aws.amazon.com/forecast/latest/dg/metrics.html#metrics-wQL>
A more concise doc.
upvoted 1 times

SANDEEP_AWS

1 month, 2 weeks ago

PDF Pg 23
upvoted 1 times

[Removed]

9 months ago

C is correct
upvoted 2 times

An energy company has wind turbines, weather stations, and solar panels that generate telemetry data. The company wants to perform predictive maintenance on these devices. The devices are in various locations and have unstable internet connectivity.

A team of data scientists is using the telemetry data to perform machine learning (ML) to conduct anomaly detection and predict maintenance before the devices start to deteriorate. The team needs a scalable, secure, high-velocity data ingestion mechanism. The team has decided to use Amazon S3 as the data storage location.

Which approach meets these requirements?

- A. Ingest the data by using an HTTP API call to a web server that is hosted on Amazon EC2. Set up EC2 instances in an Auto Scaling configuration behind an Elastic Load Balancer to load the data into Amazon S3.
- B. Ingest the data over Message Queuing Telemetry Transport (MQTT) to AWS IoT Core. Set up a rule in AWS IoT Core to use Amazon Kinesis Data Firehose to send data to an Amazon Kinesis data stream that is configured to write to an S3 bucket.
- C. Ingest the data over Message Queuing Telemetry Transport (MQTT) to AWS IoT Core. Set up a rule in AWS IoT Core to direct all MQTT data to an Amazon Kinesis Data Firehose delivery stream that is configured to write to an S3 bucket.
- D. Ingest the data over Message Queuing Telemetry Transport (MQTT) to Amazon Kinesis data stream that is configured to write to an S3 bucket.

Correct Answer: C

Reference:
<https://aws.amazon.com/blogs/industries/real-time-operational-monitoring-of-renewable-energy-assets-with-aws-iot/>

Community vote distribution

C (100%)

  **siju13**

Highly Voted 

 11 months, 4 weeks ago

Selected Answer: C

Answer is C.

B, D wrong because Kinesis data stream cannot write to S3 directly.

upvoted 11 times

  **edvardo**

Highly Voted 

 12 months ago

A. Not enough.

C. Correct. Check <https://docs.aws.amazon.com/firehose/latest/dev/writing-with-iot.html>

B, D: Wrong. KDS doesn't write directly to S3

upvoted 7 times

  **blanco750**

Most Recent 

 1 month, 1 week ago

Selected Answer: C

C is correct

upvoted 2 times

A retail company collects customer comments about its products from social media, the company website, and customer call logs. A team of data scientists and engineers wants to find common topics and determine which products the customers are referring to in their comments. The team is using natural language processing (NLP) to build a model to help with this classification.

Each product can be classified into multiple categories that the company defines. These categories are related but are not mutually exclusive. For example, if there is mention of "Sample Yogurt" in the document of customer comments, then "Sample Yogurt" should be classified as "yogurt," "snack," and "dairy product."

The team is using Amazon Comprehend to train the model and must complete the project as soon as possible.

Which functionality of Amazon Comprehend should the team use to meet these requirements?

Correct Answer: A

Community vote distribution

B (100%)

-   **spaceexplorer**

Highly Voted 

 1 year ago

Selected Answer: B

The answer is B. In multi-label mode, individual classes represent different categories, but these categories are not mutually exclusive while individual classes are mutually exclusive in multi-class mode

upvoted 12 times
-   **Ivandrigo**

Highly Voted 

 12 months ago

B - In simple language, it is tagging

upvoted 9 times
-   **Chelseajcole**

Most Recent 

 2 months ago

Selected Answer: B

In multi-label classification, individual classes represent different categories, but these categories are somehow related and are not mutually exclusive. As a result, each document has at least one class assigned to it, but can have more. For example, a movie can simply be an action movie, or it can be an action movie, a science fiction movie, and a comedy, all at the same time.

In multi-class classification, each document can have one and only one class assigned to it. The individual classes are mutually exclusive. For example, a movie can be classed as a documentary or as science fiction, but not both at the same time.

upvoted 3 times
-   **Peeking** 4 months, 3 weeks ago

Selected Answer: B

<https://docs.aws.amazon.com/comprehend/latest/dg/prep-classifier-data-multi-label.html>

upvoted 1 times
-   **Shailendraa** 7 months, 3 weeks ago

12-sep exam

upvoted 2 times
-   **matteocal** 9 months, 1 week ago

Selected Answer: B

B. Multi-label:

<https://docs.aws.amazon.com/comprehend/latest/dg/prep-classifier-data-multi-label.html>

upvoted 2 times
-   **Morsa** 9 months, 3 weeks ago

Selected Answer: B

Written in the technical guide related but not mutually exclusive

upvoted 1 times

A data engineer is using AWS Glue to create optimized, secure datasets in Amazon S3. The data science team wants the ability to access the ETL scripts directly from Amazon SageMaker notebooks within a VPC. After this setup is complete, the data science team wants the ability to run the AWS Glue job and invoke the SageMaker training job.

Which combination of steps should the data engineer take to meet these requirements? (Choose three.)

- A. Create a SageMaker development endpoint in the data science team's VPC.
- B. Create an AWS Glue development endpoint in the data science team's VPC.
- C. Create SageMaker notebooks by using the AWS Glue development endpoint.
- D. Create SageMaker notebooks by using the SageMaker console.
- E. Attach a decryption policy to the SageMaker notebooks.
- F. Create an IAM policy and an IAM role for the SageMaker notebooks.

Correct Answer: ADF

Reference:

<https://aws.amazon.com/blogs/machine-learning/access-amazon-s3-data-managed-by-aws-glue-data-catalog-from-amazon-sagemaker-notebooks/>

Community vote distribution

BCF (100%)

 **spaceexplorer** Highly Voted 1 year ago

Selected Answer: BCF

BCF; This tutorial doc says so: <https://docs.aws.amazon.com/glue/latest/dg/dev-endpoint-tutorial-sage.html>
upvoted 7 times

 **Chelseajcole** Most Recent 2 months ago

Selected Answer: BCF

In the AWS Glue console, choose Dev endpoints to navigate to the development endpoints list.
Select the check box next to the name of a development endpoint that you want to use, and on the Action menu, choose Create SageMaker notebook.
Fill out the Create and configure a notebook page as follows:
Enter a notebook name.
Under Attach to development endpoint, verify the development endpoint.
Create or choose an AWS Identity and Access Management (IAM) role.
upvoted 2 times

 **tgaos** 10 months, 3 weeks ago

Selected Answer: BCF

BCF is correct
upvoted 3 times

A data engineer needs to provide a team of data scientists with the appropriate dataset to run machine learning training jobs. The data will be stored in Amazon S3. The data engineer is obtaining the data from an Amazon Redshift database and is using join queries to extract a single tabular dataset. A portion of the schema is as follows:

TransactionTimestamp (Timestamp)
CardName (Varchar)
CardNo (Varchar)

The data engineer must provide the data so that any row with a CardNo value of NULL is removed. Also, the TransactionTimestamp column must be separated into a TransactionDate column and a TransactionTime column. Finally, the CardName column must be renamed to NameOnCard.

The data will be extracted on a monthly basis and will be loaded into an S3 bucket. The solution must minimize the effort that is needed to set up infrastructure for the ingestion and transformation. The solution also must be automated and must minimize the load on the Amazon Redshift cluster.

Which solution meets these requirements?

- A. Set up an Amazon EMR cluster. Create an Apache Spark job to read the data from the Amazon Redshift cluster and transform the data. Load the data into the S3 bucket. Schedule the job to run monthly.
- B. Set up an Amazon EC2 instance with a SQL client tool, such as SQL Workbench/J, to query the data from the Amazon Redshift cluster directly Export the resulting dataset into a file. Upload the file into the S3 bucket. Perform these tasks monthly.
- C. Set up an AWS Glue job that has the Amazon Redshift cluster as the source and the S3 bucket as the destination. Use the built-in transforms Filter, Map, and RenameField to perform the required transformations. Schedule the job to run monthly.
- D. Use Amazon Redshift Spectrum to run a query that writes the data directly to the S3 bucket. Create an AWS Lambda function to run the query monthly.

Correct Answer: C

Community vote distribution

C (100%)

ystotest

Highly Voted

5 months ago

Selected Answer: C

agreed with C

upvoted 11 times

Jerry84

Most Recent

3 months, 2 weeks ago

Selected Answer: C

<https://docs.aws.amazon.com/glue/latest/dg/aws-glue-programming-python-transforms.html>

upvoted 2 times

A machine learning (ML) specialist wants to bring a custom training algorithm to Amazon SageMaker. The ML specialist implements the algorithm in a Docker container that is supported by SageMaker.

How should the ML specialist package the Docker container so that SageMaker can launch the training correctly?

- A. Specify the server argument in the ENTRYPOINT instruction in the Dockerfile.
- B. Specify the training program in the ENTRYPOINT instruction in the Dockerfile.
- C. Include the path to the training data in the docker build command when packaging the container.
- D. Use a COPY instruction in the Dockerfile to copy the training program to the /opt/ml/train directory.

Correct Answer: B

Community vote distribution

B (100%)

  **VinceCar**

Highly Voted 

 5 months ago

Selected Answer: B

<https://docs.aws.amazon.com/sagemaker/latest/dg/your-algorithms-training-algo-dockerfile.html>
upvoted 5 times

  **wjohnny**

Most Recent 

 4 months, 2 weeks ago

Selected Answer: B

<https://docs.aws.amazon.com/sagemaker/latest/dg/adapt-training-container.html>

In Step 2, it is mentioned to use this instruction on dockerfile:

Defines train.py as script entrypoint
ENV SAGEMAKER_PROGRAM train.py
upvoted 4 times

An ecommerce company wants to use machine learning (ML) to monitor fraudulent transactions on its website. The company is using Amazon SageMaker to research, train, deploy, and monitor the ML models.

The historical transactions data is in a .csv file that is stored in Amazon S3. The data contains features such as the user's IP address, navigation time, average time on each page, and the number of clicks for each session. There is no label in the data to indicate if a transaction is anomalous.

Which models should the company use in combination to detect anomalous transactions? (Choose two.)

- A. IP Insights
- B. K-nearest neighbors (k-NN)
- C. Linear learner with a logistic function
- D. Random Cut Forest (RCF)
- E. XGBoost

Correct Answer: AC

Community vote distribution

AD (76%)

AC (18%)

6%

- GauravLahotiML

Highly Voted

5 months, 1 week ago

Selected Answer: AD

AD are the right answer

upvoted 5 times
- SANDEEP_AWS

Most Recent

1 month, 3 weeks ago

Selected Answer: CD

Can't be A, as we don't have data in the format expected for IP Insights algorithm(<https://docs.aws.amazon.com/sagemaker/latest/dg/ip-insights-training-data-formats.html>).

upvoted 1 times
- blanco750

1 month, 1 week ago

It expects CSV format and the question mentions data is in CSV format so IP Insights is correct

upvoted 1 times
- Jerry84

3 months, 2 weeks ago

Selected Answer: AD

A and D are correct.

A. IP Insights for Pattern recognition.

D. Random Cut Forest (RCF) for Anomaly detection

B,C,E are normally Supervised learning algorithm which are against the wordings "There is no label ..."

upvoted 1 times
- Peeking

4 months, 3 weeks ago

Selected Answer: AD

C is not part of the answer. IP insight because the data contain IP address. RCF because the data is unlabeled and anomaly is being detected for fraud.

upvoted 3 times
- Amit11011996

5 months ago

Selected Answer: AD

AD are correct

upvoted 1 times
- ystotest

5 months ago

Selected Answer: AD

apprently AD

upvoted 3 times
- GauravLahotiML

5 months, 1 week ago

Selected Answer: AC

AC is the correct answer to detect anomalies

upvoted 3 times

A healthcare company is using an Amazon SageMaker notebook instance to develop machine learning (ML) models. The company's data scientists will need to be able to access datasets stored in Amazon S3 to train the models. Due to regulatory requirements, access to the data from instances and services used for training must not be transmitted over the internet.

Which combination of steps should an ML specialist take to provide this access? (Choose two.)

- A. Configure the SageMaker notebook instance to be launched with a VPC attached and internet access disabled.
- B. Create and configure a VPN tunnel between SageMaker and Amazon S3.
- C. Create and configure an S3 VPC endpoint Attach it to the VPC.
- D. Create an S3 bucket policy that allows traffic from the VPC and denies traffic from the internet.
- E. Deploy AWS Transit Gateway Attach the S3 bucket and the SageMaker instance to the gateway.

Correct Answer: AC

Community vote distribution

AC (80%)

CD (20%)

  **dunhill** Highly Voted 5 months ago

I think the answer is CD.
upvoted 7 times

  **angus** 1 month, 2 weeks ago

A is wrong. SageMaker notebook does not need to have internet access disabled.
upvoted 1 times

  **VinceCar** 4 months, 3 weeks ago

Agree. The setting should be relevant to S3 and VPC, not the notebook.
upvoted 3 times

  **drcok87** 2 months ago

if notebook is not within vpc, then having s3 bucket policy to allow traffic only from vpc will block notebook to get data from s3.
A
C
upvoted 1 times

  **Amit11011996** Highly Voted 5 months ago

Selected Answer: AC

A and C seems fine
upvoted 6 times

  **cox1960** Most Recent 1 day, 12 hours ago

Selected Answer: CD

the requirements are about providing secure access from notebooks to S3, nothing else.
upvoted 1 times

  **ZSun** 1 week ago

the answer is C,D.
Firstly, the VPC need to connect with S3 through gateway endpoint, check "https://docs.aws.amazon.com/vpc/latest/privatelink/vpc-endpoints-s3.html"
secondly, after connection is created, we need to define the policy from s3 side. restrict access to s3 only from specified VPC or VPC endpoint.
"https://docs.aws.amazon.com/vpc/latest/privatelink/vpc-endpoints-s3.html#bucket-policies-s3"
the confusion about A is tricky. Ideally, you need to create sagemaker in private subnet with no internet access. But I assume the question "access to the data from instances and services" only requires the process from obtaining data from s3, you don't need to specify the requirement about data egress from training service(even though disable internet connection from sagemaker is crucial)
upvoted 1 times

  **blanco750** 1 month, 1 week ago

Selected Answer: AC

A and C are correct.
To disable direct internet access, you can specify a VPC for your notebook instance. By doing so, you prevent SageMaker from providing internet access to your notebook instance. As a result, the notebook instance can't train or host models unless your VPC has an interface endpoint (AWS PrivateLink) or a NAT gateway and your security groups allow outbound connections.
https://docs.aws.amazon.com/sagemaker/latest/dg/appendix-notebook-and-internet-access.html

<https://docs.aws.amazon.com/sagemaker/latest/dg/notebook-interface-endpoint.html>
D is wrong. Bucket policy cant be used to deny internet access. It can only enforce access from VPC or VPC endpoint
upvoted 2 times

  **ZSun** 1 week ago
your statement "Bucket policy cant be used to deny internet access" is completely wrong, you can either specify "Allow" or "Deny" in bucket policy, check "<https://docs.aws.amazon.com/vpc/latest/privatelink/vpc-endpoints-s3.html#bucket-policies-s3>"
You can create a bucket policy that restricts access to a specific endpoint by using the aws:sourceVpce condition key.
upvoted 1 times

  **Chelseajcole** 2 months ago

Selected Answer: CD

You can use Amazon S3 bucket policies to control access to buckets from specific virtual private cloud (VPC) endpoints, or specific VPCs. This section contains example bucket policies that can be used to control Amazon S3 bucket access from VPC endpoints.

Notebook doesn't need to be created within Vpc
upvoted 1 times

  **Maaayaaa** 1 day, 18 hours ago
The question requires to access to the data from instances and services used for training must not be transmitted over the internet. So the traffic has to go through the VPC endpoints, thus the notebook has to live in the VPC.
upvoted 1 times

A machine learning (ML) specialist at a retail company is forecasting sales for one of the company's stores. The ML specialist is using data from the past 10 years. The company has provided a dataset that includes the total amount of money in sales each day for the store. Approximately 5% of the days are missing sales data.

The ML specialist builds a simple forecasting model with the dataset and discovers that the model performs poorly. The performance is poor around the time of seasonal events, when the model consistently predicts sales figures that are too low or too high.

Which actions should the ML specialist take to try to improve the model's performance? (Choose two.)

- A. Add information about the store's sales periods to the dataset.
- B. Aggregate sales figures from stores in the same proximity.
- C. Apply smoothing to correct for seasonal variation.
- D. Change the forecast frequency from daily to weekly.
- E. Replace missing values in the dataset by using linear interpolation.

Correct Answer: *BE*

Community vote distribution

AE (67%)

CE (33%)

- dunhill

Highly Voted

5 months ago

I think the answer is BC

upvoted 8 times
- BoroJohn

Highly Voted

4 months, 2 weeks ago

BC - <https://docs.aws.amazon.com/forecast/latest/dg/aws-forecast-recipe-ets.html>

upvoted 7 times
- MIlb

Most Recent

4 weeks, 1 day ago

Selected Answer: AE

A to improve model in seasonal periods

E to fill missing data

upvoted 2 times
- blanco750

1 month, 1 week ago

Selected Answer: CE

I would go for C and E

C is quite obvious I think

E Linear interpolation is a technique to fill the missing data <https://towardsdatascience.com/4-techniques-to-handle-missing-values-in-time-series-data-c3568589b5a8>

upvoted 1 times
- Amit11011996

2 months ago

why not C and D?

upvoted 1 times
- Chelseajcole

2 months ago

Maybe the sales event can last longer than a week?

upvoted 1 times

A newspaper publisher has a table of customer data that consists of several numerical and categorical features, such as age and education history, as well as subscription status. The company wants to build a targeted marketing model for predicting the subscription status based on the table data.

Which Amazon SageMaker built-in algorithm should be used to model the targeted marketing?

- A. Random Cut Forest (RCF)
- B. XGBoost
- C. Neural Topic Model (NTM)
- D. DeepAR forecasting

Correct Answer: C

Community vote distribution

B (92%)8%

- hichemck**

Highly Voted

5 months ago

Selected Answer: B

B is correct. IMO
A - No, Random cut forest is for anomaly detection
B - Yes, exactly was XGBoost is good for. Binary classification based on a variety of input features
C - No, NTM is unsupervised. The problem states the table already has subscription status, therefore we need a supervised algorithm
D - No, DeepAR is used for time-series data

upvoted 5 times
- Peeking**

Highly Voted

4 months, 3 weeks ago

Selected Answer: B

Whether subscription status is binary or multi-class XGBoost can handle the problem in this case problem.

upvoted 5 times
- blanco750**

Most Recent

1 month, 1 week ago

Selected Answer: B

XGBoost is a popular and powerful algorithm for binary classification problems such as this one, where the goal is to predict a binary outcome (e.g. whether a customer subscribes or not). It is particularly effective when the dataset has a mix of numerical and categorical features.

upvoted 2 times
- blobfishtu**

5 months ago

The answer is B:
A. Random Cut Forest (RCF): anomaly detection
B. XGBoost: allows classification tasks like the use case in the question
C. Neural Topic Model (NTM): topic modelling
D. DeepAR forecasting: time series

upvoted 3 times
- dunhill**

5 months ago

I think the answer is B.
It looks like no time serials condition, so it may be not suitable to A and D.

upvoted 4 times
- VinceCar**

5 months ago

Selected Answer: D

D. Refer to
<https://docs.aws.amazon.com/forecast/latest/dg/aws-forecast-recipe-deeparplus.html>

upvoted 1 times
- Amit11011996**

5 months ago

can you please let us know, where in question it states that date/time column is available.
It is all about numerical or categorical columns and we need to predict subscription status which can be done by the use XG-BOOST

upvoted 2 times
- VinceCar**

4 months, 4 weeks ago

you're right. there is no time series.

upvoted 2 times

A company will use Amazon SageMaker to train and host a machine learning model for a marketing campaign. The data must be encrypted at rest. Most of the data is sensitive customer data. The company wants AWS to maintain the root of trust for the encryption keys and wants key usage to be logged.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Use AWS Security Token Service (AWS STS) to create temporary tokens to encrypt the storage volumes for all SageMaker instances and to encrypt the model artifacts and data in Amazon S3.
- B. Use customer managed keys in AWS Key Management Service (AWS KMS) to encrypt the storage volumes for all SageMaker instances and to encrypt the model artifacts and data in Amazon S3.
- C. Use encryption keys stored in AWS CloudHSM to encrypt the storage volumes for all SageMaker instances and to encrypt the model artifacts and data in Amazon S3.
- D. Use SageMaker built-in transient keys to encrypt the storage volumes for all SageMaker instances. Enable default encryption for new Amazon Elastic Block Store (Amazon EBS) volumes.

Correct Answer: D

Community vote distribution

B (100%)

- akhjhjk

Highly Voted

4 months, 4 weeks ago

Selected Answer: B

It seems to be B

upvoted 6 times
- VinceCar

Most Recent

5 months ago

why no B?

upvoted 2 times
- dunhill

5 months ago

It should be B. This question is the same as Q143.

upvoted 5 times

A data scientist is working on a model to predict a company's required inventory stock levels. All historical data is stored in .csv files in the company's data lake on Amazon S3. The dataset consists of approximately 500 GB of data The data scientist wants to use SQL to explore the data before training the model. The company wants to minimize costs.

Which option meets these requirements with the LEAST operational overhead?

- A. Create an Amazon EMR cluster. Create external tables in the Apache Hive metastore, referencing the data that is stored in the S3 bucket. Explore the data from the Hive console.
- B. Use AWS Glue to crawl the S3 bucket and create tables in the AWS Glue Data Catalog. Use Amazon Athena to explore the data.
- C. Create an Amazon Redshift cluster. Use the COPY command to ingest the data from Amazon S3. Explore the data from the Amazon Redshift query editor GUI.
- D. Create an Amazon Redshift cluster. Create external tables in an external schema, referencing the S3 bucket that contains the data. Explore the data from the Amazon Redshift query editor GUI.

Correct Answer: D

Community vote distribution

B (100%)

- dunhill

Highly Voted

5 months ago

I think the answer is B. The others are quite expensive and complicated.

upvoted 6 times
- ZSun

Most Recent

1 week, 5 days ago

The advantage of D is that Redshift, as a data warehouse, can handle large dataset(> 1TB) and complex frequent query. In this example, 500GB dataset and infrequent query(I consider this just one-time ad-hoc query, just verify the data before training.) Athena would be a much better option.

upvoted 1 times
- oso0348

Selected Answer: B

1 month ago

The option that meets these requirements with the LEAST operational overhead is option B: Use AWS Glue to crawl the S3 bucket and create tables in the AWS Glue Data Catalog. Use Amazon Athena to explore the data.

AWS Glue is a fully managed ETL service that can automatically discover and catalog metadata about data stored in various data stores, including Amazon S3. By using AWS Glue to crawl the S3 bucket, the data scientist can easily create tables in the AWS Glue Data Catalog, without needing to create or manage any infrastructure.

Amazon Athena is an interactive query service that allows querying data stored in Amazon S3 using SQL. By using Amazon Athena, the data scientist can easily explore the data using SQL, without needing to set up any infrastructure.

upvoted 3 times
- Jerry84

Selected Answer: B

3 months, 2 weeks ago

Both Glue and Athena are serverless hence cost effective.

upvoted 3 times
- Peeking

Selected Answer: B

4 months, 3 weeks ago

B is highly managed unlike other options.

upvoted 4 times
- akhjhjk

Selected Answer: B

4 months, 4 weeks ago

It seems to be B

upvoted 4 times

A geospatial analysis company processes thousands of new satellite images each day to produce vessel detection data for commercial shipping. The company stores the training data in Amazon S3. The training data incrementally increases in size with new images each day.

The company has configured an Amazon SageMaker training job to use a single ml.p2.xlarge instance with File input mode to train the built-in Object Detection algorithm. The training process was successful last month but is now failing because of a lack of storage. Aside from the addition of training data, nothing has changed in the model training process.

A machine learning (ML) specialist needs to change the training configuration to fix the problem. The solution must optimize performance and must minimize the cost of training.

Which solution will meet these requirements?

- A. Modify the training configuration to use two ml.p2.xlarge instances.
- B. Modify the training configuration to use Pipe input mode.
- C. Modify the training configuration to use a single ml.p3.2xlarge instance.
- D. Modify the training configuration to use Amazon Elastic File System (Amazon EFS) instead of Amazon S3 to store the input training data.

Correct Answer: C

Community vote distribution

B (100%)

-   **tsangckl** Highly Voted 5 months ago

Selected Answer: B

Agreed with B
<https://aws.amazon.com/blogs/machine-learning/using-pipe-input-mode-for-amazon-sagemaker-algorithms/>
upvoted 8 times
-   **oso0348** Most Recent 1 month ago

Selected Answer: B

The correct solution would be to modify the training configuration to use Pipe input mode. This will allow the training data to stream directly into the training instance as it is being consumed, rather than first being downloaded from S3 into the instance's local storage. This can help reduce storage requirements and optimize performance, while also minimizing costs. Using more or larger instances may help with processing power, but it will not address the storage issue, and may even increase costs unnecessarily. Using Amazon EFS may also be an option, but it may come with additional costs and operational overhead.
upvoted 2 times
-   **Peeking** 4 months, 3 weeks ago

Selected Answer: B

Pipe mode solves the problem without incurring extra storage cost. Data is streamed directly to the training algorithm without the need to be stored in the EBS volume.
upvoted 3 times
-   **dunhill** 5 months ago

I think the answer is B. D is incorrect because EFS is more expensive than S3.
It looks like scaling up or out is no help for storage issue. Therefore A and C are not helpful.
upvoted 3 times

A company is using Amazon SageMaker to build a machine learning (ML) model to predict customer churn based on customer call transcripts. Audio files from customer calls are located in an on-premises VoIP system that has petabytes of recorded calls. The on-premises infrastructure has high-velocity networking and connects to the company's AWS infrastructure through a VPN connection over a 100 Mbps connection.

The company has an algorithm for transcribing customer calls that requires GPUs for inference. The company wants to store these transcriptions in an Amazon S3 bucket in the AWS Cloud for model development.

Which solution should an ML specialist use to deliver the transcriptions to the S3 bucket as quickly as possible?

- A. Order and use an AWS Snowball Edge Compute Optimized device with an NVIDIA Tesla module to run the transcription algorithm. Use AWS DataSync to send the resulting transcriptions to the transcription S3 bucket.
- B. Order and use an AWS Snowcone device with Amazon EC2 Inf1 instances to run the transcription algorithm. Use AWS DataSync to send the resulting transcriptions to the transcription S3 bucket.
- C. Order and use AWS Outposts to run the transcription algorithm on GPU-based Amazon EC2 instances. Store the resulting transcriptions in the transcription S3 bucket.
- D. Use AWS DataSync to ingest the audio files to Amazon S3. Create an AWS Lambda function to run the transcription algorithm on the audio files when they are uploaded to Amazon S3. Configure the function to write the resulting transcriptions to the transcription S3 bucket.

Correct Answer: D

Community vote distribution

A (64%)

C (21%)

Other

- dunhill

Highly Voted

5 months ago

I think the answer is A.
B: Snowcone has limit with 8 TB.
C: is AWS on-premises solution, but the company wants to store these transcriptions in an Amazon S3 bucket in the AWS Cloud for model development.
D: 100 Mbps cannot handle petabytes datasync.
upvoted 10 times
- MIlb

Most Recent

4 weeks, 1 day ago

Selected Answer: A

The key is faster
Transfer speeds of up to 100gb per second
upvoted 2 times
- blanco750

1 month, 1 week ago

Selected Answer: A

It either A or C.
But C is too complicated to setup. You order rack and AWS installs that plus you need enterprise support and the biggest reason it is not possible in this case is that it requires at least 1GB connection.
The question clearly asks as soon as possible so A is the best choice in my opinion
upvoted 1 times
- pan_b

1 month, 2 weeks ago

Selected Answer: A

The key is to deliver the transcripts to S3 as early as possible. Outpost order and provisioning takes months. I would go for A as its logical to do local inference and send transcripts to S3.
upvoted 1 times
- Valcilio

1 month, 3 weeks ago

Selected Answer: A

The model needs to run on-premises to don't be necessarily upload all the audio data to after run the model and this solution can use datasync to upload the results after too, then it's a good choice!
upvoted 2 times
- Chelseajcole

2 months ago

Selected Answer: A

Outpost for use case when customer don't want to transfer their data out
upvoted 2 times

-  **Imimi** 2 months ago

It is A.

C can't be the answer, as to transfer 1PB data, it may take 1,000 days under a 100 mbps network.

upvoted 2 times
-  **Siyuan_Zhu** 2 months ago

Selected Answer: C

My vote goes to C,
A. Snowball device only stores around 80 TB of data and uploading the newly transcribed data through datasync still goes through the slow connection between on-site and AWS. Outpost seem like the only feasible solution here that can satisfy both requiriements

upvoted 1 times
-  **CertArvind** 3 months ago

Correct Answer A.
D is incorrect as it takes 1024 Days Approx to transfer Petabyte of data.

upvoted 3 times
-  **IvanHuang** 3 months, 1 week ago

Selected Answer: AD

The best solution would be option A and D.
Option A: Order and use an AWS Snowball Edge Compute Optimized device with NVIDIA Tesla modules to run the transcription algorithm. Use AWS DataSync to send the generated transcriptions to a transcription S3 bucket.
This option allows to use a device that has the necessary GPU for running the transcription algorithm and then use the AWS DataSync to send the generated transcriptions to the S3 bucket.
Option D: Use AWS DataSync to ingest the audio files to Amazon S3. Create an AWS Lambda function to run the transcription algorithm on the audio file as it is uploaded to Amazon S3. Configure the function to write the generated transcriptions to the transcriptions S3 bucket.
This option allows to automatically transcribing the audio files as they are uploaded to S3. This means that the transcriptions are ready as soon as the audio files are uploaded and eliminates the need to transcribe the audio files separately.

upvoted 1 times
-  **BTRYING** 3 months, 2 weeks ago

Selected Answer: A

[ws.amazon.com/about-aws/whats-new/2020/07/aws-snowball-edge-compute-optimized-now-available-additional-aws-regions/](https://aws.amazon.com/about-aws/whats-new/2020/07/aws-snowball-edge-compute-optimized-now-available-additional-aws-regions/)

upvoted 1 times
-  **wjohnny** 4 months, 2 weeks ago

Selected Answer: C

I guess C, given that "The company wants to store these transcriptions".
Petabytes audio data can keep in on-prem.

upvoted 2 times
-  **VinceCar** 5 months ago

Selected Answer: D

There is high-velocity networking. So A, B, C(Snowball, Snowcone, and Outputs) are not required.

upvoted 1 times
-  **VinceCar** 4 months, 3 weeks ago

should be option A. There is petabytes of records.

upvoted 1 times

A company has a podcast platform that has thousands of users. The company has implemented an anomaly detection algorithm to detect low podcast engagement based on a 10-minute running window of user events such as listening, pausing, and exiting the podcast. A machine learning (ML) specialist is designing the data ingestion of these events with the knowledge that the event payload needs some small transformations before inference.

How should the ML specialist design the data ingestion to meet these requirements with the LEAST operational overhead?

- A. Ingest event data by using a GraphQLAPI in AWS AppSync. Store the data in an Amazon DynamoDB table. Use DynamoDB Streams to call an AWS Lambda function to transform the most recent 10 minutes of data before inference.
- B. Ingest event data by using Amazon Kinesis Data Streams. Store the data in Amazon S3 by using Amazon Kinesis Data Firehose. Use AWS Glue to transform the most recent 10 minutes of data before inference.
- C. Ingest event data by using Amazon Kinesis Data Streams. Use an Amazon Kinesis Data Analytics for Apache Flink application to transform the most recent 10 minutes of data before inference.
- D. Ingest event data by using Amazon Managed Streaming for Apache Kafka (Amazon MSK). Use an AWS Lambda function to transform the most recent 10 minutes of data before inference.

Correct Answer: B

Community vote distribution

B (69%)

C (31%)

- wjohnny

Highly Voted

4 months, 2 weeks ago

Selected Answer: B

B, but it was possible to use Kinesis Data Firehose directly, insted Kinesis Data Stream

upvoted 6 times
- dunhill

Highly Voted

5 months ago

I think the answer is B.

upvoted 6 times
- blanco750

Most Recent

1 month, 1 week ago

Selected Answer: B

B is the answer. least management overhead

upvoted 3 times

blanco750

1 month, 1 week ago

And for C you have to author and build your Apache Flink application. extra work

upvoted 1 times

pan_b

1 month, 2 weeks ago

Selected Answer: C

Answer should be C. <https://aws.amazon.com/blogs/architecture/realtime-in-stream-inference-kinesis-sagemaker-flink/>

upvoted 1 times

Valcilio

1 month, 3 weeks ago

Selected Answer: C

It's C, this data requires small transformations who can be did with apache flink in kinesis data analytics. If wasn't this, then could be using Glue.

upvoted 1 times

Valcilio

1 month, 3 weeks ago

Selected Answer: C

It's C, the Glue's not necessary here because it says that is just small transformations.

upvoted 1 times

Chelseajcole

2 months ago

Using glue with pyspark you can have tumbling window but it still require coding. Kda is more a realistic answer even though the question didn't require real-time feature

upvoted 1 times

Jerry84

2 months, 1 week ago

Selected Answer: C
- <https://www.examtopics.com/exams/amazon/aws-certified-machine-learning-specialty/custom-view/>

349/406

<https://aws.amazon.com/blogs/architecture/realtime-in-stream-inference-kinesis-sagemaker-flink/>

From the link above, C makes more sense.

upvoted 1 times

  **drcok87** 2 months, 1 week ago

b <https://aws.amazon.com/blogs/machine-learning/build-a-predictive-maintenance-solution-with-amazon-kinesis-aws-glue-and-amazon-sagemaker/>

c <https://aws.amazon.com/blogs/architecture/realtime-in-stream-inference-kinesis-sagemaker-flink/>

Both can do what is being asked.

C seems possible, we transform the incoming data and directly call inference endpoint from flink app in KDA.

upvoted 1 times

  **drcok87** 2 months, 1 week ago

thinking over again between c and b:

KKDA requires us to write JAVA application to transform the data this is not least “operational overhead”, there could be updates to the app or library we will need to manage.

Glue will be easier, so may be choose B ?

upvoted 1 times

  **uninit** 3 months ago

I would choose C-

<https://docs.aws.amazon.com/kinesisanalytics/latest/java/how-it-works.html>

We can use tumbling window to process last 10 minutes of data -

https://docs.amazonaws.cn/en_us/kinesisanalytics/latest/java/examples-tumbling.html

I am not sure how a tumbling window algorithm would be implemented in Glue.

upvoted 2 times

A company wants to predict the classification of documents that are created from an application. New documents are saved to an Amazon S3 bucket every 3 seconds. The company has developed three versions of a machine learning (ML) model within Amazon SageMaker to classify document text. The company wants to deploy these three versions to predict the classification of each document.

Which approach will meet these requirements with the LEAST operational overhead?

- A. Configure an S3 event notification that invokes an AWS Lambda function when new documents are created. Configure the Lambda function to create three SageMaker batch transform jobs, one batch transform job for each model for each document.
- B. Deploy all the models to a single SageMaker endpoint. Treat each model as a production variant. Configure an S3 event notification that invokes an AWS Lambda function when new documents are created. Configure the Lambda function to call each production variant and return the results of each model.
- C. Deploy each model to its own SageMaker endpoint. Configure an S3 event notification that invokes an AWS Lambda function when new documents are created. Configure the Lambda function to call each endpoint and return the results of each model.
- D. Deploy each model to its own SageMaker endpoint. Create three AWS Lambda functions. Configure each Lambda function to call a different endpoint and return the results. Configure three S3 event notifications to invoke the Lambda functions when new documents are created.

Correct Answer: C

Community vote distribution

B (73%)

C (27%)

- Amit11011996

Highly Voted

5 months ago

Selected Answer: B

yes, It seems to be 'B'

upvoted 8 times
- VinceCar

5 months ago

agreed, shadow testing is supported on SageMaker.

<https://aws.amazon.com/cn/blogs/aws/new-for-amazon-sagemaker-perform-shadow-tests-to-compare-inference-performance-between-ml-model-variants/>

upvoted 4 times
- Maaayaaa

1 day, 12 hours ago

Not sure if the shadow testing fits into the purpose here. "The company has developed three versions of a machine learning (ML) model within Amazon SageMaker to classify document text."

upvoted 1 times
- cox1960

Most Recent

1 day, 10 hours ago

Selected Answer: B

B - the reason is not shadow testing since it is not named and does not require client logic. The reason is that it is possible to target a model

<https://docs.aws.amazon.com/sagemaker/latest/dg/invoke-multi-model-endpoint.html>

upvoted 1 times
- blanco750

1 month, 1 week ago

Selected Answer: B

B it is which involves using single endpoint for multiple model versions

upvoted 2 times
- pan_b

1 month, 2 weeks ago

Selected Answer: C

I think the answer should be C. As there is no production version of the model identified, all the 3 models need to be invoked.

upvoted 2 times
- testtaker1984

1 month, 4 weeks ago

C, prod variant is used for traffic routing. All model needs to be invoked.

upvoted 1 times
- BTRYING

3 months ago

Selected Answer: C

C is correct

upvoted 2 times

  **dunhill** 5 months ago

I think the answer is B.

upvoted 3 times

Question #199

Topic 1

A manufacturing company needs to identify returned smartphones that have been damaged by moisture. The company has an automated process that produces 2,000 diagnostic values for each phone. The database contains more than five million phone evaluations. The evaluation process is consistent, and there are no missing values in the data. A machine learning (ML) specialist has trained an Amazon SageMaker linear learner ML model to classify phones as moisture damaged or not moisture damaged by using all available features. The model's F1 score is 0.6.

Which changes in model training would MOST likely improve the model's F1 score? (Choose two.)

- A. Continue to use the SageMaker linear learner algorithm. Reduce the number of features with the SageMaker principal component analysis (PCA) algorithm.
- B. Continue to use the SageMaker linear learner algorithm. Reduce the number of features with the scikit-learn multi-dimensional scaling (MDS) algorithm.
- C. Continue to use the SageMaker linear learner algorithm. Set the predictor type to regressor.
- D. Use the SageMaker k-means algorithm with k of less than 1,000 to train the model.
- E. Use the SageMaker k-nearest neighbors (k-NN) algorithm. Set a dimension reduction target of less than 1,000 to train the model.

Correct Answer: AE

Community vote distribution

AE (100%)

  **Peeking**

Highly Voted 

 4 months, 3 weeks ago

Selected Answer: AE

KNN can be used for dimensionality reduction through NCA (https://scikit-learn.org/stable/auto_examples/neighbors/plot_nca_dim_reduction.html#)

upvoted 8 times

  **Jerry84**

Highly Voted 

 3 months, 2 weeks ago

Selected Answer: AE

- A. Correct
- B. Incorrect. MDS is Non-linear dimensionality reduction method. <https://towardsdatascience.com/11-dimensionality-reduction-techniques-you-should-know-in-2021-dcb9500d388b>
- C. Incorrect. This is a classification problem instead of Regression.
- D. Incorrect. K-means is for Clustering(Unsupervised learning).
- E. Correct.
- upvoted 5 times

  **drcok87** 2 months, 1 week ago

E : <https://docs.aws.amazon.com/sagemaker/latest/dg/k-nearest-neighbors.html>

upvoted 1 times

  **MIlb**

Most Recent 

 4 weeks, 1 day ago

Selected Answer: AE

<https://docs.aws.amazon.com/sagemaker/latest/dg/k-nearest-neighbors.html>

upvoted 4 times

  **dunhill** 5 months ago

I think the answer is AB. k-means and k-nearest neighbors are not for reduce dimension.

upvoted 2 times

  **cox1960** 1 day, 10 hours ago

k-nn does. <https://docs.aws.amazon.com/sagemaker/latest/dg/k-nearest-neighbors.html>

upvoted 1 times

A company is building a machine learning (ML) model to classify images of plants. An ML specialist has trained the model using the Amazon SageMaker built-in Image Classification algorithm. The model is hosted using a SageMaker endpoint on an ml.m5.xlarge instance for real-time inference. When used by researchers in the field, the inference has greater latency than is acceptable. The latency gets worse when multiple researchers perform inference at the same time on their devices. Using Amazon CloudWatch metrics, the ML specialist notices that the ModelLatency metric shows a high value and is responsible for most of the response latency.

The ML specialist needs to fix the performance issue so that researchers can experience less latency when performing inference from their devices.

Which action should the ML specialist take to meet this requirement?

- A. Change the endpoint instance to an ml.t3 burstable instance with the same vCPU number as the ml.m5.xlarge instance has.
- B. Attach an Amazon Elastic Inference ml.eia2.medium accelerator to the endpoint instance.
- C. Enable Amazon SageMaker Autopilot to automatically tune performance of the model.
- D. Change the endpoint instance to use a memory optimized ML instance.

Correct Answer: A

Community vote distribution

B (100%)

- tsangckl

Highly Voted

5 months ago

Selected Answer: B

It's B
<https://aws.amazon.com/premiumsupport/knowledge-center/sagemaker-endpoint-latency/>
upvoted 9 times
- hichemck

5 months ago

false. its C. in the link you shared under High ModelLatency, it states "If an endpoint is overused, it might cause higher model latency. You can add Auto scaling to an endpoint to dynamically increase and decrease the number of instances available for an instance."
upvoted 1 times
- Peeking

4 months, 3 weeks ago

Autopilot is is not autoscaling in AWS. Autopilot is for model training, Autoscaling is during inference.
upvoted 2 times
- Maaayaaa

Most Recent

1 day, 11 hours ago

B is not correct anymore.
After April 15, 2023, new customers will not be able to launch instances with Amazon EI accelerators in Amazon SageMaker, Amazon ECS, or Amazon EC2. (<https://docs.aws.amazon.com/sagemaker/latest/dg/ei.html>)
upvoted 1 times
- MIlb

4 weeks, 1 day ago

Selected Answer: B

It's B.
Burstable instances only solves when lot of users are making inferences at the same time
upvoted 2 times
- Ajose0

2 months, 1 week ago

Selected Answer: B

The ModelLatency metric shows that the model inference time is causing the latency issue. Amazon Elastic Inference is designed to speed up the inference process of a machine learning model without needing to deploy the model on a more powerful instance. By attaching an Elastic Inference accelerator to the endpoint instance, the ML specialist can offload the compute-intensive parts of the inference process to the accelerator, resulting in faster inference times and lower latency.
upvoted 2 times
- It626

3 months, 2 weeks ago

B - <https://aws.amazon.com/premiumsupport/knowledge-center/sagemaker-endpoint-latency/>
upvoted 1 times
- Peeking

4 months, 3 weeks ago

Selected Answer: B

Elastic Inference accelerator (and AutoScaling but there is no autoscaling in the option). Be aware that Autopilot is is not autoscaling.

upvoted 3 times

An automotive company is using computer vision in its autonomous cars. The company has trained its models successfully by using transfer learning from a convolutional neural network (CNN). The models are trained with PyTorch through the use of the Amazon SageMaker SDK. The company wants to reduce the time that is required for performing inferences, given the low latency that is required for self-driving.

Which solution should the company use to evaluate and improve the performance of the models?

- A. Use Amazon CloudWatch algorithm metrics for visibility into the SageMaker training weights, gradients, biases, and activation outputs. Compute the filter ranks based on this information. Apply pruning to remove the low-ranking filters. Set the new weights. Run a new training job with the pruned model.
- B. Use SageMaker Debugger for visibility into the training weights, gradients, biases, and activation outputs. Adjust the model hyperparameters, and look for lower inference times. Run a new training job.
- C. Use SageMaker Debugger for visibility into the training weights, gradients, biases, and activation outputs. Compute the filter ranks based on this information. Apply pruning to remove the low-ranking filters. Set the new weights. Run a new training job with the pruned model.
- D. Use SageMaker Model Monitor for visibility into the ModelLatency metric and OverheadLatency metric of the model after the model is deployed. Adjust the model hyperparameters, and look for lower inference times. Run a new training job.

Correct Answer: C

Community vote distribution

C (100%)

  **Tony_1406** 2 days, 7 hours ago

Selected Answer: C

Same example here:
<https://aws.amazon.com/blogs/machine-learning/pruning-machine-learning-models-with-amazon-sagemaker-debugger-and-amazon-sagemaker-experiments/>
upvoted 1 times

  **Gaby999** 5 days, 17 hours ago

Selected Answer: B
To reduce the time required for performing inferences in autonomous cars, the automotive company should use SageMaker Debugger for visibility into the training weights, gradients, biases, and activation outputs. They can adjust the model hyperparameters and look for lower inference times. They can also use SageMaker Model Monitor for visibility into the ModelLatency metric and OverheadLatency metric of the model after the model is deployed. However, option C, which suggests computing the filter ranks based on the training outputs and applying pruning to remove the low-ranking filters, is not applicable for transfer learning models since the layers in the pre-trained model are already trained and cannot be changed. Therefore, the correct solution is B.
upvoted 1 times

  **Valcilio** 1 month, 3 weeks ago

Selected Answer: C

Even if a better machine could help, the problem is about the model, not about the general or the machine in specific.
upvoted 2 times

  **Ajose0** 2 months, 1 week ago

Selected Answer: C

Using SageMaker Debugger, the company can monitor the training process and evaluate the performance of the model by computing filter ranks based on information like weights, gradients, biases, and activation outputs.

After identifying the low-ranking filters, the company can apply pruning to remove them and set new weights.

By doing so, the company can reduce the model size and improve the inference time. Finally, a new training job with the pruned model can be run to verify the performance improvements

Not D because Model Monitor is a tool for monitoring the performance of deployed models, and it does not provide any direct feedback or insights into the model training process or ways to improve model inference time. Therefore, while Model Monitor can be useful for monitoring the performance of deployed models, it is not the best choice for evaluating and improving the performance of the models during the training phase, which is what the question is asking for.
upvoted 2 times

  **wolfsong** 2 months, 1 week ago

It's between C and D. But I think it's C.
C. <https://aws.amazon.com/blogs/machine-learning/pruning-machine-learning-models-with-amazon-sagemaker-debugger-and-amazon-sagemaker-experiments/> Everything is there.

D: <https://aws.amazon.com/premiumsupport/knowledge-center/sagemaker-endpoint-latency/> Here it says use Cloudwatch to view ModelLatency and OverheadLatency, not Model Monitor. I think Model Monitor is just for model performance i.e. drift, bias, accuracy etc.

upvoted 1 times

  **expertguru** 3 months, 2 weeks ago

The answer I guess D per below , they should have said Sagemaker model monitor using cloud watch

<https://docs.aws.amazon.com/sagemaker/latest/dg/model-monitor.html>

upvoted 1 times

  **jim20541** 4 months, 1 week ago

C,
<https://aws.amazon.com/blogs/machine-learning/pruning-machine-learning-models-with-amazon-sagemaker-debugger-and-amazon-sagemaker-experiments/>

upvoted 4 times

  **Alphacentavra** 4 months, 1 week ago

I would say 'D as a more generic approach than C. The problem can be caused not just filters.

upvoted 1 times

  **BoroJohn** 4 months, 2 weeks ago

The answer is "c" as the question is asking for evaluate and improve the performance of the models?

"<https://docs.aws.amazon.com/sagemaker/latest/dg/debugger-visualization.html>"

upvoted 2 times

  **dunhill** 5 months ago

I think the answer is D.

upvoted 2 times

A company's machine learning (ML) specialist is designing a scalable data storage solution for Amazon SageMaker. The company has an existing TensorFlow-based model that uses a train.py script. The model relies on static training data that is currently stored in TFRecord format.

What should the ML specialist do to provide the training data to SageMaker with the LEAST development overhead?

- A. Put the TFRecord data into an Amazon S3 bucket. Use AWS Glue or AWS Lambda to reformat the data to protobuf format and store the data in a second S3 bucket. Point the SageMaker training invocation to the second S3 bucket.
- B. Rewrite the train.py script to add a section that converts TFRecord data to protobuf format. Point the SageMaker training invocation to the local path of the data. Ingest the protobuf data instead of the TFRecord data.
- C. Use SageMaker script mode, and use train.py unchanged. Point the SageMaker training invocation to the local path of the data without reformatting the training data.
- D. Use SageMaker script mode, and use train.py unchanged. Put the TFRecord data into an Amazon S3 bucket. Point the SageMaker training invocation to the S3 bucket without reformatting the training data.

Correct Answer: B

Community vote distribution

D (100%)

- VinceCar

Highly Voted

5 months ago

Selected Answer: D

Should be D. TFRecord could be uploaded to S3 directly and be used as SageMaker's data source.
https://sagemaker-examples.readthedocs.io/en/latest/sagemaker_batch_transform/working_with_tfrecords/working-with-tfrecords.html#Upload-dataset-to-S3
upvoted 5 times
- BTRYING

3 months, 2 weeks ago

Then why not C

upvoted 1 times
- drcok87

2 months, 1 week ago

then how is local path a "scalable data storage solution" ?

answer is D

upvoted 1 times
- Amit11011996

Highly Voted

5 months ago

Selected Answer: D

It has to option D.
upvoted 5 times
- Ajose0

Most Recent

2 months, 1 week ago

Selected Answer: D

This option leverages SageMaker's built-in support for the TensorFlow framework and script mode. The existing train.py script can be used without any modifications. SageMaker will automatically download the training data from the specified S3 location to the instance running the training job.

This option saves development time by avoiding the need to rewrite the train.py script or reformat the training data.
upvoted 3 times

An ecommerce company wants to train a large image classification model with 10,000 classes. The company runs multiple model training iterations and needs to minimize operational overhead and cost. The company also needs to avoid loss of work and model retraining.

Which solution will meet these requirements?

- A. Create the training jobs as AWS Batch jobs that use Amazon EC2 Spot Instances in a managed compute environment.
- B. Use Amazon EC2 Spot Instances to run the training jobs. Use a Spot Instance interruption notice to save a snapshot of the model to Amazon S3 before an instance is terminated.
- C. Use AWS Lambda to run the training jobs. Save model weights to Amazon S3.
- D. Use managed spot training in Amazon SageMaker. Launch the training jobs with checkpointing enabled.

Correct Answer: C

Community vote distribution

D (100%)

Amit11011996

Highly Voted

5 months ago

Selected Answer: D

It has to be D.
With Spot training we can reduce the cost and save the model weights with checkpoint enabled.
upvoted 6 times

hichemck

5 months ago

agree. managed spot training is also cost effective
upvoted 4 times

Peeking

Highly Voted

4 months, 3 weeks ago

Selected Answer: D

<https://docs.aws.amazon.com/sagemaker/latest/dg/model-managed-spot-training.html>

Managed spot training can optimize the cost of training models up to 90% over on-demand instances. SageMaker manages the Spot interruptions on your behalf.

"Spot instances can be interrupted, causing jobs to take longer to start or finish. You can configure your managed spot training job to use checkpoints. SageMaker copies checkpoint data from a local path to Amazon S3. When the job is restarted, SageMaker copies the data from Amazon S3 back into the local path. The training job can then resume from the last checkpoint instead of restarting."
upvoted 5 times

Gaby999

Most Recent

5 days, 17 hours ago

Selected Answer: D

Use managed spot training in Amazon SageMaker. Launch the training jobs with checkpointing enabled.

Managed spot training in Amazon SageMaker can help minimize operational overhead and cost by using spot instances to perform the training. This can significantly reduce the cost of training, while still achieving the same accuracy. SageMaker provides built-in checkpointing capability, which allows saving model weights and progress to Amazon S3 periodically. This ensures that even if the spot instances are terminated, the training can resume from the last saved checkpoint. Additionally, SageMaker provides a managed service, so the ecommerce company does not need to worry about managing the infrastructure, and can focus on building and tuning their model.
upvoted 1 times

Gaby999

5 days, 17 hours ago

Selected Answer: D

The ML specialist should choose option D, which provides the training data to SageMaker with the least development overhead. This option involves putting the TFRecord data into an Amazon S3 bucket and pointing the SageMaker training invocation to the S3 bucket without reformatting the training data. Using SageMaker script mode is a convenient way to execute training scripts without any modification. Since the training script train.py already works with TFRecord data, it can be used as is without any changes. By storing the data in S3 and accessing it from there, the specialist can take advantage of SageMaker's built-in data distribution and parallelization capabilities, which can significantly speed up training. Rewriting the train.py script or using additional services like AWS Glue or Lambda would add unnecessary complexity and increase development overhead.
upvoted 1 times

Ajose0

2 months, 1 week ago

Selected Answer: D

Managed spot training in Amazon SageMaker provides a cost-effective way to run large machine learning workloads.

With managed spot training, the training jobs are executed using Amazon EC2 Spot instances, which can significantly reduce the cost of training.

Additionally, by launching training jobs with checkpointing enabled, the work done up to the last checkpoint is saved to Amazon S3. This ensures that the training job can be resumed from the last checkpoint in case of instance failure or termination. This minimizes the risk of data loss and avoids the need for retraining the model from scratch. Using Amazon SageMaker also reduces the operational overhead required to set up and manage the training environment.

upvoted 3 times

A retail company uses a machine learning (ML) model for daily sales forecasting. The model has provided inaccurate results for the past 3 weeks. At the end of each day, an AWS Glue job consolidates the input data that is used for the forecasting with the actual daily sales data and the predictions of the model. The AWS Glue job stores the data in Amazon S3.

The company's ML team determines that the inaccuracies are occurring because of a change in the value distributions of the model features. The ML team must implement a solution that will detect when this type of change occurs in the future.

Which solution will meet these requirements with the LEAST amount of operational overhead?

- A. Use Amazon SageMaker Model Monitor to create a data quality baseline. Confirm that the emit_metrics option is set to Enabled in the baseline constraints file. Set up an Amazon CloudWatch alarm for the metric.
- B. Use Amazon SageMaker Model Monitor to create a model quality baseline. Confirm that the emit_metrics option is set to Enabled in the baseline constraints file. Set up an Amazon CloudWatch alarm for the metric.
- C. Use Amazon SageMaker Debugger to create rules to capture feature values Set up an Amazon CloudWatch alarm for the rules.
- D. Use Amazon CloudWatch to monitor Amazon SageMaker endpoints. Analyze logs in Amazon CloudWatch Logs to check for data drift.

Correct Answer: A

Community vote distribution

A (92%)

8%

  **hichemck**

Highly Voted 

 5 months ago

Selected Answer: A

A is correct.
"If the statistical nature of the data that your model receives while in production drifts away from the nature of the baseline data it was trained on, the model begins to lose accuracy in its predictions. Amazon SageMaker Model Monitor uses rules to detect data drift and alerts you when it happens."
<https://docs.aws.amazon.com/sagemaker/latest/dg/model-monitor-data-quality.html>
upvoted 8 times

  **Ajose0**

Most Recent 

 2 months, 1 week ago

Selected Answer: A

A is correct.

The best solution to meet the requirements is to use Amazon SageMaker Model Monitor to create a data quality baseline. The ML team can set up a data quality baseline to detect when the input data to the model has drifted significantly from the historical distribution of the data. When data drift occurs, the Model Monitor emits a metric that can trigger an alarm in Amazon CloudWatch. The ML team can use this alarm to investigate and take corrective action.

Option B is incorrect because model quality baseline monitors model performance, not the input data quality.

Option C is incorrect because Amazon SageMaker Debugger is used to debug machine learning models and to identify problems with model training, not data quality.

Option D is incorrect because Amazon CloudWatch does not provide any features to monitor data drift in the input data used for the machine learning model.
upvoted 2 times

  **drcok87** 2 months, 1 week ago

Selected Answer: A

Data monitor
<https://docs.aws.amazon.com/sagemaker/latest/dg/model-monitor-data-quality.html>
Properties of independent variables changes due to seasonality, customer preferences
Model monitor
<https://docs.aws.amazon.com/sagemaker/latest/dg/model-monitor-model-quality.html>
Concept of what is spam email, changes over time
"The company's ML team determines that the inaccuracies are occurring because of a change in the value distributions of the model features."
They know model features that is data for model input is changing so we monitor data
<https://pair-code.github.io/what-if-tool/learn/tutorials/features-overview/>

a
upvoted 1 times

  **drcok87** 2 months, 1 week ago

Data monitor
<https://docs.aws.amazon.com/sagemaker/latest/dg/model-monitor-data-quality.html>
Properties of independent variables changes due to seasonality, customer preferences
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They know model features that is data for model input is changing so we monitor data
<https://pair-code.github.io/what-if-tool/learn/tutorials/features-overview/>

a
upvoted 1 times

  **dunhill** 5 months ago
I think the answer is B. Data quality can be monitored via model monitor model quality baseline.
upvoted 1 times

  **VinceCar** 5 months ago

Selected Answer: B

B. Since it's "a change in the value distributions of the model features".

upvoted 1 times

  **tsangckl** 5 months ago
What is the difference of ans A and B?
upvoted 1 times

  **pass3in3mon** 5 months ago
model quality baseline vs data quality baseline
upvoted 1 times

A machine learning (ML) specialist has prepared and used a custom container image with Amazon SageMaker to train an image classification model. The ML specialist is performing hyperparameter optimization (HPO) with this custom container image to produce a higher quality image classifier.

The ML specialist needs to determine whether HPO with the SageMaker built-in image classification algorithm will produce a better model than the model produced by HPO with the custom container image. All ML experiments and HPO jobs must be invoked from scripts inside SageMaker Studio notebooks.

How can the ML specialist meet these requirements in the LEAST amount of time?

- A. Prepare a custom HPO script that runs multiple training jobs in SageMaker Studio in local mode to tune the model of the custom container image. Use the automatic model tuning capability of SageMaker with early stopping enabled to tune the model of the built-in image classification algorithm. Select the model with the best objective metric value.
- B. Use SageMaker Autopilot to tune the model of the custom container image. Use the automatic model tuning capability of SageMaker with early stopping enabled to tune the model of the built-in image classification algorithm. Compare the objective metric values of the resulting models of the SageMaker AutopilotAutoML job and the automatic model tuning job. Select the model with the best objective metric value.
- C. Use SageMaker Experiments to run and manage multiple training jobs and tune the model of the custom container image. Use the automatic model tuning capability of SageMaker to tune the model of the built-in image classification algorithm. Select the model with the best objective metric value.
- D. Use the automatic model tuning capability of SageMaker to tune the models of the custom container image and the built-in image classification algorithm at the same time. Select the model with the best objective metric value.

Correct Answer: B

Community vote distribution

B (50%)

C (45%)

5%

- VinceCar

Highly Voted

4 months, 4 weeks ago
- Selected Answer: B

Autopilot is the faster. "Amazon SageMaker Autopilot experiments are now up to 2x faster in Hyperparameter Optimization training mode" . Refer to https://aws.amazon.com/about-aws/whats-new/2022/11/amazon-sagemaker-autopilot-experiments-2x-faster-hyperparameter-optimization-training-mode/?nc1=h_ls

upvoted 6 times
- Alexkats_87

3 months, 1 week ago
- It seems that Autopilot doesn't support image data (image classification), so B will be incorrect in this case <https://docs.aws.amazon.com/sagemaker/latest/dg/autopilot-datasets-problem-types.html#autopilot-datasets>

upvoted 2 times
- rrshah83

Highly Voted

4 months ago
- Selected Answer: C

<https://docs.aws.amazon.com/sagemaker/latest/dg/experiments.html>

upvoted 5 times
- cox1960

Most Recent

18 hours, 3 minutes ago
- Selected Answer: C

we have 2 algorithms already and to experiment. Autopilot selects algorithms.

upvoted 1 times
- injohe

1 week ago
- Option D:

What can be easier than calling Hyperparameter Job for both custom container and built-in algorithm using the same command inside SageMaker Studio, probably nothing. So, let's look into proofs why this is possible. Remember, we have to find the possible solution which can be achieved in the LEAST amount of development time.

1. It's possible to Tune Multiple Algorithms with Hyperparameter Optimization to Find the Best Model. We only need to provide job settings that apply to all of the algorithms to be tested with list of training definitions. <https://docs.aws.amazon.com/sagemaker/latest/dg/multiple-algorithm-hpo.html>

2. You can use SageMaker AMT with built-in algorithms, custom algorithms, or SageMaker pre-built containers for machine learning frameworks. (Ctrl + F).
<https://docs.aws.amazon.com/sagemaker/latest/dg/automatic-model-tuning.html>
upvoted 1 times

  **injoho** 1 week ago

Eliminating other options:

A - Performing custom HPO is possible but time-consuming.
B - Autopilot is a possible, great, and in terms of development time very fast solution. But pay attention, we are asked to compare two specific algorithms (custom algorithm and built-in image classification algorithm) and not run Autopilot which will evaluate on its own list of algorithms which one is better/worse. We want to compare our two algorithms!
C - Manually run multiple jobs and log experiments into SageMaker Experiments for custom algorithm and then compare with AMT Job for built-in algorithm seems like a lot of redundant work compared to solution D.

upvoted 2 times

  **MIlb** 4 weeks, 1 day ago

Selected Answer: D

Option D is the most effective in time because applies automatic model tuning (AMT) for all (built-in and custom)

upvoted 1 times

  **MIlb** 4 weeks, 1 day ago

Autopilot can work with images

upvoted 1 times

  **MIlb** 4 weeks, 1 day ago

Sorry, can't

upvoted 1 times

  **pan_b** 1 month, 2 weeks ago

Selected Answer: C

Image classification is not a supported problem type. <https://docs.aws.amazon.com/sagemaker/latest/dg/autopilot-datasets-problem-types.html>
C should be the answer as Experiments can be triggered from Notebook and support HPO

upvoted 1 times

  **Valcilio** 1 month, 3 weeks ago

Selected Answer: C

It's C, even if you can transform the images to parquet or csv, it's will take some time turning this process more slowly than just use the other options.

upvoted 1 times

  **Chelseajcole** 2 months ago

Selected Answer: C

Autopilot used for picking model for you, this is comparing built in vs custom, it is experiment job

upvoted 2 times

  **Siyuan_Zhu** 2 months ago

Selected Answer: B

B
You can easily convert image files to parquet files to be trained. I think B is correct.
C -- Does not mention the tuning and comparison of the built-in model, therefore it does not meet the requirements

upvoted 2 times

  **Ajose0** 2 months, 1 week ago

Selected Answer: B

B is the best option.

SageMaker Autopilot provides a fully managed automated machine learning solution that automates the end-to-end machine learning process. It automatically tunes the model's hyperparameters and creates candidate models that match the given objective metric. In this case, the ML specialist can use Autopilot to tune the model of the custom container image and then use the automatic model tuning capability of SageMaker to tune the model of the built-in image classification algorithm. They can then compare the objective metric values of the resulting models of the SageMaker Autopilot AutoML job and the automatic model tuning job and select the model with the best objective metric value.

This approach is likely to be the least time-consuming because it allows for parallelized experimentation and optimization across two models.

upvoted 3 times

  **wolfsong** 2 months, 1 week ago

Autopilot doesn't support image data, only csv and parquet. So can't be B. See <https://docs.aws.amazon.com/sagemaker/latest/dg/autopilot-automate-model-development.html>

upvoted 1 times

  **drcok87** 2 months, 2 weeks ago

auto pilot is for someone who is not familiar with ML to use this tool for gaining insights into their problem; this service will automatically choose the solution i.e., ML algorithm to solve the problem.
In this case however, the team has already trained a model that too with a custom container; these folks are experts in what they do and just look

for HPO, another term for HPO is automatic model tuning <https://docs.aws.amazon.com/sagemaker/latest/dg/automatic-model-tuning.html>

c
upvoted 2 times

  **drcok87** 2 months, 1 week ago
Refer to the phrase "You can use SageMaker AMT with built-in algorithms, custom algorithms, or SageMaker pre-built containers for machine learning frameworks." on <https://docs.aws.amazon.com/sagemaker/latest/dg/automatic-model-tuning.html>

I knew its Automatic Model Tuning for the question, but seems like this can be used for both built-in and custom algorithm/containers.

I'm changing my answer to D
upvoted 3 times

  **BoroJohn** 4 months, 2 weeks ago
The answer is "c" <https://docs.aws.amazon.com/sagemaker/latest/dg/experiments.html> Amazon SageMaker Autopilot doesn't support Image Classification problem. It supports only binary classification or regression. <https://docs.aws.amazon.com/sagemaker/latest/dg/autopilot-datasets-problem-types.html>
upvoted 4 times

  **dunhill** 5 months ago
I think the answer is C.
upvoted 4 times

  **dunhill** 4 months, 3 weeks ago
Per the link that VinceCar provided, B might be right. I am not very sure, but I agree that view point.
upvoted 2 times

A company wants to deliver digital car management services to its customers. The company plans to analyze data to predict the likelihood of users changing cars. The company has 10 TB of data that is stored in an Amazon Redshift cluster. The company's data engineering team is using Amazon SageMaker Studio for data analysis and model development. Only a subset of the data is relevant for developing the machine learning models. The data engineering team needs a secure and cost-effective way to export the data to a data repository in Amazon S3 for model development.

Which solutions will meet these requirements? (Choose two.)

- A. Launch multiple medium-sized instances in a distributed SageMaker Processing job. Use the prebuilt Docker images for Apache Spark to query and plot the relevant data and to export the relevant data from Amazon Redshift to Amazon S3.
- B. Launch multiple medium-sized notebook instances with a PySpark kernel in distributed mode. Download the data from Amazon Redshift to the notebook cluster. Query and plot the relevant data. Export the relevant data from the notebook cluster to Amazon S3.
- C. Use AWS Secrets Manager to store the Amazon Redshift credentials. From a SageMaker Studio notebook, use the stored credentials to connect to Amazon Redshift with a Python adapter. Use the Python client to query the relevant data and to export the relevant data from Amazon Redshift to Amazon S3.
- D. Use AWS Secrets Manager to store the Amazon Redshift credentials. Launch a SageMaker extra-large notebook instance with block storage that is slightly larger than 10 TB. Use the stored credentials to connect to Amazon Redshift with a Python adapter. Download, query, and plot the relevant data. Export the relevant data from the local notebook drive to Amazon S3.
- E. Use SageMaker Data Wrangler to query and plot the relevant data and to export the relevant data from Amazon Redshift to Amazon S3.

Correct Answer: AC

Community vote distribution

CE (100%)

- VinceCar

Highly Voted

5 months ago

Selected Answer: CE

C and E. No secure control is in option A.

upvoted 5 times
- drcok87

Most Recent

2 months, 2 weeks ago

e is obvious choice:
c <https://aws.amazon.com/blogs/big-data/using-the-amazon-redshift-data-api-to-interact-from-an-amazon-sagemaker-jupyter-notebook/>

upvoted 2 times
- solution123

2 months, 3 weeks ago

Selected Answer: CE

CE

Option A: Launching multiple medium-sized instances in a distributed SageMaker Processing job and using the prebuilt Docker images for Apache Spark to query and plot the relevant data is a possible solution, but it may not be the most cost-effective solution as it requires spinning up multiple instances.

Option B: Launching multiple medium-sized notebook instances with a PySpark kernel in distributed mode is another solution, but it may not be the most secure solution as the data would be stored on the instances and not in a centralized data repository.

Option D: Using AWS Secrets Manager to store the Amazon Redshift credentials and launching a SageMaker extra-large notebook instance is a solution, but the block storage requirement that is slightly larger than 10 TB could be costly and may not be necessary.

upvoted 4 times
- BoroJohn

4 months, 2 weeks ago

C & E seems right - <https://docs.aws.amazon.com/sagemaker/latest/dg/data-wrangler.html>

upvoted 3 times

A company is building an application that can predict spam email messages based on email text. The company can generate a few thousand human-labeled datasets that contain a list of email messages and a label of "spam" or "not spam" for each email message. A machine learning (ML) specialist wants to use transfer learning with a Bidirectional Encoder Representations from Transformers (BERT) model that is trained on English Wikipedia text data.

What should the ML specialist do to initialize the model to fine-tune the model with the custom data?

- A. Initialize the model with pretrained weights in all layers except the last fully connected layer.
- B. Initialize the model with pretrained weights in all layers. Stack a classifier on top of the first output position. Train the classifier with the labeled data.
- C. Initialize the model with random weights in all layers. Replace the last fully connected layer with a classifier. Train the classifier with the labeled data.
- D. Initialize the model with pretrained weights in all layers. Replace the last fully connected layer with a classifier. Train the classifier with the labeled data.

Correct Answer: D

Community vote distribution

D (100%)

  **blanco750** 1 month, 1 week ago

Selected Answer: D

D seems correct
upvoted 1 times

  **rrshah83** 4 months ago

Selected Answer: D

D is a best practice
upvoted 3 times

  **BoroJohn** 4 months, 2 weeks ago

Is B correct? - <https://www.analyticsvidhya.com/blog/2020/07/transfer-learning-for-nlp-fine-tuning-bert-for-text-classification/>

Freeze the entire architecture – We can even freeze all the layers of the model and attach a few neural network layers of our own and train this new model. Note that the weights of only the attached layers will be updated during model training.

upvoted 2 times

  **dunhill** 5 months ago

I think the answer is D.
upvoted 4 times

A company is using a legacy telephony platform and has several years remaining on its contract. The company wants to move to AWS and wants to implement the following machine learning features:

- Call transcription in multiple languages
- Categorization of calls based on the transcript
- Detection of the main customer issues in the calls
- Customer sentiment analysis for each line of the transcript, with positive or negative indication and scoring of that sentiment

Which AWS solution will meet these requirements with the LEAST amount of custom model training?

- A. Use Amazon Transcribe to process audio calls to produce transcripts, categorize calls, and detect issues. Use Amazon Comprehend to analyze sentiment.
- B. Use Amazon Transcribe to process audio calls to produce transcripts. Use Amazon Comprehend to categorize calls, detect issues, and analyze sentiment
- C. Use Contact Lens for Amazon Connect to process audio calls to produce transcripts, categorize calls, detect issues, and analyze sentiment.
- D. Use Contact Lens for Amazon Connect to process audio calls to produce transcripts. Use Amazon Comprehend to categorize calls, detect issues, and analyze sentiment.

Correct Answer: D

Community vote distribution

C (92%)

8%

- tsangckl

Highly Voted

5 months ago

I think C. Contact Lens can do all.

https://aws.amazon.com/connect/contact-lens/

upvoted 7 times
- solution123

Highly Voted

2 months, 3 weeks ago

Selected Answer: C

The correct answer is C. Use Contact Lens for Amazon Connect to process audio calls to produce transcripts, categorize calls, detect issues, and analyze sentiment.

Contact Lens is a fully managed service that provides advanced analytics for customer service interactions in Amazon Connect. It includes call transcription, sentiment analysis, and issue detection, which meets all the requirements in the question. Using a single service like Contact Lens will reduce the complexity of integrating multiple AWS services and also minimize the need for custom model training.

While Amazon Transcribe and Amazon Comprehend are also valuable AWS services, they are not designed specifically for customer service interactions and may require additional configuration and custom model training to meet the specific requirements listed in the question. You can see detail in <https://www.udemy.com/course/aws-certified-machine-learning-specialty-2023/>

upvoted 5 times
- cox1960

Most Recent

17 hours, 49 minutes ago

Selected Answer: B

"several years remaining on its contract" means no Amazon Connect.

upvoted 1 times
- drcok87

2 months, 2 weeks ago

c <https://docs.aws.amazon.com/connect/latest/adminguide/analyze-conversations.html>

upvoted 1 times
- BoroJohn

4 months, 2 weeks ago

Company has several years remaining on its contract. Does this change the solution approach? That means they cant use Amazon Connect. So Contact Lens may not be applicable here. If that assumption is true, B should be the answer.

upvoted 4 times
- It626

3 months, 2 weeks ago

Having the same though. If the company needs to remain in the legacy telephony platform, the answer would be B. Otherwise, C.

upvoted 1 times
- shubhary25

4 months, 3 weeks ago

Selected Answer: C

Contact Lens is an out of the box feature for Amazon Connect that leverages Amazon Transcribe to generate call transcripts and Amazon Comprehend to apply natural language processing (NLP) on these transcripts, with no coding required.
<https://aws.amazon.com/connect/contact-lens/#:~:text=Contact%20Lens%20is%20an%20out,transcripts%2C%20with%20no%20coding%20required>.

upvoted 3 times

  **akhjhjk** 5 months ago

Selected Answer: C

c seems to be correct

upvoted 4 times

  **dunhill** 5 months ago

I think the answer is D. Using Amazon Comprehend to categorize calls, detect issues, and analyze sentiment.

upvoted 1 times

  **dunhill** 4 months, 3 weeks ago

I correct myself. It should be C.
tsangckl and akjhjk are right.

upvoted 1 times

A finance company needs to forecast the price of a commodity. The company has compiled a dataset of historical daily prices. A data scientist must train various forecasting models on 80% of the dataset and must validate the efficacy of those models on the remaining 20% of the dataset.

How should the data scientist split the dataset into a training dataset and a validation dataset to compare model performance?

- A. Pick a date so that 80% of the data points precede the date. Assign that group of data points as the training dataset. Assign all the remaining data points to the validation dataset.
- B. Pick a date so that 80% of the data points occur after the date. Assign that group of data points as the training dataset. Assign all the remaining data points to the validation dataset.
- C. Starting from the earliest date in the dataset, pick eight data points for the training dataset and two data points for the validation dataset. Repeat this stratified sampling until no data points remain.
- D. Sample data points randomly without replacement so that 80% of the data points are in the training dataset. Assign all the remaining data points to the validation dataset.

Correct Answer: B

Community vote distribution

A (91%)

9%

- blanco750

1 month, 1 week ago

Selected Answer: A

For time series data, it is important to split the dataset chronologically, with the training dataset containing the earlier dates and the validation dataset containing the later dates

upvoted 3 times
- SANDEEP_AWS

1 month, 3 weeks ago

Selected Answer: A

A, As it's a time series problem.

upvoted 3 times
- Valcilio

1 month, 3 weeks ago

Selected Answer: A

It's a timeseries problem, then the splitting needs to be made by date.

upvoted 1 times
- Ajose0

2 months, 1 week ago

Selected Answer: A

Option A is the recommended approach where the training dataset contains historical prices that precede a certain date, and the validation dataset contains prices that occur after that date. This ensures that the model is trained on past data and evaluated on future data, which is more representative of real-world performance.

Option D is NOT the recommended approach for time series data because it ignores the time aspect of the data. Randomly sampling data points without considering the time sequence can result in data leakage and poor model performance.

upvoted 3 times
- Adi_09

2 months, 2 weeks ago

a since this is time series problem

upvoted 2 times
- drcok87

2 months, 2 weeks ago

a <https://towardsdatascience.com/time-series-from-scratch-train-test-splits-and-evaluation-metrics-4fd654de1b37>

upvoted 2 times
- HP0510

2 months, 2 weeks ago

Selected Answer: D

Because it randomly selects data points for both the training and validation datasets, ensuring that the samples are representative of the entire dataset and reducing the chances of overfitting. By randomly sampling without replacement, the data scientist can avoid any biases in the selection of data points and ensure that the training and validation datasets are independent.

upvoted 1 times
- merki

2 months, 3 weeks ago

I think it should be A!

upvoted 1 times

A retail company wants to build a recommendation system for the company's website. The system needs to provide recommendations for existing users and needs to base those recommendations on each user's past browsing history. The system also must filter out any items that the user previously purchased.

Which solution will meet these requirements with the LEAST development effort?

- A. Train a model by using a user-based collaborative filtering algorithm on Amazon SageMaker. Host the model on a SageMaker real-time endpoint. Configure an Amazon API Gateway API and an AWS Lambda function to handle real-time inference requests that the web application sends. Exclude the items that the user previously purchased from the results before sending the results back to the web application.
- B. Use an Amazon Personalize PERSONALIZED_RANKING recipe to train a model. Create a real-time filter to exclude items that the user previously purchased. Create and deploy a campaign on Amazon Personalize. Use the GetPersonalizedRanking API operation to get the real-time recommendations.
- C. Use an Amazon Personalize USER_PERSONALIZATION recipe to train a model. Create a real-time filter to exclude items that the user previously purchased. Create and deploy a campaign on Amazon Personalize. Use the GetRecommendations API operation to get the real-time recommendations.
- D. Train a neural collaborative filtering model on Amazon SageMaker by using GPU instances. Host the model on a SageMaker real-time endpoint. Configure an Amazon API Gateway API and an AWS Lambda function to handle real-time inference requests that the web application sends. Exclude the items that the user previously purchased from the results before sending the results back to the web application.

Correct Answer: C

Community vote distribution

C (74%)

B (26%)

- MIlb

3 weeks, 6 days ago

Selected Answer: B

B it,s the correct answer

upvoted 2 times
- blanco750

1 month, 1 week ago

Selected Answer: C

C looks the right choice

upvoted 3 times
- pan_b

1 month, 2 weeks ago

Selected Answer: C

Its C, User Personalizationis recommended for user interaction scenarios

https://docs.aws.amazon.com/personalize/latest/dg/native-recipe-new-item-USER_PERSONALIZATION.html

upvoted 3 times
- Valcilio

1 month, 3 weeks ago

Selected Answer: C

It's C, User Personalization is recommended for self-user user case.

upvoted 2 times
- Chelseajcole

1 month, 3 weeks ago

Selected Answer: C

The User-Personalization (aws-user-personalization) recipe is optimized for all personalized recommendation scenarios. It predicts the items that a user will interact with based on Interactions, Items, and Users datasets. When recommending items, it uses automatic item exploration.

upvoted 2 times
- GiyeonShin

2 months, 1 week ago

Selected Answer: C

Option B: https://docs.aws.amazon.com/personalize/latest/dg/native-recipe-search.html

"With Personalized-Ranking, you must manually create a new solution version (retrain the model) to reflect updates to your catalog and update the model with your user's most recent behavior."

Option B has a disadvantage to update the catalog in retail company.

So, Option C has the less effort to oprate than Option B

upvoted 4 times

  **Ajose0** 2 months, 1 week ago

Selected Answer: B

Option B is a better fit for the given requirements since it specifically mentions the need to filter out items that the user has previously purchased.

The PERSONALIZED_RANKING recipe in Amazon Personalize is designed to provide personalized recommendations while allowing for exclusion of previously purchased items using a filter.

In contrast, the USER_PERSONALIZATION recipe in option C is designed to provide personalized recommendations without the ability to filter out previously purchased items.

Therefore, option B is the best choice for meeting the given requirements with the least development effort.

upvoted 3 times

  **damaldon** 2 months, 3 weeks ago

Answer is C

https://docs.aws.amazon.com/personalize/latest/dg/native-recipe-new-item-USER_PERSONALIZATION.html

upvoted 4 times

  **damaldon** 2 months, 3 weeks ago

Answer is C

<https://docs.aws.amazon.com/personalize/latest>

upvoted 3 times

A bank wants to use a machine learning (ML) model to predict if users will default on credit card payments. The training data consists of 30,000 labeled records and is evenly balanced between two categories. For the model, an ML specialist selects the Amazon SageMaker built-in XGBoost algorithm and configures a SageMaker automatic hyperparameter optimization job with the Bayesian method. The ML specialist uses the validation accuracy as the objective metric.

When the bank implements the solution with this model, the prediction accuracy is 75%. The bank has given the ML specialist 1 day to improve the model in production.

Which approach is the FASTEST way to improve the model's accuracy?

- A. Run a SageMaker incremental training based on the best candidate from the current model's tuning job. Monitor the same metric that was used as the objective metric in the previous tuning, and look for improvements.
- B. Set the Area Under the ROC Curve (AUC) as the objective metric for a new SageMaker automatic hyperparameter tuning job. Use the same maximum training jobs parameter that was used in the previous tuning job.
- C. Run a SageMaker warm start hyperparameter tuning job based on the current model's tuning job. Use the same objective metric that was used in the previous tuning.
- D. Set the F1 score as the objective metric for a new SageMaker automatic hyperparameter tuning job. Double the maximum training jobs parameter that was used in the previous tuning job.

Correct Answer: A

Community vote distribution

C (75%)

B (25%)

  **Ahmedhadi_** 1 week, 2 days ago

Selected Answer: C

C is the correct answer because it uses the results from past HPO jobs and builds upon them to improve accuracy.
upvoted 2 times

  **mawsman** 2 weeks ago

Selected Answer: C

I go with C - warm start, A is not supported on XGBoost, and other options will start tuning from scratch and might be just as bad as the initial tuning job. We only have 1 day, so more tuning with existing job to inform the new trainging job is the only option here
<https://docs.aws.amazon.com/sagemaker/latest/dg/automatic-model-tuning-warm-start.html>
upvoted 1 times

  **MIlb** 1 month ago

C is the correct answer.
You can't use Incremental training on Xgboost algorithm <https://docs.aws.amazon.com/sagemaker/latest/dg/incremental-training.html>
upvoted 1 times

  **MIlb** 3 weeks, 1 day ago

It appears in 2023-April-3
upvoted 1 times

  **SANDEEP_AWS** 1 month, 2 weeks ago

Selected Answer: B

Since ROC-AUC is presumed to be one of the best for a binary classification. Hence option B. Option A -- Incremental training is suited wherein the training dataset gets updated frequently.
upvoted 1 times

  **drcok87** 2 months, 1 week ago

<https://docs.aws.amazon.com/sagemaker/latest/dg/automatic-model-tuning-warm-start.html>
<https://francesca-donadoni.medium.com/training-an-xgboost-model-for-pricing-analysis-using-aws-sagemaker-55d777708e52>

<https://docs.aws.amazon.com/sagemaker/latest/dg/incremental-training.html>

C
upvoted 3 times

  **drcok87** 2 months, 1 week ago

also it cannot be a: "Only three built-in algorithms currently support incremental training: Object Detection - MXNet, Image Classification - MXNet, and Semantic Segmentation Algorithm." from <https://docs.aws.amazon.com/sagemaker/latest/dg/incremental-training.html>
upvoted 1 times

Question #212

Topic 1

A data scientist has 20 TB of data in CSV format in an Amazon S3 bucket. The data scientist needs to convert the data to Apache Parquet format.

How can the data scientist convert the file format with the LEAST amount of effort?

- A. Use an AWS Glue crawler to convert the file format.
- B. Write a script to convert the file format. Run the script as an AWS Glue job.
- C. Write a script to convert the file format. Run the script on an Amazon EMR cluster.
- D. Write a script to convert the file format. Run the script in an Amazon SageMaker notebook.

Correct Answer: D

Community vote distribution

B (100%)

  **GiyeonShin** 2 months, 1 week ago

Selected Answer: B

Option B.
A - Glue crawler creates Glue Data Catalog from S3 buckets. It can be used to query by athena.
C, D - not serverless and not generally used for etl.
upvoted 2 times

  **Ajose0** 2 months, 1 week ago

Selected Answer: B

AWS Glue is a fully-managed ETL service that makes it easy to move data between data stores. AWS Glue can be used to automate the conversion of CSV files to Parquet format with minimal effort. AWS Glue supports reading data from CSV files, transforming the data, and writing the transformed data to Parquet files.

Option A is incorrect because AWS Glue crawler is used to infer the schema of data stored in S3 and create AWS Glue Data Catalog tables.

Option C is incorrect because while Amazon EMR can be used to process large amounts of data and perform data conversions, it requires more operational effort than AWS Glue.

Option D is incorrect because Amazon SageMaker is a machine learning service, and while it can be used for data processing, it is not the best option for simple data format conversion tasks.
upvoted 2 times

  **drcok87** 2 months, 2 weeks ago

in sagemaker notebook, you'd have to write python code but question is asking for something easy so i choose option b
<https://blog.searce.com/convert-csv-json-files-to-apache-parquet-using-aws-glue-a760d177b45f>
upvoted 2 times

  **Jerry84** 2 months, 2 weeks ago

From you link, A(Glue crawler) Should be correct.
upvoted 1 times

  **drcok87** 2 months, 1 week ago

crawler just creates the data catalog (schema), it does not actually converts the data to another format. As per details in that article, you are creating a job where source is schema created by crawler and destination is output s3 where we store formatted data.
upvoted 3 times

A company is building a pipeline that periodically retrains its machine learning (ML) models by using new streaming data from devices. The company's data engineering team wants to build a data ingestion system that has high throughput, durable storage, and scalability. The company can tolerate up to 5 minutes of latency for data ingestion. The company needs a solution that can apply basic data transformation during the ingestion process.

Which solution will meet these requirements with the MOST operational efficiency?

- A. Configure the devices to send streaming data to an Amazon Kinesis data stream. Configure an Amazon Kinesis Data Firehose delivery stream to automatically consume the Kinesis data stream, transform the data with an AWS Lambda function, and save the output into an Amazon S3 bucket.
- B. Configure the devices to send streaming data to an Amazon S3 bucket. Configure an AWS Lambda function that is invoked by S3 event notifications to transform the data and load the data into an Amazon Kinesis data stream. Configure an Amazon Kinesis Data Firehose delivery stream to automatically consume the Kinesis data stream and load the output back into the S3 bucket.
- C. Configure the devices to send streaming data to an Amazon S3 bucket. Configure an AWS Glue job that is invoked by S3 event notifications to read the data, transform the data, and load the output into a new S3 bucket.
- D. Configure the devices to send streaming data to an Amazon Kinesis Data Firehose delivery stream. Configure an AWS Glue job that connects to the delivery stream to transform the data and load the output into an Amazon S3 bucket.

Correct Answer: A

Community vote distribution

A (86%)

14%

- DS2021

1 month ago

Selected Answer: D

It is D

upvoted 1 times
- avland

4 days, 3 hours ago

Glue can't read from Firehose.

It's A.

upvoted 1 times
- DS2021

1 month, 1 week ago

It is C

upvoted 1 times
- ParkXD

3 weeks, 1 day ago

Option C uses AWS Glue, which can perform data transformation and load data into S3 buckets. However, Glue may not be the most efficient option for this use case, as it requires setting up a Glue job, which can introduce additional latency.

upvoted 3 times
- ParkXD

3 weeks, 1 day ago

Option A uses Amazon Kinesis data stream, which is optimized for high throughput, durable storage, and scalability.

upvoted 1 times
- sevosevo

1 month, 2 weeks ago

Why not C?

upvoted 1 times
- Valcilio

1 month, 3 weeks ago

Selected Answer: A

Firehose can take just at a maximum of 5 minutes, then it's the best solution for transformations.

upvoted 2 times
- GiyeonShin

2 months, 1 week ago

Selected Answer: A

A general architecture for (near)real - time ingesting & processing data:
Kinesis Data Streams - Kinesis Data Firehose - (If needs etl, lambda) - S3(Redshift, ...)

upvoted 2 times

  **Ajose0** 2 months, 1 week ago

Selected Answer: A

This solution provides a highly scalable and efficient way to ingest streaming data from devices with high throughput and durable storage by using Amazon Kinesis Data Streams and Amazon Kinesis Data Firehose.

By configuring an AWS Lambda function to transform the data during the ingestion process, the solution also applies basic data transformation with low latency. Additionally, Amazon S3 provides highly durable and scalable storage for the transformed data, which can be easily accessed by downstream processes such as machine learning model training.

upvoted 2 times

  **wolfsong** 2 months, 1 week ago

A:
<https://docs.aws.amazon.com/firehose/latest/dev/data-transformation.html>

upvoted 2 times

A retail company is ingesting purchasing records from its network of 20,000 stores to Amazon S3 by using Amazon Kinesis Data Firehose. The company uses a small, server-based application in each store to send the data to AWS over the internet. The company uses this data to train a machine learning model that is retrained each day. The company's data science team has identified existing attributes on these records that could be combined to create an improved model.

Which change will create the required transformed records with the LEAST operational overhead?

- A. Create an AWS Lambda function that can transform the incoming records. Enable data transformation on the ingestion Kinesis Data Firehose delivery stream. Use the Lambda function as the invocation target.
- B. Deploy an Amazon EMR cluster that runs Apache Spark and includes the transformation logic. Use Amazon EventBridge (Amazon CloudWatch Events) to schedule an AWS Lambda function to launch the cluster each day and transform the records that accumulate in Amazon S3. Deliver the transformed records to Amazon S3.
- C. Deploy an Amazon S3 File Gateway in the stores. Update the in-store software to deliver data to the S3 File Gateway. Use a scheduled daily AWS Glue job to transform the data that the S3 File Gateway delivers to Amazon S3.
- D. Launch a fleet of Amazon EC2 instances that include the transformation logic. Configure the EC2 instances with a daily cron job to transform the records that accumulate in Amazon S3. Deliver the transformed records to Amazon S3.

Correct Answer: C

Community vote distribution

A (100%)

  **Valcilio** 1 month, 3 weeks ago

Selected Answer: A

Firehose can use lambda functions to do data transformations!
upvoted 3 times

  **Ajose0** 2 months, 1 week ago

Selected Answer: A

A is the best option for this use case.

By creating an AWS Lambda function that can transform the incoming records and enabling data transformation on the ingestion Kinesis Data Firehose delivery stream, the company can transform the data with minimal operational overhead.

The Lambda function can be the invocation target for Kinesis Data Firehose, so that data is transformed as it is ingested.

This approach is serverless and scalable, and it does not require the company to manage any additional infrastructure.
upvoted 3 times

  **wolfsong** 2 months, 1 week ago

A: Lambda as invocation target just means that the function will invoke in response to the Firehose stream. See following:
<https://docs.aws.amazon.com/lambda/latest/dg/lambda-services.html>
<https://docs.aws.amazon.com/lambda/latest/dg/services-kinesisfirehose.html>
note: invocationid in Firehose message event.
upvoted 2 times

  **drcok87** 2 months, 1 week ago

a - seems to be an easy to manage solution however the phrase "Use the Lambda function as the invocation target." confuses me a bit.
upvoted 2 times

A sports broadcasting company is planning to introduce subtitles in multiple languages for a live broadcast. The commentary is in English. The company needs the transcriptions to appear on screen in French or Spanish, depending on the broadcasting country. The transcriptions must be able to capture domain-specific terminology, names, and locations based on the commentary context. The company needs a solution that can support options to provide tuning data.

Which combination of AWS services and features will meet these requirements with the LEAST operational overhead? (Choose two.)

- A. Amazon Transcribe with custom vocabularies
- B. Amazon Transcribe with custom language models
- C. Amazon SageMaker Seq2Seq
- D. Amazon SageMaker with Hugging Face Speech2Text
- E. Amazon Translate

Correct Answer: *BE*

Community vote distribution

BE (83%)

AE (17%)

-   **MIlb** 1 month ago

Selected Answer: BE

Comentary context is custom language
upvoted 2 times
-   **Valcilio** 1 month, 3 weeks ago

Selected Answer: BE

It's BE, custom languages is for domain-specific speech like terminologies, custom vocabulary is for words liken nouns.
upvoted 3 times
-   **angus** 1 month, 2 weeks ago

<https://docs.aws.amazon.com/transcribe/latest/dg/custom-language-models.html>
upvoted 1 times
-   **GiyeonShin** 2 months ago

Selected Answer: AE

Two sub-processes are needed: Speech to Text and Text to Text.
We can consider Amazon Transcribe for Speech2Text. If we use custom language models or SageMaker, we would need to gather our own data to train or retrain models. for less effort, (a) option is better than (b) option.
Then, Amazon Translate can be used to translate transcription to other language: (e) option
upvoted 1 times
-   **drcok87** 2 months ago

option A cannot capture "commentary context".
B and E should be correct.
upvoted 2 times
-   **drcok87** 2 months, 1 week ago

b: because The transcriptions must be able to capture domain-specific terminology, names, and locations based on the "commentary context"
e: managed service
upvoted 2 times

A data scientist at a retail company is forecasting sales for a product over the next 3 months. After preliminary analysis, the data scientist identifies that sales are seasonal and that holidays affect sales. The data scientist also determines that sales of the product are correlated with sales of other products in the same category.

The data scientist needs to train a sales forecasting model that incorporates this information.

Which solution will meet this requirement with the LEAST development effort?

- A. Use Amazon Forecast with Holidays featurization and the built-in autoregressive integrated moving average (ARIMA) algorithm to train the model.
- B. Use Amazon Forecast with Holidays featurization and the built-in DeepAR+ algorithm to train the model.
- C. Use Amazon SageMaker Processing to enrich the data with holiday information. Train the model by using the SageMaker DeepAR built-in algorithm.
- D. Use Amazon SageMaker Processing to enrich the data with holiday information. Train the model by using the Gluon Time Series (GluonTS) toolkit.

Correct Answer: B

Community vote distribution

B (89%)

11%

  **blanco750** 1 month, 1 week ago

Selected Answer: B

B is correct
upvoted 2 times

  **Valcilio** 1 month, 3 weeks ago

Selected Answer: B

DeepAR accepts exogenous regressors different from ARIMA and can understand seasonal effects, ARIMA can't do it too.
upvoted 2 times

  **Chelseajcole** 1 month, 3 weeks ago

Selected Answer: B

It is deepAR
upvoted 2 times

  **oso0348** 1 month, 4 weeks ago

Selected Answer: A

Option B is a good choice, as the DeepAR+ algorithm is specifically designed for forecasting in time series data with seasonality and long-term dependencies. However, it may require more development effort compared to the ARIMA algorithm.
upvoted 1 times

  **Ajose0** 2 months, 1 week ago

Selected Answer: B

Amazon Forecast is an AWS service that uses machine learning to build accurate time-series forecasts. It provides several built-in algorithms that support holiday featurization, and the DeepAR+ algorithm can handle the seasonality and correlation with other products with minimal development effort. With Amazon Forecast, the data scientist can easily configure the forecast horizon, select the appropriate forecast frequency, and configure the model training to incorporate the available historical data. Using Amazon SageMaker Processing to enrich the data with holiday information may require more development effort and does not offer the same level of automation and integration as Amazon Forecast.
upvoted 2 times

  **drcok87** 2 months, 2 weeks ago

b
<https://docs.aws.amazon.com/forecast/latest/dg/holidays.html>
<https://docs.aws.amazon.com/whitepapers/latest/time-series-forecasting-principles-with-amazon-forecast/appendix-a-faqs.html>
upvoted 2 times

A company is building a predictive maintenance model for its warehouse equipment. The model must predict the probability of failure of all machines in the warehouse. The company has collected 10,000 event samples within 3 months. The event samples include 100 failure cases that are evenly distributed across 50 different machine types.

How should the company prepare the data for the model to improve the model's accuracy?

- A. Adjust the class weight to account for each machine type.
- B. Oversample the failure cases by using the Synthetic Minority Oversampling Technique (SMOTE).
- C. Undersample the non-failure events. Stratify the non-failure events by machine type.
- D. Undersample the non-failure events by using the Synthetic Minority Oversampling Technique (SMOTE).

Correct Answer: D

Community vote distribution

B (100%)

blt23

Highly Voted

2 months, 3 weeks ago

Selected Answer: B

oversample the minority class

upvoted 6 times

oso0348

Most Recent

1 month, 4 weeks ago

Selected Answer: B

B. Oversample the failure cases by using the Synthetic Minority Oversampling Technique (SMOTE).

Since the number of failure cases is relatively small, oversampling the failure cases using techniques like SMOTE can help balance the class distribution and prevent the model from being biased towards the majority class. SMOTE creates synthetic samples for the minority class by interpolating new samples between existing ones. This will help improve the model's accuracy in predicting failure cases. Adjusting class weights (A) or undersampling (C, D) may not be as effective in this scenario.

upvoted 2 times

Ajose0

2 months, 1 week ago

Selected Answer: B

The data provided is imbalanced, with only 100 failure cases out of 10,000 event samples. Therefore, it is important to address this imbalance to improve the accuracy of the predictive maintenance model.

upvoted 2 times

A company stores its documents in Amazon S3 with no predefined product categories. A data scientist needs to build a machine learning model to categorize the documents for all the company's products.

Which solution will meet these requirements with the MOST operational efficiency?

- A. Build a custom clustering model. Create a Dockerfile and build a Docker image. Register the Docker image in Amazon Elastic Container Registry (Amazon ECR). Use the custom image in Amazon SageMaker to generate a trained model.
- B. Tokenize the data and transform the data into tabular data. Train an Amazon SageMaker k-means model to generate the product categories.
- C. Train an Amazon SageMaker Neural Topic Model (NTM) model to generate the product categories.
- D. Train an Amazon SageMaker Blazing Text model to generate the product categories.

Correct Answer: B

Community vote distribution

C (67%)

D (22%)

11%

  **Ajose0**

Highly Voted 

 2 months, 1 week ago

Selected Answer: C

C. Train an Amazon SageMaker Neural Topic Model (NTM) model to generate the product categories.

The task is to build a machine learning model to categorize documents for all the company's products. Among the given options, training an Amazon SageMaker Neural Topic Model (NTM) model would be the most efficient and effective solution.

An NTM model can identify topics in text data and group similar documents into specific categories, making it a suitable model for document categorization. With an NTM model, the data scientist would not need to define product categories beforehand, as the model would automatically group similar documents into topics. This saves time and resources compared to the other options.

upvoted 6 times

  **mawsman**

Most Recent 

 2 weeks ago

Selected Answer: D

Assign pre-defined categories to documents in a corpus: categorize books in a library into academic disciplines - BlazingText algorithm

<https://docs.aws.amazon.com/sagemaker/latest/dg/algos.html>

upvoted 2 times

  **mawsman** 2 weeks ago

Reading it again - C

upvoted 3 times

  **oso0348** 1 month, 4 weeks ago

Selected Answer: B

Option C is wrong because it suggests using a Neural Topic Model (NTM) to categorize documents. While NTM can be used to discover the underlying topics in a corpus of documents, it may not be the most efficient solution for categorizing documents for specific products. NTM is more suited for unsupervised learning problems where the goal is to discover the underlying themes or topics of the document corpus.

In this scenario, the data scientist needs to categorize documents based on predefined product categories. Therefore, a supervised learning algorithm like a text classification model would be more suitable. Amazon SageMaker Blazing Text algorithm provides an efficient and scalable solution for text classification problems.

upvoted 1 times

  **zshafi13** 1 month, 3 weeks ago

"no predefined product categories" -> unsupervised learning, C.

upvoted 3 times

  **drcok87** 2 months, 2 weeks ago

No predefined product category: topic modeling with NTM or LDA (Organize a set of documents into topics (not known in advance): tag a document as belonging to a medical category based on the terms used in the document.)

Predefined product category: topic modeling with blazing text (categorize books in a library into academic disciplines)

c

upvoted 1 times

A sports analytics company is providing services at a marathon. Each runner in the marathon will have their race ID printed as text on the front of their shirt. The company needs to extract race IDs from images of the runners.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Use Amazon Rekognition.
- B. Use a custom convolutional neural network (CNN).
- C. Use the Amazon SageMaker Object Detection algorithm.
- D. Use Amazon Lookout for Vision.

Correct Answer: C

Community vote distribution

A (100%)

-  **Valcilio** 1 month, 3 weeks ago

Selected Answer: A

It's A, rekgonition has methods to do text detection.

upvoted 3 times
-  **angus** 1 month, 2 weeks ago

<https://aws.amazon.com/rekognition/>

upvoted 2 times
-  **GiyeonShin** 2 months, 1 week ago

Selected Answer: A

OCR : Amazon Rekognition

upvoted 2 times
-  **Ajose0** 2 months, 1 week ago

Selected Answer: A

Rekognition's Text Detection feature allows you to easily extract text from images and videos without the need to create a custom model or perform complex training. It's a fully managed service that provides accurate and scalable text detection, recognition, and analysis capabilities. Additionally, Rekognition provides a simple API and SDKs for integrating text detection functionality into your applications.

<https://docs.aws.amazon.com/rekognition/latest/dg/text-detection.html?pg=ln&sec=ft>

upvoted 1 times
-  **drcok87** 2 months, 2 weeks ago

a - LEAST operational overhead

upvoted 1 times

A manufacturing company wants to monitor its devices for anomalous behavior. A data scientist has trained an Amazon SageMaker scikit-learn model that classifies a device as normal or anomalous based on its 4-day telemetry. The 4-day telemetry of each device is collected in a separate file and is placed in an Amazon S3 bucket once every hour. The total time to run the model across the telemetry for all devices is 5 minutes.

What is the MOST cost-effective solution for the company to use to run the model across the telemetry for all the devices?

- A. SageMaker Batch Transform
- B. SageMaker Asynchronous Inference
- C. SageMaker Processing
- D. A SageMaker multi-container endpoint

Correct Answer: C

Community vote distribution

A (86%)

14%

- blanco750

1 month, 1 week ago

Selected Answer: A

Key point is data is collected every hour. seems like a batch solution is most cost effective

upvoted 2 times
- pan_b

1 month, 2 weeks ago

Selected Answer: A

I will go with A. The Async inference seems promising but the size of telemetry file is not known.

As per <https://docs.aws.amazon.com/sagemaker/latest/dg/inference-cost-optimization.html>

"Use batch inference for workloads for which you need inference for a large set of data for processes that happen offline (that is, you don't need a persistent endpoint). You pay for the instance for the duration of the batch inference job". As you pay for the batch job duration, cost should not be an issue with Batch transform.

"Use asynchronous inference for asynchronous workloads that process up to 1 GB of data (such as text corpus, image, video, and audio) that are latency insensitive and cost sensitive. With asynchronous inference, you can control costs by specifying a fixed number of instances for the optimal processing rate instead of provisioning for the peak. You can also scale down to zero to save additional costs."

upvoted 2 times
- oso0348

1 month, 4 weeks ago

Selected Answer: A

Based on the requirements and constraints given in the scenario, the MOST cost-effective solution for the company to use to run the model across the telemetry for all the devices is SageMaker Batch Transform.

SageMaker Batch Transform is a cost-effective solution for performing offline inference, as it allows for large amounts of data to be processed at a lower cost compared to real-time inference. In this case, the telemetry data for each device is collected hourly and can be processed in batches using SageMaker Batch Transform. This can help to reduce the cost of inference, as the data is not being processed in real-time and can be processed offline.

upvoted 2 times
- Siyuan_Zhu

2 months ago

Selected Answer: B

B -- based on what Drcock87 said, as well as this: "Amazon SageMaker Asynchronous Inference is a new capability in SageMaker that queues incoming requests and processes them asynchronously. Compared to Batch Transform Asynchronous Inference provides immediate access to the results of the inference job rather than waiting for the job to complete"

upvoted 1 times
- drcok87

2 months ago

I still think its A because:

- "The 4-day telemetry of each device is collected in a separate file and is placed in an Amazon S3 bucket once every hour." Which means this is use-case where data is available upfront for inferencing.

- Also, unlike async the batch transform does not keep an active endpoint all the time. async is similar to real-time inference, used when you need inference right-away; the question is not asking for real-time inference.

upvoted 2 times
- drcok87

2 months, 2 weeks ago

Real-Time Inference is suitable for workloads where payload sizes are up to 6MB and need to be processed with low latency requirements in the order of milliseconds or seconds.

Serverless Inference: Serverless inference is ideal when you have intermittent or unpredictable traffic patterns.

Batch transform is ideal for offline predictions on large batches of data that is available upfront.

We are introducing Amazon SageMaker Asynchronous Inference, a new inference option in Amazon SageMaker that queues incoming requests and processes them asynchronously. This option is ideal for inferences with large payload sizes (up to 1GB) and/or long processing times (up to 15 minutes) that need to be processed as requests arrive. Asynchronous inference enables you to save on costs by autoscaling the instance count to zero when there are no requests to process, so you only pay when your endpoint is processing requests.

a
upvoted 2 times

  **drcok87** 2 months ago

Real-time inference is suitable for workloads where payload sizes are up to 6MB and need to be processed with low latency requirements in the order of milliseconds or seconds. Batch transform is ideal for offline predictions on large batches of data that is available upfront. The new asynchronous inference option is ideal for workloads where the request sizes are large (up to 1GB) and inference processing times are in the order of minutes (up to 15 minutes). Example workloads for asynchronous inference include running predictions on high resolution images generated from a mobile device at different intervals during the day and providing responses within minutes of receiving the request.
upvoted 1 times

A company wants to segment a large group of customers into subgroups based on shared characteristics. The company’s data scientist is planning to use the Amazon SageMaker built-in k-means clustering algorithm for this task. The data scientist needs to determine the optimal number of subgroups (k) to use.

Which data visualization approach will MOST accurately determine the optimal value of k?

- A. Calculate the principal component analysis (PCA) components. Run the k-means clustering algorithm for a range of k by using only the first two PCA components. For each value of k, create a scatter plot with a different color for each cluster. The optimal value of k is the value where the clusters start to look reasonably separated.
- B. Calculate the principal component analysis (PCA) components. Create a line plot of the number of components against the explained variance. The optimal value of k is the number of PCA components after which the curve starts decreasing in a linear fashion.
- C. Create a t-distributed stochastic neighbor embedding (t-SNE) plot for a range of perplexity values. The optimal value of k is the value of perplexity, where the clusters start to look reasonably separated.
- D. Run the k-means clustering algorithm for a range of k. For each value of k, calculate the sum of squared errors (SSE). Plot a line chart of the SSE for each value of k. The optimal value of k is the point after which the curve starts decreasing in a linear fashion.

Correct Answer: A

Community vote distribution

D (100%)

- Amit11011996

Highly Voted

2 months, 3 weeks ago

Selected Answer: D

The Answer is D

upvoted 5 times
- oso0348

Most Recent

1 month, 4 weeks ago

Selected Answer: D

The Sum of square shows variation within each cluster

upvoted 3 times
- Ajose0

2 months, 1 week ago

Selected Answer: D

D. Run the k-means clustering algorithm for a range of k. For each value of k, calculate the sum of squared errors (SSE). Plot a line chart of the SSE for each value of k. The optimal value of k is the point after which the curve starts decreasing in a linear fashion.

The sum of squared errors (SSE) measures the total variation within each cluster, and the optimal value of k is typically the point where the SSE begins to level off or decrease sharply. Plotting the SSE against the number of clusters (k) allows the data scientist to identify the optimal number of clusters based on where the SSE curve starts decreasing linearly.

upvoted 3 times
- drcok87

2 months, 2 weeks ago

d <https://towardsdatascience.com/explain-ml-in-a-simple-way-k-means-clustering-e925d019743b>

upvoted 2 times

A data scientist at a financial services company used Amazon SageMaker to train and deploy a model that predicts loan defaults. The model analyzes new loan applications and predicts the risk of loan default. To train the model, the data scientist manually extracted loan data from a database. The data scientist performed the model training and deployment steps in a Jupyter notebook that is hosted on SageMaker Studio notebooks. The model's prediction accuracy is decreasing over time.

Which combination of steps is the MOST operationally efficient way for the data scientist to maintain the model's accuracy? (Choose two.)

- A. Use SageMaker Pipelines to create an automated workflow that extracts fresh data, trains the model, and deploys a new version of the model.
- B. Configure SageMaker Model Monitor with an accuracy threshold to check for model drift. Initiate an Amazon CloudWatch alarm when the threshold is exceeded. Connect the workflow in SageMaker Pipelines with the CloudWatch alarm to automatically initiate retraining.
- C. Store the model predictions in Amazon S3. Create a daily SageMaker Processing job that reads the predictions from Amazon S3, checks for changes in model prediction accuracy, and sends an email notification if a significant change is detected.
- D. Rerun the steps in the Jupyter notebook that is hosted on SageMaker Studio notebooks to retrain the model and redeploy a new version of the model.
- E. Export the training and deployment code from the SageMaker Studio notebooks into a Python script. Package the script into an Amazon Elastic Container Service (Amazon ECS) task that an AWS Lambda function can initiate.

Correct Answer: BC

Community vote distribution

AB (100%)

  **MIlb** 1 month ago

Selected Answer: AB

<https://aws.amazon.com/blogs/machine-learning/automate-model-retraining-with-amazon-sagemaker-pipelines-when-drift-is-detected/>
upvoted 2 times

  **Valcilio** 1 month, 3 weeks ago

Selected Answer: AB

Retrain the model when the accuracy is decreasing is the most recommended way to take of your models.
upvoted 2 times

  **oso0348** 1 month, 4 weeks ago

Selected Answer: AB

The MOST operationally efficient way for the data scientist to maintain the model's accuracy would be to choose options A and B:

A. Use SageMaker Pipelines to create an automated workflow that extracts fresh data, trains the model, and deploys a new version of the model.

Using SageMaker Pipelines allows the data scientist to automate the entire workflow from data extraction to model deployment. This ensures that the model is trained and deployed on the latest data automatically without the need for manual intervention. The data scientist can set up the pipeline to run on a schedule or trigger it based on certain events.

B. Configure SageMaker Model Monitor with an accuracy threshold to check for model drift. Initiate an Amazon CloudWatch alarm when the threshold is exceeded. Connect the workflow in SageMaker Pipelines with the CloudWatch alarm to automatically initiate retraining.
upvoted 2 times

  **Ajose0** 2 months, 1 week ago

Selected Answer: AB

Using SageMaker Pipelines to create an automated workflow that extracts fresh data, trains the model, and deploys a new version of the model is an efficient way to automate the process of model retraining and deployment.

Configuring SageMaker Model Monitor with an accuracy threshold to check for model drift and initiating an Amazon CloudWatch alarm when the threshold is exceeded is an efficient way to monitor the accuracy of the deployed model and initiate retraining when necessary. This approach helps to maintain the accuracy of the model over time.
upvoted 2 times

  **Jerry84** 2 months, 2 weeks ago

Selected Answer: AB

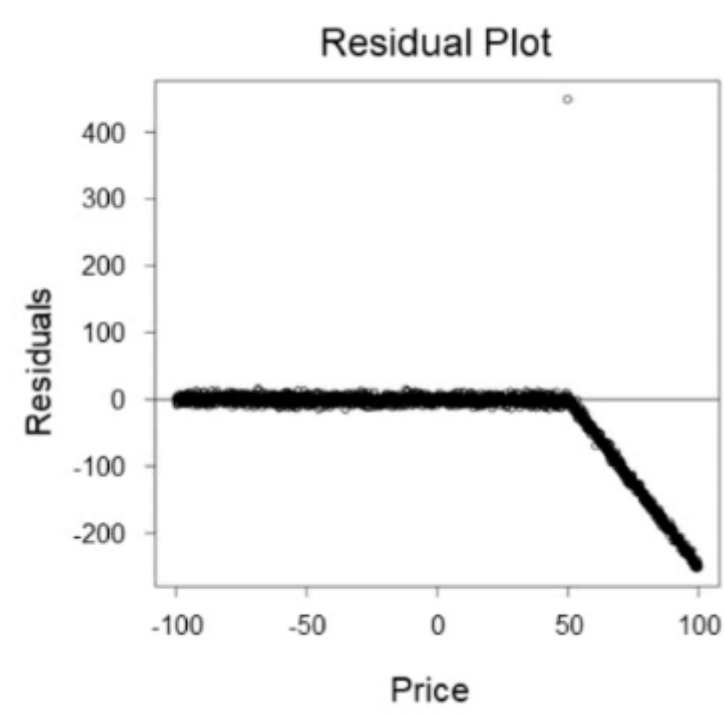
<https://aws.amazon.com/blogs/machine-learning/automate-model-retraining-with-amazon-sagemaker-pipelines-when-drift-is-detected/>
upvoted 2 times

  **drcok87** 2 months, 2 weeks ago

B first and then A

upvoted 1 times

A retail company wants to create a system that can predict sales based on the price of an item. A machine learning (ML) engineer built an initial linear model that resulted in the following residual plot:



Which actions should the ML engineer take to improve the accuracy of the predictions in the next phase of model building? (Choose three.)

- A. Downsample the data uniformly to reduce the amount of data.
- B. Create two different models for different sections of the data.
- C. Downsample the data in sections where Price < 50.
- D. Offset the input data by a constant value where Price > 50.
- E. Examine the input data, and apply non-linear data transformations where appropriate.
- F. Use a non-linear model instead of a linear model.

Correct Answer: ACE

Community vote distribution

BEF (67%)CDE (33%)

- MIlb 1 month ago

The linear model doesn't capture the data complexity

upvoted 1 times
- MIlb 1 month ago

Then, BEF

upvoted 3 times
- MIlb 3 weeks, 1 day ago

It appears on 2023-April-03

upvoted 1 times
- stjokerli 1 month, 2 weeks ago

Selected Answer: BEF

As per wolfsong said

upvoted 4 times
- Chelseajcole 2 months ago

Selected Answer: CDE

Two models , add a constant or in-put data transformation

upvoted 1 times
- Chelseajcole 2 months ago

Bde should be the answer

upvoted 1 times
- Ajose0 2 months, 1 week ago

Selected Answer: CDE

The residual plot shows that the linear model is not fitting the data well, with a clear pattern indicating that the model is underfitting. To improve the accuracy of the predictions, the ML engineer should take the following actions:

C. Downsample the data in sections where Price < 50: This could be an option since there seems to be a higher variance in the residuals in the region where Price < 50.

D. Offset the input data by a constant value where Price > 50: This could be an option since there seems to be a systematic bias in the residuals in the region where Price > 50.

E. Examine the input data, and apply non-linear data transformations where appropriate: This is necessary since the residual plot shows that the linear model is not capturing the non-linear relationships in the data.

upvoted 1 times

  **wolfsong** 2 months, 1 week ago

A good residual plot is a flat line at $y = 0$. So...

- Not sure if D is right. If you offset by a constant value, you're just moving the plot up or down. You'd have to add a term like $K \cdot \text{Price}$, where price > 50 and $K > 0$, for you to flatten that curve beyond Price > 50.

- Also unsure about C. The variance looks fairly good for Price < 50 as it's mostly around zero which is what you want. The problem is the residual value at Price > 50 which goes way off.

I'd go with B, E & F:

E: obvious

F: use non-linear model instead as it will remove the kink in the plot

B: Not an answer I like, but if you can't use a nonlinear model, you need to use a piecewise-linear model that separates the data in two.

Something like this: <https://towardsdatascience.com/piecewise-linear-regression-model-what-is-it-and-when-can-we-use-it-93286cfee452>

upvoted 9 times

  **GiyeonShin** 2 months, 1 week ago

If D is the answer of this question, Isn't B the another answer too?

Suppose that the initial linear model means $y = aX + b$, then D means $y = a(X - C) + b \rightarrow y = aX + b'$ (when Price > 50)

I think that D means we would use two different linear models for different sections (Price = 50) of the data.

upvoted 1 times

A data scientist at a food production company wants to use an Amazon SageMaker built-in model to classify different vegetables. The current dataset has many features. The company wants to save on memory costs when the data scientist trains and deploys the model. The company also wants to be able to find similar data points for each test data point.

Which algorithm will meet these requirements?

- A. K-nearest neighbors (k-NN) with dimension reduction
- B. Linear learner with early stopping
- C. K-means
- D. Principal component analysis (PCA) with the algorithm mode set to random

Correct Answer: B

Community vote distribution

A (71%)

C (29%)

- blanco750

1 month, 1 week ago

Selected Answer: A

KNN to reduce dimensionality which may help reduce memory utilisation.
upvoted 2 times
- Valcilio

1 month, 3 weeks ago

Selected Answer: A

It's A, needs to reduce the dimensionality of the dataset.
upvoted 1 times
- Chelseajcole

2 months ago

Selected Answer: A

They want less features
upvoted 2 times
- Imimi

2 months ago

I will go with A.

C is not valid, as K-means is a clustering algorithm that can group similar data points together. However, it does not perform classification, and it is not clear how it addresses the memory cost and similarity search requirements mentioned in the question.
upvoted 1 times
- Amit11011996

2 months ago

Selected Answer: C

It should be C,
because it is an unsupervised classification problem.
upvoted 1 times
- Ajose0

2 months, 1 week ago

Selected Answer: A

option A suggests using the k-nearest neighbors (k-NN) algorithm with dimension reduction. The k-NN algorithm can be used for classification tasks and dimension reduction can help reduce memory costs. Additionally, k-NN can be used for finding similar data points.

K-NN is a simple algorithm that works well with high-dimensional data and can find similar data points.
upvoted 2 times
- blanco750

1 month, 1 week ago

agree. By reducing the number of dimensions, you may achieve comparable analysis results using less memory and in a shorter amount of time.
upvoted 1 times
- drcok87

2 months, 1 week ago

<https://ealizadeh.com/blog/knn-and-kmeans/#:~:text=unsupervised%20learning%20algorithm.-,K%20in%20K%20Means%20refers%20to%20the%20number%20of%20clusters,using%20different%20values%20for%20K.>
KNN does use a lot of memory because in lazy learning it stores (memorizes) the training dataset. AWS SageMaker however has an improved version of this algorithm.
Because the question does not mention we have labels, we cannot use supervised learning

K-means : unsupervised and helps us to classify different vegetables based on their many features. This also find "similar data points for each test data point"

C

upvoted 1 times

  **Jerry84** 2 months, 2 weeks ago

Selected Answer: A

"Training with the k-NN algorithm has three steps: sampling, dimension reduction, and index building. Sampling reduces the size of the initial dataset so that it fits into memory. For dimension reduction, the algorithm decreases the feature dimension of the data to reduce the footprint of the k-NN model in memory and inference latency."

" The main objective of k-NN's training is to construct the index. The index enables efficient lookups of distances between points whose values or class labels have not yet been determined and the k nearest points to use for inference."

<https://docs.aws.amazon.com/sagemaker/latest/dg/k-nearest-neighbors.html>

upvoted 3 times

  **Aninina** 2 months, 3 weeks ago

Selected Answer: C

should be C

upvoted 3 times

A data scientist is training a large PyTorch model by using Amazon SageMaker. It takes 10 hours on average to train the model on GPU instances. The data scientist suspects that training is not converging and that resource utilization is not optimal.

What should the data scientist do to identify and address training issues with the LEAST development effort?

- A. Use CPU utilization metrics that are captured in Amazon CloudWatch. Configure a CloudWatch alarm to stop the training job early if low CPU utilization occurs.
- B. Use high-resolution custom metrics that are captured in Amazon CloudWatch. Configure an AWS Lambda function to analyze the metrics and to stop the training job early if issues are detected.
- C. Use the SageMaker Debugger vanishing_gradient and LowGPUUtilization built-in rules to detect issues and to launch the StopTrainingJob action if issues are detected.
- D. Use the SageMaker Debugger confusion and feature_importance_overweight built-in rules to detect issues and to launch the StopTrainingJob action if issues are detected.

Correct Answer: D

Community vote distribution

C (100%)

Amit11011996

Highly Voted

2 months, 3 weeks ago

It has to be C.

upvoted 5 times

oso0348

Most Recent

1 month, 4 weeks ago

Selected Answer: C

Answer is C

The best option for the data scientist to identify and address training issues with the least development effort is option C: Use the SageMaker Debugger vanishing_gradient and LowGPUUtilization built-in rules to detect issues and to launch the StopTrainingJob action if issues are detected.

SageMaker Debugger is a tool that helps to debug machine learning training processes. It provides several built-in rules that can detect and diagnose common issues that can occur during training. In this case, the data scientist suspects that the training is not converging and that resource utilization is not optimal. The vanishing_gradient and LowGPUUtilization rules can help to identify these issues.

upvoted 3 times

Ajose0

2 months, 1 week ago

Selected Answer: C

C. Use the SageMaker Debugger vanishing_gradient and LowGPUUtilization built-in rules to detect issues and to launch the StopTrainingJob action if issues are detected.

The SageMaker Debugger is a built-in tool that helps with debugging and profiling machine learning models trained in SageMaker. In this scenario, the data scientist suspects that there are issues with the training process, so using the SageMaker Debugger is the most appropriate solution. The vanishing_gradient and LowGPUUtilization built-in rules can detect common training issues such as a vanishing gradient problem or low GPU utilization, which could affect the training convergence and resource utilization. By launching the StopTrainingJob action if issues are detected, the training job can be stopped early, which can help to save resources and time. This approach requires the least development effort, as it is built-in to SageMaker and does not require the data scientist to create custom metrics or configure CloudWatch alarms.

upvoted 3 times

Jerry84

2 months, 2 weeks ago

Selected Answer: C

<https://docs.aws.amazon.com/sagemaker/latest/dg/debugger-built-in-rules.html>

upvoted 4 times

A bank wants to launch a low-rate credit promotion campaign. The bank must identify which customers to target with the promotion and wants to make sure that each customer's full credit history is considered when an approval or denial decision is made.

The bank's data science team used the XGBoost algorithm to train a classification model based on account transaction features. The data science team deployed the model by using the Amazon SageMaker model hosting service. The accuracy of the model is sufficient, but the data science team wants to be able to explain why the model denies the promotion to some customers.

What should the data science team do to meet this requirement in the MOST operationally efficient manner?

- A. Create a SageMaker notebook instance. Upload the model artifact to the notebook. Use the `plot_importance()` method in the Python XGBoost interface to create a feature importance chart for the individual predictions.
- B. Retrain the model by using SageMaker Debugger. Configure Debugger to calculate and collect Shapley values. Create a chart that shows features and SHapley. Additive explanations (SHAP) values to explain how the features affect the model outcomes.
- C. Set up and run an explainability job powered by SageMaker Clarify to analyze the individual customer data, using the training data as a baseline. Create a chart that shows features and SHapley Additive explanations (SHAP) values to explain how the features affect the model outcomes.
- D. Use SageMaker Model Monitor to create Shapley values that help explain model behavior. Store the Shapley values in Amazon S3. Create a chart that shows features and SHapley Additive explanations (SHAP) values to explain how the features affect the model outcomes.

Correct Answer: D

Community vote distribution

C (75%)

B (25%)

  **Gaby999** 5 days, 17 hours ago

Selected Answer: B
Retrain the model by using SageMaker Debugger. Configure Debugger to calculate and collect Shapley values. Create a chart that shows features and SHapley Additive explanations (SHAP) values to explain how the features affect the model outcomes.

While A, C, and D are all options for explaining the model's behavior, the most efficient way to meet the bank's requirements is to use SageMaker Debugger to calculate and collect

S
Shapley values for each prediction. This allows the data science team to easily explain why the model denied the promotion to certain customers. SageMaker Debugger also provides built-in integration with SageMaker Studio, which enables data scientists to visualize the Shapley values and other debugging information through a user-friendly interface.
upvoted 2 times

  **chotoc** 3 weeks, 1 day ago

I think it's D. Model monitor automatically integrated with Clarify
upvoted 1 times

  **oso0348** 4 weeks, 1 day ago

Selected Answer: B
B. Retrain the model by using SageMaker Debugger. Configure Debugger to calculate and collect Shapley values. Create a chart that shows features and Shapley Additive explanations (SHAP) values to explain how the features affect the model outcomes would be the most operationally efficient way to meet the requirement of explaining why the model denies the promotion to some customers.
upvoted 2 times

  **MIlb** 1 month ago

Selected Answer: C
Clarify is the best solution. The key is training data
<https://www.amazonaws.cn/en/sagemaker/clarify/>
upvoted 2 times

  **daidaidai** 1 day, 7 hours ago

Amazon SageMaker Clarify provides machine learning developers with greater visibility into their training data and models so they can identify and limit bias and explain predictions.
I think B is correct.
upvoted 1 times

  **blanco750** 1 month, 1 week ago

Selected Answer: C

Its between B and C
SageMaker Clarify is used to promote transparency and accountability in machine learning models. Thats what we are looking for why model denies promotion to some customers
upvoted 2 times

  **sevosevo** 1 month, 2 weeks ago

Selected Answer: C

"Explain individual model predictions

Customers and internal stakeholders both want transparency into how models make their predictions. SageMaker Clarify integrates with SageMaker Experiments to show you the importance of each model input for a specific prediction. Results can be made available to customer-facing employees so that they have an understanding of the model’s behavior when making decisions based on model predictions."

<https://www.amazonaws.cn/en/sagemaker/clarify/>
upvoted 2 times

A company has hired a data scientist to create a loan risk model. The dataset contains loan amounts and variables such as loan type, region, and other demographic variables. The data scientist wants to use Amazon SageMaker to test bias regarding the loan amount distribution with respect to some of these categorical variables.

Which pretraining bias metrics should the data scientist use to check the bias distribution? (Choose three.)

- A. Class imbalance
- B. Conditional demographic disparity
- C. Difference in proportions of labels
- D. Jensen-Shannon divergence
- E. Kullback-Leibler divergence
- F. Total variation distance

Correct Answer: *ACF*

Community vote distribution

DEF (60%)

ABC (40%)

- cox1960

16 hours, 1 minute ago

Selected Answer: DEF

answers "How different are the distributions for loan application outcomes for different demographic groups?"
upvoted 1 times
- Ahmedhadi_

5 days, 23 hours ago

Selected Answer: DEF

<https://docs.aws.amazon.com/sagemaker/latest/dg/clarify-measure-data-bias.html>
upvoted 2 times
- MIlb

1 month ago

Selected Answer: ABC

D. Jensen-Shannon divergence and E. Kullback-Leibler divergence are post-training bias metrics that measure the distance between two probability distributions. They are not pretraining bias metrics and cannot be used to check the bias distribution of the dataset.

F. Total variation distance is a post-training bias metric that measures the difference between two probability distributions. It is not a pretraining bias metric and cannot be used to check the bias distribution of the dataset.
- Send a message...

upvoted 2 times
- injohe

6 days, 17 hours ago

The are all pretraining metrics
<https://docs.aws.amazon.com/sagemaker/latest/dg/clarify-measure-data-bias.html>
upvoted 2 times
- jackzhao

1 month, 1 week ago

Agree with austinoy, answer should be DEF.
upvoted 3 times
- sevoesevo

1 month, 2 weeks ago

<https://docs.aws.amazon.com/sagemaker/latest/dg/clarify-measure-data-bias.html>
upvoted 1 times
- austinoy

1 month, 1 week ago

based on the link shouldn't DEF be the answers?
upvoted 1 times

A retail company wants to use Amazon Forecast to predict daily stock levels of inventory. The cost of running out of items in stock is much higher for the company than the cost of having excess inventory. The company has millions of data samples for multiple years for thousands of items. The company’s purchasing department needs to predict demand for 30-day cycles for each item to ensure that restocking occurs.

A machine learning (ML) specialist wants to use item-related features such as "category," "brand," and "safety stock count." The ML specialist also wants to use a binary time series feature that has "promotion applied?" as its name. Future promotion information is available only for the next 5 days.

The ML specialist must choose an algorithm and an evaluation metric for a solution to produce prediction results that will maximize company profit.

Which solution will meet these requirements?

- A. Train a model by using the Autoregressive Integrated Moving Average (ARIMA) algorithm. Evaluate the model by using the Weighted Quantile Loss (wQL) metric at 0.75 (P75).
- B. Train a model by using the Autoregressive Integrated Moving Average (ARIMA) algorithm. Evaluate the model by using the Weighted Absolute Percentage Error (WAPE) metric.
- C. Train a model by using the Convolutional Neural Network - Quantile Regression (CNN-QR) algorithm. Evaluate the model by using the Weighted Quantile Loss (wQL) metric at 0.75 (P75).
- D. Train a model by using the Convolutional Neural Network - Quantile Regression (CNN-QR) algorithm. Evaluate the model by using the Weighted Absolute Percentage Error (WAPE) metric.

Correct Answer: D

Community vote distribution

C (80%)

A (20%)

- mawsman

2 weeks ago

Selected Answer: A

A retail company wants to use Amazon Forecast to predict daily stock levels of inventory. The cost of running out of items in stock is much higher for the company than the cost of having excess inventory. The company has millions of data samples for multiple years for thousands of items. The company’s purchasing department needs to predict demand for 30-day cycles for each item to ensure that restocking occurs.

upvoted 1 times
- blanco750

1 month, 1 week ago

Selected Answer: C

WQL is particularly useful when there are different costs for underpredicting and overpredicting. By setting the weight (τ) of the wQL function, you can automatically incorporate differing penalties for underpredicting and overpredicting.

upvoted 4 times
- austinoy

1 month, 1 week ago

I'll go with, C?

<https://docs.aws.amazon.com/forecast/latest/dg/aws-forecast-algo-cnnqr.html>

<https://docs.aws.amazon.com/forecast/latest/dg/metrics.html#metrics-wQL>

upvoted 3 times
- sevosevo

1 month, 2 weeks ago

<https://docs.aws.amazon.com/forecast/latest/dg/metrics.html>

<https://docs.aws.amazon.com/forecast/latest/dg/aws-forecast-recipe-arima.html>

<https://docs.aws.amazon.com/forecast/latest/dg/aws-forecast-algo-cnnqr.html>

upvoted 1 times

An online retail company wants to develop a natural language processing (NLP) model to improve customer service. A machine learning (ML) specialist is setting up distributed training of a Bidirectional Encoder Representations from Transformers (BERT) model on Amazon SageMaker. SageMaker will use eight compute instances for the distributed training.

The ML specialist wants to ensure the security of the data during the distributed training. The data is stored in an Amazon S3 bucket.

Which combination of steps should the ML specialist take to protect the data during the distributed training? (Choose three.)

- A. Run distributed training jobs in a private VPC. Enable inter-container traffic encryption.
- B. Run distributed training jobs across multiple VPCs. Enable VPC peering.
- C. Create an S3 VPC endpoint. Then configure network routes, endpoint policies, and S3 bucket policies.
- D. Grant read-only access to SageMaker resources by using an IAM role.
- E. Create a NAT gateway. Assign an Elastic IP address for the NAT gateway.
- F. Configure an inbound rule to allow traffic from a security group that is associated with the training instances.

Correct Answer: *BEF*

Community vote distribution

ACD (55%)

ACF (45%)

- MIlb

Highly Voted

1 month ago

Selected Answer: ACF

I'm not agree with E because assigns read-only access to Sagemaker

upvoted 5 times
- blanco750

Most Recent

1 month, 1 week ago

A C and F look correct

upvoted 2 times
- blanco750

1 month, 1 week ago

Configure the VPC Security Group

In distributed training, you must allow communication between the different containers in the same training job. To do that, configure a rule for your security group that allows inbound connections between members of the same security group. For EFA-enabled instances, ensure that both inbound and outbound connections allow all traffic from the same security group. For information, see Security Group Rules in the Amazon Virtual Private Cloud User Guide.

upvoted 2 times
- oso0348

1 month, 1 week ago

Selected Answer: ACD

A - Running the training jobs in a private VPC will ensure that the data is transmitted over an encrypted channel. Enabling inter-container traffic encryption will encrypt data that is transmitted between containers. This will help protect the data during the distributed training.

C - Creating an S3 VPC endpoint will provide a secure and private connection between the VPC and the S3 bucket. Network routes, endpoint policies, and S3 bucket policies can be configured to further secure the data during the distributed training.

D - Granting read-only access to SageMaker resources by using an IAM role will ensure that the data is only accessed by the necessary resources during the distributed training. This will help prevent unauthorized access to the data.

upvoted 4 times
- sevosevo

1 month, 2 weeks ago

Selected Answer: ACD

A,B pretty sure; D best guess.

https://docs.aws.amazon.com/sagemaker/latest/dg/train-encrypt.html

https://docs.aws.amazon.com/sagemaker/latest/dg/train-vpc.html

upvoted 2 times
- austino

1 month, 1 week ago

My best "guess" is ACF

upvoted 5 times

An analytics company has an Amazon SageMaker hosted endpoint for an image classification model. The model is a custom-built convolutional neural network (CNN) and uses the PyTorch deep learning framework. The company wants to increase throughput and decrease latency for customers that use the model.

Which solution will meet these requirements MOST cost-effectively?

- A. Use Amazon Elastic Inference on the SageMaker hosted endpoint.
- B. Retrain the CNN with more layers and a larger dataset.
- C. Retrain the CNN with more layers and a smaller dataset.
- D. Choose a SageMaker instance type that has multiple GPUs.

Correct Answer: C

Community vote distribution

A (100%)

-   **oso0348** 1 month, 1 week ago

Selected Answer: A

Use Amazon Elastic Inference on the SageMaker hosted endpoint would be the most cost-effective solution for increasing throughput and decreasing latency. Amazon Elastic Inference is a service that allows you to attach GPU-powered inference acceleration to Amazon SageMaker hosted endpoints and EC2 instances. By attaching an Elastic Inference accelerator to the SageMaker endpoint, you can achieve better performance with lower costs than using a larger, more expensive instance type.

upvoted 2 times
-   **sevosevo** 1 month, 2 weeks ago

Selected Answer: A

"cost efficient" therefore A based on slide 20: <https://pages.awscloud.com/rs/112-TZM-766/images/AL-ML%20for%20Startups%20-%20Select%20the%20Right%20ML%20Instance.pdf>

upvoted 1 times

An ecommerce company is collecting structured data and unstructured data from its website, mobile apps, and IoT devices. The data is stored in several databases and Amazon S3 buckets. The company is implementing a scalable repository to store structured data and unstructured data. The company must implement a solution that provides a central data catalog, self-service access to the data, and granular data access policies and encryption to protect the data.

Which combination of actions will meet these requirements with the LEAST amount of setup? (Choose three.)

- A. Identify the existing data in the databases and S3 buckets. Link the data to AWS Lake Formation.
- B. Identify the existing data in the databases and S3 buckets. Link the data to AWS Glue.
- C. Run AWS Glue crawlers on the linked data sources to create a central data catalog.
- D. Apply granular access policies by using AWS Identity and Access Management (IAM). Configure server-side encryption on each data source.
- E. Apply granular access policies and encryption by using AWS Lake Formation.
- F. Apply granular access policies and encryption by using AWS Glue.

Correct Answer: ADE

Community vote distribution

ACE (80%)ACD (20%)

- MIlb** 1 month ago

Selected Answer: ACE

<https://docs.aws.amazon.com/lake-formation/latest/dg/what-is-lake-formation.html>

upvoted 3 times
- MIlb** 1 month ago

Lake Formation provides a single place to manage access controls for data in your data lake. You can define security policies that restrict access to data at the database, table, column, row, and cell levels. These policies apply to IAM users and roles, and to users and groups when federating through an external identity provider. You can use fine-grained controls to access data secured by Lake Formation within Amazon Redshift Spectrum, Athena, AWS Glue ETL, and Amazon EMR for Apache Spark. Whenever you create IAM identities, make sure to follow IAM best practices.

upvoted 1 times
- DS2021** 1 month, 1 week ago

Selected Answer: ACE

ACE looks good

upvoted 3 times
- jackzhao** 1 month, 1 week ago

I will choose ACE

<https://aws.amazon.com/blogs/big-data/build-secure-encrypted-data-lakes-with-aws-lake-formation/>

upvoted 3 times
- GiorgioGss** 1 month, 1 week ago

ACE looks legit.

upvoted 2 times
- blanco750** 1 month, 1 week ago

Selected Answer: ACE

ACE looks correct

upvoted 2 times
- austino** 1 month, 1 week ago

I'll go with, ACE?

upvoted 2 times
- sevo** 1 month, 2 weeks ago

Selected Answer: ACD

A: <https://docs.aws.amazon.com/lake-formation/latest/dg/what-is-lake-formation.html>

C: <https://docs.aws.amazon.com/lake-formation/latest/dg/upgrade-glue-lake-formation.html>

D: <https://docs.aws.amazon.com/lake-formation/latest/dg/what-is-lake-formation.html>

upvoted 2 times

Question #232

Topic 1

A machine learning (ML) specialist is developing a deep learning sentiment analysis model that is based on data from movie reviews. After the ML specialist trains the model and reviews the model results on the validation set, the ML specialist discovers that the model is overfitting.

Which solutions will MOST improve the model generalization and reduce overfitting? (Choose three.)

A. Shuffle the dataset with a different seed.

B. Decrease the learning rate.

C. Increase the number of layers in the network.

D. Add L1 regularization and L2 regularization.

E. Add dropout.

F. Decrease the number of layers in the network.

Correct Answer: CEF

Community vote distribution

DEF (100%)

oso0348

4 weeks, 1 day ago

Selected Answer: DEF

To improve the generalization of the deep learning sentiment analysis model and reduce overfitting, the following three solutions can be implemented:

Add Dropout: Dropout is a regularization technique that randomly drops out (sets to zero) a certain percentage of nodes in the neural network during each training epoch. This helps to prevent overfitting and improve generalization.

Add L1 and L2 Regularization: L1 and L2 regularization are techniques used to add a penalty to the loss function of the neural network, which helps to prevent overfitting. L1 regularization adds a penalty based on the absolute value of the weights, while L2 regularization adds a penalty based on the squared value of the weights.

Decrease the number of layers in the network: Deep neural networks with too many layers can be prone to overfitting. Reducing the number of layers in the network can help to prevent overfitting and improve generalization.

upvoted 3 times

MIlb

1 month ago

Selected Answer: DEF

We don't have to touch learning rate because the model is overfitting

upvoted 2 times

blanco750

1 month, 1 week ago

Selected Answer: DEF

DEF are correct

upvoted 2 times

sevosevo

1 month, 2 weeks ago

Selected Answer: DEF

A: possible but unlikely for movie reviews

B: wrong https://www.google.com/url?sa=t&rct=j&q=&esrc=s&source=web&cd=&cad=rja&uact=8&ved=2ahUKEwi31N_10eX9AhWYQ0EAHXDFCAwQFnoECA8QAw&url=https%3A%2F%2Fdeepchecks.com%2Fquestion%2Fdoes-learning-rate-affect-overfitting%2F&usg=AOvVaw19RT-u_XyEe8FG_10R6aFC

C: wrong because would increase complexity and potentially overfitting

D: correct

E: correct

F: correct

upvoted 4 times

https://www.examtopycs.com/exams/amazon/aws-certified-machine-learning-specialty/custom-view/

400/406

An online advertising company is developing a linear model to predict the bid price of advertisements in real time with low-latency predictions. A data scientist has trained the linear model by using many features, but the model is overfitting the training dataset. The data scientist needs to prevent overfitting and must reduce the number of features.

Which solution will meet these requirements?

- A. Retrain the model with L1 regularization applied.
- B. Retrain the model with L2 regularization applied.
- C. Retrain the model with dropout regularization applied.
- D. Retrain the model by using more data.

Correct Answer: A

Community vote distribution

A (100%)

-   **jackzhao** 1 month, 1 week ago

A is correct

upvoted 2 times
-   **GiorgioGss** 1 month, 1 week ago

Selected Answer: A

L1 shrinks the less important feature’s coefficient to zero thus, removing some feature altogether. So, this works well for feature selection in case we have a huge number of features.

upvoted 2 times
-   **blanco750** 1 month, 1 week ago

Selected Answer: A

Yes L1 for feature reduction

upvoted 2 times
-   **sevosevo** 1 month, 2 weeks ago

Selected Answer: A

https://www.google.com/url?sa=t&rct=j&q=&esrc=s&source=web&cd=&cad=rja&uact=8&ved=2ahUKEwim5YGd0-X9AhXPYcAKHSAIAkoQFnoECA0QAQ&url=https%3A%2F%2Fdocs.aws.amazon.com%2Fmachine-learning%2Flatest%2Fdvg%2Fmodel-fit-underfitting-vs-overfitting.html&usg=AOvVaw2jwLt-J0jRSWeiDyjEzI_S

upvoted 3 times

A credit card company wants to identify fraudulent transactions in real time. A data scientist builds a machine learning model for this purpose. The transactional data is captured and stored in Amazon S3. The historic data is already labeled with two classes: fraud (positive) and fair transactions (negative). The data scientist removes all the missing data and builds a classifier by using the XGBoost algorithm in Amazon SageMaker. The model produces the following results:

- True positive rate (TPR): 0.700
- False negative rate (FNR): 0.300
- True negative rate (TNR): 0.977
- False positive rate (FPR): 0.023
- Overall accuracy: 0.949

Which solution should the data scientist use to improve the performance of the model?

- A. Apply the Synthetic Minority Oversampling Technique (SMOTE) on the minority class in the training dataset. Retrain the model with the updated training data.
- B. Apply the Synthetic Minority Oversampling Technique (SMOTE) on the majority class in the training dataset. Retrain the model with the updated training data.
- C. Undersample the minority class.
- D. Oversample the majority class.

Correct Answer: C

Community vote distribution

A (100%)

-   **blanco750** 1 month, 1 week ago

Selected Answer: A

SMOTE for minority class for unbalanced data

upvoted 4 times
-   **sevosevo** 1 month, 2 weeks ago

Selected Answer: A

<https://www.google.com/url?sa=t&rct=j&q=&esrc=s&source=web&cd=&cad=rja&uact=8&ved=2ahUKEwjLkqfb1OX9AhXkQ0EAHYIVDq0QFnoECBMQAw&url=https%3A%2F%2Ftowardsdatascience.com%2F5-smote-techniques-for-oversampling-your-imbalance-data-b8155bdbe2b5&usg=AOvVaw1FdrxDEbLDjNhacXn3d-Tu>

upvoted 2 times

A company is training machine learning (ML) models on Amazon SageMaker by using 200 TB of data that is stored in Amazon S3 buckets. The training data consists of individual files that are each larger than 200 MB in size. The company needs a data access solution that offers the shortest processing time and the least amount of setup.

Which solution will meet these requirements?

- A. Use File mode in SageMaker to copy the dataset from the S3 buckets to the ML instance storage.
- B. Create an Amazon FSx for Lustre file system. Link the file system to the S3 buckets.
- C. Create an Amazon Elastic File System (Amazon EFS) file system. Mount the file system to the training instances.
- D. Use FastFile mode in SageMaker to stream the files on demand from the S3 buckets.

Correct Answer: D

Community vote distribution

B (53%)

D (47%)

  **mawsman**

Highly Voted 

 2 weeks ago

Selected Answer: D

When to use fast file mode:

For larger datasets with larger files (more than 50 MB per file), the first option is to try fast file mode, which is more straightforward to use than FSx for Lustre because it doesn't require creating a file system, or connecting to a VPC. Fast file mode is ideal for large file containers (more than 150 MB), and might also do well with files more than 50 MB.

<https://docs.aws.amazon.com/sagemaker/latest/dg/model-access-training-data.html#model-access-training-data-best-practices>
upvoted 6 times

  **MIlb**

Most Recent 

 1 month ago

Selected Answer: B

Option D, FastFile mode, streams files on demand from S3 buckets to the training instance, which can be efficient for small datasets but may not be optimal for large datasets. Moreover, this solution does not provide a file system that is optimized for high performance, and it may require additional development effort to set up

upvoted 3 times

  **MIlb** 4 weeks, 1 day ago

B because we have 200TB

<https://saturncloud.io/blog/using-aws-sagemaker-input-modes-amazon-s3-efs-or-fsx/>

upvoted 2 times

  **blanco750** 1 month, 1 week ago

Selected Answer: D

Fast File Mode combines the ease of use of the existing File Mode with the performance of Pipe Mode. This provides convenient access to data as if it was downloaded locally, while offering the performance benefit of streaming the data directly from Amazon S3.

No code change required or no lengthy setup

upvoted 3 times

  **oso0348** 1 month, 1 week ago

Selected Answer: B

The solution that meets the requirements of the company is B, which involves creating an Amazon FSx for Lustre file system and linking it to the S3 buckets. Amazon FSx for Lustre is a fully managed, high-performance file system optimized for compute-intensive workloads, such as machine learning training. It is designed to provide low latencies and high throughput for processing large data sets, and it can directly access data from S3 buckets without any data movement or copying. This solution requires minimal setup and provides the shortest processing time since the data can be accessed in parallel by multiple instances.

upvoted 4 times

  **austinoy** 1 month, 1 week ago

I will go with D

<https://sagemaker.readthedocs.io/en/stable/api/utility/inputs.html>

upvoted 3 times

  **sevosevo** 1 month, 2 weeks ago

Selected Answer: B

<https://aws.amazon.com/blogs/machine-learning/speed-up-training-on-amazon-sagemaker-using-amazon-efs-or-amazon-fsx-for-lustre-file-systems/>

upvoted 3 times

Question #236

Topic 1

An online store is predicting future book sales by using a linear regression model that is based on past sales data. The data includes duration, a numerical feature that represents the number of days that a book has been listed in the online store. A data scientist performs an exploratory data analysis and discovers that the relationship between book sales and duration is skewed and non-linear.

Which data transformation step should the data scientist take to improve the predictions of the model?

- A. One-hot encoding
- B. Cartesian product transformation
- C. Quantile binning
- D. Normalization

Correct Answer: A

Community vote distribution

C (100%)

  **jackzhao** 1 month, 1 week ago

C is correct
upvoted 3 times

  **blanco750** 1 month, 1 week ago

Selected Answer: C

C is the best answer I guess
upvoted 3 times

  **oso0348** 1 month, 1 week ago

Selected Answer: C

the correct answer is C, Quantile binning. This transformation divides the data into quantiles (equal-sized intervals) based on the values of the feature (in this case, duration) and replaces the values with the bin number. This transformation can help capture non-linear relationships between features by creating more representative categories for skewed data. The transformed data can then be used to train a non-linear regression model, such as a polynomial regression, to better predict future book sales.

upvoted 2 times

  **sevosevo** 1 month, 2 weeks ago

Selected Answer: C

<https://docs.aws.amazon.com/machine-learning/latest/dg/data-transformations-reference.html>
upvoted 4 times

A company's data engineer wants to use Amazon S3 to share datasets with data scientists. The data scientists work in three departments: Finance, Marketing, and Human Resources. Each department has its own IAM user group. Some datasets contain sensitive information and should be accessed only by the data scientists from the Finance department.

How can the data engineer set up access to meet these requirements?

- A. Create an S3 bucket for each dataset. Create an ACL for each S3 bucket. For each S3 bucket that contains a sensitive dataset, set the ACL to allow access only from the Finance department user group. Allow all three department user groups to access each S3 bucket that contains a non-sensitive dataset.
- B. Create an S3 bucket for each dataset. For each S3 bucket that contains a sensitive dataset, set the bucket policy to allow access only from the Finance department user group. Allow all three department user groups to access each S3 bucket that contains a non-sensitive dataset.
- C. Create a single S3 bucket that includes two folders to separate the sensitive datasets from the non-sensitive datasets. For the Finance department user group, attach an IAM policy that provides access to both folders. For the Marketing and Human Resources department user groups, attach an IAM policy that provides access to only the folder that contains the non-sensitive datasets.
- D. Create a single S3 bucket that includes two folders to separate the sensitive datasets from the non-sensitive datasets. Set the policy for the S3 bucket to allow only the Finance department user group to access the folder that contains the sensitive datasets. Allow all three department user groups to access the folder that contains the non-sensitive datasets.

Correct Answer: D

Community vote distribution

C (50%)

D (33%)

B (17%)

- injoho

6 days, 17 hours ago

Option C

<https://stackoverflow.com/questions/35944349/iam-aws-s3-to-restrict-to-a-specific-sub-folder>

<https://aws.amazon.com/blogs/security/how-to-restrict-amazon-s3-bucket-access-to-a-specific-iam-role/>

upvoted 2 times
- staskrocket

1 week, 1 day ago

Selected Answer: C

I will choose C

upvoted 3 times
- jackzhao

1 month, 1 week ago

I will choose C

upvoted 3 times
- blanco750

1 month, 1 week ago

Both B and D look apparently correct but they are not because in s3 bucket policy , IAM Group cant be the principal. In other words you cant give access to a User group to s3 buckets using s3 bucket policy. It can only be an IAM user or role.<https://stackoverflow.com/questions/30667678/s3-bucket-policy-how-to-allow-a-iam-group-from-another-account>

I would go for C

upvoted 4 times
- blanco750

1 month, 1 week ago

Selected Answer: D

single bucket looks a better option. Ease of management and still secure

upvoted 1 times
- blanco750

1 month, 1 week ago

Actually this is not possible. I will go for C

upvoted 2 times
- oso0348

1 month, 1 week ago

Selected Answer: D

Creating a single S3 bucket that includes two folders to separate the sensitive datasets from the non-sensitive datasets would be the best approach. The policy of the S3 bucket can be set to allow only the Finance department user group to access the folder that contains the sensitive datasets. The folder that contains non-sensitive datasets can be made available to all three department user groups. This approach will ensure that sensitive datasets are only accessible to users who need access to them.

upvoted 1 times

  **austinoy** 1 month, 1 week ago

I'll go with D
upvoted 1 times

  **austinoy** 4 weeks, 1 day ago

I stand corrected - it's C
upvoted 3 times

  **sevosevo** 1 month, 2 weeks ago

Selected Answer: B

<https://stackoverflow.com/questions/47815526/s3-bucket-policy-vs-access-control-list>
upvoted 1 times